

Airplane Flight Manual

The Airplane must be operated in compliance with the LIMITATIONS prescribed herein.

MODEL CL-600-2B16
(Serial No. 6050 and Subsequent)

21,863 kg / 48,200 lb MTOW

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CHALLENGER 650

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CHAPTER 1 - INTRODUCTION

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1. INTRODUCTION

This Airplane Flight Manual contains limitations, procedures and performance data for the operation of the CL600, Model 2B16, Challenger airplane (Serial No. 6050 and subsequent), manufactured by Bombardier Inc., Montreal, Quebec, Canada.

It is the responsibility of pilots who are qualified to operate the Challenger airplane to be entirely familiar with the information contained in this publication, to ensure that the airplane is operated at all times within the approved flight envelope.

For clarity and simplicity, the manual is written in the imperative, in order that the information and operating instructions may be presented in a positive sense and require no interpretation by the user.

Specific items requiring emphasis are expanded upon and ranked in increasing order of importance in the form of a NOTE, a CAUTION or a WARNING.

NOTE

Expands on information which is considered essential to emphasize.
Information contained in notes may also be safety related.



Provides information that may result in damage to equipment if not followed.



Emphasizes information that may result in personal injury or loss of life if not followed.

2. SCOPE OF THE CL600-2B16 CHALLENGER AIRPLANE FLIGHT MANUAL

A. Limitations

Observance of the limitations contained in the LIMITATIONS chapter of this Airplane Flight Manual is mandatory.

B. Status of Airplane Flight Manual Information

In the EMERGENCY PROCEDURES, NORMAL PROCEDURES, CONSOLIDATED CHECK LISTS, ABNORMAL PROCEDURES and PERFORMANCE chapters of this Airplane Flight Manual, checks and procedures written in the imperative or qualified by “shall” or “must” are defined as those actions which, if not observed, could result in a significant adverse effect on the safety of the airplane.

C. Emergency Procedures

Procedures given in the EMERGENCY PROCEDURES chapter of this Airplane Flight Manual are concerned with foreseeable but unusual situations in which immediate and precise crew action, as indicated within the “boxed” area at the beginning of each procedure, will substantially reduce the hazard.

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2. SCOPE OF THE CL600-2B16 CHALLENGER AIRPLANE FLIGHT MANUAL (CONT'D)

D. Normal Procedures

The NORMAL PROCEDURES chapter of this Airplane Flight Manual contains procedures and checks peculiar to the airplane and are supplementary to normal procedures common to the operation of modern transport category jet airplanes.

E. Abnormal Procedures

The ABNORMAL PROCEDURES chapter of this Airplane Flight Manual addresses foreseeable situations, usually involving a failure condition, in which the use of the normal or alternate systems can be expected to maintain an acceptable level of airplane operation.

F. Performance

The PERFORMANCE chapter of this Airplane Flight Manual contains details pertinent to the airplane configurations and conditions for operation in compliance with the applicable airworthiness performance requirements.

G. Supplements

The SUPPLEMENTS chapter of this Airplane Flight Manual consists of information which supersedes or is in addition to the basic Airplane Flight Manual.

This information may be a result of the issuance of a Supplemental Type Approval/Supplemental Type Certificate, or approved changes to AFM limitations, procedures or performance without a Supplemental Type Approval/Supplemental Type Certificate.

H. Appendices

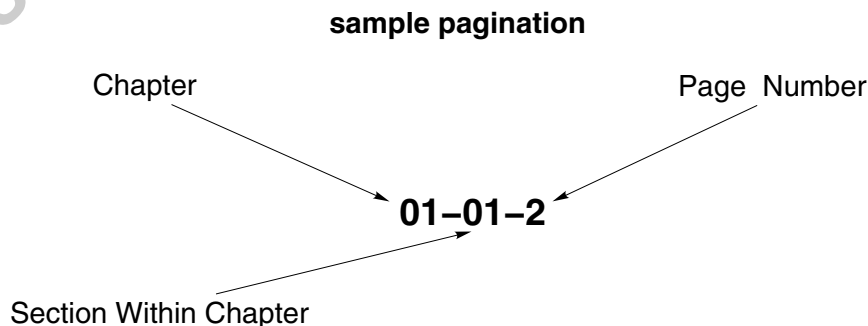
The APPENDICES chapter of this Airplane Flight Manual includes additions to the manual that are required by the installation of optional equipment or for specific operations (engine inoperative ferry, reduced thrust take-off, configuration deviation list, etc.).

3. PAGINATION

Each chapter of the Airplane Flight Manual is subdivided into sections categorized by the subject or type of material presented.

This permits issuance of small blocks of revision pages without renumbering and reprinting complete sections of the manual.

The volume is paginated with a Chapter/Section/Page numbering system as follows:



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Additional identification data carried in the margin of the page are the date of issue or revision date printed below the page number and the chapter title and subject carried in the masthead.

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3. PAGINATION (CONT'D)

Blank pages resulting from the end of a section and back-up pages for foldout illustrations or illustrations that must be arranged as facing pages are provided to avoid the impression that a page is missing. These pages are numbered and are included in the List of Effective Pages.

4. DEFINITIONS

The following definitions are for terms commonly used throughout the manual. Items not commonly used will be defined in the respective sections where they apply.

A. Airspeeds

IAS	Indicated airspeed – Airspeed indicator reading corrected for instrument error. Values in this manual assume zero instrument error.
CAS	Calibrated airspeed – Indicated airspeed corrected for static source position error.
EAS	Equivalent airspeed – Calibrated airspeed corrected for compressibility.
TAS	True airspeed – Equivalent airspeed corrected for density.
M	Indicated Mach number – Machmeter reading corrected for instrument error.
M _T	True Mach number – Machmeter reading corrected for both instrument and static source position errors.
V ₁	Take-off decision speed (formerly designated as Critical engine failure recognition speed) – The speed used as a reference at which, due to engine failure or other causes, the pilot may elect to stop or continue the take-off.
V _{1MBE}	Maximum V ₁ for brake energy – The maximum speed on the ground from which a stop can be accomplished within the energy capabilities of the brakes.
V _{1MCG}	Minimum V ₁ limited by control on the ground – The take-off decision speed following an engine failure at V _{MCG} . V ₁ may not be less than V _{1MCG} .
V ₂	Take-off safety speed – Target climb speed to be attained at or before a height of 35 feet above the runway during a continued take-off, following an engine failure.
V _A	Design maneuvering speed – The maximum speed at which application of full available aileron, rudder or elevator will not overstress the airplane.
V _{EF}	Critical engine failure speed – The speed at which, if the critical engine fails, the engine failure is recognized at V ₁ .
V _{ERC}	Enroute climb speed – The climb speed with the aircraft in the enroute (clean) configuration and the critical engine inoperative.
V _{FE}	Maximum flap extended speed – is the highest speed permissible with the wing flaps in a prescribed extended position.

4. DEFINITIONS (CONT'D)

A. Airspeeds (Cont'd)

V_{FTO}	Final take-off climb speed – The speed attained at the end of the flight path acceleration segment during a continued take-off following an engine failure, and is the climb speed scheduled for the final take-off climb.
V_{LE}	Maximum landing gear extended speed – The maximum speed at which the airplane can be safely flown with the landing gear extended.
V_{LO}	Maximum landing gear operating speed – The maximum speed at which the landing gear can be safely extended or retracted.
V_{MO}/M_{MO}	Maximum Operating Limit Speed/Mach Number – The maximum operating limit speed (airspeed or Mach number) is the speed that may not be deliberately exceeded in any regime of flight (climb, cruise, or descent) unless a higher speed is authorized for flight test or pilot training operations.
V_{MCA}	Minimum control speed, air – Minimum flight speed at which the airplane is controllable with a maximum of 5° bank, when the critical engine suddenly becomes inoperative, with the remaining engine at maximum take-off thrust.
V_{MCG}	Minimum control speed, ground – Minimum speed on the ground at which control can be maintained and the take-off continued using aerodynamic controls alone, when the critical engine suddenly becomes inoperative, with the remaining engine at maximum take-off thrust.
V_{MCL}	Minimum control speed during landing approach – Minimum flight speed at which the airplane is controllable with a maximum of 5° bank, when the critical engine suddenly becomes inoperative, with the remaining engine at maximum take-off thrust.
V_R	Rotation speed – Speed at which rotation is initiated during take-off.
V_{REF}	Approach speed – The landing reference speed at a height of 50 feet above the runway threshold in the normal landing configuration.
V_{SIG}	Stalling speed and pusher activation speed.
V_{SR}	Reference Stall Speed – The reference stall speed upon which all performance data is based.

B. Temperature

ISA	International Standard Atmospheric conditions.
OAT	Outside air temperature – The free air temperature, obtained either from in-flight temperature indications or ground meteorological sources.
SAT	Static air temperature (In-flight, SAT = OAT).
TAT	Total air temperature – Static air temperature plus adiabatic compression (ram) rise.

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4. DEFINITIONS (CONT'D)

C. Distances

Take-off Distance	The take-off distance is the longer of: - The distance from the start of the take-off roll to the point where the airplane attains a height of 35 feet above the take-off surface, with a failure of the critical engine at V_{EF}. 115% of the distance from the start of the take-off roll to the point at which the airplane attains a height of 35 feet above the take-off surface, with all engines operating.
Take-off Run	The take-off run is the longer of: - The distance from the start of the take-off roll to the mid point between lift-off and the point at which the airplane attains a height of 35 feet above the take-off surface, with a failure of the critical engine at V_{EF}. 115% of the distance from the start of the take-off roll to the mid point between lift-off and the point at which the airplane attains a height of 35 feet above the take-off surface, with all engines operating.
Accelerate – Stop Distance	The accelerate – stop distance is the longer of (a.) and (b.) as defined below: - The sum of the distances necessary to: - Accelerate the airplane from a standing start to V_{EF} with all engines operating; - Accelerate the airplane from V_{EF} to V_1 assuming the critical engine fails at V_{EF}; and - Come to a full stop from the point reached at the end of the acceleration period prescribed in paragraph (a.) 2 above, assuming that the pilot does not apply any means of retarding the airplane until that point is reached; plus - A distance equivalent to 2 seconds at constant V_1. - The sum of the distances necessary to: - Accelerate the airplane from a standing start to V_1 with all engines operating; - Come to a full stop at the end of the period prescribed in paragraph (b.) 1 above, assuming that the pilot does not apply any means of retarding the airplane until that point is reached; plus - A distance equivalent to 2 seconds at constant V_1.

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4. DEFINITIONS (CONT'D)

C. Distances (Cont'd)

Clearway	An area beyond the runway, not less than 500 feet wide, centrally located about the extended centerline of the runway, and under the control of the airport authorities. The clearway is expressed in terms of a clearway plane, extending from the end of the runway with an upward slope not exceeding 1.25% above which no object nor any terrain protrudes. However, threshold lights may protrude above the plane if their height above the end of the runway is 26 inches or less and if they are located to each side of the runway.
Stopway	An area beyond the take-off runway at least as wide as the runway and centered upon the extended centerline of the runway, able to support the airplane during an aborted take-off without causing structural damage to the airplane, and designated by the airport authorities for use in decelerating the airplane during an aborted take-off.
Actual Landing Distance	Actual landing distance is the distance from a 50 feet height at V_{REF} with the flaps in the normal landing configuration, to a full stop on a smooth, dry, level, hard-surfaced runway and ISA temperature.
Landing Field Length	The performance charts in this chapter include factors for operational rules which require the use of 60% of the available runway in determining landing field length requirements. The required landing field length on a dry runway is the actual landing distance divided by 0.6. The landing field length required under wet runway conditions is equal to 115% of the dry runway landing field length.

D. Take-Off Path

Take-Off Path	The take-off path begins from a standing start and ends at 1,500 feet above the take-off surface or at the point where transition from take-off to enroute configuration is completed, whichever is higher.
Take-Off Flight Path	The take-off flight path begins at the end of the take-off distance and at a height of 35 feet above the take-off surface, and ends at 1,500 feet above the take-off surface, or at the point where transition from take-off to enroute configuration is completed, whichever is higher. This is also known as the Gross Take-off Flight Path.
Net Take-Off Flight Path	The net take-off flight path is the gross take-off flight path diminished by 0.8% climb gradient capability (or equivalent reduction in acceleration along that part of the take-off flight path at which the airplane is accelerated in level flight). The net take-off flight path must clear all obstacles in the take-off area by at least 35 feet vertically.

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4. DEFINITIONS (CONT'D)

E. Climb Gradient

Climb Gradient	The ratio of the change in height, during a portion of a climb, to the horizontal distance traversed in the same time interval.
Gross Gradient	Climb gradient expressed as a percentage ratio, obtained using the following formula: $(\text{Change in height} \div \text{Horizontal distance travelled}) \times 100$ The gradients shown on the charts are based on true (not pressure) rates of climb.
Net Gradient	The gross gradient reduced by the required margins.
First Segment	The first segment starts from the 35-foot height and extends to the point where the landing gear is fully retracted, at a constant V_2 speed and flaps in the take-off position.
Second Segment	The second segment starts at the point where the landing gear is fully retracted up to at least 400 feet above the runway, flown at V_2 speed and flaps in the take-off position.
Acceleration Segment	The acceleration segment is the part of the take-off flight path that begins at the end of the second segment and extends horizontally over the distance required to retract flaps to zero, and accelerate to the final take-off climb speed.
Final Segment	The final segment starts from the end of the acceleration segment and extends to the end of the take-off flight path, flown at the final segment climb speed, flaps up.

F. Miscellaneous

APR	Automatic performance reserve – In the event of an engine failure during take-off, the APR system, if operating, will increase the thrust rating of the remaining engine. The APR rating must be set manually in the event of a single-engine go-around.
Centre of Gravity (c.g.)	The point at which the airplane would balance if suspended. The c.g. distance from the reference datum is found by dividing the total moment by the total weight of the airplane.
GW	Gross weight – The maximum weight to which the airplane is certified.
MLW	Maximum landing weight - The maximum airplane weight approved for the landing touchdown at a maximum descent velocity of 10 feet per second.
MTOW	Maximum take-off weight – The maximum airplane weight approved for the start of take-off or for landing at a reduced descent velocity not to exceed 6 feet per second.
MZFW	Maximum zero fuel weight.

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5. ABBREVIATIONS

The following abbreviations may be used by flight compartment displays, integrated radio tuning displays and flight management system or be found throughout the manual. Some abbreviations may also appear in lower case letters. Abbreviations having very limited usage are explained in the systems chapters where they are used.

A

A/C	Air-Conditioning	ALTS	Selected Altitude Arm/Abort (PFD/FD)
A/G	Air/Ground	AM	Amplitude Modulation
A/ICE	Anti-ice	AMB	Ambient
AP, A/P	Autopilot	AMP	Ampere(s)
A/S	Airspeed	ANNUN	Annunciator
A/SKID	Anti-skid	ANT	Antenna
ABS	Absolute	AOA	Angle of Attack
AC	Alternating Current	AP/SP	Autopilot/Stick Pusher
ACCEL	Acceleration, accelerate(d), accelerometers	APC	Auxiliary Power Control
ACM	Air Cycle Machine	APP	Approach
ACMP	Alternating Current Motor Pump/Electric Hydraulic Pump	APPROX	Approximately
		APR	Automatic Performance Reserve
ACT	Active	APU	Auxiliary Power Unit
ACU	Air-Conditioning Unit	AR	Aviation Register, Authorization Required
ADC	Air Data Computer	ARINC	Aeronautical Radio Incorporated
ADDR	Address	ASYM	Asymmetrical
ADF	Automatic Direction Finder	ATA	Air Transport Association (USA)
ADG	Air Driven Generator	ATC	Air Traffic Control
ADI	Attitude Director Indicator	ATIS	Automatic Terminal Information Service
ADS	Air Data System	ATN	Aeronautical Telecommunication Network
AFCS	Automatic Flight Control System	ATT	Attitude
AFD	Adaptive Flight Display	ATS	Autothrottle System
AFT	Afterward	AUTO	Automatic
AGL	Above Ground Level	AUTO BAL	Automatic Balance
AHC	Attitude Heading Computer	AUTO XFER	Automatic Transfer
AIL	Aileron	AUX	Auxiliary
ALIGN	Aligning, alignment	AV	Avionics
ALPHA	Alpha	AVAIL	Available
ALT	Altitude, Altimeter	AZ	Azimuth
ALT	Altitude Hold (PFD/FD)		
ALT CAP	Altitude Capture (PFD/FD)		
ALT HOLD	Altitude Hold		
ALTN	Alternate		

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5. ABBREVIATIONS (CONT'D)

B

B/AIR	Bleed Air	BK	Brake
B/C	Back Course	BLD	Bleed
B/CRS	Back Course	BOARD	Boarding
B/LEAK	Bleed Leak	BOOM	Headset microphone
BARO	Barometric	BRG	Bearing
BAT	Battery	BRKR(s)	Breaker(s)
BATT	Battery	BRT	Bright
BFO	Beat Frequency Oscillator	BTL	Bottle
BITE	Built-In Test Equipment	BYPS	Bypass

C

C	Center, Caution, Cabin	CK	Check
CAAC	Civil Aviation Authority of China	CKLST	Checklist
CAAI	Civil Aviation Administration (Israel)	CKPT	Cockpit
CAL	Calibrate	CLB	Climb
CAP	Capture	CLK	Clock
CAPT	Captain	CLSD	Closed
CAS	Calibrated Airspeed	Cm	Centimeter(s)
CAS	Crew Alerting System	CMD	Command
CAT	Category	CMPS	Compass
CAT II	Category II	CMPTR	Computer
CB, C/B	Circuit Breaker	CO ₂	Carbon Dioxide
CBP	Circuit Breaker Panel	COL	Collision
CC	Cubic Centimeter	COM	Communication
CCP	Cursor Control Panel	COMM	Communication
CCT	Circuit	COMP	Compressor, Comparator
CCW	Counter-clockwise	COMPT	Compartment
CDL	Configuration Deviation List	COND	Condition
CDP	Compressor Discharge Pressure	CONFIG	Configuration
CDU	Control Display Unit	CONN	Connection
CFM	Cubic Feet per Minute	CONT	Control, Continuous, Contactor, Controller
CG	Centre of Gravity	CONT'D	Continued
CH	Chapter, Channel	COOL	Cooling
CHAN	Channel	CORR	Correction
CHGR	Charger	CPAM	Cabin Pressure Acquisition Module
CHR	Chronograph	CPLT	Copilot
CHRT	Chart	CRS	Course
		CRT	Cathode Ray Tube

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5. ABBREVIATIONS (CONT'D)

CRZ	Cruise	CVR	Cockpit Voice Recorder
CSD	Constant Speed Drive	CW	Clockwise
CTR	Center	CYL	Cylinder

D

DA	Drift Angle	DG	Directional Gyro
DBU	Data Base Unit	DH	Decision Height
DC	Direct Current	DIFF	Differential
DCP	Display Control Panel	DIM	Dimming
DCU	Data Concentrator Unit	DIR	Direct
DECEL	Decelerate(d)	DIS	Distance (to way point), Disconnect
DECR	Decrease	DISC	Disconnect
DEFL	Defuel	DISCH	Discharge
DEG	Degree	DISP	Dispatch, Display
DEPR	Depressurize	DIST	Distance
DEPT	Departure	DME	Distance Measuring Equipment
DEST	Destination	DN	Down
DET	Detector	DOT	Department of Transport (Canada)
DEV	Deviation	DR	Door
DFDAU	Digital Flight Data Acquisition Unit		
DFDR	Digital Flight Data Recorder		

E

EAS	Equivalent Airspeed	ELEV	Elevator, Elevation
EASA	European Aviation Safety Agency	ELT	Emergency Locator Transmitter
ECS	Environmental Control System	EMER(G)	Emergency
ECU	Electronic Control Unit	ENG	Engine
EDP	Engine Driven Pump/Engine Primary Hydraulic Pump	EPC	External Power Contactor
EFIS	Electronic Flight Instrument System	EQUIP	Equipment
EGT	APU Exhaust Gas Temperature (°C)	ERP	Eye Reference Position Datum
EICAS	Engine Indication and Crew Alerting System	ESS	Essential
EL	Elevation	ET	Elapsed Time
ELEC	Electrical	ETA	Estimated Time of Arrival
		EVAC	Evacuation
		EXT	External
		EXH	Exhaust
		EXTIN	Extinguish(ed)

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5. ABBREVIATIONS (CONT'D)

F

F/CTL	Flight Controls	FIREX	Fire Extinguisher
FA	Flight Attendant	FLC	Flight Level Change
FAA	Federal Aviation Administration (USA)	FLD	Field
FAF	Final Approach Fix	FLT	Flight
FAIL	Failure	FLT DIR	Flight Director
FANS	Future Air Navigation System	FLUOR	Fluorescent
FCC	Flight Control Computer	FM	Fan Marker, Frequency Modulation
FCU	Fuel Control Unit	FMS	Flight Management System
FD, F/D	Flight Director	FPM	Feet Per Minute
FDAU	Flight Data Acquisition Unit	FREQ	Frequency
FDR	Flight Data Recorder (Digital)	FSU	File Server Unit
FECU	Flaps Electronic Control Unit	FT	Feet, Foot
FEED	Feeder	FW	Fire Wall
FF, F/F	Fuel Flow	FWD	Forward
FIRE BTL	Fire Bottle		

G

G(+/-)	Receiver Gain	GND	Ground
G/S	Glideslope	GNSS	Global Navigation Satellite System
GA	Go-Around		
GAL	Gallon(s)	GPM	Gallons Per Minute
GALY	Galley	GR	Gear
GCS	Ground Clutter Suppression	GRAV	Gravity
GCU	Generator Control Unit	GRD	Ground
GE	General Electric	GS	Ground Speed
GEN	Generator	GUIDE	Guidance
GMT	Greenwich Mean Time	GW	Gross Weight

H

HDG	Heading	HKCAD	Hong Kong Civil Aviation Department
HDG HOLD	Heading Hold	HLDR	Holder
HDG SEL	Heading Select	HOR, HORIZ	Horizontal
HEAT	Heater	HP	High Pressure
HF	High Frequency (3 – 30 mHz)	HPa	Hectopascal(s)
Hg	Mercury	HSI	Horizontal Situation Indicator
HI	High	HSTA	Horizontal Stabilizer Trim Actuator

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5. ABBREVIATIONS (CONT'D)

HSTCU	Horizontal Stabilizer Trim Control Unit	HUD	Heads-up Display
HTR	Heater	HYD	Hydraulic
		Hz	Hertz
I			
IAPS	Integrated Avionics Processing System	INCR	Increase
INBD	Inboard	IND	Indication, Indicator
I/C	Intercom, Inspection Check	INFLT	In Flight
IAS	Indicated Air Speed	INHIB	Inhibit
ICAO	International Civil Aviation Organization	INOP	Inoperative
ID	Identification	INTPH	Interphone
IDENT	Identification	INSP	Inspection
IDG	Integrated Drive Generator	INST, INSTR	Instrument(s)
IFR	Instrument Flight Rules	INT	Internal, Integral, Intersection
IFIS	Integrated Flight Information System	INTEG	Integral
IGN	Ignition	IRS	Inertial Reference System
ILS	Instrument Landing System	IRU	Inertial Reference Unit
IM	ILS Inner Marker	ISA	International Standard Atmosphere
IMC	Instrument Meteorological Conditions	ISI	Integrated Standby Instrument
IMP	Imperial	ISO	International Standards Organization
IN, in	Inch(es)	ISOL	Isolation, Isolated
IN Hg	Inches of Mercury	ITT	Engine, Inter Turbine Temperature (°C)
INBD	Inboard		

J

JAA	Joint Aviation Authorities
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K

K, KT, KTS	Knots	KIAS	Knots Indicated Airspeed
kg	Kilogram(s)	kPa	Kilopascal(s)
kHz	Kilohertz	kW	Kilowatt(s)

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5. ABBREVIATIONS (CONT'D)

L

L	Left, Landing	LNAV	Lateral Navigation
L/T	Landing/Taxi	LO	Low
LB, lb	Pound(s)	LOC	ILS Localizer
LCN	Load Classification Number	LOM, MM	Compass Locator at Outer Marker
LCV	Load Control Valve	LOP	Low Oil Pressure
LDG	Landing	LP	Low Pressure
LDG GR	Landing Gear	LPM	Liters Per Minute
LDU	Lamp Driver Unit	LPV	Localizer Precision with Vertical Guidance
LE	Leading Edge	LR	Left Rear
LG	Landing Gear	LRC	Long Range Cruise
LGC	Landing Gear Controller	LSB	Lower Side Band
LGW	Landing Gross Weight	LT(s)	Light(s)
LH	Left Hand	LW	Left Wing
LIM	Limit	LWD	Left Wing Down
LK	Leak	LWR	Lower
LN	Left Nose		

M

M	Mach Number	MFD	Multifunction Display
m	Meter	MGP	MLS Glideslope
MAA	Maximum Authorized IFR Altitude	MHz	Megahertz
MAC	Mean Aerodynamic Chord	MI	Miles
MAG	Magnetic	MI	Mach Number Indicated
MAINT	Maintenance	MIC	Microphone
MALF	Malfunction	MIL	.001 of an inch
MAN	Manual	MIN	Minimum
MAP	Ground Map (WXR)	MISC	Miscellaneous
MAX	Maximum	MISCOMP	Miscompare
MAZ	MLS Azimuth	MKR	Marker
MB	Millibars	MLG	Main Landing Gear
MCA	Minimum Crossing Altitude	MLS	Microwave Landing System
MCT	Maximum Continuous Thrust	MLW	Maximum Landing Weight
MDA	Minimum Descent Altitude	MM	ILS Middle Marker
MEA	Minimum Enroute IFR Altitude	MMEL	Master Minimum Equipment List
MECH	Mechanic	M _{MO}	Maximum Operating Speed in Mach Number
MED	Medium	MOCA	Minimum Obstruction Clearance Altitude
MEL	Minimum Equipment List		

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5. ABBREVIATIONS (CONT'D)

MOD	Module	MTG	Miles to Go
MON	Monitor	MTO	Maximum Take-off
MPH	Miles Per Hour	MTOW	Maximum Take-off Weight
MRA	Minimum Reception Altitude	MTW	Maximum Taxi Weight
MSG	Message	MZFW	Maximum Zero Fuel Weight
MSL	Mean Sea Level		

Misc

%	Percent	°C	Degrees Celsius
&	and	°F	Degrees Fahrenheit

N

N/A	Not applicable	NL	Nose Left
N ₁	Low Pressure Rotor	NLG	Nose Landing Gear
N ₂	High Pressure Rotor	NM	Nautical Mile(s)
NAM	Nautical Air Miles	No.	Number
NAV	Navigation	NOPT	No Procedure Turn Required
ND	Nose Down	NORM	Normal
NDB (ADF)	Nondirectional Beacon (Automatic Direction Finder)	NOSE	Nose Wheel
NEG	Negative	NOTAM	Notice to Airmen
NEUT	Neutral	NR	Nose Right
		NU	Nose Up

O

OAT	Outside Air Temperature	OVBD	Overboard
OUTBD	Outboard	OVHT, OH	Overheat
OBS	Observer	OVLD	Overload
OEI	One Engine Inoperative	OVSP	Overspeed
OEW	Operating Empty Weight	OVSPD	Overspeed
OH, OVHD	Overhead	OVTEMP	Overtemperature
OK	Okay	OVERTEMP	Overtemperature
OM	ILS Outer Marker	OXY, O ₂	Oxygen

P

P#6	Panel 6	PBE	Protective Breathing Equipment (Smoke Hood)
P/S	Pitot/Static	PCU	Power Control Unit
PA	Passenger Address	PF	Pilot Flying
PASS, PAX	Passenger(s)		

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5. ABBREVIATIONS (CONT'D)

PFD	Primary Flight Display	PROT	Protection
PIREP	Pilot Weather Report	PROX	Proximity
PLA	Power Lever Angle	PSEU	Proximity Sensor Electronics Unit
PLT(s)	Pilot(s)	PSI	Pounds Per Square Inch
PNF	Pilot Not Flying	PSID	Pounds Per Square Inch Differential
PNL(s)	Panel(s)	PSIG	Pounds Per Square Inch Gauge
PO	Outside Air Pressure	PSS	Proximity Sensor System
POS	Position	PT2	Engine Inlet Pressure
PPH	Pounds Per Hour	PTCT	Protect
PPM	Parts Per Million	PTT	Push To Talk
PRESS	Pressure, Pressurization	PWR	Power
PRI	Primary		
PRIM	Primary		
PROC	Procedure		

Q

QAR	Quick Access Recorder	QNH	Altimeter Setting
QFE	Local Station Pressure	QTY	Quantity
QNE	ISA, Barometric Pressure		

R

R	Right	REF(s)	Reference(s)
RA	Radio Altitude, Resolution Advisory	REFL	Refuel
RAAS	Runway Awareness and Advisory System	REV	Reverse
RAI	Registro Aeronautico Italiano (Italy)	RH	Right Hand
RAIM	Receiver Autonomous Integrity Monitoring	RMI	Radio Magnetic Indicator
RAT	Ram Air Turbine	RNP	Required Navigation Performance
RCCB	Remote Controlled Circuit Breaker	ROT	Rotation
RCDR	Recorder	RPM	Revolutions Per Minute
RCVR	Receiver	RT, R/T	Receiver-Transmitter
RDC	Remote Data Concentrator	RTE	Route
RDR	Radar	RTE DATA	Route Data
REC	Receiver, Recorder	RTO	Rejected Take-off
RCL	Recall	RUD	Rudder
RECOG	Recognition	RVR	Runway Visual Range
		RVSR	Reverser
		RW	Right Wing
		RWD	Right Wing Down
		RWY	Runway

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5. ABBREVIATIONS (CONT'D)

S

S	Status	SPD	Speed
SAAU	State Aviation Administration of Ukraine	SPKR	Speaker
SAT	Static Air Temperature	SPLR(S)	Spoiler(s)
SBAS	Satellite-Based Augmentation System	SQL	Squelch
SCAV	Scavenge	SSB	Single Side Band
SEC	Second, Secondary	STA	Station
SEL	Select, Selector	STAB	Stabilizer
SEL CAL	Selective Call	STAR	Standard Terminal Arrival Route
SENS	Sensitivity, Sensor	STAT	Status
SERV, SVCE	Service	STBY	Standby
SID	Standard Instrument Departure	STEER	Steering
SKP	Skip	SUPPL	Supply
SMKG	Smoking	SVS	Synthetic Vision System
SOV	Shutoff Valve	SW(s)	Switch(es)
SP	Stick Pusher	SYN	Synchronize
		SYNC	Synchronous
		SYS, SYST	System

T

T/C, TOC	Top Of Climb	TE	Trailing Edge
T/D, TOD	Top Of Descent	TEMP	Temperature
T/R	Thrust Reverser	TGT	Target
TACAN	UHF Tactical Air Navigation Aid	TO, T/O	Take-off
TAS	True Airspeed	TOL	Tolerance
TAT	Total Air Temperature	TRB, TURB	Turbulence
TAWS	Terrain Awareness Warning System	TRK	Track
TCAS	Traffic Alert and Collision Avoidance System	TRM	Trim
		TRU	Transformer Rectifier Unit
		TT2	Engine inlet temperature

U

UNSCHD	Unscheduled	USG	United States Gallons
USB	Upper Side Band	UTIL	Utility

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5. ABBREVIATIONS (CONT'D)

V

V	Volt	VERT	Vertical
V _{LOF}	Lift-Off Speed	VFR	Visual Flight Rules
V _{MC}	Minimum Control Speed with the Critical Engine Inoperative	VG	Vertical Gyro
V _{MO} /M _{MO}	Maximum Operating Limit Speed/Mach number	VGP	Vertical Glide Path
V _R	Rotation Speed	VHF	Very High Frequency (30 – 300 mHz)
V _T	Target Airspeed	VIB	Vibration
V ₁	Take-Off Decision Speed (formerly Denoted as Critical Engine Failure Speed)	VMC	Visual Meteorological Conditions
V ₂	Take-Off Safety Speed	VNAV	Vertical Navigation
V _{2min}	Minimum Take-Off Safety Speed	VOL	Volume
V/S	Vertical Speed	VOLT	Voltage
V-MDA	Vertical Minimum Descent Altitude	VOR	VHF Omnidirectional Range Station
		VORTAC	VOR and TACAN co-located
		VSI	Vertical Speed Indicator

W

WARN	Warning	WIND	Window
W/C	Wind Component	WOW	Weight-On-Wheels
W/S	Wind Shear	WPT(s)	Waypoint(s)
W/W	Wheel Well	WSHLD	Windshield
WGT	Weight	WX	Weather
WHLS	Wheels	WXR	Weather Radar

X

X	Cross Transfer	XPNDR	Transponder
XFER, XFR	Transfer	XTK	Cross Track
XFLOW	Cross Flow	XWC	Cross Wind Component
XMIT	Transmit		

Y

YD, Y/D	Yaw Damper
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5. ABBREVIATIONS (CONT'D)

Z

ZFW Zero Fuel Weight

6. REGISTRY CODES

The following registry codes are utilized throughout the Airplane Flight Manual to identify data that is only applicable to a particular certification authority:

AR	Aviation Registry (Russian)
CAAI	Civil Aviation Authority of Israel
EASA	European Aviation Safety Agency
FAA	Federal Aviation Administration (US)
SAAU	State Aviation Administration of Ukraine

7. SERVICE BULLETINS

Referenced throughout the manual are airplane serial numbers and Service Bulletins shown in the following list:

Service Bulletin	Title
SB 650-23-007	Datalink System, Introduction of Datalink 2000+ Program
SB 650-23-008	Future Air Navigation System, Upload of Encryption Application Keys
SB 650-34-005	Radio Altimeter System, Installation of a Second Radio Altimeter
SB 650-34-007	Flight Management System, Installation of the FAA Vspeed Database
SB 650-34-008	Terrain Awareness and Warning System, Introduction of Steep Approach
SB 650-34-009	Flight Management System, Introduction of RNP AR 0.3 Capability
SB 650-34-010	Terrain Awareness and Warning System, Introduction of Runway Awareness and Advisory System
SB 650-34-012	Flight Management System, Add Maximum Take-Off (MTO) and Reduced Take-Off Thrust Mode Capability
SB 650-34-015	Flight Management System, Correct the Erroneous Display of the Single Engine Enroute Climb Speed (V_{ERC})
SB 650-73-001	Fuel System, Russian Fuel Sampling Program

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1. GENERAL

The limitations included in this chapter contain items peculiar to the CL600, model 2B16 Challenger airplane (Serial No. 6050 and subsequent).

Observance of these limitations is mandatory.

2. KINDS OF AIRPLANE OPERATION

The airplane is certified for operations in the following conditions, when the equipment and instruments required by the airworthiness and operating regulations are approved, installed and in an operable condition:

- Day and night
- VFR and IFR
- Flight in icing conditions

The airplane is certified for ditching when the safety equipment, specified by the applicable regulations, is installed.

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CHALLENGER 650	LIMITATIONS Structural Weight	02-02-1
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1. STRUCTURAL WEIGHT LIMITATION

Maximum taxi and ramp weight:	21,909 kg (48,300 lb)
Maximum take-off weight:	21,863 kg (48,200 lb)
Maximum landing weight:	17,237 kg (38,000 lb)
Maximum zero fuel weight:	14,515 kg (32,000 lb)
Minimum flight weight:	11,794 kg (26,000 lb)
Minimum operating empty weight:	10,047 kg (22,150 lb)

NOTE

1. The maximum take-off weight (MTOW) and/or maximum landing weight (MLW) may be further limited due to performance considerations.
2. The maximum landing weight (MLW) is further limited when landing at airport elevations above 10,000 feet (Refer to Chapter 7; SUPPLEMENT 3 – OPERATION AT AIRPORT ELEVATIONS UP TO 14,000 FEET) .

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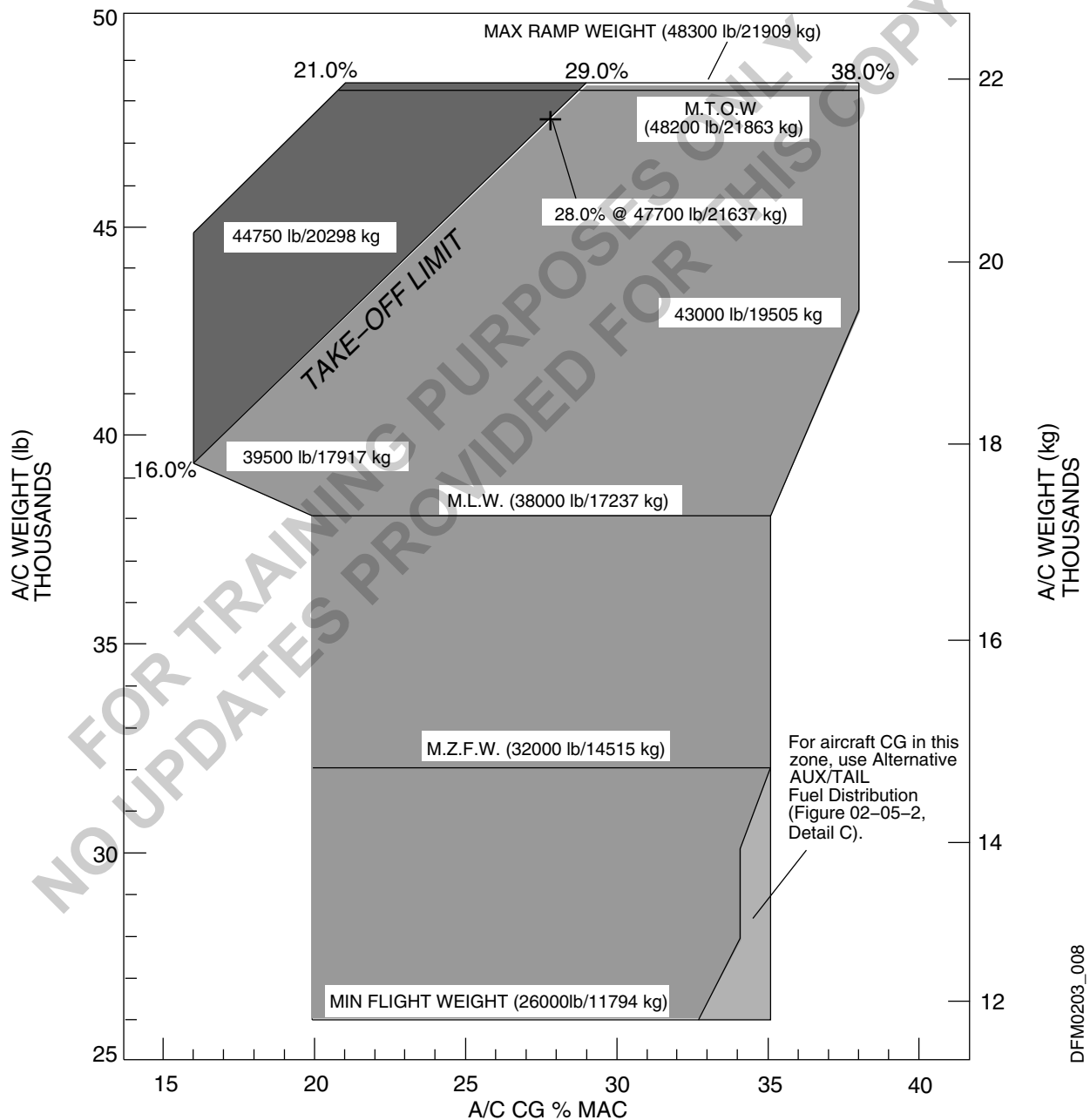
CHALLENGER 650	LIMITATIONS Centre of Gravity	02-03-1
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1. CENTRE OF GRAVITY LIMITS (MTOW 48,200 LB)

The maximum permissible centre of gravity (CG) limits are as shown in Figure 02-03-1.

The airplane including interior payload, passengers and fuel (refer to Figure 02-05-2) must be loaded such that the airplane weight and centre of gravity are maintained within the specified limits (including variations due to fuel consumption, passenger movement, retraction of landing gear, etc.).

The stabilizer trim setting must be in accordance with Chapter 4; NORMAL PROCEDURES – FLIGHT CONTROLS – Before Take-Off.



Centre of Gravity Limits (MTOW 48,200 lb)
Figure 02-03-1

CHALLENGER 650	LIMITATIONS Centre of Gravity	02-03-2
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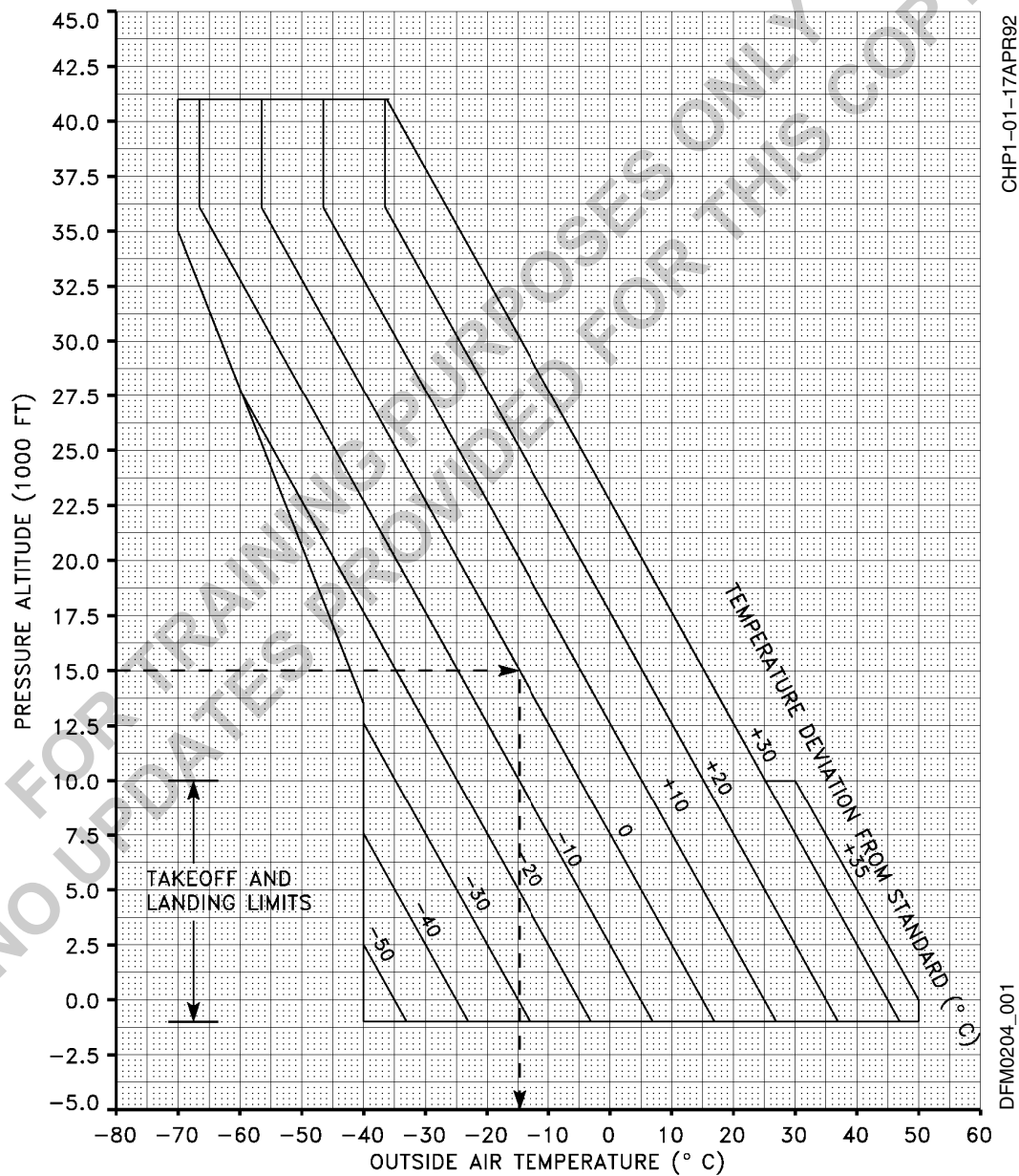
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1. ALTITUDE AND TEMPERATURE OPERATING LIMIT

The altitude and temperature operating limit is as shown in Figure 02-04-1.

- Maximum airport pressure altitude for take-off and landing is 10,000 feet.
- Maximum operating altitude is 41,000 feet.
- Maximum ambient air temperature approved for take-off and landing is ISA + 35°C.
- Minimum ambient temperature approved for take-off is -40°C (-40°F).
- At altitudes above 25,000 feet, a safety harness must be worn by at least one pilot.



Altitude and Temperature Operating Limits
Figure 02-04-1

CHALLENGER 650	LIMITATIONS Operating Limitations	02-04-2
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2. TAKE-OFF

A. Rotation Rate and Pitch Attitude

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

B. Pitch Trim

WARNING

Failure to set the pitch trim appropriate to the computed centre of gravity may result in excessive rotation rate at take-off.

Pitch trim must be set according to the airplane's computed centre of gravity.

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3. COLD WEATHER OPERATIONS

WARNING

Even small amounts of frost, ice, snow or slush on the wing leading edges and forward upper wing surface may adversely change the stall speeds, stall characteristics and the protection provided by the stall protection system, which may result in loss of control on take-off.

Take-off is prohibited with frost, ice, snow or slush adhering to any critical surface (wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets) except that take-off is permitted with frost adhering to:

- the upper surface of the fuselage; and/or
- the underside of the wing, that is caused by cold soaked fuel,

in accordance with the instructions provided in the Flight Crew Operating Manual, Volume 1, Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PRE-FLIGHT PREPARATION – External Safety Inspection.

Although the upper fuselage is not defined as a critical surface, it must nonetheless be deiced to remove contamination, other than allowable frost, anytime the wing and tail surfaces require deicing.

NOTE

Comprehensive procedures for operating in cold weather are provided in the Flight Crew Operating Manual, Volume 1, Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION.

In addition to a visual check, a tactile check of the wing leading edge, wing forward upper surface and wing rear upper surface is required during the External Walkaround inspection to determine that the wing is free from frost, ice, snow or slush when:

- the outside air temperature (OAT) is 5°C (41°F) or less, or
- the wing fuel temperature is 0°C (32°F) or less; or
- the atmospheric conditions have been conducive to frost formation.

NOTE

Ice and frost may continue to adhere to wing surfaces for some time even at outside air temperatures above 5°C (41°F).

CHALLENGER 650	LIMITATIONS Operating Limitations	02-04-4
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4. OPERATION IN ICING CONDITIONS

During cold weather operations, the flight crew must ensure that the airplane fuselage, wings and tail surfaces are free from ice, snow or frost.

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WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margin provided by the stall protection computer.

A. Cowl Anti-ice System

(1) Ground Operations

The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).

The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.

Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The engine cowl anti-ice system must be on:

At or above 22,000 feet:

- when ice is indicated by the ice detection system, or
- when in icing conditions, if an ice detector has failed.

Below 22,000 feet:

- when in icing conditions, or
- when ice is indicated by the ice detection system.

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4. OPERATION IN ICING CONDITIONS (CONT'D)

B. Wing Anti-ice System

(1) Ground Operations

NOTE

Icing conditions exist on the ground when the OAT is 5°C (41°F) or below and:

1. visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
2. the runway is wet or contaminated, or
3. in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.

NOTE

1. If wing anti-ice is selected ON during taxi, L (R) WING A/ICE caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and WING A/ICE ON or WING/COWL A/ICE ON advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
2. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.

The wing anti-ice system must be selected and confirmed ON for take-off, when the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

NOTE

If wing anti-ice is selected ON for take-off, the cowl anti-ice must also be selected ON.

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.

Refer to the Flight Crew Operating Manual, Volume 1, Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PHASE OF FLIGHT PROCEDURES.

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4. OPERATION IN ICING CONDITIONS (CONT'D)

B. Wing Anti-ice System (Cont'd)

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WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected ON, if required, just prior to thrust increase for take-off.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The wing anti-ice system must be on:

At or above 22,000 feet:

- when ice is indicated by the ice detection system, or
- when in icing conditions, if an ice detector has failed.

Below 22,000 feet:

- when in icing conditions, or
- when ice is indicated by the ice detection system.

C. Super-Cooled Large Droplet Icing

Continued operation in areas where super-cooled large droplet (SLD) icing conditions exist is prohibited.

SLD icing conditions are indicated by ice accretion on the flight compartment side windows.

- The wing anti-icing system must be ON in SLD icing conditions.
- The cowl anti-icing system must be ON in SLD icing conditions.
- Leave icing conditions when side window icing occurs.

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5. ENHANCED TAKE-OFF PROCEDURES AND WINTER OPERATIONS TRAINING

Take-offs conducted where the OAT is 5°C (41°F) or below, are subject to the following training limitations:

- the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off, if the OAT is 5°C (41°F) or below, or
- the Aircraft Commander has successfully completed specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

Additionally, no take-off shall be conducted by the First Officer where the OAT is 5°C (41°F) or below, unless that pilot has successfully completed the specific training within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

6. RUNWAY SLOPES

The maximum runway slopes approved for take-off and landing are:

+2% (uphill)

–2% (downhill)

7. TAILWIND CONDITIONS

The maximum tailwind component approved for take-off and landing is 10 knots.

8. APPROACHES

Operations and limitations with a glidepath angle that exceeds 3.5 degrees are described in the following supplements:

- SUPPLEMENT 11 – APPROACHES BELOW 4.5 DEGREES
- SUPPLEMENT 12 – STEEP APPROACHES UP TO 5.5 DEGREES

9. MINIMUM FLIGHT CREW

The minimum flight crew is one pilot and one copilot.

10. MAXIMUM OCCUPANTS

The total number of occupants, including no more than nineteen passengers, must not exceed the lesser of the following:

- Twenty-two or,
- The number for which seating accommodation approved for take-off and landing is provided.

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11. MAXIMUM CROSSWIND COMPONENT

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The maximum crosswind component for take-off and landing is 24 knots (12 meters/sec).

SAAU Certified Airplanes

The maximum crosswind component for take-off and landing on a dry runway is 24 knots (12 meters/sec).

The maximum crosswind component for take-off and landing on a wet runway with water depth no more than 3.00 mm (0.125 inch) is defined in the following table for different values of the coefficient of friction:

REPORTED COEFFICIENT OF FRICTION	MAXIMUM CROSSWIND COMPONENT
0.3 (poor braking)	10 knots (5 meters/sec)
0.4 (average braking)	15 knots (8 meters/sec)
0.5 (good braking, equivalent to dry)	24 knots (12 meters/sec)

The maximum crosswind component for take-off and landing on a contaminated runway is 10 knots (5 meters/sec).

Operation on runways with a coefficient of friction less than 0.3 is prohibited.

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CHALLENGER 650	LIMITATIONS Power Plant	02-05-1
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1. ENGINES

Type: General Electric CF34-3B, quantity two.

2. ENGINE INDICATIONS

The engine limit display markings on EICAS must be used to determine compliance with the maximum/minimum Limits and precautionary ranges. If EICAS markings show more conservative Limits than those specified below, the limit markings on the EICAS must be used.

INDICATION	RED (MAX/MIN LIMITS)	AMBER (CAUTION RANGE)	GREEN (NORMAL RANGE)
N₁ % RPM	98.6	—	0 to 98.5
ITT °C:			
— APR not operating	900 (for first 2 min.)	—	0 to 900
	884 (for next 3 min.)	—	0 to 884
— APR operating	928 (for first 2 min.)	—	0 to 928
	900 (for next 3 min.)	—	0 to 900
— Maximum Continuous Thrust (MCT)	900	—	0 to 900
N₂ % RPM:			
— wing anti-ice on	99.3	0 to 77.9	78 to 99.2
— wing anti-ice off	99.3	—	0 to 99.2
OIL TEMP °C	163	155 to 162	—40 to 154
OIL PRESS psi	0 to 25	116 to 156	26 to 115

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3. ENGINE OPERATING LIMITS

A. Engine Operating Limits Table

CONDITION	CORE RPM N ₂ %	FAN RPM N ₁ %	ITT °C
Start	20		903 (50 seconds)
Acceleration			930 (5 seconds)
Max Continuous	99.2	98.6	899
Normal Take-Off	98.3	96.2	884 (5 minutes)* 900 (2 minutes out of 5 total transient)*
Maximum Take-Off (APR operating)	99.4	98.6	899 (5 minutes)* 928 (2 minutes out of 5 total transient)*
* Transient limits			

NOTE

1. The take-off, go-around and maximum continuous thrust N₁ values for the CF34-3B engine are presented on the appropriate engine thrust setting charts contained in Chapter 6; PERFORMANCE – THRUST SETTINGS of this manual.
2. If above 40,000 feet, one air-conditioning unit or cowl anti-ice must be selected on for each engine.
3. Engine to engine N₂ split at ground idle power should not be greater than 2% N₂.

B. Airplane Cold Soak

Before the first flight of a day, when the airplane is cold-soaked at an ambient temperature of –30°C (–22°F) or below for more than 8 hours, the engines must be motored for 60 seconds, and fan rotation must be verified, before the engine start is initiated.

Thrust reversers must be actuated until the deploy and stow cycles are less than 5 seconds.

C. Oil Temperature

Minimum for starting	–40°C
Maximum Continuous	+155°C
Maximum Permissible	+163°C (15 minutes maximum)

D. Oil Pressure

Steady state idle	25 psi minimum
Take-off power	45 psi minimum
Maximum continuous	115 psi maximum
Maximum transient	
After cold start	156 psi (130 psi at idle, 10 minutes maximum)

NOTE: Engine must remain at idle until oil pressure returns to normal range.

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4. CONTINUOUS ENGINE IGNITION

Continuous engine ignition must be used during the following:

- Take-offs and landings on contaminated runways;
- Take-offs with high crosswind components;
- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence;
- Flight in the vicinity of thunderstorms.

5. AUTOTHROTTLE (ATS)

- ATS operation is restricted to two-engine operation only.
- ATS operation is prohibited during category II approach operations.
- Crew must confirm engine power is set to appropriate take-off N_1 limit prior to reaching 80 KIAS during take-off.
- At first indication of stall (stall buffet, stick shaker, or stick pusher), the crew must disengage the ATS and set thrust levers as required.
- ATS operation with APR selected on FMS performance THRUST LIMIT page is prohibited.
- If a **DISENG'D** or **FAIL** message occurs on the ATS MSD, the crew must position the thrust levers as required. Press either thrust lever ATS DISC switch to cancel ATS message.
- Selection of AFCS FLC mode following go-around mode is prohibited, unless the altitude preselector is set higher than current altitude when ATS is engaged.
- The AFM engine limits must not be exceeded when manually entering FMS thrust target (TGT) limit. The ATS will set engine thrust to this target without regard to engine operating limits.
- ATS operation during take-off with thrust target (TGT) selected on FMS performance THRUST LIMIT page is prohibited.
- ATS operation with maximum continuous thrust (MCT) selected as the FMS performance thrust limit is prohibited.
- If both mode status displays (MSDs) fail, the crew must disengage the ATS and consider the system inoperative.
- N_1 thrust limit data is required and must be available from the FMS for ATS operation.
- The radio altimeter must be valid for the ATS retard mode operation. If radio altitude is unavailable or invalid, the ATS must be disengaged prior to reaching 100 feet AGL.
- Use of the ATS during touch and go landings is prohibited. ATS must be disengaged if system engages during touch and go landings.
- Use of the ATS during go-around is prohibited.
- If flaps are selected to less than 45° for landing, ATS must be disengaged prior to reaching 100 feet AGL.

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6. AUTOMATIC PERFORMANCE RESERVE (APR)

If take-off performance is predicated upon the use of APR, the APR system must be verified operative prior to take-off.

The APR system must be selected off, if an APR INOP caution message is displayed on EICAS.

APR OFF Performance must be used if an APR INOP caution message is displayed.

7. STARTER CRANKING LIMITS (GROUND AND AIR)

The starter must not be used if indicated N₂ rpm exceeds 55%.

A. Engine Start

Normal engine start – Three consecutive engine start cycles with 5 minutes cooling between additional cycles.

B. Associated Conditions

At initiation of thrust lever movement from SHUT OFF to IDLE:

(1) ITT must be 120°C or less for all ground starts.

(2) ITT must be 90°C or less for all air starts.

C. Dry Motoring Cycle

Dry motoring is performed with ignition OFF and thrust levers at SHUT OFF.

Dry motoring may be used for engine ground starts and engine airstarts.

START	MAXIMUM TIME ON	FOLLOWED BY
1	90 seconds	5 minutes off
2 and subsequent	30 seconds	5 minutes off

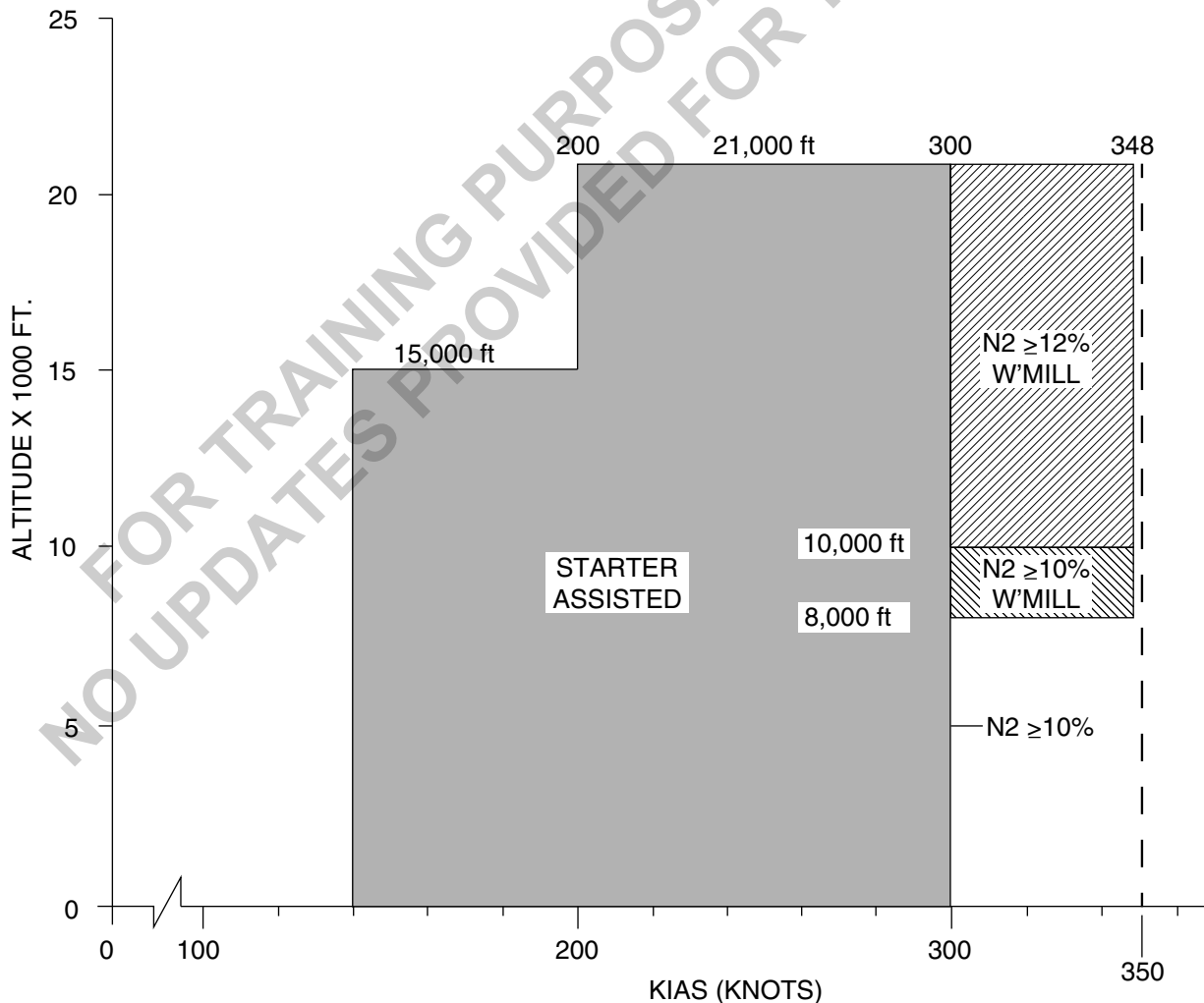
8. ENGINE RELIGHT

Engine starting in-flight is only permitted within the envelope defined in Figure 02-05-1:

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8. ENGINE RELIGHT (CONT'D)

RELIGHT TYPE	ENVELOPE (FIGURE 02-05-1)
Windmilling	Altitude from 21,000 to 10,000 feet: Airspeed from 300 to 348 KIAS, and from 12 to 55% N₂. Altitude from 10,000 to 8,000 feet: Airspeed from 300 to 348 KIAS, and from 10 to 55% N₂. Altitude from sea level to 8,000 feet: Airspeed 300 KIAS, and from 10 to 55% N₂. NOTE N₂ should be stable or increasing.
Starter-assisted Cross Bleed	Altitude from 21,000 feet to 15,000 feet: Airspeed from 200 KIAS up to 300 KIAS, and from 0 to 55% N₂. Altitude from 15,000 feet to sea level: Airspeed from 140 KIAS up to 300 KIAS, and from 0 to 55% N₂.



Engine Air Start Envelope (In Flight)
Figure 02-05-1

CHALLENGER 650	LIMITATIONS Power Plant	02-05-6
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9. FUEL

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 182 kg (400 lb) for take-off and taxi.

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 363 kg (800 lb) in flight.

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 182 kg (400 lb) for landing.

Fuel remaining in a tank when the appropriate fuel quantity indicator reads zero is not usable.

Based upon a fuel density of 6.75 lb/US gallon, the maximum usable fuel load achieved by pressure refueling for each tank is given below:

Left main tank	2,205 kg (4,860 lb)
Right main tank	2,205 kg (4,860 lb)
Auxiliary tank	3,251 kg (7,168 lb)
Tail tank	1,411 kg (3,112 lb)
 Total	 9,072 kg (20,000 lb)

To determine approximate maximum usable fuel load achieved by gravity refueling, reduce weight by 7%.

Normal AUX/TAIL Fuel Distribution for Take-Off:

- Auxiliary tank quantity must be full, or at least 2.65 times the tail tank quantity, for take-off (as shown in Figure 02-05-2, Detail A, Normal Fuel Distribution for Take-Off).

OR

Provisional AUX/TAIL Fuel Distribution for Forward CG Limit Take-Off:

- Where the normal fuel distribution would cause the airplane centre of gravity to fall ahead of the forward limit for take-off, the provisional fuel distribution may be applied to bring the airplane centre of gravity within the permissible forward CG limit (as shown in Figure 02-05-2, Detail B, Provisional Fuel Distribution for Take-Off). This fuel distribution is permitted only if the resulting CG remains ahead of 34% MAC.

OR

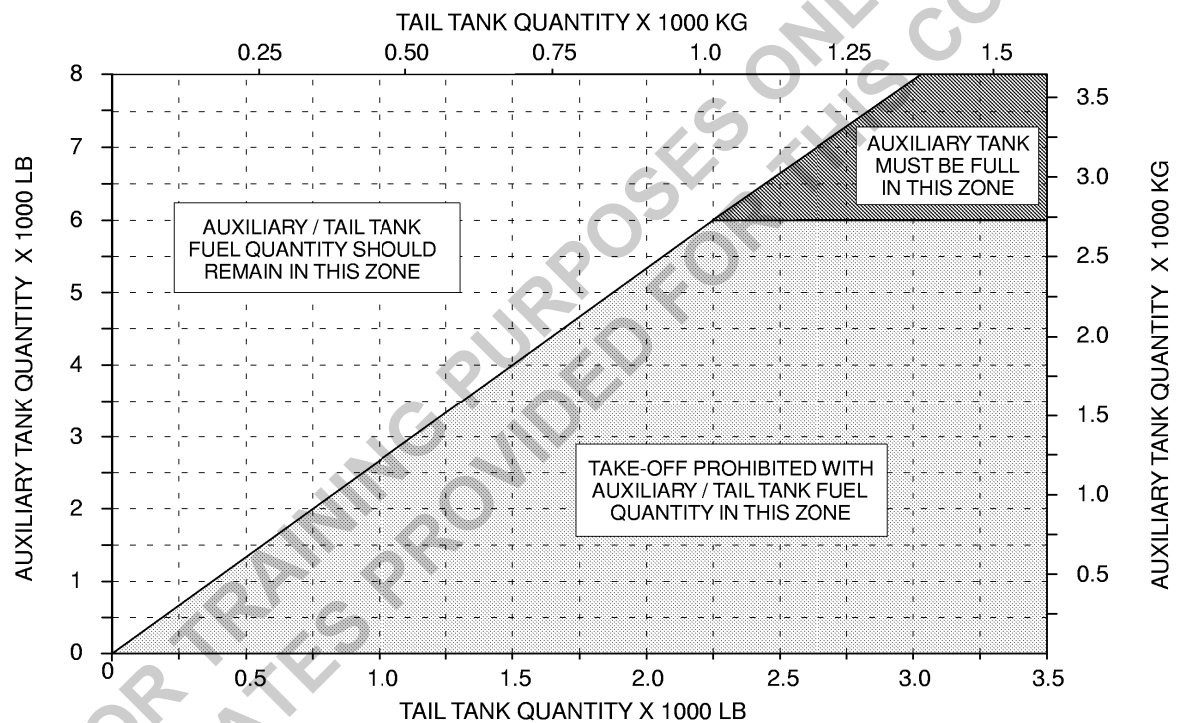
Alternative AUX/TAIL Fuel Distribution for Aft CG Limit at ZFW:

- If the aircraft CG at ZFW is within the envelope aft lower zone (CG between 32,000 lb/35%, 30,000 lb/34%, 28,000 lb/34%, 26,000 lb/32.5% and 26,000 lb/35%) an alternative manual fuel distribution may be applied. Auxiliary tank quantity must be full, or at least 2.95 times the tail tank quantity (as shown in Figure 02-05-2, Detail C, Alternative Fuel Distribution for Take-Off).

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9. FUEL (CONT'D)



Detail A – Normal Fuel Distribution for Take-Off

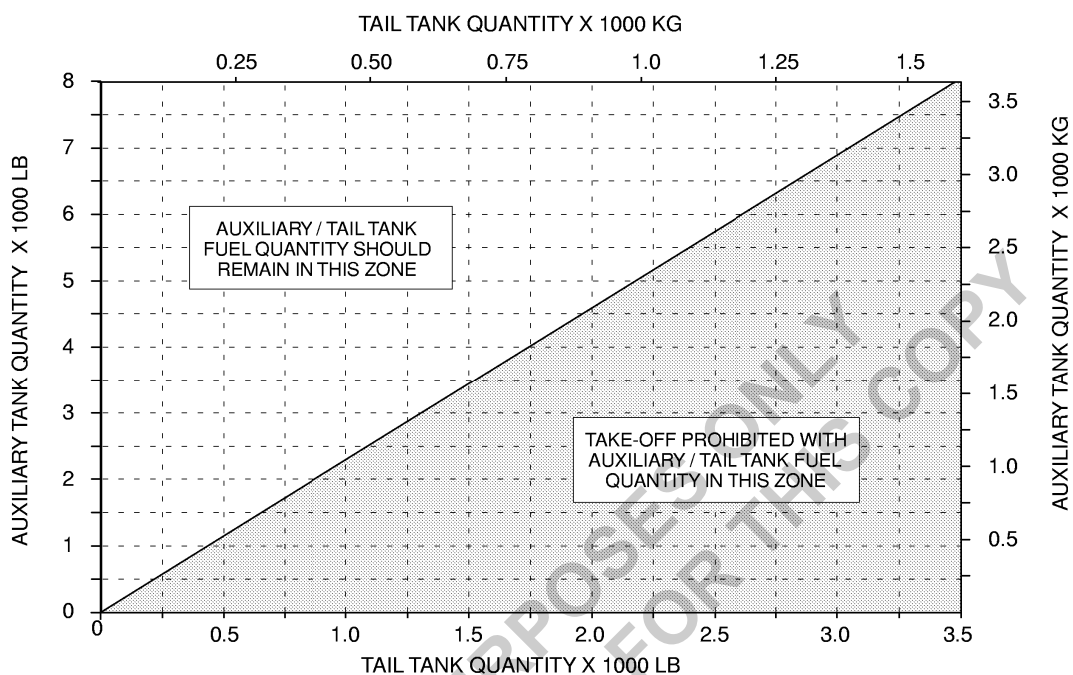
NOTE: Based on a fuel density of 6.75 lb/US gallon the auxiliary tank must be full for tail tank quantities greater than 1227 kg (2705lb)

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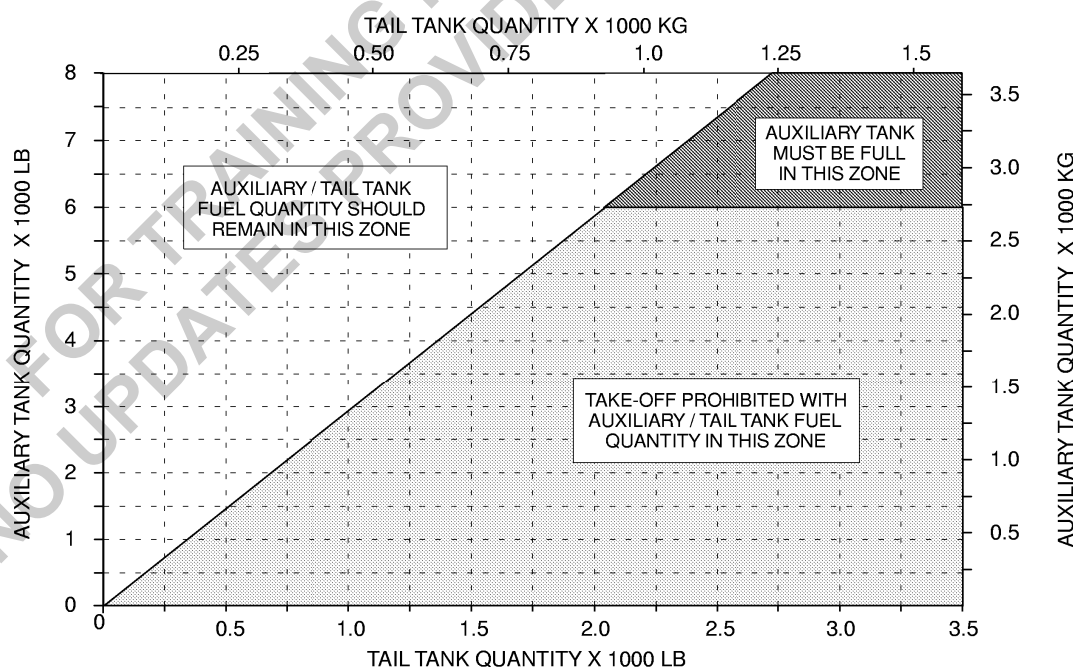
Auxiliary Tank Quantity/Tail Tank Quantity – Relation Verses Centre of Gravity (Page 1 of 2)
Figure 02-05-2

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9. FUEL (CONT'D)



Detail B – Provisional Fuel Distribution for Take-Off



Detail C – Alternative Fuel Distribution for Take-Off

NOTE: Based on a fuel density of 6.75 lb/US gallon the auxiliary tank must be full for tail tank quantities greater than 1102 kg (2430 lb)

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Auxiliary Tank Quantity/Tail Tank Quantity – Relation Verses Centre of Gravity (Page 2 of 2)
Figure 02-05-2

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9. FUEL (CONT'D)

Take-off with up to 230 kg (500 lb) of fuel in the auxiliary tank is permitted, provided that there is at least 690 kg (1,500 lb) of fuel in each wing tank and no fuel in the tail tank.

Take-off with more than 230 kg (500 lb) of fuel in the auxiliary tank is permitted, provided that the minimum quantity of fuel in each wing tank is as per Figure 02-05-3.

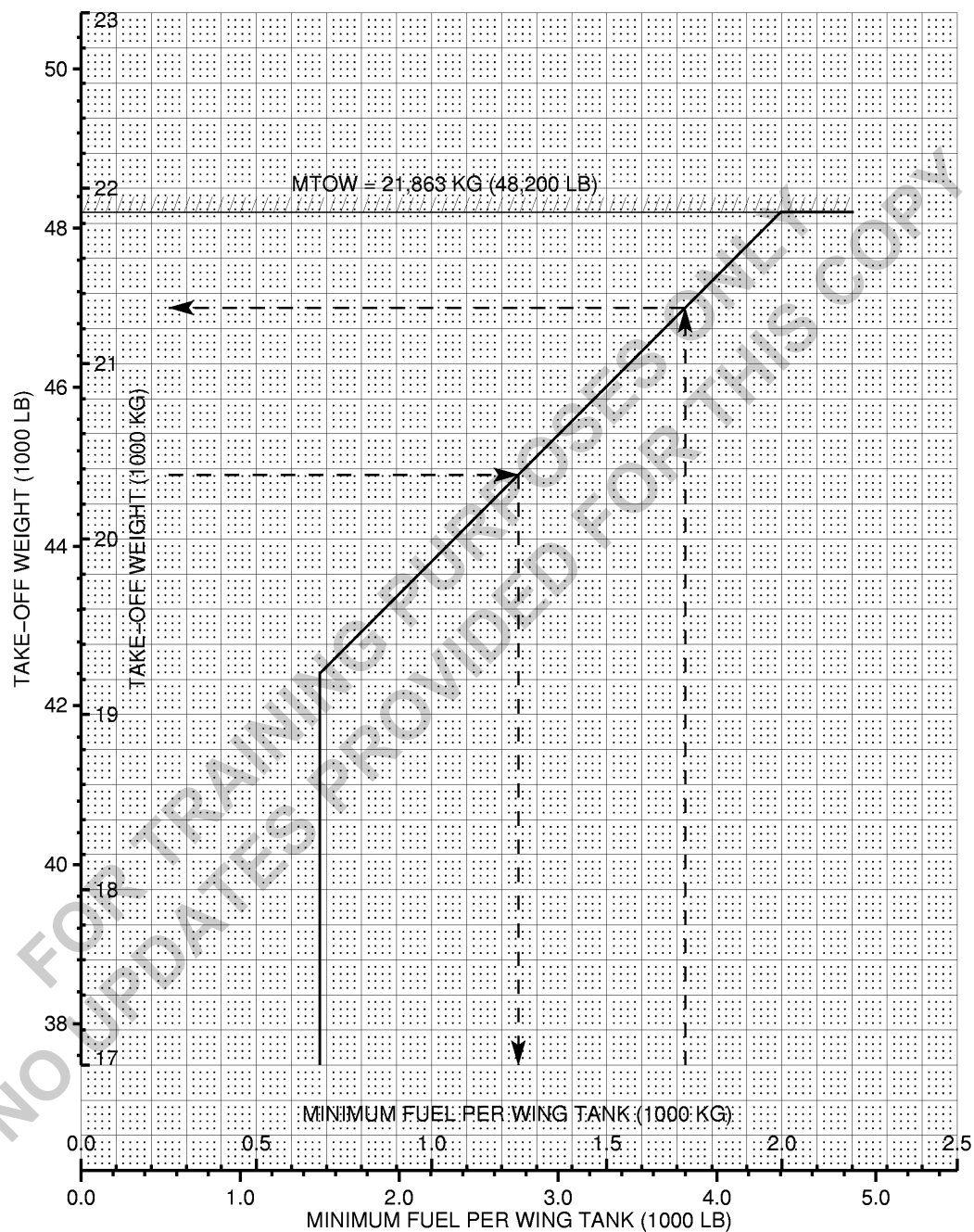
- For example, if there is at least 1,724 kg (3,800 lb) of fuel in each wing tank, then according to Figure 02-05-3, the take-off weight cannot exceed 21,319 kg (47,000 lb).
- Conversely, if the take-off weight is for example 20,366 kg (44,900 lb), then according to Figure 02-05-3, there must be at least 1,247 kg (2,750 lb) of fuel in each wing tank.

The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10° nose up.

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9. FUEL (CONT'D)



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Minimum Wing Tank Fuel Quantity for Take-Off
Figure 02-05-3

CHALLENGER 650	LIMITATIONS Power Plant	02-05-11
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9. FUEL (CONT'D)

A. Fuel Temperature

Take-off with engine fuel temperature indications below 5°C (41°F) is prohibited.

Take-off with bulk fuel temperature indications below the limits stated is prohibited.

During flight, bulk fuel temperature must remain above the applicable bulk fuel freezing point.

FUEL	BULK FUEL TAKE-OFF LIMIT	BULK FUEL FREEZING POINT
ASTM D 1655 JET A	-30°C	-40°C
ASTM D 1655 JET A1	-37°C	-47°C
ASTM D 6615 JET B	-40°C	-50°C
MIL-DTL-5624 JP 4	-48°C	-58°C
MIL-DTL-5624 JP 5	-36°C	-46°C
MIL-DTL-83133 JP 8	-37°C	-47°C
GB 6537-2006 No. 3 JET	-37°C	-47°C
Russian TS-1	-43°C	[†]
Russian RT	-40°C	[†]
Ukrainian TS-1	-42°C	-52°C
Ukrainian RT	-42°C	-52°C
[†] Russian TS-1 (TC-1 in Russian script) and RT fuels with a freeze point of not higher than -50°C are approved for use when the ground level OAT is not below -30°C during the 24 hours before departure. TS-1 fuel with a freeze point of not higher than -60°C and RT fuel with a freeze point of not higher than -55°C, for use in low temperature regions, are available at the operator's request.		

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		Rev 10, Dec 04/17

9. FUEL (CONT'D)

B. Fuel Grades

Fuels conforming to any of the following specifications are approved for use.

Mixing of fuels is permitted.

CANADIAN	AMERICAN	BRITISH	CHINESE	RUSSIAN/ UKRAINIAN
KEROSENE TYPE				
CAN 2 - 3.23	ASTM D1655 JET A	—	—	—
CAN 2 - 3.23	ASTM D1655 JET A1	DEF STAN 91-91	GB 6537-2006 No. 3 JET	TS-1 [††] or RT
—	MIL-DTL-83133 JP 8	DEF STAN 91-87	—	—
—	MIL-DTL-5624 JP 5	DEF STAN 91-86	—	—
WIDE CUT TYPE				
CAN 2 - 3.22	ASTM D6615 JET B	D. ENG. RD. 2486	—	—
CAN 2 - 3.22	MIL-DTL-5624 JP 4	D. ENG. RD. 2454	—	—
[††] When using TS-1 fuel (TC-1 in Russian script), engine fuel system components must be inspected in compliance with SB 650-73-001.				

NOTE

ASTM D1655, DEF STAN 91-91 and CAN 2 - 3.23 jet fuels whose origin is ASTM D7566 (Aviation Turbine Fuel Containing Synthesized Hydrocarbons) and are re-identified as Jet A or Jet A1 fuels are considered equivalent to fossil based fuels.

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9. FUEL (CONT'D)**C. Fuel Additives**

The following additives, used individually or in combination, are approved:

(1) ANTI-ICING

Anti-icing additives to the latest revision of specifications MIL-I-27686E or MIL-DTL-85470B or any direct equivalent at a concentration of 0.10 to 0.15% by volume.



Do not add unblended PRIST additive directly into the fuel tank, as this may damage fuel tank components.

Alcohol-based anti-icing additives of Fuel System Icing Inhibitors (FSII) type Fluid I and Fluid I-M should not be used continuously. Russian and Ukrainian fuel containing (or pre-mixed with) alcohol-based additives of anti-icing Fuel System Icing Inhibitor (FSII) type Fluid I and Fluid I-M should not be used continuously. Contact engine manufacturer for details of impact on a reduced component inspection program if continuous use has been determined.

(2) BIOCIDES

SOHIO Biobor JF biocide additive at a concentration not in excess of 270 parts per million (20 parts per million elemental boron) for the initial dose to prevent the growth of micro-organisms. A maintenance dose of 135 parts per million should be used thereafter.

Kathon FP 1.5 biocide additive at a concentration not in excess of 100 parts per million for the initial dose to prevent the growth of micro-organisms. A maintenance dose of 50 parts per million should be used thereafter.

(3) ANTI-STATIC

Stadis 450 anti-static additive at a concentration of 3 g/m³ for the initial dose. Maximum dosage should not exceed 5 g/m³.

For Russian fuels, Sigbol static dissipater additive may be used in concentrations of less than 0.0005% by weight.

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9. FUEL (CONT'D)

C. Fuel Additives (Cont'd)

(4) CORROSION INHIBITOR

Corrosion inhibitors listed below are approved, by the concentrations indicated, for hydro-treated fuels only. It is recommended that corrosion inhibitors, conforming to MIL-I-25017, be blended with the fuel to provide lubricity. The corrosion inhibitor must be added after water removal and downstream of any clay filters (these processes remove the inhibitor).

- Appollo PRI-19, at a maximum concentration of 23 g/m³;
- Octel DCI-4A, at a maximum concentration of 23 g/m³;
- Octel DCI-6A, at a maximum concentration of 9 g/m³;
- Hitec E-515, at a maximum concentration of 21 g/m³;
- Hitec E-580, at a maximum concentration of 23 g/m³;
- Nalco 5403, at a maximum concentration of 23 g/m³;
- Nalco 5405, at a maximum concentration of 11 g/m³;
- Tolad 245, at a maximum concentration of 34 g/m³;

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9. FUEL (CONT'D)**D. Fuel Jettison**

Fuel jettison must only be carried out with flaps set to 0.

Jettisoning of fuel in known lightning conditions is prohibited.

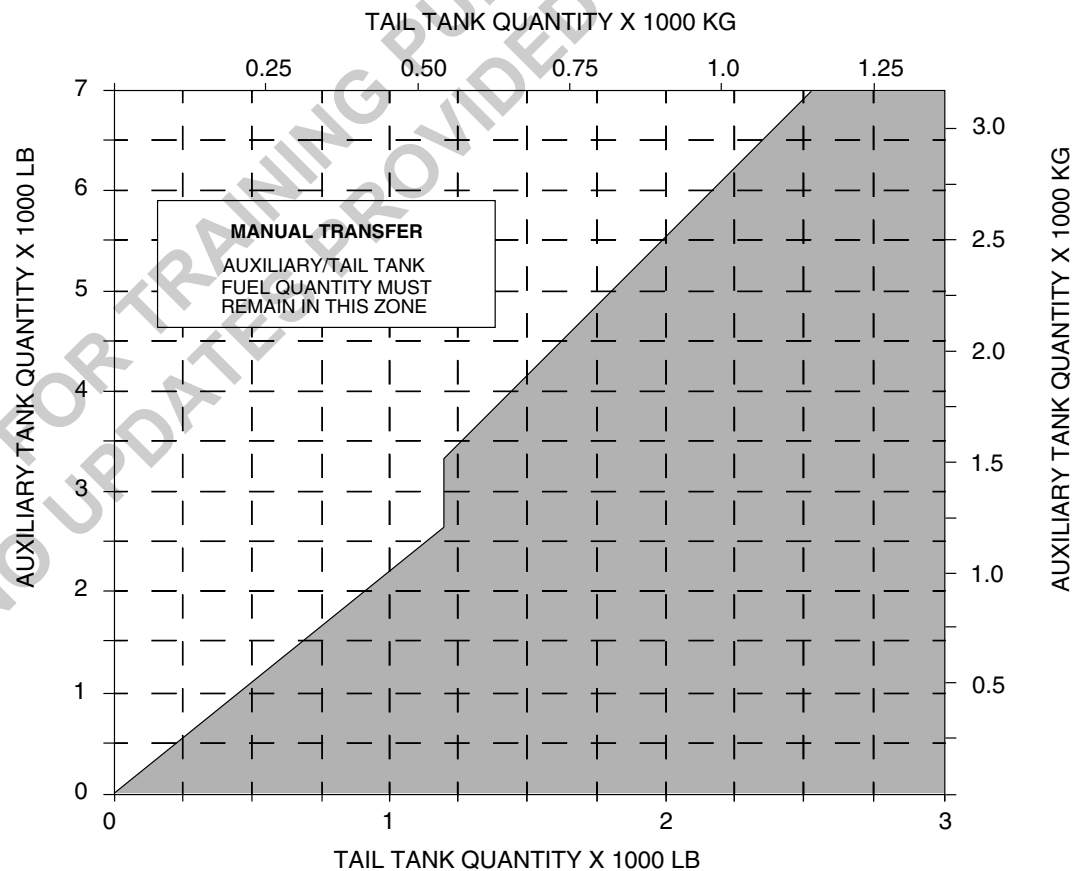
E. Fuel Transfer

Fuel transfer (left or right tank to auxiliary tank/gravity) must be off for take-off.

During normal operation, the tail tank transfer system must not be used as a centre of gravity control device; it must remain selected to automatic mode.

During manual transfer operation – Refer to Figure 02-05-4:

- Auxiliary tank quantity must be at least 2.75 times the tail tank quantity, if tail tank quantity is equal to or greater than 544 kg (1,200 lb).
- Auxiliary tank quantity must be at least 2.2 times the tail tank quantity, if tail tank quantity is less than 544 kg (1,200 lb).
- Manual tail tank fuel transfer is not permitted with more than 3,175 kg (7,000 lb) of fuel in auxiliary tank.
- Auxiliary tank quantity must be continuously monitored for overfill during manual tail tank fuel transfer.



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Manual Fuel Transfer – Auxiliary Tank Quantity vs Tail Tank Quantity
Figure 02-05-4

CHALLENGER 650	LIMITATIONS Power Plant	02-05-16
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10. OIL GRADES

Refer to appropriate maintenance or servicing manual for approved oil grades.

11. OIL CONSUMPTION

Maximum oil consumption on each engine is 189 milliliters per hour (6.4 ounces per hour or 0.05 US gallons per hour).

12. OIL LEVEL

The maximum duration without engine oil servicing is 16 operating hours.

13. OIL REPLENISHMENT SYSTEM

The engine oil level should be checked between 15 minutes to 2 hours after engine shutdown. The engines must be motored if the replenishment period is exceeded. Maximum refill allowable is 1.9 liters (2 US quarts) without dry motoring the engine.

14. AUXILIARY POWER UNIT

A. Type

36-150 (CL)

B. Maximum RPM: 110%

The APU overspeed control will automatically shutdown the APU at 107% rpm.

C. Maximum EGT: 731°C

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14. AUXILIARY POWER UNIT (CONT'D)**D. Starting:**

- (1) Minimum ambient temperature for starting a cold soaked APU on the ground is -40°C .
- (2) The following APU start cycles are permitted:
 - (a) Using airplane batteries on the ground, or for normal in-flight start:
 - Three start attempts, each of 30 seconds continuous cranking
 - Followed by a 20-minute off-time
 - Followed by two further attempts each of 30 seconds continuous cranking.
 - (b) Using ground power:
 - Two start attempts, each of 15 seconds continuous cranking
 - Followed by a 20-minute off-time
 - Followed by two further attempts each of 15 seconds continuous cranking.
 - (c) If, in either case (2)(a) or (2)(b), a successful start is not obtained, a further start must not be attempted for a period of at least 35 minutes.
- (3) Maximum EGT:
974°C (not to be exceeded under any operating conditions).
- (4) Hung start:
If not greater than 30% rpm, within 60 seconds.
- (5) APU starting and operation is permitted within the following operating envelope:
 - Temperature – Refer to Figure 02-04-1.
 - Altitude and Airspeed – Refer to Figure 02-05-5.
- (6) In-flight start:
In-flight starting is guaranteed at altitudes up to 20,000 feet.

CHALLENGER 650	LIMITATIONS Power Plant	02-05-18
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14. AUXILIARY POWER UNIT (CONT'D)

E. APU Bleed Air

- (1) Engine-start during ground operations up to 15,000 feet:
 - (a) No bleed air extraction limitation. Each engine may be started using the APU as a bleed air source.
 - (b) If both engines are to be started using the APU bleed air, then the operating engine's thrust must not exceed 70% N₂.
- (2) Engine-start during flight:
 - (a) During single engine operations, APU bleed air extraction for an engine start is not permitted.
 - (b) During double engine failure conditions, APU bleed air extraction for engine starts is permitted.
Inflight APU main engine starts have been demonstrated to 15,000 feet.
- (3) Air Conditioning:
 - (a) Bleed air extraction from the APU is not permitted above 15,000 feet.

F. APU Generator

The maximum permissible load on the APU generator in flight is 30kVA.

G. APU Indications

The APU limit display markings on the EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified below, the limit markings on the EICAS should be used.

INDICATION		RED (MAX LIMITATIONS)	AMBER (CAUTION RANGE)	GREEN (NORMAL RANGE)
APU EGT °C	start scale	974 and above	—	0 to 973
	normal scale	731 and above	680 to 730	0 to 679
APU RPM %		110 and above	105 to 109	0 to 104

14. AUXILIARY POWER UNIT (CONT'D)

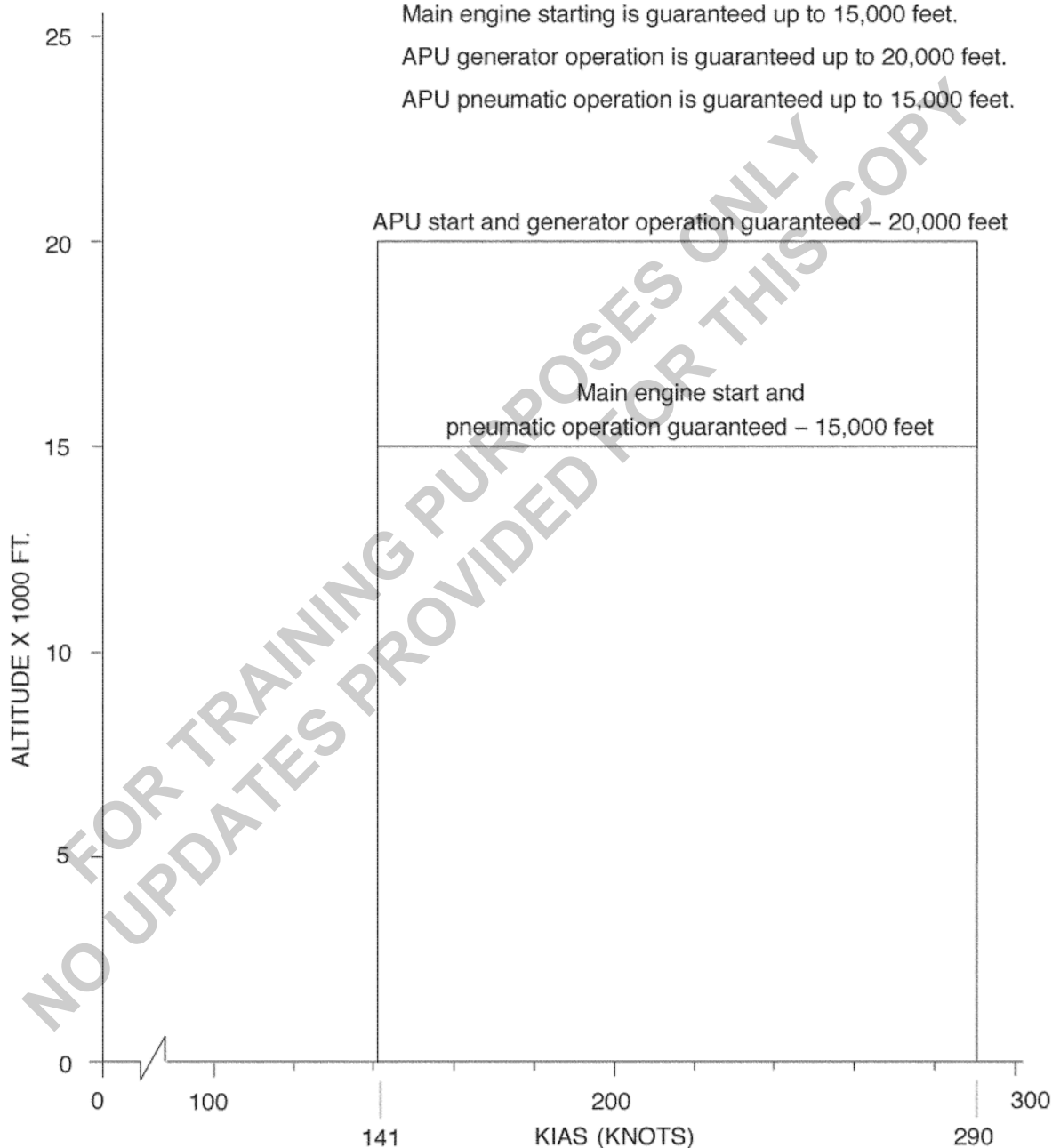
APU Operation is permitted up to 20,000 feet.

APU in-flight starting is guaranteed up to 20,000 feet.

Main engine starting is guaranteed up to 15,000 feet.

APU generator operation is guaranteed up to 20,000 feet.

APU pneumatic operation is guaranteed up to 15,000 feet.



APU In-Flight Envelope
Figure 02-05-5

CHALLENGER ⁶⁵⁰	LIMITATIONS Power Plant	02-05-20
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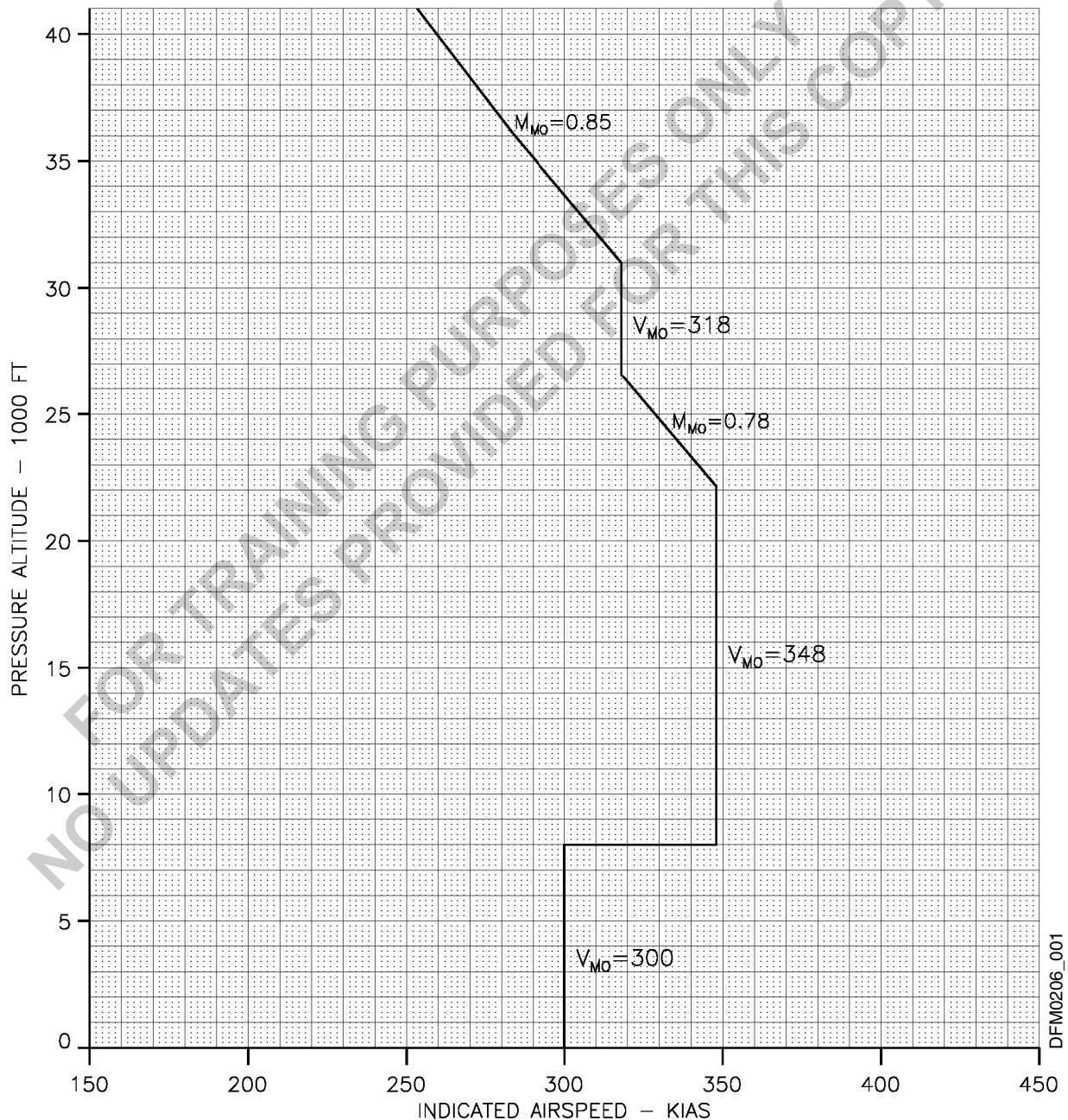
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1. MAXIMUM OPERATING SPEED AND MACH NUMBER

Maximum operating limit speeds as given in Figure 02-06-1, must not be deliberately exceeded in any regime of flight (climb, cruise or descent), unless a higher speed is specifically authorized for flight test or training operations.

NOTE

Overspeed alerting in the flight deck is based on baro-corrected altitude. This is applicable to both the primary flight displays and to the integrated standby instrument.



Maximum Operating Speed and Mach Number
Figure 02-06-1

CHALLENGER 650	LIMITATIONS Operating Speeds	02-06-2
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2. RVSM MAXIMUM OPERATING SPEED

The maximum cruise mach number during flight in RVSM airspace is 0.83.

3. FLAPS EXTENDED SPEED

The maximum speeds at which the flaps may be extended are:

Flaps to 20 degrees: 231 KIAS

Flaps to 30 degrees: 197 KIAS

Flaps to 45 degrees: 189 KIAS

4. MAXIMUM LANDING GEAR OPERATING SPEED

The maximum airspeed at which it is safe to extend the landing gear is 197 KIAS / 0.6 MI.

The maximum airspeed at which it is safe to retract the landing gear is 197 KIAS / 0.6 MI.

5. MAXIMUM LANDING GEAR EXTENDED SPEED

The maximum airspeed at which the airplane may be flown with the landing gear extended and locked is 250 KIAS / 0.7 MI.

Flight at altitudes above 20,000 feet with the landing gear extended is prohibited.

6. TIRE LIMIT SPEED

The tire limit speed is 182 knots ground speed.

7. TURBULENCE PENETRATION SPEED

Maximum air speed for turbulence penetration is 280 KIAS or 0.75 Mach, whichever is lower.

8. MINIMUM OPERATING LIMIT SPEED

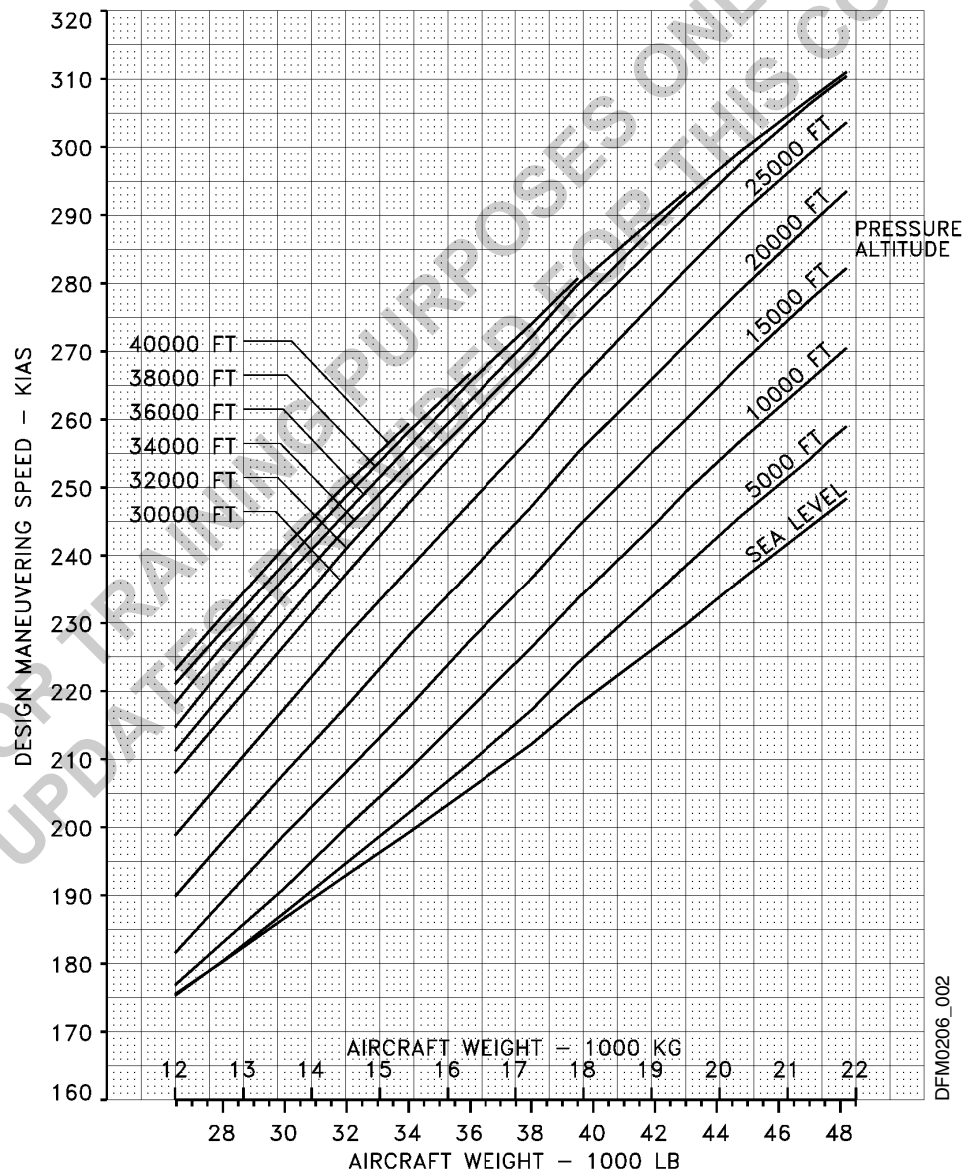
Intentional speed reduction below the onset of stall warning, as defined by stick shaker operation, is prohibited unless a lower speed is specifically authorized for flight test or training operations.

9. DESIGN MANEUVERING SPEED

Full application of rudder and aileron controls as well as maneuvers that involve angles of attack near the stall, must be confined to speeds below V_A . Values of V_A are given in Figure 02-06-2, for varying pressure altitudes and airplane weights.



Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw (e.g. large side slip angles), as they may cause structural failure at any speed, including below V_A .



Design Maneuvering Speeds
Figure 02-06-2

CHALLENGER ⁶⁵⁰	LIMITATIONS Operating Speeds	02-06-4
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CHALLENGER 650	LIMITATIONS Maneuvering Loads	02-07-1
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1. MANEUVERING LIMIT LOAD FACTORS

These load factors limit the permissible angles of bank in turns and the severity of pull-up and push-over maneuvers:

- Flaps up: -1.0 G to 2.5 G.
- Flaps down: 0.0 G to 2.0 G.

2. SIDE SLIP MANEUVERS

Avoid unnecessary and large side-slip maneuvers during low speed high altitude cruise.

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CHALLENGER ⁶⁵⁰	LIMITATIONS Maneuvering Loads	02-07-2
		Oct 27/15

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CHALLENGER 650	LIMITATIONS Systems Limitations	02-08-1 Rev 13, Aug 23/18
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1. AIR-CONDITIONING AND PRESSURIZATION

The maximum relief differential pressure is 9.2 psi.

The maximum negative differential pressure is -0.5 psi.

During taxi, take-off and landing, the pressure differential must not exceed 0.2 psi.

The airplane must be completely depressurized prior to opening any of the airplane doors.

2. AUTOMATIC FLIGHT CONTROL SYSTEM

WARNING

During the ALTS CAP mode the flight guidance/autopilot may not maintain the selected airspeed. If an engine failure occurs during a climb while in ALTS CAP or just before ALTS CAP, the airspeed may rapidly decay below the safe operating speed. Crew intervention may be required to maintain safe operating speed and to avoid loss of control.

The minimum autopilot engage height after take-off is 320 feet AGL.

The minimum autopilot use height for visual and non-precision approaches is 320 feet AGL.

The minimum autopilot use height for precision approaches (ILS) and LPV approaches, with a glidepath angle of 3.5 degrees or less, is 80 feet AGL.

It is prohibited to select the XFR button on the flight control panel while simultaneously making inputs on a FMS CDU.

3. BLEED AIR SYSTEMS

The bleed air 10th stage valves must be closed for take-off and landing if the engine cowl and/or wing anti-ice systems have been selected on.

If above 40,000 feet, one air-conditioning unit or cowl anti-ice system must be selected on for each engine.

4. ELECTRICAL SYSTEMS

A. Permissible Loads on AC System

Individual AC generator loading must not exceed the following values:

ALTITUDE (FEET)	LOAD LIMITATION (KVA)	
	MAIN GENERATOR (EACH)	APU GENERATOR
0 to 20,000	30	30
20,001 to 35,000	30	0
35,001 and above	25	0

B. Permissible Loads on DC Systems

The maximum permissible continuous load on each TRU is 100 amps.

C. Circuit Breaker Reset (In Flight)

A circuit breaker must not be reset or cycled (i.e., opened or closed) unless doing so is consistent with explicit procedures specified in the Airplane Flight Manual or unless, in the judgement of the Pilot-in-Command, that resetting or cycling of the circuit breaker is necessary for the safe completion of the flight.

5. FLIGHT CONTROLS – LIFT/DRAG DEVICES

A. Flaps

Enroute use of flaps is prohibited.

Flight with flaps extended at altitudes above 15,500 feet is prohibited.

B. Flight Spoilers

Flight below an altitude of 300 feet AGL with flight spoilers extended is prohibited.

To ensure adequate maneuver margins, flight spoilers must not be extended in flight at airspeeds below the recommended approach speed plus 10 KIAS (refer to Chapter 6; PERFORMANCE – LANDING PERFORMANCE).

6. STALL PROTECTION SYSTEM

Both stall protection system pusher switches must remain on for all phases of flight.

The stall protection test indicator must only be used for stall protection system functional test purposes.

The PFD normalized AOA indicator is advisory only, it does not replace the airspeed indicator as primary speed instrument, and it does not replace the stick shaker as primary stall warning.

The PFD normalized AOA indicator shall not be used during take-off.

7. THRUST REVERSERS

Thrust reversers are approved for ground use only.

The thrust reversers are intended for use during full stop landings. Do not attempt a go-around maneuver after deployment of the thrust reversers.

Backing the airplane with the use of reverse thrust is prohibited.

Take-off with any of the following thrust reverser icons or EICAS messages displayed is prohibited:

- REV icon on N_1 gauge,
- L (R) REV UNLOCKED caution message, and
- L (R) REV UNSAFE caution message.

During landing, application of maximum reverse thrust is not permitted at airspeeds below 60 KIAS. Below 60 KIAS, reverse thrust must be reduced to 60% N_1 or less.

The maximum demonstrated crosswind component approved for use of reverse thrust is 24 knots [at 33 feet (10 meters) tower height]. This value was demonstrated on a dry runway, and is considered limiting.

8. TAXI AND LANDING LIGHTS

The taxi lights must be selected off whenever the airplane is stationary.

NOTE

During prolonged taxiing, the nose landing lights may be used in place of the taxi lights, if necessary, at the pilot's discretion.

The left and right landing lights must be selected off during taxi or when the airplane is stationary.

The left and right landing lights must not be used on the ground, except for the take-off roll and landing.

9. WHEEL BRAKE COOLING LIMITATIONS

Brake cooling times (established in accordance with the procedures in Chapter 6; PERFORMANCE – TURN-AROUND TIME) must be observed between a landing or a low-energy rejected take-off (RTO) and a subsequent take-off, to ensure that sufficient brake energy is available to bring the airplane to a complete stop, if the subsequent take-off is rejected.

If a fusible plug releases, the wheels, brakes and tires and the anti-skid speed sensors must be inspected in accordance with the procedure in the time limits/maintenance checks, provided in the CL-650 Airplane Maintenance Manual, and any damage rectified before the next take-off.

10. CONFIGURATION DEVIATION LIST

If the aircraft is to be operated with certain secondary airframe and/or any nacelle parts missing, operation must be in accordance with the limitations specified in the basic Airplane Flight Manual, and as amended by the Configuration Deviation List (Appendix 1).

11. ELECTRICAL/AVIONICS EQUIPMENT

During ground operation at ambient temperatures above 40°C (104°F), operation of electrical/avionics equipment must be limited to 30 minutes, unless at least one air conditioning pack is operating and the passenger door is closed.

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CHALLENGER 650	LIMITATIONS Navigation Systems Limitations	02-09-1
		Rev 15, Mar 25/19

1. FLIGHT MANAGEMENT SYSTEM

The flight management system (FMS) must be operated in accordance with the latest revision of the following:

- Airplane Flight Manual, and
- Pro Line 21 Advanced Flight Management System Operators Guide (part number 523-0822377 at the latest applicable revision).

If a difference exists between the AFM and the Operators Guide, the AFM shall take precedence. The Pro Line 21 Advanced Flight Management System Operators Guide must be immediately available to the flight crew whenever navigation is predicated on the use of the FMS.

The FMS is approved for use with software version FMC HSCD 946-2720-110, or later applicable revision.

On airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015:

The FMS is approved for use with software version FMC HSCD 946-2720-102, or later applicable revision.

The following performance database, part number 096-5752-101, must be verified to be current and valid.

On airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015:

The following performance database, part number 096-5752-001, must be verified to be current and valid.

The following thrust and VERC database, part number 096-5753-101, must be verified to be current and valid.

On airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015:

The following thrust database, part number 096-5753-001, must be verified to be current and valid.

The following V speed database must be verified to be current and valid:

	ALL REGISTRIES EXCEPT FAA	FAA REGISTRY
Airplanes 6147 and Subs and airplanes incorporating Service Bulletin 650-34-012 or 650-34-015	part number 096-5750-101	part number 096-5751-101
Airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015	part number 096-5750-002	part number 096-5751-002

AR Certified Airplanes:

The FMS V speed data is advisory only.

The FMS V speed data does not need to be verified with the AFM, provided both flight management computers are available.

The FMS V speed data is the primary means of information, provided two active flight management computers are available.

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1. FLIGHT MANAGEMENT SYSTEM (CONT'D)

When FMS 3 is selected to 1 or 2 for take-off, manual V speed data (cyan) must be used for take-off.

On airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015:

The AFM must be used for the V_{ERC} data. The use of the V_{ERC} data on the THRUST LIMIT page is prohibited.

On airplanes 6147 and subs and airplanes incorporating Service Bulletin 650-34-012 or 650-34-015:

The V_{ERC} data shown on the THRUST LIMIT page is advisory information only and must be cross checked with the AFM.

The FMS thrust setting data is the primary means of information, provided two air data computers and two flight management computers are available.

FMS database information must be verified as being current. Waypoints must be checked for accuracy prior to use.

It is prohibited to load pilot routes from the PILOT ROUTE LIST during flight.

It is prohibited to upload flight plans from ROUTE MENU (FPLN RECALL) during flight.

It is prohibited to select FPLN WIND UPDATE from the ROUTE MENU.

It is prohibited to tune a localizer on the TUNE page using the identifier only.

It is prohibited to initiate a NAV RADIO SELF TEST from either the NAV radio tuning page on the FMS CDU or the MFD nav radio menu during flight.

It is prohibited to use the FMS as a navigation source when it is not receiving suitable navigation information from the following:

- One GNSS; or
- Two DMEs; or
- One VOR / DME; or
- One IRS.

FMS navigation (FMS selected as NAV Source) and display of magnetic heading is prohibited in Canadian Northern Domestic Airspace and at the following locations:

- Latitudes greater than 82°N
- Latitudes greater than 73°N with a longitude between 80°W and 170°W
- Latitudes greater than 82°S
- Latitudes greater than 60°S with a longitude between 120°E and 160°E.

It is prohibited to load FMS procedures that are referenced to True Heading into the flightplan.

During periods of dead reckoning, indicated by the FMS DR annunciation, the FMS shall not be utilized as the primary source of navigation.

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1. FLIGHT MANAGEMENT SYSTEM (CONT'D)

BRNAV and PRNAV operations with the following message are prohibited:

- IRS ONLY (on CDU)

When conducting RNP operations the pilot must verify the PFD CDI scale annunciation is appropriate for the RNP leg as follows:

- RNP APCH CDI Scale 0.3 nm
- BASIC RNP-1 CDI Scale 1.0 nm
- RNP-2 CDI Scale 2.0 nm
- RNP-4 CDI Scale 4.0 nm (selected by pilot)
- RNP-10 CDI Scale 4.0 nm (selected by pilot).

When conducting BASIC RNP-1 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 200 NM.

When conducting RNP-2 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 400 NM.

When conducting RNP-4 or RNP-10 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 500 NM.

Pilot defined waypoints (e.g. along track offset) must not be used to create individual legs shorter than the limiting length.

When conducting non-AR RNP operations, pilots should maintain CDI lateral deviation less than ± 1 dot deviation except brief deviation in excess of ± 1 dot deviation during waypoint sequencing is permitted.

RNP operations with any of the following FMS messages annunciated, are prohibited:

- NO APPR (on PFD)
- UNABLE RNP (on PFD)

VERTICAL NAVIGATION

Vertical guidance deviation information is an aid to assist pilots to comply with altitude restrictions and plan descent profiles. When using vertical guidance the pilot must use the primary barometric altimeter to ensure compliance with altitude restrictions, particularly during instrument approach operations.

Use of VNAV vertical guidance is prohibited when the barometric altitude is corrected to the landing field elevation (QFE).

Use of VNAV vertical guidance is prohibited with a single ADC source when the 3rd FMS is reverted in.

It is prohibited to select MCRZ from the FMS VNAV CRUISE page.

Use of VFLC (FLC with VNAV) is prohibited during climb when the airspeed reference is displayed as Mach.

It is prohibited to select VNAV following a go-around during the missed approach procedure.

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1. FLIGHT MANAGEMENT SYSTEM (CONT'D)



Errors to VNAV defined paths may occur because of coding errors in the navigation data base and because of altimetry errors. The actual VNAV path may deviate significantly below the intended VNAV path in very cold temperatures; the temperature compensation function is used to correct for nonstandard temperatures. Flight crew must coordinate use of temperature compensation with ATC prior to employing this function.

Database coding of VNAV altitudes for approach waypoints may result in VNAV paths continuing below Minimum Descent Altitude (MDA), Decision Altitude (DA), or Decision Height (DH), or ending at an altitude too high to continue a safe descent to landing. VNAV paths (often called pseudo glide paths) are not equivalent to an ILS glideslope.

Position along an approach must be verified prior to commencement of VNAV descent as displayed by the FMS. The required visual reference must be obtained prior to commencing descent below published MDAs, DAs, or DHs.

PERFORMANCE

The FMS will automatically display the displaced thresholds of runways provided that they are identified in the loaded active cycle FMS navigation database. If RUNWAY LENGTH is manually entered then the FMS will not compute headwind/crosswind; these must be manually entered.

Airplane operation must not be predicated upon performance data; ETE/ETA and fuel remaining. FMS range, fuel management information and ETE/ETA is advisory only.

APPROACH

Instrument approaches must be accomplished only in accordance with instrument approach procedures that are contained in the FMS database.

NOTE

Vectored ILS approaches are exempt from this limitation as they do not require nor do they rely on FMS guidance. ILS back course approaches also are exempt if they are vectored.

The flight crew must verify any altitude/speed constraints on the last waypoint of any entered STAR procedure and/or the first waypoint of any entered approach procedure. The flight crew may correct any such altitude/speed constraint(s) as required to agree with current chart procedures.

It is prohibited to conduct LPV approach operations with FMS2 as pilot navigation source and FMS1 as copilot navigation source.

Use of pilot defined FMS approaches, in instrument meteorological conditions, is prohibited.

Use of FMS instrument approaches is prohibited unless RNP APPR or LPV APPR is annunciated on the PFD.

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1. FLIGHT MANAGEMENT SYSTEM (CONT'D)

RNP operations down to LPV minimum when any of the following messages are present, are prohibited:

- Red boxed “VNV” flag (vertical deviation scale on PFD)
- NO APPR (on PFD)
- EFIS MISCOMPARE (CAS message)
- EFIS COMP INOP (CAS message)
- LPV NOT AVAIL (on CDU)

RNP operations down to LPV minimum must not be commenced or continued unless all required airborne equipment are operating satisfactorily.

The required airborne equipment are:

- Dual PFD
- Two GNSS sensors must be available to commence an approach and one GNSS sensor is required to continue an approach.
- Dual FMC.

Use of VNAV vertical guidance for V-MDA type approach between the final approach fix and the missed approach fix is prohibited.

FMS as a primary navigation source beyond the FAF is prohibited for ILS, LOC and LOCBC approaches.

FMS as a supplementary navigation source beyond the FAF is permitted when APPR or REF APPR is annunciated on the PFD if the navigation aid referenced in the approach procedure is displayed and monitored by the flight crew during the approach.

NOTE

For the FMS approaches for which GNSS is not authorized as a primary navigation source (i.e. GNSS does not appear in the approach title), APPR will be annunciated on the PFD if the GNSS sensors are deselected; otherwise, REF APPR will be annunciated.

The FMS with inputs from GNSS may only be used for approach guidance if the reference coordinate data system for the instrument approach is WGS84 or NAD83.

GNSS sensors must be deselected when on approach in airspace not referenced to WGS84 or NAD83.

APPROVAL STATUS

The approvals listed in the OPERATIONAL CAPABILITIES do not constitute operational approval. Some operations require additional crew qualification, communication and surveillance equipage in order to obtain operational approval from regulatory authorities.

2. NAVIGATION RECEIVER

It is prohibited to conduct ILS approach operations with LOC2 as pilot navigation source and LOC1 as copilot navigation source.

It is prohibited to navigate using VOR or LOC as Nav source when displaying true heading mode.

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3. TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

Navigation must not be predicated upon the use of the terrain awareness display.

TAWS TERRAIN OFF must be selected when using QFE operations, and GPS is not available.

4. RUNWAY AWARENESS AND ADVISORY SYSTEM (RAAS)

RAAS advisories are not intended for navigation purposes.

Pilots should not deviate from ATC clearances or otherwise correct aircraft position based solely on RAAS advisories.

Pilots may not use RAAS advisories as a replacement for NOTAM or ATIS information.

RAAS Short and Insufficient Runway functions do not take into account aircraft performance factors such as aircraft weight, wind, runway conditions, slope, air temperature and altitude of airport. The runway length settings for these functions were determined from a review of takeoff and landing performance and are considered to be acceptable guidelines for RAAS purposes. This information is intended to enhance the flight crew awareness by identifying that the runway length available may be marginal for the current conditions.

Data on newly constructed runways, changes to length of existing runways, or the rare change of runway identifier may not necessarily be included in the RAAS runway database.

RAAS does not have knowledge of where painted hold lines are located at airports. In some cases, the RAAS approaching runway advisory may be issued after an aircraft has crossed a hold line.

The RAAS system is a situational awareness tool to the flight crew and should not be used as the primary source of determining field position, runway length, field performance, etc. The absence of RAAS aural and/or visual messages such as "short runway" or "long landing" does not guarantee that the aircraft can be safely operated from or onto the runway.

Altimeter Monitor–Below Transition Altitude function, is not available at airports defined as QFE in the EGPWS database.

5. INTEGRATED FLIGHT INFORMATION SYSTEM

Adequate back-up documentation for IFIS must be immediately available to the flight crew.

IFIS functions must not be used unless the related databases (i.e., charts, airspace, airways, geographic, political, graphical weather) incorporate the current update cycle.

The IFIS does not include enroute charts.

The aircraft symbol on the electronic charts is advisory and is presented for increased situational awareness. It is not to be used for navigation.

The aircraft symbol displayed on airport diagram charts during ground operations is advisory and is presented for increased situational awareness. It is not to be used for precision maneuvering on the ground.

The display of geo-political boundaries, airspace and airways on the MFD (enhanced map overlays) is for enhanced situational awareness only, and use for navigation is prohibited.

All weather products satellite linked to the aircraft are advisory.

It is not possible to display landscape formatted approach charts in split view.

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6. SYNTHETIC VISION SYSTEM (SVS)

TAWS alerts must take priority over the image depicted on SVS.

Use of the SVS for airplane navigation is prohibited.

Use of the SVS is prohibited when the barometric altitude is corrected to the landing field elevation (QFE).

When conducting an approach in airspace not referenced to WGS-84 or NAD-83, the SVS shall be deselected prior to passing the FAF.

When conducting an approach to a runway not entered in the FMS flight plan, the SVS shall be deselected, prior to passing the FAF.

The SVS Airport/Runway database must incorporate the current update cycle. If not, the SVS shall be deselected prior to take-off and prior to passing the FAF.

The SVS does not render a video image above latitude 85N and below latitude 85S. The SVS shall be deselected prior reaching these latitudes.

The SVS is intended as a situational awareness enhancement feature and shall not be the sole source of reference by the pilot for any of the following:

- Continuation of an approach, landing or takeoff below the minima published for that approach.
- Execution of an approach, landing or takeoff in low visibility conditions.
- As a replacement for natural vision where there are specific visual requirements to be achieved.
- As a replacement for the basic attitude, flight guidance or navigation displays and information.
- The performance of any specific flight related task.
- The flight crew making any judgment, decision or action that is solely based on the synthetic vision display.
- The application of any operational credit.

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7. DATALINK

The datalink is approved for the transmission and receipt of messages that will not create an unsafe condition if the message is improperly received. An unsafe condition may exist if:

- The message, or part of the message, is delayed or not received,
- The message is delivered to the wrong recipient, or
- The message content is corrupted.

Crew action-based messages, such as pre-departure clearance, oceanic clearance, digital automatic terminal information service, weight and balance, take-off data (speeds, trim settings, runway distances) are prohibited, unless the approved operational procedures are used to verify that the message is received by the intended recipient, that the message is valid, and that the content is not corrupted.

All weather products data linked to the aircraft are advisory.

On airplanes incorporating Service Bulletin 650-23-007:

ATN B1 CPDLC installation is limited to providing a supplementary means of communication; voice shall remain the primary means of communication. Use of ATN B1 CPDLC should be avoided during periods of high pilot workload and critical phases of flight where pilot-ATC communications are time critical. This AFM entry does not, by itself, constitute an operational approval where such an approval is required.

On EASA registered airplanes:

ATN B1 CPDLC shall only be used for routine datalink exchanges during en-route operations in upper airspace.

On airplanes incorporating Service Bulletin 650-23-008:

FANS 1/A+ datalink system must be operated in accordance with ATN and FANS Interlocked CPDLC Applications Operators Guide (part number 523-0822435 at the latest applicable revision).

The crew must ensure that the FMS is updated with the proposed flight plan and its associated constraints and procedures when receiving a route clearance via CPDLC.

Before executing any flight plan modification received via CPDLC, the crew must ensure that the entire content of the flight plan has been inserted.

Use of FANS INSERT MOD is prohibited within 5NM of sequencing a waypoint.

It is prohibited to use the WX DEV TO from the FANS ROUTE REQ 1/2 page.

It is prohibited to use the DIVERT TO from the EMERGENCY MSG 1/2 page using the FANS REQUEST page.

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1. GENERAL

Procedures given in this chapter of the Airplane Flight Manual are concerned with foreseeable but unusual situations in which immediate and precise crew action, as indicated by the items within the “boxed” area at the beginning of each procedure, will substantially reduce the risk of disaster.

In addition to the failure indications referenced in the following procedures, the master warning system and appropriate switch/lights operate where applicable.

The emergency procedures contained within this chapter have been developed and recommended by Bombardier and approved by Transport Canada for use in the operation of the CL-600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent). These procedures are provided as guidance, and should not be construed as prohibiting the development of equivalent Transport Canada-approved procedures.

2. LANDING DISTANCE FACTORS

Landing distance factors are provided herein with and without the use of thrust reversers. No operation should be predicated on the use of thrust reversers.

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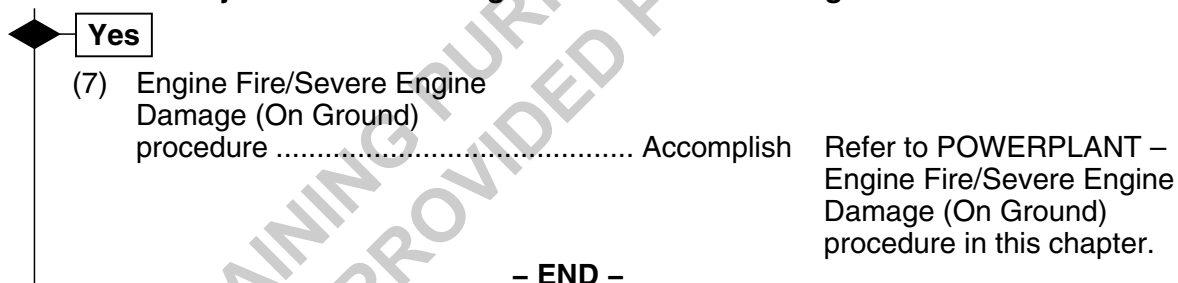
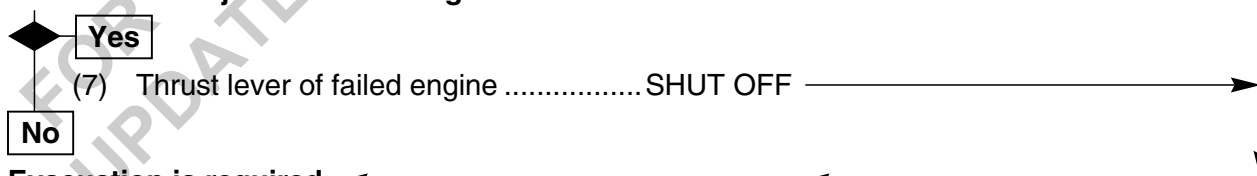
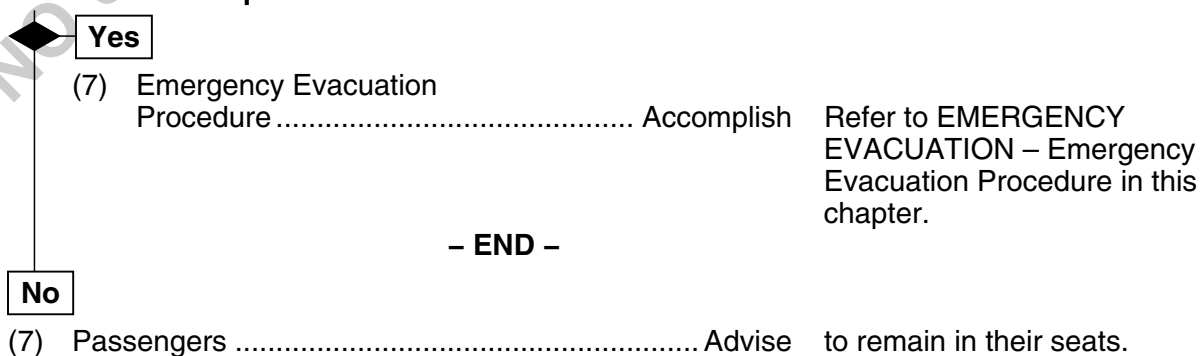
	CL-650 Airplane Flight Manual PSP 650-1	
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1. REJECTED TAKE-OFF**A. Rejected Take-Off Before Achieving V_1**

Rejection of the take-off must be initiated no later than V_1 .

Simultaneously:

- | | | |
|---|-------------|--|
| (1) Take-off | Discontinue | immediately. |
| (2) Thrust levers..... | IDLE | |
| (3) ATS DISC switch | Depress | |
| (4) Wheel brakes | Apply | maximum braking, until a safe stop on runway is assured. |
| (5) FLIGHT SPOILER lever | Select | to MAX. |
| (6) Thrust reverser(s)
[operating engine(s)] | Apply | maximum; consistent with directional control. |

Take-off was rejected due to an engine fire or severe damage:**Take-off was rejected due to engine failure:****Evacuation is required:**

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1. REJECTED TAKE-OFF (CONT'D)

A. Rejected Take-Off Before Achieving V_1 (Cont'd)

After a rejected take-off, procedures associated with cooling of wheels, brakes and tires must be observed (Refer to Chapter 6; PERFORMANCE – TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires).

————— **END** —————

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1. POWER PLANT

NOTE

In the event of a lightning strike, simultaneous fire/overheat indications could occur. Abnormal values of secondary symptoms may be used to determine if an actual fire exists.

A. Engine Fire/Severe Engine Damage (In Flight)

Indication: L (R) ENG FIRE PUSH light on, L (R) ENGINE FIRE warning message and fire bell on.

During take-off after achieving V_1 :

- ◆ **Yes**
- (1) Take-off Continue
- (2) Airplane Rotate at V_R to an initial take-off attitude of 10 degrees, and adjust pitch, as required, to achieve an airspeed of V_2 at 35 feet.

When positive rate of climb is achieved:

- (3) Landing gear Retract
- (4) Airspeed Maintain at V_2 to engine-out level-off height (Refer to Chapter 6; PERFORMANCE – OBSTACLE CLEARANCE).

NOTE

If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$ KIAS).

◆ **No**

After take-off and at a safe altitude:

Affected engine:

- (1) ATS DISC switch Depress
- (2) Thrust lever Confirm and IDLE
- (3) Thrust lever Confirm and SHUT OFF
- (4) ENG FIRE PUSH switch/light Press in
- Check the following:
- BOTTLE ARMED PUSH TO DISCH lights on,
 - L (R) ENG SOV CLSD advisory message on and
 - HYD SOV 1 (2) CLSD advisory message on.

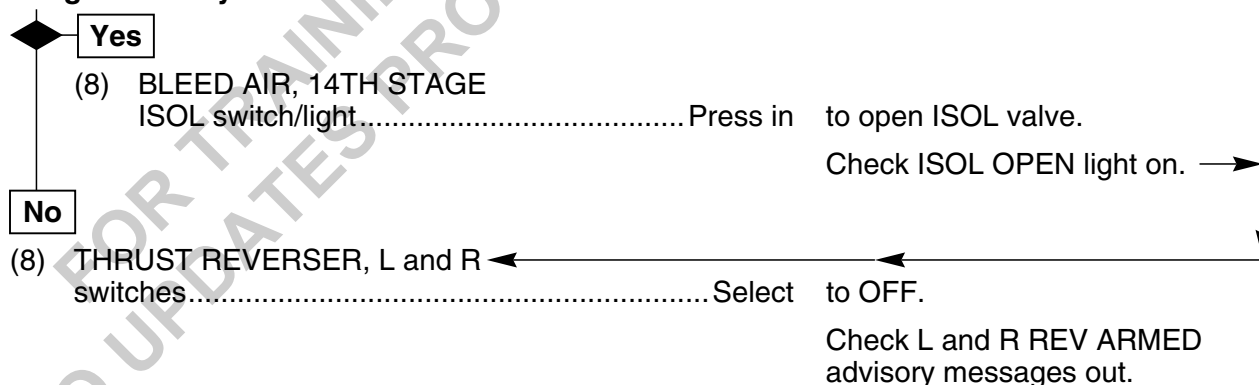
1. POWER PLANT (CONT'D)

A. Engine Fire/Severe Engine Damage (In Flight) (Cont'd)

- (5) FUEL, L (R) BOOST PUMP switch/light.....Press out to disarm affected boost pump.
Check L (R) FUEL PUMP caution message on.
- If warning persists (after 10 sec):**
- (6) BOTTLE ARMED PUSH TO DISCH switch/light.....Press in and hold until light goes out, to discharge firex bottle.
Check ENG BOTTLE 1 (2) LO caution message on.
- If warning persists (after an additional 30 sec):**
- (7) Other BOTTLE ARMED PUSH TO DISCH switch/light.....Press in and hold until light goes out, to discharge other firex bottle.
Check ENG BOTTLE 1 (2) LO caution message on.

After take-off (if applicable):

Wing anti-ice system selected on:



At not less than 400 feet AGL or the engine-out level off height:

- (9) Thrust Set thrust to maximum continuous thrust (MCT) (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (10) Airspeed Accelerate in level flight to V_{FTO} .
- (11) FLAPS Select to UP at a speed of not less than $V_{FTO} - 5$ KIAS.
- (12) Climb Initiate at V_{FTO} .
- (13) CAS messages..... Review



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1. POWER PLANT (CONT'D)

A. Engine Fire/Severe Engine Damage (In Flight) (Cont'd)

At a safe altitude:

- (14) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– POWER PLANT – In-Flight
Engine Failure/Shutdown.

NOTE

Do not operate the APU in flight if engine rotor burst damage is suspected.

————— **END** —————

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1. POWER PLANT (CONT'D)

B. Engine Fire/Severe Engine Damage (On Ground)

Indication:
L (R) ENG FIRE PUSH light on, L (R) ENGINE FIRE warning message and fire bell on.

(1) Parking brake Set

Affected engine:

(2) Thrust lever SHUT OFF

(3) ENG FIRE PUSH switch/light Press in

(4) FUEL, L and R BOOST PUMP switch/lights Press out

If warning persists (after 10 sec):

(5) BOTTLE 1 and 2 ARMED PUSH TO DISCH switch/lights Press in

Check the following:

- BOTTLE ARMED PUSH TO DISCH lights on,
- L (R) ENG SOV CLSD advisory message on and
- HYD SOV 1 (2) CLSD advisory message on.

to disarm both boost pumps.

Check L and R FUEL PUMP caution messages on.

and hold until lights go out, to discharge firex bottles.

Check ENG BOTTLE 1 and 2 LO caution messages on.

(6) Emergency Evacuation Procedure Accomplish

Refer to EMERGENCY EVACUATION – Emergency Evacuation Procedure in this chapter.

————— END —————

CHALLENGER 650	EMERGENCY PROCEDURES Power Plant	03-03-5 Rev 9, Jun 16/17
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1. POWER PLANT (CONT'D)

C. Double Engine Failure

Indication: Engine failure indicated on EICAS; N₁, N₂, ITT and fuel indications.

- (1) IGNITION, CONT switch/light..... Press in Check IGNITION A/B advisory message on.
- (2) Airspeed 240 KIAS minimum

- (3) Engine instruments..... Verify N₁, N₂, ITT

If engines continue to run-down:

- (4) Thrust levers (both) SHUT OFF
- (5) ADG manual deploy handle Pull and leave extended.
- (6) AC ELECTRICAL page Check ESS BUS is powered.

When ADG power is established:

- (7) STAB TRIM, CH 2 switch Press in to engage STAB TRIM CH 2.
- (8) Oxygen masks (if required) Don Set to 100%.
- (9) Crew communication Establish Set MASK/BOOM switch to MASK.
- (10) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (11) APU (if available)..... Start when at or below 20,000 feet.
- (12) AC POWER, APU GEN switch..... Select to ON.
- (13) AIR-CONDITIONING, L and R PACK switch/lights Press out to select both packs off.
Check L and R PACK OFF lights on.
- (14) ANTI-ICE, WING switch Select to OFF.
- (15) ANTI-ICE, L and R COWL switch/lights Press out to select cowl anti-ice off.

WARNING

To avoid thermal seizure, increase airspeed as required to maintain at least 4% N₂ indicated. If the N₂ is allowed to drop to 0%, the engine may thermally seize and may not rotate even at higher airspeeds or with starter engagement.



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1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

NOTE

- Between 21,000 feet and 10,000 feet, a minimum of 12% N_2 is necessary for a windmill relight.
- At 10,000 feet and below, a minimum of 10% N_2 is necessary for a windmill relight.
- Acceleration to V_{MO} is recommended to attain the necessary N_2 for a windmill relight.
- The altitude loss when accelerating from 240 KIAS to V_{MO} , could be more than 5,000 feet.
- A push-over to as steep as 15° nose down may be required.

Windmill relight is feasible:

Yes

(16) Glide airspeed Increase to V_{MO}



The resulting high rate of descent (up to 7800 fpm) will cause a significant reduction of available gliding distance.

(17) Descent Initiate to 21,000 feet or below, at V_{MO} .

(18) FUEL, L and R BOOST PUMP switch/lights Confirm ON

At 21,000 feet and below, attempt to start both engines at the same time.

When ITT is $\leq 90^\circ\text{C}$ and N_2 is $\geq 12\%$ (between 21,000 feet and 10,000 feet) or N_2 is $\geq 10\%$ (10,000 feet and below):

(19) IGNITION, CONT switch/light Confirm ON

(20) Thrust levers (both) IDLE

(21) Engine indications Monitor engine acceleration, ITT and oil pressure.

At least one engine relights within 25 seconds (stable IDLE within 2 minutes):

Yes

(22) Thrust lever(s) As required

Operating engine(s):

(23) Applicable AC POWER, GEN switch(es) Select to OFF/RESET then ON.

Check applicable GEN 1 OFF and/or GEN 2 OFF caution message(s) out.

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

- (24) Applicable
AIR-CONDITIONING,
PACK switch/light(s) Press in to restart pack(s).

Check applicable PACK OFF
light(s) out.

NOTE

Use only one air-conditioning pack during single
engine operations, unless the APU is the bleed
source.

- (25) IGNITION, CONT
switch/light Press out to select continuous ignition
off.

Check IGNITION A/B advisory
message out.

Re-establish normal power:

- (26) ADG manual deploy
handle Stow
- (27) ADG, PWR TXFR
OVERRIDE switch Press to override.
- (28) CAS messages Review
- (29) Affected systems Restore

Only one engine is operating:

Yes

- (30) In-Flight Engine
Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– POWER PLANT – In-Flight
Engine Failure/Shutdown.

– END –

No

- (30) No further action required.

– END –

No

- (22) Thrust levers (both) SHUT OFF

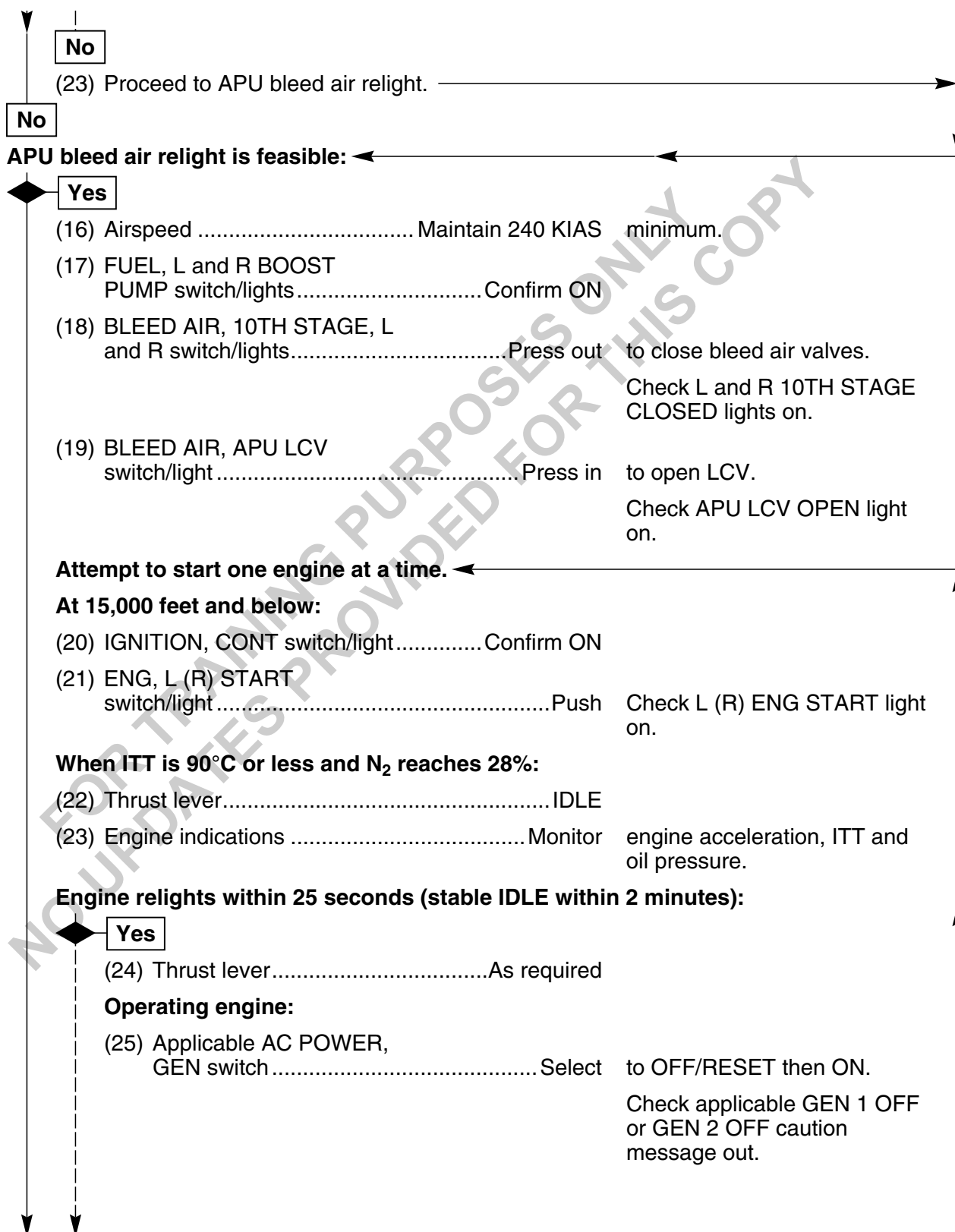
Another windmill relight attempt is feasible:

Yes

- (23) Airspeed Maintain V_{MO}
- (24) Wait 30 seconds, then repeat windmill relight procedure.

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)



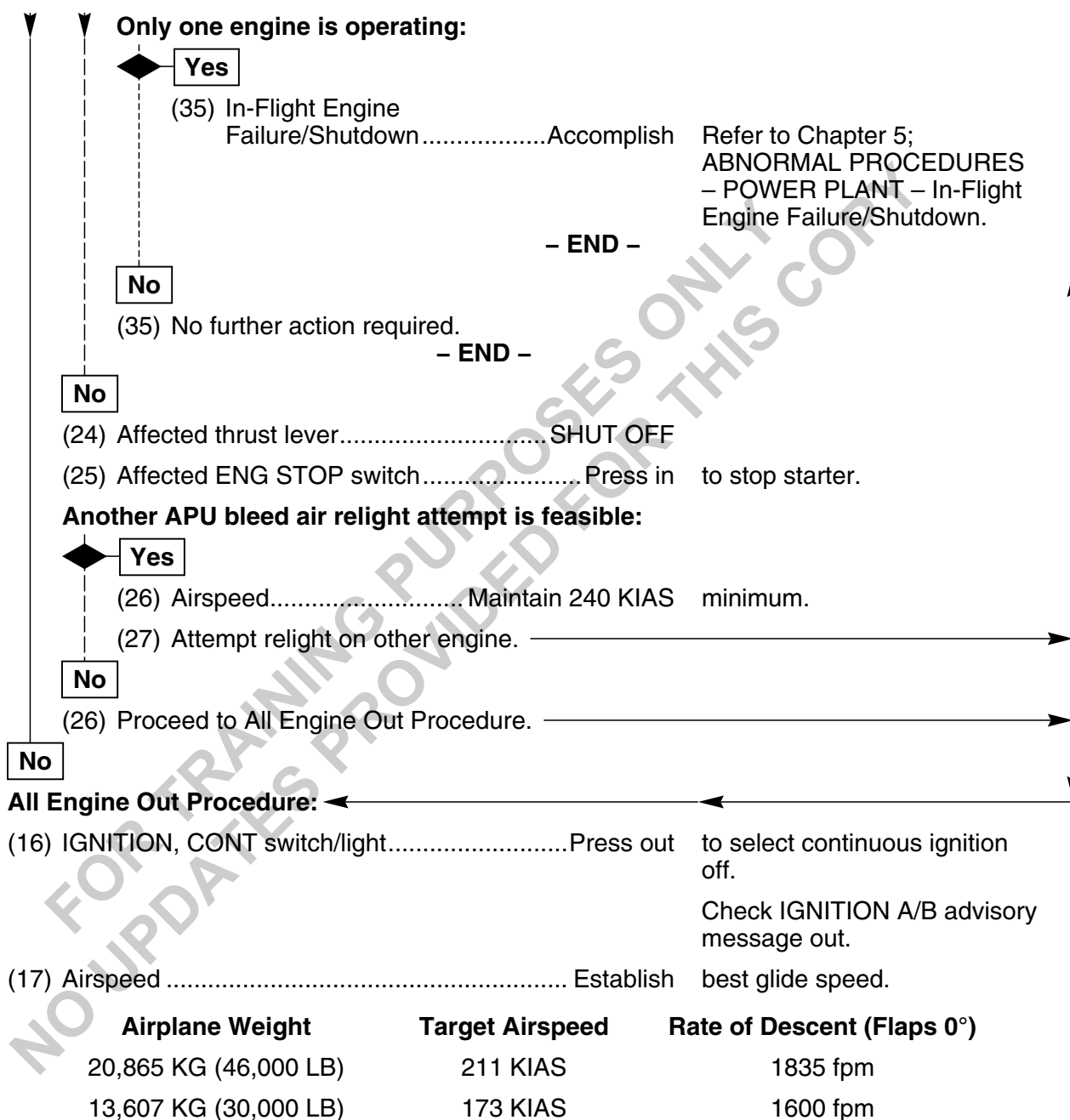
1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

- (26) BLEED AIR, APU LCV switch/light Press out to close LCV.
Check APU LCV OPEN light out.
- (27) Applicable BLEED AIR, 10TH STAGE switch/light Press in to open bleed air valve.
Check applicable 10TH STAGE CLOSED light out.
- (28) Applicable AIR-CONDITIONING, PACK switch/light Press in to restart pack.
Check applicable PACK OFF light out.
- NOTE**
Use only one air-conditioning pack during single engine operations, unless the APU is the bleed source.
- (29) IGNITION, CONT switch/light Press out to select continuous ignition off.
Check IGNITION A/B advisory message out.
- Re-establish normal power:**
- (30) ADG manual deploy handle Stow
- (31) ADG, PWR TXFR OVERRIDE switch Press to override.
- (32) CAS messages Review
- (33) Affected systems Restore
- (34) Other engine Start Refer to Chapter 5; ABNORMAL PROCEDURES – AIR START.

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

**NOTE**

1. Target glide speed provides a range of approximately 2.5 NM for each 1,000 feet altitude.
2. Refer to Flight Planning and Cruise Control Manual, Chapter 6: Special Operational Data – Descent with Two Engines Inoperative, for detailed descent (time/distance) performance data.

1. POWER PLANT (CONT'D)

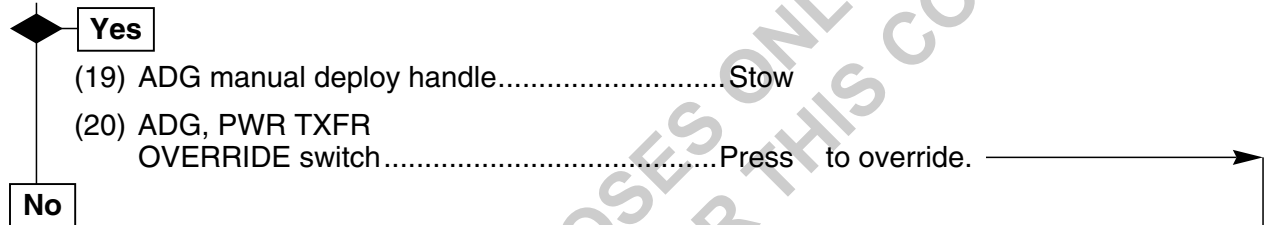
C. Double Engine Failure (Cont'd)

(18) Descent and approachPlan

NOTE

Recommended approach is a descending 360° turn starting at 5,000 feet above the landing area.

APU GEN available:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights
- SPS altitude compensation
- Auto throttle system

At 10,000 feet MSL or lowest safe altitude:

- (19) AIR-CONDITIONING,
RAM-AIR switch/light.....Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (20) TAWS WARNING, FLAPS OFF
switch/light.....Press in to mute flap aural warning.
- (21) TAWS WARNING, RAAS OFF
switch/light (if installed)Press in Check OFF light on.



1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

- (22) Approach start point Establish at 5,000 feet AGL over landing area.
- (23) Turn (20° to 30° bank angle) Start over landing area.
- (24) Target airspeed Maintain
- (25) Downwind leg Establish at 2,500 feet AGL abeam landing area and 180° from final approach heading.
- (26) FLAPS Select to 20° for landing.
- (27) LDG GEAR lever DN prior to turning final.
- (28) Final approach Establish at 1,500 feet AGL.
- (29) Final approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (30) Actual landing distance Increase as applicable by factor given below:

APU generator available	Emergency power only
1.55 (55%)	3.20 (220%)

When time and conditions permit:

- (31) Ditching and Forced Landing procedures Review Refer to DITCHING AND FORCED LANDING procedures in this chapter.

————— **END** —————

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		Oct 27/15

1. POWER PLANT (CONT'D)

D. Jet Pipe Overheat

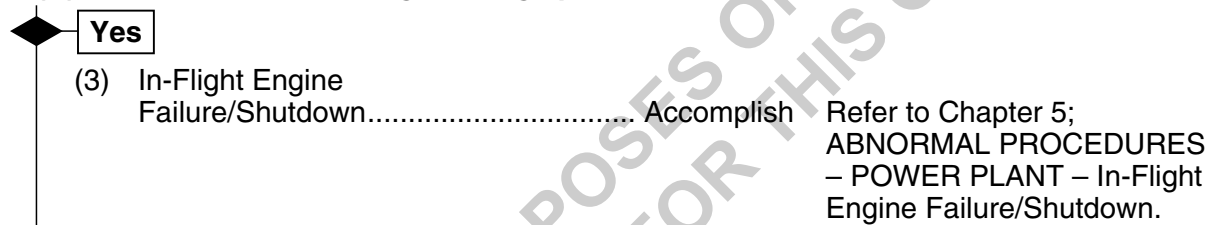
Indication: L (R) JETPIPE OVHT warning message and “JETPIPE OVERHEAT” aural on.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard slowly until warning message goes out.

L (R) JETPIPE OVHT warning message persists:



No

(3) No further action required.

END

CHALLENGER 650	EMERGENCY PROCEDURES Power Plant	03-03-14
		Oct 27/15

1. POWER PLANT (CONT'D)

E. Engine Oil Pressure Low

Indication: L (R) ENG OIL PRESS warning message, "ENGINE OIL" aural on, or left or right engine oil pressure is below 25 psi.

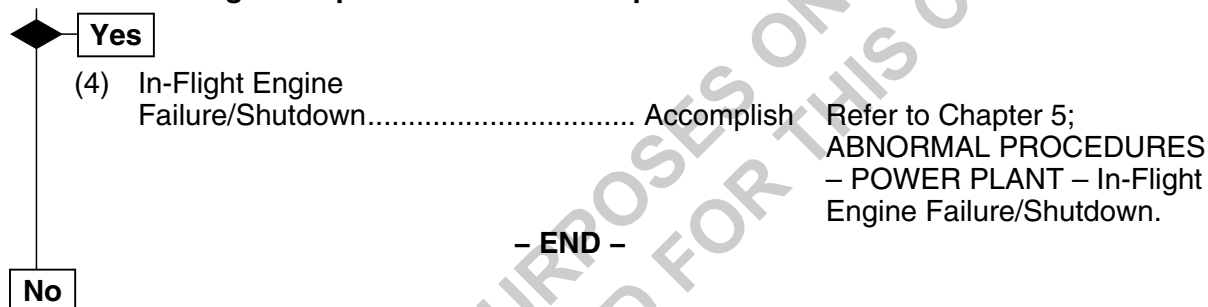
(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and IDLE

(3) Oil pressure Check

**L (R) ENG OIL PRESS warning message is on
and affected engine oil pressure is below 25 psi:**



1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment

Indication: L (R) REV UNLOCKED caution message.

On ground:

Yes



If the L or R REV UNLOCKED caution message is on during ground operations, ensure that the affected thrust lever is at IDLE. Do not cycle the thrust reverser(s). Maintenance inspection is required.

- END -

No

In flight:

Affected engine:

- (1) Thrust lever Confirm and IDLE
- (2) ATS DISC switch Depress
- (3) THRUST REVERSER, EMER
STOW switch/light Press in

Note that:

- L (R) REV UNLOCKED caution message will remain, and
- Stow indicated by improved airplane handling and normal operation of affected thrust lever.

- (4) Airspeed Not more than 200 KIAS.

Anti-ice is on:

Yes

- (5) ANTI-ICE, WING switch Select to OFF.

- (6) Affected ANTI-ICE, COWL
switch/light Press out to select cowl anti-ice off.

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No

Thrust reverser stows:

Yes

CAUTION

Limit airspeed to not more than 200 KIAS.

(5) Thrust levers..... As required

(6) THRUST REVERSER, L and
R switches Select to OFF.Check L and R REV ARMED
advisory messages out.**CAUTION**

Do not arm thrust reverser.

(7) Land at the nearest suitable airport.

Anti-ice is required:

Yes

(8) ANTI-ICE, WING switch..... Select to NORM.

(9) Affected ANTI-ICE,
COWL switch/light Press in to select cowl anti-ice on.

No

Prior to landing:

(8) TAWS WARNING, FLAPS
OFF switch/light..... Press in to mute flap aural warning.(9) TAWS WARNING,
RAAS OFF switch/light
(if installed) Press in Check OFF light on.

(10) FLAPS Select to 20° for landing.

(11) Final approach speed Not less than V_{REF} (Flaps 45°) + 22 KIAS.

(12) Actual landing distance..... Increase by a factor of 1.70 (70%).

- END -

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No

Thrust lever locked at idle, and continued airplane buffet and yaw:

Affected engine:

- (5) Thrust lever Confirm and
SHUT OFF

CAUTION

Do not relight engine.

- (6) Land at the nearest suitable airport.
- (7) FUEL, TAIL TANK TRANSFER,
MANUAL OVRD switch/light Press out if MANUAL light on.
- (8) Affected FUEL, L (R) BOOST
PUMP switch/light Press out to disarm affected boost
pump.
Check L (R) FUEL PUMP
caution message on.
- (9) Fuel system Check and monitor the following:
- Fuel quantity, and
 - Fuel balance.

Anti-ice is required:

Yes

- (10) BLEED AIR, 14TH STAGE
ISOL switch/light Press in to open ISOL valve.
Check 14TH ISOL OPEN light
on.
- (11) ANTI-ICE, WING switch Select to NORM.
- (12) Leave icing conditions (if possible).

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No

- (10) HYDRAULIC pump switch.....Select applicable pump to ON. ←
- If left engine shut down, turn 1B ON. Check that HYD 1 LO PRESS caution message out.
 - If right engine shut down, turn 2B ON. Check that HYD 2 LO PRESS caution message out.
- (11) AIR-CONDITIONING, L (R) PACK switch/lightPress out to shutdown affected pack.
Check L (R) PACK OFF light on.
- (12) BLEED AIR, 10TH STAGE L (R) switch/lightPress out to close affected shut-off valve.
Check L (R) CLOSED light on.

NOTE

Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th-stage bleed source.

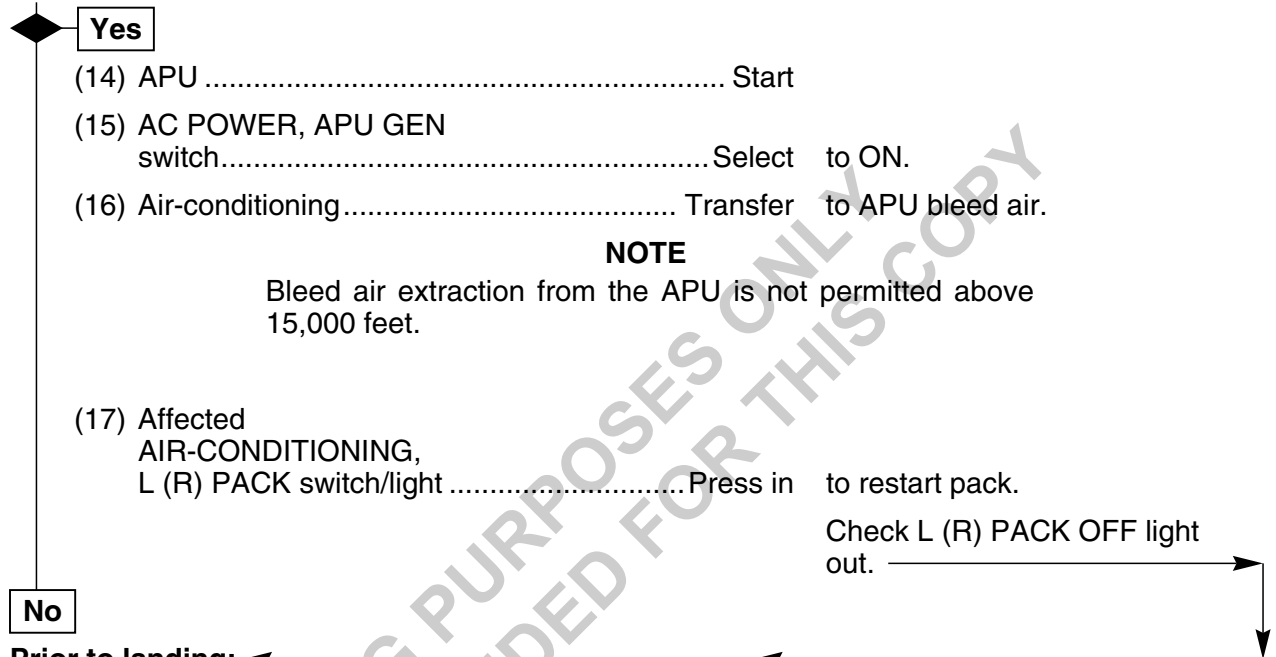
- (13) AC POWER, GEN 1 (2) switch.....Select affected generator to OFF.
Check GEN 1 (2) OFF caution message on.



1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

Below 20,000 feet and APU is available:



Prior to landing:

- (14) TAWS WARNING, FLAPS OFF
switch/light Press in to mute flap aural warning.
- (15) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (16) FLAPS Select to 20° for landing.
- (17) Final approach speed Not less than V_{REF} (Flaps 45°) + 22 KIAS.
- (18) Actual landing distance Increase by a factor of 1.70 (70%).

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

END

1. POWER PLANT (CONT'D)

G. Post-Shutdown Engine Tail Pipe Fire

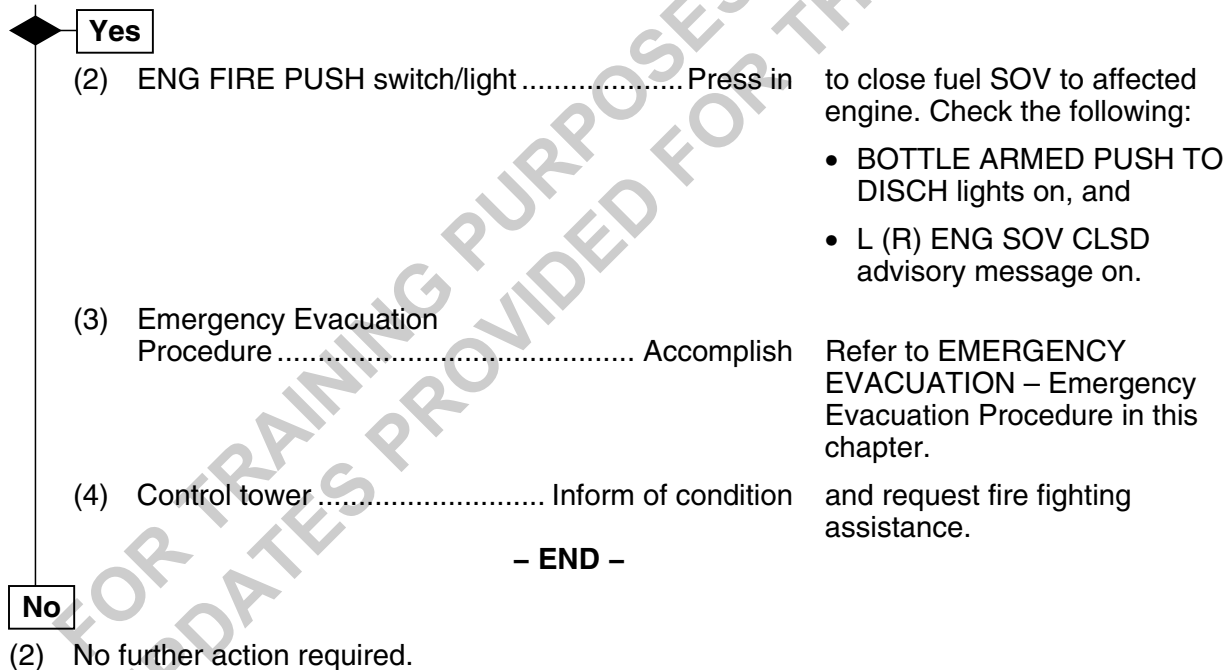
Indication: Engine ITT exceeds 350°C and ITT is rising rapidly, or if flame is reported at engine tailpipe.

Affected engine:

- (1) Dry motor..... Accomplish until ITT is reduced below 150°C or starter limit.

NOTE

The APU or ground air source (if available) may be used to dry motor the engine.

Tail pipe fire indications persist:**END**

1. POWER PLANT (CONT'D)

H. N₁/N₂ Overspeed

Indication: ENGINE warning message. Engine N₁ or N₂ readout indicates engine limits being exceeded.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard until engine responds.

Engine limits continue to be exceeded:

Yes

(3) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

– END –

No

(3) Both thrust levers Retard to below 79% N₂.
(4) Both ENGINE CONTROL ENG SPEED switches OFF to prevent asymmetric thrust.

NOTE

The APR system is inoperative when the ENG SPEED switches are selected OFF.

(5) Both thrust levers Advance

NOTE

Thrust lever controls are more sensitive to changes.

Engine accelerates to desired power:

Yes

(6) Engine indications Monitor

NOTE

Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

– END –

No

(6) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

— END —

1. POWER PLANT (CONT'D)

I. Engine Overtemperature

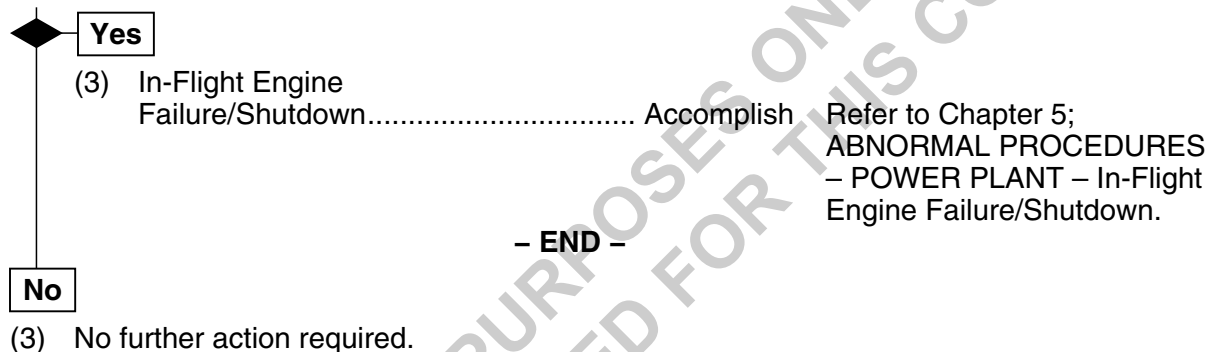
Indication: ENGINE warning message. Engine ITT readout indicates engine limits being exceeded.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard until ITT below limit range.

ITT limits continue to be exceeded:



– END –

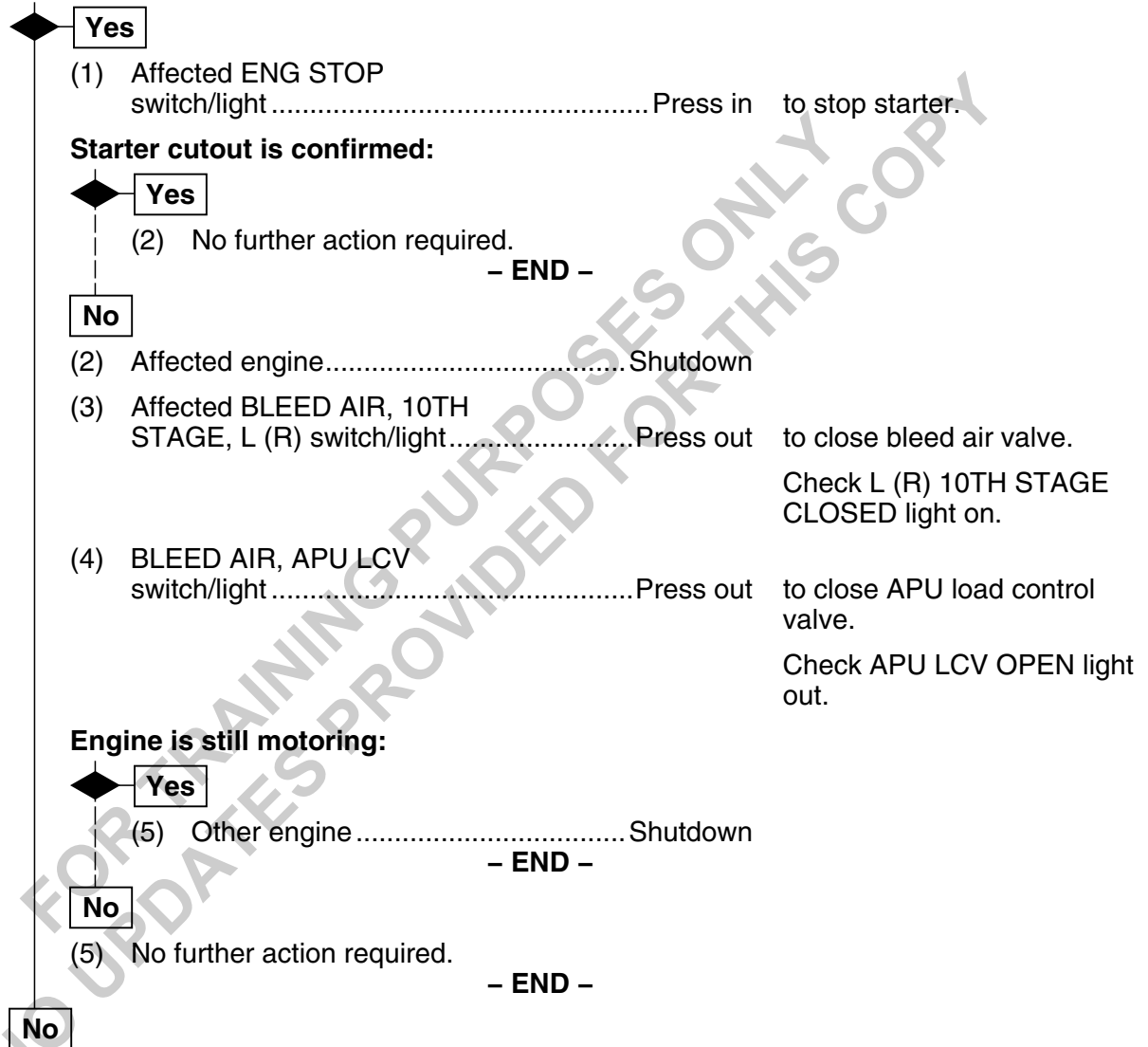
END

1. POWER PLANT (CONT'D)

J. Engine Starter Fails to Cutout

Indication: L (R) ENGINE START SOV warning message. L (R) ENGINE START light and IGNITION A (B) advisory message on after 55% N₂.

On ground:



In-flight:

- (1) Affected ENG STOP switch/light Press in to stop starter.
- (2) IGNITION, A (B) switch/light Press out to disarm applicable ignition system.
Check ARM light out at applicable switch/light.
- (3) Affected ENG START circuit breaker (L-1L10) (R-1L9) Open



1. POWER PLANT (CONT'D)

J. Engine Starter Fails to Cutout (Cont'd)

Engine is still motoring:



Yes

(4) Affected 10 ST B/AIR SOV
circuit breaker (L-4C11)
(R-2F11) Open

(5) Land at the nearest suitable airport.
- END -

No

(4) No further action required.

————— END —————

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1. POWER PLANT (CONT'D)

K. Uncommanded Acceleration

Indication: Engine accelerates to thrust settings other than as selected with thrust lever.

(1) ATS DISC switch Depress

On ground, affected engine:

(2) Thrust lever Confirm and IDLE then SHUT OFF.

During flight, affected engine:

(2) Thrust lever Confirm and IDLE Check engine response.

CAUTION

The ATS **SPEED** mode N_1 synchronization will attempt inappropriate deceleration/acceleration of the remaining engine to maintain synchronization until disengaged.

Engine responds:

Yes

(3) Both thrust levers Advance to below 79% N_2 .

(4) Both ENGINE CONTROL
ENG SPEED switches OFF

NOTE

The APR system is inoperative when the ENG SPEED switches are selected OFF.

(5) Both thrust levers Advance

CAUTION

Do not change power settings abruptly as engine response is degraded.

(6) Engine indications Monitor

NOTE

Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

– END –

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		Rev 2, Mar 11/16

1. POWER PLANT (CONT'D)

K. Uncommanded Acceleration (Cont'd)

No

- (3) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– POWER PLANT – In-Flight
Engine Failure/Shutdown.

————— END —————

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CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-1 Oct 27/15
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1. SMOKE/FIRE/FUMES

A. Smoke/Fire/Fumes Procedure

WARNING

Time is critical during smoke/fire/fumes emergencies. The flight crew should consider an immediate landing anytime the situation cannot be controlled.

CAUTION

Passenger oxygen masks should not be deployed when performing smoke or fire procedures.

- (1) Oxygen masks/smoke goggles
(if required) Don Set to 100% and EMERGENCY.
- (2) Crew communication Establish Set MASK/BOOM switch to MASK.
- (3) Land at the nearest suitable airport.

CAUTION

Continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke or fumes have dissipated.

WARNING

Anytime smoke or fumes becomes the greatest threat, accomplish the Smoke or Fumes Removal Procedure. Refer to Smoke or Fumes Removal Procedure in this section.

Time and conditions permit and source of smoke/fire/fumes can be quickly identified and eliminated:

Yes

- (4) Appropriate procedure Accomplish
 - Air-Conditioning..See SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM
 - Electrical.....See SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM
 - CabinSee SOURCE IS IDENTIFIED AS CABIN

– END –

CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-2
		Oct 27/15

1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

No

- (4) Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

– END –

SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM

APU is supplying bleed air to the packs:

Yes

- (1) Air-conditioning Transfer to engine bleeds. →

No

- (1) AIR-CONDITIONING, L PACK switch/light Press out to shutdown left pack.
Check L PACK OFF light on.

Monitor and determine if smoke/fire/fumes persists.

Smoke/fire/fumes persists:

Yes

- (2) AIR-CONDITIONING, L PACK switch/light Press in Check L PACK OFF light out.
- (3) AIR-CONDITIONING, R PACK switch/light Press out to shutdown right pack.
Check R PACK OFF light on.

Monitor and determine if smoke/fire/fumes persists.

Smoke/fire/fumes persists:

Yes

- (4) AIR-CONDITIONING, R PACK switch/light Press in Check R PACK OFF light out. →

No

If required:

- (2) Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

– END –



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1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM

(1) AC and DC electrical loads Monitor

Source of electrical smoke/fire/fumes is positively identified:

◆ **Yes**

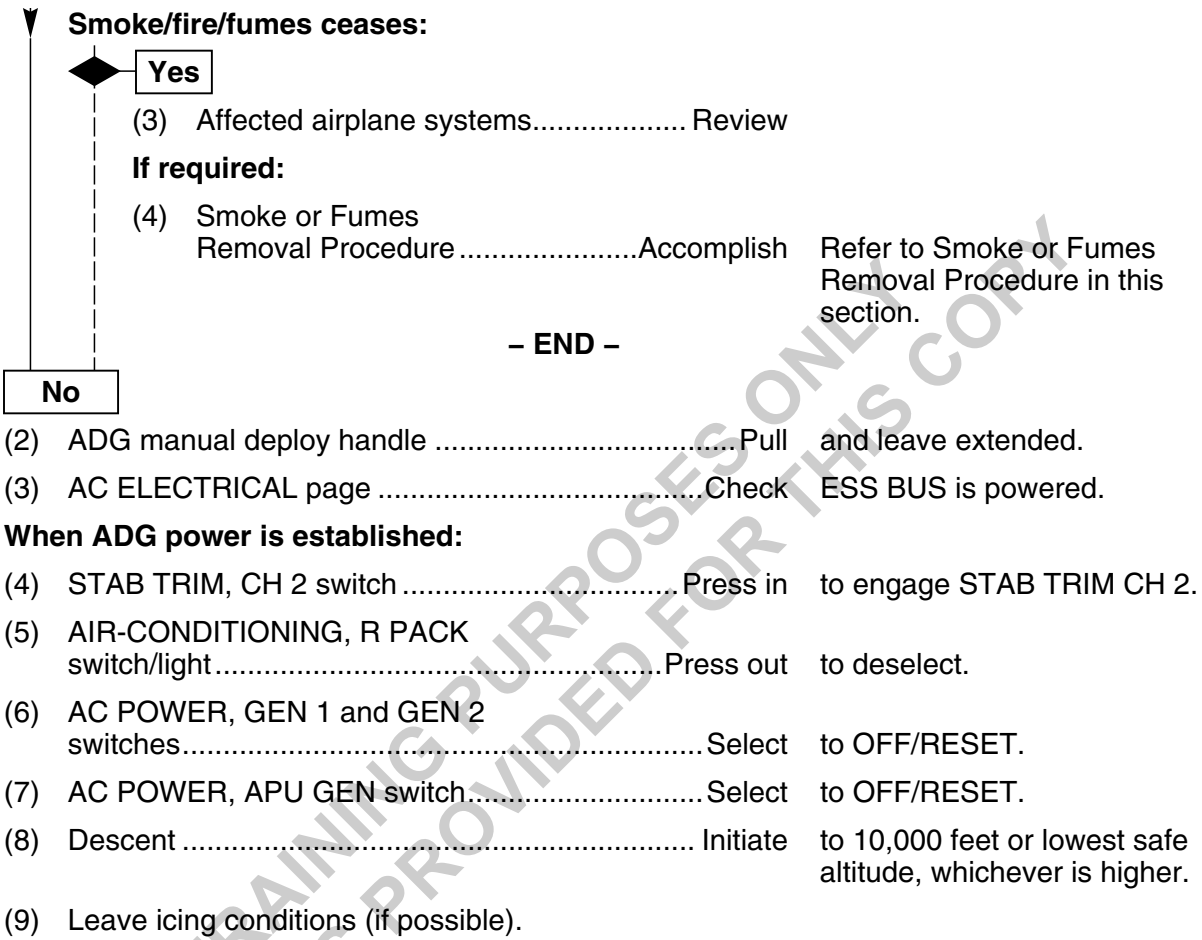
WARNING

The table that follows provides guidance on how to isolate each electrical bus. Select only the steps required to isolate the positively identified bus. Do not use the table in an attempt to identify the source bus by trial and error.

(2) Affected electrical system..... Isolate See table below:

IDENTIFIED BUS	APPLICABLE CIRCUIT BREAKER/SWITCH
AC BUS 1 and DC BUS 1	AC ESS XFER..... PRESS IN LEFT AUTOXFER PRESS IN GEN 1 SWITCH OFF
AC BUS 2 and DC BUS 2	RIGHT AUTOXFER..... PRESS IN GEN 2 SWITCH OFF
DC BUS 1 CONTROL	FEED CB (1E8) OPEN
DC BUS 2 CONTROL	FEED CB (2E8) OPEN
AC ESS BUS	115 VAC ESS FEED CBs (1B5 and 2B8) OPEN
DC ESS BUS	RCCB CONT, FEED 1 and FEED 2 DC ESS CBs (2M6, 2M8 and 2M10) OPEN
DC BATT BUS	RCCB CONT MAIN and APU BATT CBs (2M4 and 2M5) and FEED 1 and FEED 2 BATT BUS CBs (2M7 and 2M9) OPEN
DC EMER BUS	APU BATT FEED CB (1S7) and EMERG BUS FEED CB (1M1) OPEN
AC UTLY 1 and 2 DC UTLY 1 and 2	AC/DC UTILITY PRESS IN

1. SMOKE/FIRE/FUMES (CONT'D)
A. Smoke/Fire/Fumes Procedure (Cont'd)

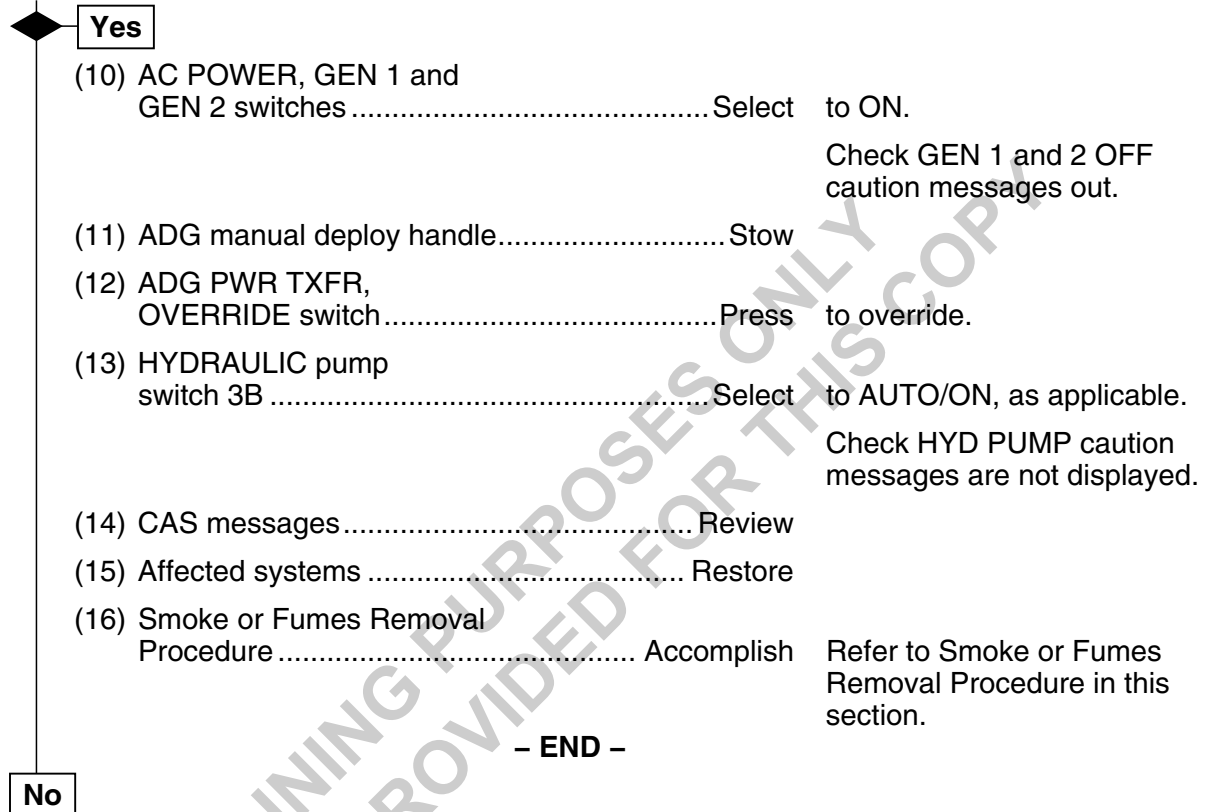


CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-5
		Oct 27/15

1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

Smoke/fire/fumes persists:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights
- SPS altitude compensation
- Auto throttle system



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CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-6
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1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

NOTE

1. With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.
2. Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin Δpressure on EICAS.
3. Flaps will operate (extend or retract) at half speed.

(10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

(11) Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

— END —

SOURCE IS IDENTIFIED AS CABIN

- (1) Designated crew member Advise to isolate and extinguish source of fire or smoke, and to secure the area as necessary.
- (2) AC/DC UTILITY switch/light Press in to isolate AC and DC utility busses.
Check OFF light on.
- (3) Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

————— END —————

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CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-7
		Oct 27/15

1. SMOKE/FIRE/FUMES (CONT'D)

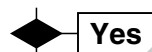
B. Smoke or Fumes Removal Procedure

- (1) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
- (2) Passenger oxygen Select to CLOSED.
- (3) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select manual control.
Check PRESS CONT
MANUAL light on.
- (4) CABIN PRESSURIZATION,
MAN ALT selector Select UP and hold.
- (5) CABIN PRESSURIZATION,
MAN RATE control Select to maximum INCR.
- (6) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press in Check the following:
 - EMER DEPRESS ON light on, and
 - EMER DEPRESS caution message on.

NOTE

To assist in smoke clearance, the ram air vent may be used. Maintain at least 250 KIAS to clear smoke using the ram air vent.

Smoke/fire/fumes is uncontrolled:



- (7) Land immediately at the nearest suitable airport.



Dependant upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

- (8) Emergency Evacuation
Procedure Accomplish Refer to EMERGENCY
EVACUATION – Emergency
Evacuation Procedure in this
chapter.

– END –

CHALLENGER 650	EMERGENCY PROCEDURES Smoke or Fire	03-04-8
		Oct 27/15

1. SMOKE/FIRE/FUMES (CONT'D)

B. Smoke or Fumes Removal Procedure (Cont'd)

No

If required:

(7) Smoke/Fire/Fumes Procedure Continue Refer to Smoke/Fire/Fumes Procedure in this section.

————— **END** —————

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CHALLENGER 650	EMERGENCY PROCEDURES Air-Conditioning and Pressurization	03-05-1
		Oct 27/15

1. AIR-CONDITIONING AND PRESSURIZATION

A. Emergency Descent Procedure

(1) Oxygen masks.....	Don	set to 100%.
(2) Crew communication	Establish	set MASK/BOOM switch to MASK.
(3) PASS SIGNS, NO SMKG and SEAT BLTS switches	Select	both to ON.
(4) Passenger oxygen.....	Deploy	
(5) Descent	Initiate	to 10,000 feet or lowest safe altitude, whichever is higher.
(6) Thrust levers.....	IDLE	
(7) ATS DISC switch	Depress	
(8) FLIGHT SPOILER lever	Select	to MAX.
		Check EICAS indicates deployment.

- (9) Airspeed Do not exceed
 V_{MO}/M_{MO} .



If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.



Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- (10) Oxygen and masks..... As required

NOTE

1. Closing the doors on the mask stowage compartments and pressing RESET will stop the flow of oxygen to the masks.
2. If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

- (11) Passenger oxygen..... Select to CLOSED then NORMAL to reset passenger oxygen.

————— **END** —————

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1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Uncontrollable Loss of Cabin Pressure

Indication: CABIN ALT warning message and “CABIN PRESSURE” aural on.

(1) Oxygen masks..... Don

set to 100%.

(2) Crew communication Establish

set MASK/BOOM switch to MASK.

(3) PASS SIGNS, NO SMKG and SEAT BLTS switches Select

both to ON.

(4) Passenger oxygen..... Deploy

(5) Descent Initiate

to 10,000 feet or lowest safe altitude, whichever is higher.

(6) Thrust levers..... IDLE

(7) ATS DISC switch Depress

(8) FLIGHT SPOILER lever Select

to MAX.

Check EICAS indicates deployment.

- (9) Airspeed Do not exceed
- V_{MO}/M_{MO}.



If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.



Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- (10) Crew oxygen masks As required

NOTE

1. Closing the doors on the crew mask stowage compartments and pressing RESET will stop the flow of oxygen to the crew masks.
2. If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

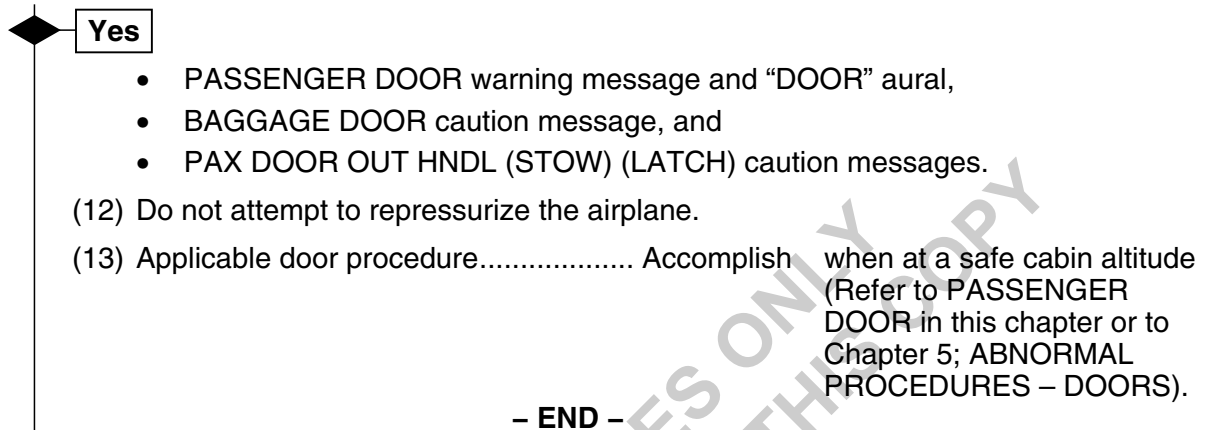
- (11) Passenger oxygen..... Select
- to CLOSED then NORMAL to reset passenger oxygen.

CHALLENGER 650	EMERGENCY PROCEDURES Air-Conditioning and Pressurization	03-05-3
		Oct 27/15

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Uncontrollable Loss of Cabin Pressure (Cont'd)

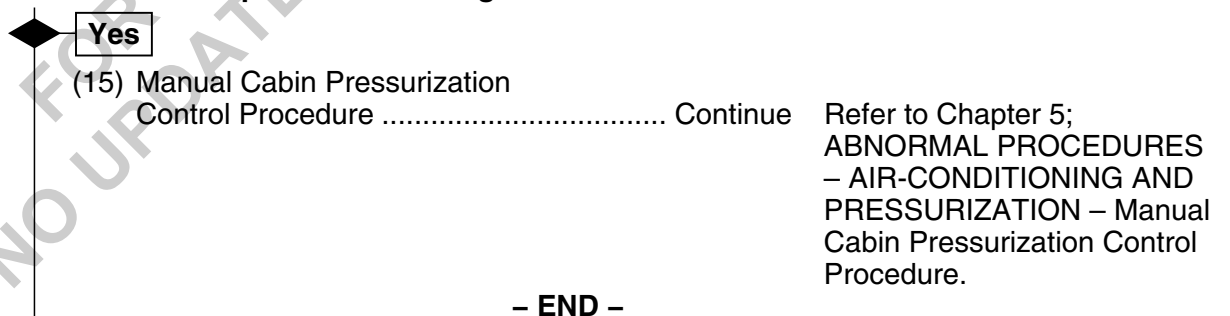
Accompanied by any door/hatch message or unsafe condition:



Attempt to regain control of cabin pressurization:

- (12) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual mode.
Check PRESS CONT MANUAL light on.
- (13) CABIN PRESSURIZATION, MAN ALT selector Select DN and hold to decrease cabin altitude.
- (14) CABIN PRESSURIZATION, MAN RATE switch Select to maximum INCR.

Control of cabin pressurization regained:



- (15) Unpressurized Flight Procedure Accomplish if required (Refer to Chapter 5; ABNORMAL PROCEDURES – AIR-CONDITIONING AND PRESSURIZATION – Unpressurized Flight Procedure).

————— **END** —————

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1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

C. High Rate of Cabin Pressurization or Over-Pressurization

Indication: DIFF PRESS warning message.
Differential pressure indicated as greater than 9.0 psi.

- (1) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select Manual mode.
Check PRESS CONT MANUAL light on.
- (2) CABIN PRESSURIZATION, MAN
ALT selector Select UP and hold.
- (3) CABIN PRESSURIZATION, MAN
RATE switch..... As required to increase cabin altitude and
reduce differential pressure.

DIFF PRESS warning message persists:

- Yes

- (4) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
 - (5) AIR-CONDITIONING, L (R)
PACK switch/light Press out to shutdown one pack.
Check L (R) PACK OFF light on.

DIFF PRESS warning message persists:

- Yes

- (6) AIR-CONDITIONING, L
and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.

DIFF PRESS warning message persists:

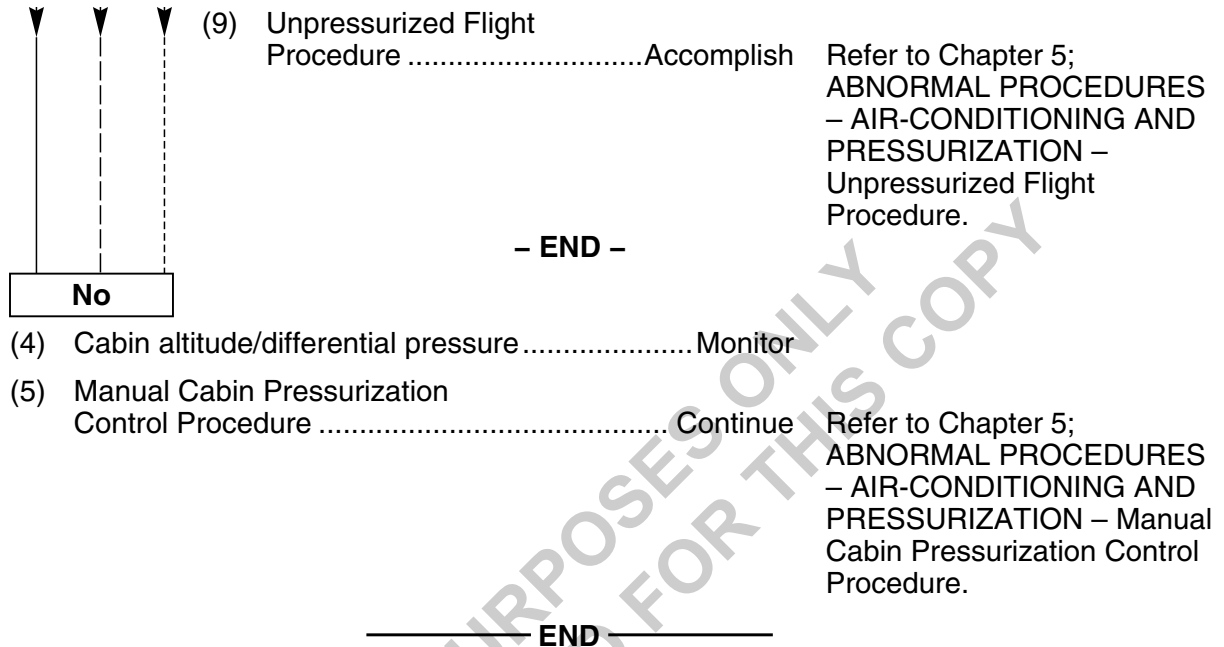
- Yes

- (7) CABIN
PRESSURIZATION,
EMER DEPRESS
switch/light Press in to depressurize the airplane.
Check EMER DEPRESS caution message on.
 - (8) Descent..... Initiate to the lowest possible altitude.

CHALLENGER 650	EMERGENCY PROCEDURES Air-Conditioning and Pressurization	03-05-5
		Oct 27/15

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

C. High Rate of Cabin Pressurization or Over-Pressurization (Cont'd)



CHALLENGER ⁶⁵⁰	EMERGENCY PROCEDURES Air-Conditioning and Pressurization	03-05-6
		Oct 27/15

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CHALLENGER 650	EMERGENCY PROCEDURES Automatic Flight Control System	03-06-1
		Oct 27/15

1. AUTOMATIC FLIGHT CONTROL SYSTEM

NOTE

After generator transfers, AFCS channel 2 will momentarily fail and the following must be manually re-engaged:

- Yaw damper 2 (YD 2), and
- Autopilot (if it was coupled before the power interruption, and operating on AFCS channel 2).

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or "CAVALRY CHARGE" aural and/or flashing AP (red) message on the primary flight displays.

- (1) Autopilot Disconnect using AP/SP DISC switch on the control wheel, or the AP DISC switch on the FCP.
- (2) AFCS SEL 1 (2) switch (centre pedestal)..... Select to operative AFCS channel.
Check AFCS 1 (2) INOP status message on.
- (3) Autopilot Re-engage

Autopilot warning persists:

- Yes**
- (4) Autopilot..... Disconnect using AP/SP DISC switch on the control wheel, or the AP DISC switch on the FCP.
 - (5) Manual control Resume
— END —
- No**
- (4) No further action required.

— END —

CHALLENGER 650	EMERGENCY PROCEDURES Automatic Flight Control System	03-06-2
		Oct 27/15

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

B. AFCS Messages Failure

Indication: AFCS MSGS FAIL warning message and/or flashing AP (red) message on the primary flight displays.

- (1) Autopilot Disconnect using AP/SP DISC switch on the control wheel, or the AP DISC switch on the FCP.

NOTE

Do not use the autopilot when an AFCS MSGS FAIL warning message is displayed.

- (2) Manual control..... Resume

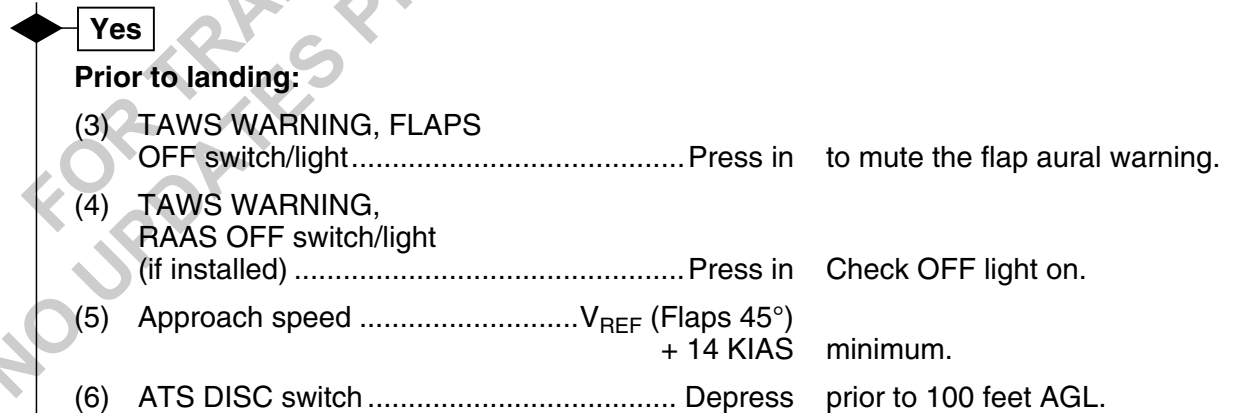
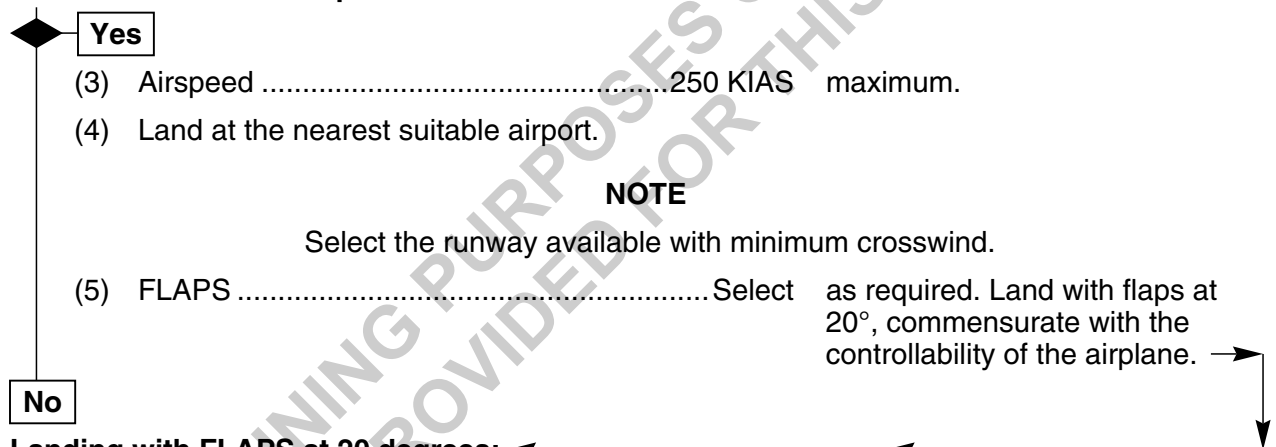
END

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Uncommanded Yaw Motion

Indication:	Abnormal/uncommanded change in yaw attitude. Continuous or intermittent rudder "kicks". Sustained oscillation ("Dutch Roll").	
(1) Controls	Assume	manual control and counter aircraft motion using handwheel inputs.
(2) YAW DAMPER, DISC button	Press in	to disconnect both yaw dampers.

Uncommanded motion persists:



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Uncommanded Yaw Motion (Cont'd)

- ▼ (7) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

No

- (3) No further action required.

— END —

CHALLENGER 650	EMERGENCY PROCEDURES Auxiliary Power Plant	03-07-1
		Oct 27/15

1. AUXILIARY POWER PLANT

A. APU Fire

Indication: APU FIRE PUSH light on, APU FIRE warning message and fire bell on.

(1) APU FIRE PUSH switch/light Press in Check the following:

- BOTTLE ARMED PUSH TO DISCH light on, and
- APU SOV CLSD advisory message on.

After 5 seconds and APU fire warning persists:

(2) BOTTLE ARMED PUSH TO DISCH switch/light..... Press in and hold until light out, to discharge firex bottle.

Check the following:

- BOTTLE ARMED PUSH TO DISCH light out, and
- APU BOTTLE LO caution message on.

APU fire warning persists during flight:

◆ Yes

(3) Land at the nearest suitable airport. →

No

APU is supplying bleed air for air-conditioning: ←

◆ Yes

(3) Air-conditioning Transfer to engine bleeds. →

No

(3) APU Shutdown ←

(4) Do not restart APU.

————— **END** —————

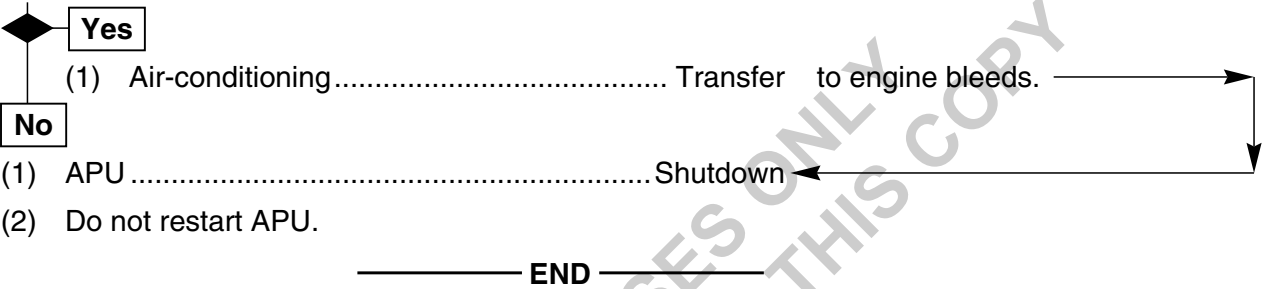
	CL-650 Airplane Flight Manual PSP 650-1	
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1. AUXILIARY POWER PLANT (CONT'D)

B. APU Overspeed

Indication: APU OVERSPEED warning message and “APU” aural on, APU RPM gauge and readout indicate speed equal to or greater than 110%, and APU automatically shuts down at 107%.

APU is supplying bleed air for air-conditioning:



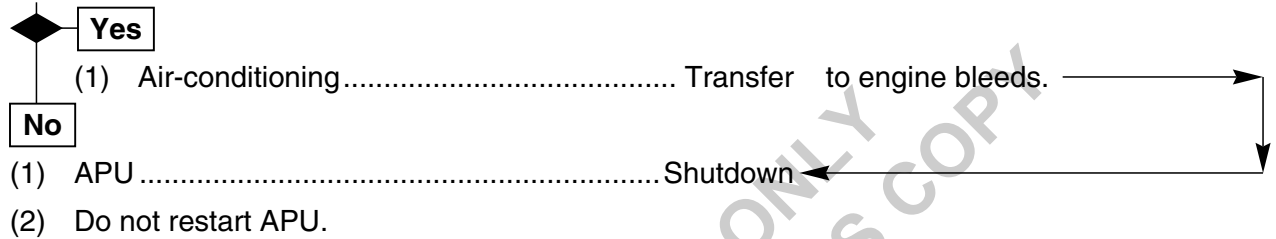
CHALLENGER 650	EMERGENCY PROCEDURES Auxiliary Power Plant	03-07-3
		Oct 27/15

1. AUXILIARY POWER PLANT (CONT'D)

C. APU Overtemperature

Indication: APU OVERTEMP warning message and “APU” aural on, APU EGT gauge and readout indicate temperature greater than 732°C, and APU shuts down automatically.

APU is supplying bleed air for air-conditioning:



————— **END** —————

CHALLENGER 650	EMERGENCY PROCEDURES Auxiliary Power Plant	03-07-4
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CHALLENGER 650	EMERGENCY PROCEDURES Bleed Air Leaks	03-08-1
		Oct 27/15

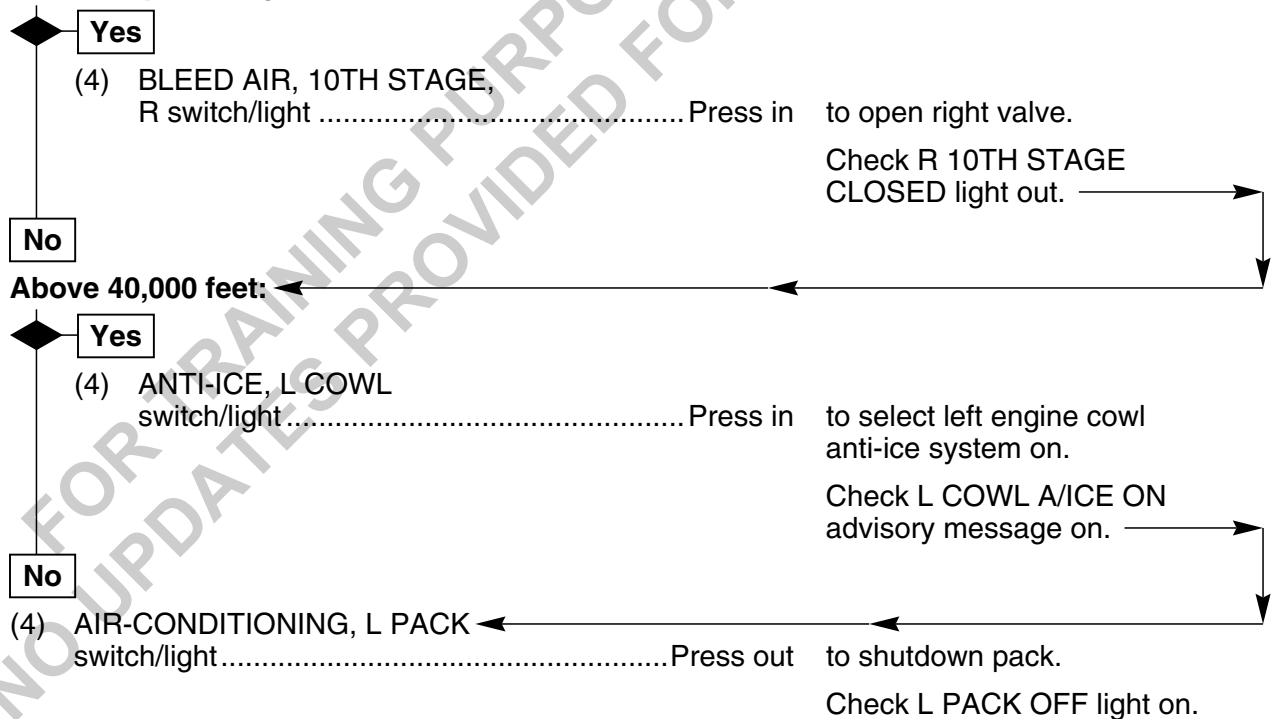
1. BLEED AIR LEAKS

A. 10TH Stage, Left Bleed Duct Failure

Indication: L 10TH DUCT warning message and “BLEED AIR DUCT” aural on:

- (1) BLEED AIR, 10TH STAGE, L switch/light Press out to close bleed air valve.
Check L 10TH STAGE CLOSED light on.
- (2) BLEED AIR, APU LCV switch/light Press out to close APU load control valve.
Check APU LCV OPEN light out.
- (3) BLEED AIR, 10TH STAGE ISOL switch/light Press out to close ISOL valve.
Check 10TH ISOL OPEN light out.

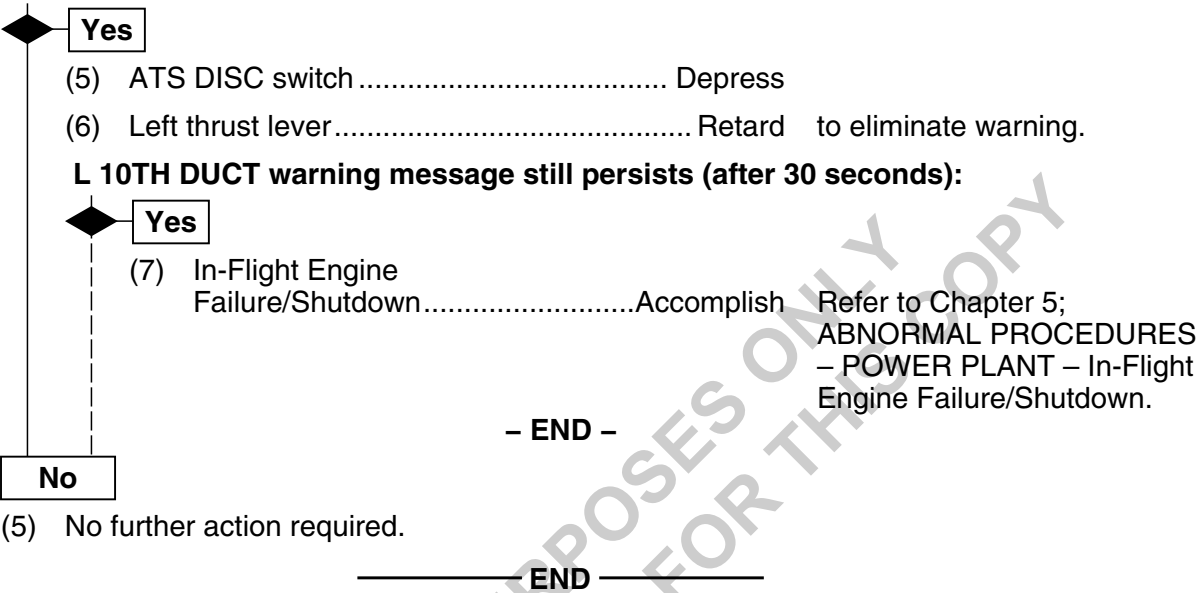
APU was providing bleed air:



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1. BLEED AIR LEAKS (CONT'D)
A. 10TH Stage, Left Bleed Duct Failure (Cont'd)

Warning message persists:



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		Oct 27/15

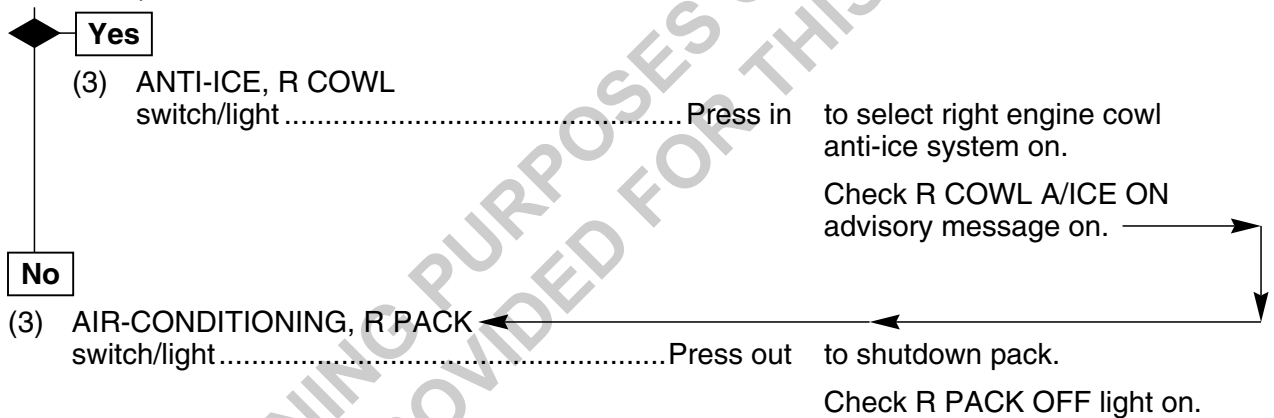
1. BLEED AIR LEAKS (CONT'D)

B. 10TH Stage, Right Bleed Duct Failure

Indication: R 10TH DUCT warning message and "BLEED AIR DUCT" aural on:

- (1) BLEED AIR, 10TH STAGE, R switch/light Press out to close bleed air valve.
Check R 10TH STAGE CLOSED light on.
- (2) BLEED AIR, 10TH STAGE ISOL switch/light Press out to close ISOL valve.
Check 10TH ISOL OPEN light out.

Above 40,000 feet:



Warning message persists:

- Decision: **Yes**
- (4) ATS DISC switch Depress
 - (5) Right thrust lever Retard to eliminate warning.

R 10TH DUCT warning message still persists (after 30 seconds):

- Decision: **Yes**
- (6) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.
- END -

- Decision: **No**
- (4) No further action required.
- END -

1. BLEED AIR LEAKS (CONT'D)

C. 14TH Stage, Left/Right Bleed Duct Failure

Indication: L (R) 14TH DUCT warning message and “BLEED AIR DUCT” aural on.

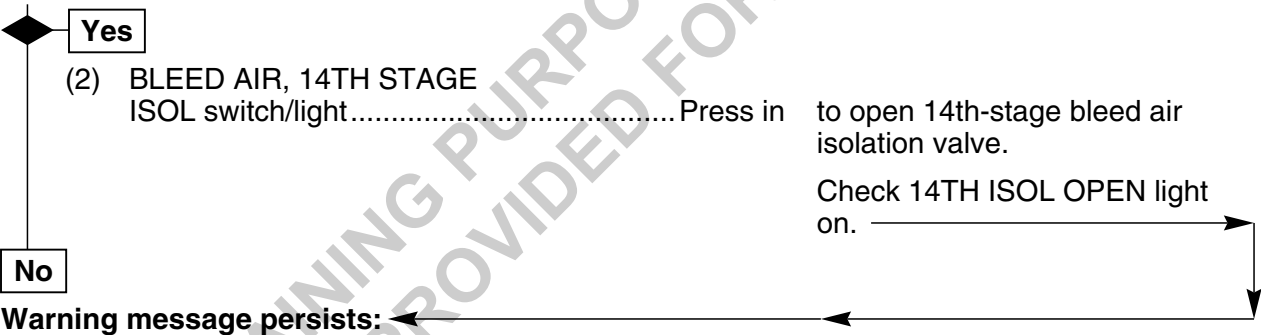
Affected engine:

- (1) BLEED AIR, 14TH STAGE L (R) switch/light.....Press out to close bleed air valve on affected engine.
- Check L (R) 14TH STAGE CLOSED light on.

NOTE

1. Avoid icing conditions.
2. Cowl anti-icing will not be available on the affected engine.
3. The thrust reverser is inoperative on the affected engine.

Wing anti-ice required:



– END –

————— END —————

CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-1
		Oct 27/15

1. DITCHING AND FORCED LANDING

TO ENSURE A SAFE WATER LANDING:	TO ENSURE A SAFE FORCED LANDING:
• Approach speed..... V_{REF}	• Approach speed..... V_{REF}
• Descent rate (if thrust available)..... 200 – 300 fpm	• Descent rate (if thrust available)..... 200 – 300 fpm
• Landing gear Retracted	• Landing gear As required
• Flaps setting..... 45 degrees	• Flaps setting..... 45 degrees
• IF FLAPS 45 IS NOT POSSIBLE, USE THE MOST EXTENDED FLAP SETTING AVAILABLE. • FINAL APPROACH SPEED: • FLAPS 0: $V_{REF} + 30$ KTS • FLAPS 20: $V_{REF} + 14$ KTS • FLAPS 30: $V_{REF} + 7$ KTS	

A. Ditching or Forced Landing Imminent

This procedure is intended for use if an immediate airplane ditching is required.

PRELIMINARY

(1) AIR-CONDITIONING, L and R PACK switch/lights	Press out	to ensure minimum differential pressure within airplane.
(2) CABIN PRESSURIZATION, EMER DEPRESS switch/light	Press in	Prior to 200 feet AGL, to drive outflow valves open. Check EMER DEPRESS caution message on.

JUST BEFORE CONTACT

- (1) Thrust levers..... SHUT OFF
- (2) APU, LH and RH ENG FIRE
PUSH switch/lights (3)..... Press in
- Check all three BOTTLE
ARMED – PUSH TO DISCH
lights on.
- Check the following advisory
messages on:
- L and R ENG SOV CLSD,
 - APU SOV CLSD, and
 - HYD SOV 1 and 2 CLSD.
- (3) Water (or terrain) Contact
- with minimum forward speed,
but not less than stick shaker
speed, and at minimum sink
rate.



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-2
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

A. Ditching or Forced Landing Imminent (Cont'd)

AFTER CONTACT

When the airplane has stopped:

- (1) BOTTLE ARMED – PUSH TO DISCH switch/lights (3) Press in and hold until lights go out, to discharge firex bottles.
Check ENG BOTTLE 1 and 2 LO and APU BOTTLE LO caution messages on.
- (2) Door and overwing exit..... Open
- (3) Passenger evacuation..... Initiate
- (4) BATT MASTER switch Select to OFF, to prevent CVR erasure.
- (5) Airplane Abandon with survival equipment.

————— **END** —————

B. Planned Ditching

This procedure is intended for use where sufficient time is available.

PRELIMINARY

- (1) Descent Plan to reduce fuel to a minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach.
- (2) Crew Alert and brief type of emergency, time available, airplane attitude after ditching and exits available for ditching.
- (3) Air Traffic Control Notify
- (4) MFD or CDU, ATC/TCAS tuning Select to transponder code 7700.
- (5) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (6) Loose Equipment Secure and stow all flight compartment loose equipment.
- (7) TAWS circuit breaker (1B13)..... Open to silence TAWS aural warnings.
Check TAWS FAIL status message on.



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-3
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

- (8) AUDIO WARNING switchesDISABLE to silence EICAS (DCU) aural alerts.
Check DCU 1 and 2 AURAL INOP status messages on.
- (9) Survival equipmentCheck
- (10) Life vest, harness and belts On and tightened.

NOTE

Life vests should be donned, but not inflated, until outside of the airplane. Light plugs should be removed only if ditching at night.

- (11) Shoulder harness reel control Up to lock shoulder harness.
- (12) LDG GEAR lever Select UP for ditching.
- (13) FLAPS Select as required.

APPROACH

If possible, ditch in the vicinity of rescue vessels, near coastlines or islands.

At approximately 2,000 feet:

- (1) Sea conditions and wind direction Determine
- (2) Ditching heading Establish Ditching heading should be:
- Wind speed <15 kts; contact parallel to swells.
 - Wind speed between 15 and 45 kts; compromise between wind and swell.
 - Wind speed >45 kts; ditch into the wind.
- (3) Descent rate/approach speed Establish
- (4) AIR-CONDITIONING, L and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (5) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual control.
Check PRESS CONT MANUAL light on.
- (6) CABIN PRESSURIZATION, MAN ALT selector Select UP and hold.
- (7) CABIN PRESSURIZATION, MAN RATE control Select to MAX INCR.



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-4
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

When the airplane is completely depressurized:

- (8) BLEED AIR, 10TH STAGE L and R switch/lights Press out to close both valves.
Check L and R 10TH STAGE CLOSED lights on.
- (9) CABIN PRESSURIZATION, MAN ALT selector Select DN and hold to fully close outflow valves (approx 60 seconds).
- (10) FUEL, L and R BOOST PUMP switch/lights Press out to shutdown both boost pumps.
Check L and R FUEL PUMP ON advisory messages out.
- (11) APU (if not essential) Shut down

At 500 feet (Radio Altitude):

- (12) Crew Alert that ditching is imminent.
- (13) Radio Transmit Final Position.
- (14) BRACE FOR IMPACT ORDER over PA system : "BRACE – BRACE – BRACE".
- (15) EMER LTS switch Select to ON.
Check EMER LTS ON status message on.
- (16) LANDING LTS, LEFT and RIGHT switches Select to STDY.
- (17) LANDING LTS, NOSE switch Select to ON.

JUST BEFORE CONTACT

- (1) Thrust levers SHUT OFF
- (2) APU, LH and RH ENG FIRE PUSH switch/lights (3) Press in
Check all three BOTTLE ARMED – PUSH TO DISCH lights on.
Check the following advisory messages on:
- L and R ENG SOV CLSD,
 - APU SOV CLSD, and
 - HYD SOV 1 and 2 CLSD.



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-5
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

- (3) Water..... Contact with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

AFTER WATER CONTACT

When the airplane has stopped:

- (1) BOTTLE ARMED – PUSH TO DISCH switch/lights (3) Press in and hold until lights go out, to discharge firex bottles.
Check ENG BOTTLE 1 and 2 LO and APU BOTTLE LO caution messages on.
- (2) Door and overwing exit..... Open
- (3) Passenger evacuation..... Initiate
- (4) BATT MASTER switch Select to OFF, to prevent CVR erasure.
- (5) Airplane Abandon with survival equipment.

END

CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-6
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing

This procedure is intended for use where sufficient time is available.

PRELIMINARY

- (1) Descent Plan to reduce fuel to a minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach.
- (2) Crew Alert and brief type of emergency, time available, airplane attitude after landing and exits available.
- (3) Air Traffic Control Notify
- (4) MFD or CDU, ATC/TCAS tuning Select to transponder code 7700.
- (5) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (6) Loose Equipment Secure and stow all flight compartment loose equipment.
- (7) TAWS circuit breaker (1B13) Open to silence TAWS aural warnings.
Check TAWS FAIL status message on.
- (8) AUDIO WARNING switches DISABLE to silence EICAS (DCU) aural alerts.
Check DCU 1 and 2 AURAL INOP status messages on.
- (9) Survival equipment Check
- (10) Crew harness and belts On and tightened.
- (11) Shoulder harness reel control Up to lock shoulder harness.
- (12) LDG GEAR lever Select as required.
- (13) FLAPS Select as required.

APPROACH

At approximately 2,000 feet:

- (1) Landing area conditions and wind direction Determine
- (2) Descent rate/approach speed Establish
- (3) AIR-CONDITIONING, L and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-7
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing (Cont'd)

- (4) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual control.

Check PRESS CONT MANUAL light on.

- (5) CABIN PRESSURIZATION, MAN ALT selector Select UP and hold.

- (6) CABIN PRESSURIZATION, MAN RATE control Select to MAX INCR.

When the airplane is completely depressurized:

- (7) BLEED AIR, 10TH STAGE L and R switch/lights Press out to close both valves.

Check L and R 10TH STAGE CLOSED lights on.

- (8) FUEL, L and R BOOST PUMP switch/lights Press out to shutdown both boost pumps.

Check L and R FUEL PUMP ON advisory messages out.

- (9) APU (if not essential) Shut down

At 500 feet (Radio altitude):

- (10) Crew Alert that forced landing is imminent.

- (11) Radio Transmit Final Position.

- (12) BRACE FOR IMPACT ORDER over PA system ; "BRACE – BRACE – BRACE".

- (13) EMER LTS switch Select to ON.

Check EMER LTS ON status message on.

- (14) LANDING LTS, LEFT and RIGHT switches Select to STDY.

- (15) LANDING LTS, NOSE switch Select to ON.

JUST BEFORE CONTACT

- (1) Thrust levers SHUT OFF



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CHALLENGER 650	EMERGENCY PROCEDURES Ditching and Forced Landing	03-09-8
		Oct 27/15

1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing (Cont'd)

- (2) APU, LH and RH ENG FIRE
 PUSH switch/lights (3) Press in Check all three BOTTLE ARMED – PUSH TO DISCH lights on.
- Check the following advisory messages on:
- L and R ENG SOV CLSD,
 - APU SOV CLSD, and
 - HYD SOV 1 and 2 CLSD.
- (3) Terrain Contact with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

AFTER CONTACT

When the airplane has stopped:

- (1) BOTTLE ARMED – PUSH TO DISCH switch/lights (3) Press in and hold until lights go out, to discharge firex bottles.
- Check ENG BOTTLE 1 and 2 LO and APU BOTTLE LO caution messages on.
- (2) Door and overwing exit Open
- (3) Passenger evacuation Initiate
- (4) BATT MASTER switch Select to OFF, to prevent CVR erasure.
- (5) Airplane Abandon with survival equipment.

———— **END** ————

CHALLENGER 650	EMERGENCY PROCEDURES Emergency Evacuation	03-10-1
		Oct 27/15

1. EMERGENCY EVACUATION

A. Emergency Evacuation Procedure

This procedure contains the steps required to prepare for passenger evacuation on land, and is initiated when a condition potentially endangering life or physical well-being of passengers and crew exists.

Pilot:

- (1) Parking brake Set
- (2) Evacuation..... Command
- (3) GND SPOILERS switch Select to DISARM.
- (4) FLIGHT SPOILER lever Select to RETRACT, to fair spoilers with wing surface.
Check EICAS indicates spoilers retracted.
- (5) Thrust levers..... SHUT OFF
- (6) APU, LH and RH ENG FIRE
PUSH switch/lights (3)..... Press in Check all three BOTTLE ARMED PUSH TO DISCH lights on.
Check the following advisory messages on:
 - L and R ENG SOV CLSD,
 - APU SOV CLSD, and
 - HYD SOV 1 and 2 CLSD.
- (7) BATT MASTER switchOFF to prevent CVR erasure.

Copilot: On evacuation command

- (8) ATCNotify of emergency conditions and of intention to evacuate.
- (9) AIR-CONDITIONING, L and R
PACK switch/lightsPress out to ensure minimum differential pressure within airplane.
- (10) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press in to drive outflow valves open.
Check EMER DEPRESS caution message on.
- (11) EMER LTS switch Select to ON.
Check EMER LTS ON status message on.

- (12) Exits..... Open



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CHALLENGER 650	EMERGENCY PROCEDURES Emergency Evacuation	03-10-2
		Oct 27/15

1. EMERGENCY EVACUATION (CONT'D)

A. Emergency Evacuation Procedure (Cont'd)

(13) Passenger evacuation Render assistance and direct passengers away from the airplane.

(14) Airplane Abandon by any suitable exit.

————— **END** —————

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1. ELECTRICAL

A. Loss of All Normal Electrical Power

Indication: ADG automatically deployed and EMER PWR ONLY warning message.

OR

AC ESS BUS caution message and no generator on-line.

If ADG did not deploy:

- (1) ADG manual deploy handle Pull and leave extended.
- (2) AC ELECTRICAL page Check ESS BUS is powered.

When ADG power is established:

- (3) STAB TRIM, CH 2 switch Press in to engage STAB TRIM CH 2.
- (4) Engine instruments Verify and check N₁, N₂ and ITT indications.

Both engines confirmed failed:

Yes

- (5) Double Engine Failure procedure Accomplish Refer to POWER PLANT – Double Engine Failure procedure in this chapter.

– END –

No

- (5) AIR-CONDITIONING, R PACK switch/light Press out to deselect.
- (6) Thrust levers Position manually as required.
- (7) Cabin altitude Monitor
- (8) Descent As required If an immediate descent is required, refer to AIR-CONDITIONING AND PRESSURIZATION – Emergency Descent procedure in this chapter.
- (9) AC POWER, GEN 1 and GEN 2 switches Select to OFF/RESET then ON.
Check GEN 1 and 2 OFF caution messages out.



1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

Any generator comes on-line:

Yes

Re-establish normal power:

No

Below 20,000 feet and APU is available:

Yes

(10) APU Start

(11) AC POWER, APU GEN
switch Select to ON.

Check APU GEN OFF caution
message out.

APU generator comes on-line:

Yes

(12) ADG manual deploy
handle Stow

(13) ADG PWR TXFR,
OVERRIDE switch Press to override.

(14) HYDRAULIC pump
switch 3B Select to AUTO/ON, as applicable.
Check HYD PUMP caution
messages are not displayed.

(15) CAS messages Review

(16) Affected systems Restore

- END -

No

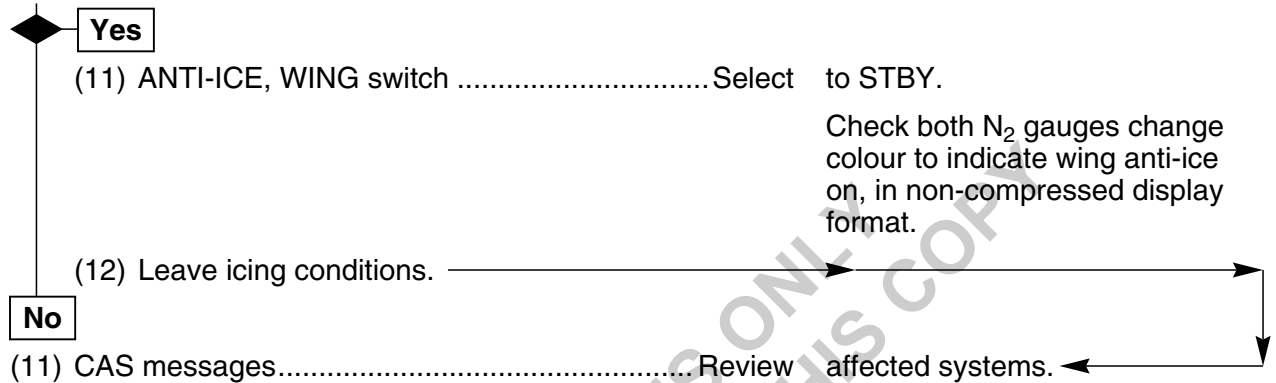
(10) Descent Continue to 10,000 feet or lowest safe
altitude (if required).

CHALLENGER 650	EMERGENCY PROCEDURES Electrical	03-11-3
		Rev 9, Jun 16/17

1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

Wing anti-ice is required:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights.
- SPS altitude compensation
- Auto throttle system

NOTE

1. With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.
2. Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin Δpressure on EICAS.
3. Flaps will operate (extend or retract) at half speed.

(12) Land at the nearest suitable airport.



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1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

NOTE

If the ADG is not operating properly, and the airplane is on battery power only, then all electrical power may be lost after 30 minutes.

Hydraulic pump 3B will be inoperative when on battery power only.

Prior to landing:

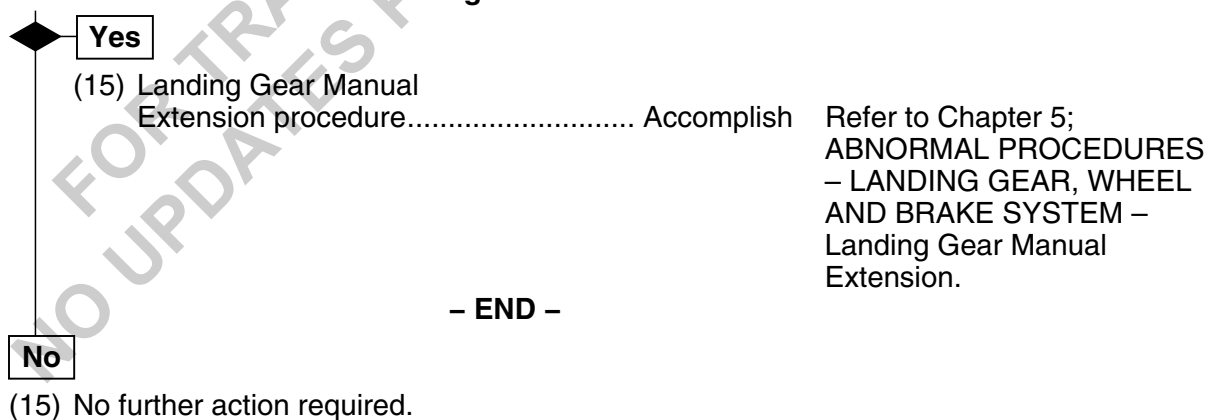
- (13) ADG manual deploy handle Pull
- (14) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

HYD 3 LO PRESS caution message is on:



END

CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-1
		Rev 5, Aug 30/16

1. FLIGHT CONTROLS

A. Aileron System Jammed

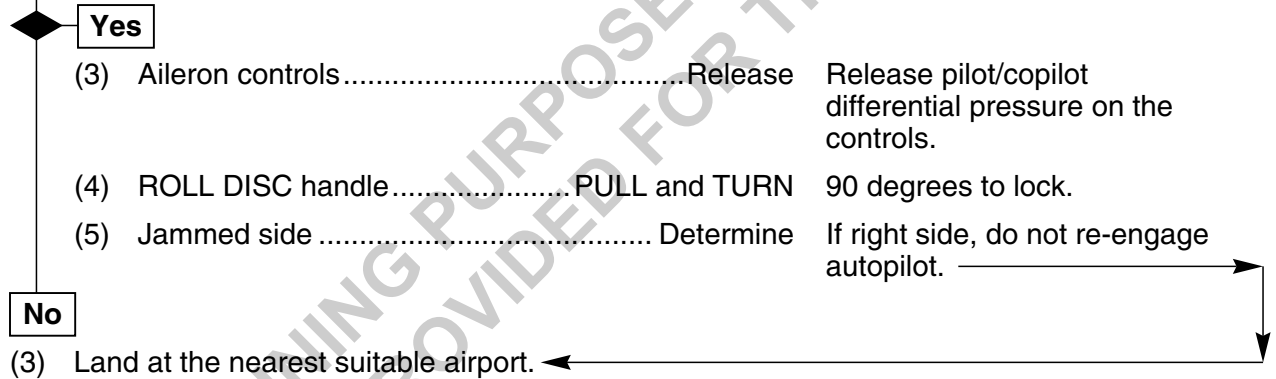
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- | | | |
|---------------------------|------------|--|
| (1) Autopilot | Disconnect | if coupled. |
| (2) Aileron controls..... | Overpower | Attempt to overpower jammed condition. |
| | | Attempt to use aileron trim to alleviate forces on controls, if jam has occurred away from neutral position. |

Jam persists:



NOTE

Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:

- | | | |
|--|---|---------------------------------|
| (4) TAWS WARNING, FLAPS OFF switch/light..... | Press in | to mute the flap aural warning. |
| (5) TAWS WARNING, RAAS OFF switch/light (if installed) | Press in | Check OFF light on. |
| (6) FLAPS | Select | to 20° for landing. |
| (7) Approach speed | V _{REF} (Flaps 45°)
+ 14 KIAS | minimum. |
| (8) ATS DISC switch..... | Depress | prior to 100 feet AGL. |



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.



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1. FLIGHT CONTROLS (CONT'D)

A. Aileron System Jammed (Cont'd)

- (9) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— **END** —————

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CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-3 Rev 5, Aug 30/16
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1. FLIGHT CONTROLS (CONT'D)

B. Elevator System Jammed

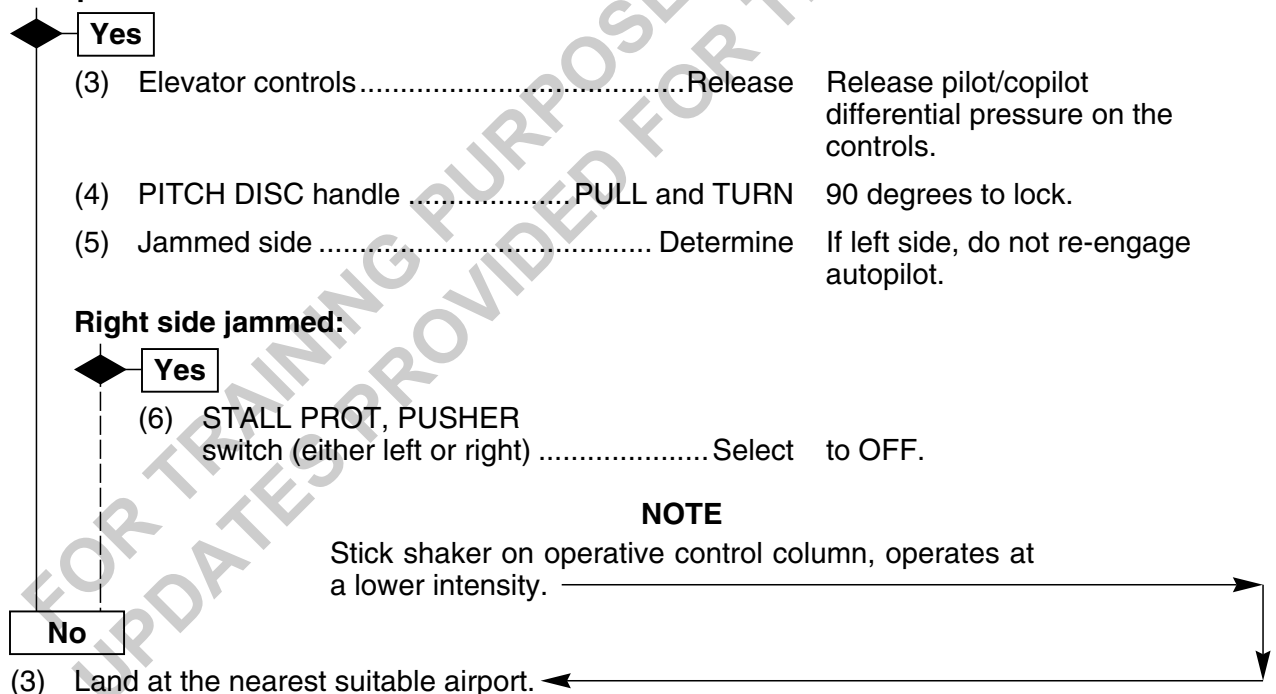
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.
- (2) Elevator controls..... Overpower Attempt to overpower jammed condition.
Use stabilizer trim to compensate, if elevator jam has occurred away from the trimmed position.

Jam persists:



NOTE

Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (4) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (5) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (6) FLAPS Select to 20° for landing.



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1. FLIGHT CONTROLS (CONT'D)

B. Elevator System Jammed (Cont'd)

- (7) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (8) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (9) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-5
		Rev 5, Aug 30/16

1. FLIGHT CONTROLS (CONT'D)

C. Rudder System Jammed

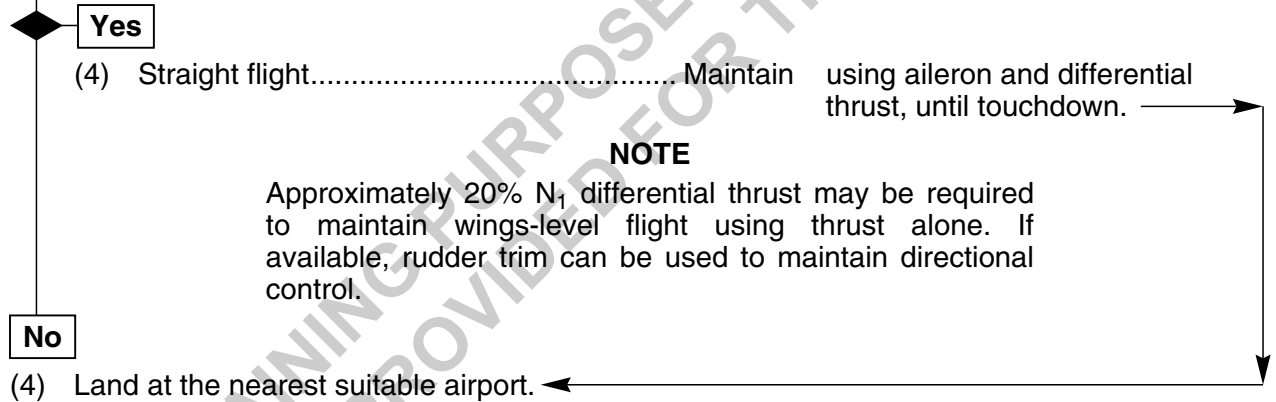
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.
- (2) YD DISC pushbutton Press to disengage yaw damper.
Check YAW DAMPER caution message on.
- (3) Rudder pedals Overpower Attempt to overpower jammed condition.

Jam persists:



NOTE

Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (5) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (7) FLAPS Select to 20° for landing.
- (8) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (9) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.



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1. FLIGHT CONTROLS (CONT'D)

C. Rudder System Jammed (Cont'd)

(10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

NOTE

After touchdown, use nose wheel steering and, if necessary, differential braking to maintain directional control, as the airplane will turn in the direction of the jammed rudder.

— **END** —

CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-7
		Oct 27/15

1. FLIGHT CONTROLS (CONT'D)

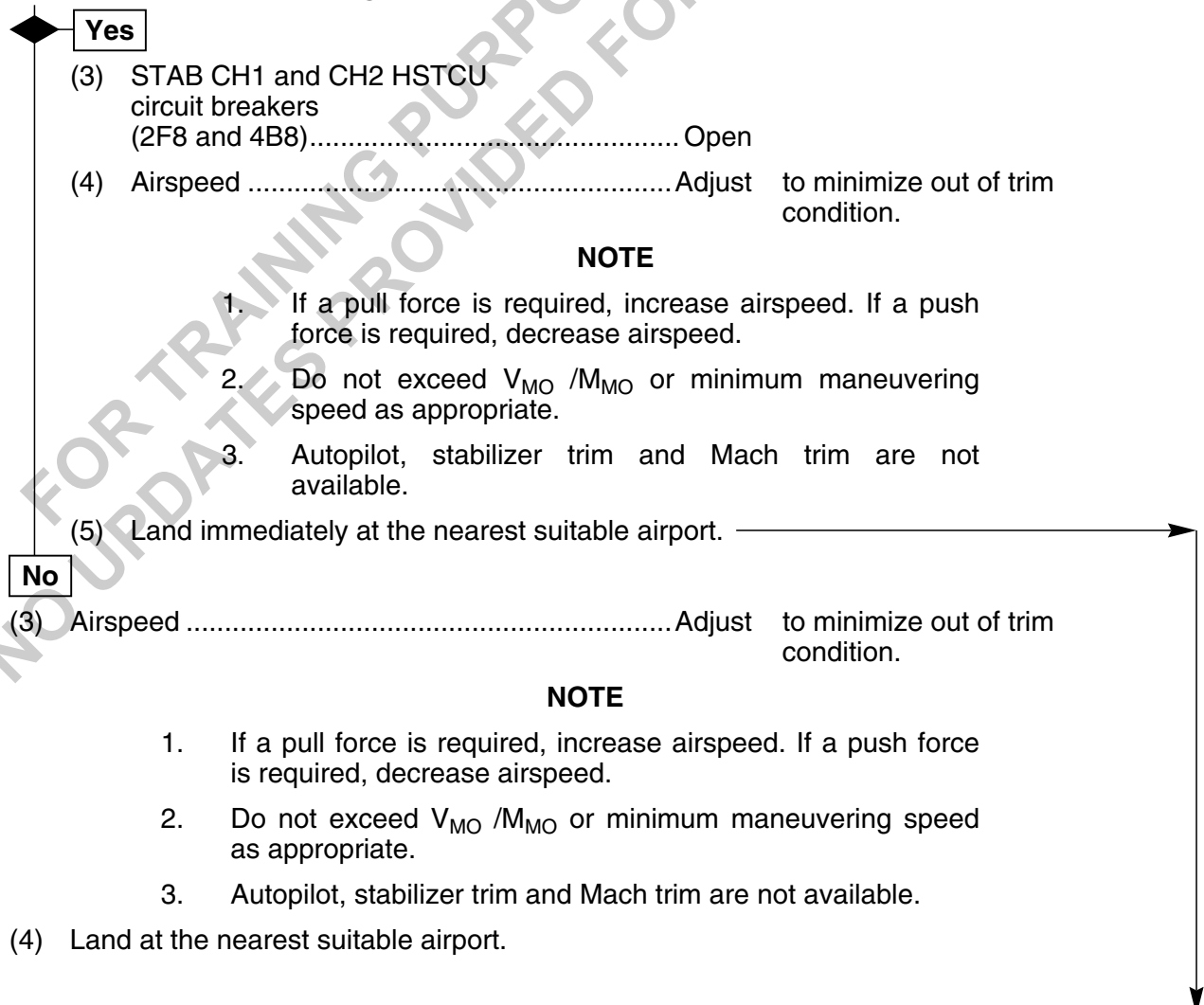
D. Stabilizer Trim Runaway

Indication:	Abnormal/uncommanded change in pitch attitude. Stabilizer trim changes without pilot input. Higher or lower than normal and increasing or decreasing pitch control forces.	
(1) Control column	Assume	manual control and override runaway.
(2) Pilot's or copilot's STAB TRIM DISC switch	Press	hold and release.

NOTE

- Flight spoilers will reduce airspeed and provide nose up pitch.
- Reducing thrust will reduce airspeed and provide nose up pitch.

STAB TRIM caution message is out or uncommanded stabilizer motion continues:



1. FLIGHT CONTROLS (CONT'D)

D. Stabilizer Trim Runaway (Cont'd)

Prior to landing: ←

- (5) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (7) Recommended FLAPS Select to 20° for landing.



Selection of Flaps 20 will increase nose down pitching moment slightly, and therefore should be selected just prior to approach in the event of a full nose down stabilizer position. Selection of Flaps 20 significantly reduces landing field length compared to Flaps 0.

If control forces are excessive, some relief may be gained by adjusting airspeed or flap position. Reduced flap settings generally reduce pull forces. Increased flap settings generally reduce push forces.

- (8) Final approach speed V_{REF} (Flaps 45°)
+ ΔV_{REF} for applicable flap setting.
- (9) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase by factor given below:

Flaps Setting (Degrees)	ΔV_{REF} (KTS)	ALD Factor	
		Without Thrust Reversers	With Thrust Reversers
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)



1. FLIGHT CONTROLS (CONT'D)

D. Stabilizer Trim Runaway (Cont'd)

CAUTION

Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

WARNING

In the event of a go-around, increasing thrust will increase nose down forces. Increase thrust slowly to avoid excessive pitch force.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— **END** —————

CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-10
		Oct 27/15

1. FLIGHT CONTROLS (CONT'D)

E. Stall Recovery

Indication:	Aircraft buffet, uncommanded roll, stick shaker activated and/or stall warbler on.	
(1) Autopilot	Disconnect	if coupled.
(2) Pitch attitude.....	Lower nose	to reduce angle of attack.
(3) ATS DISC switch	Depress	
(4) Thrust levers.....	Advance	to MAX POWER.
(5) Roll attitude	Wings level	
(6) FLIGHT SPOILER lever	Select	to RETRACT.

After airspeed increases and stall warning is extinguished:

- (7) Pitch attitude.....Adjust to minimize altitude loss.
- (8) Thrust levers and airplane configurationAdjust as required.

NOTE

1. It is essential that the AOA be immediately reduced, even if this means a loss of altitude.
2. Avoid abrupt or aggressive pitch control inputs during recovery. Inappropriate recovery inputs can result in a secondary stall.
3. Height loss resulting from high AOA recovery, especially at cruise altitude and/or low initial thrust conditions, can be significant .

————— **END** —————

CHALLENGER 650	EMERGENCY PROCEDURES Flight Controls	03-12-11
		Rev 11, Mar 01/18

1. FLIGHT CONTROLS (CONT'D)

F. Overspeed Warning

Indication: Overspeed warning clacker on.

(1) Airspeed Reduce until less than V_{MO}/M_{MO} or V_{FE} .

————— **END** —————

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CHALLENGER ⁶⁵⁰	EMERGENCY PROCEDURES Flight Controls	03-12-12
		Rev 11, Mar 01/18

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CHALLENGER 650	EMERGENCY PROCEDURES Fuel System	03-13-1
		Oct 27/15

1. FUEL SYSTEM

A. Low Fuel

Indication: The EICAS fuel quantities turn amber when either main tank is less than 230 kg (500 lb), or the total fuel quantity is less than 460 kg (1,000 lb).

- (1) AirplaneNot more than 5°
nose up
- (2) Fuel balance.....Check
- (3) Land at the nearest suitable airport.

NOTE

Do not attempt go-around. The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10° nose up.

————— **END** —————

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CHALLENGER 650	EMERGENCY PROCEDURES Ice and Rain Protection	03-14-1
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1. ICE AND RAIN PROTECTION

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

A. Wing Leading Edge Overheat

Indication: WING OVHT warning message and "WING OVERHEAT" aural on.

(1) ANTI-ICE, WING switch Select to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message out.

After 40 seconds, WING OVHT warning message persists:

Yes

(2) BLEED AIR, 14TH STAGE, L and R switch/lights Press out to close shutoff valves.

Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(3) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.

(4) Leave icing conditions.

If required:

(5) Ice Dispersal Procedure Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – ICE AND RAIN PROTECTION – Ice Dispersal Procedure.

– END –

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1. ICE AND RAIN PROTECTION (CONT'D)

A. Wing Leading Edge Overheat (Cont'd)

No

(2) ANTI-ICE, WING switch Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message on.

L HEAT or R HEAT light is out and/or L (R) WING A/ICE caution message is on:

Yes

(3) BLEED AIR, 14TH STAGE, ISOL switch/light Press in to open isolation valve.

Check 14TH STAGE ISOL OPEN light on. →

No

WING OVHT warning message re-occurs: ←

Yes

(3) ANTI-ICE, WING switch Select to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message out.

(4) BLEED AIR, 14TH STAGE, L and R switch/lights Press out to close shutoff valves.

Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(5) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.



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1. ICE AND RAIN PROTECTION (CONT'D)

A. Wing Leading Edge Overheat (Cont'd)

▼ (6) Leave icing conditions.

If required:

(7) Ice Dispersal Procedure Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– ICE AND RAIN
PROTECTION – Ice Dispersal
Procedure.

– END –

No

(3) No further action required.

————— **END** —————

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1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure

Indication: ANTI-ICE DUCT warning message and “ANTI-ICE DUCT” aural on.

(1) ANTI-ICE, WING switch Select to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message out.

(2) Leave icing conditions.

ANTI-ICE DUCT warning message persists (after 30 seconds):

Yes

(3) BLEED AIR, 14TH STAGE, L and R switch/lights Press out to close shutoff valves.

Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(4) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.

After leaving icing conditions:

Ice accumulation is observed on the heated portion of the wing leading edge:

Yes

(5) Airspeed Increase to V_{MO}/M_{MO} to disperse ice, if possible.

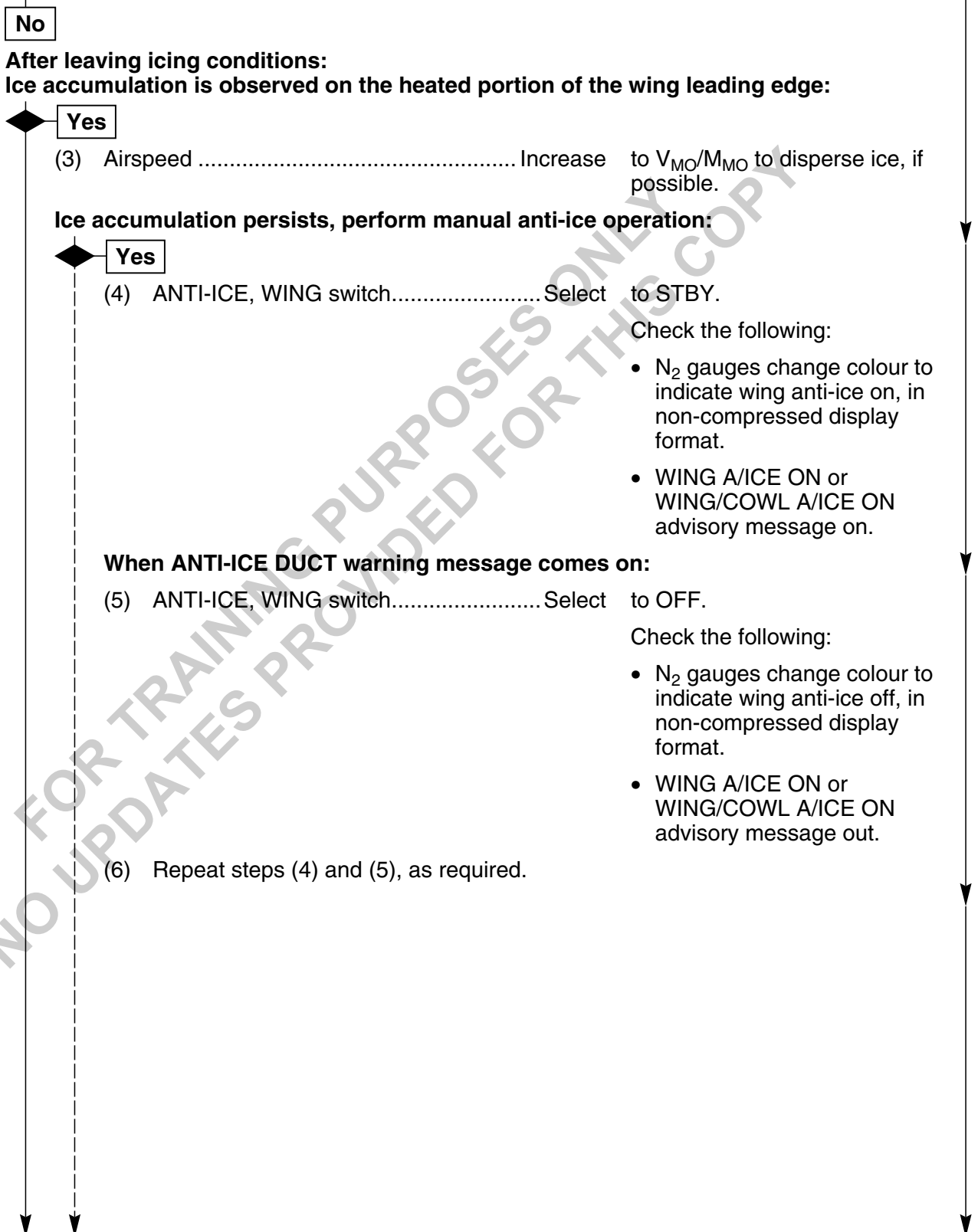
No

(5) No further action required.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)



1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)

Impossible to remove ice from the wing leading edge: ←

Yes

Assure adequate stall margin:

- (7) Maneuvering speed Not less than
200 KIAS

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics, or the warning margins provided by the stall protection system.

Prior to landing:

- (8) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (9) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (10) FLAPS Select to 20° for landing.
- (11) Approach speed..... V_{REF} (Flaps 45°) + 27 KIAS
- (12) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

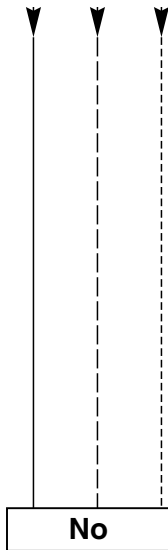
ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (13) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)



At 50 feet AGL:

(14) Thrust levers IDLE and do not prolong the flare.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

(3) No further action required.

- END -

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1. INSTRUMENTS SYSTEM

A. Unreliable Airspeed

Indication:

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed.
- Large airspeed difference between PFDs and/or ISI.
- Loss of multiple airspeed indication.
- EFIS MISCOMP and EFIS COMP INOP caution messages on and/or amber IAS flag.



Respect stall warning and stick pusher.

- (1) ATS DISC switch Depress
- (2) Autopilot Disconnect
- (3) FDs Deselect

Above minimum safe altitude:

- (4) Proceed to step (7).

Below minimum safe altitude:

- (4) Pitch/ N_1 15 degrees/TO
- (5) Go-around Initiate Clean up and climb above minimum safe altitude.

When above 15,000 feet:

- (6) Pitch/ N_1 10 degrees/CLB



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

When above minimum safe altitude:

- (7) Pitch/N₁ Set as per Table A to achieve approximate level flight.

TABLE A – LEVEL FLIGHT CRUISE – FLAP 0									
			WEIGHT kg (lb)						Expected Indicated Airspeed
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	KIAS MACH
ALTITUDE (1,000 ft)	40	Pitch (°)	1.0	1.5	2.0	2.5			210 to 250
		N ₁ (%)	83.0	84.6	87.2	89.9			
		AOA	0.4	0.4	0.5	0.5			
	35	Pitch (°)	1.5	2.0	2.5	2.5	3.0		230 to 270
		N ₁ (%)	81.8	82.7	83.9	85.7	87.9		
		AOA	0.4	0.4	0.4	0.4	0.5		
	30	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	240 to 290
		N ₁ (%)	79.9	80.4	81.3	82.3	83.8	84.7	
		AOA	0.3	0.4	0.4	0.4	0.4	0.4	
	25	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	240 to 290
		N ₁ (%)	76.3	76.7	77.5	78.3	79.5	80.3	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	
	20	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	240 to 290
		N ₁ (%)	72.9	73.3	74.0	74.9	75.9	76.7	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	
	15	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	240 to 290
		N ₁ (%)	70.5	70.8	71.5	72.3	73.3	74.0	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	205 to 235
		N ₁ (%)	58.0	58.9	60.2	61.7	63.1	63.9	
		AOA	0.4	0.4	0.5	0.5	0.6	0.6	
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	200 to 230
		N ₁ (%)	54.8	55.7	57.0	58.3	59.7	60.5	
		AOA	0.4	0.4	0.4	0.5	0.5	0.5	

If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.



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1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

EFIS Comparator Failure and EFIS Parameter Mismatch procedures for EFIS MISCOMP and EFIS COMP INOP caution messages and/or amber IAS flag are contained within this procedure.

- (8) Autopilot or FD Engage basic PTCH/ROLL.

NOTE

1. Autopilot/yaw damper should only be engaged once N_1 is set.
2. Cross check AOA after setting pitch attitude.
3. Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Adjust pitch to stop climb/descent.
4. Disregard low speed cue, flight path vector and airspeed trend vector as these may be erroneous.
5. If overspeed warning sounds, select the affected AUDIO WARNING switch to mute aural and disregard.
6. A nuisance NOSE DOOR OPEN warning message can be disregarded.
7. Altitude, Mode C and TCAS may be in error up to 600 feet. Vertical speed indications may also be unreliable.
8. Groundspeed indication may be unreliable.
9. FMS cruise time and fuel predictions may be incorrect.
10. Mach trim may be inoperative or operate erratically. Disengage Mach trim if erratic stabilizer motion is observed.
11. Using the FD/autopilot in PTCH, ALT, HDG and ROLL modes will help reduce workload.



Do not use FLC mode or ATS.

- (9) When the airplane is stabilized, proceed to **TROUBLESHOOTING** within this section to regain use of PFD/airspeed if possible.



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1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

TROUBLESHOOTING

- (1) ADC source selector Select to NORM.
- (2) Both PFDs and standby airspeed indicator Compare airspeed.
- (3) Determine which of the following conditions apply:
 - If no indicated airspeed is considered reliable, proceed to **CONDITION D**.
 - If both PFDs agree and the indicated airspeed is considered reliable, proceed to **CONDITION A**.
 - If one PFD and the ISI agree and the indicated airspeed is considered reliable, proceed to **CONDITION B**.
 - If only one indicated airspeed is considered reliable, proceed to **CONDITION C**.

WARNING

An airspeed should only be considered reliable if it is consistent with pitch, thrust, external noise and the expected airspeed from Table A.

CONDITION A	If both PFDs agree and the indicated airspeed is consistent with pitch, thrust, external noise, and the expected airspeed from Table A:
--------------------	--

- (4) Use airspeed/altitude information per normal operations.
- (5) Accomplish remaining CAS messages.

————— **END** —————

CONDITION B	If one PFD and the ISI agree and the indicated airspeed is consistent with pitch, thrust, external noise, and the expected airspeed from Table A:
--------------------	--

- (4) Autopilot Disconnect
- (5) FDs Deselect
- (6) ADC source selector Select to reliable side.
- (7) FDs Engage if desired.
- (8) Autopilot Engage if desired.
- (9) Use airspeed/altitude information per normal operations.
- (10) Accomplish remaining CAS messages.

————— **END** —————

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1. INSTRUMENTS SYSTEM (CONT'D)

CONDITION C	If only one indicated airspeed is consistent with pitch, thrust, external noise, and the expected airspeed from Table A:
--------------------	---

WARNING

Do not rely on only one airspeed unless it is in agreement with the expected airspeed column in Table A.

ISI is identified as reliable airspeed source:

Yes

- (4) Use airspeed/altitude information per normal operations.

NOTE

Heading and ILS/VOR course deviation indicator information on the PFD are reliable and will aid situational awareness.

- (5) Accomplish remaining CAS messages.

— **END** —

No

One PFD is identified as reliable airspeed source:

- (4) Autopilot Disconnect
- (5) FDs Deselect
- (6) ADC source selector Select to reliable side.
- (7) FDs Engage if desired.
- (8) Autopilot Engage if desired.
- (9) Use airspeed/altitude information per normal operations.
- (10) Accomplish remaining CAS messages.

— **END** —

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1. INSTRUMENTS SYSTEM (CONT'D)

CONDITION D	If no indicated airspeed is consistent with pitch, thrust, external noise, and the expected airspeed from Table A:
--------------------	---

- (4) Airspeed indications Disregard
- (5) Pitch/ N_1 Maintain
- (6) Land at the nearest suitable airport.

NOTE

- If environmental conditions (heavy rain, icing, ice crystals, volcanic ash, etc) are suspected to have caused unreliable airspeed, exit conditions as soon as possible, including descent, if required.
- Plan to recover to an airfield with an ILS, LNAV or VNAV under Visual Meteorological Conditions if possible.
- The following may be unreliable:
 - TAWS
 - Windshear detection
 - Landing gear warning horn
 - Flap overspeed
 - NOSE DOOR OPEN warning message



If at any time down to and including minimum safe altitude, it is suspected that a reliable airspeed source may be available, restart the Unreliable Airspeed procedure in order to confirm airspeed.

- (7) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.25 (25%)	1.20 (20%)

- (8) When ready for descent, proceed to **RECOVERY TO AN AIRFIELD** within this section.

————— **END** —————

1. INSTRUMENTS SYSTEM (CONT'D)

RECOVERY TO AN AIRFIELD

When ready to commence descent:

- (1) ANTI-ICE, WING switch As required
- (2) ANTI-ICE, L and R COWL
switch/lights As required
- (3) Pitch Adjust per Table B and continue to
adjust pitch during descent.

TABLE B - DESCENT - FLAPS 0 - IDLE									
Altitude (ft)	0	5,000	10,000	15,000	20,000	25,000	30,000	35,000	40,000
Pitch (°)	-1.0	-1.0	-1.0	-1.0	-2.0	-2.0	-2.0	-1.5	-1.5
AOA	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4

- (4) Thrust levers..... IDLE

NOTE

- With anti-ice ON, increase N_1 if required to obtain the minimum N_2 of 78%.
 - For an N_1 increase of 5%, add 0.5° to the pitch from Table B.
 - For an N_1 increase of 10%, add 1° to the pitch from Table B.
 - For an N_1 increase of 15%, add 2° to the pitch from Table B.



Do not select flaps to 20° until directed later in the procedure. The range of airspeeds possible in Table B could result in a significant flap overspeed and flap damage. The aircraft must be leveled and slowed down before flaps can be extended.

- (5) Descent Check Accomplish
- (6) In-Range Check Accomplish

NOTE

Plan to intercept a long final to allow time to re-configure for approach/landing and set pitch/ N_1 .



1. INSTRUMENTS SYSTEM (CONT'D)

If intermediate level off is required during the descent:

- (7) Pitch/ N_1 Set as per Table C, then adjust pitch to stop climb/descent.

TABLE C – LEVEL FLIGHT CRUISE – FLAP 0									
			WEIGHT kg (lb)						Expected Indicated Airspeed
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	KIAS MACH
ALTITUDE (1,000 ft)	35	Pitch (°)	1.5	2.0	2.5	2.5	3.0		230 to 270
		N_1 (%)	81.8	82.7	83.9	85.7	87.9		
		AOA	0.4	0.4	0.4	0.4	0.5		
	30	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	240 to 290
		N_1 (%)	79.9	80.4	81.3	82.3	83.8	84.7	
		AOA	0.3	0.4	0.4	0.4	0.4	0.4	
	25	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	240 to 290
		N_1 (%)	76.3	76.7	77.5	78.3	79.5	80.3	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	
	20	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	240 to 290
		N_1 (%)	72.9	73.3	74.0	74.9	75.9	76.7	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	
	15	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	240 to 290
		N_1 (%)	70.5	70.8	71.5	72.3	73.3	74.0	
		AOA	0.4	0.4	0.4	0.4	0.5	0.5	

If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.

For continued descent:

- (8) Pitch Adjust per Table D and continue to adjust pitch during descent.

TABLE D - DESCENT - FLAPS 0 - IDLE									
Altitude (ft)	0	5,000	10,000	15,000	20,000	25,000	30,000	35,000	40,000
Pitch (°)	-1.0	-1.0	-1.0	-1.0	-2.0	-2.0	-2.0	-1.5	-1.5
AOA	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4



1. INSTRUMENTS SYSTEM (CONT'D)

At initial approach altitude:

- (9) Pitch/ N_1 Set as per Table E, then adjust pitch to stop climb/descent.

TABLE E – LEVEL FLIGHT APPROACH – FLAP 0									
			WEIGHT kg (lb)						Expected Indicated Airspeed
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	KIAS
ALTITUDE (1,000 ft)	15	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	205 to 240
		N ₁ (%)	61.1	62.2	63.6	65.3	67.0	68.0	
		AOA	0.4	0.5	0.5	0.6	0.6	0.6	
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	205 to 235
		N ₁ (%)	58.0	58.9	60.2	61.7	63.1	63.9	
		AOA	0.4	0.4	0.5	0.5	0.6	0.6	
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	200 to 230
		N ₁ (%)	54.8	55.7	57.0	58.3	59.7	60.5	
		AOA	0.4	0.4	0.4	0.5	0.5	0.5	
	0	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	200 to 230
		N ₁ (%)	51.9	52.7	53.9	55.2	56.5	57.3	
		AOA	0.3	0.4	0.4	0.5	0.5	0.5	
If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.									



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1. INSTRUMENTS SYSTEM (CONT'D)

NOTE

Allow sufficient time for stabilization prior to re-configuring the airplane due to the pitch change as flaps are extended from 0 to 20.

When stabilized:

- (10) FLAPS Select to 20°.
- (11) Pitch/ N_1 Set as per Table F, then adjust pitch to stop climb/descent.

TABLE F – LEVEL FLIGHT APPROACH – FLAPS 20 – GEAR UP						
	WEIGHT kg (lb)					
	11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)
Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5
N_1 (%)	61.5	62.3	63.5	64.9	66.6	67.6
AOA	0.3	0.3	0.4	0.5	0.5	0.5

NOTE

Heading and ILS/VOR course deviation indicator information on the PFD are reliable and will aid situational awareness.

- (12) Final course Intercept

Upon intercept of final course (long final):

- (13) LDG GEAR lever Select to DN.
- (14) FLAPS Select to 30°.
- (15) Pitch/ N_1 Set as per Table G, then adjust pitch to stop climb/descent.

TABLE G – LEVEL FLIGHT APPROACH – FLAPS 30 – GEAR DN						
	WEIGHT kg (lb)					
	11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)
Pitch (°)	-2.0	-1.0	0.5	1.5	3.0	3.5
N_1 (%)	71.2	72.0	73.0	74.0	75.1	75.9
AOA	0.3	0.4	0.4	0.5	0.6	0.6



Do not descend for final approach with flaps 30°. The aircraft must remain level with gear down/flaps 30° or a flap overspeed and flap damage could result. Flaps 45° is selected at the start of the descent on the glideslope/glidepath.



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1. INSTRUMENTS SYSTEM (CONT'D)

At glideslope capture or descent point:

(16) FLAPS Select to 45°.

(17) Pitch/N₁ Re-adjust as follows:

TABLE H – APPROACH – FLAPS 45 – GEAR DN – 3° G/S								
			WEIGHT kg (lb)					
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)
ALTITUDE (ft)	15,000	Pitch (°)	–3.0	–3.0	–3.0	–3.0	–3.0	–3.0
		N ₁ (%)	65.5	67.7	71.2	74.9	77.3	76.7
		AOA	0.5	0.5	0.5	0.5	0.5	0.5
	10,000	Pitch (°)	–2.5	–2.5	–2.5	–2.5	–2.5	–2.5
		N ₁ (%)	60.8	62.5	66.3	69.8	72.8	74.6
		AOA	0.5	0.5	0.5	0.5	0.5	0.5
	5,000	Pitch (°)	–2.0	–2.0	–2.0	–2.0	–2.0	–2.0
		N ₁ (%)	55.9	57.9	61.4	65.0	67.9	69.4
		AOA	0.6	0.6	0.6	0.6	0.6	0.6
	0	Pitch (°)	–2.0	–1.0	–1.0	–1.0	–1.0	–1.0
		N ₁ (%)	52.5	53.4	56.7	59.9	62.8	64.2
		AOA	0.6	0.6	0.6	0.6	0.6	0.6

This table results in airspeeds equal to V_{REF} ±10 kts.

(18) Pitch Set to maintain glideslope/glidepath.

(19) N₁ Set to maintain AOA.

(20) Before Landing Check Accomplish

NOTE

In the event of a go-around, perform the go-around in accordance with the boxed actions at the beginning of this procedure.

END

CHALLENGER 650	EMERGENCY PROCEDURES Instruments System	03-15-12
		Rev 5, Aug 30/16

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CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-1
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM

A. Gear UP Disagree

Indication: GEAR DISAGREE warning message and “GEAR DISAGREE” aural on and/or

One or more landing gear (box) symbols (nose, left or right main gear) display red hash marks (not safe), with LDG GEAR lever selected UP.

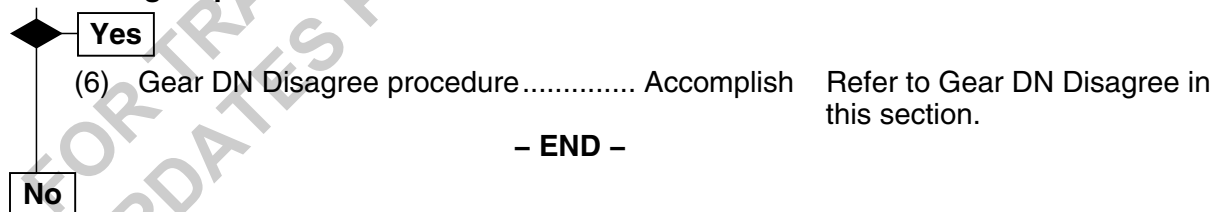
- (1) Airspeed Reduce to not more than:
 - Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.
- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LANDING GEAR MANUAL RELEASE handle PULL to full extension.

NOTE

The NOSE DOOR OPEN warning message may be displayed when manual gear extension is accomplished.

- (4) LDG GEAR lever Select to DN.
- (5) Landing gear indication Check for three DN indications.

Gear disagree persists:



Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing.

- (6) No further action required.

————— **END** —————

	CL-650 Airplane Flight Manual PSP 650-1	
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1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Gear DN Disagree

Indication: GEAR DISAGREE warning message and “GEAR DISAGREE” aural on and/or

One or more landing gear (box) symbols (nose, left or right main gear) display red hash marks (not safe), with LDG GEAR lever selected DN.

- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.

• Flaps 20° – 197 KIAS.

• Flaps 30° – 197 KIAS.

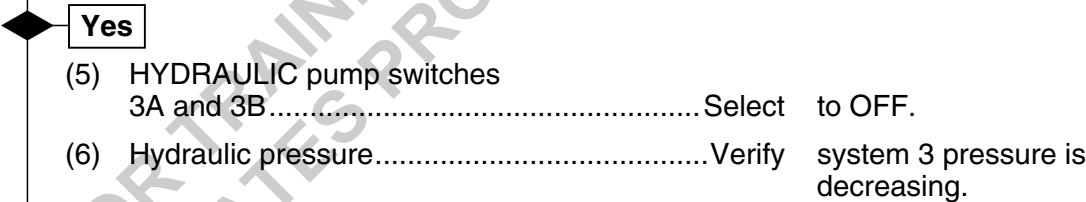
• Flaps 45° – 189 KIAS.
- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LANDING GEAR MANUAL RELEASE handle PULL to full extension.

NOTE

The NOSE DOOR OPEN warning message may be displayed when manual gear extension is accomplished.

- (4) Landing gear indicationCheck for three DN indications.

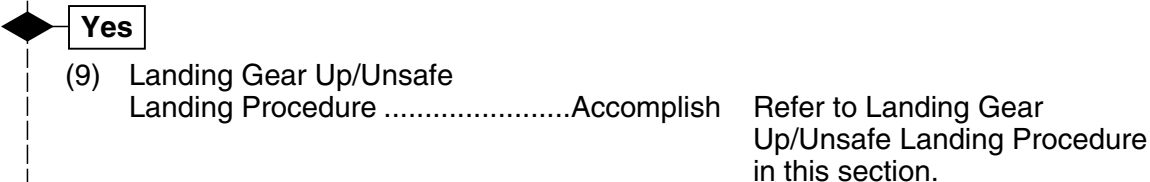
Gear disagree persists:



When hydraulic system 3 pressure is less than 200 psi:

- (7) HYDRAULIC pump switches 3A and 3B Select to ON.
- (8) Landing gear indicationCheck for three DN indications.

Gear disagree persists:



- END -

CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-3
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Gear DN Disagree (Cont'd)

No



Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing.

(5) No further action required.

————— **END** —————

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CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-4
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

C. Landing Gear Lever Jammed in the UP Position

- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.

- (2) HYDRAULIC pump switches 2B and 3B Select to ON.

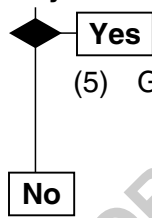
- (3) LANDING GEAR MANUAL RELEASE handle PULL to full extension.

NOTE

1. The NOSE DOOR OPEN warning message may be displayed when manual gear extension is accomplished.
2. The GEAR DISAGREE warning message will appear with the LDG GEAR lever jammed in the UP position if any landing gear is not in its uplock.

- (4) Landing gear indication Check for three DN indications.

Any landing gear fails to lock in the down position:

- 
Yes
 (5) Gear DN Disagree procedure Accomplish Refer to Gear DN Disagree in this section.

– END –

No

Prior to landing:

- (5) NOSE STEER switch Select to OFF.



Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering will not be available upon landing.

- (6) No further action required.

————— **END** —————

CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-5
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Landing Gear Up/Unsafe Landing Procedure

NOTE

1. IF ONE MAIN LANDING GEAR IS UP OR UNSAFE, hold applicable wing up for as long as possible. Maintain directional control with rudder and nosewheel steering (if considered safe). When wing touches the ground, apply asymmetrical braking for directional control.
2. IF NOSE LANDING GEAR IS UP OR UNSAFE, relocate passengers (if possible) to obtain aft CG. Trim stabilizer nose-up after touchdown. Gently lower the nose before elevator effectiveness is lost.
3. IF ALL WHEELS ARE UP OR UNSAFE, perform a nose high attitude touchdown, but do not reduce touchdown speed below stick shaker speed.
4. IF BOTH MAIN LANDING GEAR CANNOT BE LOCKED, consideration should be given to landing with all wheels up.

PRELIMINARY

- (1) Descent Plan to reduce fuel to the minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach.

PREPARATION

- (1) Crew Alert and brief type of emergency, time available, airplane attitude after landing and exits available for use.
- (2) ATC Notify
- (3) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (4) Loose equipment Secure stow all loose equipment in the flight compartment.
- (5) TAWS circuit breaker (1B13) Open to silence TAWS aural warnings.
Check TAWS FAIL status message on.
- (6) AUDIO WARNING switches DISABLE to silence DCU aural alerts.
Check DCU 1 and 2 AURAL INOP status messages on.
- (7) Shoulder harness Tightened and locked.



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CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-6
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Landing Gear Up/Unsafe Landing Procedure (Cont'd)

APPROACH

- (1) AIR-CONDITIONING, L and R
PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select manual mode.
Check PRESS CONT MANUAL light on.
- (3) CABIN PRESSURIZATION, MAN
ALT selector Select UP and hold.
- (4) CABIN PRESSURIZATION, MAN
RATE switch As required

When depressurized:

- (5) BLEED AIR, L and R 10TH
STAGE BLEED switch/lights Press out to close both SOVs.
Check L and R 10TH STAGE CLOSED lights on.
- (6) FLAPS Select to 45°.
- (7) EMER LTS switch Select to ON.
Check EMER LTS ON status message on.
- (8) Brace for impact Order over the PA system,
"BRACE-BRACE-BRACE".

BEFORE IMPACT

- (1) Airplane attitude Maintain a nose high attitude.
- (2) Ground/landing field Contact with minimum forward speed, but not less than stick shaker speed, and at a minimum sink rate.

AFTER LANDING

- (1) Thrust levers SHUT OFF



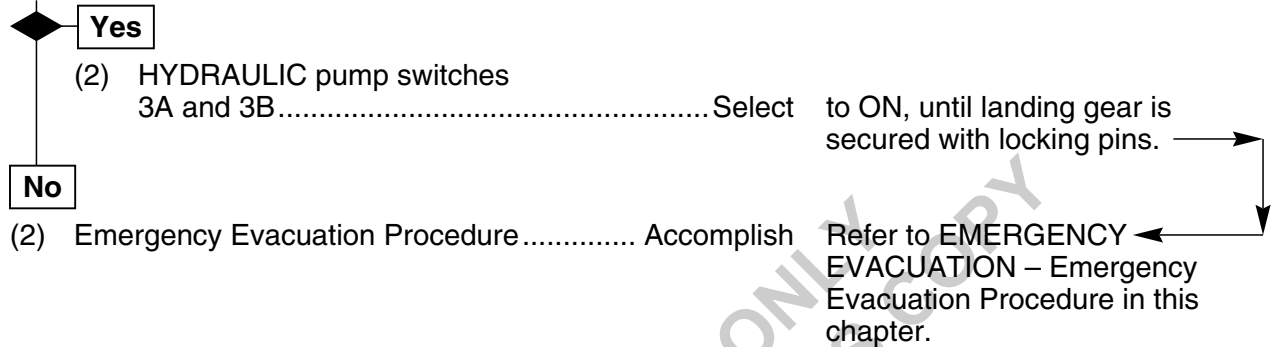
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CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-7
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Landing Gear Up/Unsafe Landing Procedure (Cont'd)

Any gear extended:



————— **END** —————

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

E. Main Landing Gear Bay Overheat Warning

Indication: MLG BAY OVHT warning message and "GEAR BAY OVERHEAT" aural on.

- (1) Airspeed Reduce to 197 KIAS.
- (2) LDG GEAR lever Select to DN.

MLG BAY OVHT warning message persists:

Yes

(3) LDG GEAR lever Leave DN

(4) Airspeed Do not exceed 250 KIAS

(5) Land at the nearest suitable airport.

— END —

No

(3) Flight..... Continue with gear down, for minimum of 5 minutes.

(4) LDG GEAR lever Select to UP.

— END —

CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-9
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

F. Nose Door Fails Open

Indication: NOSE DOOR OPEN warning message and “NOSE DOOR” aural on.

- (1) Airspeed Reduce to prevent structural damage to nose gear doors, as follows:
 - Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.
- (2) LDG GEAR lever Select to DN.
- (3) Land at the nearest suitable airport.

NOTE

Use of LANDING GEAR MANUAL RELEASE handle will cause NOSE DOOR OPEN warning message to be displayed.

— **END** —

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1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

G. Parking Brake On

Indication: PARKING BRAKE warning message, and “CONFIG BRAKES” aural on.

On ground (both engines N₁ greater than 70%):

- ☒ **Yes**
 (1) Thrust levers Retard
 (2) Parking brake As required
 – END –

☐ **No**

In flight:

- (1) Parking brake handle Check released.

PARKING BRAKE warning message persists:

☒ **Yes**

CAUTION

Inboard and outboard anti-skid systems are inoperative. Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

– END –

☐ **No**

- (2) No further action required.

————— **END** —————

CHALLENGER 650	EMERGENCY PROCEDURES Landing Gear, Wheel and Brake System	03-16-11
		Rev 5, Aug 30/16

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

H. Excessive Asymmetry or Loss of Braking

If excessive asymmetry or loss of braking is observed:

- (1) Wheel brakesRelease momentarily.
- (2) ANTI SKID switch.....Select to OFF.
- (3) Wheel brakes Re-apply as required.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

— END —

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		Rev 5, Aug 30/16

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CHALLENGER 650	EMERGENCY PROCEDURES Passenger Door	03-17-1
		Rev 5, Aug 30/16

1. PASSENGER DOOR

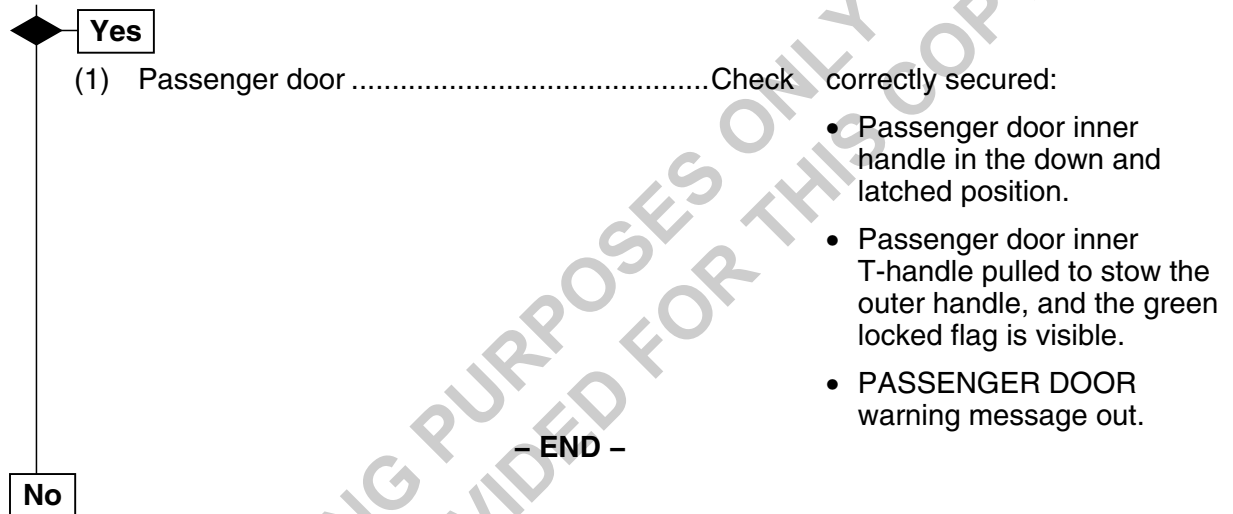
A. Passenger Door Failure

Indication: PASSENGER DOOR warning message and “DOOR” aural on (when engines are running).

NOTE

The passenger door warning comes on whenever two or more of the six door/handle sensors indicate that the door is open.

On ground:



In flight:

- (1) CABIN PRESSURIZATION, EMER DEPRESS switch/light Press in to depressurize the airplane.
- Check the following:
- EMER DEPRESS caution message on, and
 - CABIN ALT caution message on at 8,500 feet cabin altitude.
 - CABIN ALT warning message and “CABIN PRESSURE” aural on at 10,000 feet cabin altitude.



Do not attempt to pressurize the airplane.

- (2) Land at the nearest suitable airport.



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CHALLENGER 650	EMERGENCY PROCEDURES Passenger Door	03-17-2
		Rev 5, Aug 30/16

1. PASSENGER DOOR (CONT'D)

A. Passenger Door Failure (Cont'd)

- (3) Emergency Descent Procedure Initiate to safe altitude.
- Refer to AIR-CONDITIONING
AND PRESSURIZATION –
Emergency Descent
Procedure in this chapter.

————— **END** —————

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1. AURAL/VISUAL WARNING SYSTEM

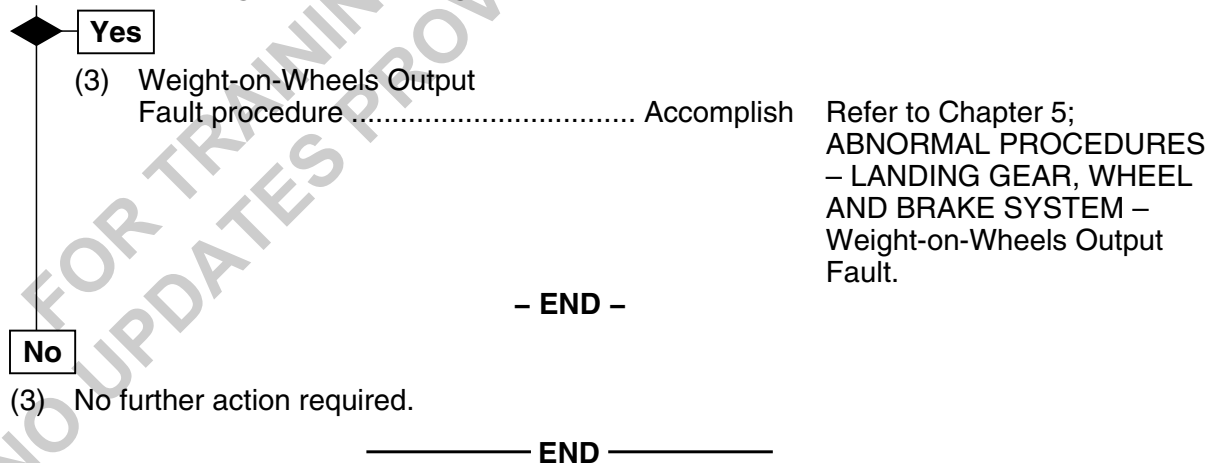
A. Configuration Warning

Indication: The following configuration warning messages and aural warnings come on when advancing thrust levers to take-off power (greater than 70% N_1):

WARNING MESSAGE	AURAL WARNING
CONFIG AP	"CONFIG AUTOPILOT"
CONFIG FLAPS	"CONFIG FLAPS"
CONFIG SPOILERS	"CONFIG SPOILERS"
CONFIG STAB	"CONFIG TRIM"
PARKING BRAKE	"CONFIG BRAKES"

- (1) Take-off Reject
- (2) Airplane configuration Confirm
- Check the position of the following items:
- Autopilot;
 - Flaps;
 - Spoilers;
 - Stabilizer; and
 - Parking brake.

Erroneous configuration messages and aural warnings occur:



CHALLENGER 650	EMERGENCY PROCEDURES Aural/Visual Warning System	03-18-2
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CHALLENGER 650	NORMAL PROCEDURES Introduction	04-01-1
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1. GENERAL

The normal operating procedures included in this chapter contain items peculiar to the CL-600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent). These checks and procedures are defined as being fundamental to the normal safe operation of the airplane and are supplementary to normal procedures common to the operation of any modern transport jet airplane.

These normal procedures have been developed and recommended by Bombardier and approved by Transport Canada for use in the operation of the Challenger airplane. These procedures are provided as guidance, and should not be construed as prohibiting the development of equivalent Transport Canada-approved procedures.

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CHALLENGER 650	NORMAL PROCEDURES Power Plant	04-02-1
		Rev 4, Jun 09/16

1. POWER PLANT

A. Power Plant – First Flight of the Day

- (1) BATT MASTER switchSelect to ON.
- (2) Engine and Jetpipe, Fire Detection System Test Procedure..... Accomplish Refer to FIRE PROTECTION SYSTEM in this chapter.
- (3) Engine Firex Monitor System Test Procedure..... Accomplish Refer to FIRE PROTECTION SYSTEM in this chapter.

B. Engine Vibration Test – First Flight of the Day

- (1) ENGINE CONTROL, VIB switchSelect to TEST. Check the following:
 - VIB icon displayed on N₂ gauges or adjacent to N₂ readout.
 - FAN VIB readout of 3.5 ±0.1 mils (amber) or VIB icon displayed on N₁ gauges in compressed display format.

C. Before Starting Engines (On Ground)

NOTE

At ambient temperatures of –30°C (–22°F) and below, the engines must be motored for 60 seconds and fan rotation must be assured before engine start is initiated (Refer to Chapter 2; LIMITATIONS – POWER PLANT).

- (1) BATT MASTER switchCheck selected to ON.
- (2) FUEL, L and R BOOST PUMP switch/lights Press in to turn on both boost pumps. Check the following:
 - L and R BOOST PUMP ON lights on.
 - L and R FUEL PUMP ON advisory messages on.
- (3) BLEED AIR, 14TH STAGE L (R) switch/light..... Press in to select the applicable 14th-stage SOV open.

NOTE

Both L and R 14th-stage shutoff valves must remain open throughout the normal operation of the airplane.

- (4) IGNITION A (B) switch/light..... Press in to arm applicable ignition system.
Check ARM light on at applicable switch/light.



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1. POWER PLANT (CONT'D)

C. Before Starting Engines (On Ground) (Cont'd)

- (5) Thrust levers.....Check both selected to SHUT OFF.

D. Starting Engine(s) Using APU Air Supply



If both engines are to be started using APU bleed air, then the operating engine's thrust must not exceed 70% N_2 .

With APU operating:

- (1) BLEED AIR, APU LCV switch/light.....Press in to open load control valve.
Check APU LCV OPEN light on.
- (2) L (R) ENG START switch/lightPress to start applicable engine.
Check the following:
- Applicable START light comes on.
 - IGNITION A (B), ON light comes on.
 - IGNITION A (B) advisory message comes on.
 - Check that there is positive bleed air pressure.
 - Check that N_1 increases.

When N_2 reaches 20% RPM and ITT is below 120°C:

- (3) Left or right thrust leverAdvance applicable lever to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



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1. POWER PLANT (CONT'D)

D. Starting Engine(s) Using APU Air Supply (Cont'd)

- (4) Engine instruments.....Monitor Check the following:
- Starter disengages at 55% N_2 .
 - IGNITION A (B) advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N_1 stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].

CAUTION

When operating on the ground with one engine operating, a fuel imbalance may occur with both boost pumps ON. This fuel imbalance may result in fuel spillage if fuel tanks are full. Select the inoperative engine boost pump to INOP. If a fuel imbalance occurs on the ground during single engine operation, open the respective L TO AUX or R TO AUX XFLOW valve to balance the fuel tanks.

If the other engine is to be started using APU bleed air:

- (5) Thrust lever of operating engine.....Check set to IDLE.
- (6) 10th stage bleed air pressure.....Check that positive pressure is available.
- (7) L (R) ENG START switch/lightPress to start applicable engine.
- Check the following:
- Applicable START light comes on.
 - IGNITION A (B), ON light comes on.
 - IGNITION A (B) advisory message comes on.
 - Check that N_1 increases.

When N_2 reaches 20% RPM and ITT is below 120°C:

- (8) Thrust lever of starting engine.....Advance to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



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1. POWER PLANT (CONT'D)

D. Starting Engine(s) Using APU Air Supply (Cont'd)

- (9) Engine instruments.....Monitor Check the following:
- Starter disengages at 55% N₂.
 - IGNITION A (B) advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N₁ stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].
- (10) APU EGT indications.....Check within normal limits.
- (11) Idle speed.....Check N₂ variation between engines is within 2%. If greater, notify maintenance personnel.

E. Starting Engine Using External Air Supply

NOTE

AC power is required for this procedure.

- (1) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.
- (2) External air Connected Verify 45 psi minimum bleed air pressure on EICAS.

NOTE

The external air source must be capable of supplying 45 psi manifold pressure at the initiation of start, however, lower pressure units, dependent upon altitude and flow, have been proven satisfactory.

- (3) L (R) ENG START switch/lightPress to start applicable engine.
Check the following:
- Applicable START light comes on.
 - IGNITION A (B), ON light comes on.
 - IGNITION A (B) advisory message comes on.
 - Check that there is positive bleed air pressure.
 - Check that N₁ increases.



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1. POWER PLANT (CONT'D)

E. Starting Engine Using External Air Supply (Cont'd)

When N_2 reaches 20% RPM and ITT is below 120°C:

- (4) Left or right thrust lever Advance applicable lever to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.

- (5) Engine instruments..... Monitor Check the following:

- Starter disengages at 55% N_2 .
- IGNITION A (B) advisory message goes out.
- ITT is 450 – 600°C (max. during start is 900°C).
- N_1 stabilizes and does not drop below 22% RPM.
- Check that oil pressure is within normal range [above 25 psi (174 kPa)].

CAUTION

When operating on the ground with one engine operating, a fuel imbalance may occur with both boost pumps ON. This fuel imbalance may result in fuel spillage if fuel tanks are full. Select the inoperative engine boost pump to INOP. If a fuel imbalance occurs on the ground during single engine operation, open the respective L TO AUX or R TO AUX XFLOW valve to balance the fuel tanks.

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1. POWER PLANT (CONT'D)

F. Starting Engine Using Engine Cross Bleed Air



This procedure requires the use of higher than normal thrust settings on the ground. Ensure that intake and exhaust areas of the operating engine are secure.

When starting engines in close quarters, consideration should be given to the effects of jet blast.

- (1) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.
- (2) BLEED AIR, 10TH STAGE L (R) switch/light.....Press in to open 10th stage SOV for operating engine.
Check L (R) 10TH STAGE CLOSED light out.
- (3) SUMMARY pageSelect and monitor bleed pressure.
- (4) Thrust lever of operating engine.....Advance to achieve a minimum of 60 psi bleed air pressure (approximately 85% N₂).
- (5) L (R) ENG START switch/lightPress to start applicable engine.
Check the following:
 - Applicable START light comes on.
 - IGNITION A (B), ON light comes on.
 - IGNITION A (B) advisory message comes on.
 - Check that N₁ increases.

When N₂ reaches 20% RPM and ITT is below 120°C:

- (6) Thrust lever of starting engine.....Advance to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



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1. POWER PLANT (CONT'D)

F. Starting Engine Using Engine Cross Bleed Air (Cont'd)

- (7) Engine instruments.....
Monitor

Check the following:
 - Starter disengages at 55% N₂.
 - IGNITION A (B) advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N₁ stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].
- (8) Thrust lever of first engine
Select

to IDLE.
- (9) Idle speed.....
Check

N₂ variation between engines is within 2%. If greater, notify maintenance personnel.
- (10) BLEED AIR, 10TH STAGE L and R switch/lights
Press in

to open both 10th stage SOVs.

Check L and R 10TH STAGE CLOSED lights out.

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1. POWER PLANT (CONT'D)

G. Failure to Start

Light-Off:

Light-off, as indicated by a rise in ITT, is **normally** achieved within 10 seconds from thrust lever operation. Starter motor operation continues until automatic disengagement occurs at 55% N₂ (normally within 20 – 40 seconds), or if the start is aborted by selecting STOP.

Failure to Start:

At ambient temperatures of greater than 15°C (59°F), the start must be aborted if the engine fails to start (no light-off) **after 25 seconds** from thrust lever movement to IDLE.

After 60 seconds of start operation and the engine has not reached starter cutout speed, the start should be aborted by selecting STOP.

At temperatures below 15°C (59°F), the start sequence may exceed 60 seconds.

- (1) Thrust lever of starting engine Select to SHUT OFF.
- (2) IGNITION Disarm

If one engine is operating:

- (3) Inoperative engine FUEL,
L (R) BOOST PUMP switch/light Press out to disarm boost pump.
Check L (R) FUEL PUMP
caution message on.
- (4) Dry motor until ITT is below 120°C
(maximum 90 seconds) Accomplish Observe starter limits (Refer to
Chapter 2; LIMITATIONS –
POWER PLANT).

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1. POWER PLANT (CONT'D)

H. After Starting Engines



If N₂ is 57% or below and OAT is above -20°C (-4°F), do not accelerate above idle.

NOTE

1. Do not accelerate engine until oil pressure has stabilized below 116 psi.
 2. A high vortex noise may occur during stationary engine run-up to high power in crosswind or tailwind conditions. This phenomenon is not harmful to the engines, and may be ignored.
- (1) IGNITION A (B) switch/light.....Press out to disarm applicable ignition system.
Check applicable ARM light out.
- (2) ANTI-ICE, WING switch As required
- (3) ANTI-ICE, COWL L and R switch/lights As required
- (4) 10TH STAGE Bleed air As required

I. ATS Check

NOTE

1. Prior to all sensors becoming valid, and upon completion of the ATS self-test, the ATS Mode Status Display (MSD) will display an amber **FAIL** message. The **FAIL** message can be cancelled by depressing either ATS DISC switch. The ATS system will perform an automatic self-test after all sensors become valid following power-up.
2. Satisfactory self-test will result in illumination of the two green LED annunciators on the ATS Control Panel and scrolling **SELFTEST** message in each MSD. At the completion of the self-test, the green LED annunciators will extinguish and both MSDs will blank.
3. The LED displays in the MSDs and the ATS Control can be tested by holding the LAMP TEST switch in position 1 or 2. The MSD will display the scrolling **SELFTEST** message described above. The ATS system will perform a system self-test at this time also. If the ATS is engaged, the lamp test and self-test are inhibited.

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1. POWER PLANT (CONT'D)

J. APR Check

- (1) ENGINE CONTROL, ENG
SPEED switchesSelect both to ON.
- (2) ENGINE CONTROL, right APR
switchSelect to TEST 1 and TEST 2, in
turn.
Check the following:
 - APR TEST 1 OK advisory
message on.
 - APR TEST 2 OK advisory
message on.

NOTE

APR TEST 1 and 2 OK advisory messages go out after 10 seconds.

- (3) ENGINE CONTROL, left APR
switchSelect to ARM.

NOTE

At 79% N_1 during take-off, the APR ARM advisory message should
come on.

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1. POWER PLANT (CONT'D)

K. Anti-Ice Cowl Check – First Flight of the Day

- (1) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice on.
Check the following:
- COWL A/ICE ON advisory message on.
 - ITT gauges indicate rise in temperature.

NOTE

It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

On airplanes with Supplemental Ground Wing Anti-Icing:

Forty-five seconds after selecting cowl anti-ice on:

- (a) CAS messages Review Check the following:
- COWL A/ICE ON advisory message remains on.
 - L and R COWL A/ICE caution messages remain out.

NOTE

The COWL L and R switch/lights must be kept pressed in for a minimum of 45 seconds for a valid test. After 45 seconds, it may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

- (2) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.
Check the following:
- COWL A/ICE ON advisory message out.
 - ITT gauges indicate drop in temperature.

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1. POWER PLANT (CONT'D)

L. Thrust Reverser Check – First Flight of the Day

- (1) THRUST REVERSER, L switch Select to ARMED.
Check L REV ARMED advisory message on.
- (2) Left thrust reverse lever Deploy Check the following:
- REV icon appears on left N₁ gauge only.
 - R REV UNLOCKED caution message out.
 - No amber UNLK annunciator illuminates on the center console GND SPOILERS/THRUST REVERSER panel.
- (3) Left thrust reverse lever Stow Check L REV UNLOCKED caution message out.
- (4) THRUST REVERSER, R switch Select to ARMED.
Check R REV ARMED advisory messages on.
- (5) Right thrust reverse lever Deploy Check the following:
- REV icon appears on right N₁ gauge only.
 - L REV UNLOCKED caution message out.
 - No amber UNLK annunciator illuminates on the center console GND SPOILERS/THRUST REVERSER panel.
- (6) Right thrust reverse lever Stow Check R REV UNLOCKED caution message out.

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1. POWER PLANT (CONT'D)

M. Before Take-Off

If take-off performance is predicated upon the use of APR, a test must be carried out as detailed in step (6).

- (1) Minimum N_1 for take-off..... Establish using take-off thrust setting charts for prevailing OAT/altitude and anti-ice/bleed configuration (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).

NOTE

Continuous engine ignition must be used for:

- Take-offs and landings on contaminated runways;
- Take-off with high crosswind components;
- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence; or
- Flight in the vicinity of thunderstorms.

- (2) THRUST REVERSER, L and R switches..... Select both to ARMED.
Check the following:
- L and R REV ARMED advisory messages on.
 - LH and RH UNLKD lights out.
 - L and R REV UNLOCKED caution messages out.

- (3) ENGINE CONTROL, left APR switch Check selected to ARM.

- (4) Take-off Thrust Limit Select in the FMS performance THRUST LIMIT page.
Confirm N_1 thrust limit is posted on the EICAS N_1 displays.

- (5) TOGA push switch Depress Confirm N_1 TO message on MSD.

NOTE

After anti-ice checks and MSD is blank, the ATS may post an amber **FAIL** message on the MSDs when TOGA is selected prior to take-off. Depressing either ATS DISC switches on either thrust lever will clear the **FAIL** message and display **N1 TO** on the MSDs.



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1. POWER PLANT (CONT'D)

M. Before Take-Off (Cont'd)

- (6) Thrust levers.....Advance both to above 79% N_1 with wheel brakes applied, or during the initial take-off roll, as appropriate.
- Check the following:
- APR ARM advisory message is on.
 - No failure messages are displayed.
 - ATS engages.
 - The green LED annunciators illuminate on the ATS Control Panel.
 - Thrust levers advance to take-off N_1 thrust limit computed by FMS.
 - Ensure take-off thrust is achieved by 80 KIAS.
 - Above 80 KIAS, confirm N_1 HOLD mode is displayed on MSD.



If an APR INOP caution message is displayed, select the ENGINE CONTROL, APR switch to OFF and use APR-OFF performance. Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – APR Failure.

NOTE

1. APR OFF performance must be used if an APR INOP caution message is displayed.
2. If either ENG SPEED switch or the APR switch is OFF, the APR system will not operate.
3. Under normal temperature conditions, a minimum of two minutes at IDLE is recommended prior to operation at thrust levels required for take-off.
4. To perform take-off without ATS, depress either ATS DISC switch to cancel N_1 TO mode.

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1. POWER PLANT (CONT'D)

N. High Crosswind Take-Off

- (1) Thrust levers..... Set thrust to take-off thrust minus 5% N_1 .
Check APR ARM advisory message comes on when above 79% N_1 .
- (2) Thrust levers..... Advance to achieve take-off N_1 by the time 60 KIAS is reached.

O. After Take-Off

At a safe altitude:

- (1) Thrust levers..... Adjust to appropriate climb N_1 thrust setting.

NOTE

At 320 feet barometric altitude above the take-off elevation, the ATS will re-engage the thrust lever servos, the **N1 HOLD** message is replaced by **N1 TO** message, and the green ATS control annunciators illuminate.

- (2) THRUST REVERSER, L and R switches..... Select to OFF, to disarm thrust reversers.
Check L and R REV ARMED advisory messages out.

P. Climb

- (1) Thrust setting..... Select CLB on FMS performance THRUST LIMIT page.

NOTE

When a climb of less than 2,000 feet to level off is made, it is recommended to use VS or PITCH mode on the FCC and set airspeed reference (speed bug) at or above current airspeed, to ensure ATS will transition to **SPEED** mode and minimize airspeed/altitude overshoot after level off.

Q. Cruise

- (1) Airspeed Set desired cruise airspeed using AFCS airspeed reference (speed bug).
- (2) Thrust setting..... Select desired thrust setting on FMS.

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1. POWER PLANT (CONT'D)

R. Descent

(1) Descent Initiate

In AFCS FLC Mode:

(2) Thrust levers.....Manually control after 4 second thrust reduction.

In Non-FLC Mode:

(2) Airspeed Set desired airspeed using AFCS airspeed reference (speed bug).

S. Approach

(1) Thrust setting.....Select TO or TGT before transferring bleed air to APU.

(2) Airspeed Set desired airspeed using AFCS airspeed reference (speed bug).

T. Before Landing

At 100 feet AGL:

(1) ATS MSDCheck green SPEED message flashes. If message is not displayed, disengage ATS and manually control thrust.

Above 50 feet AGL:

(2) THRUST REVERSER, L and R switches.....Select to ARMED.

Check the following:

- L and R REV ARMED advisory messages on.
- L and R REV UNSAFE caution messages out.
- L and R REV UNLOCKED caution messages out.



Do not arm thrust reverser if L (R) REV UNSAFE caution message is displayed.

1. POWER PLANT (CONT'D)**U. Go-Around**

- (1) ATSDisengage
- (2) Thrust setting..... Set manually to TO.



ATS must be disengaged as thrust levers are advanced to go-around. If ATS is not disengaged and the pilot overrides the ATS as the thrust levers are advanced, the ATS may advance the thrust levers to the forward stop, causing an engine exceedance.

V. After Landing

During landing, at airports where runway structural repair or debris is known to exist, use thrust reversers with extreme caution, to preclude the possibility of foreign object damage (FOD) from occurring.

After touchdown, when nose wheel is on ground and spoilers deployed:

- (1) Thrust reverse levers..... Lift and pull back.

When interlocks release (REV icons inside N₁ gauges are green):

- (2) Thrust reverse levers..... Pull back to required reverse thrust setting.

When speed falls to 60 KIAS:

- (3) Thrust reverse levers..... Select to not more than 60% N₁ reverse thrust or STOW.

Check REV icons (on EICAS) go out when thrust reversers are stowed.

NOTE

1. Reverse thrust idle is 25 to 30% N₁.
2. Thrust reverse (piggy-back) levers must be pushed fully down in order to do the following;
 - Achieve thrust reverser stow, and
 - Ensure that the main thrust lever mechanical lock is disengaged.

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1. POWER PLANT (CONT'D)

W. Engine Shutdown

Prior to engine shutdown:

- (1) Engines Operate at IDLE for a minimum of two (2) minutes, to dissipate heat and stabilize internal operating temperatures.

NOTE

Taxi time at 70% N₂ or below may be included in the 2-minute cool down period.

- (2) FUEL, L and R BOOST PUMP switch/lights Press out to disarm/shutdown both fuel pumps.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R BOOST PUMP INOP lights on.
- L and R FUEL PUMP ON advisory messages out.
- L and R FUEL PUMP caution messages on.

After the two minute cool-down period:

- (3) Thrust reverse levers..... Confirm stowed.

NOTE

The engine may be shutdown with the following indications present, however maintenance action is required after shutdown to stow the reverser prior to a subsequent start:

- REV icon in the N₁ gauge
- L or R REV UNLOCKED caution message displayed.

- (4) Thrust levers..... Set to SHUT OFF.

- (5) Engine instruments..... Check that both engines have shutdown.

Check that N₂, fuel flow and ITT are decreasing.

NOTE

1. ITT should not exceed 350°C.
2. Due to an engine control system anomaly, an APR CMD SET caution message may be displayed upon engine shutdown. When this occurs, the message should be disregarded.

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1. FLIGHT COMPARTMENT AND CABIN

NOTE

1. The cabin pressurization system MAN ALT selector is a three (3) position switch. The switch is spring loaded to mid position/center.
2. On the first flight of the day, ensure that the L (R) PACK OFF lights are on before the packs are selected on.
3. The cabin pressurization system must not be operated to -1,500 feet when the system is in manual mode.
4. If above 40,000 feet, one air-conditioning unit or cowl anti-ice must be selected on for each engine.
5. To prevent avionics overheat, ground operation of electrical/avionics equipment is limited to 30 minutes at ambient temperatures above 40°C (104°F), unless at least one air-conditioning pack is selected on, and the passenger door is closed.

A. Bleed Leak Detection Test – First Flight of the Day

Prior to selecting any bleeds, check 10th-stage and 14th-stage bleed leak detection systems as follows:

- (1) BLEED AIR, DUCT MON switch Set to TEST. Check that:
- L and R DUCT FAIL lights (10th and 14th) are on.
 - L and R 10TH DUCT warning messages on.
 - L and R 14TH DUCT warning messages on.
 - “BLEED AIR DUCT” aural on.
 - ANTI-ICE DUCT warning message on.
 - DUCT TEST OK advisory message on.
- (2) BLEED AIR, DUCT MON switch Set to LOOP A. Check that:
- No warning messages come on.
 - LOOP A TEST OK advisory message on.
- (3) BLEED AIR, DUCT MON switch Set to LOOP B. Check that:
- No warning messages come on.
 - LOOP B TEST OK advisory message on.



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CHALLENGER 650	NORMAL PROCEDURES Air-Conditioning and Pressurization	04-03-2
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1. FLIGHT COMPARTMENT AND CABIN (CONT'D)

A. Bleed Leak Detection Test – First Flight of the Day (Cont'd)

- (4) BLEED AIR, DUCT MON switch Set to NORM.
Check that no warning messages come on.

B. Air-Conditioning on Ground Using APU

The APU is used to provide air-conditioning on the ground, as follows:

- (1) APU Operating Check APU AVAIL light on.
- (2) BLEED AIR, 10TH STAGE, L and R switch/lights Check pressed out.
Check L and R 10TH STAGE CLOSED lights on.
- (3) BLEED AIR, APU LCV switch/light Press in to open load control valve.
Check APU LCV OPEN light on.
- (4) BLEED AIR, 10TH STAGE, ISOL switch/light Press in to open isolation valve.
Check 10TH ISOL OPEN light on.
- (5) AIR-CONDITIONING, L and R PACK switch/lights Press in to turn on packs.
Check L and R PACK OFF lights out.
- (6) AIR-CONDITIONING, CKPT TEMP and CABIN TEMP selectors Set mode and temperature as desired.

C. Before Take-Off

- (1) AIR-CONDITIONING, L and R PACK switch/lights Press in to turn on packs.
Check L and R PACK OFF lights out.

If the engine bleed is desired for air-conditioning during take-off, transfer bleeds as follows:

- (2) BLEED AIR, 10TH STAGE R switch/light Press in to open right valve.
Check R 10TH CLOSED light out.



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1. FLIGHT COMPARTMENT AND CABIN (CONT'D)

C. Before Take-Off (Cont'd)

- (3) BLEED AIR, 10TH STAGE ISOL
switch/light Check pressed out, to ensure isolation valve is closed.
Check the following:
- 10TH ISOL OPEN light out.
 - APU LCV OPEN light on.
- (4) BLEED AIR, 10TH STAGE L
switch/light Press in to open left valve and close APU load control valve.
Check the following:
- L 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (5) BLEED AIR, APU LCV switch/light Press out to match APU load control valve closed position.

Pressurization controls set-up:

- (6) CABIN PRESSURIZATION,
PRESS CONT switch/light Check pressed out, thus enabling automatic operation.
Check MANUAL light out.

NOTE

If pressurization system is in manual mode (PRESS CONT switch/light indicates MANUAL), select EMER DEPRESS switch/light off and operate MAN ALT regulator and MAN RATE knob as required after take-off (Refer to Chapter 5; ABNORMAL PROCEDURES – AIR-CONDITIONING AND PRESSURIZATION).

- (7) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Check pressed out.
Check ON light out.
- (8) CABIN PRESSURIZATION, LDG
ALT selector Set the following:
- Destination airfield altitude.
 - Desired cabin pressure rate of change.
 - Barometric pressure.

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1. FLIGHT COMPARTMENT AND CABIN (CONT'D)

D. After Take-Off

To transfer bleed air for air-conditioning from the APU to the engine bleeds:

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

- (1) BLEED AIR, 10TH STAGE R
switch/light Press in to open right valve.
Check the following:
 - R 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (2) BLEED AIR, 10TH STAGE ISOL
switch/light Press out to close isolation valve.
Check the following:
 - 10TH ISOL OPEN light out.
 - APU LCV OPEN light on.
- (3) BLEED AIR, 10TH STAGE L
switch/light Press in to open left valve and close APU load control valve.
Check the following:
 - L 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (4) BLEED AIR, APU LCV switch/light Press out to match APU load control valve closed position.

E. Descent

- (1) CABIN PRESSURIZATION, LDG
ALT selector Check selected to the following:
 - Destination airfield altitude.
 - Destination airfield barometric pressure.
 - Desired cabin pressure descent rate.
 Check FAULT lamp out.

To transfer bleed air for air-conditioning from the engine bleeds to the APU:

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

When 10TH stage bleeds are closed, CLB and CRZ N_1 data is not available and will cause the ATS to fail unless TO or TGT is selected prior to closing bleeds.



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CHALLENGER 650	NORMAL PROCEDURES Air-Conditioning and Pressurization	04-03-5
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1. FLIGHT COMPARTMENT AND CABIN (CONT'D)

E. Descent (Cont'd)

- (2) APU Start Refer to the AUXILIARY POWER UNIT section in this chapter.
- (3) BLEED AIR, APU LCV switch/light..... Press in to preselect load control valve to the open position.
- (4) BLEED AIR, 10TH STAGE L switch/light..... Press out to select left valve closed.
Check the following:
- APU LCV OPEN light on.
 - L 10TH CLOSED light on.
- (5) BLEED AIR, 10TH STAGE ISOL switch/light..... Press in to open isolation valve.
Check the following:
- 10TH ISOL OPEN light on.
 - APU LCV OPEN light out.
- (6) BLEED AIR, 10TH STAGE R switch/light..... Press out to select right valve closed.
- R 10TH CLOSED light on.
 - APU LCV OPEN light on.

NOTE

10TH stage engine bleeds must be closed for landing with cowl and/or wing anti-ice system on (Refer to Chapter 2; LIMITATIONS – SYSTEM LIMITATIONS).



If the pressurization system is in manual mode (PRESS CONT switch/light indicates MANUAL), EMER DEPRESS must be selected ON after landing, to depressurize the airplane. The EMER DEPRESS ON light will not come on until 30 seconds after weight-on-wheels.

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2. UNPRESSURIZED TAKE-OFF AND LANDING

The procedures below are recommended when performing a take-off or landing without using engine and APU bleeds for pressurization.

A. Before Take-Off

- (1) AIR-CONDITIONING, RAM AIR
switch/light Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (2) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press in Check the following:
 - EMER DEPRESS ON light on, and
 - EMER DEPRESS caution message on.
- (3) AIR-CONDITIONING, L and R
PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (4) BLEED AIR, 10TH STAGE,
L and R switch/lights Press out Check L and R 10TH STAGE
CLOSED lights on.

B. After Take-Off

At a safe altitude (not below 1,500 feet AGL):

- (1) BLEED AIR, 10TH STAGE,
L and R switch/lights Press in Check L and R 10TH STAGE
CLOSED lights out.

NOTE

If an engine failure occurs, delay selecting the packs on until after obstacle clearance is assured.

- (2) AIR-CONDITIONING, R PACK
switch/light Press in to start right air-conditioning
pack.
Check R PACK OFF light out.

NOTE

A pressure bump may occur.

- (3) AIR-CONDITIONING, RAM AIR
switch/light Press out to close ram air valve.
Check RAM AIR OPEN light off.



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2. UNPRESSURIZED TAKE-OFF AND LANDING (CONT'D)

B. After Take-Off (Cont'd)

- (4) AIR-CONDITIONING, L PACK
switch/light Press in to start left air-conditioning pack.
Check L PACK OFF light out.
- (5) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press out Check the following:
- EMER DEPRESS ON light out, and
 - EMER DEPRESS caution message out.

C. Landing

- (1) Airplane altitude Descend to 10,000 feet maximum or lowest safe altitude, whichever is higher.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Check pressed out, thus enabling automatic operation.
Check MANUAL light out.

At the turn to final approach:

- (3) AIR-CONDITIONING, L and R
PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (4) BLEED AIR, 10TH STAGE,
L and R switch/lights Press out Check L and R 10TH STAGE CLOSED lights on.
- (5) AIR-CONDITIONING, RAM AIR
switch/light Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (6) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press in Check the following:
- EMER DEPRESS ON light on, and
 - EMER DEPRESS caution message on.

NOTE

Avoid high rates of descent for passenger comfort.

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CHALLENGER 650	NORMAL PROCEDURES Auxiliary Power Unit	04-04-1
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1. AUXILIARY POWER UNIT

CAAI Certified Airplanes

NOTE

APU must be operating during take-off and landing, and at altitudes below 10,000 feet AGL (Refer to Chapter 2; LIMITATIONS – POWER PLANT).

A. External Walkaround

- (1) APU exhaust portsCheck clear of any obstructions or contaminants.
- (2) APU drain portsCheck for evidence of spillage.

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CHALLENGER 650	NORMAL PROCEDURES Auxiliary Power Unit	04-04-2
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1. AUXILIARY POWER UNIT (CONT'D)

B. Starting the APU on Ground or in Flight – First Start of the Day

NOTE

The APU, PWR FUEL switch/light must be selected off between each APU start. The APU, PWR FUEL switch/light should be firmly pressed once only. If pressed twice, an immediate shutdown may occur.

- (1) BATT MASTER switch Select to ON.
- (2) APU Fire Detection Systems Test
Procedure Accomplish Refer to FIRE PROTECTION SYSTEM in this chapter.
- (3) Firex Monitor System Test
Procedure Accomplish Refer to FIRE PROTECTION SYSTEM in this chapter.
- (4) AC POWER, APU GEN switch Select to OFF.
- (5) BLEED AIR, APU LCV switch/light Check pressed out.
- (6) APU, PWR FUEL switch/light Press in Check EICAS page – APU readouts displayed.

NOTE

During in flight starts at high altitudes, the APU EGT may momentarily spike into the amber temperature zone and then immediately return to normal green values. This is normal operation and no crew action is required.

- (7) APU, START/STOP switch/light Press in and check the following:
 - START light on.
 - 35 to 45% rpm – START light out.
 - 95% rpm + 4 seconds (approx) – AVAIL light on.
 - 100% rpm + 15 seconds (approx) – APU GEN OFF caution message on.

After APU RPM and EGT have stabilized:

- (8) AC POWER, APU GEN switch Set as required (Refer to Chapter 2; LIMITATIONS – POWER PLANT).
- (9) BLEED AIR, APU LCV switch/light Set as required.

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CHALLENGER 650	NORMAL PROCEDURES Auxiliary Power Unit	04-04-3
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1. AUXILIARY POWER UNIT (CONT'D)

C. Starting the APU on Ground or in Flight – Subsequent Starts

NOTE

The APU, PWR FUEL switch/light must be selected off between each APU start. The APU, PWR FUEL switch/light should be firmly pressed once only. If pressed twice, an immediate shutdown may occur.

- (1) BATT MASTER switch Select to ON.
- (2) AC POWER, APU GEN switch Select to OFF.
- (3) BLEED AIR, APU LCV switch/light Check pressed out.
- (4) APU, PWR FUEL switch/light Press in Check EICAS page – APU readouts displayed.

NOTE

During in flight starts at high altitudes, the APU EGT may momentarily spike into the amber temperature zone and then immediately return to normal green values. This is normal operation and no crew action is required.

- (5) APU, START/STOP switch/light Press in and check the following:
 - START light on,
 - 35 to 45% rpm – START light out.
 - 95% rpm + 4 seconds (approx) – AVAIL light on.
 - 100% rpm + 15 seconds (approx) – APU GEN OFF caution message on.

After APU RPM and EGT have stabilized:

- (6) AC POWER, APU GEN switch Set as required (Refer to Chapter 2; LIMITATIONS – POWER PLANT).
- (7) BLEED AIR, APU LCV switch/light Set as required.

CHALLENGER 650	NORMAL PROCEDURES Auxiliary Power Unit	04-04-4
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1. AUXILIARY POWER UNIT (CONT'D)

D. Stopping the APU

- (1) AC POWER, APU GEN switch.....Select to OFF/RESET.
Check APU GEN OFF caution message on.
- (2) APU, START/STOP switch/light.....Press out to shutdown the APU.
Check AVAIL light out.

If on ground:

- (3) L and R PACK switch/lights.....Press out
- (4) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.

When rpm decreases to below 50% and EGT to below 350°C (662°F):

- (5) APU, PWR FUEL switch/light.....Press out and check the following:
- APU GEN OFF caution message out.
 - EICAS page – APU readouts out (after approx 60 sec).

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CHALLENGER 650	NORMAL PROCEDURES Automatic Flight Control System	04-05-1
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1. AUTOMATIC FLIGHT CONTROL SYSTEM

NOTE

- During a descent, when autopilot is in flight level change or vertical speed mode with the flight director engaged, the flight director may command excursions beyond V_{MO} / M_{MO} . Care should be taken not to exceed V_{MO} / M_{MO} .
- After generator transfers, if AFCS 2 is selected, the following must be manually re-engaged:
 - Yaw damper 2, and
 - Autopilot (if it was coupled before power interruption).

A. Yaw Dampers – Before Take-Off

- (1) YD 1 and YD 2 switch/lights Press in to engage both.

Check the following:

- YAW DAMPER caution message out,
- YD 1 INOP status message out, and
- YD 2 INOP status message out.

B. Autopilot – After Take-Off

At an altitude not less than 320 feet AGL:

- (1) AP switch/light Press in to engage.

C. Autopilot – During Flight

- (1) Autopilot operation Monitor Check for significant discrepancy between L and R elevator position.

D. AFCS – Before Landing



To avoid possible false localizer captures, do not arm approach until airplane is close to, or established on, an inbound localizer course.

- (1) Autopilot Disengage at an altitude not less than 320 feet AGL.

If performing an autopilot-coupled ILS approach:

- (1) Autopilot Disengage at an altitude not less than 80 feet AGL.
- (2) Flight Directors Ensure both displayed.

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CHALLENGER 650	NORMAL PROCEDURES Doors	04-06-1
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1. DOORS

NOTE

The main passenger door can be opened, closed, or used for static open operation up to 60 knots windspeed.

A. Passenger Door

- (1) Passenger door Shut and visually ensure that four roll-over catch witness marks are correctly aligned, and inner handle is correctly stowed in the down and latched position.
Check UNLOCKED (red) flag in the stair riser changes to a LOCKED (green) flag, when the door inner T-handle is stowed.
- (2) Passenger door, inner T-handle Stowed Check PAX DOOR CLOSED advisory message on.

B. Emergency Exits and Hatches (During Walkaround Inspection)

- (1) Airplane servicing/maintenance doors Check shut
- (2) Overwing exit Check secure
- (3) Aft equipment bay door Check shut
- (4) Baggage bay door Check shut

First flight of the day, if passenger door is closed and locked:

- (5) Passenger door outer handle trigger plate Push and as the outer handle comes out of the housing, check that the vent flap (flapper door) springs open.

First flight of the day, if passenger door is fully open:

- (5) Passenger door vent flap (flapper door) Check for spring tension by pulling door vent flap outboard, to verify that spring has not disconnected.

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CHALLENGER 650	NORMAL PROCEDURES Electrical	04-07-1
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1. ELECTRICAL

A. Before Normal APU Start

- (1) BATT MASTER switchSelect to ON.
- (2) DC ELECTRICAL pageSelect and check that battery voltage is sufficient for APU start.
- Check that:
- APU BATT above 22 VDC.
 - MAIN BATT above 22 VDC.
- (3) AC POWER, APU GEN switch.....Select to OFF/RESET.

B. Before Normal Engine Start

- (1) AC POWER, GEN 1 and GEN 2 switches.....Check selected to OFF/RESET.

C. After Starting Engines

- (1) AC POWER, GEN 1 and GEN 2 switches.....Select to ON.
- Check GEN 1 and 2 OFF caution messages out.
- (2) AC ELECTRICAL pageCheck generator voltage, load and frequency readouts normal.
- (3) DC ELECTRICAL pageCheck the following:
- Voltage readouts are normal.
 - DC load readouts are normal.
 - TIE symbol indicates DC tie open.
- (4) EICAS page.....Check the following:
- AC BUS 1 (2) caution messages out.
 - AC 1 (2) AUTOXFER caution messages out.
 - DC BUS 1 (2) caution messages out.
- (5) ADG AUTO DEPLOY CONTROL switchSelect to UNIT.
- Check TEST light comes on for two (2) seconds, then goes out.

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CHALLENGER 650	NORMAL PROCEDURES Electrical	04-07-2
		Oct 27/15

1. ELECTRICAL (CONT'D)

D. Before Normal Engine Shutdown

- (1) AC POWER, GEN 1 and GEN 2 switches.....Select to OFF/RESET.
Check GEN 1 and 2 OFF caution messages on.

E. Before Normal APU Shutdown

- (1) AC POWER, APU GEN switch.....Select to OFF/RESET.

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CHALLENGER 650	NORMAL PROCEDURES Electronic Flight Instruments	04-08-1
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1. ELECTRONIC FLIGHT INSTRUMENTS

NOTE

During airplane operations when the flight compartment and cabin temperatures are above 30°C (86°F), the air-conditioning packs should be operating, and the passenger door closed, in order to maintain display temperatures within a range that will maximize cockpit equipment service life.

A. Clock – Before Starting Engines

- (1) GPS clock..... Check/Set to correct local time.

B. Altimeters – Before Starting Engines

- (1) EFIS altimeters (2) and Integrated Standby Instrument altimeter (1) Set to departure field elevation.



Reduced vertical separation with terrain may be encountered in mountainous terrain if altimeter is incorrectly set and LNAV is armed for departure.

C. Altimeters – Before Landing

- (1) EFIS altimeters (2) and Integrated Standby Instrument altimeter (1) Set to destination airport barometric setting.

NOTE

Destination airport barometric setting should also be set at the cabin pressurization system panel.

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CHALLENGER 650	NORMAL PROCEDURES Fire Protection System	04-09-1
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1. FIRE PROTECTION SYSTEM

A. Engine, Jetpipe and APU Fire Detection Systems Test – First Flight of the Day

Check fire detection systems as follows:

- (1) FIRE DET switch
(centre pedestal)

.....Select to WARN.

Hold switch in this position for 10 seconds and check:

- Fire bell on.
- “JETPIPE OVERHEAT” aural on.

The following warning messages are on:

- L and R ENG FIRE.
- APU FIRE.
- L and R JETPIPE OVHT.

The following switch/lights are on:

- LH and RH ENG FIRE PUSH.
- APU FIRE PUSH.
- BOTTLE 1 and 2 ARMED PUSH TO DISCH.
- APU, BOTTLE ARMED PUSH TO DISCH.

All indications go out when switch released.

- (2) FIRE DET switch
(centre pedestal)

.....Select to FAIL.

Hold the switch in this position and check that the following caution messages are displayed:

- L and R FIRE FAIL.
- APU FIRE FAIL.
- L and R JETPIPE OVHT FAIL.

Messages go out when switch released.

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CHALLENGER 650	NORMAL PROCEDURES Fire Protection System	04-09-2
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1. FIRE PROTECTION SYSTEM (CONT'D)

B. Engine and APU Firex Monitor System Test – First Flight of the Day

Check firex squibs as follows:

(1) FIREX MONITOR, ENG BOTTLE

1 and 2 switches.....Select to TEST and hold.

Check the following advisory messages on:

- L ENG SQUIB 1 OK.
- L ENG SQUIB 2 OK.
- R ENG SQUIB 1 OK.
- R ENG SQUIB 2 OK.

(2) FIREX MONITOR, ENG BOTTLE

1 and 2 switches.....Release Check that the messages given above for both engine bottle squibs go out.

(3) FIREX MONITOR, APU BOTTLE

switchSelect to TEST and hold for 5 seconds.

Check the following advisory messages on:

- APU SQUIB 1 OK.
- APU SQUIB 2 OK.

(4) FIREX MONITOR, APU BOTTLE

switchRelease Check the following advisory messages stay on momentarily and then go out:

- APU SQUIB 1 OK.
- APU SQUIB 2 OK.

CHALLENGER 650	NORMAL PROCEDURES Flight Controls	04-10-1
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1. FLIGHT CONTROLS

A. Ground and Flight Spoilers, Before Take-Off – First Flight of the Day

To check ground spoiler operation:

- (1) GND SPOILERS switchSelect to DISARM.
Verify that both ground spoiler panels are retracted.
- (2) GND SPOILERS switchSelect to ARM.
Verify that both ground spoiler panels deploy to maximum travel.
- (3) GND SPOILERS switchSelect to AUTO.
Verify that both ground spoiler panels are retracted.

To check flight spoiler operation:

- (1) FLIGHT SPOILER leverSelect to MAX.
Verify that flight spoiler panels deploy.

NOTE

When flight spoilers are fully deployed, (on the ground and with throttles at IDLE), ground spoilers will deploy if selected to AUTO.

- (2) FLIGHT SPOILER leverSelect to RETRACT.
Verify that flight spoiler panels retract.

B. Flight Controls Trim Systems, Before Flight – First Flight of the Day

To check stabilizer trim system:

- (1) STAB TRIM, CH 1 and MACH TRIM switch/lightsPress in
Check STAB CH 2 INOP status message on.
Check for independent CH 1 operation from both stabilizer trim switches, in both directions, until trim clacker is activated.
- (2) Copilot's STAB TRIM DISC switchPress
Check the following:
- STAB TRIM caution message on.
 - MACH TRIM caution message on.
 - Stabilizer trim is inoperative.



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CHALLENGER 650	NORMAL PROCEDURES Flight Controls	04-10-2
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1. FLIGHT CONTROLS (CONT'D)

B. Flight Controls Trim Systems, Before Flight – First Flight of the Day (Cont'd)

- (3) STAB TRIM, CH 2 and MACH TRIM switch/lights Press in Check STAB CH 1 INOP status message on.
Check for independent CH 2 operation from both stabilizer trim switches, in both directions, until trim clacker is activated.
- (4) Pilot's STAB TRIM DISC switch Press Check the following:
- STAB TRIM caution message on.
 - MACH TRIM caution message on.
 - Stabilizer trim is inoperative.
- (5) STAB TRIM, CH 1 and CH 2 switch/lights Press in to engage CH 1 and CH 2.
Check the following:
- STAB CH 1 and CH 2 INOP status messages out.
 - STAB TRIM caution message out.
- (6) MACH TRIM switch/light Press in to engage Mach trim.
Check MACH TRIM caution message out.
- To check aileron and rudder trim systems:**
- (7) Aileron and rudder trims Check for free movement in both directions.
Set at neutral for take-off.

CHALLENGER 650	NORMAL PROCEDURES Flight Controls	04-10-3
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1. FLIGHT CONTROLS (CONT'D)

C. Before Take-Off

With both engines running, check flight control systems as follows:

- (1) Ailerons Operate over full travel range.
Check the following:
- Full free movement.
 - Corresponding surface displacement on flight control page.
 - Ensure that no hydraulic messages come on.
- (2) Elevators Operate over full travel range.
Check the following:
- Full free movement.
 - Corresponding surface displacement on flight control page.
 - Ensure that no hydraulic messages come on.
 - ELEVATOR SPLIT caution message out.
- (3) FLAPS Set for take-off.
Check corresponding surface displacement on flight control page.
- (4) NOSE STEER switch Select to OFF.
Check STEERING INOP caution message comes on.
- (5) Rudder Operate over full travel range.
Check the following:
- Full free movement.
 - Corresponding surface displacement on flight control page.
- (6) NOSE STEER switch Select to ARMED.
Check STEERING INOP caution message out.



CHALLENGER 650	NORMAL PROCEDURES Flight Controls	04-10-4
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1. FLIGHT CONTROLS (CONT'D)

C. Before Take-Off (Cont'd)

Set trims to neutral for take-off:

(7) Aileron and rudder trimsCheck Set at neutral for take-off.

(8) Stabilizer trim..... Set for take-off.

EICAS STABILIZER TRIM SETTING FOR TAKE-OFF	
C.G. Range (%MAC)	Trim Setting
16.0 – 16.2	7.2
16.3 – 17.2	7.0
17.3 – 18.3	6.8
18.4 – 19.3	6.6
19.4 – 20.3	6.4
20.4 – 21.3	6.2
21.4 – 22.4	6.0
22.5 – 23.4	5.8
23.5 – 24.4	5.6
24.5 – 25.5	5.4
25.6 – 26.5	5.2
26.6 – 27.5	5.0
27.6 – 28.5	4.8
28.6 – 29.6	4.6
29.7 – 30.6	4.4
30.7 – 31.6	4.2
31.7 – 32.6	4.0
32.7 – 33.7	3.8
33.8 – 34.7	3.6
34.8 – 36.5	3.4
36.6 – 38.0	3.2

NOTE

Trim data from the FMS should be rounded down to the next lowest multiple of 0.2 trim units.

D. During Flight

When operating flight spoilers, do not release FLIGHT SPOILER lever until normal extension or retraction of spoilers is ascertained.

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CHALLENGER 650	NORMAL PROCEDURES Navigation Systems	04-11-1
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1. FLIGHT MANAGEMENT SYSTEM (FMS)

A. Prior to FMS Use

- (1) FMS positionAuto initialize

NOTE

1. The engine bleed selection used for the FMS thrust setting computations is based upon the sensed setting of the airplane engine bleed cockpit switches, unless manually overridden by the flight crew on the THRUST LIMIT CDU page.

The engine bleed selection used for the FMS V speed computations is based upon the default setting (as selected on the DEFAULTS 5/5 CDU page) or upon manual selection, by the flight crew, for the anticipated take-off or approach configuration.

The engine bleed setting on the TAKEOFF REF and APPROACH REF CDU pages represents the planned setting and not the actual/sensed airplane setting.

2. In case of multiple obstacles, each obstacle clearance calculation is done separately. For each obstacle, obstacle height and obstacle distance from reference zero should be entered in the FMS. The FMS computes and displays the geometric gross level-off height and the required climb gradient for obstacle clearance. The highest gross level-off height and the greatest required climb gradient of all obstacles should be retained. Note that an obstacle may be the most limiting for the required climb gradient while another is the most limiting for the gross level-off height.

The most limiting gross level-off height and required climb gradient should be entered in the FMS. The FMS computes and displays the take-off weight limited by obstacle clearance, which is used to calculate the maximum take-off weight, to ensure all obstacles will be cleared.

B. After Periods of Dead Reckoning (DR) Navigation

- (1) FMS position information.....Check for accuracy.

C. Go-Around

NOTE

VNAV may be selected and used normally following the sequencing of the last waypoint on the missed approach procedure, or upon entering a new flight plan and activating a leg that is not part of the missed approach procedure.

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CHALLENGER 650	NORMAL PROCEDURES Navigation Systems	04-11-2
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2. INERTIAL REFERENCE SYSTEM (IRS)

A. Before Starting Engines

NOTE

The IRS automatically aligns using GPS position (normal IRS alignment time is approximately 7 minutes). If GPS is not available, it will align using the position entered to initialize the FMS.

Following a successful initial alignment, do not manually re-enter the position in the FMS CDU either immediately before, or immediately after airplane motion. If this has occurred, a complete normal shutdown and restart of the airplane and IRSs must be completed. Failure to do so may lead to a total IRU loss during subsequent flight legs.

3. GLOBAL POSITIONING SYSTEM (GPS)

A. Before Starting Engines

- (1) FMS Review and disposition messages.

NOTE

For any SBAS covered airspace (US RNAV routes, SIDs, and STARs), it is not required to run a pre-flight prediction of RAIM availability along the intended route if no satellites are NOTAM'd as out of service over the intended route.

For non-SBAS covered airspace, certain operations may require that an appropriate prediction program be used prior to departure for RAIM availability along the intended route.

CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-1
		Oct 27/15

1. FUEL

NOTE

Standard/normal fuel system operation is tank to engine, with electric boost pumps armed, crossflow and transfer valves closed.

A. External Walkaround

- (1) NACA ventsCheck to ensure the following:
 - Clear of any obstructions or contaminants, and
 - No evidence of fuel spillage or leakage.
- (2) Vent relief valvesCheck for evidence of fuel spillage and leakage from:
 - Lower wing, and
 - Vertical stabilizer (left-hand side).
- (3) Drains/drain valvesCheck for evidence of fuel spillage and leakage.

B. Before Starting Engines

- (1) FUEL, L and R BOOST PUMP switch/lights Press in to turn both boost pumps on.
Check the following:
 - L and R BOOST PUMP ON lights on,
 - L and R BOOST PUMP INOP lights out, and
 - L and R FUEL PUMP ON advisory messages on.

Check gravity crossflow operation:

- (2) FUEL, GRAVITY XFLOW switch/light Press in to open gravity crossflow valve.
Check GRAVITY XFLOW OPEN light on.



Do not leave the fuel system in manual or gravity crossflow while the airplane is on the ground (Uneven ground surface may create fuel imbalance).



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CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-2
		Oct 27/15

1. FUEL (CONT'D)

B. Before Starting Engines (Cont'd)

- (3) FUEL, GRAVITY XFLOW
switch/light Press out to close gravity crossflow valve.

Check GRAVITY XFLOW
OPEN light out.

Check that all fuel crossflow has been disabled:

- (4) FUEL, L TO AUX XFLOW and
R TO AUX XFLOW switch/lights Check pressed out, thus disabling auxiliary tank crossflow.

Check L TO AUX and R TO
AUX XFLOW ON lights out.

- (5) FUEL, GRAVITY XFLOW
switch/light Check pressed out, thus disabling gravity crossflow.

Check GRAVITY XFLOW ON
light out.

- (6) FUEL, TAIL TANK TRANSFER,
MANUAL OVRD switch/light Check pressed out.

Check MANUAL light out.

Check that fuel dump has been disabled:

- (7) FUEL, DUMP switch/light Check pressed out.

Check DUMP OPEN light out.

Check fuel temperature:

- (8) Fuel temperature Check the following:

- BULK FUEL TEMP caution message not displayed.
- SUMMARY page – Fuel temperatures (bulk and engine) normal.

CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-3
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1. FUEL (CONT'D)

C. After Starting One Engine – First Flight of the Day

To verify fuel system check valve operation:

- (1) FUEL, L and R BOOST PUMP

switch/lights Press out to turn both boost pumps off.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R BOOST PUMP INOP lights on.
- L and R FUEL PUMP ON advisory messages out.
- FUEL LO PRESS caution message is annunciated for the non-operating engine.

After 2 minutes, if FUEL LO PRESS caution message is still not displayed for the non-operating engine:

- (2) Dry motor..... Accomplish for maximum of 30 seconds on the non-started engine.

Applicable FUEL LO PRESS caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of the FUEL LO PRESS caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.



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CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-4
		Oct 27/15

1. FUEL (CONT'D)

C. After Starting One Engine – First Flight of the Day (Cont'd)

Before starting other engine:

(3) FUEL, L and R BOOST PUMP

switch/lights Press in to turn both boost pumps on.

Check the following:

- L and R BOOST PUMP ON lights on.
- L and R BOOST PUMP INOP lights out.
- L and R FUEL PUMP ON advisory messages on.
- All FUEL LO PRESS caution messages are out.

D. Before Take-Off

(1) FUEL, L and R BOOST PUMP

switch/lights Check pressed in.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R FUEL PUMP ON advisory messages out.

E. During Cruise

NOTE

1. During gravity crossflow operations, a steady heading sideslip must be used to ensure proper operation and to accelerate gravity crossflow.
2. To enhance gravity crossflow operation, fly one wing low slip (on the side with less fuel).
3. During uncoordinated flight, fuel indications are inaccurate. Accurate fuel indications will be obtained after 30 seconds when level and stabilized in coordinated flight.

If crossflow operation is required:

(1) FUEL, L (R) TO AUX XFLOW

switch/lights Press in/press out as required, to transfer/balance fuel load.

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CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-5
		Oct 27/15

1. FUEL (CONT'D)

F. Before Engine Shutdown – First Flight of the Day

To complete check of fuel system check valve operation:

- (1) Applicable engine that was started first (in para 1.C.).....Shutdown
- (2) FUEL, L and R BOOST PUMP switch/lightsPress out to turn both boost pumps off.
Check the following:
 - L and R BOOST PUMP ON lights out.
 - L and R BOOST PUMP INOP lights on.
 - L and R FUEL PUMP ON advisory messages out.
 - FUEL LO PRESS caution message is annunciated for the engine that now has been shutdown.

After 2 minutes, if FUEL LO PRESS caution message is still not displayed for the non-operating engine:

- (3) Dry motor..... Accomplish for maximum of 30 seconds on the non-operating engine.
Applicable FUEL LO PRESS caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of the FUEL LO PRESS caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.

CHALLENGER 650	NORMAL PROCEDURES Fuel	04-12-6
		Oct 27/15

1. FUEL (CONT'D)

G. After Engine Shutdown – Last Flight of the Day

To verify fuel system crossflow valve operation:

- (1) FUEL, L TO AUX XFLOW
switch/light Press in to open left crossflow valve.
Check L TO AUX XFLOW ON light on.
- (2) FUEL, L TO AUX XFLOW
switch/light Press out to close left crossflow valve.
Check L TO AUX XFLOW ON light out.
- (3) FUEL, R TO AUX XFLOW
switch/light Press in to open right crossflow valve.
Check R TO AUX XFLOW ON light on.
- (4) FUEL, R TO AUX XFLOW
switch/light Press out to close right crossflow valve.
Check R TO AUX XFLOW ON light out.

H. Refueling/Defueling (Flight Compartment Panel) (If installed)

A second refuel/defuel control panel, located in the flight compartment on the copilot's bulkhead, has control priority over the external refuel/defuel control panel. Upon powering up the refuel/defuel control panel in the flight compartment, power to the external refuel/defuel control panel is automatically cut.

- (1) Mode Selector Rotary Control TEST Check that the following series of indications are displayed:
 - SOV CL lights go out in sequence, then
 - SOV OP lights come on in sequence, then
 - HIGH LEVEL DETECTOR lights come on, then
 - SOV OP lights go out and SOV CL lights come on.

NOTE

Refuel pressure is required to perform the test.

During the refueling/defueling operation, check for fuel spillage from the vent relief valves, water drain valves and the NACA vents.

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CHALLENGER 650	NORMAL PROCEDURES Hydraulic Power	04-13-1
		Oct 27/15

1. HYDRAULIC POWER

WARNING

Due to automatic hydraulic system start-up during flaps selection, ensure that all control surfaces and the hydraulically-operated nose door area are clear before actuating the flaps selector, or before turning on the hydraulic 3A pump.

NOTE

When ambient temperatures are greater than 40°C (104°F), avoid prolonged ground operations with the hydraulic system operating.

A. Before Start

- (1) HYDRAULIC pump switch 3A Check selected to OFF.
- (2) HYDRAULIC pump switch 3B Select to ON.

Check the following:

- HYD 3 LO PRESS caution message out.
- 3B pump pressure is greater than 1800 psi on HYDRAULIC synoptic page.
- 3A pump outline is unpressurized (white) on HYDRAULIC synoptic page.

- (3) HYDRAULIC pump switches 1B and 2B Select to ON.

Check the following:

- HYD 1 LO PRESS and HYD 2 LO PRESS caution messages out.
- 1B and 2B pump pressure is greater than 1800 psi on HYDRAULIC synoptic page.

- (4) HYDRAULIC pump switches 1B, 3B and 2B Select to AUTO.

- (5) HYDRAULIC pump switch 3A Select to ON.

B. Before Take-Off

- (1) HYDRAULIC pump switch 3A Check selected to ON.

Check no HYD 3 LO PRESS caution message displayed.

- (2) FLAPS Select to 20°.

Check no hydraulic system caution messages displayed.

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CHALLENGER 650	NORMAL PROCEDURES Hydraulic Power	04-13-2
		Oct 27/15

1. HYDRAULIC POWER (CONT'D)

C. After Take-Off

NOTE

Hydraulic pump 3A switch must remain set to ON.

D. Before Landing

(1) FLAPS Select to 20°.

Check no hydraulic system
caution messages displayed.

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CHALLENGER 650	NORMAL PROCEDURES Ice and Rain Protection	04-14-1
		Rev 6, Nov 28/16

1. ICE AND RAIN PROTECTION

NOTE

During operations with continuous ignition selected, igniter status can be verified as follows:

IGNITION SYSTEM STATUS	APPLICABLE ADVISORY MESSAGE
AT LEAST ONE IGNITER IN USE ON EACH ENGINE	IGNITION A/B
AT LEAST ONE IGNITER IN USE	IGNITION A OR IGNITION B

A. External Walkaround

Visually check the following:

- (1) Engine cowl Check the following:
- Cowl anti-ice blowout plug is flush with skin.
 - Cowl leading edge for evidence of discoloration or deformation.

B. Wing Anti-Ice Switch Test – First Flight of the Day

On airplanes with Supplemental Ground Wing Anti-Icing:

- (1) ANTI-ICE, WING switch Check selected to OFF.
Check WING A/ICE ON advisory message out.
- (2) SUPP GND WING ANTI-ICE, SYS switch/light Check pressed out.
- (3) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice on.
Check L and R COWL A/ICE caution messages out.

NOTE

If L or R COWL A/ICE caution message comes on immediately, wing anti-ice switch has failed and maintenance is required.

- (4) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.

C. Ice Detector Test – First Flight of the Day

- (1) ANTI-ICE, TEST switch Select to DET.
Check ICE caution message on.

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CHALLENGER 650	NORMAL PROCEDURES Ice and Rain Protection	04-14-2
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1. ICE AND RAIN PROTECTION (CONT'D)

D. Windshield/Window Heat Test – Before Each Flight

- (1) ANTI-ICE, WSHLD/WIND L and R switches Select to HI.
- (2) ANTI-ICE, WSHLD/WIND, TEST switch Press and hold. Check the following:
 - L and R WSHLD HEAT OK advisory messages on.
 - L and R WINDOW HEAT OK advisory messages on.
- (3) ANTI-ICE, WSHLD/WIND, L and R switches Select to LOW, unless icing, precipitation or other conditions that could cause fogging or misting to occur.

E. Wing Overheat Detection System – Before Each Flight

With the engines not running, check the operation of the wing overheat detection system as follows:

- (1) ANTI-ICE, WING switch Select to OFF.
- (2) ANTI-ICE, TEST switch Select to WING.
Check the following:
 - WING OVHT warning message on.
 - “WING OVERHEAT” aural.
 - WING L HEAT and R HEAT lights on.
 - WING A/ICE TEST OK advisory message on.

F. 14TH Stage Valve Check – Before Each Flight (Engines Running)

- (1) BLEED AIR, 14TH STAGE L switch/light Press out Check L CLOSED light on.
- (2) BLEED AIR, 14TH STAGE R switch/light Press out Check R CLOSED light on.
- (3) BLEED AIR, 14TH STAGE L and R switch/lights Press in Check L and R CLOSED lights out.

CHALLENGER 650	NORMAL PROCEDURES Ice and Rain Protection	04-14-3
		Rev 6, Nov 28/16

1. ICE AND RAIN PROTECTION (CONT'D)

G. Wing Anti-Ice Check – Before Each Flight (Engines Running)

Check the operation of the wing anti-ice system as follows:

- (1) BLEED AIR, 14TH STAGE L and R switch/lightsCheck pressed in (SOVs open).
Check L and R CLOSED lights out.
- (2) ANTI-ICE, WING switchSelect to NORM, to open both wing anti-icing valves.
Check the following:
- Left and right engines' ITT increases.
 - N₂ gauges change colour to indicate wing anti-ice mode, in non-compressed display format.
 - L and R WING A/ICE caution messages come on briefly, then go out.
If messages do not go out, advance thrust levers to not more than 75% N₂.
 - WING A/ICE ON advisory message on.
 - WING L HEAT and R HEAT lights on.
- (3) ANTI-ICE, WING switchSelect to OFF.
Check the following:
- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
 - WING A/ICE ON advisory message out.

1. ICE AND RAIN PROTECTION (CONT'D)

H. Wing Anti-Ice/Isolation Valve – First Flight of the Day (Engines Running)

With both engines running, check the operation of the 14th-stage isolation valve and wing anti-ice normal mode as follows:

- (1) BLEED AIR, 14TH STAGE L and R switch/lights Check pressed in (SOVs open).
Check L and R CLOSED lights out.
- (2) ANTI-ICE, WING switch Select to NORM, to open both wing anti-icing valves.
Check the following:
 - Left and right engines' ITT increases.
 - N₂ gauges change colour to indicate wing anti-ice mode, in non-compressed display format.
 - L and R WING A/ICE caution messages come on briefly, then go out.
If messages do not go out, advance thrust levers to not more than 75% N₂.
 - WING A/ICE ON advisory message on.
 - WING L HEAT and R HEAT lights on.
- (3) BLEED AIR, 14TH STAGE ISOL switch/light Press in to open isolation valve.
Check ISOL OPEN light on.
- (4) BLEED AIR, 14TH STAGE ISOL switch/light Press out to close isolation valve.
Check ISOL OPEN light out.
- (5) ANTI-ICE, WING switch Select to OFF.
Check the following:
 - N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
 - WING A/ICE ON advisory message out.

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		Rev 6, Nov 28/16

1. ICE AND RAIN PROTECTION (CONT'D)

I. Supplemental Ground Wing Anti-Ice Check (if installed) – First Flight of the Day (Engines Running)

With both engines running, check the operation of the supplemental ground wing anti-ice as follows:

- (1) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice on.
Check the following:
 - COWL A/ICE ON advisory message on.
 - L and R COWL A/ICE caution messages out.
- Forty-five seconds after selecting cowl anti-ice on:**
- (2) CAS messages..... Review Check COWL A/ICE ON advisory message remains on.

NOTE

It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

- (3) SUPP GND WING ANTI-ICE, SYS switch/light..... Check pressed out.
- (4) SUPP GND WING ANTI-ICE, TEST switch/light..... Press in Check the following:
 - TEST ON light comes on.
 - Within 2 minutes, check TEST OK light comes on.
- (5) SUPP GND WING ANTI-ICE, TEST switch/light..... Press out Check all lights out.
- (6) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.
Check COWL A/ICE ON advisory message out.

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		Rev 6, Nov 28/16

1. ICE AND RAIN PROTECTION (CONT'D)

J. Cowl Anti-Ice – Before Flight (Engines Running)

- (1) ANTI-ICE, COWL L and R
switch/lightsSelect as required (See Note).

NOTE

1. The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below, and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).
2. The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below, when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.
3. Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.

On airplanes with Supplemental Ground Wing Anti-Icing:

4. If the cowl anti-ice is selected ON, wait a minimum of 45 seconds to ensure that L and R COWL A/ICE caution messages remain out. It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

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CHALLENGER 650	NORMAL PROCEDURES Ice and Rain Protection	04-14-7 Rev 6, Nov 28/16
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1. ICE AND RAIN PROTECTION (CONT'D)

I K. Wing Anti-Ice – Before Flight (Engines Running)

(1) ANTI-ICE, WING switchSelect as required (See Notes).

If selected OFF, check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- WING A/ICE ON advisory message out.

NOTE

1. The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.
2. If wing anti-ice is selected ON during taxi, L (R) WING A/ICE caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and WING A/ICE ON or WING/COWL A/ICE ON advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
3. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.
4. The wing anti-ice system must be selected and confirmed ON for take-off, when the OAT is 5°C (41°F) or below and:
 - (a) visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
 - (b) the runway is wet or contaminated, or
 - (c) in the presence of any precipitation (such as rain, snow, sleet or ice crystals).
5. If wing anti-ice is selected ON for take-off, the cowl anti-ice must also be selected ON.
6. When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.
7. Refer to the Flight Crew Operating Manual, Volume 1, Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PHASE OF FLIGHT PROCEDURES.



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1. ICE AND RAIN PROTECTION (CONT'D)

K. Wing Anti-Ice – Before Flight (Engines Running) (Cont'd)

AR Certified Airplanes

WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

NOTE

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected ON, if required, just prior to thrust increase for take-off.

L. Supplemental Ground Wing Anti-Ice Check (if installed) – Before Take-Off

- (1) SUPP GND WING ANTI-ICE, TEST and SYS switch/lights.....Check pressed out.
Check TEST ON, TEST OK and OFF lights out.
- (2) SUPP GND WING ANTI-ICE, TEMP lightsCheck lights out.

1. ICE AND RAIN PROTECTION (CONT'D)

M. Icing Conditions During Flight

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

WARNING

Delaying the use of cowl anti-ice may result in engine ice ingestion, resulting in possible severe engine damage and/or flameout.

Delaying the use of wing anti-ice may result in small amounts of ice accumulation on the wing leading edge.

Small accumulations of ice on the wing leading edge can change the stall characteristics, the stall speed and the stall margin provided by the stall protection system.

**At or above 22,000 feet, when ice is indicated by the ice detection system,
OR**

Below 22,000 feet, when in icing conditions, or when ice is indicated by the ice detection system,

Accomplish the following:

- (1) ThrustCheck N₂ RPM 78% minimum.
- (2) ANTI-ICE, COWL L and R
switch/lightsPress in to select cowl anti-ice on.
Check COWL A/ICE ON
advisory message on.



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1. ICE AND RAIN PROTECTION (CONT'D)

M. Icing Conditions During Flight (Cont'd)

(3) ANTI-ICE, WING switchSelect to NORM.

Check the following:

- L and R WING A/ICE caution messages may come on briefly, then out.
- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- WING/COWL A/ICE ON advisory message on.
- WING L HEAT and R HEAT lights on.

NOTE

When wing and/or cowl anti-ice is selected at or above 30,000 feet, the ATS will display a **FAIL** message and disengage. Data from the FMS is not valid and the crew should review the aircraft performance. Target (TGT) is available on the FMS performance THRUST LIMIT page, but caution should be used not to exceed the engine ITT limits.

If WING/COWL A/ICE ON advisory message is out
OR

If WING L HEAT or R HEAT light is out:

(4) Thrust Increase until the following come on:

- WING/COWL A/ICE ON advisory message.
- WING L HEAT and R HEAT lights.

If WING/COWL A/ICE ON advisory message remains out,
OR

If WING L HEAT or R HEAT light remains out:

(5) Leave icing conditions.



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1. ICE AND RAIN PROTECTION (CONT'D)**M. Icing Conditions During Flight (Cont'd)****If required:**

- (6) ANTI-ICE, WSHLD/WIND L and R
switches.....Select to HI.

NOTE

1. Flaps should not be extended in icing conditions, except as required for take-off, approach and landing. However, if flaps are deployed in icing conditions for extended periods or in severe icing, light to moderate buffet may be encountered. No handling difficulties will result, and normal landing flaps can be used. If the buffet is alleviated by reducing the flap setting, a landing in this configuration may be made at the discretion of the pilot. Refer to Chapter 5; ABNORMAL PROCEDURES – FLIGHT CONTROLS – Flaps Failure, for factors to be applied to approach speed and landing distance.
2. Higher than normal engine vibration, of a fluctuating nature, may be experienced in icing conditions. If the FAN VIB gauge indicates greater than 2.7 mils or if amber VIB icon is displayed in the N₁ gauge in compressed display format, refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT.
3. During single engine operation in icing conditions with wing anti-ice system on, open bleed air isolation valve to prevent asymmetric anti-icing. N₂ RPM in excess of 78% may be required to provide sufficient bleed air to prevent L and R WING A/ICE caution messages.
4. Use of cowl and wing anti-ice systems affects engine and airplane performance. Refer to Chapter 6; PERFORMANCE, for applicable factors.
5. If the airplane flies out of icing conditions and the ice detector system senses no ice present, the ICE message will go out (after 60 seconds).

CHALLENGER 650	NORMAL PROCEDURES Ice and Rain Protection	04-14-12
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1. ICE AND RAIN PROTECTION (CONT'D)

N. Before Landing

- (1) ANTI-ICE, WING switch and L and R COWL switchesCheck status.

NOTE

Use of cowl and wing anti-ice systems affects engine and airplane performance. Refer to Chapter 6; PERFORMANCE, for applicable factors.

O. After Landing

If no icing, precipitation or other conditions that could cause fogging or misting exist:

- (1) ANTI-ICE, WSHLD/WIND L and R switches.....Select as required.

P. During Engine Shutdown

- (1) ANTI-ICE, WING switchSelect to OFF.

Check the following:

- N₂ gauges indicate wing anti-ice off.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message out.

- (2) ANTI-ICE, COWL, L and R switch/lightsPress out to select cowl anti-ice off.

Check the following:

- COWL A/ICE ON advisory message out.
- Drop in ITT.

CHALLENGER 650	NORMAL PROCEDURES Landing Gear, Wheels and Brakes	04-15-1
		Oct 27/15

1. LANDING GEAR, WHEELS AND BRAKES

A. Main Landing Gear Bay Overheat Detection – First Flight of the Day

Check the MLG overheat detection system as follows:

- (1) MLG BAY OVHT TEST, OVHT

switch Select to OVHT and hold.

Check the following:

- MLG BAY OVHT warning message on.
- “GEAR BAY OVERHEAT” aural message on.

Release switch and check that aural and warning messages go out.

- (2) MLG BAY OVHT TEST, FAIL

switch Select to FAIL and hold.

Check MLG OVHT FAIL caution message on.

Release switch and check that caution message goes out.

B. Brakes and Anti-Skid System – Before Each Flight

Check the wheel brake and anti-skid systems as follows:

- (1) PARKING BRAKE Select to off.

Check PARKING BRAKE ON advisory message out.

- (2) ANTI SKID, ARMED switch Select to ARMED.

- (3) ANTI SKID, TEST switch Select to TEST momentarily, and check A/SKID IN TEST advisory message on.

After approximately 6 seconds, check A/SKID IN TEST advisory message out.

NOTE

Anti-skid system is inoperative if the A/SKID IN TEST advisory message does not come on during test.

Anti-skid system is inoperative if A/SKID INBD or A/SKID OUTBD caution messages come on during test.

- (4) PARKING BRAKE Select to on.

Check PARKING BRAKE ON advisory message on.

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CHALLENGER 650	NORMAL PROCEDURES Landing Gear, Wheels and Brakes	04-15-2
		Rev 10, Dec 04/17

1. LANDING GEAR, WHEELS AND BRAKES (CONT'D)

C. Brakes and Anti-Skid System – Before Landing

Check the wheel brake and anti-skid systems as follows:

- (1) LDG GEAR leverSelect to DN, and check for proper indication.

NOTE

To avoid nuisance A/SKID INBD and A/SKID OUTBD caution messages, wait 5 seconds after landing gear is down and locked before checking anti-skid system.

- (2) ANTI SKID, ARMED switchCheck selected to ARMED.

- (3) ANTI SKID, TEST switchSelect to TEST momentarily, and check A/SKID IN TEST advisory message on.

After approximately 3 seconds, check A/SKID IN TEST advisory message out.

NOTE

Anti-skid system is inoperative if the A/SKID IN TEST advisory message does not come on during test.

Anti-skid system is inoperative if A/SKID INBD or A/SKID OUTBD caution messages come on during test.

- (4) NOSE STEER switchCheck selected to ARMED.

NOTE

When the nose wheel steering system is armed in flight, the tiller should be centered.

CHALLENGER 650	NORMAL PROCEDURES Landing Gear, Wheels and Brakes	04-15-3
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1. LANDING GEAR, WHEELS AND BRAKES (CONT'D)

D. Procedures for Operating from Wet, Snow Covered or Slush Covered Surfaces

When operating from wet, snow covered or slush covered runways or taxiways, or following overnight parking in known icing conditions, the following steps are applicable in order to prevent freezing of the wheel brakes:

- (1) During taxi, perform the following number of firm brake applications from 25 knots to 10 knots. Perform the prescribed number of brake applications during the last mile/kilometer of taxi prior to and not including the final stop or snub before take-off. Do not drag the brakes.

AIRPLANE WEIGHT	NUMBER OF BRAKE APPLICATIONS
Greater than 42,000 lb	5
36,000 to 42,000 lb	6
Less than 36,000 lb	7

NOTE

1. Use of symmetric braking is recommended to ensure uniform brake heating.
2. Warming of the brakes will preclude the chance of water saturated brakes freezing at altitude and being locked for landing touchdown.
- (2) During take-off, unless weather conditions or performance requirements prohibit, delay retraction of the landing gear until excess water, snow or slush is thrown off by wheel rotation and/or slip stream force.
- (3) When landing, carry out a positive landing to ensure initial wheel spin-up and brakeout of frozen brakes, if icing accumulation on the wheels and brakes is suspected.
- (4) During the landing roll and subsequent taxi, use the brakes to prevent progressive build-up of ice on the wheels and brakes.

Following take-off or landing on water, snow or slush covered runways and taxiways, tires should be inspected for flat-spotting prior to the next flight.



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CHALLENGER 650	NORMAL PROCEDURES Landing Gear, Wheels and Brakes	04-15-4
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1. LANDING GEAR, WHEELS AND BRAKES (CONT'D)

D. Procedures for Operating from Wet, Snow Covered or Slush Covered Surfaces (Cont'd)

AR Certified Airplanes

WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

NOTE

1. When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected on, if required, just prior to thrust increase for take-off.
2. The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below, and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).
3. The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.
4. Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

E. Cooling of Wheels, Brakes and Tires

Brake cooling times, derived from Chapter 6; PERFORMANCE – TURN-AROUND TIME, must be observed between a landing or a low energy rejected take-off (RTO) and a subsequent take-off.

F. Nose Wheel Steering – On Ground

CAUTION

Always make sure that the steering is within the nominal operating range of $\pm 55^\circ$ before re-cycling the arming switch.

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CHALLENGER 650	NORMAL PROCEDURES Lighting	04-16-1
		Oct 27/15

1. LIGHTING

A. Emergency Lighting System – First Flight of the Day

With the airplane powered, check the emergency lighting system as follows:

- (1) EMER LTS switchSelect to OFF.

Check EMER LTS OFF caution message on.

- (2) EMER LTS switchSelect to ON.

Check the following:

- Emergency lights are on, and
- EMER LTS ON status message on.

- (3) EMER LTS switchSelect to ARM.

Check the following:

- Emergency lights are off, and
- EMER LTS ON status message out.

B. Before Take-Off

- (1) Cockpit lightingSelect as required.

- (2) External lightsSelect as required.



The left and right landing lights must not be used on the ground, except for the take-off roll and landing.

CHALLENGER 650	NORMAL PROCEDURES Lighting	04-16-2
		Oct 27/15

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CHALLENGER 650	NORMAL PROCEDURES Oxygen System	04-17-1
		Oct 27/15

1. CREW OXYGEN

A. Oxygen System

Check the oxygen system as follows:

- (1) Crew oxygen system pressureCheck the following:
- Oxygen supply, pressure and quantity are sufficient for flight, and
 - OXY LO PRESS caution message out.
- (2) Crew oxygen masksCheck that the masks are fully operational. Check the following:
- Flow indicator indicates oxygen flow,
 - 100% is operational, and
 - Oxygen flow sound is picked up by mask microphone.
- (3) Passenger oxygen.....Select to NORMAL.

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CHALLENGER 650	NORMAL PROCEDURES Stall Protection	04-18-1
		Oct 27/15

1. STALL PROTECTION

WARNING

Both stall protection system pusher switches must remain on for all phases of flight.

A. Stall Protection System – First Flight of the Day

NOTE

To ensure a successful test, all items within the following test must be verified to have occurred in the sequence indicated.

Both SPS TEST switches must be operated simultaneously.

- (1) Stabilizer trim switch.....Select to NOSE DN and hold until trim is 0 units.
- (2) FLIGHT CONTROLS page.....Select
- (3) STALL PROT, PUSHER switches.....Select to ON.
- (4) SPS TEST, L and R switches.....Select TEST and hold.

Check STALL FAIL caution message comes on.

During test:

- (5) SPS TEST INDICATORS.....Verify that both test indicator needles sweep counter clockwise past the red region, and then sweep clockwise back to the red region, and remain there.

Check the following:

- IGNITION A, B and CONT ON lights come on.
- IGNITION A and B advisory messages come on.

within the yellow region:

- Stick shakers are activated.

within the red region:

- Stick pusher is activated.
- STALL icon is displayed on both PFDs (flashes for 5 seconds then steady).
- Stall aural warning sounds.

While in stick pusher mode:

- (a) Pilot's AP/SP DISC switch.....Press Stick pusher is de-activated.
- (b) Pilot's AP/SP DISC switch.....Release Stick pusher is re-activated.
- (c) Copilot's AP/SP DISC switch.....Press Stick pusher is de-activated.



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CHALLENGER 650	NORMAL PROCEDURES Stall Protection	04-18-2
		Oct 27/15

1. STALL PROTECTION (CONT'D)

A. Stall Protection System – First Flight of the Day (Cont'd)

NOTE

During copilot's AP/SP DISC test, the control column will take several seconds to go back to neutral position. Copilot should pull (override) control column, and note diminished feel force loads while pressing the AP/SP DISC switch.

- (d) Copilot's AP/SP DISC switch.....Release Stick pusher is re-activated.
- (e) FLIGHT CONTROLS page.....Check right elevator surface pointer indicates full travel or copilot control column is against forward stop.
- (f) G SWITCH.....Select TEST and hold.
Stick pusher is de-activated.
- (g) G SWITCH.....Release Stick pusher is re-activated.
- (6) SPS TEST, L and R switches.....Release Check the following:
- STALL FAIL caution message out.
 - Stick pusher de-activated.
 - Stick shakers de-activated.
 - IGNITION A, B and CONT ON lights out.
 - IGNITION A and B advisory messages out.
 - PFD STALL icons out.
 - Stall aural warning stops.

CHALLENGER 650	NORMAL PROCEDURES Aural/Visual Warning System	04-19-1
		Oct 27/15

1. AURAL/VISUAL WARNING SYSTEM

A. Lamp Test – First Flight of the Day

- (1) LAMP TEST switchSelect to 1 and hold.
Check all switch/lights come on.
Release switch and check all switch/lights go out.
- (2) LAMP TEST switchSelect to 2 and hold.
Check all switch/lights come on.
Release switch and check all switch/lights go out.

B. Aural Warning System – First Flight of the Day

- (1) AURAL WARN TEST switchSelect to 1 and release.
Check that at least one tone, chime or voice message is heard in the audio system. It is NOT necessary to listen to all aural.
- (2) AURAL WARN TEST switchSelect to 2 and release.
Check that at least one tone, chime or voice message is heard in the audio system. It is NOT necessary to listen to all aural.

NOTE

Re-selection of the TEST switch during the test will cancel the aural messages.

C. Before Take-Off

- (1) Crew alerting system (CAS)Check that T/O CONFIG OK advisory (green) message on.
- (2) EICAS page.....Check that stabilizer trim setting is in the correct position (refer to FLIGHT CONTROLS – Before Take-Off in this chapter).

D. Before Landing

- (1) EICAS page.....Check that stabilizer trim setting is in the correct range.

CHALLENGER 650	NORMAL PROCEDURES Aural/Visual Warning System	04-19-2 Rev 11, Mar 01/18
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2. TERRAIN AWARENESS AND WARNING SYSTEM, AND WINDSHEAR DETECTION AND ALERTING SYSTEM

A. Before Flight

If airport is not in TAWS database, or if GPS is not available during QFE operations:

- (1) TAWS WARNING, TERRAIN OFF
switch/light Press in Check TAWS WARNING,
TERRAIN OFF light on.

B. During Climb

If Terrain Warning was selected OFF before flight:

- (1) TAWS WARNING, TERRAIN OFF
switch/light Press out Check TAWS WARNING,
TERRAIN OFF light out.

C. During Descent

Within 15 NM of approach if any of the following conditions apply:

- Runway is less than 3,500 feet in length, or
- Airport is not in the TAWS database, or
- Intended approach is not compatible with TAWS terrain awareness alerting, or
- QFE operation with no GPS available.

- (1) TAWS WARNING, TERRAIN OFF
switch/light Press in Check TAWS WARNING,
TERRAIN OFF light on.

CHALLENGER 650	NORMAL PROCEDURES Aural/Visual Warning System	04-19-3 Rev 11, Mar 01/18
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2. TERRAIN AWARENESS AND WARNING SYSTEM, AND WINDSHEAR DETECTION AND ALERTING SYSTEM (CONT'D)

D. TAWS Warning During Flight

TAWS Warnings are indicate as follows:

- “TERRAIN, TERRAIN, PULL UP”, “OBSTACLE, OBSTACLE, PULL UP” and “PULL UP” aural.
 - PULL UP warning message on the PFDs.
- (1) ATS DISC switch Depress
 - (2) Autopilot Disconnect
 - (3) Thrust levers..... Advance to maximum take-off thrust.
 - (4) FLIGHT SPOILER lever Select to RETRACT.
 - (5) Immediately initiate a smooth pull up.
 - (6) Landing gear Retract
 - (7) Maintain climb until warning ceases and it is confirmed that a safe ground clearance exists.

E. TAWS Caution During Flight

TAWS cautions are indicated as follows:

- “CAUTION TERRAIN”, “CAUTION OBSTACLE”, “TERRAIN”, “TOO LOW TERRAIN”, “TOO LOW GEAR”, “TOO LOW FLAPS”, “SINK RATE”, “DON’T SINK” OR “GLIDESLOPE” aural.
 - GND PROX message on the PFDs.
- (1) Airplane flight path or configuration Correct

CHALLENGER 650	NORMAL PROCEDURES Aural/Visual Warning System	04-19-4
		Rev 11, Mar 01/18

2. TERRAIN AWARENESS AND WARNING SYSTEM, AND WINDSHEAR DETECTION AND ALERTING SYSTEM (CONT'D)

F. Windshear Warning

Windshear warnings are indicated as follows:

- (Siren) “WINDSHEAR, WINDSHEAR, WINDSHEAR” aural.
- WINDSHEAR warning message on PFD.

NOTE

The autopilot (if engaged) will automatically disconnect, and the flight director bars will be removed within two (2) seconds after a windshear warning.

During take-off, approach or landing:

- | | | | |
|-----|----------------------------|---------|---|
| (1) | ATS DISC switch | Depress | |
| (2) | Thrust levers | Advance | to maximum take-off thrust. |
| (3) | FLIGHT SPOILER lever | Select | to RETRACT. |
| (4) | Airplane | Rotate | smoothly to go-around/take-off pitch attitude and allow airspeed to decrease, if necessary. |

NOTE

Stick shaker must be respected. Reduce pitch attitude sufficient to stop shaker.

If airplane is descending:

- | | | | |
|-----|------------------------------|----------|---|
| (5) | Pitch attitude | Increase | smoothly and in small increments, reducing airspeed as necessary to stop descent. |
| (6) | Airplane configuration | Maintain | until positively out of windshear condition. |
- Do not change flap, gear or trim position until WINDSHEAR warning message is removed and above 1,500 feet AGL.

WARNING

If ground or obstacle contact is imminent, use all available engine thrust.

CHALLENGER 650	NORMAL PROCEDURES Aural/Visual Warning System	04-19-5 Rev 11, Mar 01/18
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2. TERRAIN AWARENESS AND WARNING SYSTEM, AND WINDSHEAR DETECTION AND ALERTING SYSTEM (CONT'D)

G. Windshear Caution

Windshear cautions are indicated as follows:

- WINDSHEAR caution message on PFD.

During take-off, approach or landing:

- (1) Thrust levers/TOGA switch Advance/press Manually advance thrust levers to maximum take-off thrust while simultaneously pressing either take-off or go-around (TOGA) switch and ATS DISC switch.
- (2) Normal go-around Initiate

3. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

A. Resolution Advisory During Flight

Resolution advisories are indicated as follows:

- Vertical maneuver commands (red/green band) on vertical speed indicator,
- Threat level symbology (red square symbol) on multi-function display, including closure rates and relative altitude, and
- Vertical maneuver voice aural.



Compliance with a TCAS resolution advisory is necessary unless the pilot considers the maneuver unsafe to accomplish, or unless the pilot has accurate information about the cause of the resolution advisory and can maintain safe separation from a nearby airplane.



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3. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS) (CONT'D)

A. Resolution Advisory During Flight (Cont'd)

During flight:

- (1) Autopilot Disconnect
- (2) Applicable maneuver Accomplish immediately, adjusting engine thrust as required.

If an additional maneuver is indicated (e.g. a reversal), it also must be accomplished immediately.

NOTE

- 1. For a non-crossing RA, the vertical speed should be accurately adjusted to comply with the RA. This avoids negating the effectiveness of a co-ordinated maneuver by the intruder aircraft.
- 2. For a crossing RA, non-compliance by one aircraft can result in reduced vertical separation with the need to achieve safe horizontal separation by visual means.



Once an RA maneuver has been initiated, the maneuver must be continued until a "CLEAR OF CONFLICT" aural comes on.

If stabilized in approach:

- (1) Autopilot Disconnect
- (2) Go-around Initiate

If RA is accompanied with a TAWS or STALL warning:

- (1) Applicable RA maneuver Discontinue
- (2) Applicable recovery procedure (TAWS or STALL) Accomplish

When TCAS "CLEAR OF CONFLICT" aural comes on:

- (3) Airplane Re-position to assigned altitude.

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		Rev 11, Mar 01/18

3. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS) (CONT'D)

B. Traffic Advisory During Flight

Traffic advisories are indicated as follows:

- Threat level symbology on Primary Flight Display and Multi-Function Display, including closure rate and relative altitude symbols, and
- “TRAFFIC – TRAFFIC” voice aural.

NOTE

The pilot should not initiate evasive maneuvers using information from the traffic display only, or on a traffic advisory (TA) without visually sighting the traffic.

These displays and advisories are intended only for assistance in visually locating the traffic, and lack the flight path trends necessary for use in evasive maneuvering. However, while climbing or descending, modest changes in vertical speed based on traffic display information is not considered evasive maneuvering.

- (1) Display Control Panel or Cursor Control Panel, TFC button.....Push to select TCAS traffic overlay.

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		Rev 11, Mar 01/18

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CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-1
		Oct 27/15

1. CONSOLIDATED CHECKLISTS

A. Introduction

The following checklists comprise a consolidation of the systems and equipment procedures which are defined in this chapter. The checklists are supplementary to those checks arising from basic airmanship and procedural practices common to the operation of any modern transport category jet airplane.

Check items have been abbreviated and reference must be made to the systems sections within this chapter for the definitive check procedure.

B. Normal Power-Up Check

- (1) Airplane and Flight Pubs.....On board
- (2) Circuit breakers..... Closed
- (3) Cabin and flight compartment emergency equipment Check
- (4) Hydraulic pump switches OFF
- (5) NOSE STEER switch..... OFF
- (6) LDG GEAR lever DN
- (7) FLIGHT SPOILER lever..... Retracted
- (8) FLAPS lever/position Check/Matched
- (9) ADG manual releaseStowed
- (10) BATT MASTER switch..... ON
- (11) DC electrical page Check min 22 volts
- (12) Firex monitor system test (first flight of the day) Complete
- (13) Fire detection systems test (first flight of the day) Complete
- (14) NAV lights switch ON
- (15) APU (if required) Start

After APU indications stabilize, select APU GEN to ON.

If APU and AC external power are not available:

- (15) Engine Start Procedure using Battery and External Air..... Complete, go to step (22)
- (16) AC and DC electrics Established
- (17) Lamp test (first flight of the day) Complete
- (18) Bleed leak detection test (first flight of the day) Complete
- (19) APU LCV switch/light..... OPEN
- (20) 10TH STAGE ISOL switch/light..... OPEN
- (21) L and R PACK switch/lightsPress in
- (22) All FMS POS INIT

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-2
		Rev 10, Dec 04/17

1. CONSOLIDATED CHECKLISTS (CONT'D)

C. External Walkaround

- (1) APU exhaust ports Check
- (2) APU drain ports Check
- (3) Airplane servicing/maintenance doors Check closed
- (4) Overwing exit Check secure
- (5) Aft equipment bay door Check closed
- (6) Passenger door vent flap (first flight of the day) Check
- (7) NACA vents Check
- (8) Vent relief valves Check
- (9) Drains/drain valves Check
- (10) Engine cowl Check
- (11) NLG door safety switch FLT/CLOSE DOOR

NOTE

For remainder of walkaround inspection items, refer to Flight Crew Operating Manual, PSP 605-6, Volume 1, Chapter 4; NORMAL PROCEDURES – AIRPLANE PREPARATION – External Walkaround.

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-3
		Rev 6, Nov 28/16

1. CONSOLIDATED CHECKLISTS (CONT'D)

D. Systems Check

- (1) Seat and harness.....Adjust
- (2) Rudder pedalsAdjust

When nosewheel area is confirmed clear:

- (3) Hydraulic pump 3A switch ON
- (4) Nosewheel door..... Close
- (5) Oxygen system check..... Complete
- (6) Audio warning switch/lights..... Guarded/Off
- (7) Cockpit voice recorder.....Test
- (8) Electrical power panel..... Check
- (9) APU panel..... As required
- (10) Hydraulic panel Check/AUTO/ON
- (11) Lighting panels..... As required
- (12) Start/Ignition panel..... Check
- (13) Fuel panel Check
- (14) Air conditioning panel Check
- (15) ELTARM
- (16) Cabin pressurization panel Check
- (17) Bleed air panel..... Check
- (18) Anti-ice panel..... Check

On airplanes with Supplemental Ground Wing Anti-Icing:

- (19) Wing anti-ice switch test (first flight of the day)..... Complete
- (20) Ice detector test (first flight of the day)..... Complete
- (21) Passenger signs ON
- (22) Emergency lights (first flight of the day)..... Check/ARM
- (23) Standby compass Check
- (24) Glareshield..... Check
- (25) Air Data/Altimeter..... Set
- (26) Side panels Check/Set
- (27) EFIS..... Check
- (28) EICAS Check
- (29) ISI Check
- (30) GPS clock Check/Set
- (31) SUPP GND WING ANTI-ICE panel (if installed)..... Check



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1. CONSOLIDATED CHECKLISTS (CONT'D)

D. Systems Check (Cont'd)

- (32) HORN switch Guarded
- (33) ANTI SKID switch ARMED
- (34) MLG overheat detection system check
(first flight of the day) Complete
- (35) TAWS WARNING, TERRAIN OFF switch/light As required
- (36) TAWS WARNING, FLAPS OFF switch/light Press out
- (37) TAWS WARNING, RAAS OFF switch/light (if installed) Press out
- (38) PITCH and ROLL DISC handles In/Stowed
- (39) GND SPOILERS switch AUTO
- (40) THRUST REVERSER, L and R switches ARMED
- (41) FLIGHT SPOILER lever Retracted
- (42) Thrust levers Checked/SHUT OFF
- (43) FLAPS selector Check
- (44) Engine vibration check (first flight of the day) Complete
- (45) ENG SPEED, L and R switches ON
- (46) Aural warning system check (first flight of the day) Complete
- (47) Stall protection system check (first flight of the day) Complete
- (48) DCP As required
- (49) CCP As required
- (50) Radios/Avionics Set
- (51) Flight control trim systems check (first flight of the day) Complete
- (52) Yaw dampers Engage
- (53) CDU and MFD TUNE INHIBIT switches Press out
- (54) Reversionary selectors NORM
- (55) AFCS selector Set
- (56) IRS reversionary selectors As required
- (57) FMS 3 selector (if installed) As required
- (58) LANDING GEAR MANUAL RELEASE handle Stowed
- (59) ADG handle Stowed
- (60) ADG AUTO DEPLOY CONTROL switch LAMP

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-5
		Oct 27/15

1. CONSOLIDATED CHECKLISTS (CONT'D)

E. Before Start Check

- (1) Gravity crossflow check Complete
- (2) Windshield/window heat test Complete
- (3) Wing overheat detection system test Complete
- (4) Pressurization controls Set
- (5) Altimeters Set
- (6) EFIS Set for departure
- (7) CAS Check
- (8) Brakes and anti-skid system check Complete
- (9) All FMS Set
- (10) FMS messages Check
- (11) Radios/Nav aids Set for departure
- (12) Take-off data Set
- (13) Take-off briefing Complete

F. Start Check

- (1) Passenger door Check closed
- (2) Fuel quantity and balance Check
- (3) BEACON switch (if installed) ON
- (4) L and R BOOST PUMP switch/lights ON
- (5) BLEED AIR switch/lights Set for start
- (6) IGNITION, A (B) switch/light ARM
- (7) Engines START

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-6
		Oct 27/15

1. CONSOLIDATED CHECKLISTS (CONT'D)

G. APU Assisted Engine Start Procedure

With the APU operating:

- (1) APU LCV switch/light..... Check OPEN
- (2) L (R) ENG START switch/light..... Press

When N₂ reaches 20% RPM and ITT is below 120°C:

- (3) Left or right thrust lever..... Advance to IDLE
- (4) Engine instruments Monitor
- (5) Fuel system check valve operation
(first flight of the day) Verify



When operating with one engine, select inoperative engine boost pump to INOP.

If the other engine is to be started using APU bleed air:

- (6) Thrust lever (of operating engine)..... Check at IDLE
- (7) 10th stage bleed air pressure Check
- (8) L (R) ENG START switch/light..... Press

When N₂ reaches 20% RPM and ITT is below 120°C:

- (9) Thrust lever (of starting engine)..... Advance to IDLE
- (10) Engine instruments Monitor
- (11) APU EGT indications Check
- (12) Idle speed Check

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-7
		Oct 27/15

1. CONSOLIDATED CHECKLISTS (CONT'D)

H. Engine Start Procedure using Battery and External Air

- (1) Normal Power-Up Check As instructed
- (2) APU LCV switch/light Press out
- (3) External air Connected
- (4) External cart pressure Confirm 45 psi minimum
- (5) R BOOST PUMP switch/light Press in
- (6) IGNITION B switch/light ON
- (7) R ENG START switch/light Press

When N₂ reaches 20% RPM and ITT is below 120°C:

- (8) Right thrust lever Advance to IDLE
- (9) Engine instruments Monitor
- (10) GEN 2 switch ON
- (11) AC and DC electrical page Check
- (12) Lamp test (first flight of the day) Complete
- (13) Fuel system check valve operation
(first flight of the day) Verify



When operating with one engine, select inoperative engine boost pump to INOP.

- (14) External air Disconnect
- (15) Bleed leak detection test (first flight of the day) Complete
- (16) R 10TH STAGE switch/light Press in
- (17) R PACK switch/light Press in
- (18) Normal Power-Up Check Complete

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-8
		Rev 9, Jun 16/17

1. CONSOLIDATED CHECKLISTS (CONT'D)

I. Engine Start Procedure using External Air (with AC electrical power)

- (1) APU LCV switch/light..... Press out
- (2) External air..... Connected
- (3) 10th stage bleed air pressure Check 45 psi minimum
- (4) L (R) ENG START switch/light..... Press

When N₂ reaches 20% RPM and ITT is below 120°C:

- (5) Left or right thrust lever Advance to IDLE
- (6) Engine instruments Monitor
- (7) Fuel system check valve operation
(first flight of the day) Verify



When operating with one engine, select inoperative engine boost pump to INOP.

- (8) Other engine Start

NOTE

When starting the other engine using cross-bleed, accomplish steps (9) to (11) first.

- (9) Applicable GEN switch ON
- (10) External AC power (if connected) Disconnect
- (11) External air..... Disconnect

J. Cross-Bleed Engine Start Procedure

- (1) APU LCV switch/light..... Press out
- (2) L (R) 10TH STAGE switch/light (of operating engine) Press in
- (3) Thrust lever (of operating engine)..... Advance
- (4) 10th stage bleed air pressure Check 60 psi minimum
- (5) L (R) ENG START switch/light..... Press

When N₂ reaches 20% RPM and ITT is below 120°C:

- (6) Thrust lever (of starting engine)..... Advance to IDLE
- (7) Engine instruments Monitor
- (8) Thrust lever (of first engine) IDLE
- (9) Idle speed Check
- (10) 10TH STAGE switch/lights Press in

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CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-9
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1. CONSOLIDATED CHECKLISTS (CONT'D)

K. Failure to Start Procedure

- (1) Thrust lever (of starting engine) SHUT OFF
- (2) Ignition Off

If one engine is operating:

- (3) Inoperative engine, L (R) BOOST PUMP switch/light Press out
- (4) Dry motor until ITT is below 120°C
(maximum 90 seconds) Accomplish

L. After Engine Start Check

- (1) GEN 1 and 2 switches ON
- (2) AC and DC electrical pages Check
- (3) APU GEN switch As required
- (4) APU As required
- (5) IGNITION A (B) switch/light Press out
- (6) L and R PACK switch/lights Press in
- (7) 14TH stage valve check Complete
- (8) Wing anti-ice check Complete
- (9) Wing anti-ice/isolation valve check
(first flight of the day) Complete
- (10) Anti-Ice cowl check (first flight of the day) Complete
- (11) Supplemental ground wing anti-ice check (if installed)
(first flight of the day) Complete
- (12) Wing and cowl anti-ice As required
- (13) ATS Check
- (14) Flight controls Check
- (15) Ground spoilers check (first flight of the day) Complete
- (16) Flight spoilers check (first flight of the day) Complete
- (17) NOSE STEER switch ARMED
- (18) APR check Complete
- (19) ADG AUTO DEPLOY CONTROL switch UNIT

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-10
		Rev 6, Nov 28/16

1. CONSOLIDATED CHECKLISTS (CONT'D)

M. Taxi Check

- (1) Transponder/TCAS As required
- (2) Brakes Check
- (3) FLAPS 20°
- (4) Thrust reversers check (first flight of the day) Complete
- (5) Aileron and rudder trims Neutral
- (6) Stabilizer trim Set for take-off
- (7) Flight instruments Check/Set
- (8) Fuel balance Check
- (9) Wing and cowl anti-ice As required

N. Take-Off Check

- (1) Ignition As required
- (2) Wing and cowl anti-ice As required
- (3) WSHLD/WIND L and R switches HI or LOW
- (4) PROBES switches ON
- (5) Cabin Advise
- (6) Cockpit lighting As required
- (7) External lights As required
- (8) Crew Alerting System (CAS) Check/Clear
- (9) Take-off data (V_1 , V_R , V_2 and distance) Check displayed
- (10) SUPP GND WING ANTI-ICE panel (if installed) Check
- (11) Weather radar As required
- (12) Transponder/TCAS Set

O. After Take-Off Check

- (1) LDG GEAR lever UP
- (2) FLAPS Set to 0°
- (3) THRUST REVERSER, L and R switches OFF
- (4) CAS Check/Clear
- (5) Ignition As required
- (6) Wing and cowl anti-ice As required

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-11
		Rev 9, Jun 16/17

1. CONSOLIDATED CHECKLISTS (CONT'D)

P. Climb Check

- (1) Pressurization Check
- (2) Air-conditioning Transfer to engine bleeds
- (3) Climb thrust Set
- (4) APU GEN switch OFF/RESET
- (5) APU Shutdown
- (6) TAWS WARNING, TERRAIN OFF switch/light Check/Pressed out
- (7) Passenger signs As required
- (8) External lights As required

At transition level:

- (9) Altimeters Set

Q. Cruise Check

- (1) Wing and cowl anti-ice As required
- (2) Cruise thrust Set
- (3) Pressurization Check
- (4) Fuel Monitor

R. Descent Check

- (1) ATIS Copy
- (2) CAS Check/Clear
- (3) TAWS WARNING, TERRAIN OFF switch/light As required
- (4) Landing data Set
- (5) Approach briefing Complete
- (6) Shoulder harness Secure
- (7) Pressurization Check/Set
- (8) Wing and cowl anti-ice As required
- (9) WSHLD/WIND L and R switches HI or LOW
- (10) Fuel quantity and balance Check

At transition level:

- (11) Altimeters Set/Cross-check

CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-12
		Rev 9, Jun 16/17

1. CONSOLIDATED CHECKLISTS (CONT'D)

S. In Range Check

(1) APU (if required) Start

After APU indications stabilize, select APU GEN as required.

If APU is available:

(2) Air-conditioning Transfer to APU bleed air

(3) External lights As required

(4) Passenger signs As required

T. Before Landing Check

(1) Ignition As required

(2) Wing and cowl anti-ice As required

(3) Cabin Advise

(4) Passenger signs ON

(5) LDG GEAR lever DN/Indicating

(6) Brake and anti-skid system check Complete

(7) THRUST REVERSER, L and R switches ARMED

(8) FLAPS As required

(9) ATS As required

U. Go-Around Procedure



If unreliable airspeed is suspected, disregard all reference to TOGA.
Do not use TOGA switch.

(1) ATS DISC Depress

(2) Thrust levers/TOGA switch Advance/Press

(3) FLIGHT SPOILER lever (if extended) RETRACT

(4) Airplane Rotate

(5) Pitch attitude Adjust

(6) FLAPS 20°

When a positive rate of climb is achieved:

(7) LDG GEAR lever UP

At a safe altitude (not below 400 feet AGL):

(8) FLAPS 0°

(9) THRUST REVERSER, L and R switches OFF

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CHALLENGER 650	NORMAL PROCEDURES Consolidated Checklists	04-20-13
		Rev 9, Jun 16/17

1. CONSOLIDATED CHECKLISTS (CONT'D)

V. After Landing Check

- (1) FLAPS As required
- (2) FLIGHT SPOILER lever..... Retracted
- (3) Transponder/TCAS As required
- (4) PROBES switches OFF
- (5) Wing and cowl anti-ice As required
- (6) Windshield/window heat As required
- (7) Ignition Off
- (8) External Lights As required
- (9) LANDING LTS OFF
- (10) Weather radar STANDBY
- (11) Trims Reset

W. Shutdown Check

- (1) Parking brake As required
- (2) Thrust reversers Stow
- (3) Wing and cowl anti-ice OFF
- (4) WSHLD/WIND L and R switches OFF
- (5) APU Start
- After APU indications stabilize, select APU GEN to ON.
- (6) GEN 1 and 2 switches OFF/RESET
- (7) Engine that started first Shutdown
- (8) L and R BOOST PUMP switch/lights Press out
- (9) Fuel system check valve operation
(first flight of the day) Complete
- (10) Other engine Shutdown
- (11) Fuel system crossflow valve operation
(last flight of the day) Check
- (12) Passenger signs OFF
- (13) BEACON switch (if installed) OFF
- (14) NOSE STEER switch OFF

On airplanes incorporating Service Bulletin 650-23-008:

- (15) ATN Log OFF
- (16) Transponder/TCAS STBY

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1. CONSOLIDATED CHECKLISTS (CONT'D)

X. Securing Check

- (1) Parking brake..... As required
- (2) THRUST REVERSER, L and R switches OFF
- (3) External lights OFF
- (4) EMER LTS switch..... OFF
- (5) Hydraulic pump switches OFF
- (6) APU GEN switch..... OFF/RESET
- (7) APU Shutdown
- (8) L and R PACK switch/lights OFF
- (9) APU LCV switch/light..... Press out
- (10) 10TH STAGE ISOL switch/light..... Press out
- (11) BATT MASTER switch..... OFF
- (12) MISC LTS switches OFF

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CHALLENGER 650	NORMAL PROCEDURES Severe Turbulence Penetration	04-21-1
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1. SEVERE TURBULENCE PENETRATION

A. Autopilot

The autopilot should be disconnected.

B. Airspeed

Turbulence penetration speed is 280 KIAS or 0.75 M_I, whichever is lower.

Severe turbulence may cause large and rapid variations in airspeed. Do not chase airspeed.

C. Attitude

Maintain wings level with desired pitch attitude. Use the PFD attitude indicator as the primary instrument. In extreme windshear, large attitude changes may occur. Do not use sudden or large control inputs.

D. Stabilizer

Maintain control of the airplane with elevators. After establishing the trim setting for penetration speed, avoid using the stabilizer trim for controlling attitude.

E. Altitude

The penetration altitude should be an altitude which provides adequate maneuvering margins. Large altitude variations are possible in severe turbulence. Sacrifice altitude to maintain desired attitude and airspeed. Refer to Chapter 6; PERFORMANCE – GENERAL – Flight Capabilities – Maneuvering Capabilities.

F. Thrust

Make initial thrust setting for target airspeed. Change thrust only in case of extreme airspeed changes.

G. Thunderstorms

Continuous engine ignition must be used during the following:

- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence;
- Flight in the vicinity of thunderstorms.

If freezing rain, hail or ice crystals are evident, engine anti-ice should be selected on.

H. Maneuvering Capability

A maneuvering capability chart is presented in Chapter 6; PERFORMANCE – GENERAL – Flight Capabilities – Maneuvering Capabilities.

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1. GENERAL

The abnormal operating procedures included in this chapter contain items peculiar to the CL 600, model 2B16 Challenger airplane (Serial No. 6050 and subsequent). These checks address foreseeable situations, usually involving a failure condition, in which the use of the normal or alternate systems can be expected to maintain an acceptable level of airworthiness.

In addition to the failure indications referenced in the following abnormal procedures, the master caution system operates where applicable.

Unless otherwise specified in these abnormal procedures, the landing configuration of the airplane will be landing gear down and flaps 45.

These procedures have been developed and recommended by Bombardier and approved by Transport Canada for use in the operation of the Challenger airplane. These procedures are provided as guidance and should not be construed as prohibiting the development of equivalent Transport Canada-approved procedures.

2. LANDING DISTANCE FACTORS

Landing distance factors are provided herein with and without the use of thrust reversers. No operation should be predicated on the use of thrust reversers.

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1. POWER PLANT

A. Reverser Unsafe to Arm

Indication: L (R) REV UNSAFE caution message.



Do not arm affected thrust reverser during flight.

- (1) Affected THRUST REVERSER
ARM switchOFF L (R) REV ARMED advisory message out.



Use remaining thrust reverser carefully upon landing.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-2
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1. POWER PLANT (CONT'D)

B. In-Flight Engine Failure/Shutdown

Indication: L (R) ENGINE caution message.

Accomplish an engine shutdown only when flight conditions permit.

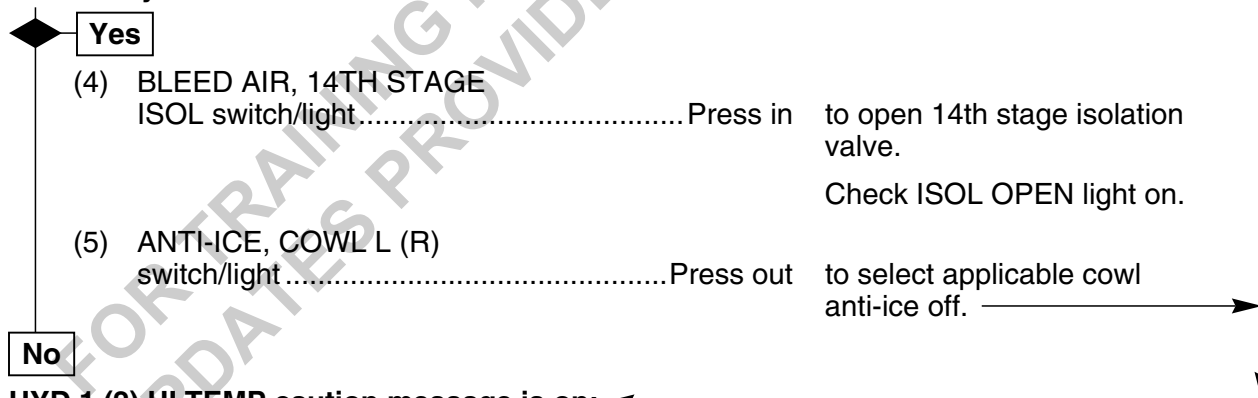
Shut down the engine if any of the following malfunction symptoms/indications occur:

- Excessive or uncontrollable power changes, or
- Indications outside the engine operating limitations, or
- Unsuccessful restart attempts after an unexplained flameout, or
- Any other condition that indicates advisability of engine shutdown.

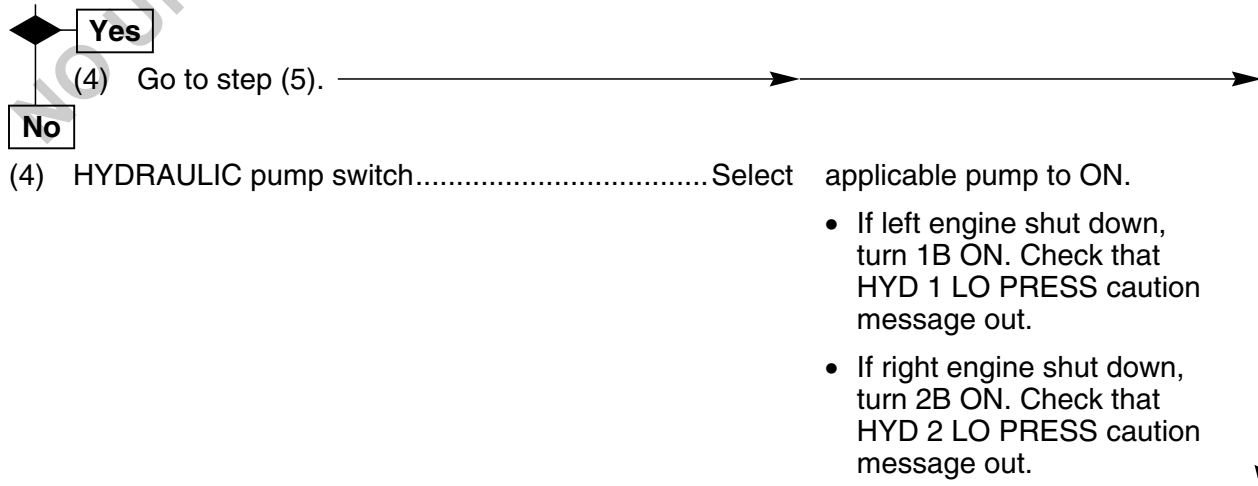
Affected engine:

- (1) Thrust lever Confirm and IDLE
- (2) Thrust lever Confirm and SHUT OFF
- (3) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press out if MANUAL light on, and if rotor burst is suspected.

Anti-ice system is on:



HYD 1 (2) HI TEMP caution message is on:



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1. POWER PLANT (CONT'D)

B. In-Flight Engine Failure/Shutdown (Cont'd)

- (5) AIR-CONDITIONING, L (R) PACK switch/light Press out to shutdown affected pack.
Check L (R) PACK OFF light on.
- (6) BLEED AIR, 10TH STAGE L (R) switch/light Press out to close affected shut-off valve.
Check L (R) CLOSED light on.

NOTE

Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th-stage bleed source.

- (7) AC POWER, GEN 1 (2) switch Select affected generator to OFF.
Check GEN 1 (2) OFF caution message on.

Below 20,000 feet and APU is available:

- Yes**
- (8) APU Start
- (9) AC POWER, APU GEN switch Select to ON.
- (10) Air-conditioning Transfer to APU bleed air.

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

- (11) Affected AIR-CONDITIONING, L (R) PACK switch/light Press in to restart pack.
Check L (R) PACK OFF light out.
- No**
- (8) Fuel system Check and monitor the following:
- Fuel quantity, and
 - Fuel balance.
- (9) FUEL, L (R) BOOST PUMP switch/light Press out to disarm affected boost pump.
Check L (R) FUEL PUMP caution message on.



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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-4
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1. POWER PLANT (CONT'D)

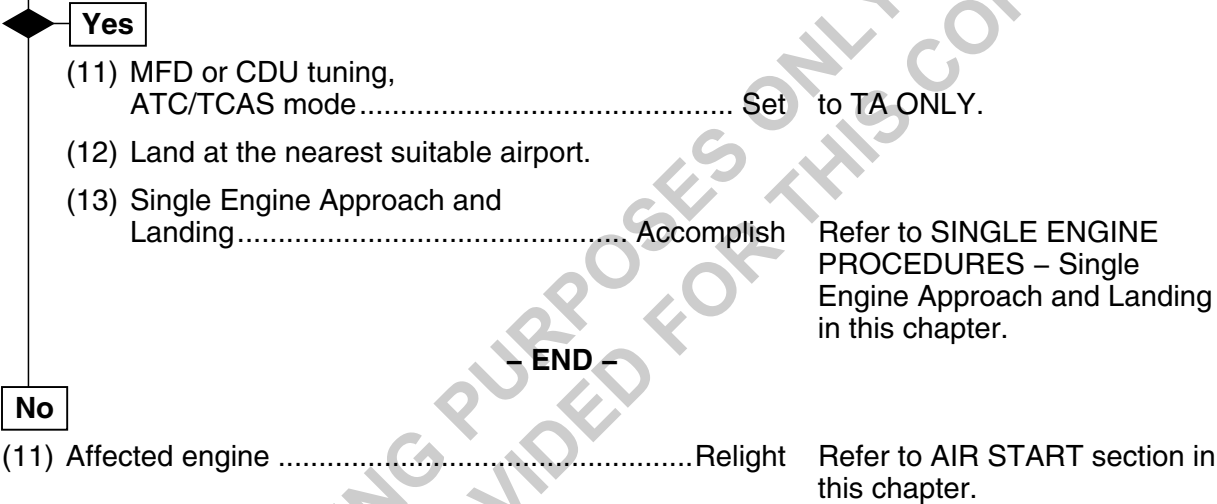
B. In-Flight Engine Failure/Shutdown (Cont'd)

(10) Leave icing conditions (if possible).

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

Engine damage suspected or intentional shutdown:



NOTE

1. Do not attempt to relight an engine that is suspected to be not serviceable.
2. Relight engine using starter-assisted cross bleed whenever possible.

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-5
		Oct 27/15

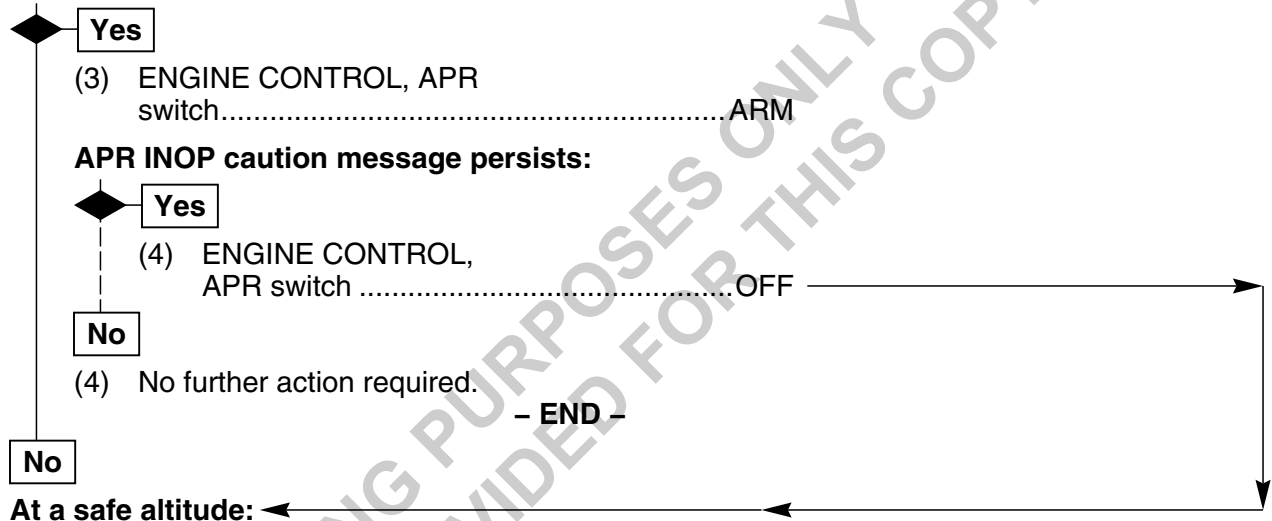
1. POWER PLANT (CONT'D)

C. APR Failure

Indication: APR INOP caution message on.

- (1) ENGINE CONTROL, APR switch.....OFF
- (2) ENGINE CONTROL, both ENG
SPEED switchesCheck ON and monitor engine indications.

Engine indications are normal:



At a safe altitude:

- (3) ATS DISC switch..... Depress
- (4) Both thrust levers Retard to below 79% N_2 .
- (5) ENGINE CONTROL, both ENG
SPEED switchesOFF to prevent asymmetric thrust.

NOTE

Use APR-off performance. Refer to Chapter 6; PERFORMANCE.

- (6) Thrust levers.....Advance as required.

NOTE

Thrust lever controls are more sensitive to changes.

Due to the possibility of N_1 overspeed, monitor the N_1 if more than idle reverse thrust is used after landing. Do not exceed 90% N_1 reverse thrust.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-6
		Oct 27/15

1. POWER PLANT (CONT'D)

D. Inadvertent APR Command

Indication: APR CMD SET caution message. (APR CMD SET caution message may be accompanied by APR operation on one or both engines.)

On the ground:

◆ **Yes**

(1) Do not take off.

– END –

No

In flight:

- (1) Thrust levers.....Adjust to within normal thrust range.
- (2) ENGINE CONTROL, APR switch.....OFF
- (3) Engine performanceMonitor during approach.

————— END —————

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-7
		Oct 27/15

1. POWER PLANT (CONT'D)

E. Uncommanded Deceleration or Abnormal Engine Oscillation/Fluctuation

Indication: Engine thrust does not agree with thrust lever selection.

- (1) ATS DISC switch Depress



The ATS SPEED mode N_1 synchronization will attempt inappropriate deceleration/acceleration of the remaining engine, to maintain synchronization until disengaged.

NOTE

To prevent engine overspeed when selecting ENG SPEED switches OFF, N_2 indications should be referenced instead of N_1 .

Affected engine:

- (2) Thrust lever Adjust to below 79% N_2 or IDLE (whichever occurs first).
- (3) ENGINE CONTROL, ENG SPEED switch OFF
- (4) Thrust lever Advance



Do not change power settings abruptly as engine response is degraded.

- (5) Engine indications Monitor

Engine responds:



NOTE

- To minimize asymmetric thrust, both ENG SPEED switches should be selected OFF prior to approach and landing.
- Exercise caution to prevent overspeed during thrust reverse use, when operating with ENG SPEED switches OFF.
- The APR system is inoperative when the ENG SPEED switches are selected OFF.

— END —



- (6) In-Flight Engine Failure/Shutdown Accomplish Refer to In-Flight Engine Failure/Shutdown procedure in this section.

————— END —————

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-8
		Oct 27/15

1. POWER PLANT (CONT'D)

F. N₁ Fan Vibration

Indication: FAN VIB readout greater than 2.7 Mils or
VIB icon on N₁ gauge in compressed display format.



It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, abnormal increase in ITT, ITT exceedance, or abnormal engine vibration is felt through the airframe.

NOTE

1. If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.
2. An ITT increase of more than 10°C from stable, matched N₁ power is an abnormal increase.

(1) ATS DISC switch Depress

Icing conditions:



Yes

Affected engine:

(2) Thrust lever Adjust N₂ to above 80%, and N₁ to above 70%.

Do not exceed maximum continuous thrust setting.

Maintain thrust settings until vibration is reduced.

NOTE

1. Vibration may increase during the ice shedding process.
2. Vibrations may momentarily indicate full scale during the ice shedding process.
3. Thrust settings up to and including maximum continuous may be required to shed ice.

(3) Engine indications Monitor ITT.

No

Affected engine:

(2) Thrust lever Adjust to a setting that maintains vibration level within normal range.

(3) Engine Monitor



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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-9
		Oct 27/15

1. POWER PLANT (CONT'D)

F. N₁ Fan Vibration (Cont'd)

Vibration persists and the condition indicates advisability of an engine shutdown: ←

Yes

(4) Affected engine.....Shutdown

Refer to In-Flight Engine Failure/Shutdown procedure in this section.

— END —

No

(4) No further action required.

— END —

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-10
		Oct 27/15

1. POWER PLANT (CONT'D)

G. N₂ Core Vibration

Indication: VIB icon on N₂ gauge or adjacent to N₂ readout.



It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, high ITT, or abnormal engine vibration is felt through the airframe.

NOTE

N₂ VIB icon indicates core vibration is greater than 1.7 Mils.

(1) ATS DISC switch Depress

Affected engine:

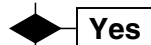
(2) Thrust lever Adjust to a setting that maintains vibration level within normal range.

NOTE

If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.

(3) Engine indications Monitor

Vibration can be controlled or reduced:



(4) No further action required.

— END —



(4) Affected engine Shutdown Refer to In-Flight Engine Failure/Shutdown procedure in this section.

————— END —————

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-11
		Oct 27/15

1. POWER PLANT (CONT'D)

H. Engine Hot Start

Indication: HOT icon displayed on ITT gauge or adjacent to ITT readout and/or readout indicates 900°C or greater.

Affected engine:

- (1) Thrust lever SHUT OFF

(2) IGNITION switch/lights (all) Press out
- to turn off all ignition. ON lights go out. IGNITION A and B advisory messages and CONT IGNITION light out.

If one engine is operating:

- (3) Inoperative engine FUEL, L (R) BOOST PUMP switch/light..... Press out

(4) Dry motor..... Accomplish
- to disarm boost pump.

Check L (R) FUEL PUMP caution message on.

until ITT is reduced to below 120°C or starter limit, whichever comes first.

END

I. Engine DCU Parameter Disagreement

Indication: L (R) ENG MISCOMP caution message. Affected engine's (either left or right engine) DCU parameters disagree. (N₁ and N₂ by 5%, and ITT by 40°C).

- (1) Multifunction Displays (both) Select

(2) Affected engine's parameters (either left or right) Cross-check

(3) Affected engine Maintain
- to EICAS page.

to find disagreement.

within limits, using reliable EICAS page.

END

CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-12
		Oct 27/15

1. POWER PLANT (CONT'D)

J. Thrust Lever Inoperative

Indication: Affected thrust lever inoperative.

At a safe altitude, before commencing the approach:

Affected engine:

- (1) ENG FIRE PUSH switch/light Press in to shutdown affected engine.
BOTTLE ARMED PUSH TO DISCH lights and L (R) ENG SOV CLSD advisory message on.

NOTE

Ground spoilers may not be available upon landing.

- (2) Actual landing distance Increase by a factor of 1.60 (60%), for a flaps 20° landing without the use of reverse thrust.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- (3) Affected engine Shutdown Refer to In-Flight Engine Failure/Shutdown procedure in this section.

————— **END** —————

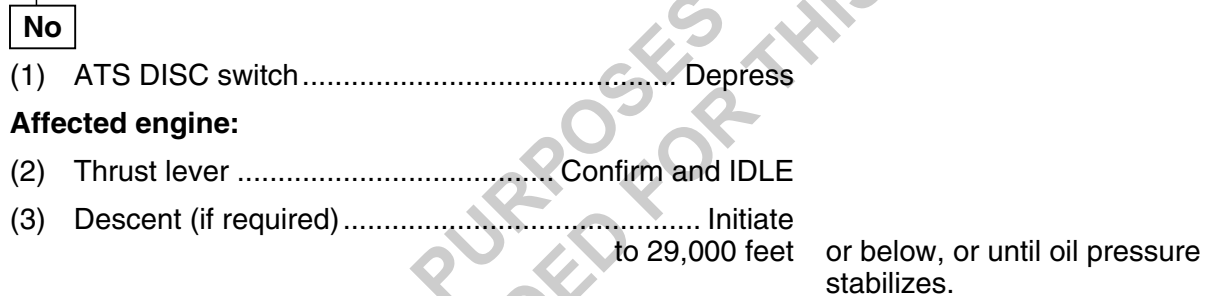
CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-13
		Oct 27/15

1. POWER PLANT (CONT'D)

K. Fluctuating Engine Oil Pressure

Indication: Left or right oil pressure readout fluctuates in excess of ± 5 psi and gradually decreases, possibly with increasing oil temperature.

Oil pressure is below 25 psi:



When affected engine oil pressure stabilizes:

(4) Thrust lever Adjust as required.

NOTE

1. If necessary, keep descending to a lower altitude to maintain a stable oil pressure, as cruise power setting is resumed.
2. If oil pressure fluctuations in excess of ± 5 psi resume, select affected thrust lever to IDLE, and descend to a lower altitude.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-14
		Oct 27/15

1. POWER PLANT (CONT'D)

L. ATS Failure

Indication: Amber FAIL message displayed in ATS MSD.

Flashing FAIL message:

- ◆ **Yes**
- (1) ATS DISC switch Depress
 - (2) Thrust levers Position manually as required.

NOTE

The ATS may be re-engaged by depressing the ATS push button switch on the ATS Control Panel. If the fault is still present, the ATS flashing amber FAIL message will reappear, and ATS will not re-engage.

— END —

No

Steady FAIL message:

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

NOTE

The steady amber FAIL message is displayed if the MSD is not receiving data from the ATS computer. Depress the ATS DISC switch to confirm disengagement of the ATS. Do not re-engage the ATS.

— END —

M. Dual ATS MSD Failure

Indication: Loss of displayed data in both ATS Mode Status Displays.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

NOTE

In the event of a dual MSD failure causing the displays to go blank, the crew must depress an ATS DISC switch to confirm ATS disengagement, and consider the system inoperative.

— END —

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-15
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1. POWER PLANT (CONT'D)

N. ATS Uncommanded Disengage

Indication: Uncommanded flashing amber DISENG'D message in ATS MSD, and ATS disengages.

- (1) ATS DISC switch Depress to cancel message.
- (2) Thrust levers Position manually as required.

NOTE

The ATS will automatically disengage and illuminate the flashing amber DISENG'D message in the MSD, if an N_1 split of 13% or greater is detected, or an N_1 greater than 98.4% is detected, or if a Thrust Reverser unlocks. After fault correction, the ATS can be re-engaged by depressing the ATS push button on the ATS control panel.

The ATS will disengage if the N_1 value is removed from the EICAS (no active selection present). This could happen when the bleeds are changed from ENGINES to APU during the IN RANGE check.

————— **END** —————

O. ATS Overspeed

Indication: Flashing amber V_{MO} LIMIT or M_{MO} LIMIT message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

————— **END** —————

P. ATS Thrust Limited

Indication: Flashing amber SPEED message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually and re-adjust attitude as required.

————— **END** —————

Q. ATS Low Speed

Indication: Flashing amber AOA LIMIT message displayed in ATS MSD.

- (1) Airspeed Set AFCS airspeed reference (speed bug) higher than AOA limit.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Power Plant	05-02-16
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1. POWER PLANT (CONT'D)

R. ATS Radio Altimeter Input Failure

Indication: Lack of RETARD mode at 50 ft. or
Early RETARD mode with green RETARD or flashing green SPEED
message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Single Engine Procedures	05-03-1
		Oct 27/15

1. SINGLE ENGINE PROCEDURES

A. Engine Failure During Take-Off After Achieving V_1

- (1) Take-off Continue
- (2) Airplane Rotate at V_R to an initial take-off attitude of 10 degrees, and adjust pitch, as required, to achieve an airspeed of V_2 at 35 feet.

When positive rate of climb is achieved:

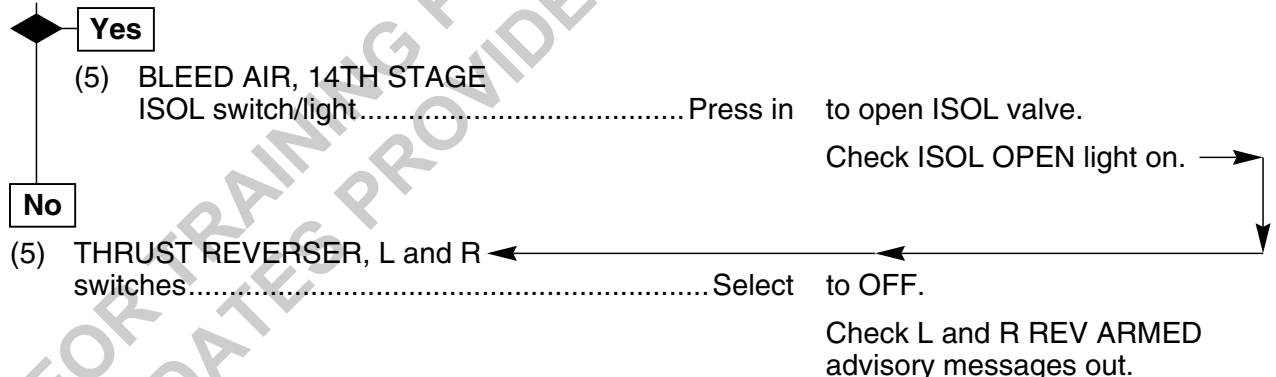
- (3) Landing gear Retract
- (4) Airspeed Maintain at V_2 to engine-out level-off height (Refer to Chapter 6; PERFORMANCE – OBSTACLE CLEARANCE).

NOTE

If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$ KIAS).

After take-off:

Wing anti-ice system selected on:



At not less than 400 feet AGL or the engine-out level off height:

- (6) Thrust Set thrust to maximum continuous thrust (MCT) (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (7) Airspeed Accelerate in level flight to V_{FTO} .
- (8) FLAPS Select to UP at a speed of not less than $V_{FTO} - 5$ KIAS.
- (9) Climb Initiate at V_{FTO} .
- (10) CAS messages Review



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CHALLENGER 650	ABNORMAL PROCEDURES Single Engine Procedures	05-03-2 Rev 13, Aug 23/18
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1. SINGLE ENGINE PROCEDURES (CONT'D)

A. Engine Failure During Take-Off After Achieving V₁ (Cont'd)

At a safe altitude:

- (11) In-Flight Engine Failure/Shutdown Accomplish Refer to POWER PLANT – In-Flight Engine Failure/Shutdown procedure in this chapter.

————— **END** —————

B. Engine Failure in Climb During ALTS CAP

- (1) Autopilot/FD Disconnect
- (2) Pitch attitude Adjust to maintain the required single engine operating airspeed.

NOTE

Vertical FD commands may be used and autopilot re-engaged after FCC mode changes to ALTS at desired altitude.

- (3) In-Flight Engine Failure/Shutdown Accomplish Refer to POWER PLANT – In-Flight Engine Failure/Shutdown procedure in this chapter.

————— **END** —————

C. Engine Failure During Approach

- (1) Autopilot Disconnect if coupled approach.
- (2) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (3) Operating engine Increase thrust as required.
- (4) FLIGHT SPOILER lever Select to RETRACT.
Check EICAS indicates spoilers retracted.
- (5) FLAPS Select to 20° for landing.
- (6) Airspeed Increase to V_{REF} (Flaps 45°) + 14 KIAS.
- (7) Airplane Retrim and continue approach, or go-around at pilot's discretion (Refer to Single Engine Go-Around, in this section).
If required, autopilot may be re-engaged, if above 800 feet AGL.



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1. SINGLE ENGINE PROCEDURES (CONT'D)

C. Engine Failure During Approach (Cont'd)

If continuing the approach:

- (8) Air-conditioning..... Transfer to APU bleed air.
- (9) Go-Around thrust..... Determine applicable single engine climb thrust (APR rating) (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (10) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute flap aural warning.
- (11) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (12) Final approach speed..... Maintain V_{REF} (Flaps 45°) + 14 KIAS.
- (13) Actual landing distance Increase by a factor of 1.55 (55%), for a flap 20° landing without the use of reverse thrust.

NOTE

Use remaining thrust reverser carefully upon landing.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

After landing:

- (14) Affected engine Shutdown

————— END —————

1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing

DURING DESCENT

- (1) ATIS Copy
- (2) CAS Check/Clear Verify and analyze any messages, and clear (if any).
- (3) TAWS WARNING, TERRAIN OFF switch/light As required
- (4) TAWS WARNING, FLAPS OFF switch/light Press in to mute flap aural warning.
- (5) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (6) Landing data Set
- (7) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a flaps 20° landing without the use of reverse thrust.
- (9) Approach briefing Complete
- (10) Shoulder harnesses Secured
- (11) CABIN PRESSURIZATION, LDG ALT selector Check and set.
- (12) ANTI-ICE, COWL L and R switch/lights Select as required.
- (13) ANTI-ICE, WING switch Select as required.
- (14) ANTI-ICE, WSHLD/WIND L and R switches Select to HI or LOW, as required.
- (15) Fuel quantity and balance Check

At transition level:

- (16) EFIS altimeters (2) and Integrated Standby Instrument altimeter (1) Set to destination airport barometric setting.

WHEN IN RANGE

- (1) APU Start Ensure that operating limitations/envelope is considered.
- (2) Air-conditioning Transfer to APU bleed air.

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

- (3) Go-Around thrust..... Determine applicable single engine climb thrust (APR rating) (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (4) Lights..... As required Select RECOG TAXI light on.
- (5) PASS SIGNS, NO SMKG switch..... Select to ON.
- (6) PASS SIGNS, SEAT BLTS switch Select to ON.

BEFORE LANDING

- (1) IGNITION, CONT switch/light..... As required
- (2) ANTI-ICE, WING switch Select as required.
- (3) ANTI-ICE, COWL L and R switch/lights Select as required.
- (4) Cabin Advise not less than five minutes prior to landing.
- (5) LDG GEAR lever Select to DN.
- (6) Brake and anti-skid system Check
- (7) Operative THRUST REVERSER, L (R) switch Select to ARMED.

CAUTION

Do not arm thrust reverser if L (R) REV UNSAFE caution message is displayed.

- (8) FLAPS..... Select to 20° for landing. Verify proper indication.
- (9) Final approach speed..... Maintain V_{REF} (Flaps 45°) + 14 KIAS.
- (10) Actual landing distance Increase by a factor of 1.55 (55%), for a flaps 20° landing without the use of reverse thrust.

NOTE

If required, use remaining thrust reverser carefully upon landing.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

1. SINGLE ENGINE PROCEDURES (CONT'D)

E. Single Engine Go-Around

NOTE

- The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
- Single engine go-around capability is not assured with flaps greater than 20°.

- (1) Single Engine Climb Thrust (APR rating)..... Set immediately (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).

FAA Certified Airplanes

- (1) Normal Take-off Thrust Set immediately (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).

- (2) TOGA push switch Depress

- (3) FLIGHT SPOILER lever (If extended) Select to RETRACT.

Check EICAS indicates spoilers retracted.

- (4) Airplane Rotate smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB).

- (5) FLAPS..... Select to 20°.

When a positive rate of climb is achieved:

- (6) LDG GEAR lever Select to UP.

At a safe altitude (not below 400 feet AGL):

- (7) FLAPS Select to 0° at a speed of not less than $V_{FTO} - 5$ KIAS.

- (8) THRUST REVERSER, L and R switches..... Select to OFF.

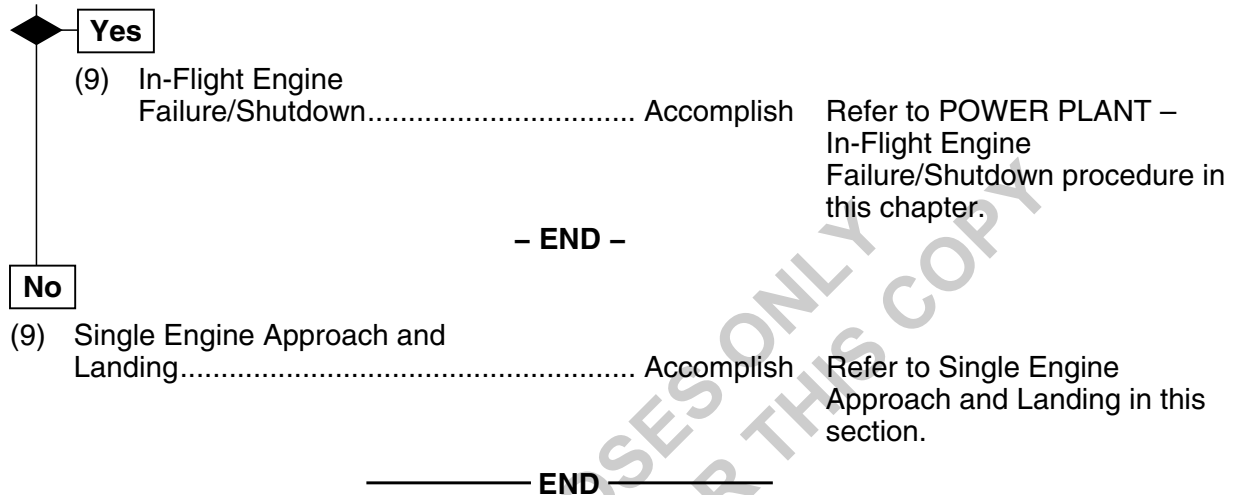
Check L and R REV ARMED advisory messages out.



1. SINGLE ENGINE PROCEDURES (CONT'D)

E. Single Engine Go-Around (Cont'd)

Go-around was due to engine failure during approach:



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-1
		Oct 27/15

1. AIR START

A. Engine Cross Bleed – Air Start Procedure

NOTE

Use cross bleed for in-flight air starts whenever possible.
Refer to Chapter 2; LIMITATIONS – POWER PLANT for starter cranking limits.

Affected engine:

- (1) Thrust leverCheck affected lever SHUT OFF.
- (2) FUEL, L (R) BOOST PUMP switch/light.....Check the following:
 - PUMP ON lights on.
 - L and R FUEL PUMP ON advisory messages on.
- (3) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.

Operating engine:

- (4) BLEED AIR, 10TH STAGE, L (R) switch/light.....Press in to open operative shut-off valve.
Check L (R) CLOSED light out.
- (5) Bleed air pressure 60 psi Adjust operating engine's thrust lever to attain 60 psi **minimum**.
- (6) Start envelope Establish

ENGINE CROSS BLEED AIR START ENVELOPE			
ALTITUDE	AIRSPEED	ITT	% N ₂
21,000 feet to 15,000 feet	200 KIAS up to 300 KIAS	90°C or less	0 to 55
15,000 feet to SL	140 KIAS up to 300 KIAS		

When ready to start:

- (7) IGNITION , CONT switch/light.....Press in for continuous ignition.
Check the following:
 - CONT ON, A and B lights on.
 - IGNITION A/B advisory message on.



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-2
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1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

Affected engine:

- (8) ENG, L (R) START switch/lightPress to cross bleed-air start affected engine.
Check L (R) START light on.

When ITT is 90°C or less and N₂ is 28%:

- (9) Thrust leverSelect to IDLE when within start envelope.

NOTE

If ITT is above 90°C, dry motor to lower ITT to 90°C before moving throttle to IDLE.

- (10) Engine indicationsMonitor carefully.

NOTE

If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, press associated engine STOP switch/light, wait 20 seconds and repeat procedure.

Engine start is successful:

◆ Yes

When the engine is stabilized at Flight Idle:

- (11) Thrust levers Set as required.
(12) IGNITION, CONT switch/lightPress out to select continuous ignition off.
Check IGNITION A/B advisory message out.
(13) BLEED AIR, 10TH STAGE, L (R) switch/lightPress in to open affected shut-off valve.
Check L (R) CLOSED light out.
(14) BLEED AIR, APU LCV switch/light As required
(15) AIR-CONDITIONING, L and R PACK switch/lightsPress in to start both packs.
Check L and R PACK OFF lights out.
(16) ANTI-ICE, COWL, L and R switches As required



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-3
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1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

- ▼
- (17) BLEED AIR, 14TH STAGE, ISOL switch/light.....Press out to close isolation valve.
Check 14TH ISOL OPEN light out.
- (18) AC POWER, GEN 1 (2) switch.....Select affected generator ON.
Check GEN 1 (2) OFF caution message out.
- (19) Affected HYDRAULIC, pump switch.....AUTO Check the following:
- HYD EDP 1A and HYD PUMP 2B caution messages out, or
 - HYD EDP 2A and HYD PUMP 1B caution messages out.
- (20) TAWS WARNING, FLAPS OFF switch/light.....Press out Check OFF light out.
- END –
- No**
- (11) IGNITION, CONT switch/light.....Press out to select continuous ignition off.
Check IGNITION A/B advisory message out.
- (12) BLEED AIR, 10TH STAGE L (R) switch/light.....Press out to select affected 10th-stage SOV closed.
Check L (R) 10TH CLOSED light on.
- (13) BLEED AIR, APU LCV switch/light..... As required
- (14) AIR-CONDITIONING, L (R) PACK switch/light.....Press in to turn on operative pack.
Check L (R) PACK OFF light out.
- (15) MFD or CDU tuning, ATC/TCAS mode Set to TA ONLY.
- (16) Land at the nearest suitable airport.



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-4
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1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

- (17) Single Engine Approach and Landing..... Accomplish Refer to SINGLE ENGINE PROCEDURES – Single Engine Approach and Landing in this chapter.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-5
		Oct 27/15

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure

NOTE

1. At thrust lever advance, N_2 should be stable or increasing.
2. Windmill airstart efficiency is enhanced by attaining the highest practical airspeed and N_2 within the relight envelope.
3. 300 knots is a target airspeed only, and may be exceeded to obtain the minimum required N_2 .

Affected engine:

- (1) Thrust lever Check affected lever SHUT OFF.
- (2) FUEL, L (R) BOOST PUMP switch/light Check the following:
 - PUMP ON lights on.
 - L and R FUEL PUMP ON advisory messages on.
- (3) BLEED AIR, 10TH STAGE, L (R) switch/light Press out to close affected shut-off valve.
Check 10TH CLOSED light on.
- (4) BLEED AIR, 10TH STAGE, ISOL switch/light Press out to close isolation valve.
Check 10TH ISOL OPEN light out.
- (5) BLEED AIR, APU LCV switch/light Press out to close load control valve.
Check APU LCV OPEN light out.
- (6) Start envelope Establish

WINDMILLING AIR START ENVELOPE			
ALTITUDE	AIRSPEED	ITT	% N_2
21,000 feet to 10,000 feet	300 KIAS up to 348 KIAS	90°C or less	12 to 55
10,000 feet to 8,000 feet			10 to 55
8,000 feet to SL	300 KIAS		



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-6
		Oct 27/15

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

When ready to start:

(7) IGNITION, CONT switch/light..... Press in for continuous ignition.

Check the following:

- CONT ON, A and B lights on.
- IGNITION A/B advisory message on.

When ITT is 90°C or less and N₂ is at least 12% (10% if below 10,000 feet):

Affected engine:

(8) Thrust lever Select to IDLE when within start envelope.

(9) Engine indications Monitor carefully.

NOTE

1. If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, maintain at least 300 KIAS for 30 seconds, and repeat procedure. After ignition is achieved, monitor N₂ acceleration rate. N₂ acceleration should be uninterrupted and positive at all times. Stable idle must be achieved within 2 minutes of advancing thrust lever to IDLE.
2. During windmilling airstart, attempt to keep the flight control inputs to a minimum.

Engine start is successful:

◆ Yes

When the engine is stabilized at Flight Idle:

(10) Thrust levers..... Set as required.

(11) IGNITION, CONT switch/light..... Press out to select continuous ignition off.

Check IGNITION A/B advisory message out.

(12) BLEED AIR, 10TH STAGE, L (R) switch/light As required

(13) BLEED AIR, APU LCV switch/light As required

(14) AIR-CONDITIONING, L (R) PACK switch/light Press in to start affected pack.

Check L (R) PACK OFF light out.

(15) ANTI-ICE, COWL, L (R) switches As required

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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-7
		Oct 27/15

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

- ▼
- (16) BLEED AIR, 14TH STAGE, ISOL switch/light.....Press out to close isolation valve.
Check 14TH ISOL OPEN light out.
- (17) AC POWER, GEN 1 (2) switch.....Select affected generator ON.
Check GEN 1 (2) OFF caution message out.
- (18) Affected HYDRAULIC, pump switch.....AUTO Check the following:
- HYD EDP 1A and HYD PUMP 2B caution messages out, or
 - HYD EDP 2A and HYD PUMP 1B caution messages out.
- (19) TAWS WARNING, FLAPS OFF switch/light.....Press out Check OFF light out.
- END -
- No**
- (10) IGNITION, CONT switch/light.....Press out to select continuous ignition off.
Check IGNITION A/B advisory message out.
- (11) BLEED AIR, 10TH STAGE L (R) switch/light.....Press out to select affected 10th-stage SOV closed.
Check L (R) 10TH CLOSED light on.
- (12) BLEED AIR, APU LCV switch/light..... As required
- (13) AIR-CONDITIONING, L (R) PACK switch/light.....Press in to turn on operative pack.
Check L (R) PACK OFF light out.
- (14) MFD or CDU tuning, ATC/TCAS mode Set to TA ONLY.
- (15) Land at the nearest suitable airport.



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CHALLENGER 650	ABNORMAL PROCEDURES Air Start	05-04-8
		Oct 27/15

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

- (16) Single Engine Approach and Landing..... Accomplish Refer to SINGLE ENGINE PROCEDURES – Single Engine Approach and Landing in this chapter.

————— **END** —————

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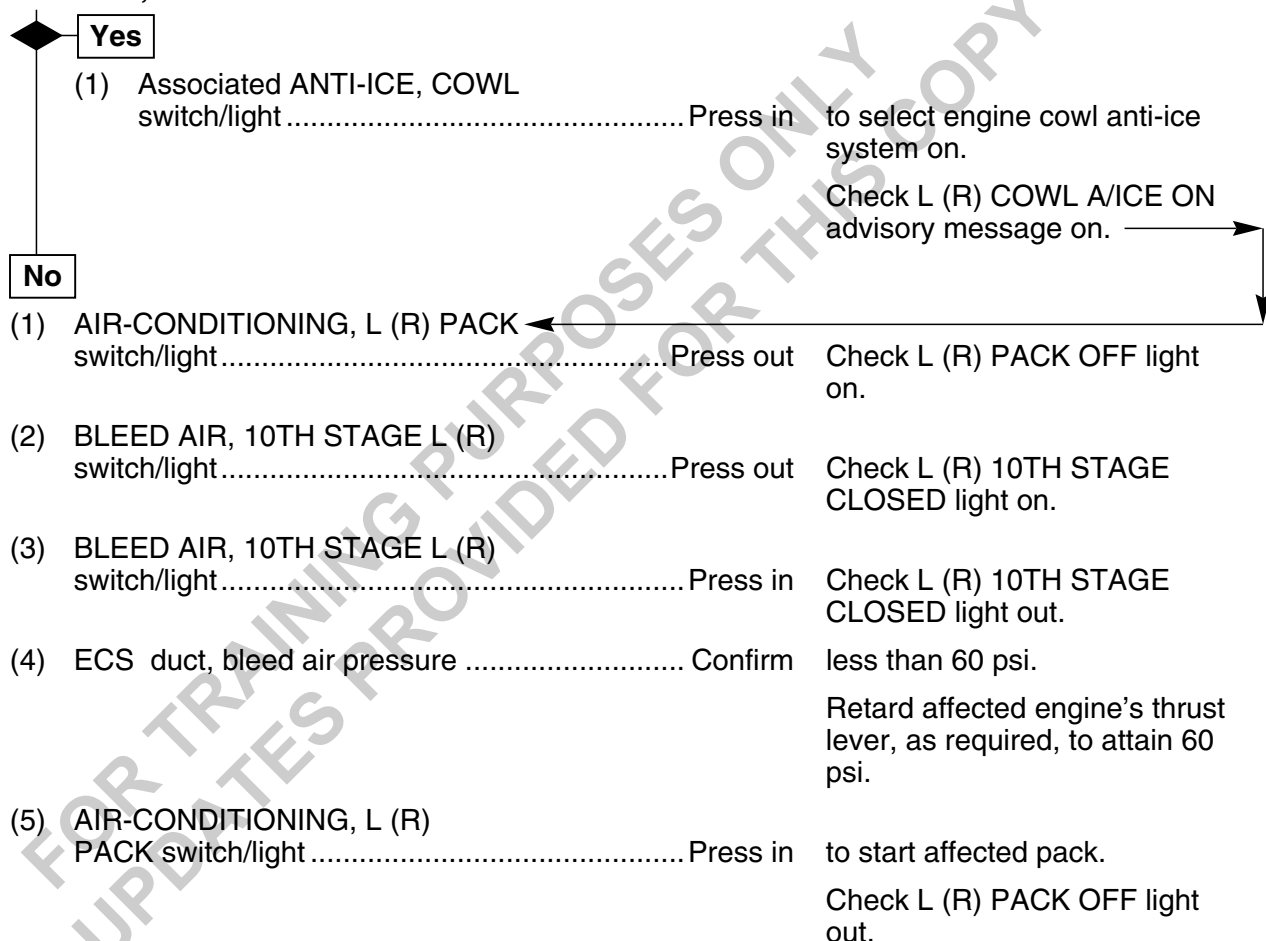
1. AIR-CONDITIONING AND PRESSURIZATION
A. Air-Conditioning Pack High Pressure

Indication: L (R) PACK HI PRESS caution message and L (R) PACK FAIL light on.

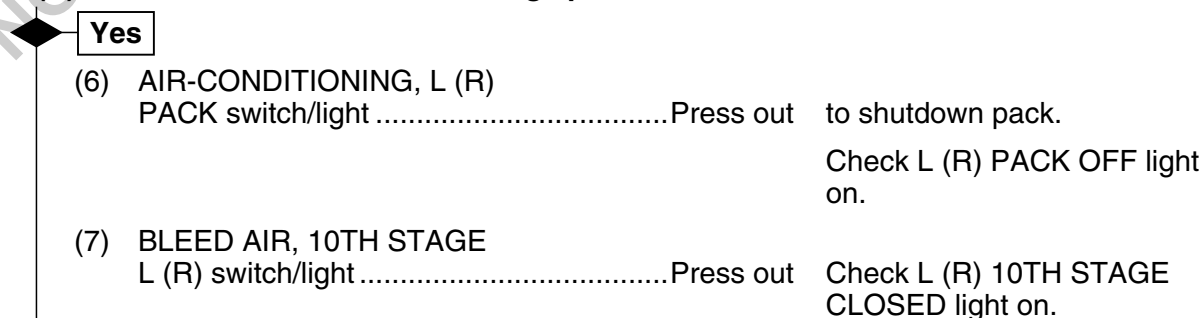
NOTE

The isolation valve, pack valve and the 10th stage bleed valve will close automatically during an overpressure condition.

Above 40,000 feet:



L (R) PACK HI PRESS caution message persists:



- END -

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

A. Air-Conditioning Pack High Pressure (Cont'd)

No

(6) No further action required.

END

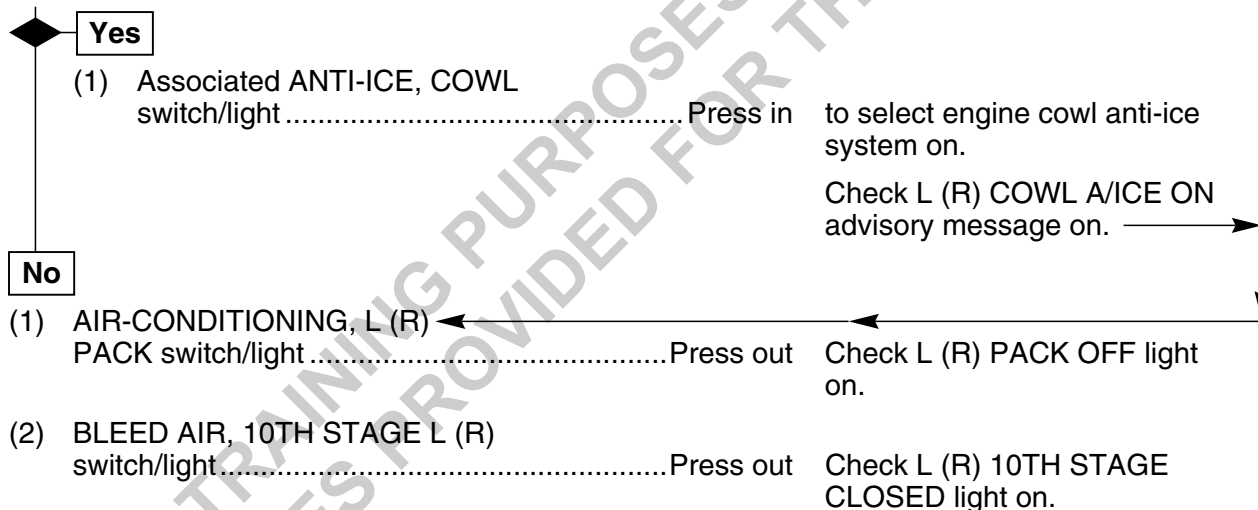
B. Air-Conditioning Pack High Temperature

Indication: L (R) PACK HI TEMP caution message and L (R) PACK FAIL light on.

NOTE

The pack valve will close automatically during an overtemperature condition.

Above 40,000 feet:



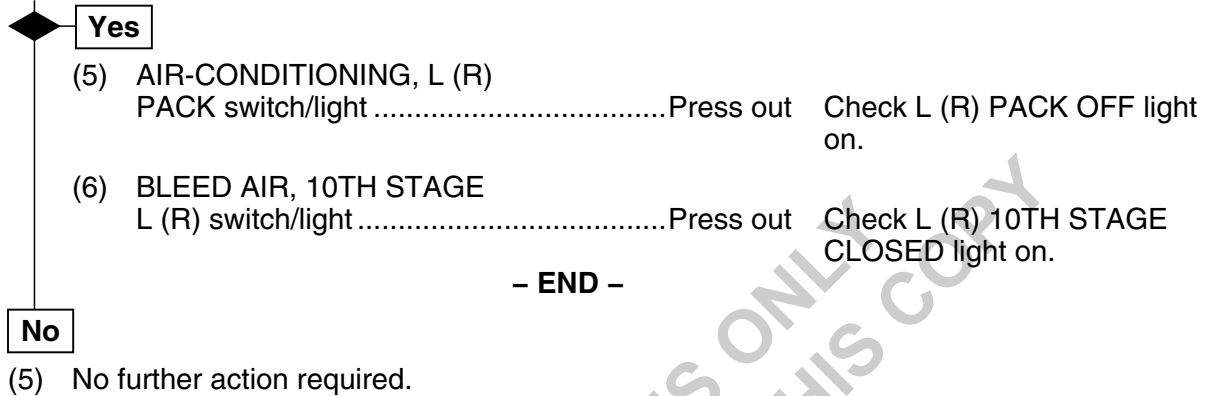
NOTE

If in manual mode with full HOT selected, decrease temperature before resetting pack.

When L (R) PACK HI TEMP caution message goes out:

- (3) BLEED AIR, 10TH STAGE L (R) switch/light Press in Check L (R) 10TH STAGE CLOSED light out.
- (4) AIR-CONDITIONING, L (R) PACK switch/light Press in to start affected pack.
Check L (R) PACK OFF light out.



1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)
B. Air-Conditioning Pack High Temperature (Cont'd)
L (R) PACK HI TEMP caution message on:

C. Automatic Pressurization System Malfunction
Indication: AUTO PRESS caution message and PRESS CONT FAIL light on.

- (1) Manual Cabin Pressurization Control Procedure Accomplish Refer to Manual Cabin Pressurization Control Procedure in this section.

NOTE

Do not attempt to reselect auto mode.

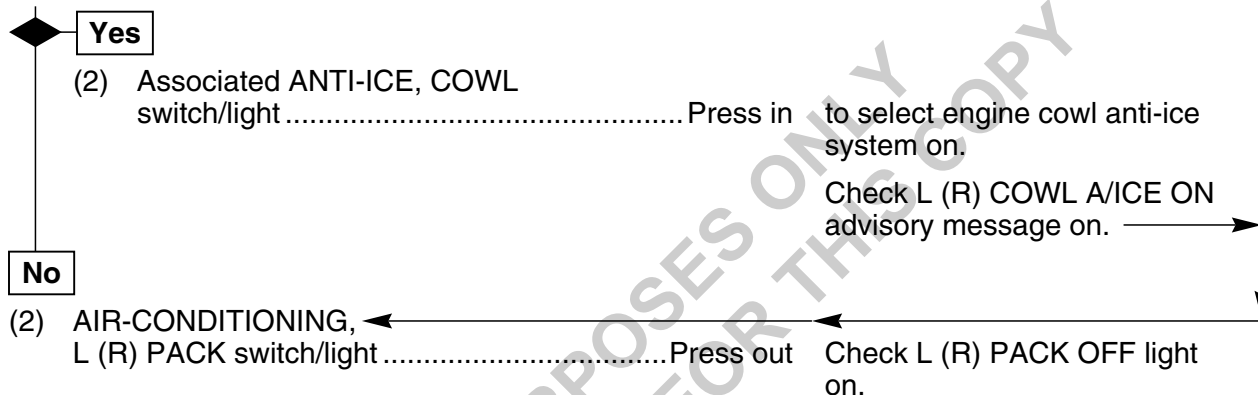
END

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)
D. Air-Conditioning Pack Pressure/Regulating Valve Failure

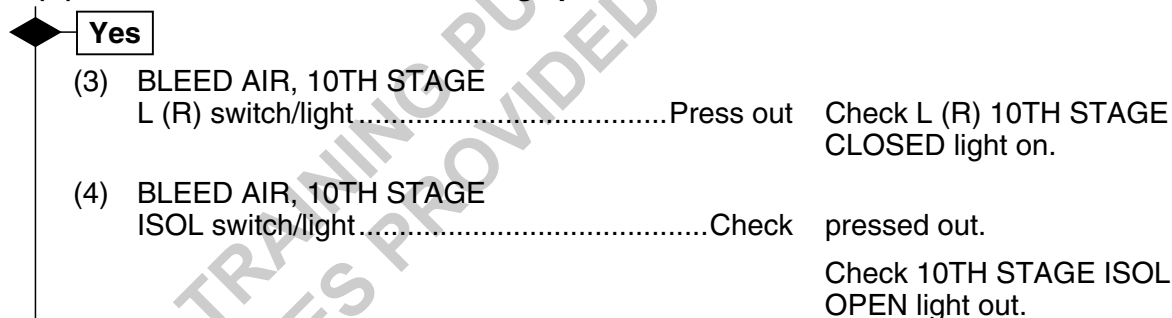
Indication: L (R) PACK NOT OFF caution message on and L (R) PACK FAIL light on.

- (1) AIR-CONDITIONING,
L (R) PACK switch/light Press in Check L (R) PACK OFF light out.

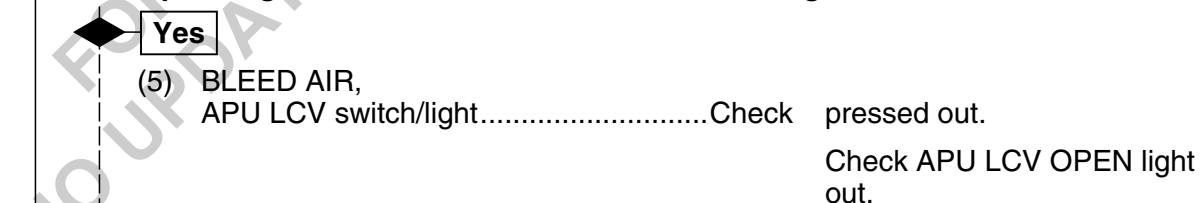
Above 40,000 feet:



L (R) PACK NOT OFF caution message persists:



APU operating and L PACK NOT OFF caution message on:



— END —

No

- (3) No further action required.

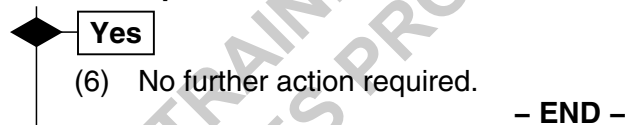
— END —

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)
E. Cabin Altitude Too High

Indication: CABIN ALT caution message on.

- (1) BLEED AIR, 10TH STAGE
L and R switch/lightsCheck pressed in.
Check L and R 10TH STAGE
CLOSED lights out.
- (2) AIR-CONDITIONING,
L and R Pack switch/lightsCheck pressed in.
Check L and R PACK OFF
lights out.
- (3) Engine thrust Increase
- (4) CABIN PRESSURIZATION,
EMER DEPRESS switch/lightCheck pressed out.
Check the following:
- EMER DEPRESS ON light out.
 - EMER DEPRESS caution message out.
- (5) CABIN PRESSURIZATION,
MAN ALT selectorCheck set to mid position/centre.

Automatic pressurization is restored:



- (6) CABIN PRESSURIZATION,
PRESS CONTPress in to select manual mode.
Check PRESS CONT
MANUAL light on.
Operate MAN ALT and MAN
RATE switches as required.
(Refer to Manual Cabin
Pressurization Control
Procedure in this section).

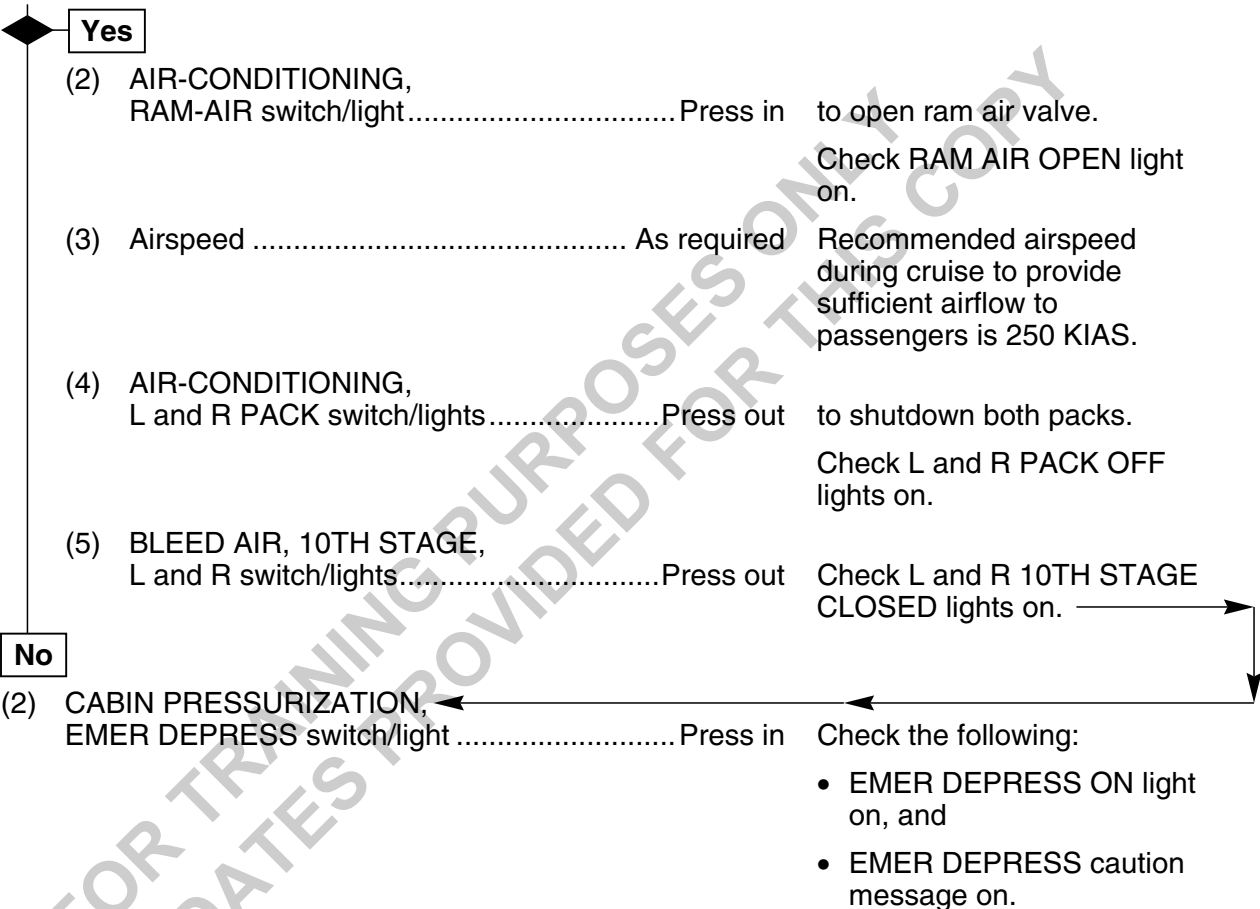
END

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

F. Unpressurized Flight Procedure

- (1) Airplane altitude..... Maintain 10,000 feet maximum or lowest safe altitude, whichever is higher.

PACKs OFF:



CHALLENGER 650	ABNORMAL PROCEDURES Air-Conditioning and Pressurization	05-05-7
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1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

G. Manual Cabin Pressurization Control Procedure

NOTE

The MAN ALT selector is a three (3) position switch.

The switch is spring-loaded to mid position/centre.

- (1) CABIN PRESSURIZATION,
MAN ALT selector Check set to mid position/centre.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select manual mode.
Check PRESS CONT
MANUAL light on.
Operate CABIN PRESS, MAN
ALT and MAN RATE switches
as required.



Do not exceed the maximum normal cabin differential pressure of 8.8 psid.

To increase cabin altitude:

- (3) CABIN PRESSURIZATION,
MAN ALT selector Select UP and hold.
- (4) CABIN PRESSURIZATION,
MAN RATE switch As required

To decrease cabin altitude:

- (3) CABIN PRESSURIZATION,
MAN ALT selector Select DN and hold.
- (4) CABIN PRESSURIZATION,
MAN RATE switch As required

To maintain cabin altitude:

When reaching target cabin altitude as per the chart below:

- (5) CABIN PRESSURIZATION,
MAN ALT selector Release Ensure selector returns to mid
position/centre.

Cruise Flight Level	180	200	220	240	260	280	300	350	400	410
Target Cabin Altitude	2100	2400	2700	3000	3400	3800	4200	5200	6600	6900

Before landing:

- (6) Cabin altitude Adjust to landing field elevation.

NOTE

Do not set cabin altitude below landing field elevation.



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CHALLENGER 650	ABNORMAL PROCEDURES Air-Conditioning and Pressurization	05-05-8
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1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

G. Manual Cabin Pressurization Control Procedure (Cont'd)

If differential pressure is not zero upon landing:

- (7) CABIN PRESSURIZATION,
MAN ALT selector Select UP and hold.
- (8) CABIN PRESSURIZATION,
MAN RATE switch Set to maximum INCR.

————— **END** —————

H. Cabin Pressure Acquisition Module Failure

Indication: CPAM FAIL caution message on.

- (1) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
- (2) Unpressurized Flight Procedure Accomplish if required (Refer to Unpressurized Flight Procedure in this section).

NOTE

EICAS pressurization system readouts (e.g., cabin altitude, rate of change, and differential pressure) are not available. EICAS cabin pressure warning and caution messages are also not available.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Automatic Flight Control System	05-06-1
		Oct 27/15

1. AUTOMATIC FLIGHT CONTROL SYSTEM

NOTE

Autopilot disconnect warnings are indicated by “Cavalry Charge” aural and flashing red AP message on primary flight display.

A. Autopilot Mistrim

Indication: AP HOLDING LWD (RWD, NOSE DOWN, NOSE UP) caution messages on.

- (1) Autopilot Monitor operation.
Retrim as required.

WARNING

An abrupt change in control force, or an out of trim situation may be experienced when disconnecting the autopilot.

SEAT BLTS should be selected ON.

Caution message persists:

- Yes**
- (2) Flight controls Hold firmly
- (3) Autopilot Disconnect
- (4) AFCS SEL 1(2) switch (centre pedestal) Select to operative AFCS channel.
Check AFCS 1 (2) INOP status message on.
- (5) Autopilot Engage and monitor operation.
Retrim as required.

Caution message persists:

- Yes**
- (6) Flight controls Hold firmly
- (7) Autopilot Disconnect and assume manual control.
- END -

No

- (2) No further action required.

- END -

END

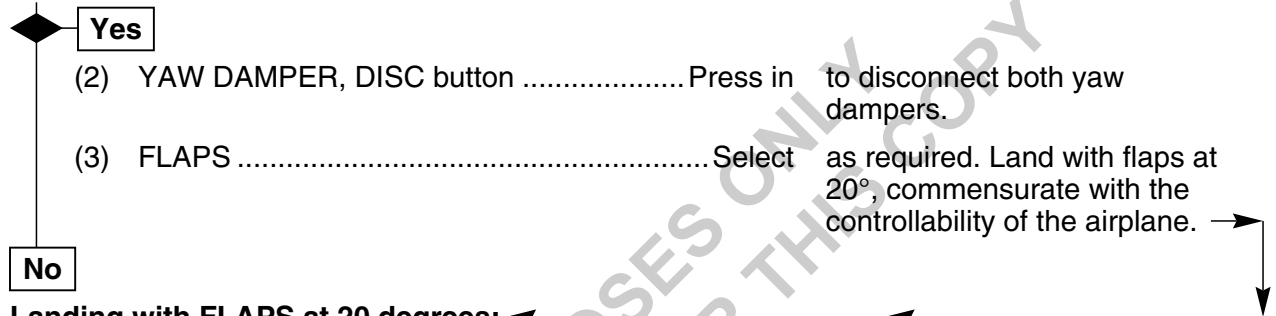
1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Double Yaw Damper Failure

Indication: YAW DAMPER caution message on. Autopilot will disengage if it was coupled.

- (1) YAW DAMPER, YD1 and YD2 switches..... Press in to engage both yaw dampers.

YAW DAMPER caution message persists:



- Decision: **Yes**
- Prior to landing:**
- (2) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (3) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (4) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (5) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)
C. Double Yaw Damper Failure (Cont'd)

- ▼ (6) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

NOTE

Select the runway available with minimum crosswind.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (2) No further action required.

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Automatic Flight Control System	05-06-5
		Oct 27/15

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

D. Single Flight Director Failed

Indication: FD 1 (2) FAIL status message on, red FD flag displayed on affected PFD, loss of flight director guidance cue on affected PFD, and autopilot may disengage.

NOTE

1. If a red FD flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to INSTRUMENTS SYSTEM – Air Data Computer Failure or Inertial Reference System Failure in this chapter.
2. Anticipate elevator out of trim situation when disconnecting the autopilot.

- (1) Flight controls Hold firmly
- (2) Autopilot Disconnect
- (3) AFCS SEL 1(2) switch
(centre pedestal) Select to operative AFCS channel.
Check AFCS 1 (2) INOP status message on.
- (4) FDs and autopilot Engage and monitor operation.
Retrim as required.

Red FD flag displayed on both PFDs:

- ☒ **Yes**
 (5) Flight control panel XFR switch Press to transfer to operable side.
 - END -

☐ **No**

Sensor failure indicated (red line across affected flight director mode annunciator):

- ☒ **Yes**
 (5) Flight control panel affected mode switch Deselect
 - END -

☐ **No**

- (5) No further action required.

————— **END** —————

CHALLENGER ⁶⁵⁰	ABNORMAL PROCEDURES Automatic Flight Control System	05-06-6
		Oct 27/15

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CHALLENGER 650	ABNORMAL PROCEDURES Auxiliary Power Unit	05-07-1
		Oct 27/15

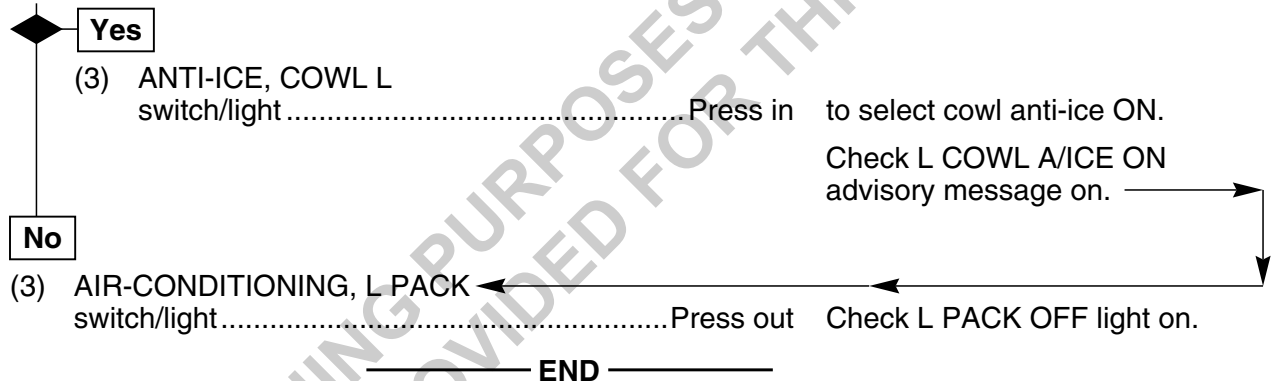
1. AUXILIARY POWER UNIT

A. APU Load Control Valve Fails to Close

Indication: APU LCV FAIL caution message and APU LCV FAIL light on.

- (1) BLEED AIR, 10TH STAGE, L
switch/light Press out to close left 10th stage bleed valve.
Check L CLOSED light on.
- (2) BLEED AIR, 10TH STAGE, ISOL
switch/light Press out to close 10th stage isolation valve.
Check 10TH ISOL OPEN light out.

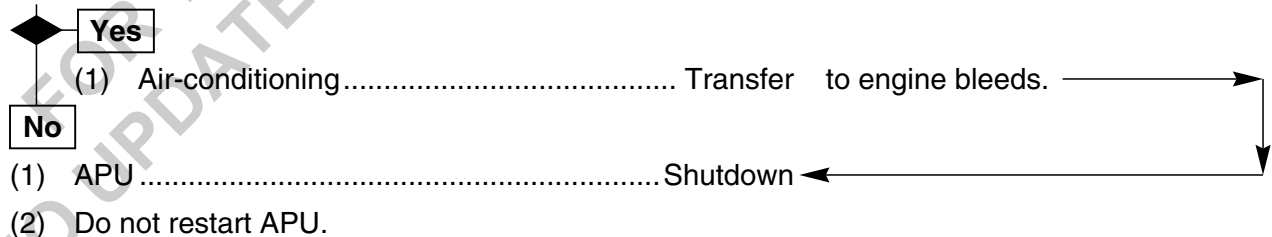
Above 40,000 feet:



B. APU Low Oil Pressure

Indication: APU OIL PRESS caution message on, and APU automatically shuts down.

APU is supplying bleed air for air-conditioning:



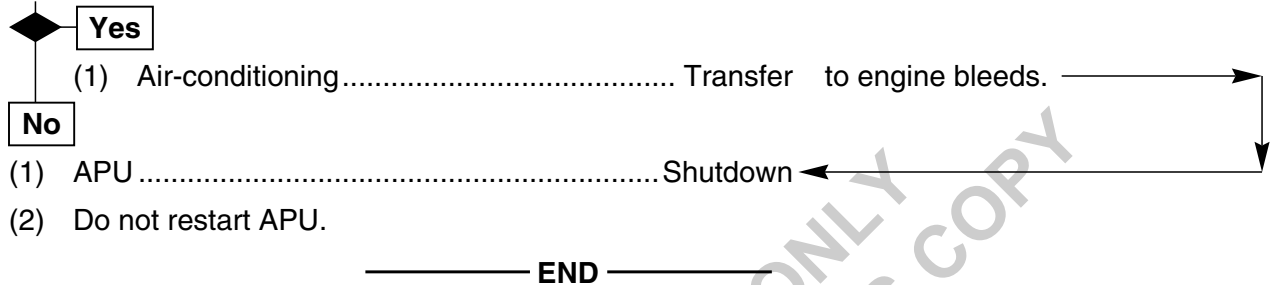
CHALLENGER 650	ABNORMAL PROCEDURES Auxiliary Power Unit	05-07-2
		Oct 27/15

1. AUXILIARY POWER UNIT (CONT'D)

C. APU High Oil Temperature

Indication: APU OIL TEMP caution message, and APU automatically shuts down.

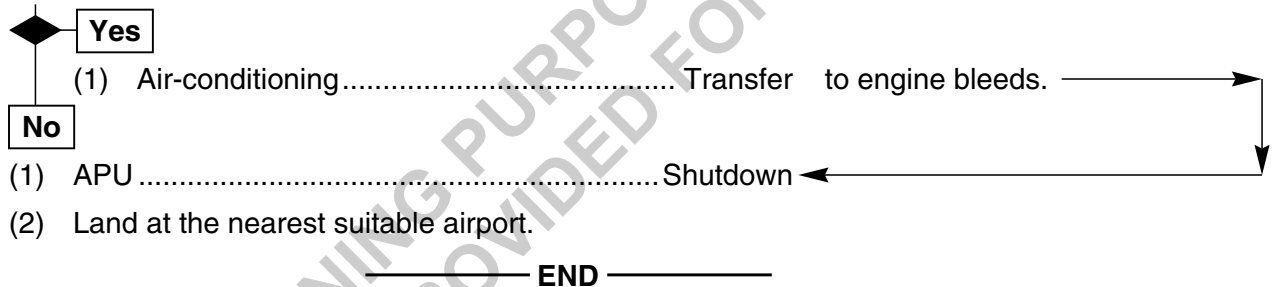
APU is supplying bleed air for air-conditioning:



D. APU Shut-Off Valve Failed to Close, After APU Shutdown Due to Fire

Indication: APU SOV caution message displayed after APU shutdown due to fire.

APU is supplying bleed air for air-conditioning:



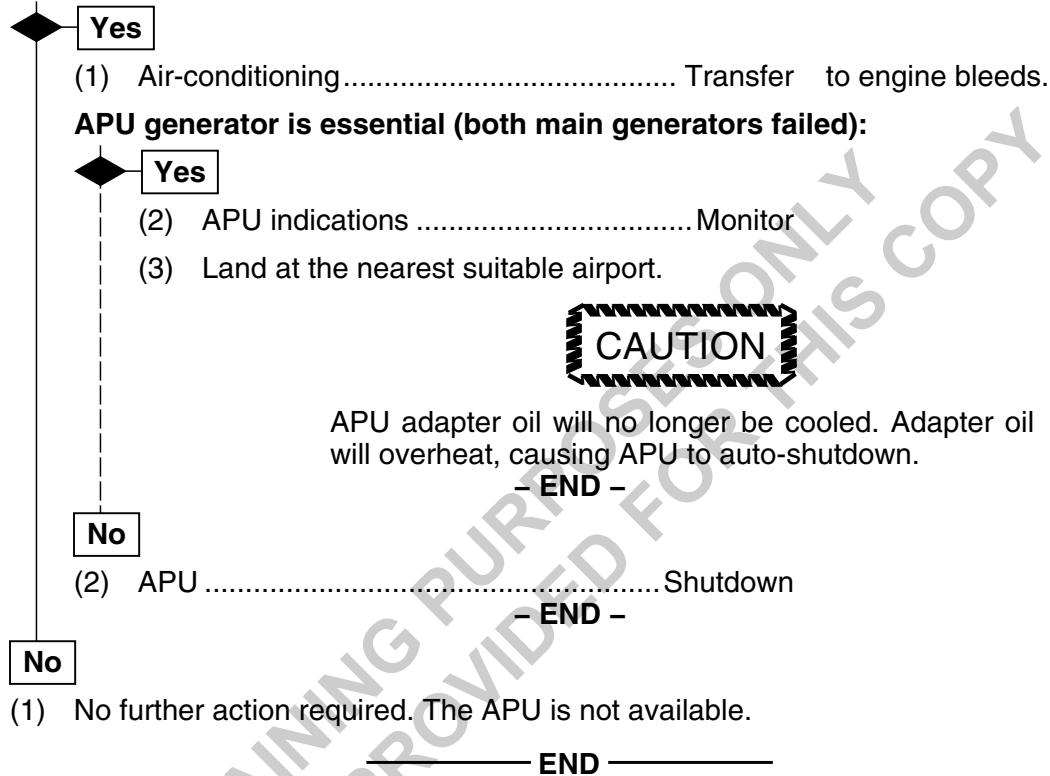
CHALLENGER 650	ABNORMAL PROCEDURES Auxiliary Power Unit	05-07-3
		Oct 27/15

1. AUXILIARY POWER UNIT (CONT'D)

E. APU Fuel Shut-Off Valve not in the Selected Position

Indication: APU SOV caution message on.

APU is operating:



F. APU Negative-G Shut-Off Valve Failed to Close After APU Shutdown Due to Fire

Indication: APU NEG-G SOV caution message on.

- (1) FUEL, L BOOST PUMP switch/light Press out to disarm affected boost pump.
- Check the following:
- L FUEL PUMP ON advisory message out, and
 - L FUEL PUMP INOP light on.
- (2) Left engine Shutdown Refer to POWER PLANT – In-Fight Engine Failure/Shutdown procedure in this chapter.

————— **END** —————

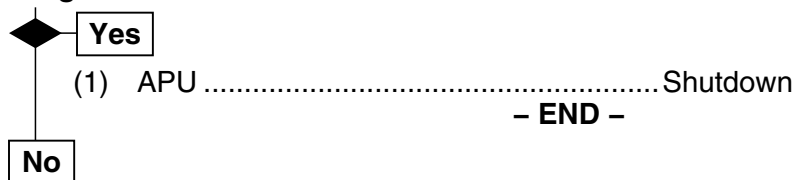
	CL-650 Airplane Flight Manual PSP 650-1	
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1. AUXILIARY POWER UNIT (CONT'D)

G. APU Pump Failure

Indication: APU PUMP caution message on and PUMP FAIL light on.

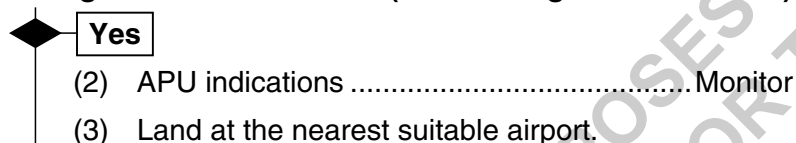
On ground:



In flight:

(1) Air-conditioning..... Transfer to engine bleeds.

APU generator is essential (both main generators failed):



APU adapter oil will no longer be cooled. Adapter oil will overheat, causing APU to auto-shutdown.

- END -



CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-1
		Oct 27/15

1. DOORS

A. Passenger Door – Outer Handle not Stowed

Indication: PAX DOOR OUT HNDL caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

◆ **Yes**

(3) Passenger door Check visually, and ensure that four roll-over catch green witness marks are correctly aligned.

All 4 witness marks are aligned:

◆ **Yes**

(4) Passenger door, inner T-handle..... Pull to stow outer handle. Green LOCKED flag-indicator visible.

When the handle is locked:

(5) Passenger door, inner T-handle..... Push in to stow inner T-handle, and then close inner T-handle access cover.

Check PAX DOOR STOW caution message out.

PAX DOOR OUT HNDL caution message persists:

◆ **Yes**

(6) Land at the nearest suitable airport.
– END –

No

(6) No further action required.
– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.



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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-2
		Oct 27/15

1. DOORS (CONT'D)

A. Passenger Door – Outer Handle not Stowed (Cont'd)

- (4) CABIN PRESSURIZATION, MAN
ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

- (5) CABIN PRESSURIZATION, MAN
RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

- (6) CABIN PRESSURIZATION, MAN
ALT selector Release Ensure selector returns to mid position/centre.
- (7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
- (8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

- (9) CABIN PRESSURIZATION, MAN
ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.
- (10) CABIN PRESSURIZATION, MAN
RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

- (11) CABIN PRESSURIZATION, MAN
ALT selector Release Ensure selector returns to mid position/centre.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-3
		Oct 27/15

1. DOORS (CONT'D)

B. Passenger Door – Inner Handle not Stowed

Indication: PAX DOOR STOW caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) Passenger door, inner handle Stow Stow inner handle by pushing it down into the latched position.

PAX DOOR STOW caution message persists:

(4) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.



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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-4
		Oct 27/15

1. DOORS (CONT'D)

B. Passenger Door – Inner Handle not Stowed (Cont'd)

At 10,000 feet or lowest safe altitude:

- (9) CABIN PRESSURIZATION, MAN
ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.
- (10) CABIN PRESSURIZATION, MAN
RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

- (11) CABIN PRESSURIZATION, MAN
ALT selector Release Ensure selector returns to mid position/centre.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-5
		Oct 27/15

1. DOORS (CONT'D)

C. Passenger Door – Not Latched

Indication: PAX DOOR LATCH caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(9) CABIN PRESSURIZATION, MAN ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.

(10) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.



	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-6
		Oct 27/15

1. DOORS (CONT'D)

C. Passenger Door – Not Latched (Cont'd)

When reaching target cabin altitude:

(11) CABIN PRESSURIZATION, MAN

ALT selectorRelease Ensure selector returns to mid position/centre.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-7
		Oct 27/15

1. DOORS (CONT'D)

D. Baggage Door – Not Closed/Locked

Indication: BAGGAGE DOOR caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(9) CABIN PRESSURIZATION, MAN ALT switch Select DN and hold to decrease cabin altitude to landing field elevation.

(10) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.



	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Doors	05-08-8
		Oct 27/15

1. DOORS (CONT'D)

D. Baggage Door – Not Closed/Locked (Cont'd)

When reaching target cabin altitude:

(11) CABIN PRESSURIZATION, MAN

ALT selectorRelease Ensure selector returns to mid position/centre.

————— **END** —————

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	CL-650 Airplane Flight Manual PSP 650-1	
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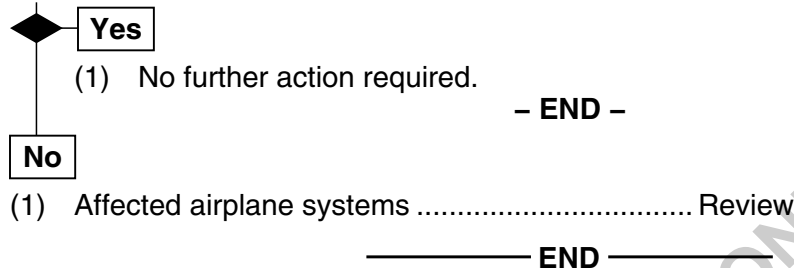
CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-1
		Oct 27/15

1. ELECTRICAL

A. AC Bus 1 Transfer Failure

Indication: AC 1 AUTOXFER caution message and associated AUTO XFER FAIL light on.

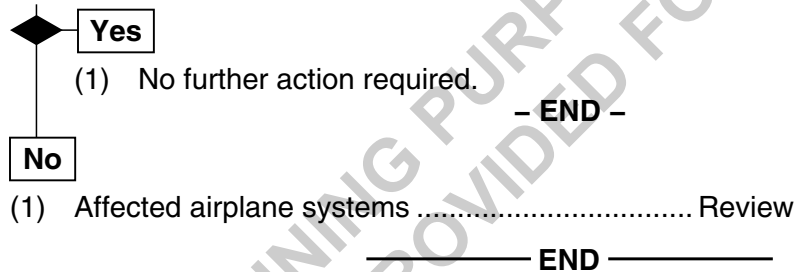
AC Bus 1 is powered:



B. AC Bus 2 Transfer Failure

Indication: AC 2 AUTO XFER caution message and associated AUTO XFER FAIL light on.

AC Bus 2 is powered:

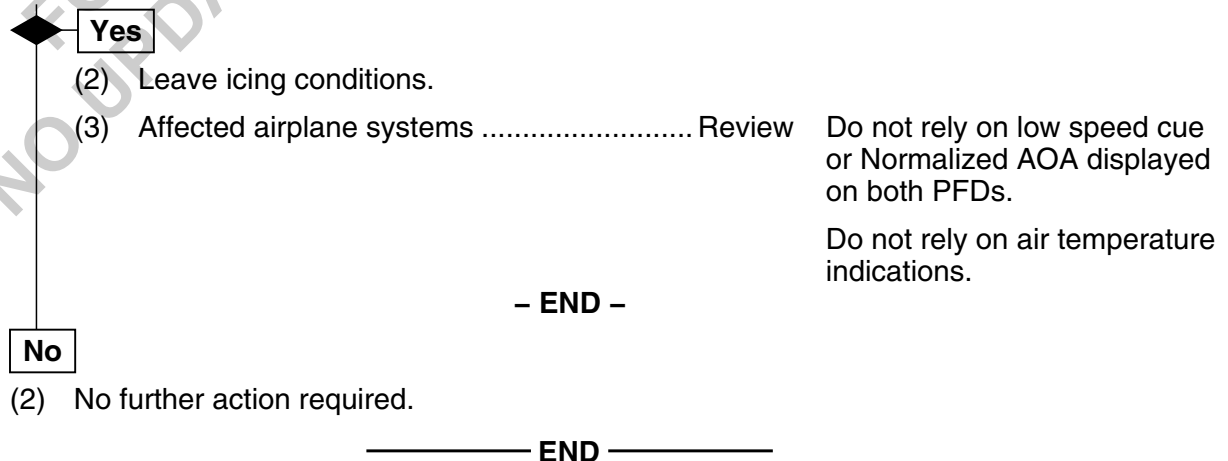


C. AC Bus 1 Failure

Indication: AC BUS 1 caution message on.

(1) AC POWER, GEN 1 switch Select to OFF/RESET then ON.

AC BUS 1 caution message persists:



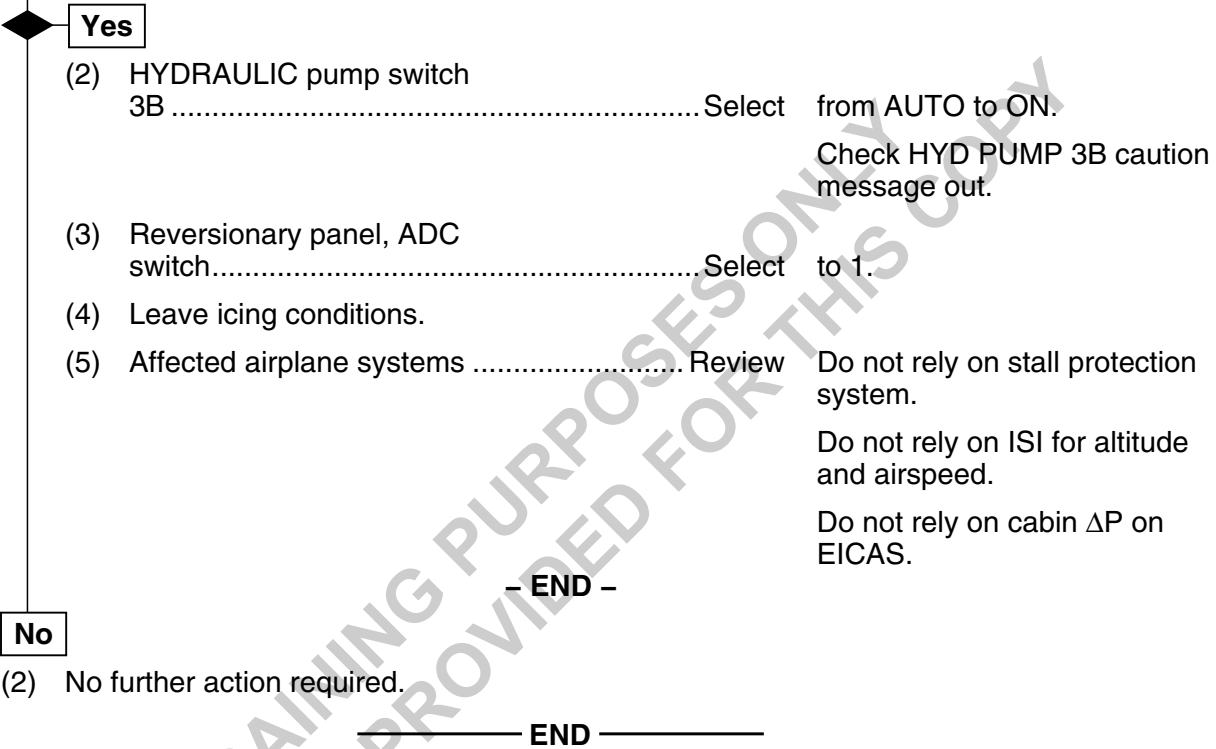
1. ELECTRICAL (CONT'D)

D. AC Bus 2 Failure

Indication: AC BUS 2 caution message on.

(1) AC POWER, GEN 2 switch Select to OFF/RESET then ON.

AC BUS 2 caution message persists:



CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-3
		Oct 27/15

1. ELECTRICAL (CONT'D)

E. AC Essential Bus Failure

Indication: AC ESS BUS caution message on.

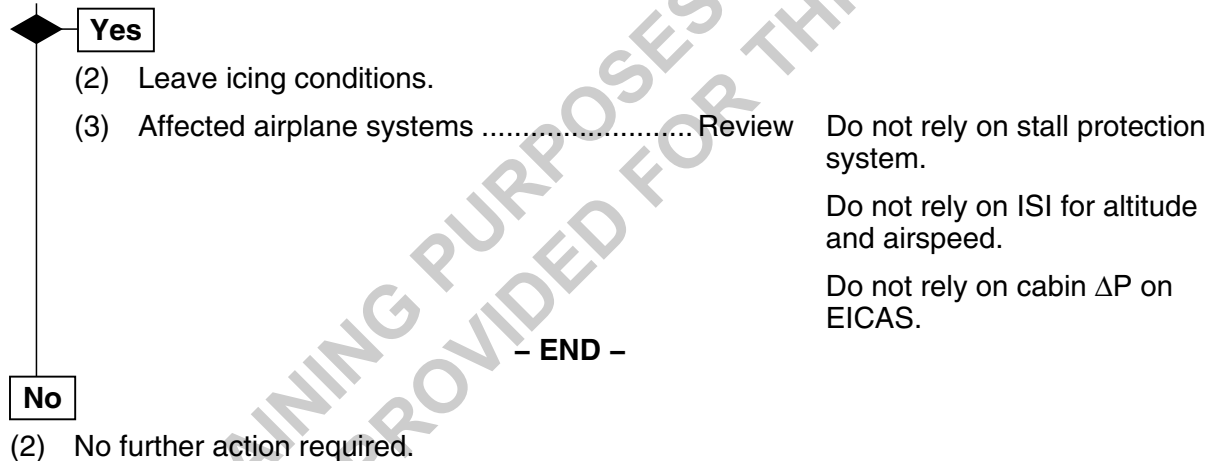
Manually transfer AC essential to alternate source:

- (1) AC POWER, AC ESS XFER switch/light Press in to manually transfer the essential bus to AC BUS 2.

Check the following:

- ALTN light on, and
- AC ESS ALTN status message on.

AC ESS BUS caution message persists:



F. ADG Inadvertent Deployment

Indication: ADG operating.

NOTE

The AC essential bus and hydraulic pump 3B can be transferred back to either main generator, or to the APU generator, if operating, by pressing the PWR TXFR OVERRIDE switch on the ADG control panel after the ADG manual deploy handle is re-stowed.

To re-establish normal operations:

- (1) ADG manual deploy handle Select to STOW (if ADG was manually deployed).
- (2) ADG PWR TXFR OVERRIDE switch Push in to override.
- (3) HYDRAULIC pump switch 3B Check selected to AUTO.

END

	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-4
		Oct 27/15

1. ELECTRICAL (CONT'D)

G. APU Battery Failure

Indication: APU BATTERY OFF caution message on. DC ELECTRICAL page – APU battery indicates less than 18 volts DC.

- (1) BATT MASTER switchCheck ON.
- (2) DC ELECTRICAL pageMonitor APU battery voltage. Assume loss of APU battery. The APU is not available.
- (3) Affected airplane systems Review

————— **END** —————

H. APU Generator Failure

Indication: APU GEN OFF caution message on.

- (1) AC POWER, APU GEN switch.....Select to OFF/RESET then ON.

APU GEN OFF caution message persists:

- ◆ **Yes**
- (2) AC POWER, APU GEN switch.....Select to OFF.
 - (3) Affected airplane systems Review
- **END** —————

No

- (2) AC ELECTRICAL pageMonitor loads on APU generator.

————— **END** —————

I. APU Generator Overload

Indication: APU GEN OVLD caution message on. AC ELECTRICAL page – APU generator load readout is greater than 35 KVA.

- (1) AC ELECTRICAL pageMonitor load on APU generator.
- (2) AC loads..... Reduce as necessary.

————— **END** —————

J. Battery Bus Failure

Indication: BATTERY BUS caution message on.

- (1) Land at the nearest suitable airport.
- (2) Affected airplane systems Review

————— **END** —————

	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-5
		Oct 27/15

1. ELECTRICAL (CONT'D)

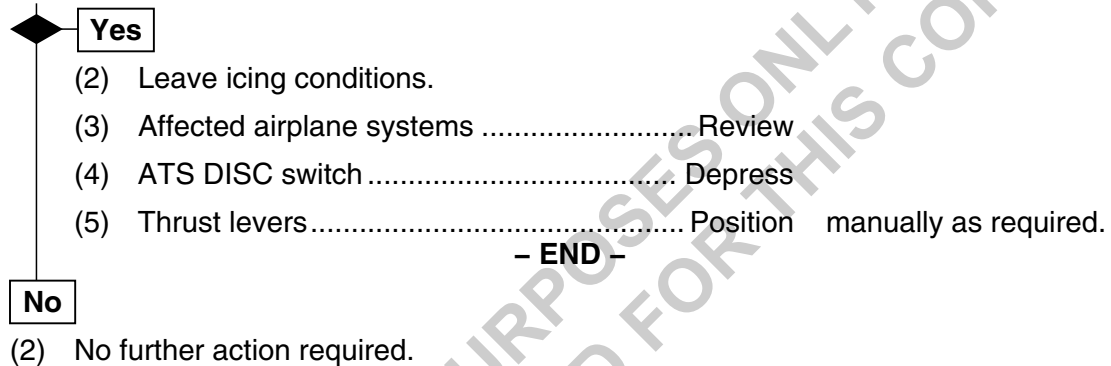
K. DC Bus 1 Failure

Indication: DC BUS 1 caution message on. DC ELECTRICAL page – DC BUS 1 inoperative.

- (1) DC POWER, MAIN BUS TIE switch/light Press in to close bus tie manually.

Check MAIN BUS TIE CLOSED light on.

DC BUS 1 caution message persists:



END

1. ELECTRICAL (CONT'D)

L. DC Bus 2 Failure

Indication: DC BUS 2 caution message on. DC ELECTRICAL page – DC BUS 2 inoperative.

- (1) DC POWER, MAIN BUS TIE switch/light Press in to close bus tie manually.

Check MAIN BUS TIE
CLOSED light on.

DC BUS 2 caution message persists:

Yes

- (2) Affected airplane systems Review

NOTE

MACH TRIM is inoperative.

- (3) Leave icing conditions as soon as possible.

- (4) Approach speed V_{REF} (Flaps 45°)
+ 10 KIAS minimum.

- (5) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

- (6) STAB TRIM channel Engage With at least one STAB TRIM channel engaged, manually trim airplane as required.

NOTE

With MACH TRIM inoperative, do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

– END –

No

- (2) No further action required.

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-7
		Oct 27/15

1. ELECTRICAL (CONT'D)

M. DC Essential Bus Failure

Indication: DC ESS BUS caution message on. DC ELECTRICAL page – DC ESS BUS inoperative.

- (1) Leave icing conditions.
- (2) STAB TRIM channel..... Engage With at least one STAB TRIM channel engaged, manually trim airplane as required.

NOTE

MACH TRIM is inoperative. Do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

- (3) L DISPLAYS switch..... Select to MFD REV.
- (4) AFCS SEL switch..... Select AFCS 2.
- (5) YAW DAMPER, YD 2 switch..... Press in to engage YD 2.

NOTE

YD 1 will be inoperative. If YD 2 does not re-engage, follow procedure for double yaw damper failure.

- (6) Affected airplane systems..... Review
- (7) Land at the nearest suitable airport.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-8
		Oct 27/15

1. ELECTRICAL (CONT'D)

N. AC Generator 1 Failure

Indication: GEN 1 OFF caution message on.

(1) AC POWER, GEN 1 switch Select to OFF/RESET then ON.

GEN 1 OFF caution message persists:

Yes

(2) AC POWER, GEN 1 switch Select to OFF.

NOTE

1. With generator 1 shutdown, HYD PUMP 2B is load shed.
2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0°.

(3) APU (if available) Start when at or below 20,000 feet.

(4) AC POWER, APU GEN switch Select to ON.

- END -

No

(2) AC ELECTRICAL page Monitor voltage, load and frequency readouts on all generators.

END

CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05-09-9 Rev 9, Jun 16/17
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1. ELECTRICAL (CONT'D)

O. AC Generator 2 Failure

Indication: GEN 2 OFF caution message on.

(1) AC POWER, GEN 2 switch Select to OFF/RESET then ON.

GEN 2 OFF caution message persists:

Yes

(2) AC POWER, GEN 2 switch Select to OFF.

NOTE

1. With generator 2 shutdown, HYD PUMP 1B is load shed.
2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0°.

(3) APU (if available) Start when at or below 20,000 feet.

(4) AC POWER, APU GEN switch Select to ON.

— END —

No

(2) AC ELECTRICAL page Monitor voltage, load and frequency readouts on all generators.

— END —

P. AC Generator Overload

Indication: GEN 1 (2) OVLD caution message on. AC ELECTRICAL page – generator load readout indicates greater than 35 KVA.

(1) AC ELECTRICAL page Monitor generator loads, and load shed as necessary.

(2) Affected airplane systems Review

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Electrical	05–09–10
		Oct 27/15

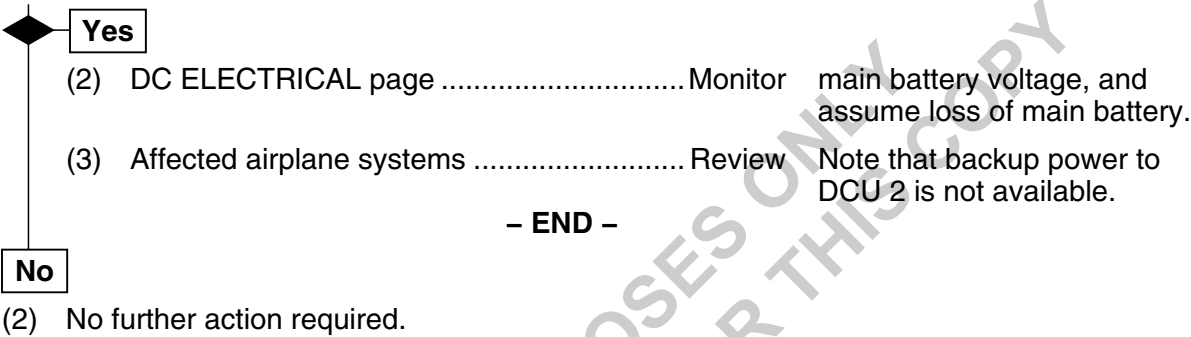
1. ELECTRICAL (CONT'D)

Q. Main Battery Failure

Indication: MAIN BATT OFF caution message on. DC ELECTRICAL page – Main battery indicates less than 18 volts DC, and main battery direct bus outline is amber.

(1) BATT MASTER switchCheck ON.

MAIN BATT OFF caution message persists:



R. DC Emergency Bus Failure

Indication: DC EMER BUS caution message on. DC ELECTRICAL page – DC emergency bus outline is amber.

- (1) Affected airplane systemsReview Engine and APU fire extinguishing systems are inoperative:
- Firex bottles are inoperative,
 - APU fuel feed shut-off valves will not close, and
 - Engine hydraulic shut-off valves will not close.
- (2) Land at the nearest suitable airport.

END

CHALLENGER 650	ABNORMAL PROCEDURES Fire Protection	05-10-1
		Oct 27/15

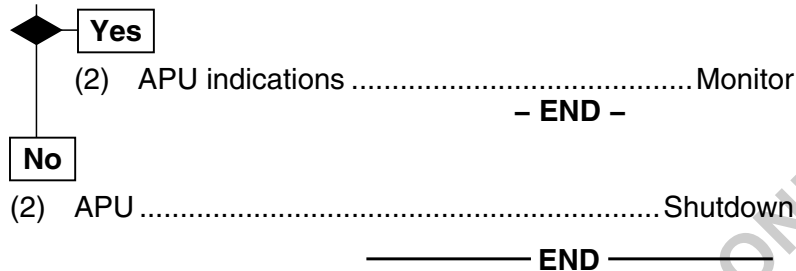
1. FIRE PROTECTION

A. APU Fire Detector Loop Failure

Indication: APU FIRE FAIL caution message on.

(1) Air-conditioning Transfer to engine bleeds.

APU generator is essential (both main generators failed):



B. Engine Fire Detector Loop Failure

Indication: L (R) FIRE FAIL caution message on.

(1) Affected engine Monitor

————— **END** —————

C. Jetpipe Overheat Detector Failure

Indication: L (R) JETPIPE OVHT FAIL caution message on.

(1) Affected engine Monitor

————— **END** —————

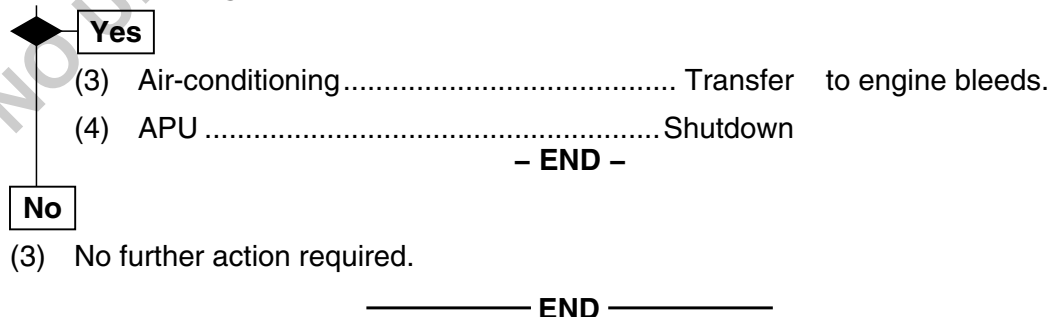
D. APU Firex Bottle Discharged

Indication: APU BOTTLE LO caution message on (without previous APU fire).

(1) FIRE DETECTION, TEST switch Select to WARN.

(2) Glareshield, APU BOTTLE switch/light Check on (confirming that bottle is charged).

APU BOTTLE light is out:



CHALLENGER 650	ABNORMAL PROCEDURES Fire Protection	05-10-2
		Oct 27/15

1. FIRE PROTECTION (CONT'D)

E. Engine Firex Bottle Discharged

Indication: ENG BOTTLE 1 (2) LO caution message on (without engine fire condition).

- (1) FIRE DETECTION, TEST switchSelect to WARN.
- (2) Glareshield, Engine BOTTLE 1
and 2 switch/lightsCheck on, confirming that engine
bottles are charged.
- (3) No further action required.

If any one engine bottle light fails to come on, assume that the affected bottle is discharged.

————— END —————

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CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05-11-1
		Oct 27/15

1. FLIGHT CONTROLS

A. Lightning Strike

Following a lightning strike, failure of the stall protection system may occur without an associated indication to the crew. Failure of the stall protection system must be assumed.

- (1) STALL PROT, PUSHER switch
(either left or right) Select to OFF.

NOTE

Stick shakers remain operative.

Prior to landing:

- (2) Approach speed V_{REF} (Flaps 45°)
+ 10 KIAS minimum, to maintain margin above stall.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

END

1. FLIGHT CONTROLS (CONT'D)

B. Stall Protection System Failure

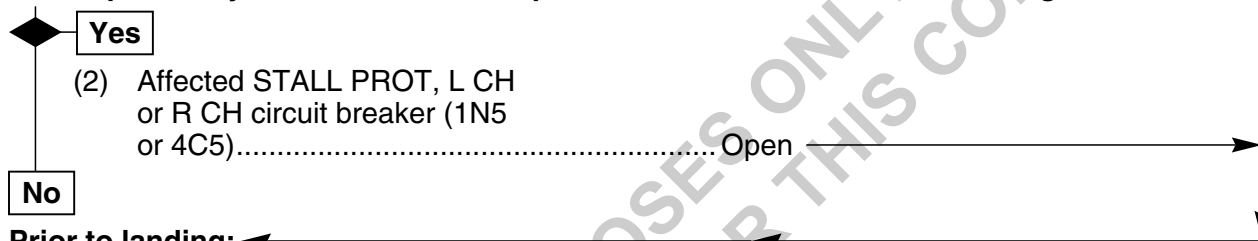
Indication: STALL FAIL caution message on.

- (1) STALL PROT, PUSHER switch
(either left or right) Select to OFF.

NOTE

Stick shakers remain operative.

Accompanied by a nuisance shaker operation and/or warbler aural warning:

**Prior to landing:**

- (2) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum, to maintain margin above stall.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

END

C. Stall Protection System Altitude Compensation Failure

Indication: ALT COMP FAIL caution message on.

NOTE

The stick pusher may be inoperative. One or both channels of the stick shaker will remain operative.

Prior to landing:

- (1) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

END

CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05-11-3
		Oct 27/15

1. FLIGHT CONTROLS (CONT'D)

D. Flight Spoilers Deployed During Flight

Indication: FLT SPLRS DEPLOY caution message on.

- (1) FLIGHT SPOILER lever Select to RETRACT.

————— **END** —————

E. Flight Spoilers Lever Jam

Indication: Flight spoilers cannot be retracted.

- (1) Approach speed V_{REF} (Flaps 45°)
+ 10 KIAS minimum.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

————— **END** —————

F. Flight Spoilers Failure to Deploy

Indication: Flight spoilers cannot be deployed.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

————— **END** —————

G. Flight Spoiler Asymmetric Extension

Indication: An uncommanded roll during flight spoiler extension.

- (1) FLIGHT SPOILER lever Retract Do not reselect in flight.
- (2) Runway Select to minimize crosswind.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

————— **END** —————

	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05-11-4
		Oct 27/15

1. FLIGHT CONTROLS (CONT'D)

H. Flight Spoiler Asymmetric Retraction

Indication: An uncommanded roll during flight spoiler retraction.

- (1) Lateral control..... Establish using ailerons, and if necessary, rudder inputs.
- (2) FLIGHT SPOILER leverRetract/adjust to eliminate roll asymmetry.
Make no further adjustments to spoilers in flight.
- (3) Runway Select to minimize crosswind.
- (4) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum.
- (5) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

————— **END** —————

I. Flight Spoilers Failure

Indication: FLT SPLRS caution message on.

On ground:

- (1) Do not take off.

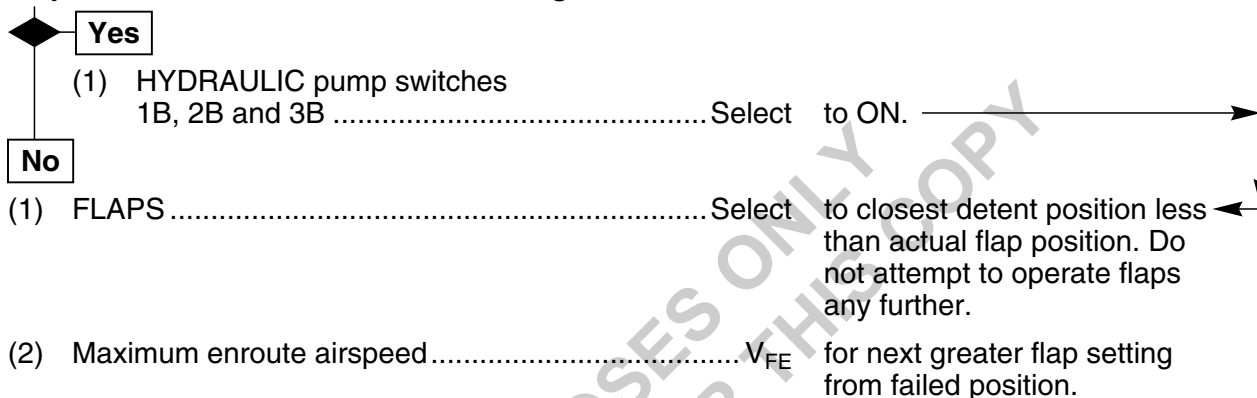
————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure

Indication: FLAPS FAIL caution message on. FLIGHT CONTROLS page displays flap outline in amber.

Flaps failure occurred at less than 8 degrees:



NOTE

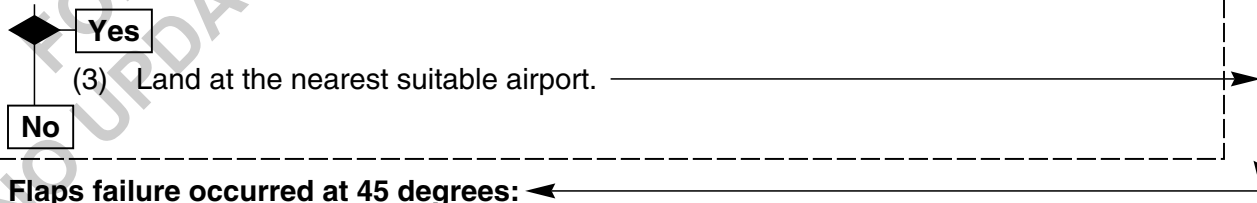
If flaps are confirmed retracted (0 degrees), reduction of cruise airspeed is not required.

CAUTION

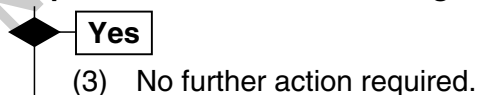
If possible, avoid prolonged flight in icing. If icing is encountered for an extended period of time, aerodynamic buffet may be experienced.

AR Certified Airplanes

Flaps failure occurred at greater than 5 degrees:



Flaps failure occurred at 45 degrees:



NOTE

If aerodynamic buffet is encountered after flying in icing conditions, reduce buffet by increasing the minimum approach speed to V_{REF} (Flaps 45°) + 5 KIAS.

– END –

1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure (Cont'd)

No

Prior to landing:

Flaps failure occurred at less than 20 degrees:

Yes

(3) Fuel balance Check

No

(3) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.

(4) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.

(5) Final approach speed V_{REF} (Flaps 45°) + ΔV_{REF} for next smaller flap setting from failed position.

(6) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(7) Thrust levers Position manually as required.

(8) Actual landing distance Increase as applicable by factor given below:

Flaps Setting (Degrees)	ΔV_{REF} (KTS)	ALD Factor (Without Thrust Reversers)	ALD Factor (With Thrust Reversers)
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)

NOTE

Accomplish a normal descent typically with a vertical speed of not more than 400 fpm at touchdown.



CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05-11-7 Rev 6, Nov 28/16
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1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure (Cont'd)



Improper landing technique during a flaps failure can generate nose gear loads sufficient to cause structural damage.

To prevent damage:

- After main gear touchdown, gently lower the nose to the runway.
- Apply brakes only after nose wheel touchdown.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

Example: If failure occurs at Flaps – 28 degrees:

Maximum Enroute Airspeed	= V_{FE} for Flaps 30 degrees
Approach & Landing Speed	= V_{REF} (Flaps 45°) + 14 KIAS
Factor Applied to Actual Landing Distance (without Thrust Reversers)	= 1.55 (55%)
Factor Applied to Actual Landing Distance (with Thrust Reversers)	= 1.50 (50%)

NOTE

If the failure occurs with the flaps at a detented flap position, use the V_{FE} , V_{REF} and landing distance factor relative to the flap position.

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

K. Ground Spoilers Failure

Indication: GND SPLRS NOT ARMED caution message on.

(1) GND SPOILERS switch Select to ARM.

GND SPLRS NOT ARMED caution message persists:

Yes

(2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

After touchdown:

(3) FLIGHT SPOILER lever Select to MAX.

— END —

No

(2) No further action required.

— END —

L. Ground Spoilers Unsafe

Indication: GND SPLRS caution message on.

On ground:

Yes

NOTE

GND SPLRS caution message may come on at taxi speeds above 16 knots.

(1) Airplane Stop

(2) Ground spoilers pre-flight check Accomplish

Refer to Chapter 4; NORMAL PROCEDURES – FLIGHT CONTROLS. Ground spoilers are unserviceable if message persists.

— END —

No

During flight:

(1) GND SPOILERS switch Select to DISARM.



1. FLIGHT CONTROLS (CONT'D)

L. Ground Spoilers Unsafe (Cont'd)

GND SPLRS caution message persists:

Yes

Prior to landing:

- (2) TAWS WARNING, FLAPS
OFF switch/light..... Press in to mute the flap aural warning.
- (3) TAWS WARNING,
RAAS OFF switch/light
(if installed) Press in Check OFF light on.
- (4) FLAPS Select to 20° for landing.
- (5) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (6) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard
thrust levers at 50 feet AGL.

- (7) Actual landing distance..... Increase as applicable by factor given
below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE

Landing distance factors for failed ground spoilers are
based upon serviceable flight spoilers.

CAUTION

With thrust reversers deployed, a nose-up pitching
tendency will occur at high power settings, particularly at
aft centre of gravity light weights. This tendency is
controllable with elevator and may be minimized by
ensuring that nose wheel touchdown is achieved and nose
down elevator applied, before selecting reverse thrust.

No

After touchdown:

- (2) GND SPOILERS switch Select to ARM.
- (3) FLIGHT SPOILER lever Select to MAX.

END

1. FLIGHT CONTROLS (CONT'D)

M. Ground Spoilers Deployed During Flight

Indication: GND SPLRS DEPLOY caution message on.

- (1) GND SPOILERS switch Select to DISARM.

Prior to landing:

- (2) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (3) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (4) FLAPS Select to 20° for landing.
- (5) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (6) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (7) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

After touchdown:

- (8) GND SPOILERS switch Select to ARM, to deploy ground spoilers.
- (9) FLIGHT SPOILER lever Select to MAX.

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

N. Stabilizer Trim Failure

Indication: STAB TRIM caution message on and both stabilizer trim are disengaged.
MACH TRIM INOP light on and MACH TRIM caution message on.

- (1) Pilot's or copilot's STAB TRIM DISC switch Press hold and release.
- (2) STAB TRIM, CH1 and CH2 switch/lights Press in to engage both channels.
Check STAB CH1 and CH2 INOP status messages out.

STAB TRIM caution message persists:

- Yes**
- (3) STAB TRIM, CH2 switch/light Press in to engage.
Check STAB CH1 INOP status message on.

STAB TRIM caution message still persists:

Yes

NOTE

1. Minimize changes to airspeed and configuration to minimize control forces or autopilot servo loads.
2. Disregard MACH TRIM caution message.

No

- (3) MACH TRIM switch/light Press in to engage. Check the following:
- MACH TRIM INOP light out.
 - MACH TRIM caution message out.

– END –

1. FLIGHT CONTROLS (CONT'D)

N. Stabilizer Trim Failure (Cont'd)

Autopilot is engaged:

Yes

(4) Autopilot..... Consider leaving engaged as long as needed to reduce workload.

AP HOLDING NOSE DOWN (NOSE UP) caution message is displayed:

Yes

(5) Airspeed..... Adjust as required to keep autopilot engaged as long as possible.

NOTE

1. If AP HOLDING NOSE DOWN is displayed, decrease airspeed. If AP HOLDING NOSE UP is displayed, increase airspeed.
2. AP HOLDING NOSE DOWN (NOSE UP) caution message will go out if the airspeed is adjusted.

AP HOLDING NOSE DOWN (NOSE UP) caution message persists:

Yes

NOTE

Anticipate an out-of-trim condition with autopilot disengagement.

(6) Autopilot..... Disconnect

No

Prior to landing:

Failure occurred with flaps at 45°:

Yes

(7) FLAPS Leave at 45°.

CAUTION

Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

- END -

1. FLIGHT CONTROLS (CONT'D)

N. Stabilizer Trim Failure (Cont'd)

No

- (7) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (8) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (9) FLAPS Select to 20° for landing.
- (10) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (11) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (12) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)



Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05–11–14
		Oct 27/15

1. FLIGHT CONTROLS (CONT'D)

O. Stabilizer Trim Malfunction

Indication: Stabilizer cannot be trimmed to the desired position. STAB CH 1 INOP status message is not displayed.

- (1) Pilot's or copilot's STAB TRIM

DISC switch Press hold and release.

Check the following:

- STAB TRIM caution message on.
- MACH TRIM caution message on.

- (2) STAB TRIM, CH2 switch/light Press in to engage.

Check STAB CH1 INOP status message on.

NOTE

Do not engage STAB TRIM CH 1.

- (3) MACH TRIM switch/light Press in to engage.

Check the following:

- MACH TRIM INOP light out, and
- MACH TRIM caution message out.

————— **END** —————

P. Mach Trim Failure

Indication: MACH TRIM caution message and MACH TRIM INOP light on.

With at least one STAB TRIM channel engaged:

- (1) MACH TRIM switch/light Press in to engage.

Check MACH TRIM INOP light out.

NOTE

With a Mach trim failure, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.

————— **END** —————

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1. FLIGHT CONTROLS (CONT'D)

Q. Aileron PCU Malfunction

Indication: AILERON PCU caution message on.

NOTE

It may be necessary to reduce airspeed to improve airplane handling.

(1) Autopilot Disconnect if coupled.

If necessary:

(2) Aileron controls..... Overpower

Difficult to maintain desired flight path:**Yes****At a safe altitude:**

(3) Aileron controls..... Release pilot/copilot differential pressure on the controls.

(4) ROLL DISC handle..... PULL and TURN 90 degrees to lock.

NOTE

1. The pilot with the operative aileron will be required to maintain a roll control input to offset the effects of the aileron PCU malfunction.
2. If the right aileron control circuit is inoperative, the autopilot system is inoperative.

No

(3) Land at the nearest suitable airport.

NOTE

Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:(4) TAWS WARNING, FLAPS OFF
switch/light..... Press in to mute the flap aural warning.(5) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.

(6) FLAPS Select to 20° for landing.

(7) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.

(8) ATS DISC switch..... Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. FLIGHT CONTROLS (CONT'D)

Q. Aileron PCU Malfunction (Cont'd)

- (9) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Flight Controls	05-11-17 Rev 5, Aug 30/16
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1. FLIGHT CONTROLS (CONT'D)

R. Elevator System Malfunction

Indication: ELEVATOR SPLIT caution message on.

NOTE

It may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.
- (2) Elevator controls..... Move slowly and monitor position on EICAS to identify malfunctioning elevator surface.

If necessary:

- (3) Elevator controls..... Overpower

Difficult to maintain desired flight path:

Yes

At a safe altitude:

- (4) Elevator controls.....Release pilot/copilot differential pressure on the controls.
- (5) PITCH DISC handlePULL and TURN 90 degrees to lock.
- (6) Jammed side Determine If left side, do not re-engage autopilot.

Right side jammed:

Yes

- (7) STALL PROT, PUSHER switch (either left or right) Select to OFF.

NOTE

Stick shaker on operative control column, operates at a lower intensity.

No

- (4) Land at the nearest suitable airport.

NOTE

Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (5) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (7) FLAPS Select to 20° for landing.



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1. FLIGHT CONTROLS (CONT'D)

R. Elevator System Malfunction (Cont'd)

- (8) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (9) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-1
		Oct 27/15

1. FUEL

A. Fuel Imbalance

Indication: FUEL IMBALANCE caution message on. Left and right fuel quantity readouts indicate amber.



If a FUEL IMBALANCE caution message is displayed:

- Do not taxi until imbalance is rectified.
- Do not land until imbalance is rectified, unless a greater emergency exists.

NOTE

1. The maximum permissible fuel imbalance is 182 kg (400 lb) for take-off, taxi, and landing.
2. The maximum permissible fuel imbalance is 363 kg (800 lb) in flight.

(1) Fuel distribution Monitor to identify appropriate action required.

Affected high tank:

(2) FUEL, L (R) TO AUX XFLOW switch/light Press in to open crossflow valve.
Check L (R) TO AUX XFLOW ON light on.

(3) Fuel tank quantity readouts Monitor

When tank quantities have equalized:

(4) FUEL, affected L (R) TO AUX XFLOW switch/light Press out to close crossflow valve.
Check L (R) TO AUX XFLOW ON light out.



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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-2
		Oct 27/15

1. FUEL (CONT'D)

A. Fuel Imbalance (Cont'd)

Fuel imbalance persists:

◆ **Yes**

- (5) FUEL, GRAVITY/XFLOW switch/light Press in to open gravity crossflow valve.
Check GRAVITY XFLOW OPEN light on.

NOTE

1. A steady heading sideslip will accelerate gravity crossflow operation. Fly one wing low slip (on the side with less fuel) to ensure proper operation and to accelerate gravity crossflow.
2. During uncoordinated flight, fuel indications are inaccurate. Accurate fuel indications will be obtained after 30 seconds when level and stabilized in coordinated flight.

On the low quantity side:

- (6) Affected engine Reduce thrust.
(7) Fuel tank quantity readouts Monitor

When main tank quantities have equalized:

- (8) FUEL, GRAVITY/XFLOW switch/light Press out to close gravity crossflow valve.
Check GRAVITY XFLOW OPEN light out.
(9) Fuel tank quantity readouts Monitor

Imbalance persists (fuel leak from wing tank is suspected):

◆ **Yes**

- (10) Fuel Leak Procedure Accomplish Refer to Fuel Leak Procedure in this section.

— END —

No

- (5) No further action required.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-3
		Oct 27/15

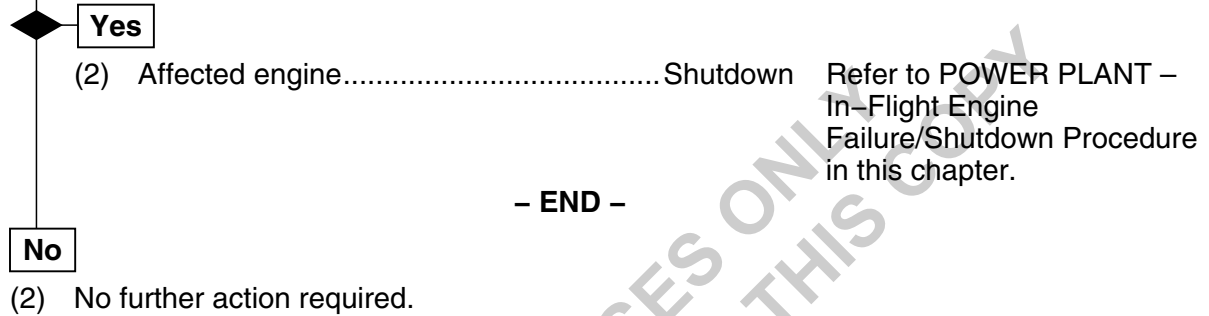
1. FUEL (CONT'D)

B. Engine Shut-Off Valve Failed Closed

Indication: L (R) ENG SOV caution message on.

- (1) Engine instruments..... Confirm engine failure using N₁, fuel flow, ITT and N₂.

Engine failure is confirmed:



C. Engine Shut-Off Valve Failed to Close After Engine Shutdown Due to Fire

Indication: L (R) ENG SOV caution message on.

- (1) Affected FUEL, BOOST PUMP switch/light..... Confirm pressed out.

Check the following:

- L (R) BOOST PUMP ON light out, and L (R) FUEL PUMP ON advisory message out,
- L (R) FUEL PUMP caution message on, and
- L (R) FUEL PUMP INOP light on.

- (2) Land at the nearest suitable airport.

END

D. Fuel Filter Impending Bypass

Indication: L (R) FUEL FILTER caution message on.

- (1) Affected engine Monitor engine indication and performance.

END

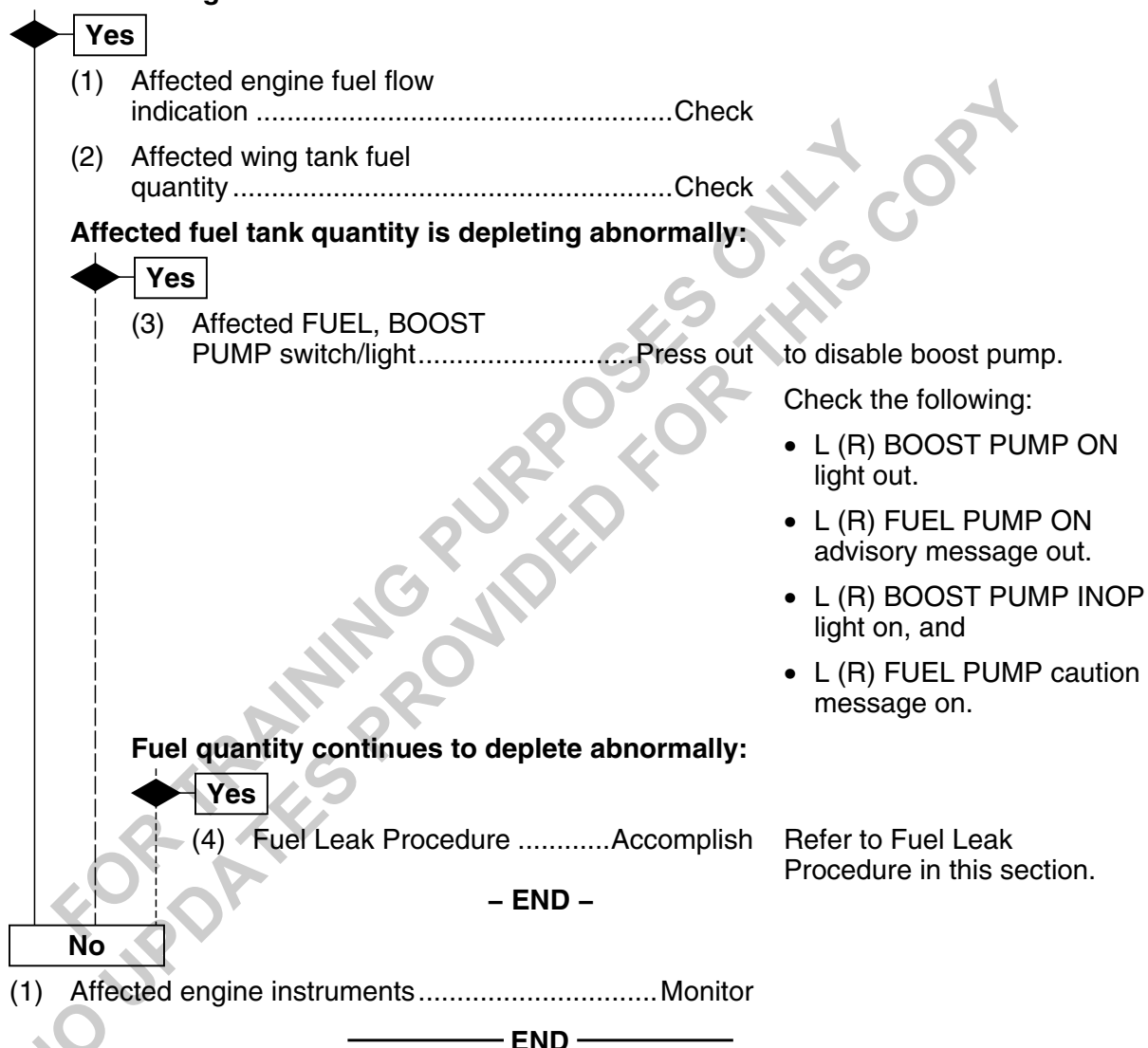
CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-4
		Oct 27/15

1. FUEL (CONT'D)

E. Low Fuel Pressure

Indication: L (R) FUEL LO PRESS caution message on.

Associated L (R) MAIN EJECTOR FAIL status message and/or L (R) SCAV EJECTOR caution message on:



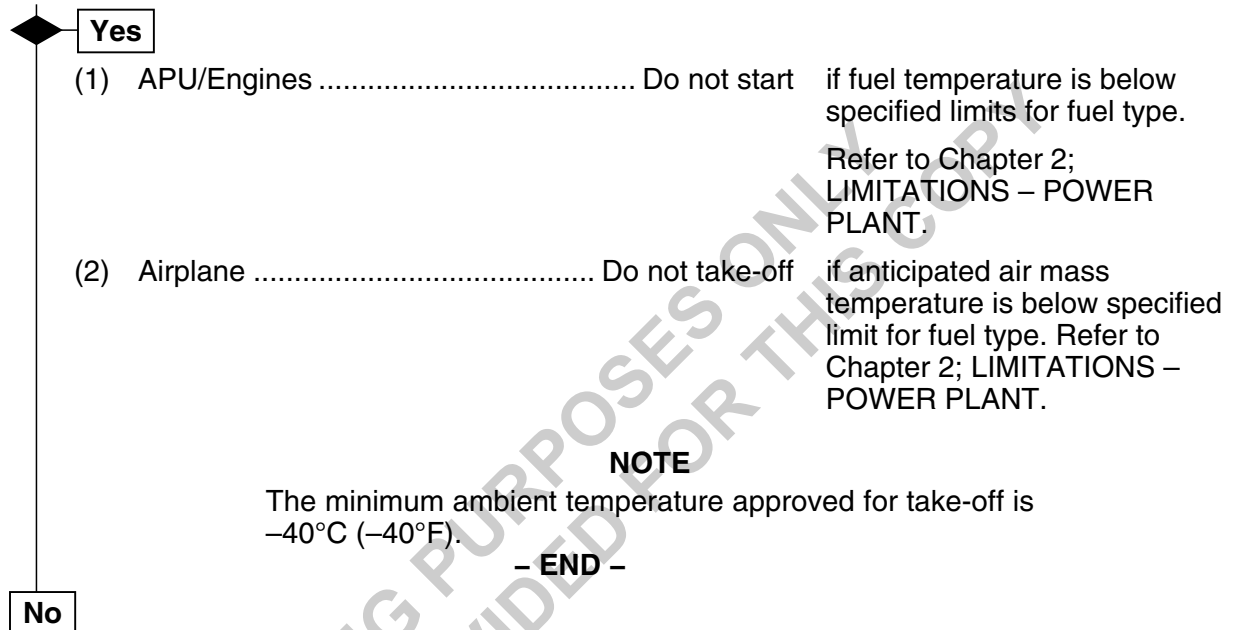
CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-5
		Oct 27/15

1. FUEL (CONT'D)

F. Bulk Fuel Temperature is Less than -37°C

Indication: BULK FUEL TEMP caution message on, and amber bulk fuel temperature readout on the SUMMARY page.

On ground:



During flight:

- (1) Airplane Operate as follows:
- Descend or deviate to warmer air mass, or
 - Increase airspeed.

— END —

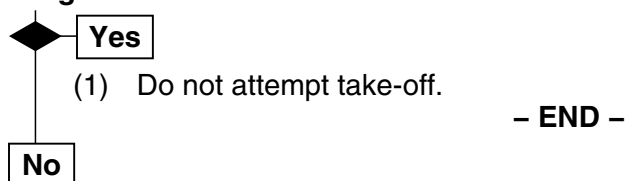
CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-6
		Oct 27/15

1. FUEL (CONT'D)

G. Fuel Temperature Less than 5°C

Indication: L (R) FUEL LO TEMP caution message on. Fuel temperature readout indicates caution range (amber).

On ground:



In flight:

(1) Affected engine instruments Monitor

NOTE

Under these conditions, the fuel feed to the engine is not preheated. L (R) FUEL FILTER caution message may come on, indicating an impending bypass.

— END —

H. Fuel Boost Pump Failure

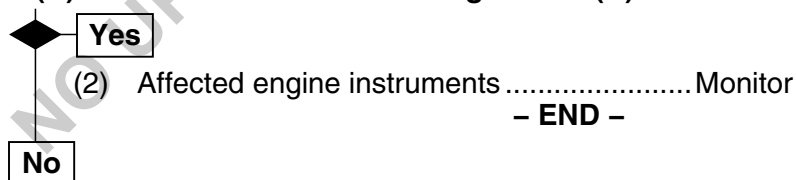
Indication: L (R) FUEL PUMP caution message and L (R) BOOST PUMP INOP light on.

(1) FUEL, L (R) BOOST PUMP switch/light Press out then press in.

Check the following:

- L (R) BOOST PUMP INOP light out.
- L (R) FUEL PUMP caution message out.

L (R) FUEL PUMP caution message and L (R) BOOST PUMP INOP light persists:



— END —

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-7
		Oct 27/15

1. FUEL (CONT'D)

I. Motive Flow Failure

Indication: L (R) SCAV EJECTOR caution message accompanied by L (R) MAIN EJECTOR FAIL status message.

(1) FUEL, L and R BOOST PUMP

switch/lightsCheck pressed in.

Check the following:

- L and R BOOST PUMP ON lights on,
- L and R FUEL PUMP ON advisory messages on, and
- L and R FUEL PUMP caution messages out.

Engine thrust is inadequate:

◆ **Yes**

(2) ATS DISC switch Depress

(3) Unaffected engineAdjust thrust to maintain flight path. →

□ **No**

(2) Fuel quantity indicatorsMonitor for abnormal fuel depletion. ←

Total fuel quantity is depleting abnormally:

◆ **Yes**

(3) Fuel Leak Procedure Accomplish Refer to Fuel Leak Procedure in this section.

— END —

□ **No**

(3) No further action required.

— END —

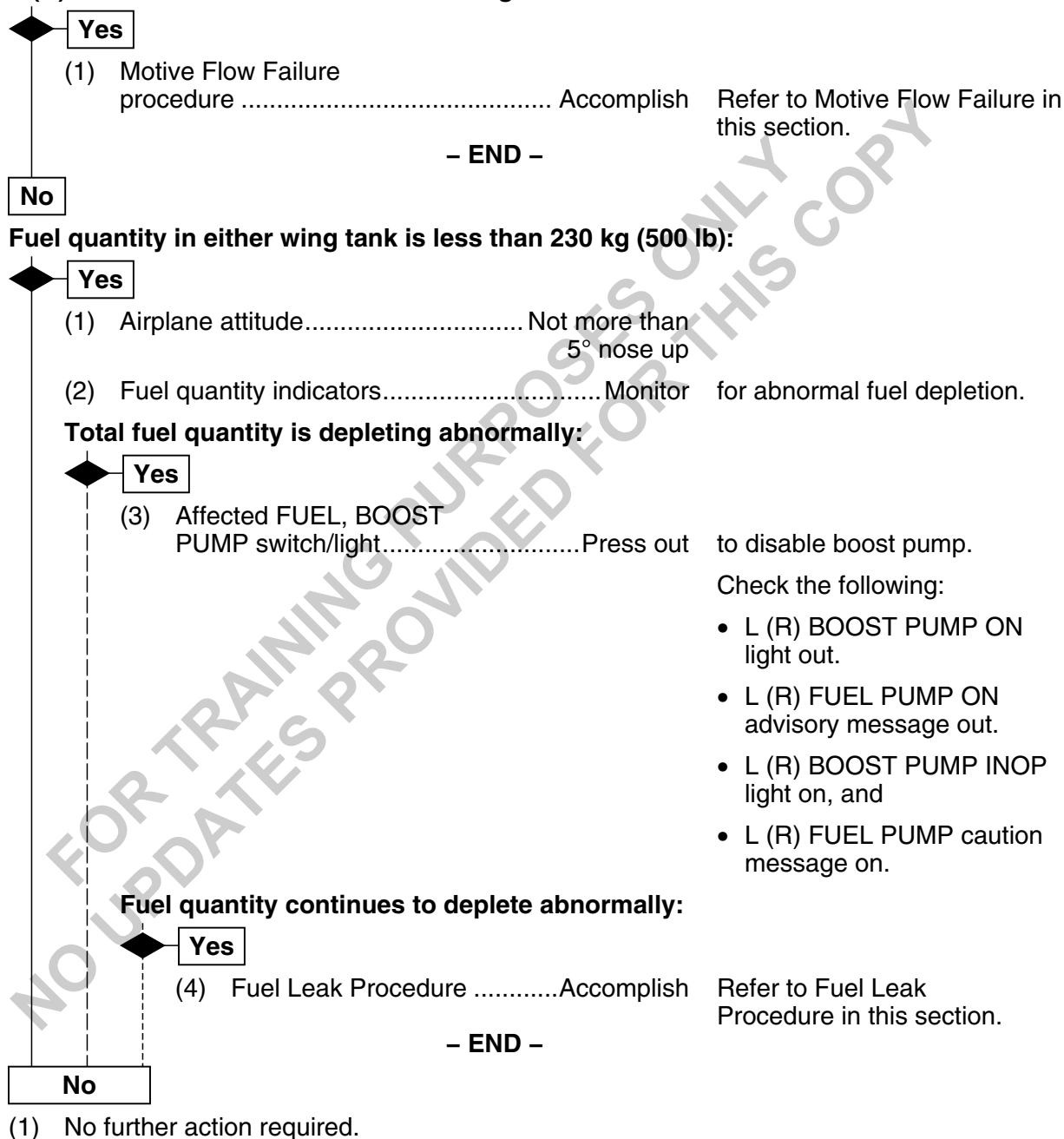
CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-8
		Oct 27/15

1. FUEL (CONT'D)

J. Scavenge Ejector Failure

Indication: L (R) SCAV EJECTOR caution message on.

L (R) MAIN EJECTOR FAIL status message on:



- END -

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-9
		Oct 27/15

1. FUEL (CONT'D)

K. Primary Tail Tank Pump Failure (Manual Mode)

Indication: PRI TAIL XFER PUMP caution message on and PRI TAIL TANK TRANSFER FAIL light on.

- (1) FUEL, SEC TAIL TANK TRANSFER switch/light Press in to start secondary pump. Check the following:
- SEC TAIL TANK TRANSFER ON light on, and
 - MANUAL OVRD, MANUAL light on.
- (2) FUEL, PRI TAIL TANK TRANSFER switch/light Press out
- (3) Fuel tank quantity readouts Monitor that secondary transfer pump maintains tail and auxiliary tank quantities normal (i.e., tail tank is depleting).

Transfer is inoperative:

- ◆ **Yes**
- (4) Airplane attitude Level flight if practical.
- (5) FUEL, DUMP switch/light Press in to jettison fuel.
Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
 2. Jettisoning of fuel in known lightning conditions is prohibited.
 3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.
- (6) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting. Tail tank fuel is unusable.
- Continue to dump fuel until level flight is achieved.
- In level flight, continue to dump fuel until tail tank quantity is 635 kg (1,400 lb).
- If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

– END –



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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-10
		Oct 27/15

1. FUEL (CONT'D)

K. Primary Tail Tank Pump Failure (Manual Mode) (Cont'd)

No

When tail tank is depleted:

- (4) FUEL, SEC TAIL TANK
TRANSFER switch/light Press out

END

L. Secondary Tail Tank Pump Failure (Manual Mode)

Indication: SEC TAIL XFER PUMP caution message on and SEC TAIL TANK
TRANSFER FAIL light on.

- (1) FUEL, PRI TAIL TANK
TRANSFER switch/light Press in to start primary pump. Check
the following:

- PRI TAIL TANK
TRANSFER ON light on.
- MANUAL OVRD, MANUAL
light on.

- (2) FUEL, SEC TAIL TANK
TRANSFER switch/light Press out

- (3) Fuel tank quantity readouts Monitor that primary transfer pump
maintains tail and auxiliary
tank quantities normal (i.e., tail
tank is depleting).

Transfer is inoperative:



Yes

- (4) Airplane attitude Level flight if practical.

- (5) FUEL, DUMP switch/light Press in to jettison fuel.
Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-11
		Oct 27/15

1. FUEL (CONT'D)

L. Secondary Tail Tank Pump Failure (Manual Mode) (Cont'd)

- ▼ (6) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.
Continue to dump fuel until level flight is achieved.
In level flight, continue to dump fuel until tail tank quantity is 635 kg (1,400 lb).
If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

– END –

No

When tail tank is depleted:

- (4) FUEL, PRI TAIL TANK
TRANSFER switch/light Press out

— END —

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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-12
		Rev 11, Mar 01/18

1. FUEL (CONT'D)

M. Tail Tank to Aux Tank Automatic Transfer Failure

Indication: AUTO TAIL XFER caution message on.

NOTE

If a level sensor fails during fuel transfer, tail tank to auxiliary tank fuel transfer may continue after the AUTO TAIL XFER caution message is displayed.

- (1) Tail tank fuel quantityCheck

Tail tank quantity is less than 90 lb and message goes out within 2 minutes:

Yes

- (2) No further action required. Disregard message.

— END —

No

- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check the following:
- MANUAL light on, and
 - AUTO TAIL XFER caution message out.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).
- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
- PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (5) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

NOTE

Do not deselect manual transfer until after landing.

— END —

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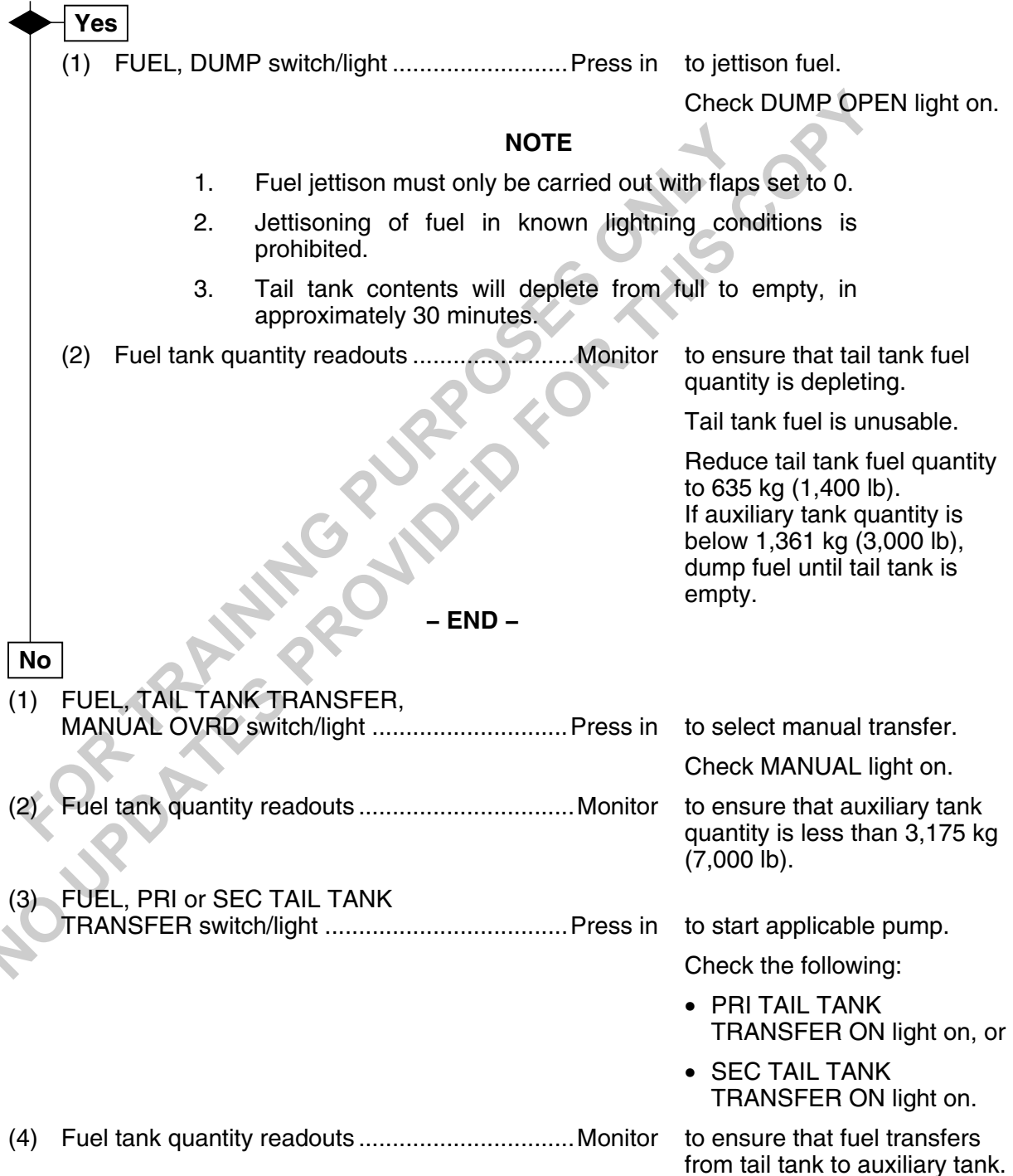
CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05–12–13
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1. FUEL (CONT'D)

N. Tail Tank Transfer is Inhibited

Indication: AUTO TAIL XFER INHIB caution message on.

Engine failure is suspected:



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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05–12–14
		Oct 27/15

1. FUEL (CONT'D)

N. Tail Tank Transfer is Inhibited (Cont'd)

When tail tank is depleted:

- (5) FUEL, PRI or SEC TAIL TANK
TRANSFER switch/light Press out to stop applicable pump.

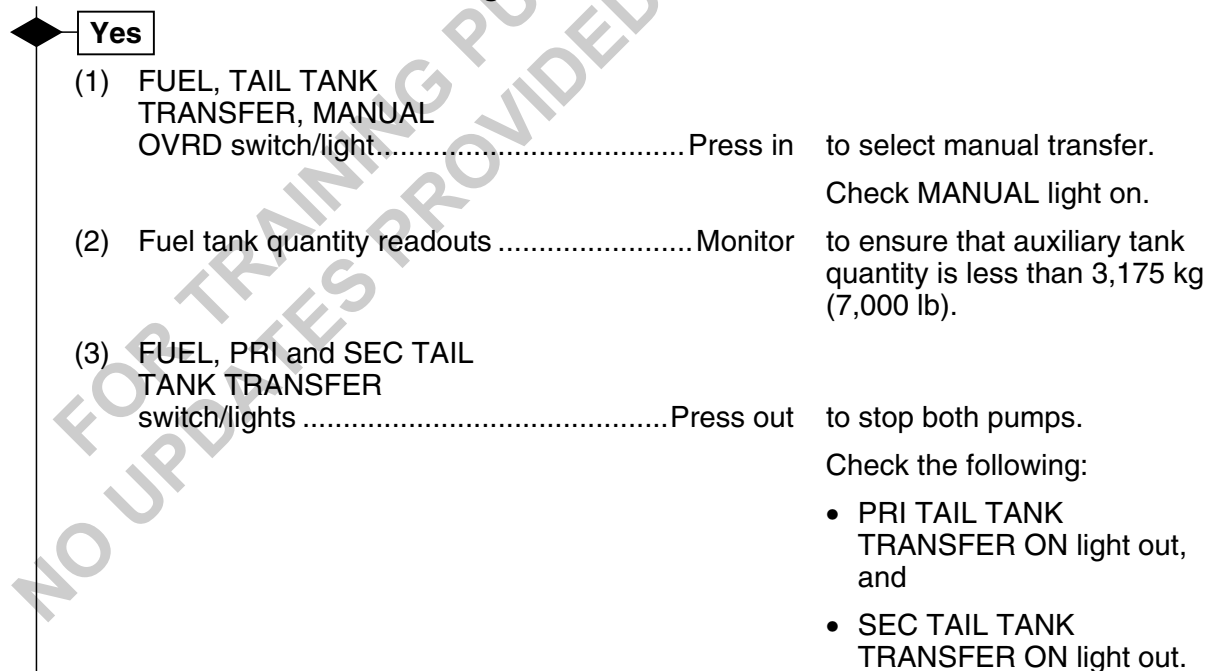
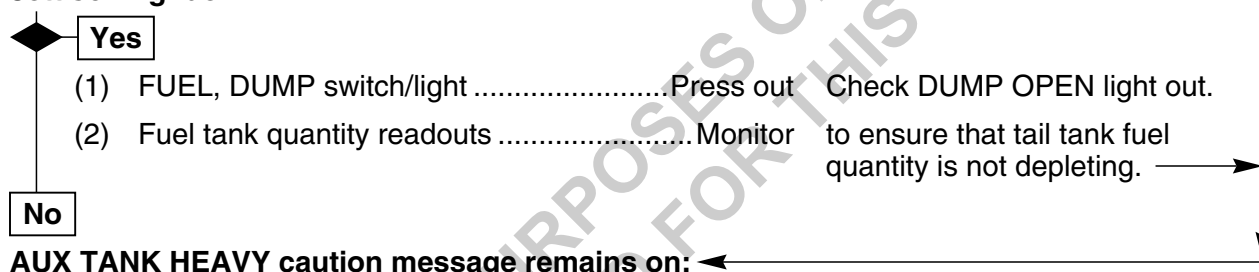
————— **END** —————

O. Excessive Fuel Load – Aux Tank

Indication: AUX TANK HEAVY caution message on.

Fuel distribution between auxiliary and tail tank is abnormal.

Jettisoning fuel:



1. FUEL (CONT'D)

O. Excessive Fuel Load – Aux Tank (Cont'd)

- (4) Fuel tank quantity readouts Monitor to ensure the following:
- Auxiliary tank fuel quantity is depleting, and
 - Tail tank fuel quantity is not depleting.

NOTE

If auxiliary tank is not depleting, auxiliary and tail tank fuel is unusable.

AUX TANK HEAVY caution message persists:**Yes****Prior to landing:**

- (5) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (7) FLAPS Select to 20° for landing.
- (8) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (9) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

– END –



1. FUEL (CONT'D)

O. Excessive Fuel Load – Aux Tank (Cont'd)

No

(1) No further action required.

END

P. Excessive Fuel Load – Tail Tank

Indication: TAIL TANK HEAVY caution message on.
Fuel distribution between auxiliary and tail tank is abnormal.

TAIL TANK TRANSFER AUTO INHIB light out:

Yes

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light.....

Press in

to select manual transfer.
Check MANUAL light on.
- (2) Fuel tank quantity readouts

Monitor

to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).
- (3) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light

Press in

to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (4) Fuel tank quantity readouts

Monitor

to ensure that tail tank fuel quantity is depleting.

**Tail tank fuel quantity is depleting and
TAIL TANK HEAVY caution message goes out:**

Yes

When tail tank is depleted:

- (5) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light

Press out

to stop applicable pump.

- END -

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-17
		Oct 27/15

1. FUEL (CONT'D)

P. Excessive Fuel Load – Tail Tank (Cont'd)

No

- (1) FUEL, DUMP switch/light Press in to jettison fuel.

Check DUMP OPEN light on.

NOTE

- Fuel jettison must only be carried out with flaps set to 0.
- Jettisoning of fuel in known lightning conditions is prohibited.
- Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- (2) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

Reduce tail tank fuel quantity to 635 kg (1,400 lb).
If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

END

Q. Tail Dump SOV Failed (SOV not in commanded position)

Indication: TAIL DUMP SOV caution message on.

FUEL DUMP selected (jettisoning);

Yes

- (1) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

Tail tank quantity is depleting:

Yes

- (2) No further action required.

– END –

No

- (2) Land at the nearest suitable airport.

– END –

No

- (1) Fuel tank quantity readouts Monitor to ensure that tail tank is not leaking.

END

	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-18
		Oct 27/15

1. FUEL (CONT'D)

R. Tail Transfer SOV Failed Closed

Indication: TAIL XFER SOV CLSD caution message on.

- (1) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is not depleting.
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).
- (4) FUEL, PRI TAIL TANK TRANSFER switch/light Press in to start primary pump.
Check PRI TAIL TANK TRANSFER ON light on.

TAIL XFER SOV CLSD caution message persists:

- ◆ **Yes**
- (5) FUEL, SEC TAIL TANK TRANSFER switch/light Press in to start secondary pump.
Check SEC TAIL TANK TRANSFER ON light on.
 - (6) FUEL, PRI TAIL TANK TRANSFER switch/light Press out
 - (7) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

Transfer is inoperative:

- ◆ **Yes**
- (8) Airplane attitude Level flight if practical.
 - (9) FUEL, DUMP switch/light Press in to jettison fuel.
Check DUMP OPEN light on.

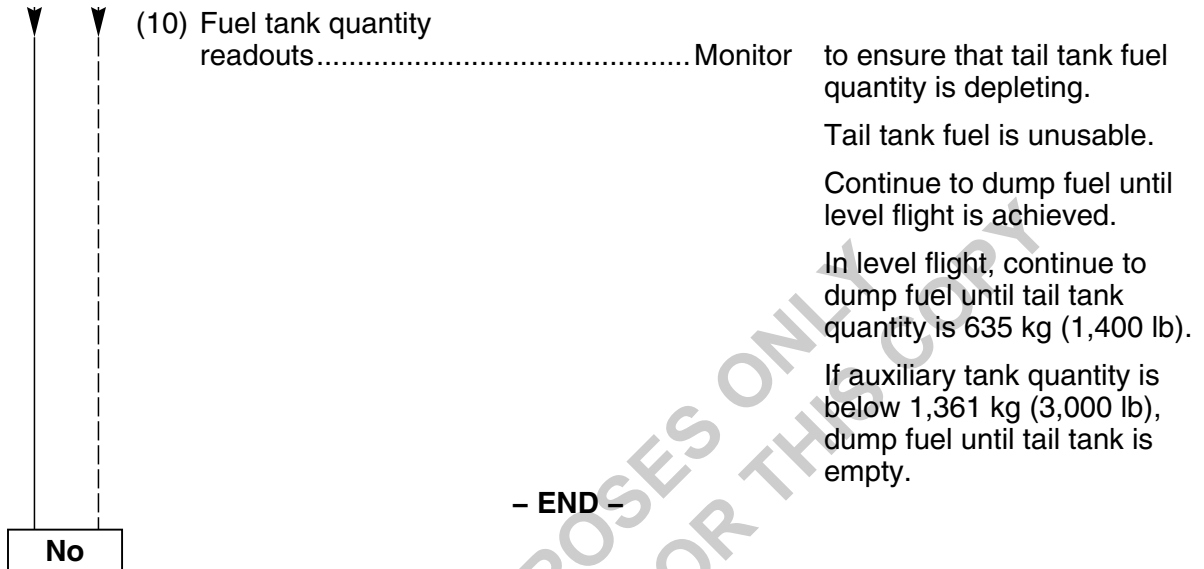
NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-19
		Oct 27/15

1. FUEL (CONT'D)

R. Tail Transfer SOV Failed Closed (Cont'd)



When tail tank is depleted:

- (5) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

— **END** —

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-20
		Oct 27/15

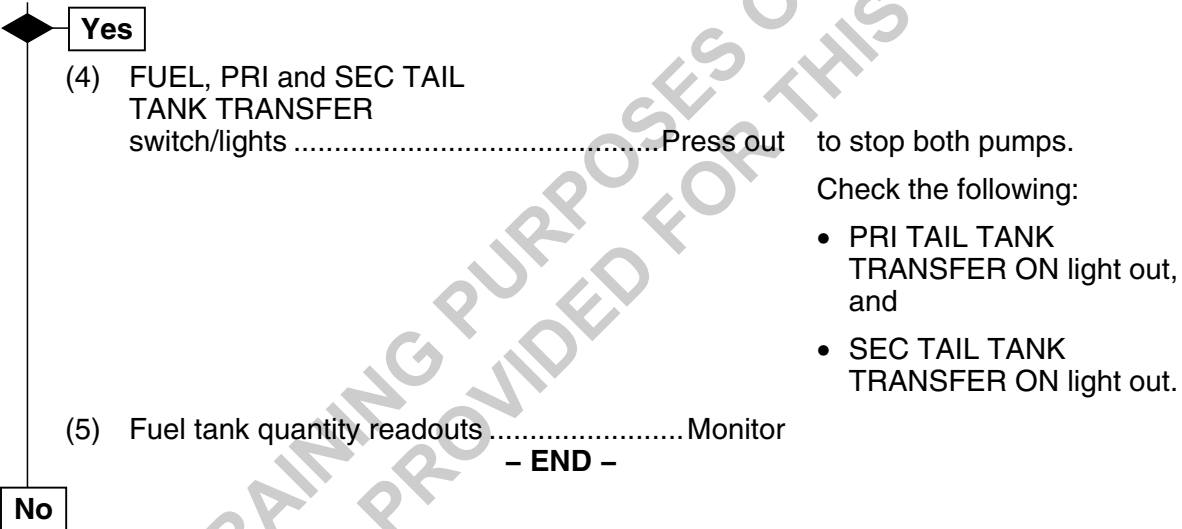
1. FUEL (CONT'D)

S. Tail Transfer SOV Failed Open

Indication: TAIL XFER SOV OPEN caution message on.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).

Excessive auxiliary tank fuel quantity:



Excessive tail tank fuel quantity:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (5) Fuel tank quantity readouts Monitor

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-21
		Oct 27/15

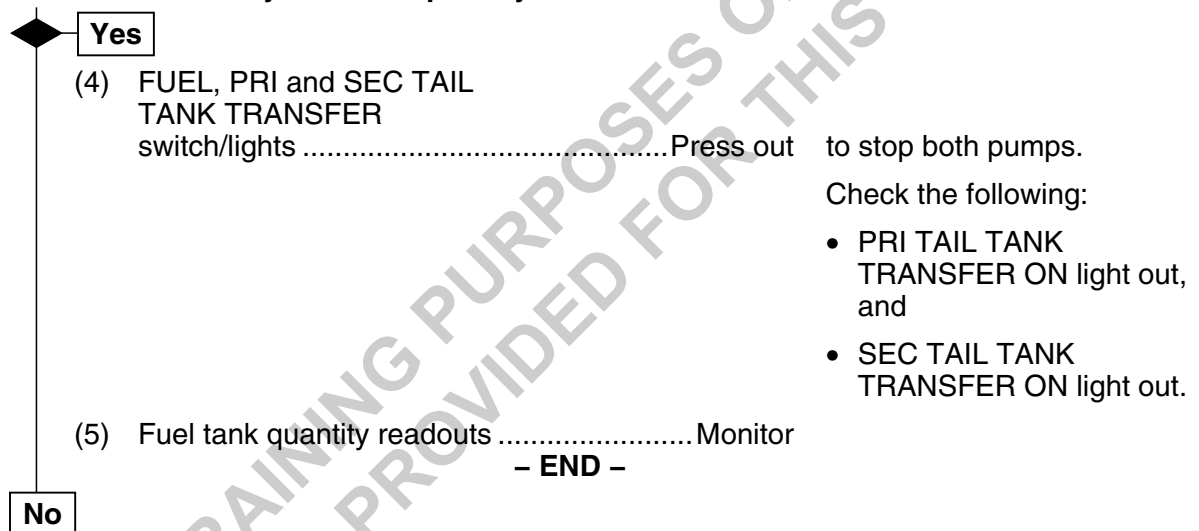
1. FUEL (CONT'D)

T. Tail Transfer SOV Failed in Middle Position

Indication: TAIL XFER SOV caution message on.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).

Excessive auxiliary tank fuel quantity:



Excessive tail tank fuel quantity:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (5) Fuel tank quantity readouts Monitor

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-22
		Oct 27/15

1. FUEL (CONT'D)

U. L/R TO AUX Crossflow Valve Open

Indication: FUEL XFLOW SOV OPEN caution message on.

- (1) Fuel tank quantity readouts Monitor until contents of left and right main tanks are equalized.

When main tanks are equalized:

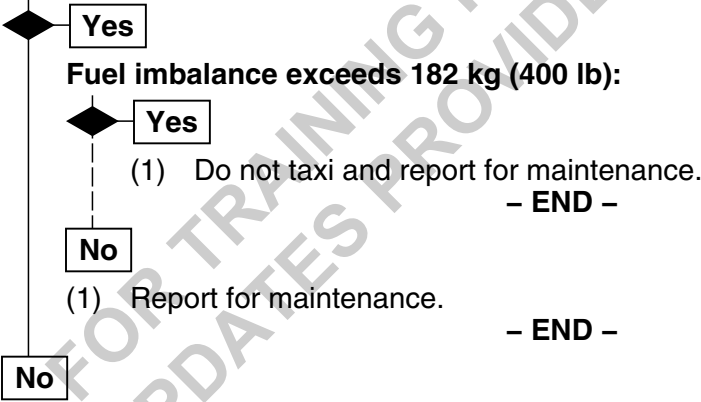
- (2) FUEL, L (R) TO AUX XFLOW switch/light Press out to close affected crossflow valve.
- Check the following:
- L (R) TO AUX XFLOW ON light out.
 - FUEL XFLOW SOV OPEN caution message out.

————— **END** —————

V. L/R TO AUX Crossflow Valve Fails to Close

Indication: Fuel crossflow continues after crossflow valve selected closed.

On ground:



Fuel imbalance exceeds 363 kg (800 lb):

- (1) Airplane attitude Level Steady heading, wings level.
- (2) FUEL, GRAVITY XFLOW switch/light Press in to open gravity crossflow valve.
- Check GRAVITY XFLOW OPEN light on.
- (3) Fuel tank quantity readouts Monitor until contents of left and right main tanks are equalized.



CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-23
		Oct 27/15

1. FUEL (CONT'D)

V. L/R TO AUX Crossflow Valve Fails to Close (Cont'd)

▼ When main tanks have equalized:

- (4) FUEL, GRAVITY XFLOW switch/light Press out to close gravity crossflow valve.
Check GRAVITY XFLOW OPEN light out.
- (5) Fuel tank quantity readouts Monitor
- END -

No

- (1) No further action required.

- END -

W. Auxiliary Tank Contents Increasing (abnormally) During L/R TO AUX Crossflow

Indication: Lateral imbalance.

- (1) Fuel tank quantity readouts Monitor and verify that lateral imbalance is corrected.
- (2) FUEL, L (R) TO AUX XFLOW switch/light Press out to close affected crossflow valve.
Check L (R) TO AUX XFLOW ON light out.

NOTE

Auxiliary tank contents may increase during L (R) TO AUX crossflow, depending upon fuel level.

- END -

X. Dump Valve Open for More than 5 Minutes

Indication: DUMP VALVE OPEN caution message on. Tail tank contents depleting rapidly.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL DUMP switch/light As required

NOTE

The tail tank contents will deplete from a full tank to empty, in approximately 30 minutes following selection.

- END -

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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-24
		Oct 27/15

1. FUEL (CONT'D)

Y. Loss of Auxiliary Tank Fuel Quantity Indication

Indication: Auxiliary tank fuel quantity indication dashed.

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (2) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (3) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

Tail tank quantity is not depleting:

Yes

- (4) FUEL, DUMP switch/light Press in to jettison fuel.
Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- END -

No

When tail tank is depleted:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— END —————

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-25
		Oct 27/15

1. FUEL (CONT'D)

Z. Loss of Wing Tank Fuel Quantity Indication

Indication: Wing tank fuel quantity indication dashed.

- (1) ENGINE CONTROL, ENG SPEED switches Check both selected to ON.
- (2) Airplane attitude Level Steady heading, wings level.
- (3) Engine fuel flow indications Match left and right fuel flow rates.
- (4) Air-conditioning Transfer to engine bleeds.
- (5) APU Shutdown if not required.

NOTE

APU operation will increase fuel burn from right wing tank, by a rate of 45 to 136 kg/hr (100 to 300 lb/hr).

- (6) Airplane trim Adjust to achieve hands off flight.
- (7) Missing fuel tank quantity Calculate using FMS fuel used page and displayed quantity indications.
- (8) Fuel distribution Monitor for wing tank fuel imbalance.

Excessive imbalance exists (more than 25% of available trim required):

Yes

Affected high tank:

- (9) FUEL, L (R) TO AUX XFLOW switch/light Press in to open crossflow valve.
Check L (R) TO AUX XFLOW ON light on.

NOTE

Cross flow transfers fuel at a rate of approximately 7 kg/min (15 lb/min) or higher.

When excessive imbalance has been corrected:

- (10) FUEL, affected L (R) TO AUX XFLOW switch/light Press out to close crossflow valve.
Check L (R) TO AUX XFLOW ON light out.

No

- (9) Fuel reserves Increase by 5%.

END

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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-26
		Oct 27/15

1. FUEL (CONT'D)

AA. Loss of Tail Tank Fuel Quantity Indication

Indication: Tail tank fuel quantity indication dashed.

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (2) FUEL, PRI and SEC TAIL TANK TRANSFER switch/lights Press in to start both pumps.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, and
 - SEC TAIL TANK TRANSFER ON light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity does not exceed 3,175 kg (7,000 lb).

When tail tank is depleted:

- (4) FUEL, PRI and SEC TAIL TANK TRANSFER switch/lights Press out
_____ **END** _____

AB. Loss of Total Fuel Quantity Indication

Indication: Total fuel quantity indication dashed.

- (1) Fuel tank quantity readouts Monitor sum of each tank vs fuel burn.
_____ **END** _____

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-27
		Oct 27/15

1. FUEL (CONT'D)

AC. Fuel Leak Procedure

Indication: Fuel leak is evident, suspected or confirmed.

- (1) Diversion may be required.

NOTE

If visibility permits, a visual check from the cabin may enable identification of the leak source.

Leak is confirmed or suspected:

- (2) Land immediately at the nearest suitable airport.



Do not delay landing while attempting to determine location of leak.

Expedite landing if tank quantity gauge indications are amber.

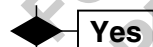
NOTE

1. The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10° nose up.
 2. Do not action FUEL IMBALANCE, L (R) FUEL LO PRESS or L (R) SCAV EJECTOR caution messages or L (R) MAIN EJECTOR FAIL status messages. Disregard fuel imbalance limitations.
- (3) Autopilot Monitor operation.

NOTE

Anticipate out of trim situation when disconnecting autopilot.

Fuel leak from engine is suspected:



On low quantity side:

- (4) Thrust lever..... Confirm and IDLE
- (5) Thrust lever..... Confirm and SHUT OFF
- (6) Affected ENG FIRE PUSH switch/light Press in to close engine fuel feed SOV.
- (7) Affected engine..... Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown Procedure in this chapter.

– END –

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CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-28
		Rev 10, Dec 04/17

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

No

Fuel leak from wing tank is suspected:

- (4) FUEL, GRAVITY/XFLOW switch/light.....Press out to close gravity crossflow valve.
Check GRAVITY XFLOW OPEN light out.

Range is a consideration:

Yes

NOTE

Consider use of asymmetric thrust to use as much fuel as possible from the leaking tank.

Leak in auxiliary tank is confirmed:

Yes

- (5) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light.....Press in to select manual transfer.
Check MANUAL light on.
- (6) Auxiliary fuel tank quantity readout.....Monitor

When auxiliary tank quantity has stabilized:

- (7) FUEL, PRI TAIL TANK TRANSFER switch/light.....Press in to start primary pump.
Check PRI TAIL TANK TRANSFER ON light on.
Ensure auxiliary and tail tank quantities remain within limits.
Refer to Figure 02-05-4.

– END –

No

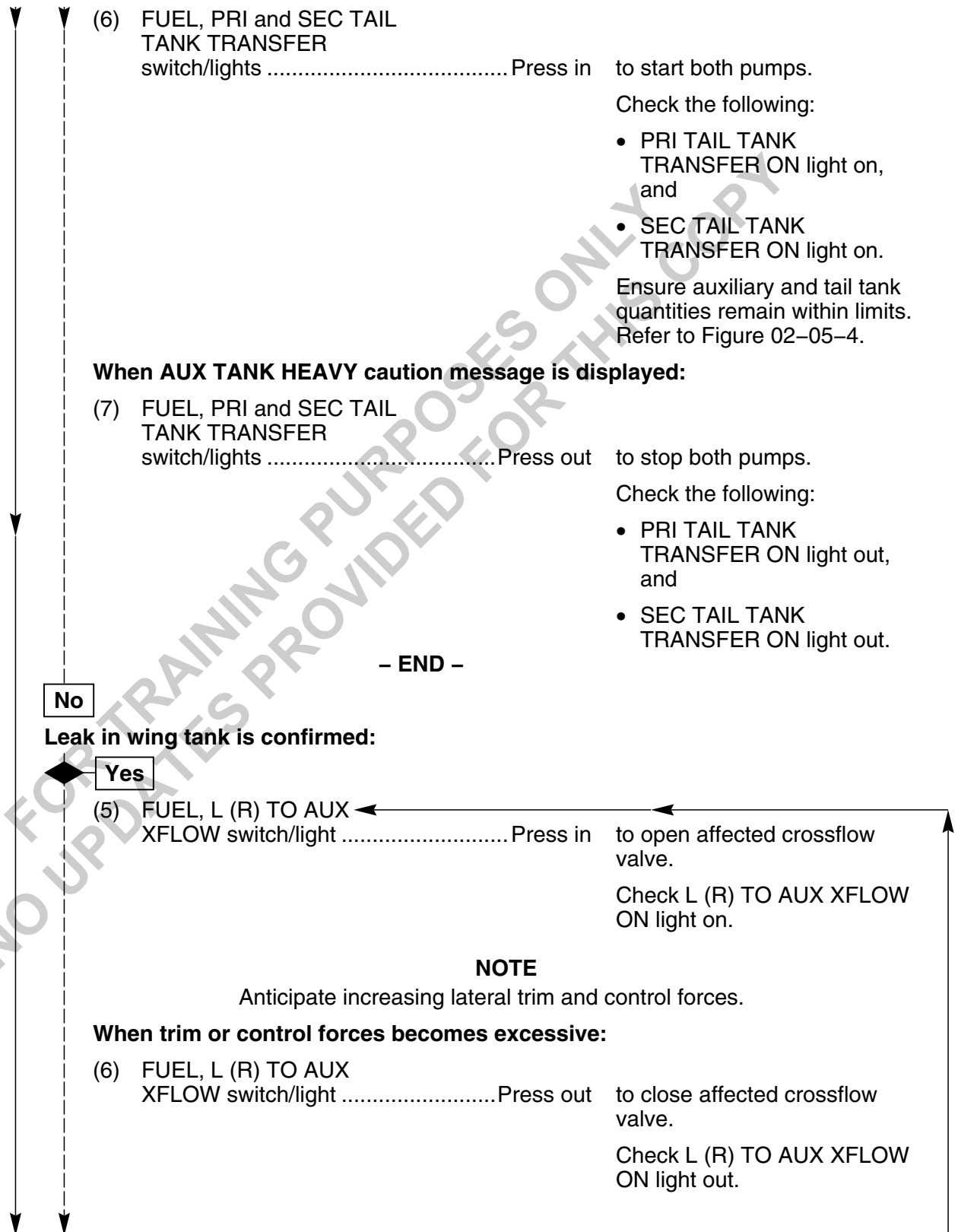
Leak in tail tank is confirmed:

Yes

- (5) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light.....Press in to select manual transfer.
Check MANUAL light on.

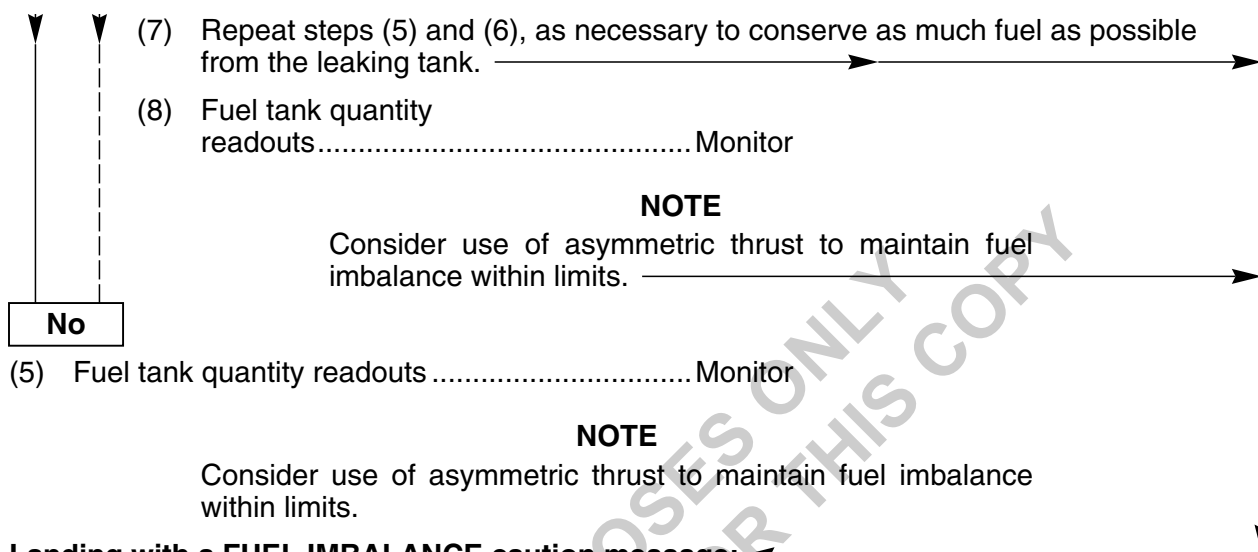
1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

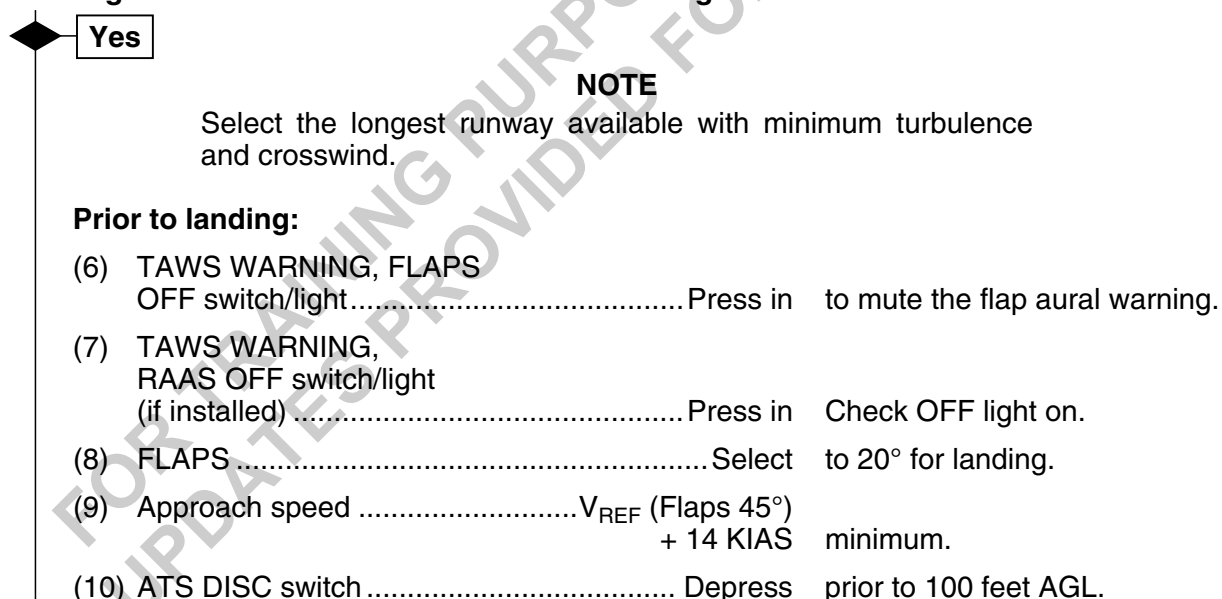


1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)



Landing with a FUEL IMBALANCE caution message:



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

▼ (11) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

No

(6) No further action required.

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Fuel	05-12-32
		Oct 27/15

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CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05-13-1 Rev 5, Aug 30/16
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1. HYDRAULIC POWER

A. System No. 1 Failure

Indication: HYD 1 LO PRESS caution message on.

- (1) HYDRAULIC pump switch 1B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 1 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (3) HYDRAULIC pump switch 1B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.

NOTE

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

- (6) ANTI SKID switch Leave ARMED.
- (7) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (8) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (9) FLAPS Select to 20° for landing.
- (10) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (11) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. HYDRAULIC POWER (CONT'D)

A. System No. 1 Failure (Cont'd)

- ▼ (12) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

No

- (3) HYDRAULIC pump switch 1B Leave at ON.
- (4) Hydraulic pressure and fluid quantity Monitor

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05-13-3
		Oct 27/15

1. HYDRAULIC POWER (CONT'D)

B. System No. 2 Failure

Indication: HYD 2 LO PRESS caution message on.

- (1) HYDRAULIC pump switch 2B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 2 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

Yes

- (3) HYDRAULIC pump switch 2B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.

NOTE

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

- (6) ANTI SKID switch Leave ARMED.
- (7) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.
- END -

No

- (3) HYDRAULIC pump switch 2B Leave at ON.
- (4) Hydraulic pressure and fluid quantity Monitor

— END —

1. HYDRAULIC POWER (CONT'D)

C. System No. 3 Failure

Indication: HYD 3 LO PRESS caution message on.

- (1) HYDRAULIC pump switch 3B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 3 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

Yes

- (3) HYDRAULIC pump switches 3A and 3B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.
- (6) ANTI SKID switch Leave ARMED.
- (7) Landing Gear Manual Extension procedure Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.
- (8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.

– END –

No

- (3) HYDRAULIC pump switch 3B Leave at ON.
- (4) Hydraulic pressure and fluid quantity Monitor

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05-13-5
		Oct 27/15

1. HYDRAULIC POWER (CONT'D)

D. System No. 1 High Temperature

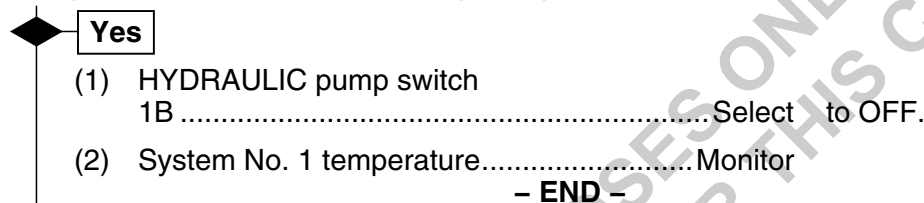
Indication: HYD 1 HI TEMP caution message on and temperature readouts greater than 96°C (205°F).

NOTE

Disregard HYD 1 HI TEMP caution message if one other hydraulic system has already failed.

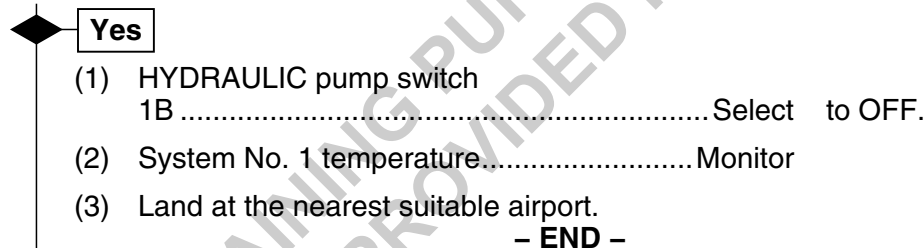
During flight:

Temperature is less than 107°C (225°F):



No

Temperature is less than 135°C (275°F):



No

- (1) HYDRAULIC pump switch 1B Select to OFF.
- (2) Left engine..... Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown in this chapter.
- (3) Inoperative systems Review Refer to HYDRAULIC page.
- (4) Actual landing distance Increase by a factor of 1.60 (60%).

— END —

CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05-13-6
		Oct 27/15

1. HYDRAULIC POWER (CONT'D)

E. System No. 2 High Temperature

Indication: HYD 2 HI TEMP caution message on and temperature readouts greater than 96°C (205°F).

NOTE

Disregard HYD 2 HI TEMP caution message if one other hydraulic system has already failed.

During flight:

Temperature is less than 107°C (225°F):

- Yes**
- (1) HYDRAULIC pump switch 2B Select to OFF.
 - (2) System No. 2 temperature..... Monitor
- END -

No

Temperature is less than 135°C (275°F):

- Yes**
- (1) HYDRAULIC pump switch 2B Select to OFF.
 - (2) System No. 2 temperature..... Monitor
 - (3) Land at the nearest suitable airport.
- END -

No

- (1) HYDRAULIC pump switch 2B Select to OFF.
- (2) Right engine Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown in this chapter.
- (3) Inoperative systems Review Refer to HYDRAULIC page.
- (4) Actual landing distance Increase by a factor of 2.25 (125%).

- END -

1. HYDRAULIC POWER (CONT'D)

F. System No. 3 High Temperature

Indication: HYD 3 HI TEMP caution message on and temperature readouts greater than 96°C (205°F).

During flight:

- (1) HYDRAULIC pump switches 3A and 3B Select to OFF.
- (2) System No. 3 temperature Monitor

Temperature is more than 107°C (225°F):

◆ Yes

- (3) Land at the nearest suitable airport. _____

No

Prior to landing: ←

- (3) HYDRAULIC pump switches 3A and 3B Select to ON.

————— END —————

G. Engine-Driven Pump Failure

Indication: HYD EDP 1A (2A) caution message on.

- (1) Associated HYDRAULIC pump switch 1B or 2B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

System quantity low:

◆ Yes

- (3) Associated HYDRAULIC pump switch 1B or 2B Select to OFF.

NOTE

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

— END —

No

- (3) No further action required.

————— END —————

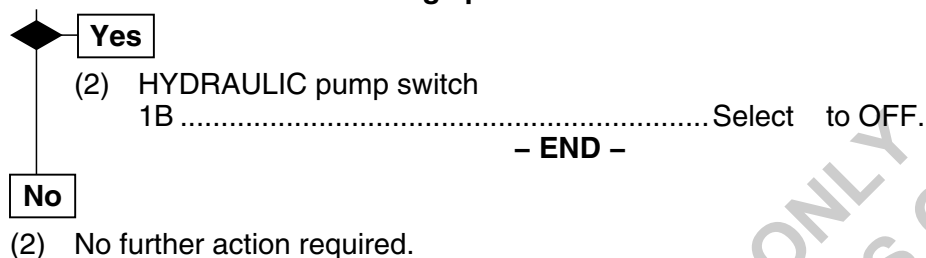
1. HYDRAULIC POWER (CONT'D)

H. Electric Pump 1B Failure

Indication: HYD PUMP 1B caution message on.

(1) HYDRAULIC pump switch 1B Select to ON.

HYD PUMP 1B caution message persists:

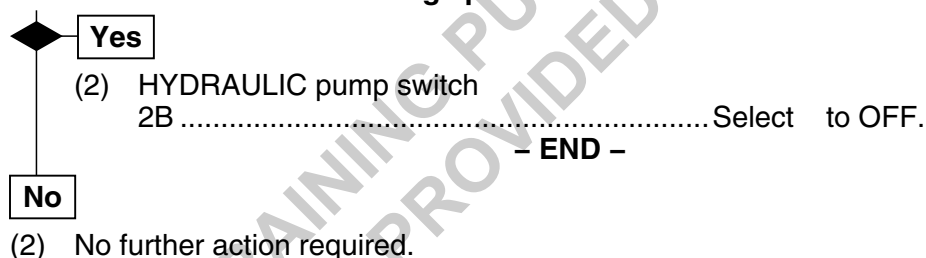


I. Electric Pump 2B Failure

Indication: HYD PUMP 2B caution message on.

(1) HYDRAULIC pump switch 2B Select to ON.

HYD PUMP 2B caution message persists:

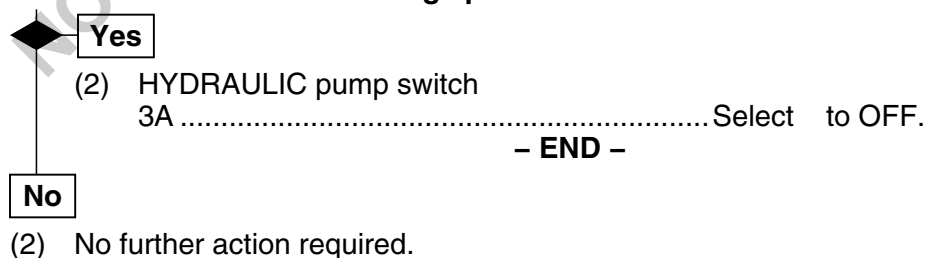


J. Electric Pump 3A Failure

Indication: HYD PUMP 3A caution message on.

(1) HYDRAULIC pump switches 3A and 3B Select both to ON.

HYD PUMP 3A caution message persists:



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		Oct 27/15

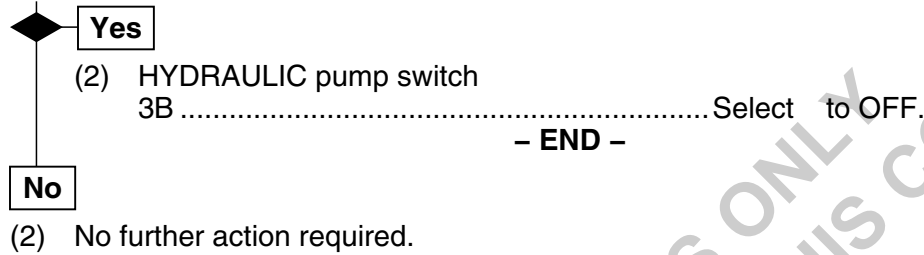
1. HYDRAULIC POWER (CONT'D)

K. Electric Pump 3B Failure

Indication: HYD PUMP 3B caution message on.

- (1) HYDRAULIC pump switches 3A
and 3B Select both to ON.

HYD PUMP 3B caution message persists:



L. Hydraulic Shut-Off Valve Failure After Engine Shutdown Due to Fire

Indication: HYD SOV 1 (2) caution message on.

- (1) Land at the nearest suitable airport.

— END —

1. HYDRAULIC POWER (CONT'D)

M. System No. 1 and No. 2 Failure

Indication: HYD 1 LO PRESS and HYD 2 LO PRESS caution messages on.

- (1) HYDRAULIC pump switches (all) Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 1 and No. 2 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (3) HYDRAULIC pump switches 1B and 2B Select to OFF, to prevent pump overheat.
 - (4) Inoperative systems Review Refer to HYDRAULIC page.
 - (5) ANTI SKID switch Leave ARMED.
 - (6) Land at the nearest suitable airport.
- Prior to landing:**
- (7) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
 - (8) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
 - (9) FLAPS Select to 20° for landing.
 - (10) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
 - (11) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. HYDRAULIC POWER (CONT'D)

M. System No. 1 and No. 2 Failure (Cont'd)

- ▼ (12) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.55 (155%)	1.95 (95%)

NOTE

Maximize the use of reverse thrust.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (3) No further action required.

— END —

1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure

Indication: HYD 2 LO PRESS and HYD 3 LO PRESS caution messages on.

- (1) Airspeed200 KIAS maximum.
- (2) HYDRAULIC pump switches (all)Select to ON.
- (3) Hydraulic pressure and fluid quantityCheck

No. 2 and No. 3 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (4) HYDRAULIC pump switches 2B, 3A and 3BSelect to OFF, to prevent pump overheat.
- (5) Inoperative systems..... Review Refer to HYDRAULIC page.
- (6) Land at the nearest suitable airport.

NOTE

1. Control is limited with No. 2 and No. 3 hydraulic systems failed.
2. Select the longest runway available with minimal crosswind and turbulence.
3. Rudder control is adequate for normal flight and should be used in coordination with aileron, if necessary, during turns.

Prior to landing:

- (7) NOSE STEER switchSelect to OFF.
- (8) ANTI SKID switchSelect to OFF.
- (9) Landing Gear Manual Extension procedure..... Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.

NOTE

1. Unless it is possible to restore use of No. 2 hydraulic system, landing gear extension relies upon gear free fall after use of gear manual extension control.
2. A sideslip, for each side, must be performed to achieve MLG down-lock. It may be required to increase airspeed to ensure down-lock during sideslip.

- (10) TAWS WARNING, FLAPS OFF switch/light.....Press in to mute the flap aural warning.

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CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05-13-13 Rev 5, Aug 30/16
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1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure (Cont'd)

- (11) TAWS WARNING,
RAAS OFF switch/light
(if installed) Press in Check OFF light on.
- (12) FLAPS Select to 20° for landing.
- (13) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (14) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

NOTE

The above double hydraulic system failure has an extremely remote probability. If it occurs, maximum use of reverse thrust must be used upon landing.

- (15) Actual landing distance Increase by a factor of 3.50 (250%).

CAUTION

Extreme caution is required during braking to avoid tire damage. Maximize use of reverse thrust.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

Upon landing:

- (16) Both engines Apply maximum reverse thrust.
- (17) Brakes Apply light to moderate braking until safe taxi speed is maintained.
- (18) Brake pressure Monitor Approximately six (6) full brake applications are possible, using brake accumulator pressure.

NOTE

INBD and OUTBD BRAKE PRESS caution messages come on at less than 1,015 psi.

– END –

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CHALLENGER ⁶⁵⁰	ABNORMAL PROCEDURES Hydraulic Power	05-13-14
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1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure (Cont'd)

No

(4) No further action required.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Hydraulic Power	05–13–15
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1. HYDRAULIC POWER (CONT'D)

O. System No. 1 and No. 3 Failure

Indication: HYD 1 LO PRESS and HYD 3 LO PRESS caution messages on.

- (1) Airspeed200 KIAS maximum.
- (2) HYDRAULIC pump switches (all)Select to ON.
- (3) Hydraulic pressure and fluid quantityCheck

No. 1 and No. 3 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (4) HYDRAULIC pump switches 1B, 3A and 3BSelect to OFF.
- (5) Inoperative systems..... Review Refer to HYDRAULIC page.
- (6) NOSE STEER switchSelect to OFF.
- (7) ANTI SKID switchLeave ARMED.
- (8) Land at the nearest suitable airport.

NOTE

1. Flight path control is limited with No. 1 and No. 3 hydraulic systems failed.
 2. Select the longest runway available with minimal crosswind and turbulence.
 3. Rudder control is adequate for normal flight and should be used in coordination with aileron during turns.
- (9) Landing Gear Manual Extension procedure..... Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.

Prior to landing:

- (10) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (11) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (12) FLAPSSelect to 20° for landing.
- (13) Approach speedV_{REF} (Flaps 45°) + 14 KIAS minimum.

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1. HYDRAULIC POWER (CONT'D)

O. System No. 1 and No. 3 Failure (Cont'd)

- ▼ (14) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (15) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.40 (140%)	1.90 (90%)

NOTE

Maximize use of reverse thrust.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

– END –

No

- (4) No further action required.

————— **END** —————

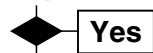
1. ICE AND RAIN PROTECTION

A. Ice Dispersal Procedure

After leaving icing conditions, and if ice accumulation is observed on the heated portion of the wing leading edge:

- (1) Airspeed Increase to V_{MO}/M_{MO} , if possible, to disperse ice.

Impossible to remove ice from the wing leading edge:



Yes

Assure adequate stall margin:

- (2) Maneuvering airspeed 200 KIAS minimum

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margins provided by the stall protection system.

Prior to landing:

- (3) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (4) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (5) FLAPS Select to 20° for landing.
- (6) Approach speed V_{REF} (Flaps 45°) + 27 KIAS
- (7) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

1. ICE AND RAIN PROTECTION (CONT'D)

A. Ice Dispersal Procedure (Cont'd)

▼ At 50 feet AGL:

- (9) Thrust levers IDLE and do not prolong the landing flare.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (2) No further action required.

— END —

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1. ICE AND RAIN PROTECTION (CONT'D)

B. Ice Detected

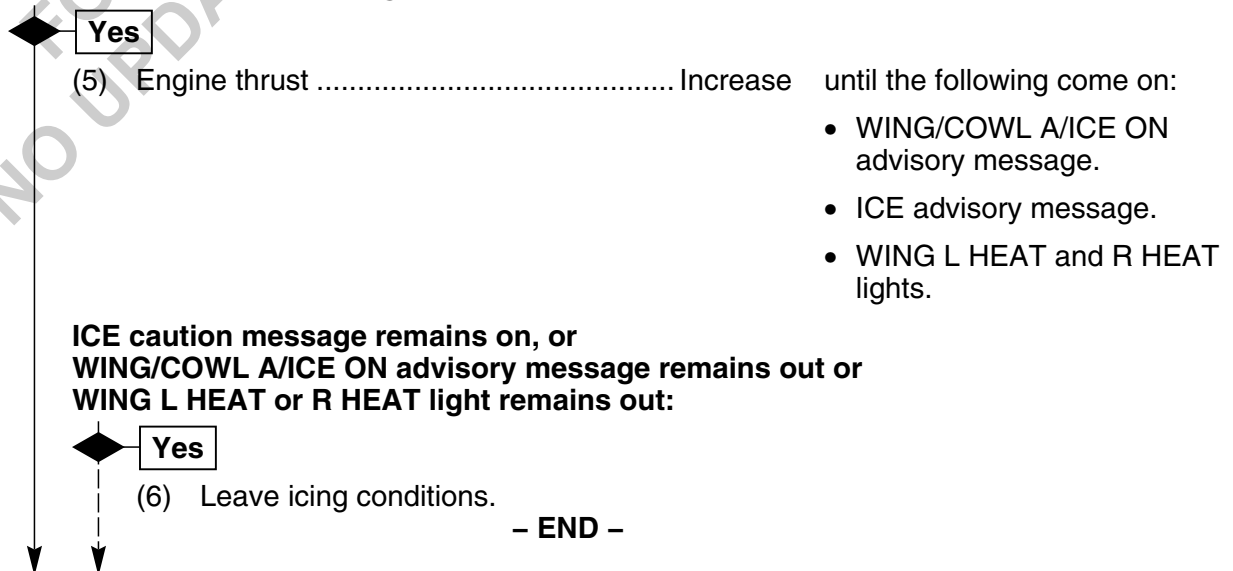
Indication: ICE caution message on.

- (1) Thrust Check N₂ RPM 78% minimum.
- (2) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice ON.
Check COWL A/ICE ON advisory message on.
- (3) ANTI-ICE, WING switch Select to NORM.
Check the following:
 - L and R WING A/ICE caution messages may come on briefly, then out.
 - N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
 - WING/COWL A/ICE ON advisory message on.
 - ICE advisory message on.
 - WING L HEAT and R HEAT lights on.

If required:

- (4) ANTI-ICE, WSHLD/WIND L and R switches Select to HI.

**ICE caution message remains on, or
WING/COWL A/ICE ON advisory message is out or
WING L HEAT or R HEAT light is out:**



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1. ICE AND RAIN PROTECTION (CONT'D)

B. Ice Detected (Cont'd)

No

(5) No further action required.

————— **END** —————

C. One Ice Detector Failed

Indication: ICE DETECTOR 1 (2) FAIL status message on.

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

When in icing conditions:

- (1) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice ON.
Check COWL A/ICE ON advisory message on.
- (2) ANTI-ICE, WING switch Select to NORM.
Check the following:
 - L and R WING A/ICE caution messages may come on briefly, then out.
 - N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
 - WING/COWL A/ICE ON advisory message on.
 - WING L HEAT and R HEAT lights on.



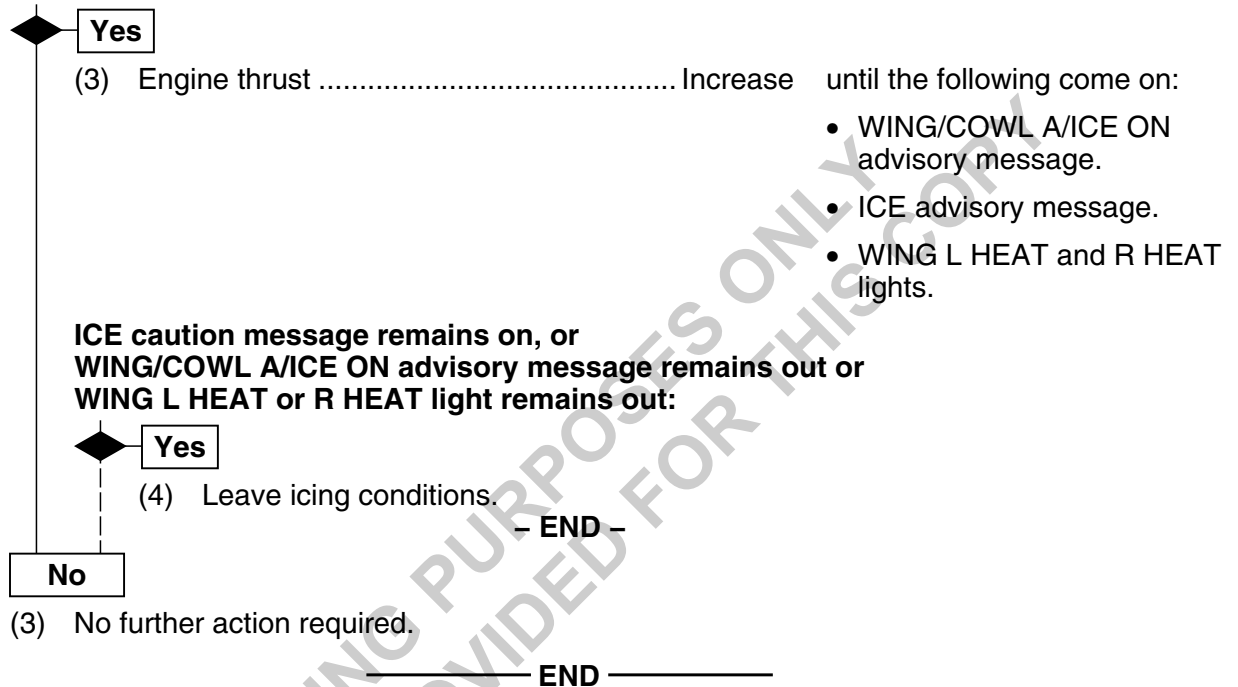
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1. ICE AND RAIN PROTECTION (CONT'D)

C. One Ice Detector Failed (Cont'd)

ICE caution message remains on, or
WING/COWL A/ICE ON advisory message is out or
WING L HEAT or R HEAT light is out:

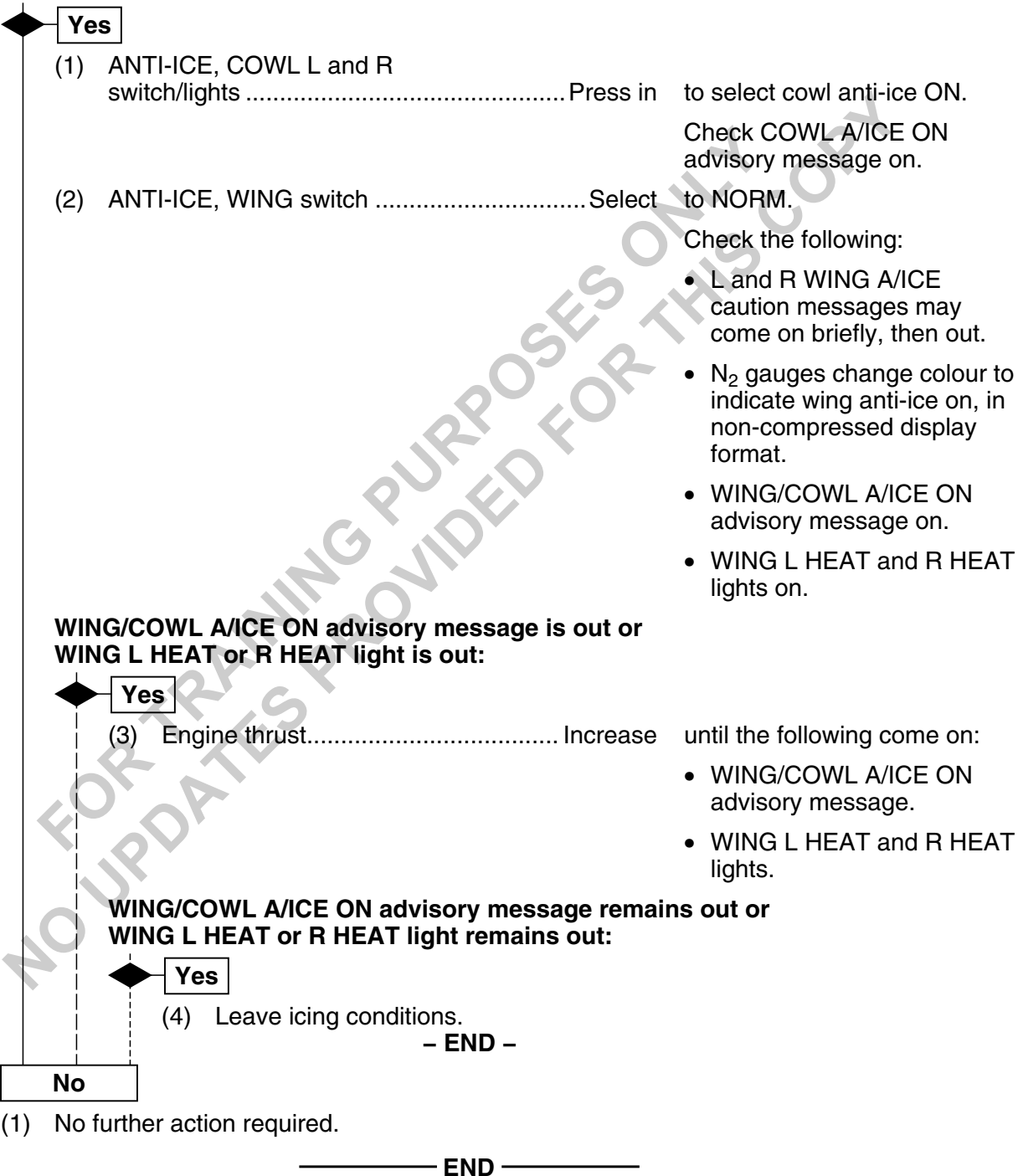


1. ICE AND RAIN PROTECTION (CONT'D)

D. Both Ice Detectors Failed

Indication: ICE DETECTORS caution message on.

Icing conditions are present or are anticipated:



CHALLENGER 650	ABNORMAL PROCEDURES Ice and Rain Protection	05-14-7
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1. ICE AND RAIN PROTECTION (CONT'D)

E. Wing Anti-Ice Sensor Failure

Indication: WING A/ICE SNSR caution message on.

(1) ANTI-ICE, WING switch Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message on.

L (R) WING A/ICE caution message displayed:

Yes

(2) BLEED AIR, 14TH STAGE ISOL switch/light Press in to open isolation valve.

Check 14TH STAGE ISOL OPEN light on.

(3) ANTI-ICE, WING switch Check at STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message on.

L (R) WING A/ICE caution message remains displayed:

Yes

(4) Leave icing conditions.

If required:

(5) Ice Dispersal Procedure Accomplish Refer to Ice Dispersal Procedure in this section.

– END –

No

(2) No further action required.

END

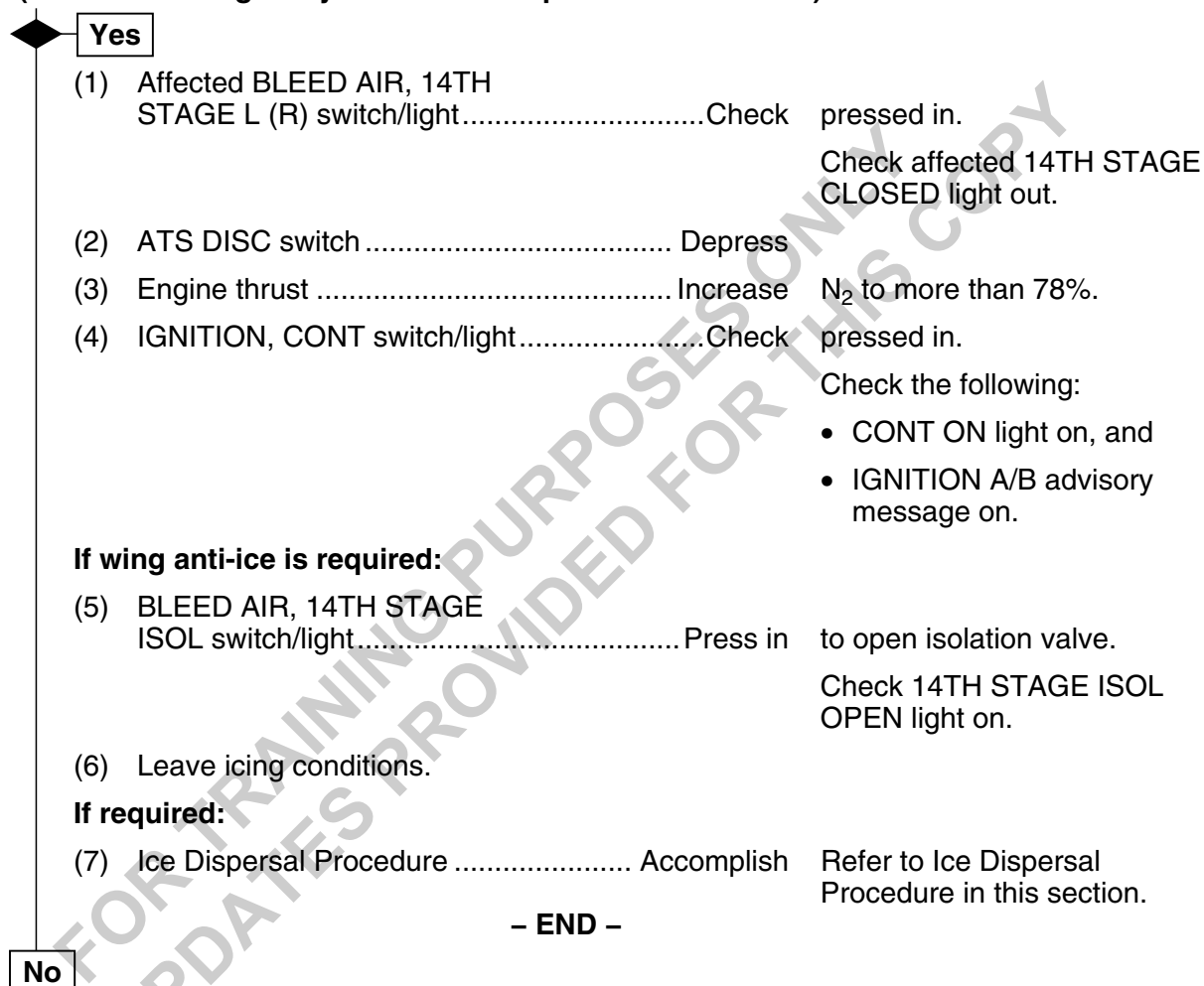
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1. ICE AND RAIN PROTECTION (CONT'D)

F. Cowl Anti-Ice Failure

Indication: L (R) COWL A/ICE caution message on.

Cowl anti-ice is selected on:
(Caution message may indicate a low pressure condition.)



Cowl anti-ice is selected off:
(Caution message may indicate a high pressure condition.)

- (1) ATS DISC switch Depress
- (2) Affected engine thrust Reduce N₂ to less than 85%.
- (3) Affected BLEED AIR, 14TH STAGE L (R) switch/light.....Press out to close shutoff valve.
Check affected 14TH STAGE CLOSED light on.

————— END —————

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1. ICE AND RAIN PROTECTION (CONT'D)

G. Wing Leading Edge Temperature and Pressure Low

Indication: L (R) WING A/ICE caution message on.

(1) Affected BLEED AIR, 14TH

STAGE L (R) switch/light.....Check pressed in.

Check affected 14TH STAGE
CLOSED light out.

(2) Engine thrust Increase

L (R) WING A/ICE caution message remains on:

◆ **Yes**

(3) BLEED AIR, 14TH STAGE

ISOL switch/light..... Press in to open isolation valve.

Check 14TH STAGE ISOL
OPEN light on.

After 30 seconds, and L (R) WING A/ICE caution message persists:

◆ **Yes**

(4) ANTI-ICE, WING switch..... Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- WING A/ICE ON or WING/COWL A/ICE ON advisory message on.

(5) Leave icing conditions.

If required:

(6) Ice Dispersal Procedure Accomplish Refer to Ice Dispersal
Procedure in this section.

— END —

◆ **No**

(3) No further action required.

— END —

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CHALLENGER 650	ABNORMAL PROCEDURES Ice and Rain Protection	05–14–10
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1. ICE AND RAIN PROTECTION (CONT'D)

H. Insufficient Heat for Wing Anti-Icing

Indication: Wing anti ice selected ON and WING L HEAT or R HEAT light goes out or WING A/ICE ON or WING/COWL A/ICE ON advisory message goes out.

(1) Thrust Increase until the following come on:

- WING A/ICE ON or WING/COWL A/ICE ON advisory message.
- WING L HEAT and R HEAT lights.

WING A/ICE ON or WING/COWL A/ICE ON advisory message remains out or WING L HEAT or R HEAT light remains out:

Yes
(2) Leave icing conditions.

– END –

No
(2) No further action required.

END

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1. ICE AND RAIN PROTECTION (CONT'D)

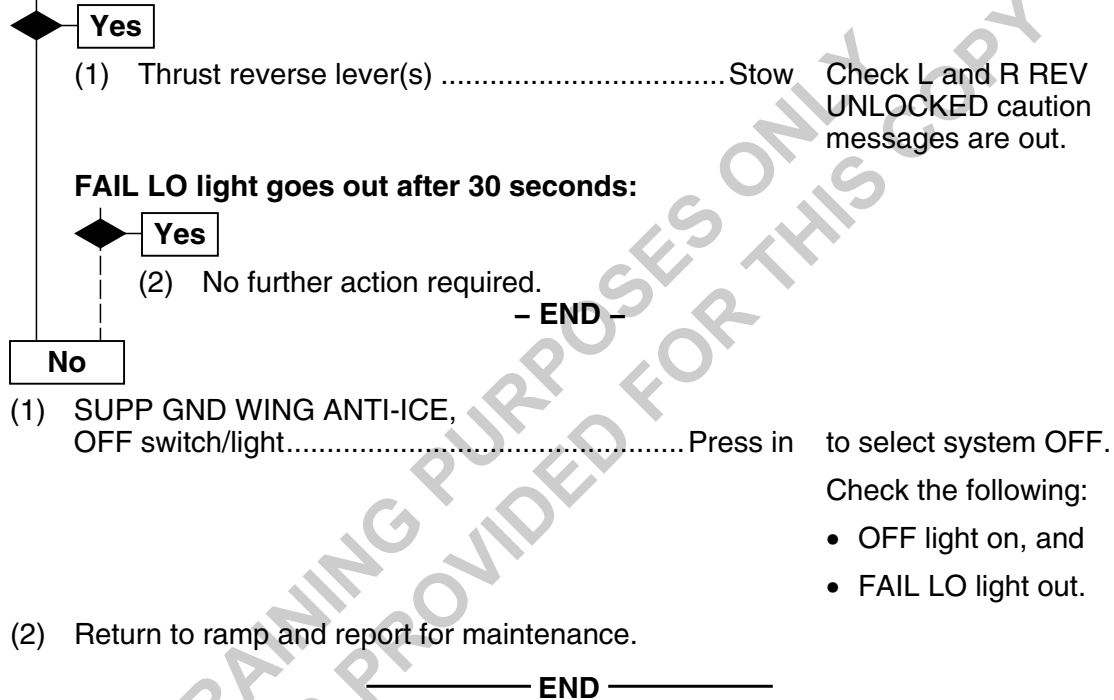
I. Insufficient Heat for Supplemental Ground Wing Anti-Icing

Indication: SUPP GND WING ANTI-ICE, FAIL LO light on.

NOTE

Extended taxi with a thrust reverser deployed may cause FAIL LO light to come on.

Thrust reverser deployed:



J. Supplemental Ground Wing Anti-Icing Overheat

Indication: SUPP GND WING ANTI-ICE, FAIL HI light on.

- (1) SUPP GND WING ANTI-ICE, OFF switch/light..... Press in to select system OFF.
Check OFF light on.
 - (2) Return to ramp and report for maintenance.
- END -

1. ICE AND RAIN PROTECTION (CONT'D)

K. Probe Heater Failures

Indication:

- L (R) AOA HEAT caution message,
- AUX AOA CASE (VANE) HEAT caution message,
- L (R) PITOT BASE (TUBE) HEAT caution message,
- L (R) STATIC HEAT caution message,
- STBY PITOT HEAT caution message,
- TAT PROBE HEAT caution message,
- L (R) PROBE HEAT OFF caution message.

- (1) ANTI-ICE, PROBES L and R switches.....Select OFF and then ON.

Applicable caution message still on:

Yes

MESSAGE	ACTION
L (R) AOA HEAT	Do not rely on stall protection system or TAWS. (2) No further action required.
AUX AOA CASE HEAT AUX AOA VANE HEAT	Do not rely on low speed cue or Normalized AOA displayed on both PFDs. (2) No further action required.
L (R) PITOT TUBE HEAT L (R) PITOT BASE HEAT	(2) Select ADC source to operative side.
L (R) STATIC HEAT	Do not rely on associated instruments: • ISI for altitude and airspeed, and • Cabin ΔP on EICAS. (2) No further action required.
STBY PITOT HEAT	Do not rely on ISI for airspeed indication. (2) No further action required.
TAT PROBE HEAT	Do not rely on air temperature indications. (2) No further action required.
L (R) PROBE HEAT OFF	During flight – No further action required.

L and R AOA HEAT caution messages persist:

Yes

- (2) Avoid icing conditions.

- (3) Stall Protection System

Failure ProcedureAccomplish Refer to FLIGHT CONTROLS
– Stall Protection System
Failure in this chapter.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

K. Probe Heater Failures (Cont'd)

No

(2) No further action required.

————— END —————

L. Windshield Heat Failure

Indication: L (R) WSHLD HEAT caution message on.

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to RESET, then LO or HI as applicable.

L (R) WSHLD HEAT caution message remains on:

Yes

(2) Leave icing conditions as soon as possible.

— END —

No

(2) No further action required.

NOTE

Moving the COCKPIT and CABIN COLD-HOT knobs to HOT may help reduce fogging.

————— END —————

M. Window Heat Failure

Indication: L (R) WINDOW HEAT caution message on.

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to RESET, then LO or HI as applicable.

NOTE

Moving the COCKPIT and CABIN COLD-HOT knobs to HOT may help reduce fogging.

————— END —————

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1. ICE AND RAIN PROTECTION (CONT'D)

N. Windshield Outer Face Ply Failure

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to OFF.
- (2) Flight Continue minimizing flight into icing conditions, precipitation or other conditions that could cause fogging or misting, whenever possible.

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Instruments System	05-15-1 Rev 5, Aug 30/16
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1. INSTRUMENTS SYSTEM

A. Primary Flight Display Failure

Indication: PFD is erratic or goes blank.

- (1) Reversionary panel, applicable
L (R) DISPLAYS switch Select to MFD REV.
- (2) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

B. Display Control Panel Failure

Indication: Amber boxed DCP flag displayed on both PFD and MFD.
DCP is inoperative. Baro, course and HDG knobs may be inoperative.

- (1) Flight control panel, XFR switch Select side corresponding to
operative display control
panel.

NOTE

When the DCP flag is displayed on the PFD, the display features normally controlled by the DCP may be controlled using the Cursor Control Panel and the PFD submenu on the MFD.

————— **END** —————

C. EFIS Comparator Failure

Indication: EFIS COMP INOP caution message.

NOTE

If message is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to Air Data Computer Failure or Inertial Reference System Failure in this section.

- (1) Affected flight instruments Monitor
- (2) Integrated Standby Instrument Use to cross-check.

NOTE

If any of the following occurs, refer to Chapter 3; EMERGENCY PROCEDURES – INSTRUMENTS SYSTEM – Unreliable Airspeed:

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed
- Large airspeed differences between PFDs and/or ISI
- Loss of multiple airspeed indication.

————— **END** —————

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1. INSTRUMENTS SYSTEM (CONT'D)

D. Multifunction Display Failure

Indication: MFD is erratic or displays stale information or goes blank.

- (1) Reversionary panel, applicable
L (R) DISPLAYS switch Select to PFD REV.
- (2) Reversionary panel, EICAS switch Select to BOTH PFD'S.
- (3) MFD TUNE INHIB switch/light Press in

————— **END** —————

E. Cursor Control Panel Failure

Indication: Cursor control panel is inoperative and CCP flag is displayed.

EICAS displayed on MFD corresponding to failed CCP:

◆ **Yes**

- (1) No further action required.

NOTE

If paging of CAS messages is required, the EICAS reversion switch must be selected to the operative CCP side.

— **END** —

◆ **No**

- (1) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

F. PFD Cross-Talk Failure

Indication: PFD X-TALK FAIL caution message on.

PFD is blank:

◆ **Yes**

- (1) Primary Flight Display Failure
procedure Accomplish Refer to Primary Flight Display
Failure procedure in this
section.

— **END** —

◆ **No**

- (1) Instruments Cross-check
- (2) Verify that PFD reference values, speed and Vspeed bugs, and selected altitude are set on both DCPs or are correctly synchronized between both PFDs.

————— **END** —————

1. INSTRUMENTS SYSTEM (CONT'D)

G. EFIS Parameter Miscompare

Indication: EFIS MISCOMP caution message on, and one of the following comparator annunciators shown on the primary flight display:

- ALT • ATT • GS • HDG • IAS
- LOC • PIT • RA • ROL

PFD	ACTION
ALT and/or IAS	(1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Centre pedestal, AIR DATA source selector switch..... Select to reliable side. If any of the following occurs, refer to Chapter 3; EMERGENCY PROCEDURES – INSTRUMENTS SYSTEM – Unreliable Airspeed: • Pitch attitude, thrust setting or external noise not consistent with indicated airspeed • Large airspeed differences between PFDs and/or ISI • Loss of multiple airspeed indication.
ATT, ROL or PIT	(1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Reversionary panel, IRS switch..... Select to reliable alternate source.
HDG	IRS 1 (2) (3) IN ATT status message displayed: Yes (1) FMSEnter present airplane heading on the IRS CONTROL page. No (1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Reversionary panel, IRS switch..... Select to reliable alternate source.
RA	(1) Radio altimeter failure procedureAccomplish Refer to Radio Altimeter Failure procedure, in this section.



1. INSTRUMENTS SYSTEM (CONT'D)

G. EFIS Parameter Miscompare (Cont'd)

PFD	ACTION
LOC or GS	If non-visual, during approach: (1) Go-aroundInitiate

————— **END** —————

H. FMS V Speed Failure

Indication: Amber VSPD flag on pilot or copilot PFD.

- (1) V speeds..... Set manually through the PFD REFS to clear the Flag. Use the AFM to determine V speeds.

NOTE

Do not use the FMS take-off and approach performance calculation function.

————— **END** —————

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1. INSTRUMENTS SYSTEM (CONT'D)

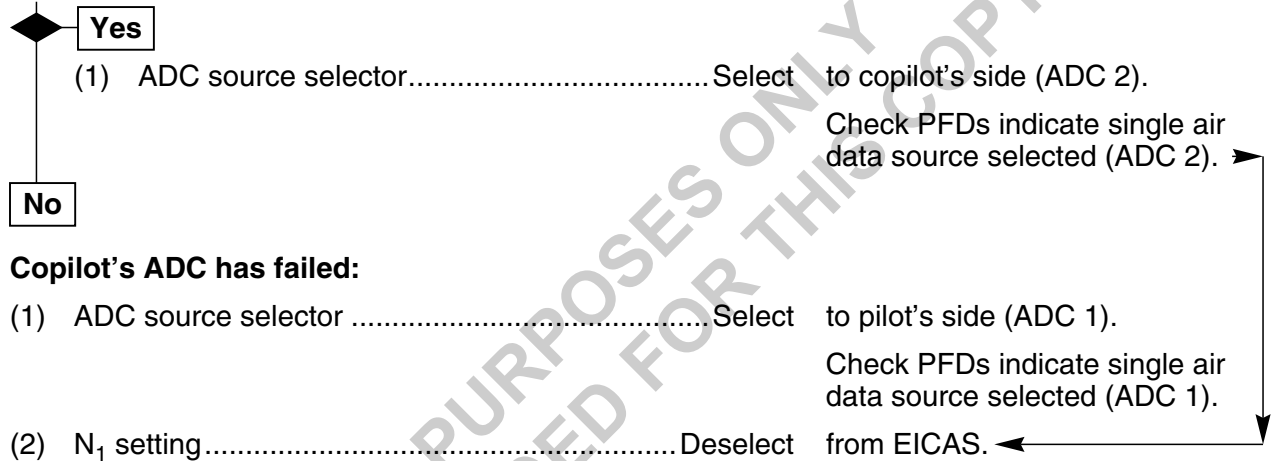
I. Air Data Computer Failure

Indication: ALT, AOA, IAS, MACH and/or VS red flags;
erratic primary flight display data.

NOTE

If FMS 3 is reverted in, use of VNAV guidance is prohibited following reversion.

Pilot's ADC has failed:



NOTE

Data from the CDU can be used if crossed checked with the AFM. TGT thrust mode may be used to set the N₁ setting on EICAS after confirmation of the data.

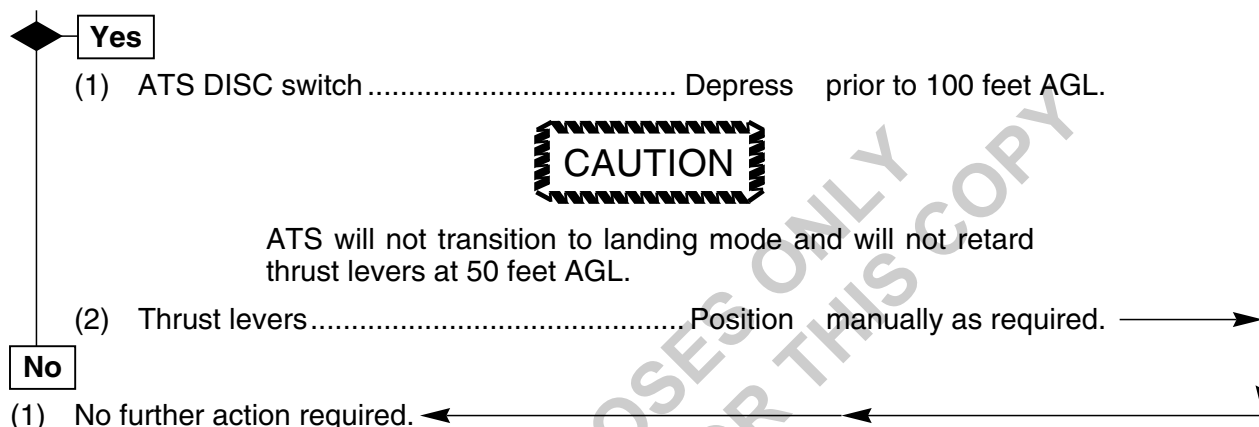
— **END** —

1. INSTRUMENTS SYSTEM (CONT'D)

J. Radio Altimeter Failure

Indication: False or intermittent radio altimeter height and/or
Amber RA miscompare flag and/or red RA fail flag on PFD.

Pilot's radio altimeter has failed:



NOTE

1. Do not rely on affected radio altitude displays.
2. TCAS, TAWS, RAAS (if installed) and windshear detection are inoperative. TCAS FAIL message on PFD or MFD and TAWS BASIC FAIL, RAAS FAIL and TAWS WINDSHEAR FAIL status messages on.

On airplanes incorporating Service Bulletin 650-34-005:

If only one radio altimeter has failed, TCAS, TAWS, RAAS and windshear detection remain operational.

3. For an intermittent or undetected radio altimeter error, the following may be incorrect or inactive:
 - Displayed radio altitudes
 - LOC/GS comparators
 - Excessive deviation alerts
 - Expanded lateral deviation scale
 - Approach minimums annunciation.
4. For an intermittent or undetected radio altimeter error, ATS RETARD mode may occur early or not occur. Refer to POWER PLANT – ATS Radio Altimeter Input Failure in this chapter.
5. For an intermittent or undetected radio altimeter error, localizer and/or glideslope AP/FD commands could be adversely affected.

— END —

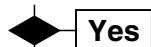
1. INSTRUMENTS SYSTEM (CONT'D)**K. Inertial Reference System Failure**

Indication: Erratic primary flight display data and/or red ATT and/or HDG failure flags.

NOTE

If an IRS is in attitude mode, a heading drift rate of up to 15 degrees per hour can occur. The FMS heading on the IRS CONTROL page should be updated periodically from the best available alternate navigation source.

Pilot's IRS has failed:



(1) IRS source selector Select to an operative IRS.

Check PFDs indicate IRS 2 single source selected (If IRS 3 is installed and selected, PFD 1 indicates IRS 3 source selected).

NOTE

If IRS 3 is not installed, pilot's flight director is inoperative, and FD 1 FAIL status message comes on.

— END —

No

Copilot's IRS has failed:

(1) IRS source selector Select to an operative IRS.

Check PFDs indicate IRS 1 single source selected (If IRS 3 is installed and selected, PFD 2 indicates IRS 3 source selected).

NOTE

If IRS 3 is not installed, copilot's flight director is inoperative, and FD 2 FAIL status message comes on.

— END —

L. Total Inertial Reference System Failure

Indication: Erratic primary flight display data and/or red ATT and/or HDG failure flags on both PFDs.

(1) ISI and standby compass Monitor

(2) Land at the nearest suitable airport.

— END —

CHALLENGER ⁶⁵⁰	ABNORMAL PROCEDURES Instruments System	05-15-8
		Rev 11, Mar 01/18

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CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-1
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM

A. Inboard Anti-Skid Failure

Indication: A/SKID INBD caution message on.

NOTE

A/SKID INBD caution message indicates anti-skid braking is lost on 1 or 2 inboard wheels, or parking brake shut-off valve has failed (PARK BRAKE SOV caution message on).



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If excessive asymmetry or loss of braking is observed:

- (2) Wheel brakes Release momentarily.
 (3) ANTI-SKID switch..... Select to OFF.
 (4) Wheel brakes Re-apply as required.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-2
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Outboard Anti-Skid Failure

Indication: A/SKID OUTBD caution message on.

NOTE

A/SKID OUTBD caution message indicates anti-skid braking is lost on 1 or 2 outboard wheels.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If excessive asymmetry or loss of braking is observed:

- (2) Wheel brakes Release momentarily.
(3) ANTI-SKID switch..... Select to OFF.
(4) Wheel brakes Re-apply as required.

END

C. Anti-Skid (Inboard and Outboard) Failure

Indication: A/SKID INBD caution message and A/SKID OUTBD caution message on.

NOTE

A/SKID INBD and A/SKID OUTBD caution messages indicate anti-skid braking is lost on 2 or more wheels.

- (1) ANTI SKID switch..... Select to OFF.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

END

CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-3
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Inboard/Outboard Brake Pressure Low

Indication: INBD (OUTBD) BRAKE PRESS caution message on.

- (1) Hydraulic pressure and fluid quantity Monitor for the affected system.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.

————— **END** —————

E. Parking Brake Shut-Off Valve Failure

Indication: PARK BRAKE SOV caution message on.

NOTE

PARK BRAKE SOV caution message indicates that the parking brake SOV is failed or is closed with the parking brake not set, possibly affecting inboard anti-skid system performance.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

————— **END** —————

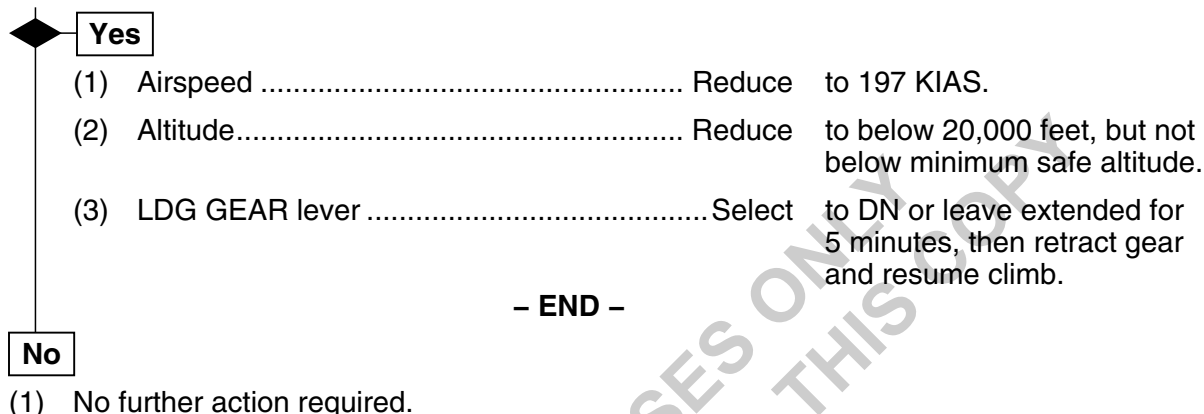
CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-4
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

F. Main Landing Gear Overheat Detector Failure

Indication: MLG OVHT FAIL caution message on.

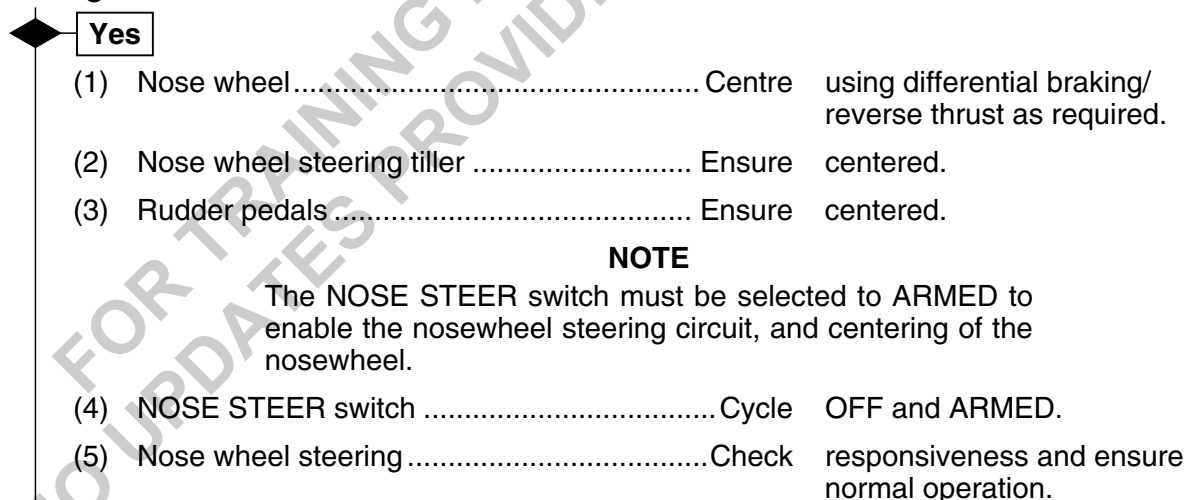
Failure occurs within 30 minutes after take-off:



G. Nose Wheel Steering System Failure

Indication: STEERING INOP caution message on.

During taxi:



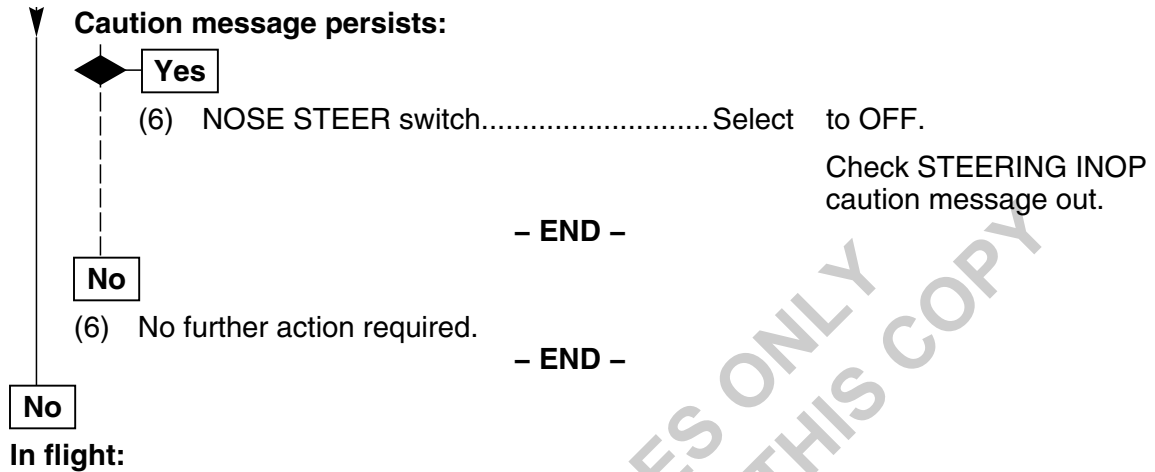
NOTE

Check nose wheel steering responsiveness by taxiing forward and checking proper aircraft response to left/right tiller inputs and rudder pedal inputs.

CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-5
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

G. Nose Wheel Steering System Failure (Cont'd)



NOTE

Centering of the nosewheel steering tiller, rudder pedals, and/or cycling the switch from OFF to ARMED will not enable centering of the nosewheel.

- (1) NOSE STEER switchSelect to OFF.
Check STEERING INOP caution message out.

NOTE

- When landing with the nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind.
- Use differential braking, rudder and engine thrust, as required, to assist directional control.

— END —

H. Weight-on-Wheels Input Fault

Indication: WOW INPUT caution message on.

- NOSE STEER switch Ensure selected to ARMED.
- No further action required.

— END —

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

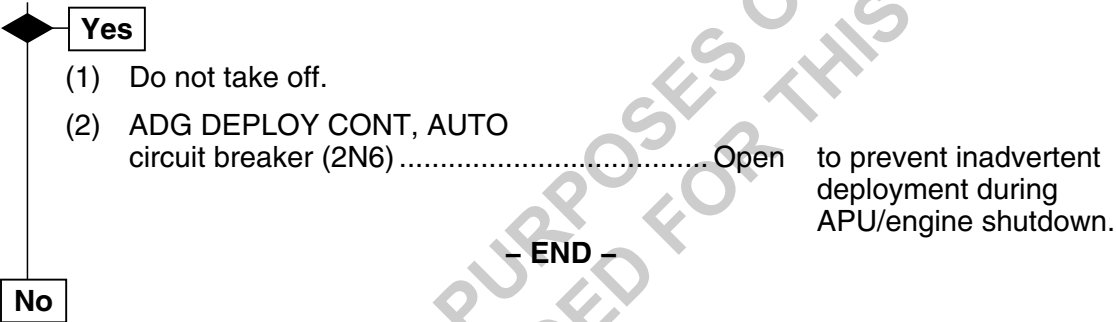
I. Weight-on-Wheels Output Fault

Indication: WOW OUTPUT caution message on.

The following may occur:

- PFD, speed trend vector inoperative.
- PFD, low speed cue comes on; HSI portion indicates TA ONLY,
- CDU/MFD indicates ATC/TCAS STANDBY,
- Inadvertent take-off configuration warnings, and
- Fuel quantity fluctuates during pitch and bank maneuvers.

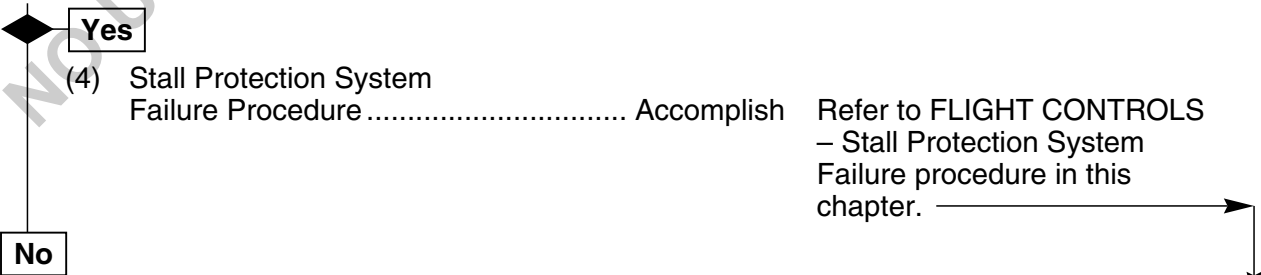
On ground:



In flight:

- (1) Manual Cabin Pressurization Control Procedure Accomplish Refer to AIR-CONDITIONING AND PRESSURIZATION – Manual Cabin Pressurization Control Procedure in this chapter.
- (2) GND SPOILERS switch Set to DISARM.
- (3) NOSE STEER switch Select to OFF.

STALL FAIL caution message is on:



After touchdown:

- (4) GND SPOILERS switch Select to ARM.

CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-7
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

I. Weight-on-Wheels Output Fault (Cont'd)

If loss of braking occurs below 35 knots:

- (5) ANTI SKID switch.....Select to OFF.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

If nose wheel steering is required:

- (6) NOSE STEER switchSelect to ARMED.

After landing:

- (7) ADG DEPLOY CONT, AUTO
circuit breaker (2N6).....Open to prevent inadvertent ADG
deployment during shutdown.

END

CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-8
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

J. Proximity Sensing System Failure (Total System Shutdown)

Indication: PROX SYS caution message on, GND SPLRS caution message on, and landing gear symbols (nose, left and right main gear) display amber dash marks (position unknown) with landing gear lever UP or DN.

NOTE

1. EICAS landing gear position indications are inoperative.
 2. Normal landing gear extension/retraction is not available.
 3. Anti-skid, ground spoilers and nose wheel steering systems are inoperative.
 4. Nose door open warning system is not available.
- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.
- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LDG GEAR lever Select to DN.
- (4) LANDING GEAR MANUAL RELEASE handle Pull to release landing gear.
Confirm gear position with control tower.
- (5) GND SPOILERS switch Set to DISARM.
- (6) ANTI SKID switch Set to OFF.



Extreme caution is required during braking to avoid tire damage or blow out. Maximize use of reverse thrust.

- (7) NOSE STEER switch Select to OFF.

NOTE

If landing with nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind. Use differential braking, rudder, and engine thrust as required during taxi.



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CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05-16-9
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

J. Proximity Sensing System Failure (Total System Shutdown) (Cont'd)

(8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.20 (120%)	1.75 (75%)

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Landing Gear, Wheel and Brake System	05–16–10
		Oct 27/15

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

K. Landing Gear Manual Extension

- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.



Leave hydraulic pump 3B OFF if pump was selected OFF due to a low or decreasing pressure, or a low quantity in system No. 3.

- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LDG GEAR lever Select to DN.
- (4) LANDING GEAR MANUAL RELEASE handle PULL to full extension.

NOTE

The NOSE DOOR OPEN warning message may be displayed when manual gear extension is accomplished.

- (5) Landing gear indication Check for three DN indications.

Three landing gear symbols indicate DN:

Yes



Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing.

- (6) No further action required.

– END –

No

Any landing gear fails to lock in the down position:

- (6) Gear DN Disagree Procedure Accomplish Refer to Chapter 3; EMERGENCY PROCEDURES – LANDING GEAR, WHEEL AND BRAKE SYSTEM – Gear DN Disagree.

————— END —————

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CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-1
		Oct 27/15

1. MISCELLANEOUS SYSTEMS

A. Emergency Lighting System Selected Off

Indication: EMER LTS OFF caution message on.

- (1) EMER LTS switchCheck status of EMER LTS.

————— **END** —————

B. Crew Oxygen Low Pressure

Indication: OXY LO PRESS caution message on.

OXY pressure readout indicates less than 825 psi:

◆ **Yes**

- (1) Descent Initiate to a safe altitude.

— **END** —

No

- (1) Crew oxygen masks Test and check pressure readout.

- (2) Oxygen quantity Monitor

OXY pressure readout drops to 0 psi (bottle(s) closed):

◆ **Yes**

- (3) Descent Initiate to a safe altitude.

— **END** —

No

- (3) No further action required.

————— **END** —————

C. Passenger Oxygen Deployed and Activated (If Installed)

Indication: PAX OXY ON caution message.

- (1) Passenger oxygen.....Check status.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-2
		Oct 27/15

1. MISCELLANEOUS SYSTEMS (CONT'D)

D. Radio Tuning Failure

Indication: Amber radio frequencies displayed and/or inability to tune radios.

NOTE

- Note all frequencies being used prior to starting procedure.
- During procedure, continue tuning and attempting communication even if frequency is displayed in amber.

(1) CDU 1 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on CDU 2 or MFD:

(2) CDU 1 TUNE INHIB switch/light..... Press out

(3) CDU 2 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on CDU 1 or MFD:

(4) CDU 1 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on MFD:

(5) CDU 1 TUNE INHIB and CDU 2
TUNE INHIB switch/lights..... Press out

(6) MFD TUNE INHIB switch/light..... Press in

If unable to restore COM tuning on at least one side:

(7) CDU 1 TUNE INHIB and CDU 2
TUNE INHIB switch/lights..... Press in to force COM 1 to 121.5.

NOTE

- To go straight to the emergency frequency (121.50 MHz), press in CDU 1 TUNE INHIB, CDU 2 TUNE INHIB and MFD TUNE INHIB.
- If 8.33 KHz channel spacing was previously tuned on VHF 1, the emergency frequency (121.50) may be displayed in amber. The amber display does not prevent VHF from operating normally on this frequency.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-3
		Oct 27/15

1. MISCELLANEOUS SYSTEMS (CONT'D)

E. Dual HF Transmission Failure

Indication: No HF transmission. Dashes displayed for HF 1 and HF 2, on CDUs and MFDs.

- (1) HF COMM 2 circuit breakers
(2L8, 2L9 and 2L10) Open
- (2) HF 1 Attempt communication.

Transmission successful:

◆ **Yes**

- (3) No further action required.

— **END** —

□ **No**

- (3) HF COMM 2 circuit breakers
(2L8, 2L9 and 2L10) Close
- (4) HF COMM 1 circuit breakers
(1K5, 1K6 and 1K7) Open
- (5) HF 2 Attempt communication.

— **END** —

F. Transponder Failure

Indication: XPDR 1 (2) FAIL caution message on and/or
Amber XPDR FAIL on CDU or MFD tuning.

- (1) MFD or CDU, ATC/TCAS
CONTROL menu Select alternate transponder.

NOTE

TCAS and ADS-B OUT will stop functioning until alternate transponder is activated.

— **END** —

G. Dual Transponder Failure

Indication: XPDR 1 & 2 FAIL caution message on.

- (1) Contact ATC on radio.

— **END** —

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CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-4 Rev 4, Jun 09/16
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1. MISCELLANEOUS SYSTEMS (CONT'D)

H. ADS-B Out Fault

Indication: ADS-B OUT FAULT caution message on.

- (1) MFD or CDU, ATC/TCAS
CONTROL menu Select alternate transponder.

————— **END** —————

I. ADS-B Out Failure

Indication: ADS-B OUT FAIL caution message on.

- (1) Contact ATC on radio.

————— **END** —————

J. Dual GPS Failure During Oceanic and Remote Operations

Indication: IRS ONLY message on PFD, or FMS DR message on PFD or MFD, or
DR EXCEEDS 5 MIN message on MFD.

- (1) GPS 1 and GPS 2 circuit breakers
(1N3 and 2H11) Open and close.

Message persists after approximately 60 seconds:

◆ **Yes**

- (2) Notify controlling agency.

— **END** —

No

- (2) No further action required.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-5 Rev 11, Mar 01/18
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1. MISCELLANEOUS SYSTEMS (CONT'D)

K. Flight Management System CDU/FMC Failure

Indication: No data available to CDU or CDU is unresponsive. Red FMS flag on PFD.

NOTE

If a red FMS flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to INSTRUMENTS SYSTEM – Air Data Computer Failure or Inertial Reference System Failure in this chapter.

- (1) Reversionary panel, FMS 3 switch
(if installed) Set to failed FMS position.
- (2) CDU 1 (2) TUNE INHIB
switch/light for affected CDU Press in

One FMS is operative:

Yes

- (3) N₁ setting Deselect from EICAS.

NOTE

1. Data from the CDU can be used if crossed checked with the AFM. TGT thrust mode may be used to set the N₁ setting on EICAS after confirmation of the data.
2. The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS 3 reversion (if installed) or an FMS cold start.

– END –

No

- (3) No further action required.

————— **END** —————

L. FSU Failure

Indication: IFIS electronic charts, MFD graphical weather and enhanced map overlays are inoperative.

FSU INOP displayed on both MFDs:

Yes

- (1) Use alternate source for electronic charts.

– END –

No

- (1) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-6 Rev 5, Aug 30/16
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1. MISCELLANEOUS SYSTEMS (CONT'D)

M. RAAS Advisories In Air or During High Speed Operations

If advisory is in conflict with expectations:

- (1) Verify position and/or distance remaining or distance available.
- (2) Contact ATC for assistance if necessary.

NOTE

If aural alerts become distracting or interfere with communications between flight crew or ATC, select the TAWS WARNING RAAS OFF switch/light OFF.

————— **END** —————

N. RAAS Advisories On Ground During Low Speed Operations

If advisory is in conflict with expectations:

- (1) Stop.
- (2) Verify position.
- (3) Contact ATC for assistance if necessary.

NOTE

If aural alerts become distracting or interfere with communications between flight crew or ATC, select the TAWS WARNING RAAS OFF switch/light OFF.

————— **END** —————

O. RAAS Inoperative

Indication: RAAS FAIL status message on.

- (1) TAWS WARNING, RAAS OFF
switch/light..... Press in Check OFF light on.

————— **END** —————

P. RAAS Not Available

Indication: RAAS NOT AVAILABLE status message on.

- (1) TAWS WARNING, RAAS OFF
switch/light..... Press in Check OFF light on.

————— **END** —————

CHALLENGER 650	ABNORMAL PROCEDURES Miscellaneous Systems	05-17-7 Rev 11, Mar 01/18
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1. MISCELLANEOUS SYSTEMS (CONT'D)

Q. EVS Heater Failure

Indication: EVS HEAT FAIL caution message.

- (1) EVS image Monitor for degradation.

————— **END** —————

R. EVS Heater Overheat

Indication: EVS HEAT OVHT caution message.

- (1) EVS image Monitor for degradation.

————— **END** —————

S. Synthetic Vision System Runway Failure

Indication: SVS RWY INOP status message on PFD.

- (1) SVS Select to OFF prior to take-off and prior to passing the final approach fix (FAF).

————— **END** —————

T. Synthetic Vision System Failure

Indication: SVS INOP status message on PFD.

- (1) SVS Select to OFF.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Aural/Visual Warning System	05-18-1
		Oct 27/15

1. AURAL/VISUAL WARNING SYSTEM

A. Loss of EICAS

Indication: MFD displaying EICAS goes blank or EICAS data is not available on selected MFD.

If display blanks, the system automatically transfers the EICAS primary page data to the operational MFD.

- (1) EICAS switch.....Select to BOTH PFD'S.

————— **END** —————

B. Data Concentrator Unit Failure

Indication: DCU 1 (2) INOP status message.

- (1) Affected AUDIO WARNING switch.....Select to DISABLE.

- (2) Other AUDIO WARNING switch.....Select to NORMAL.

————— **END** —————

C. Data Concentrator Unit Aural Failure

Indication: DCU 1 (2) AURAL INOP status message.

- (1) Affected AUDIO WARNING switch.....Select to DISABLE.

- (2) Other AUDIO WARNING switch.....Select to NORMAL.

————— **END** —————

D. EICAS Comparator Inoperative

Indication: EICAS COMP INOP caution message.

- (1) MFD displays (both)Select to EICAS page.

- (2) Engine and instruments, parameters and messagesCross-check for any anomalies.

- (3) MFD, EICAS pageUse reliable side.

- (4) Affected engine and/or systemMaintain within limits.

————— **END** —————

E. TCAS System Failure

Indication: TCAS FAIL caution message on PFD and/or MFD.

- (1) MFD or CDU tuning, ATC/TCAS modeSet to ALT ON, to select TCAS OFF.

Check TCAS OFF indicated on MFDs/PFDs.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Aural/Visual Warning System	05-18-2
		Oct 27/15

1. AURAL/VISUAL WARNING SYSTEM (CONT'D)

F. TCAS VSI Resolution Advisory Failure

Indication: NO VSI RA on PFD or MFD.

- (1) MFD or CDU tuning, ATC/TCAS
mode Set to TA ONLY.

————— **END** —————

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CHALLENGER 650	ABNORMAL PROCEDURES Other Procedures	05-19-1 Rev 14, Nov 19/18
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1. OTHER PROCEDURES

There are currently no abnormal procedures for this section.

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1. INTRODUCTION

The data in this section present the performance information of the CL600-2B16 Challenger airplane (Serial No. 6050 and subsequent), fitted with two General Electric CF34-3B engines.

The airspeeds and airplane configurations for take-off, climb and landing, as presented in this chapter, must be adhered to during the appropriate phase of flight.

All performance data are based on the following conditions:

- Minimum guaranteed engine thrust with allowance made for the installation effects, including bleed and mechanical horsepower extraction.
- Maximum nominal brake operating pressure of 3,000 psi, with anti-skid and automatic ground lift spoilers operational.
- ICAO standard atmosphere conditions (ISA), with corrections for non-standard conditions, when applicable.
- Dry, hard runway surface, unless otherwise noted.
- Flight spoilers deployed manually after touchdown or during a rejected take-off.
- Wind speed is measured at a height of 10 meters (33 feet) above the surface.
- The performance data in this chapter are not valid when:
 - The weight exceeds the maximum weight as limited by climb requirements.
 - Data is extrapolated from the values shown in the charts.
- For weights below 12,000 kg (26,455 lb) assume a weight of 12,000 kg (26,455 lb).
- For airport pressure altitudes below 0 ft, the performance may be assumed to be that appropriate to an airport altitude of 0 ft, unless information for altitudes below 0 ft is presented.

A number of performance charts presented in this chapter have correction grids with a reference line. If entering the chart from the side wherein the label "Ref." or "REFERENCE LINE" or "REF. LINE" is encountered first (usually, from the top of the chart then downwards), then the correct procedure is to go to the reference line first and then follow the guidelines (if necessary) towards the appropriate condition. Conversely, if entering the chart from the opposite direction as mentioned above, then the procedure is to go to the appropriate condition first. These procedures are clarified by the examples on the charts.

The performance data have been based on:

- A minimum V_2 speed of $1.13 V_{SR}$, and
- A minimum V_{REF} speed of $1.23 V_{SR}$.

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2. CONVERSION DATA

A. Temperature Conversion

A table to convert air temperature at a given pressure altitude into a value of temperature above or below the International Standard Atmosphere (ISA) is shown on Figure 06-01-1.

Example:

Figure 06-01-1 shows that at an atmosphere of ISA + 10°C and at an altitude of 29,000 feet, the air temperature is -32°C. Alternatively, at an altitude of 29,000 feet and an air temperature of -32°C, the atmosphere is ISA + 10°C.

Also given by Figure 06-01-1 is a conversion of temperature from °C to °F, or vice-versa.

Example:

59°F is equivalent to 15°C.

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2. CONVERSION DATA (CONT'D)

Alt. ft	ISA - 30 C		ISA - 20 C		ISA - 10 C		ISA		ISA + 10 C		ISA + 20 C		ISA + 30 C	
	C	F	C	F	C	F	C	F	C	F	C	F	C	F
0	-15.0	5.0	-5.0	23.0	5.0	41.0	15.0	59.0	25.0	77.0	35.0	95.0	45.0	113.0
1000	-17.0	1.4	-7.0	19.4	3.0	37.4	13.0	55.4	23.0	73.4	33.0	91.4	43.0	109.4
2000	-18.9	-2.1	-8.9	15.9	1.1	33.9	11.1	51.9	21.1	69.9	31.1	87.9	41.1	105.9
3000	-20.9	-5.7	-10.9	12.3	-0.9	30.3	9.1	48.3	19.1	66.3	29.1	84.3	39.1	102.3
4000	-22.9	-9.3	-12.9	8.7	-2.9	26.7	7.1	44.7	17.1	62.7	27.1	80.7	37.1	98.7
5000	-24.9	-12.8	-14.9	5.2	-4.9	23.2	5.1	41.2	15.1	59.2	25.1	77.2	35.1	95.2
6000	-26.9	-16.4	-16.9	1.6	-6.9	19.6	3.1	37.6	13.1	55.6	23.1	73.6	33.1	91.6
7000	-28.9	-20.0	-18.9	-2.0	-8.9	16.0	1.1	34.0	11.1	52.0	21.1	70.0	31.1	88.0
8000	-30.8	-23.5	-20.8	-5.5	-10.8	12.5	-0.8	30.5	9.2	48.5	19.2	66.5	29.2	84.5
9000	-32.8	-27.1	-22.8	-9.1	-12.8	8.9	-2.8	26.9	7.2	44.9	17.2	62.9	27.2	80.9
10000	-34.8	-30.7	-24.8	-12.7	-14.8	5.3	-4.8	23.3	5.2	41.3	15.2	59.3	25.2	77.3
11000	-36.8	-34.2	-26.8	-16.2	-16.8	1.8	-6.8	19.8	3.2	37.8	13.2	55.8	23.2	73.8
12000	-38.8	-37.8	-28.8	-19.8	-18.8	-1.8	-8.8	16.2	1.2	34.2	11.2	52.2	21.2	70.2
13000	-40.8	-41.4	-30.8	-23.4	-20.8	-5.4	-10.8	12.6	-0.8	30.6	9.2	48.6	19.2	66.6
14000	-42.7	-44.9	-32.7	-26.9	-22.7	-8.9	-12.7	9.1	-2.7	27.1	7.3	45.1	17.3	63.1
15000	-44.7	-48.5	-34.7	-30.5	-24.7	-12.5	-14.7	5.5	-4.7	23.5	5.3	41.5	15.3	59.5
16000	-46.7	-52.1	-36.7	-34.1	-26.7	-16.1	-16.7	1.9	-6.7	19.9	3.3	37.9	13.3	55.9
17000	-48.6	-55.6	-38.6	-37.6	-28.6	-19.6	-18.6	-1.6	-8.6	16.4	1.4	34.4	11.4	52.4
18000	-50.7	-59.2	-40.7	-41.2	-30.7	-23.2	-20.7	-5.2	-10.7	12.8	-0.7	30.8	9.3	48.8
19000	-52.7	-62.8	-42.7	-44.8	-32.7	-26.8	-22.7	-8.8	-12.7	9.2	-2.7	27.2	7.3	45.2
20000	-54.6	-66.3	-44.6	-48.3	-34.6	-30.3	-24.6	-12.3	-14.6	5.7	-4.6	23.7	5.4	41.7
21000	-56.6	-69.9	-46.6	-51.9	-36.6	-33.9	-26.6	-15.9	-16.6	2.1	-6.6	20.1	3.4	38.1
22000	-58.6	-73.5	-48.6	-55.5	-38.6	-37.5	-28.6	-19.5	-18.6	-1.5	-8.6	16.5	1.4	34.5
23000	-60.5	-77.0	-50.5	-59.0	-40.5	-41.0	-30.5	-23.0	-20.5	-5.0	-10.5	13.0	-0.5	31.0
24000	-62.5	-80.6	-52.5	-62.6	-42.5	-44.6	-32.5	-26.6	-22.5	-8.6	-12.5	9.4	-2.5	27.4
25000	-64.5	-84.2	-54.5	-66.2	-44.5	-48.2	-34.5	-30.2	-24.5	-12.2	-14.5	5.8	-4.5	23.8
26000	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7	-16.5	2.3	-6.5	20.3
27000	-68.5	-91.3	-58.5	-73.3	-48.5	-55.3	-38.5	-37.3	-28.5	-19.3	-18.5	-1.3	-8.5	16.7
28000	-70.5	-94.9	-60.5	-76.9	-50.5	-58.9	-40.5	-40.9	-30.5	-22.9	-20.5	-4.9	-10.5	13.1
29000	-72.4	-98.4	-62.4	-80.4	-52.4	-62.4	-42.4	-44.4	-32.4	-26.4	-22.4	-8.4	-12.4	9.6
30000	-74.4	-102.0	-64.4	-84.0	-54.4	-66.0	-44.4	-48.0	-34.4	-30.0	-24.4	-12.0	-14.4	6.0
31000	-76.4	-105.6	-66.4	-87.6	-56.4	-69.6	-46.4	-51.6	-36.4	-33.6	-26.4	-15.6	-16.4	2.4
32000	-78.4	-109.1	-68.4	-91.1	-58.4	-73.1	-48.4	-55.1	-38.4	-37.1	-28.4	-19.1	-18.4	-1.1
33000	-80.4	-112.7	-70.4	-94.7	-60.4	-76.7	-50.4	-58.7	-40.4	-40.7	-30.4	-22.7	-20.4	-4.7
34000	-82.4	-116.3	-72.4	-98.3	-62.4	-80.3	-52.4	-62.3	-42.4	-44.3	-32.4	-26.3	-22.4	-8.3
35000	-84.3	-119.8	-74.3	-101.8	-64.3	-83.8	-54.3	-65.8	-44.3	-47.8	-34.3	-29.8	-24.3	-11.8
36000	-86.3	-123.4	-76.3	-105.4	-66.3	-87.4	-56.3	-69.4	-46.3	-51.4	-36.3	-33.4	-26.3	-15.4
37000	-86.5	-123.7	-76.5	-105.7	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7
38000	-86.5	-123.7	-76.5	-105.7	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7
39000	-86.5	-123.7	-76.5	-105.7	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7
40000	-86.5	-123.7	-76.5	-105.7	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7
41000	-86.5	-123.7	-76.5	-105.7	-66.5	-87.7	-56.5	-69.7	-46.5	-51.7	-36.5	-33.7	-26.5	-15.7

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Temperature Conversion
Figure 06-01-1

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2. CONVERSION DATA (CONT'D)

B. Wind Component

A chart to convert wind velocity into a headwind or tailwind component is shown on Figure 06-01-2.

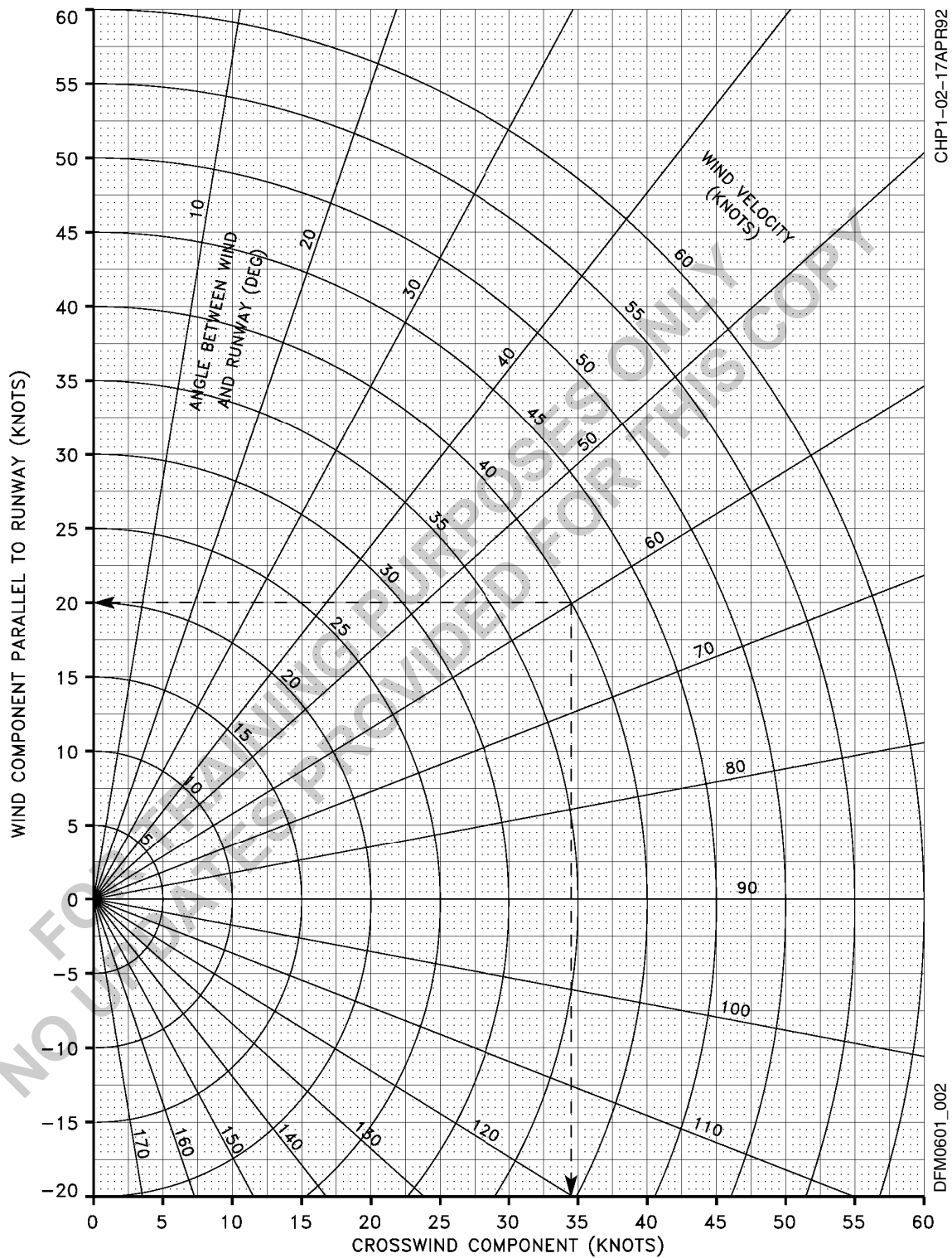
Example:

Figure 06-01-2 shows that for a wind velocity of 40 kt at an angle of 60 degrees to the runway centreline, the wind component parallel to the runway (headwind) is 20 knots, with a crosswind velocity of 34.5 knots.

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2. CONVERSION DATA (CONT'D)

Wind Component
Figure 06-01-2

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3. CALIBRATION

The position error corrections for airspeed and altitude for varying airplane speeds and altitudes, in both the landing gear extended and retracted configurations are shown on Figures 06-01-3 through 06-01-6.

Position error correction variation with airplane weight is negligible. All figures assume instrument error to be zero.

Position error corrections to altitude, airspeed and Mach number for the flaps-up configuration, are negligible for the primary flight displays.

The displayed static air temperature (SAT) is unreliable on the ground since the TAT probe gives inaccurate readings when the airplane is static or at low forward speed. In flight, the difference between OAT and indicated SAT is negligible.

Ground airspeed position errors for flaps 20° are negligible.

A. Airspeed Position Error Correction – Primary Flight Display

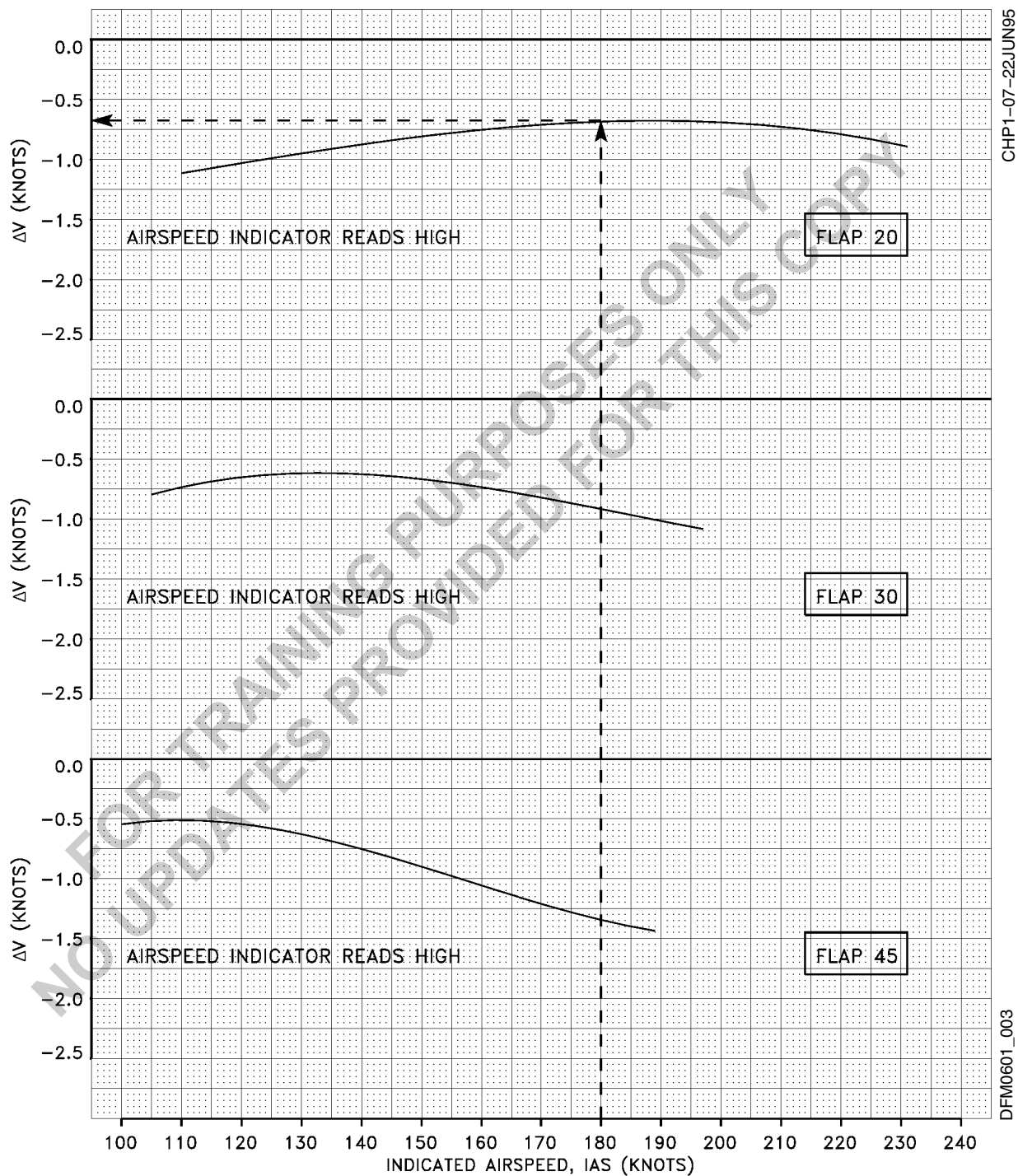
Figure 06-01-3 provides the airspeed position error correction for varying conditions of flap settings and indicated airspeed, for the primary flight display.

Example:

Indicated airspeed (IAS)	= 180 KIAS
Flaps setting	= 20°
Position correction (ΔV)	= -0.7 knot
Calibrated airspeed (CAS)	= IAS + ΔV
	= 180 + (-0.7)
	= 179.3 KCAS

3. CALIBRATION (CONT'D)

Valid for Landing Gear Extended or Retracted

Calibrated Airspeed: CAS = IAS + ΔV Airspeed Position Error Correction – Primary Flight Display
Figure 06-01-3

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3. CALIBRATION (CONT'D)

B. Altitude Position Error Correction – Primary Flight Display

Figure 06-01-4 provides the altitude position error correction for varying conditions of flap settings and indicated airspeed, for the primary flight display.

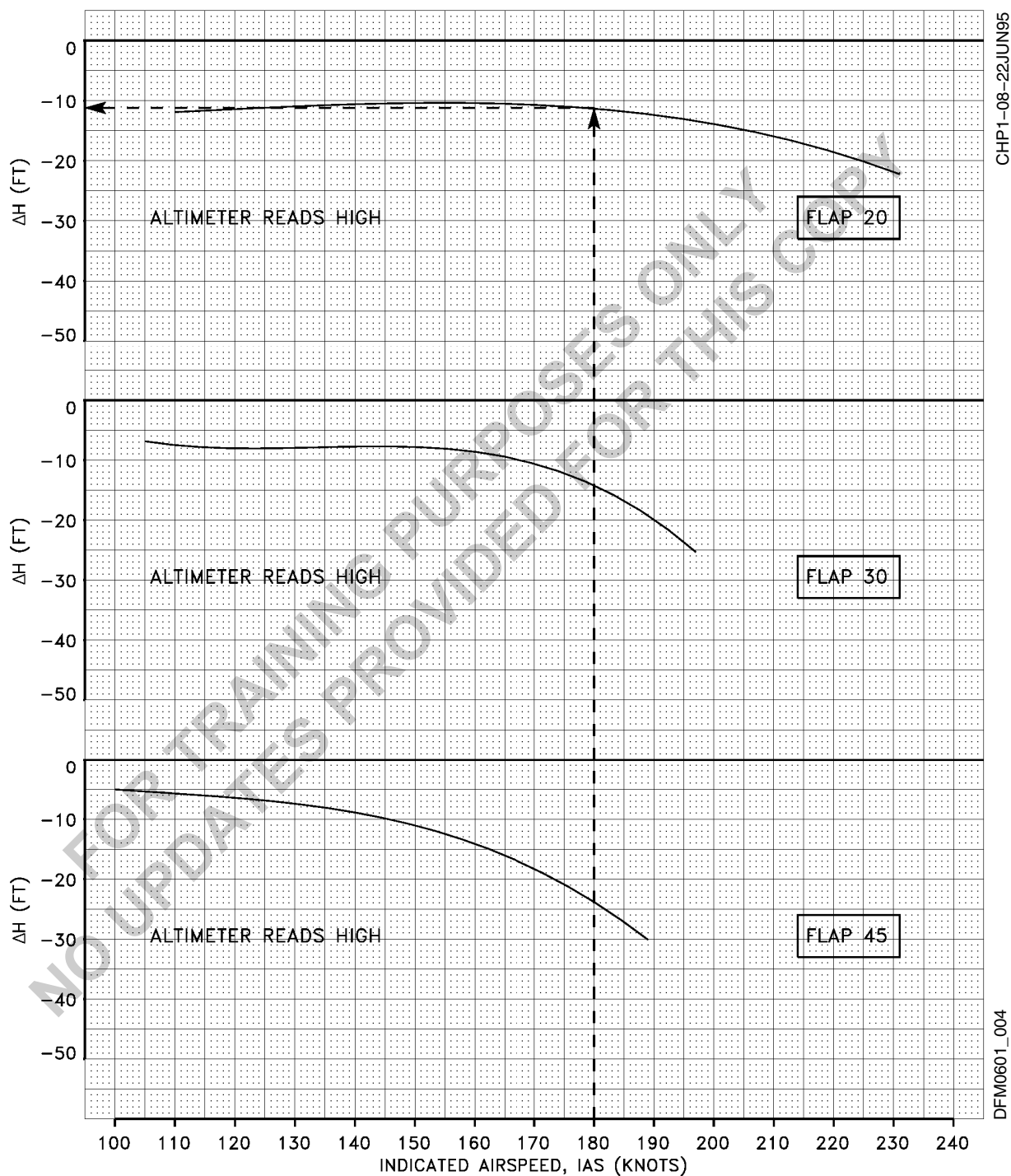
Example:

Indicated altitude (H_I)	= 1,500 feet
Indicated airspeed (IAS)	= 180 KIAS
Flaps setting	= 20°
Position correction (ΔH)	= -11 feet
True pressure altitude (H_P)	= $H_I + \Delta H$
	= 1,500 + (-11)
	= 1,489 feet

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3. CALIBRATION (CONT'D)

Valid for Landing Gear Extended or Retracted

True Pressure Altitude: $H_p = H_i + \Delta H$ 

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DFM0601_004

Altitude Position Error Correction – Primary Flight Display
Figure 06-01-4

CHALLENGER 650	PERFORMANCE General	06-01-10
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3. CALIBRATION (CONT'D)

C. Airspeed Position Error Correction – Integrated Standby Instrument (Flaps 20°, 30°, 45°)

Figure 06-01-5 provides the airspeed position error corrections for varying conditions of flap setting and varying indicated airspeed, for the Integrated Standby Instrument.

Example:

Indicated airspeed (IAS)	= 180 KIAS
Flaps setting	= 20°
Airspeed position correction (ΔV)	= 0.7 knots
Calibrated airspeed (CAS)	= IAS + ΔV
	= 180 + (0.7)
	= 180.7 KCAS

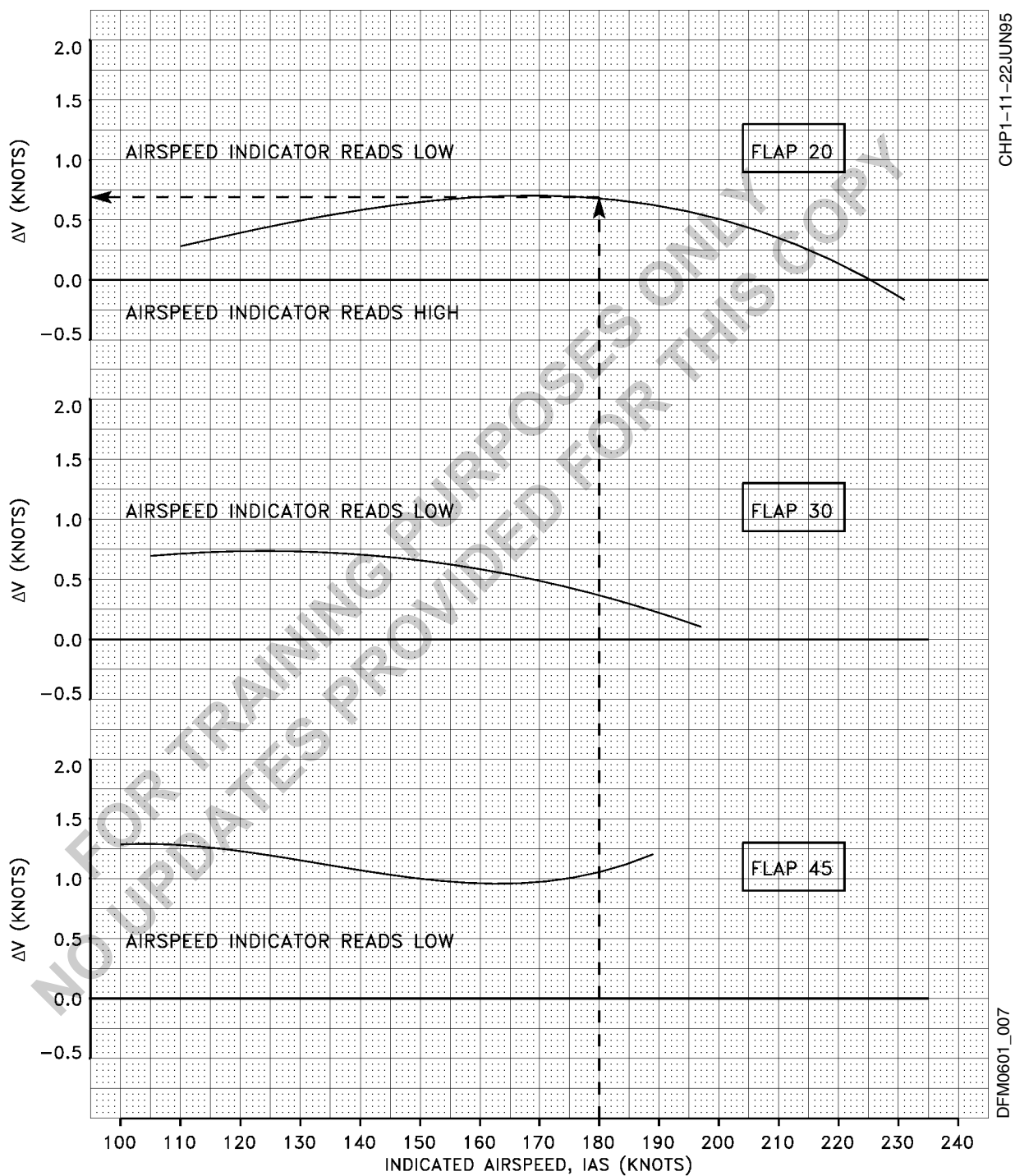
NOTE

Flaps setting of 0° (retracted) does not require airspeed position error corrections.

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3. CALIBRATION (CONT'D)

Valid for Landing Gear Extended or Retracted
Calibrated Airspeed: CAS = IAS + ΔV



Airspeed Position Error Correction – Integrated Standby Instrument (Flaps 20°, 30°, 45°)
Figure 06-01-5

CHALLENGER 650	PERFORMANCE General	06-01-12
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3. CALIBRATION (CONT'D)

D. Altitude Position Error Correction – Integrated Standby Altimeter (Flaps 20°, 30°, 45°)

Figure 06-01-6 provides the altitude position error corrections for varying conditions of flap setting and varying indicated airspeed, for the Integrated Standby Altimeter.

Example:

Indicated altitude (H_I)	= 5,000 feet
Indicated airspeed (IAS)	= 180 KIAS
Flaps setting	= 20°
Position correction (ΔH)	= 17 feet
True pressure altitude (H_p)	= $H_I + \Delta H$
	= 5,000 + (17)
	= 5,017 feet

NOTE

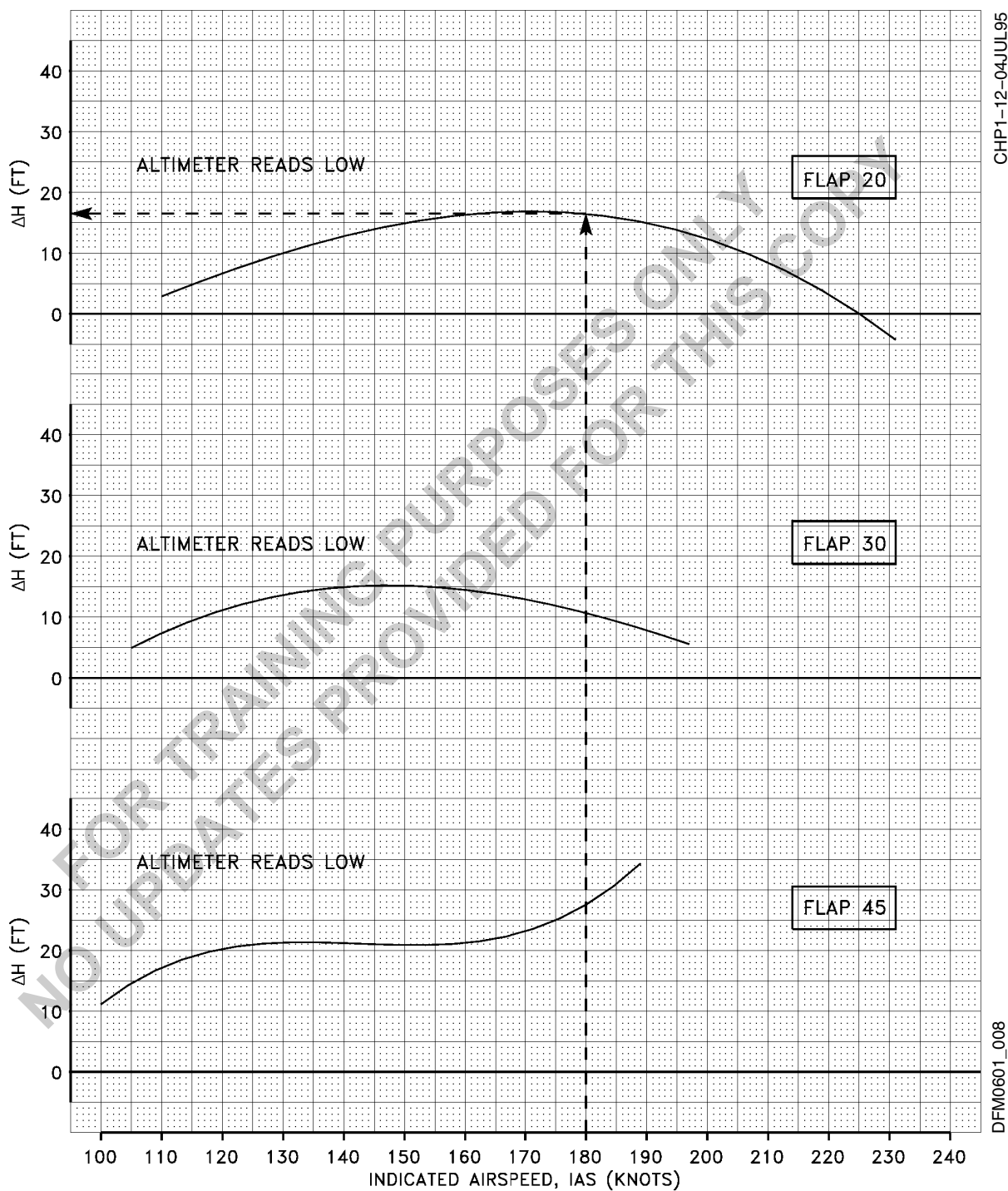
Flaps setting of 0° (retracted) does not require altitude position error corrections.

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3. CALIBRATION (CONT'D)

Valid for Landing Gear Extended or Retracted

True Pressure Altitude: $H_p = H_i + \Delta H$ Altitude Position Error Correction – Integrated Standby Altimeter (Flaps 20°, 30°, 45°)
Figure 06-01-6

CHALLENGER 650	PERFORMANCE General	06-01-14
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4. FLIGHT CAPABILITIES

A. Reference Stall Speed – Flaps 0°/Landing Gear Up

The reference stall speeds for a flaps 0° and landing gear up configuration at varying gross weights and pressure altitude, are provided in Figure 06-01-7. This chart is based upon the most forward centre of gravity limit at flaps 0°. The speeds shown are valid only for pressure altitudes at and below 15,000 feet. The reference stall speed, V_{SR} , is equal to V_{S1g} multiplied by a factor of 1.02.

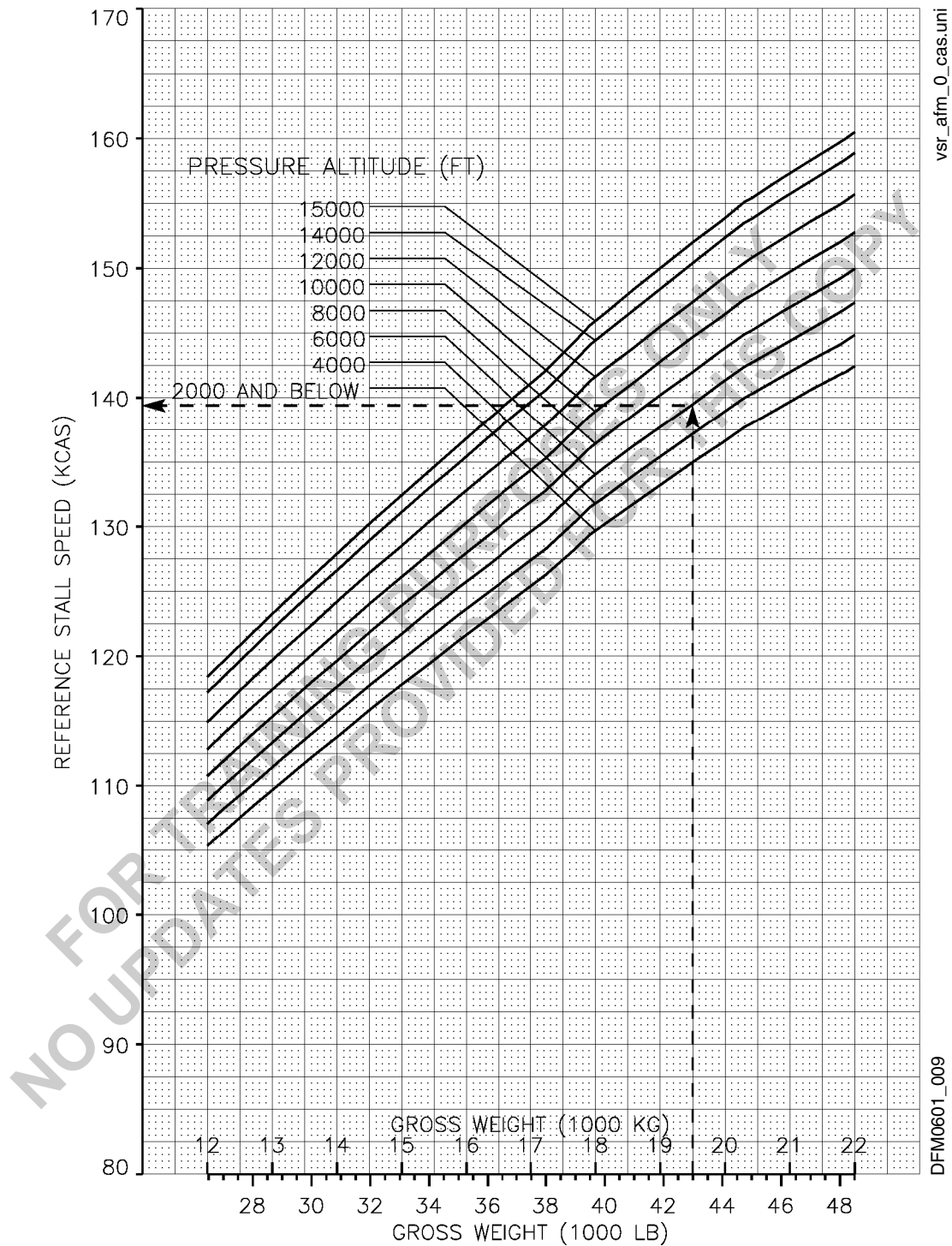
Example:

Gross weight	= 19,500 kg (42,990 lb)
Flaps setting	= 0°
Landing gear	= Up
Pressure altitude	= 6,000 feet
Reference stall speed	= 139.4 KCAS

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4. FLIGHT CAPABILITIES (CONT'D)



Reference Stall Speed – Flaps 0°/Landing Gear Up
Figure 06-01-7

CHALLENGER 650	PERFORMANCE General	06-01-16
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4. FLIGHT CAPABILITIES (CONT'D)

B. Reference Stall Speed – Flaps 20°/Landing Gear Up

The reference stall speeds for a flaps 20° and landing gear up configuration at varying gross weights and pressure altitude, are provided in Figure 06-01-8. This chart is based upon the most forward take-off centre of gravity limit at flaps 20°. The speeds shown are valid only for pressure altitudes at and below 15,000 feet.

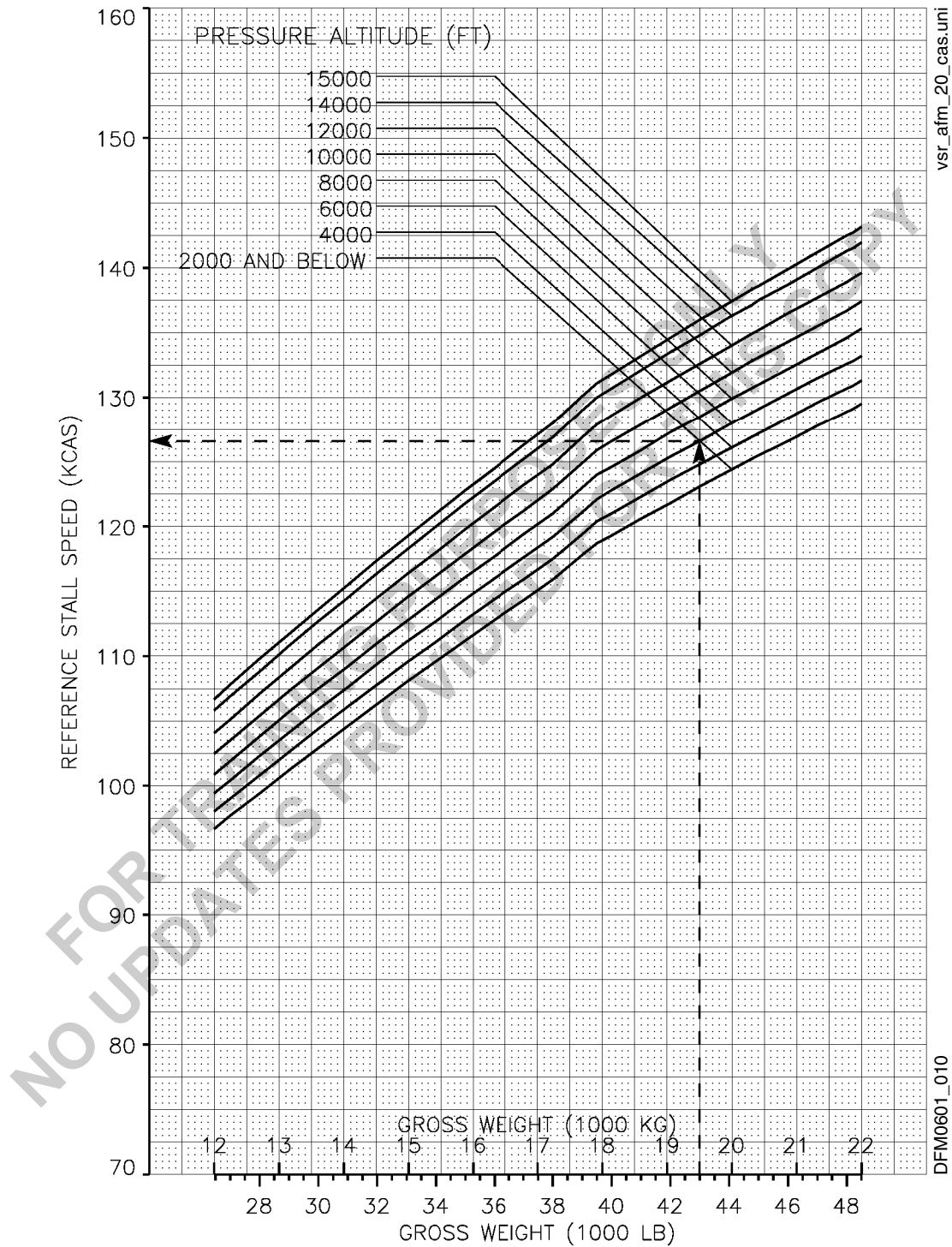
Example:

Gross weight	= 19,500 kg (42,990 lb)
Flaps setting	= 20°
Landing gear	= Up
Pressure altitude	= 6,000 feet
Reference stall speed	= 126.6 KCAS

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4. FLIGHT CAPABILITIES (CONT'D)



Reference Stall Speed – Flaps 20°/Landing Gear Up
Figure 06-01-8

CHALLENGER 650	PERFORMANCE General	06-01-18
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4. FLIGHT CAPABILITIES (CONT'D)

C. Reference Stall Speed – Flaps 30°/Landing Gear Up

The reference stall speeds for a flaps 30° and landing gear up configuration at varying gross weights and pressure altitude, are provided in Figure 06-01-9. This chart is based upon the most forward centre of gravity limit at flaps 30°. The speeds shown are valid only for pressure altitudes at and below 15,000 feet.

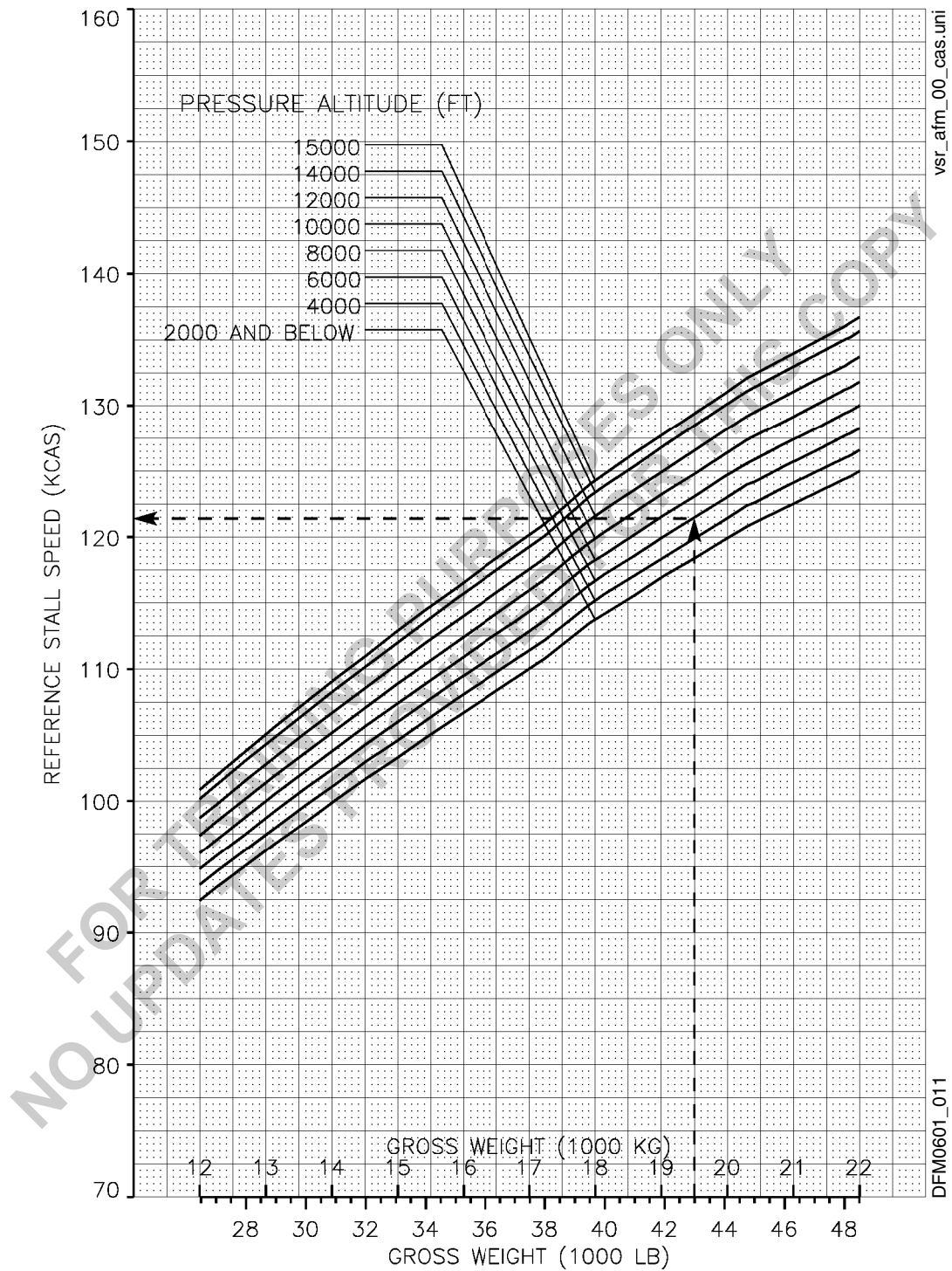
Example:

Gross weight	= 19,500 kg (42,990 lb)
Flaps setting	= 30°
Landing gear	= Up
Pressure altitude	= 6,000 feet
Reference stall speed	= 121.4 KCAS

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4. FLIGHT CAPABILITIES (CONT'D)



Reference Stall Speed – Flaps 30°/Landing Gear Up
Figure 06-01-9

CHALLENGER 650	PERFORMANCE General	06-01-20
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4. FLIGHT CAPABILITIES (CONT'D)

D. Reference Stall Speed – Flaps 45°/Landing Gear Down

The reference stall speeds for a flaps 45° and landing gear down configuration at varying gross weights and pressure altitude, are provided in Figure 06-01-10. This chart is based upon the most forward centre of gravity limit at flaps 45°. The speeds shown are valid only for pressure altitudes at and below 15,000 feet.

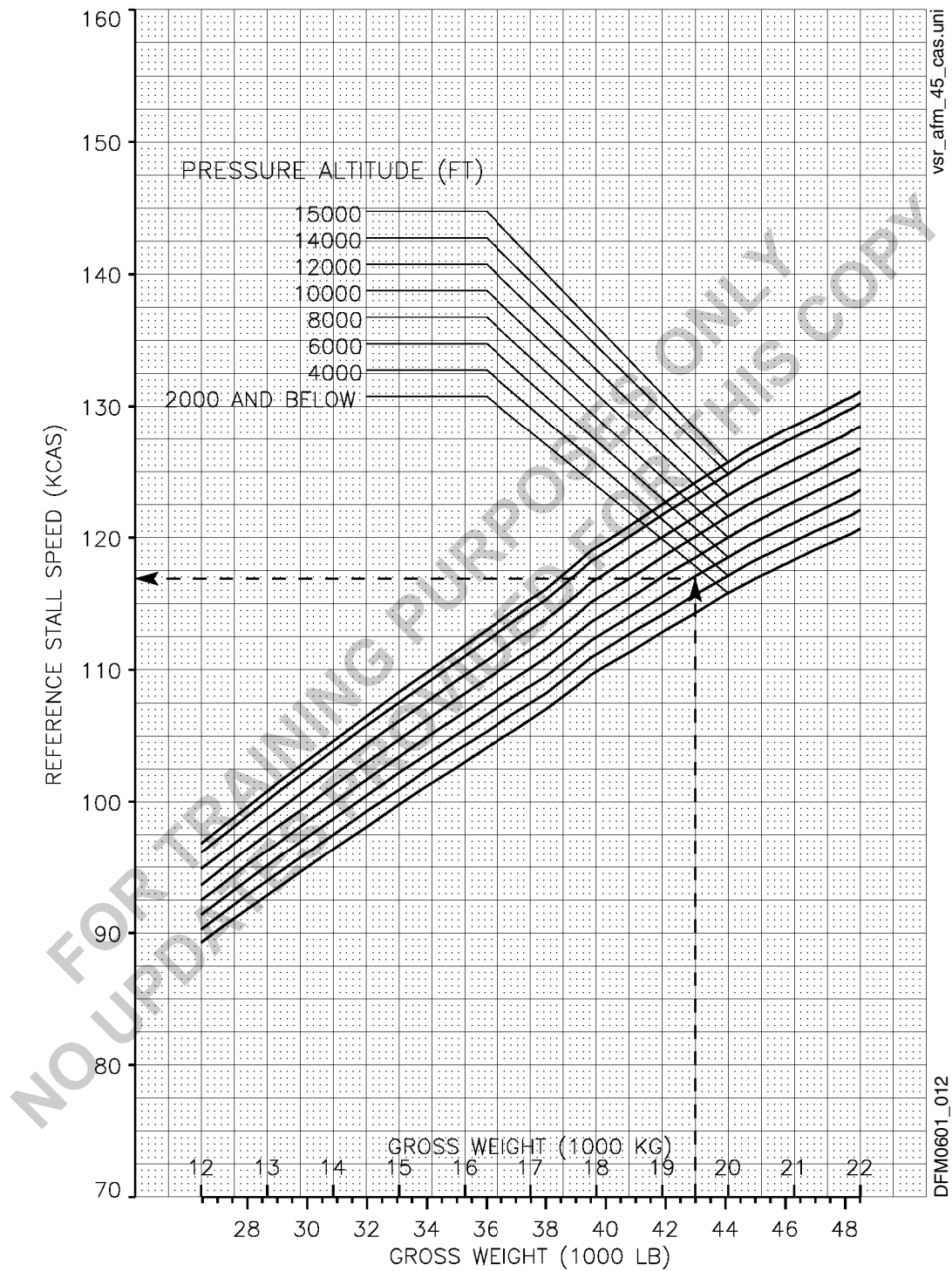
Example:

Gross weight	= 19,500 kg (42,990 lb)
Flaps setting	= 45°
Landing gear	= Down
Pressure altitude	= 6,000 feet
Reference stall speed	= 116.9 KCAS

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4. FLIGHT CAPABILITIES (CONT'D)



Reference Stall Speed – Flaps 45°/Landing Gear Down
Figure 06-01-10

CHALLENGER 650	PERFORMANCE General	06-01-22
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4. FLIGHT CAPABILITIES (CONT'D)

E. Maneuvering Capabilities

The maneuvering capabilities are shown in Figure 06-01-11. The chart shows the maneuver margin (bank angle and/or g-load factor) for a given weight, C.G., altitude and speed combination. Alternatively, for a 1.0 g load factor, the chart shows the altitude and speed margins for a given weight, C.G., and speed combination.

Maneuvering capability is defined relative to buffet onset or stick shaker activation, whichever occurs first.

Example:

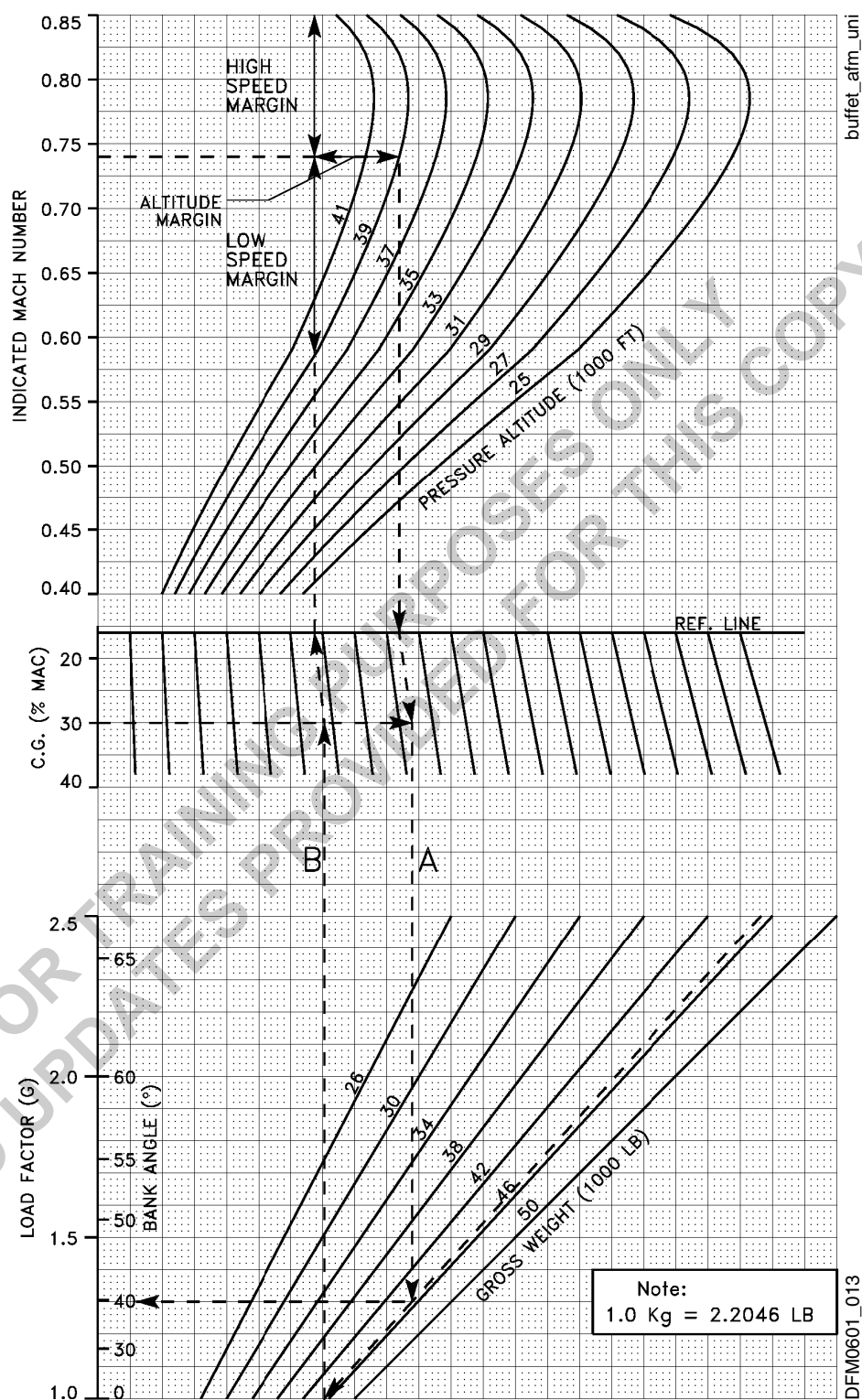
Associated conditions:

Airplane gross weight	= 20,412 kg (45,000 lb)
Centre of Gravity (C.G.)	= 30% MAC
Indicated Mach No.	= 0.74
Pressure altitude	= 39,000 feet

Example A in Figure 06-01-11, for the associated conditions mentioned above (enter the chart from the indicated mach number scale), shows that the maneuvering capability is equal to 1.3 g or a bank angle of 40°.

Example B in Figure 06-01-11, for the associated conditions mentioned above (enter the chart from the gross weight scale), shows the altitude and speed margins before buffet or stick shaker activation, for a load factor of 1.0 g.

4. FLIGHT CAPABILITIES (CONT'D)

Maneuvering Capabilities
Figure 06-01-11

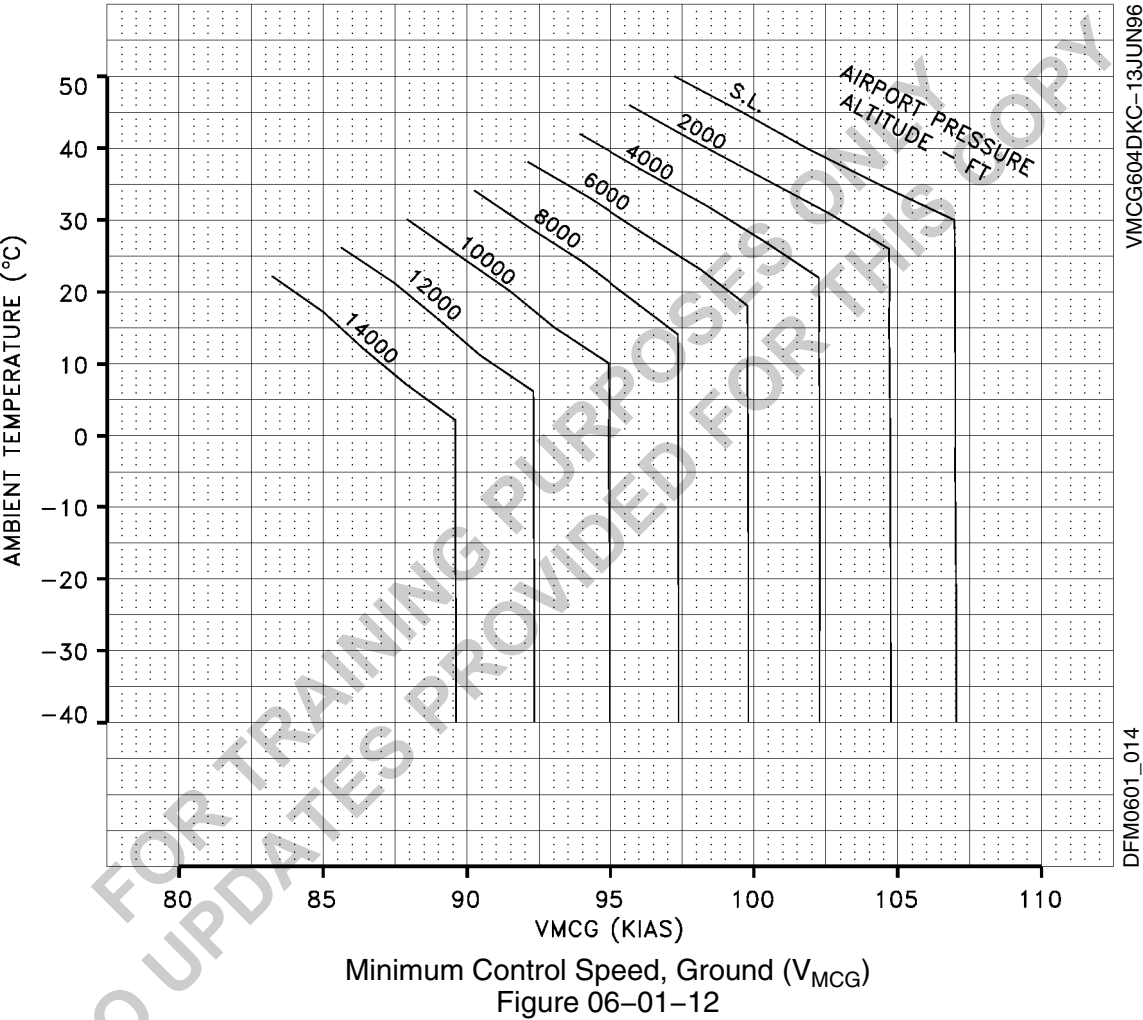
5. PERFORMANCE CONDITIONS AND CONFIGURATIONS

A. Minimum Control Speed, Air (V_{MCA})

V_{MCA} in the take-off configuration = 116 KIAS

B. Minimum Control Speed, Ground (V_{MCG})

The minimum control speed on the ground, for varying conditions of ambient temperature and airport pressure altitude, is shown on Figure 06-01-12.



C. Minimum Control Speed, Landing (V_{MCL})

V_{MCL} in the approach and landing configuration = 116 KIAS

D. Demonstrated Crosswind (Take-Off and Landing)

The maximum demonstrated crosswind component for take-off and landing [at 33 feet (10 meters) tower height] is 24 knots and is not considered limiting for take-off and landing. When using reverse thrust, this speed is limiting.

Landings have been performed at an approach speed of $V_{REF} + 1/2$ of the steady state crosswind component, plus all of the gust (regardless of direction) up to a maximum of $V_{REF} + 20$ knots.

CHALLENGER 650	PERFORMANCE General	06-01-25
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5. PERFORMANCE CONDITIONS AND CONFIGURATIONS (CONT'D)

E. Maximum Tailwind (Take-Off and Landing)

The maximum tailwind component for take-off and landing [at 33 feet (10 meters) tower height] is 10 knots.

F. Configurations and Thrusts

PHASE OF FLIGHT	ENGINE THRUST SETTING	FLAPS (DEGREES)	LANDING GEAR
Take - Off	Take-off thrust, 2 engines to V_{EF} , then 1 engine at automatic performance reserve (APR), or take-off thrust if APR is not armed.	20	Down
First Segment Climb	Automatic performance reserve (APR), or take-off thrust if APR not armed, 1 engine.	20	Down
Second Segment Climb	Same as First Segment.	20	Up
Final Take-Off Climb	Maximum continuous thrust, 1 engine	0	Up
Enroute Climb	Maximum continuous thrust, 1 engine	0	Up
Approach Climb	Single engine climb (APR rating) thrust, 1 engine, set manually. FAA Certified Airplanes Normal take-off thrust, 1 engine	20	Up
Landing Climb	Normal take-off thrust, 2 engines	45	Down
Landing	Idle, 2 engines	45	Down

Take-off performance data are presented allowing for the use of the APR system and for situations where APR is not used. The effects of 10th stage engine bleeds (for Packs) open and closed, and cowl and wing anti-ice systems on and off, are also presented.

NOTE

Flaps 30° is an intermediate flap position and no procedures or performance data are predicated on its use.

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CHALLENGER 650	PERFORMANCE Thrust Settings	06-02-1
		Oct 27/15

1. THRUST SETTINGS

A. General

This section presents normal take-off thrust settings, single engine climb thrust (APR rating) settings and single engine maximum continuous thrust (MCT) settings for various ambient temperatures, pressure altitudes and engine bleed configurations.

Normal take-off thrust settings are used during the following conditions:

- (1) Normal take-off, set statically;
- (2) Go-around with all engines operating (landing climb).

FAA Certified Airplanes:

- (1) Normal take-off, set statically;
- (2) Go-around.

Single engine climb thrust (APR rating) settings are used during the following conditions:

- (1) Manual readjustment of thrust setting during the second segment climb, for a take-off with one engine inoperative;
- (2) Go-around with one engine inoperative (approach climb).

FAA Certified Airplanes:

- (1) Manual readjustment of thrust setting during the second segment climb, for a take-off with one engine inoperative.

Single engine maximum continuous thrust (MCT) settings are used during the following conditions:

- (1) Level flight acceleration from V_2 to V_{FTO} ;
- (2) Final segment/enroute climb.

B. Thrust Setting Charts

Eleven (11) charts are presented, each for a specific configuration:

Figure 06-02-1 provides normal take-off thrust settings with zero bleed air extraction from the engines.

Figure 06-02-2 provides normal take-off thrust settings with 10th stage bleeds open/packs on.

Figure 06-02-3 provides normal take-off thrust settings with 10th stage bleeds closed and engine cowl anti-icing on.

Figure 06-02-4 provides normal take-off thrust settings with 10th stage bleeds closed and engine cowl and wing anti-icing on.

Figure 06-02-5 provides single engine (APR rating) thrust settings with zero bleed air extraction from the engines.

Figure 06-02-6 provides single engine (APR rating) thrust settings with 10th stage bleed open/one (1) pack on.

Figure 06-02-7 provides single engine (APR rating) thrust settings with 10th stage bleed closed and engine cowl anti-icing on.

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CHALLENGER 650	PERFORMANCE Thrust Settings	06-02-2
		Oct 27/15

1. THRUST SETTINGS (CONT'D)

Figure 06-02-8 provides single engine (APR rating) thrust settings with 10th stage bleed closed, and engine cowl and wing anti-icing systems on.

Figure 06-02-9 provides maximum continuous thrust settings for single engine climb with 10th stage bleed open and one (1) pack on.

Figure 06-02-10 provides maximum continuous thrust settings for single engine climb with 10th stage bleed open/one (1) pack on, and engine cowl anti-icing on.

Figure 06-02-11 provides maximum continuous thrust settings for single engine climb with 10th stage bleed open/one (1) pack on, and engine cowl and wing anti-icing on.

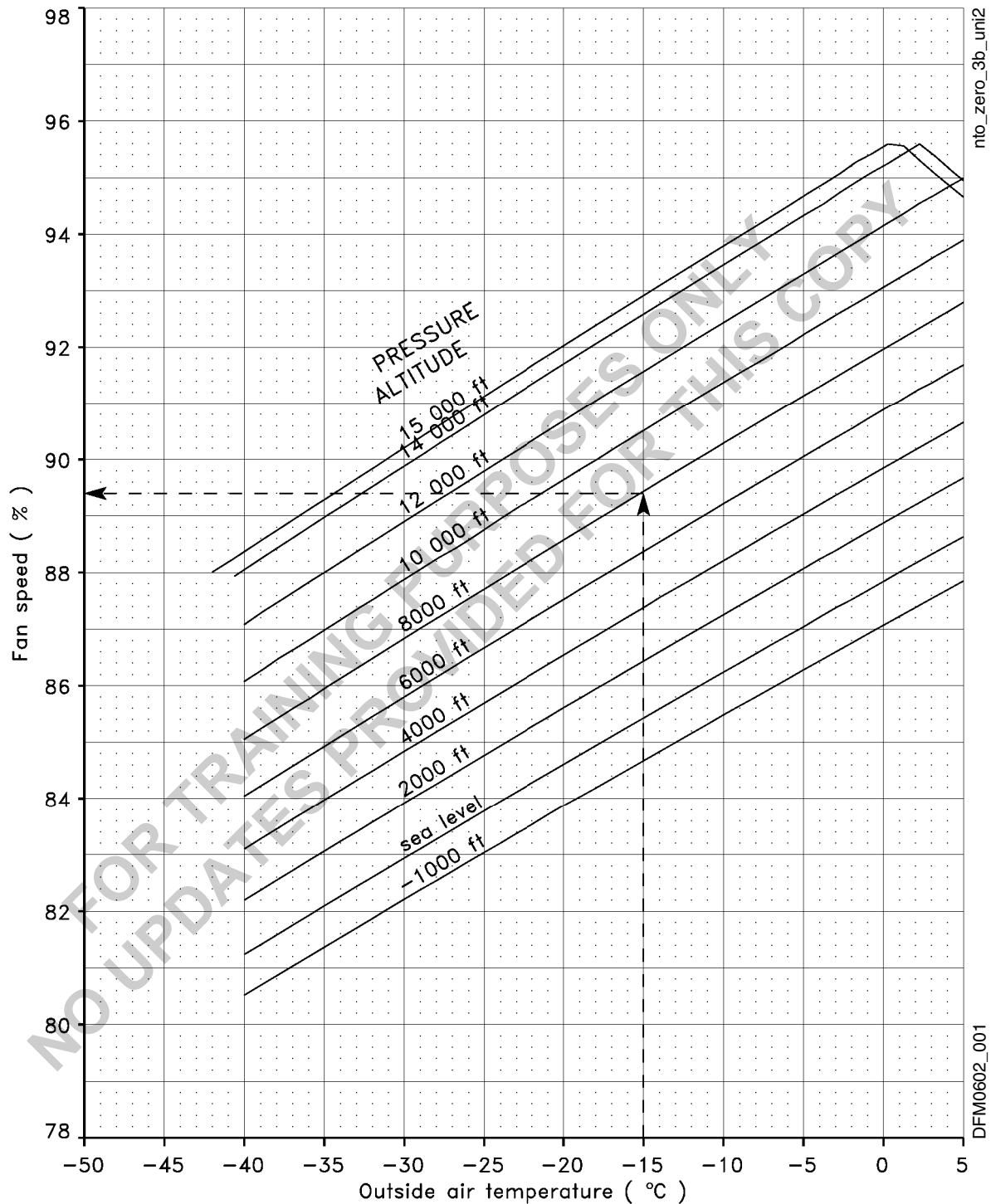
C. Normal Take-Off Thrust

Engine fan speed (N_1) settings for various conditions of outside air temperatures and pressure altitudes required to obtain the engine rated take-off thrust with two engine operation are shown on Figure 06-02-1 to Figure 06-02-4.

Example:

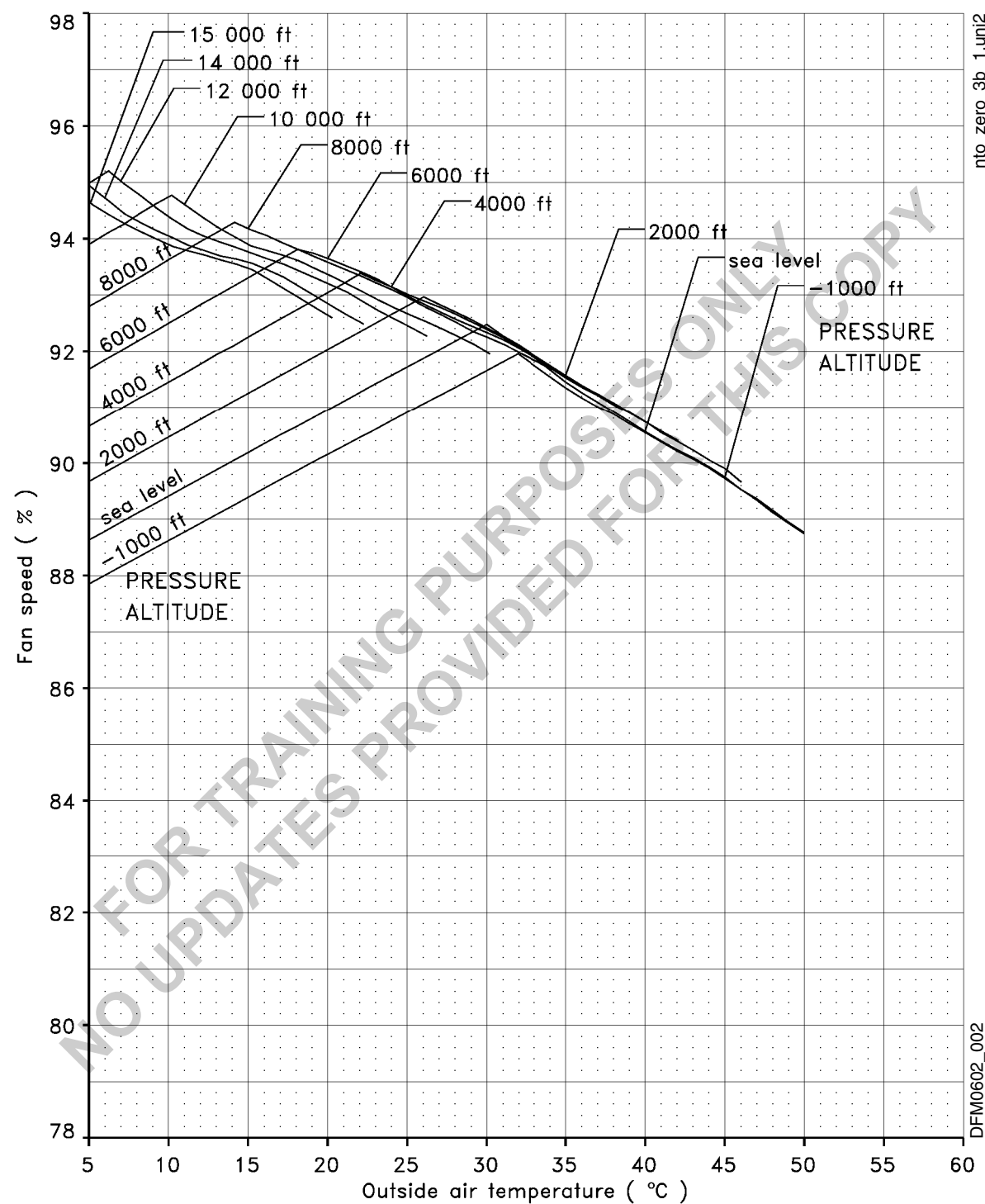
Page 1 of Figure 06-02-1 shows that for an outside air temperature of -15°C and at a pressure altitude of 8,000 feet, the engine rated normal take-off thrust with zero engine bleed will be achieved at 89.4% N_1 .

1. THRUST SETTINGS (CONT'D)



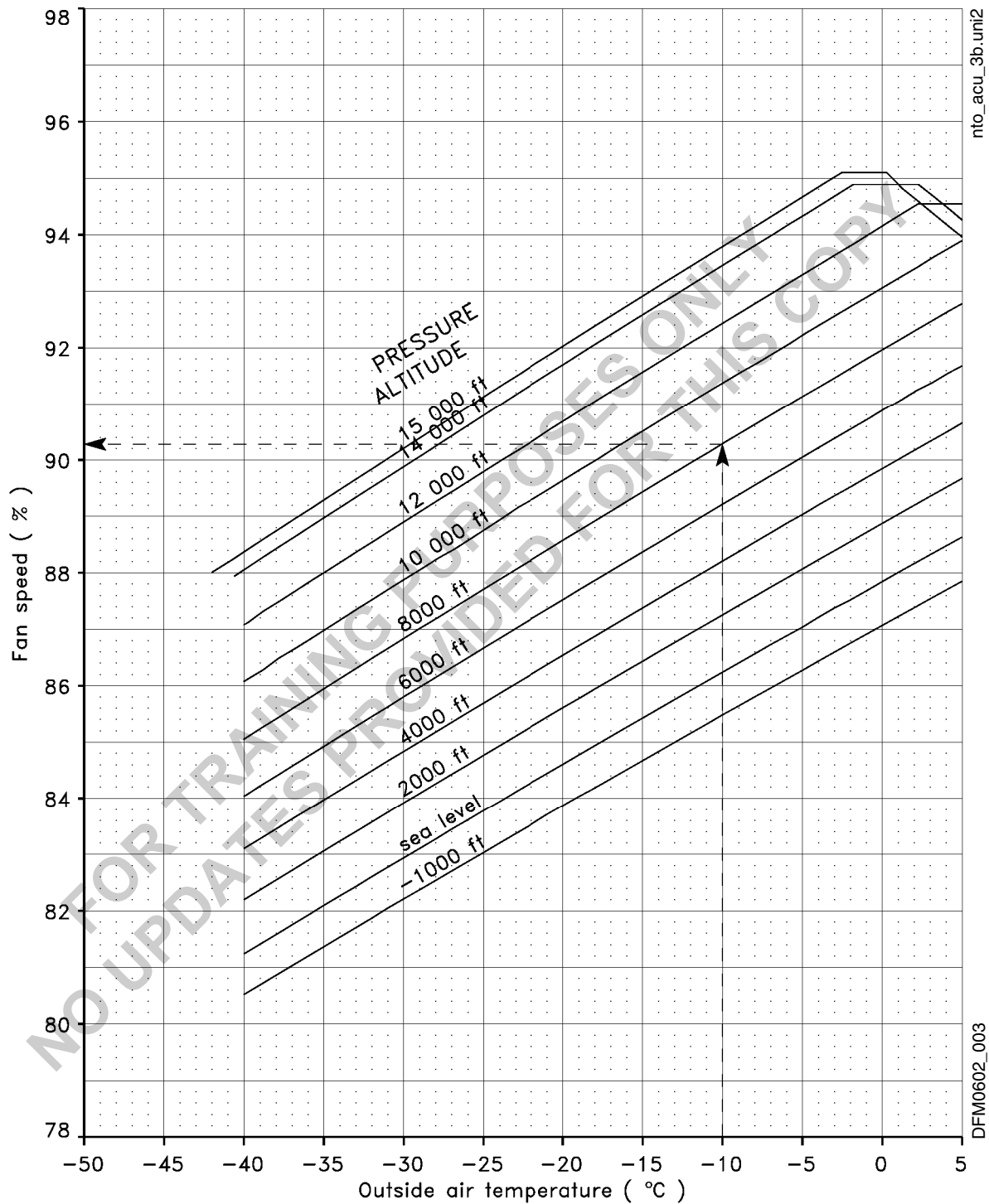
Normal Take-Off Thrust (Two Engine Operation) – Zero Engine Bleed (Page 1 of 2)
Figure 06-02-1

1. THRUST SETTINGS (CONT'D)



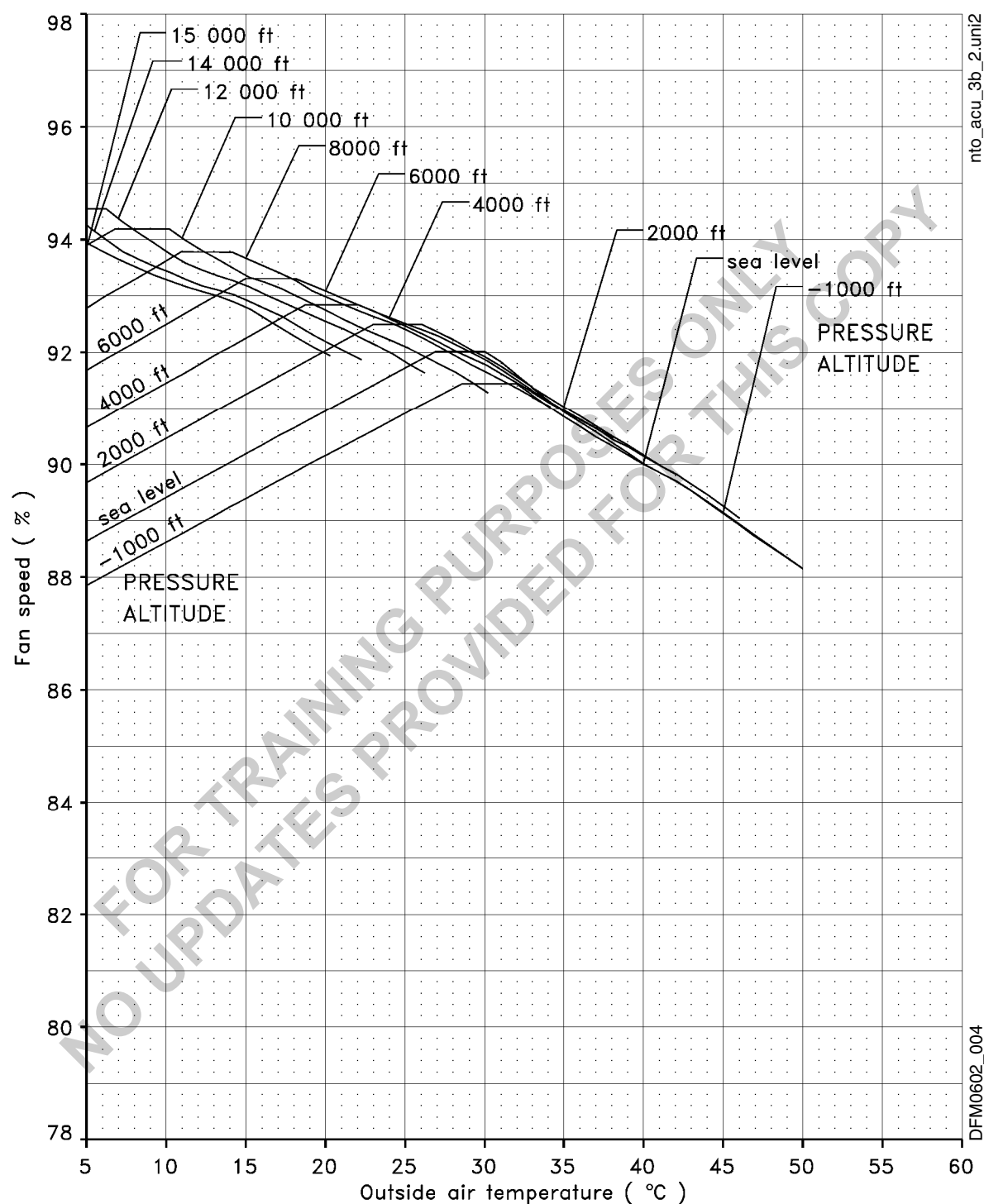
Normal Take-Off Thrust (Two Engine Operation) – Zero Engine Bleed (Page 2 of 2)
Figure 06-02-1

1. THRUST SETTINGS (CONT'D)



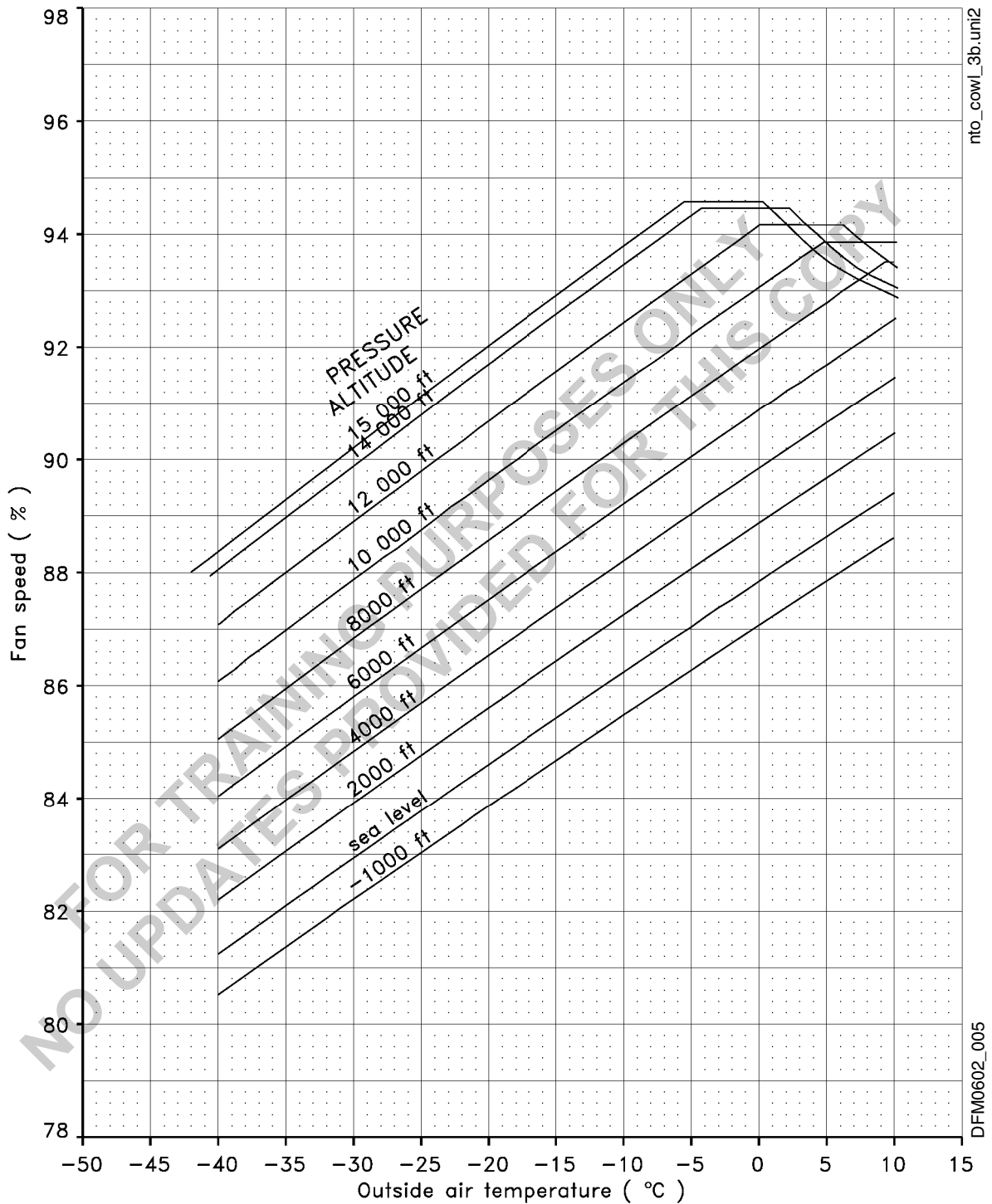
Normal Take-Off Thrust (Two Engine Operation) – 10th Stage Bleeds Open/Packs On (Page 1 of 2)
Figure 06-02-2

1. THRUST SETTINGS (CONT'D)



Normal Take-Off Thrust (Two Engine Operation) – 10th Stage Bleeds Open/Packs On (Page 2 of 2)
Figure 06-02-2

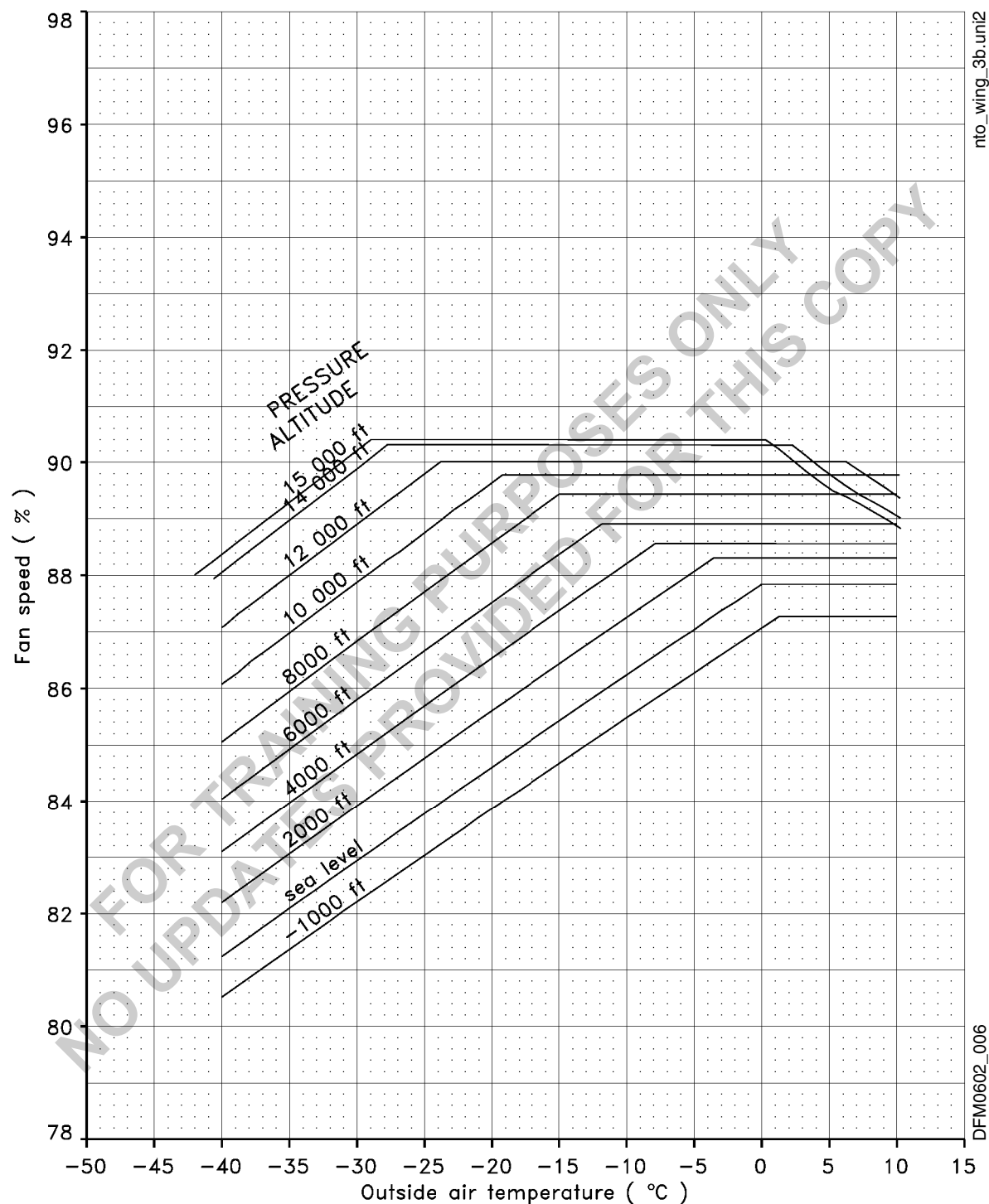
1. THRUST SETTINGS (CONT'D)



Normal Take-Off Thrust (Two Engine Operation) – 10th Stage Bleeds Closed/Engine Cowl Anti-Icing On

Figure 06-02-3

1. THRUST SETTINGS (CONT'D)



Normal Take-Off Thrust (Two Engine Operation) – 10th Stage Bleeds Closed/Engine Cowl and Wing Anti-Icing On
Figure 06-02-4

CHALLENGER 650	PERFORMANCE Thrust Settings	06-02-9
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1. THRUST SETTINGS (CONT'D)

D. Single Engine Climb Thrust (APR Rating)

Engine fan speed (N_1) settings for various ambient temperatures and pressure altitudes required to obtain the engine rated automatic performance reserve (APR rating) are shown on Figure 06-02-5 to Figure 06-02-8.

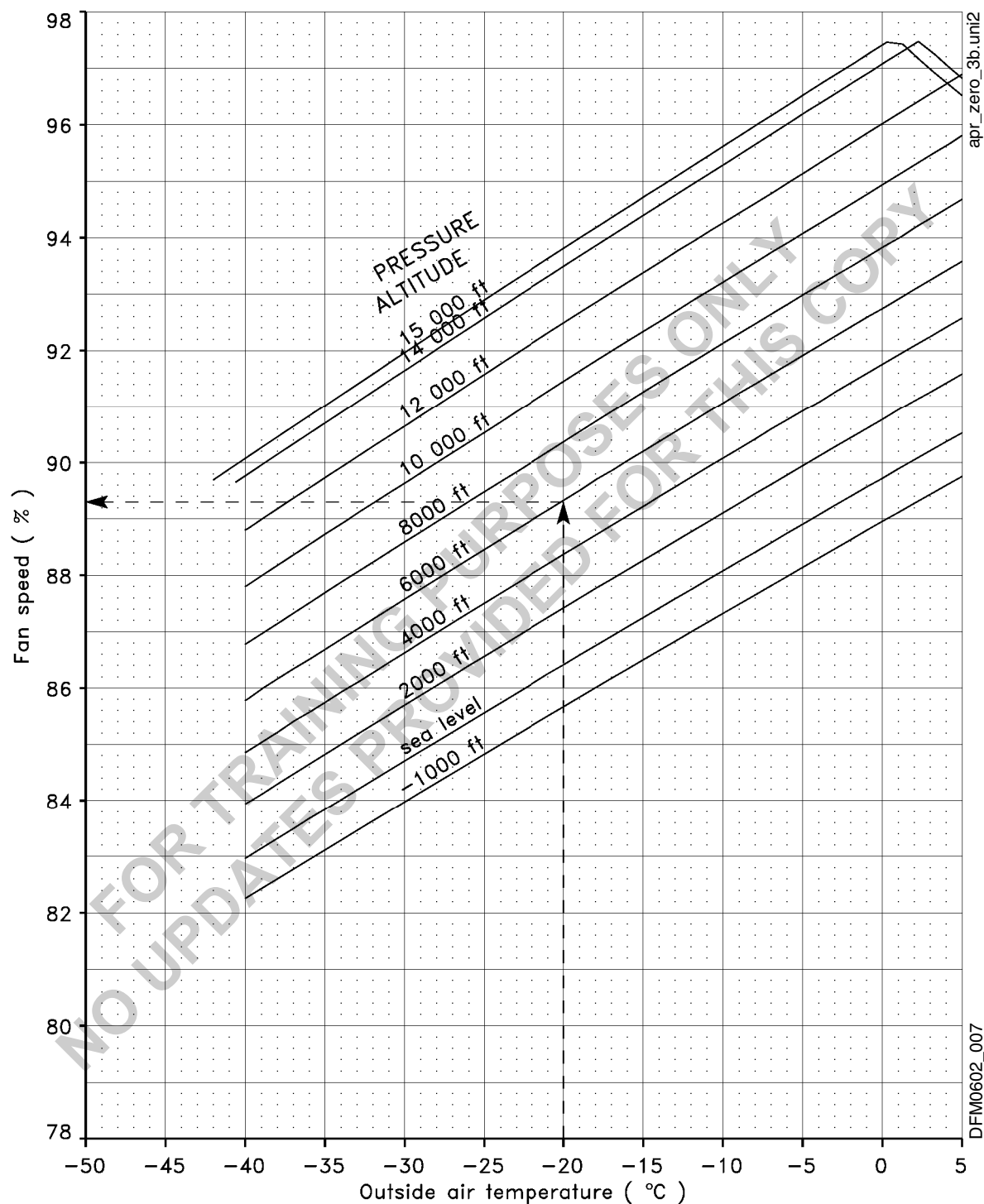
Example:

The example in page 1 of Figure 06-02-5 shows that for an outside air temperature of -20°C and at a pressure altitude of 6,000 feet, the single engine climb thrust (APR rating) with zero engine bleed, will be achieved at 89.3% N_1 .

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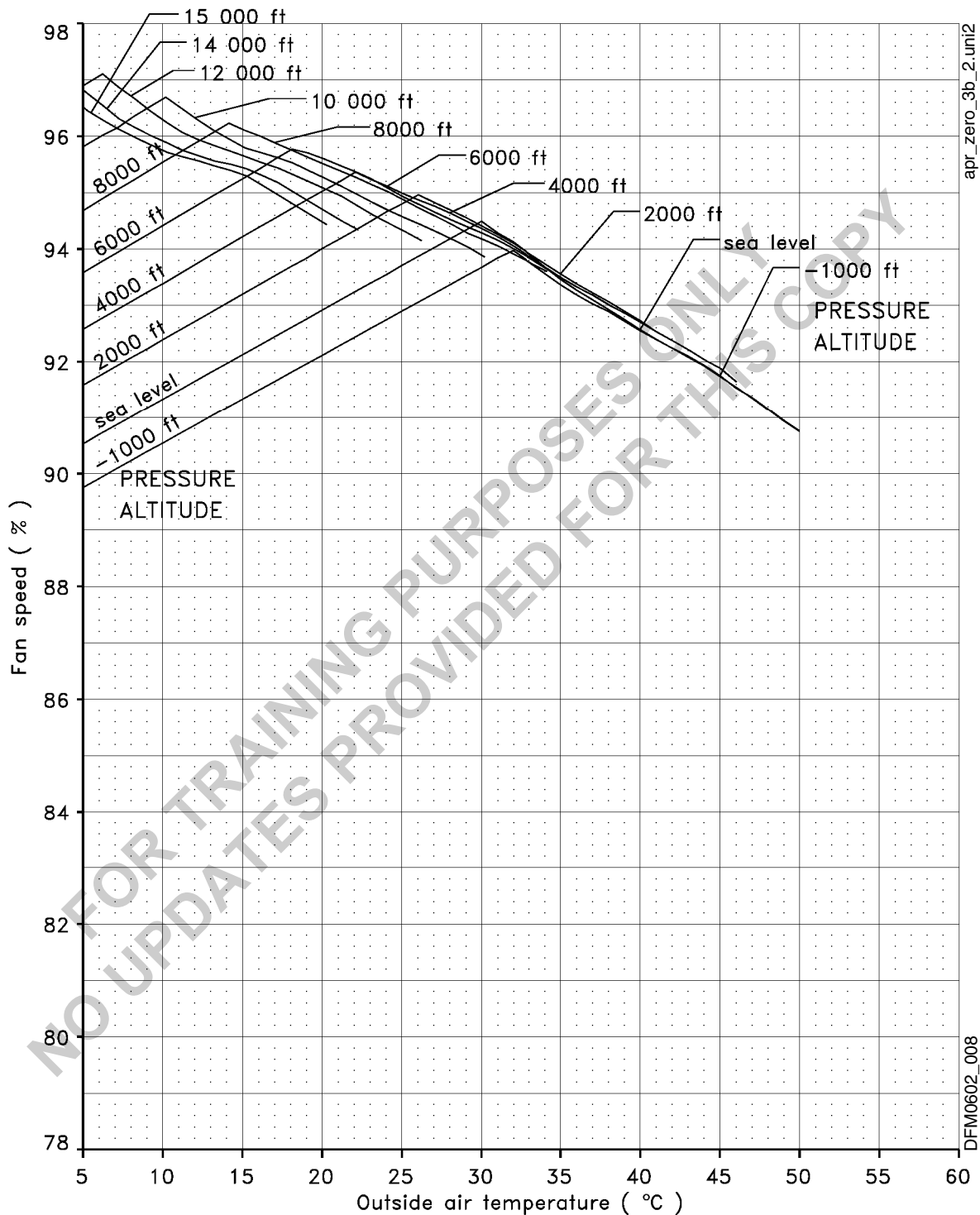
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1. THRUST SETTINGS (CONT'D)



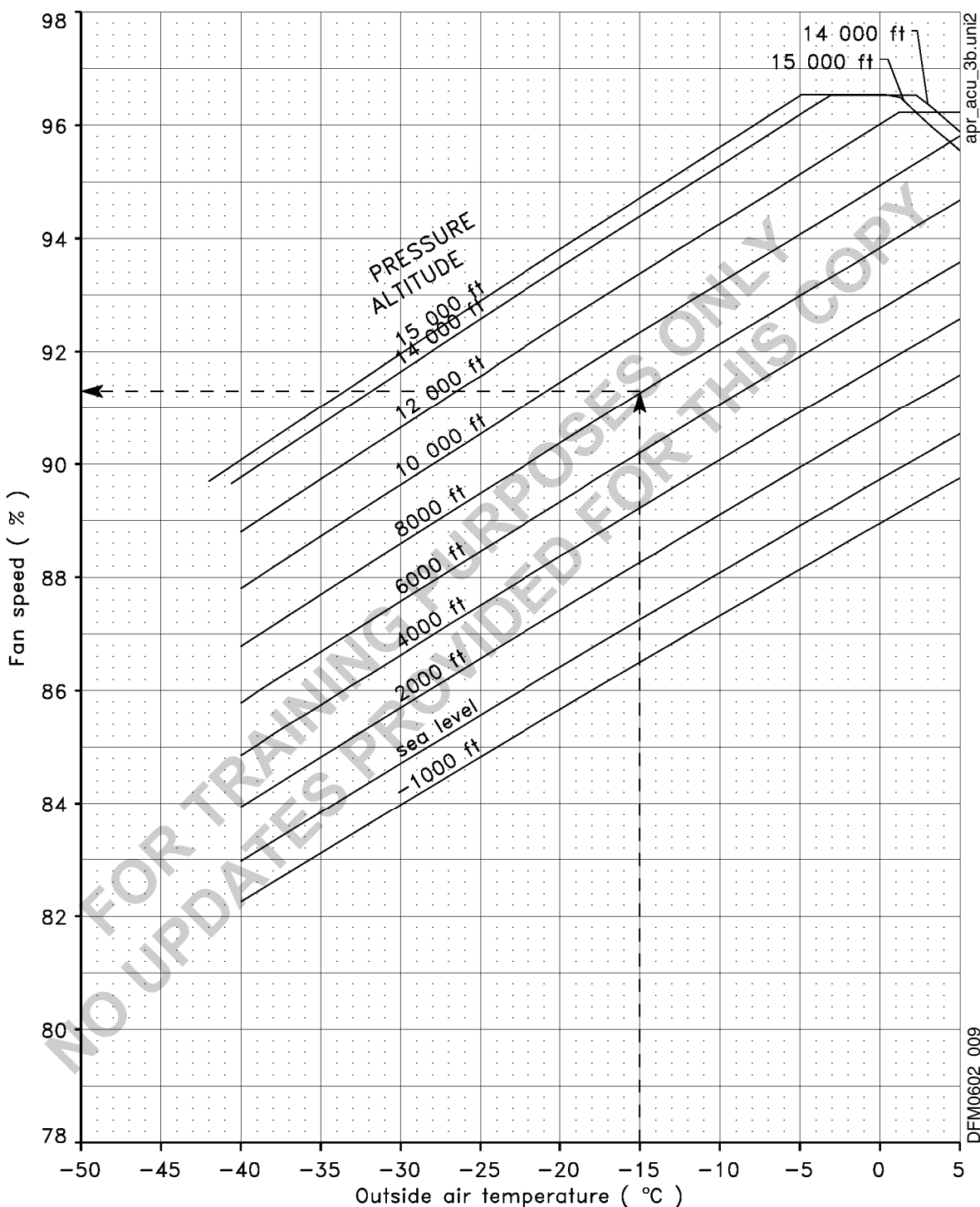
Single Engine Climb Thrust (APR Rating) – Zero Engine Bleed (Page 1 of 2)
Figure 06-02-5

1. THRUST SETTINGS (CONT'D)



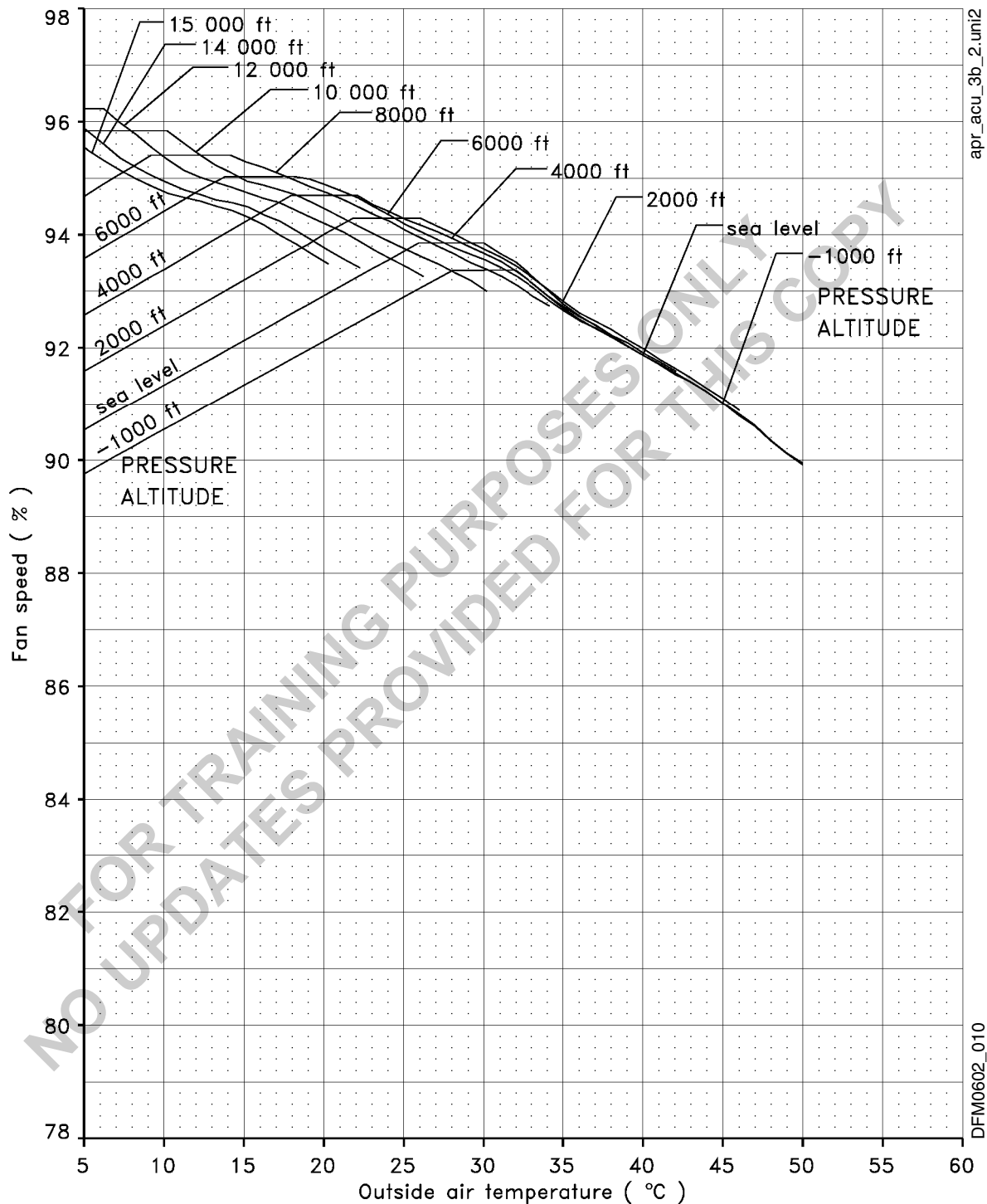
Single Engine Climb Thrust (APR Rating) – Zero Engine Bleed (Page 2 of 2)
Figure 06-02-5

1. THRUST SETTINGS (CONT'D)



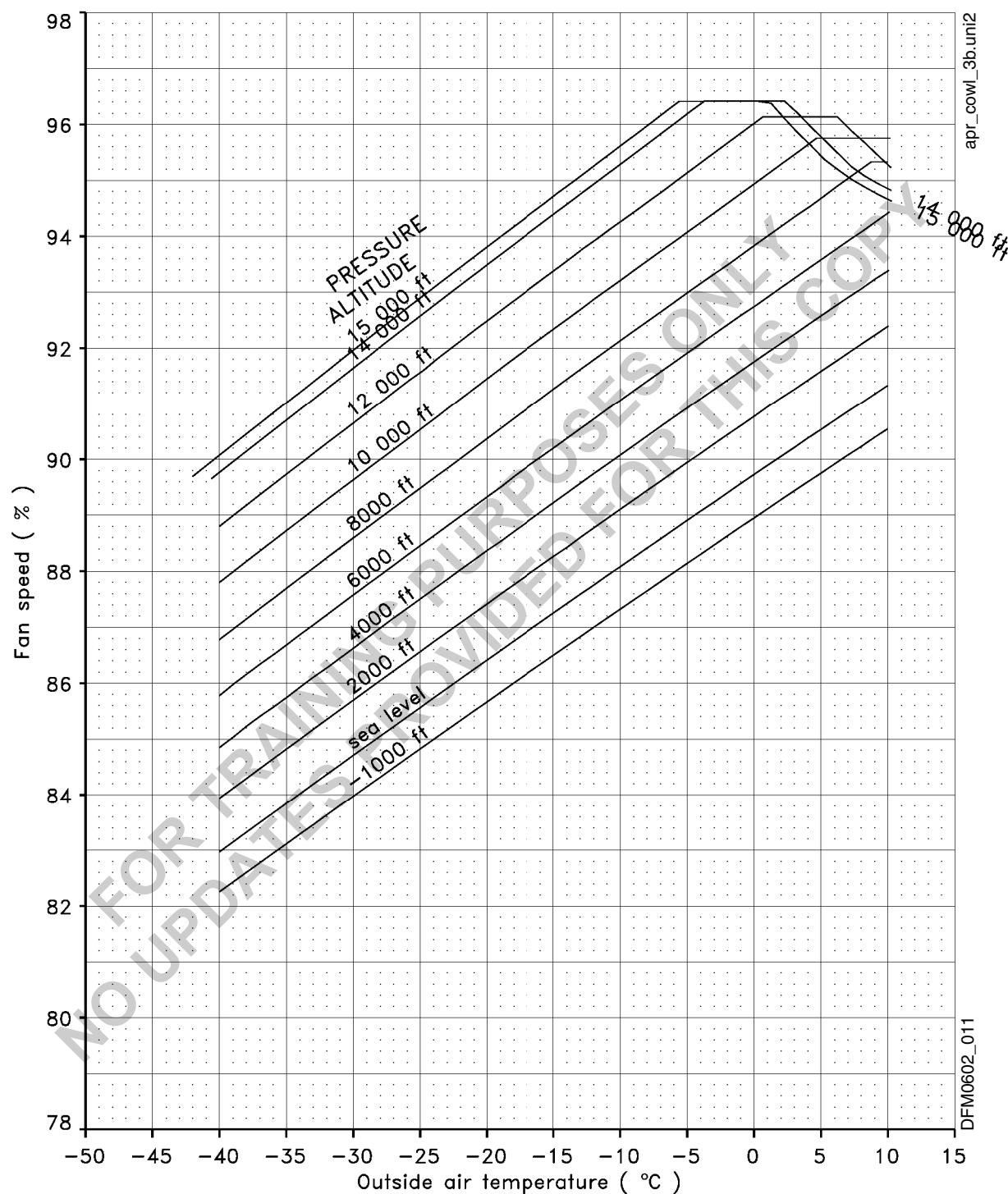
Single Engine Climb Thrust (APR Rating) – 10th Stage Bleed Open/One (1) Pack On (Page 1 of 2)
Figure 06-02-6

1. THRUST SETTINGS (CONT'D)



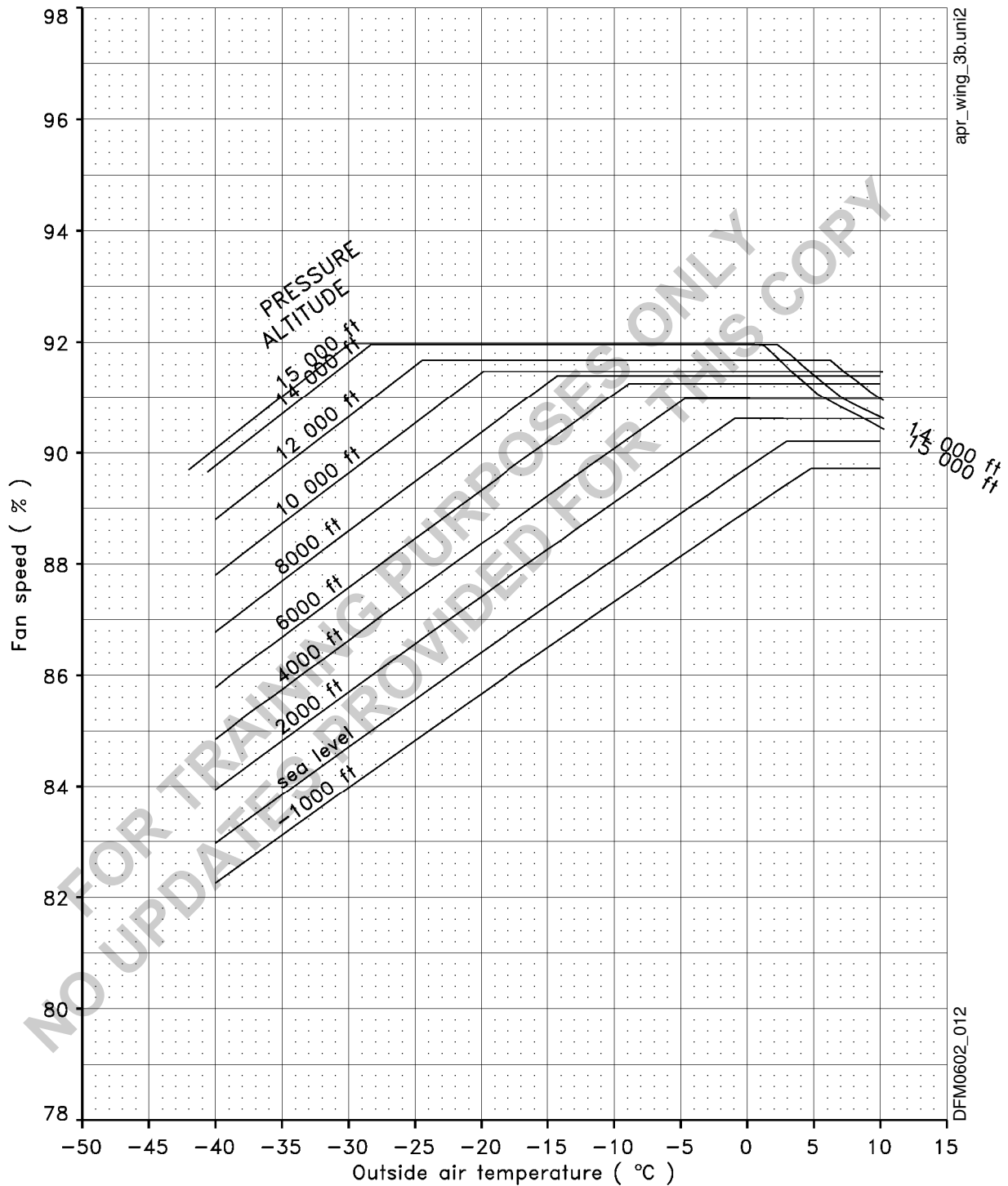
Single Engine Climb Thrust (APR Rating) – 10th Stage Bleed Open/One (1) Pack On (Page 2 of 2)
Figure 06-02-6

1. THRUST SETTINGS (CONT'D)



Single Engine Climb Thrust (APR Rating) – 10th Stage Bleeds Closed and Engine Cowl Anti-Icing On
Figure 06-02-7

1. THRUST SETTINGS (CONT'D)



Single Engine Climb Thrust (APR Rating) – 10th Stage Bleeds Closed/Engine Cowl and Wing Anti-Icing On
Figure 06-02-8

CHALLENGER 650	PERFORMANCE Thrust Settings	06-02-16
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1. THRUST SETTINGS (CONT'D)

E. Maximum Continuous Thrust, Single Engine Climb

Engine fan speed (N_1) settings for various ambient temperatures and pressure altitudes required to obtain the engine rated maximum continuous thrust (MCT) for a single engine climb, with 10th stage bleed open/one (1) pack on, are shown on Figure 06-02-9 to Figure 06-02-11.

Example:

The example in page 1 of Figure 06-02-9 shows that for an outside air temperature of -10°C and at a pressure altitude of 8,000 feet, the single engine climb thrust (maximum continuous thrust rating) with 10th stage bleed open/one (1) pack on, will be achieved at 91.7% N_1 .

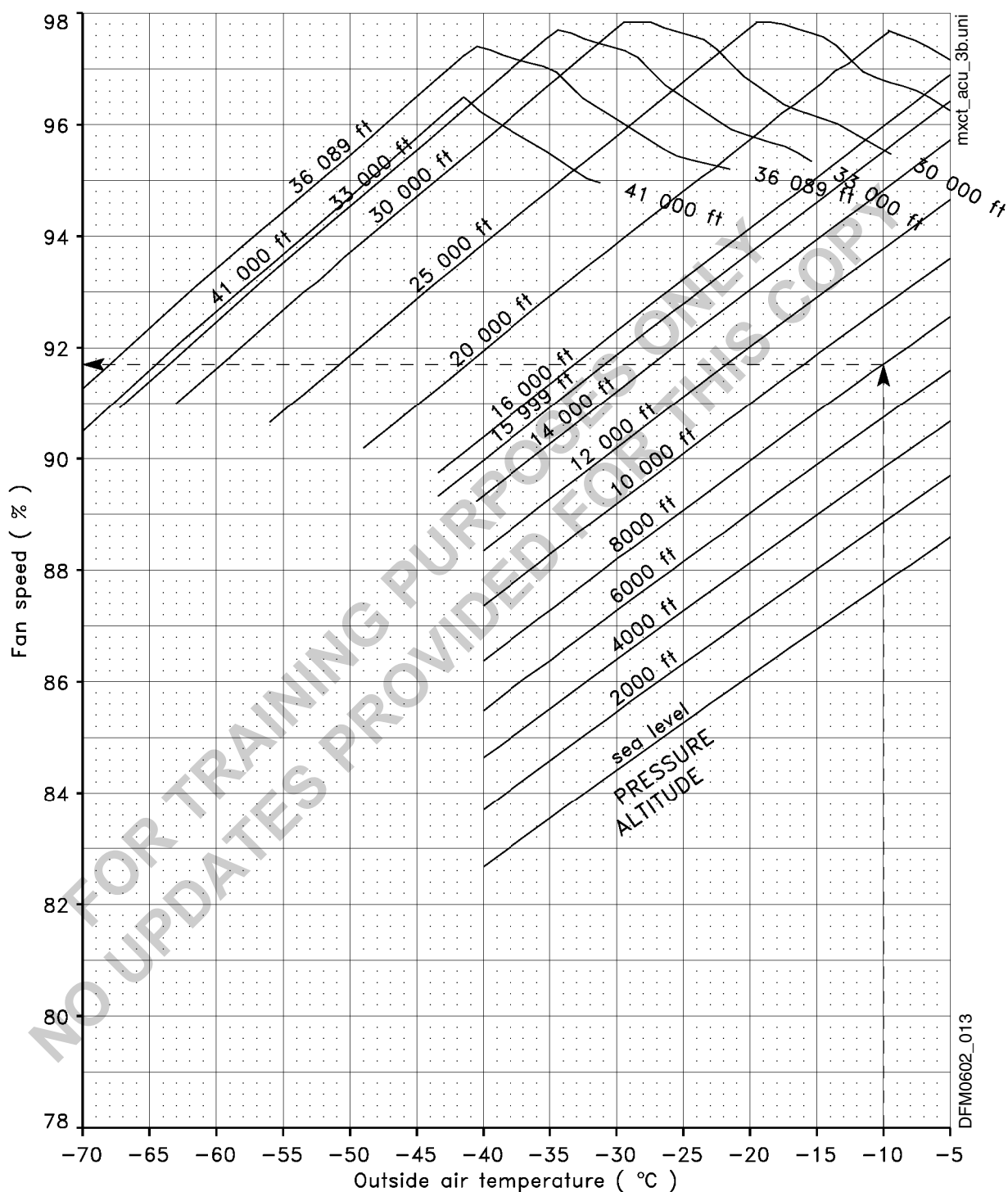
NOTE

1. Maximum continuous thrust (MCT) settings have a discontinuity at 16,000 feet. This discontinuity is due to the assumptions made to derive MCT settings.
2. At altitudes between 14,000 feet and 16,000 feet, the thrust setting curve labeled 15,999 feet must be used for interpolation.
3. At altitudes between 16,000 feet and 20,000 feet, the thrust setting curve labeled 16,000 feet must be used for interpolation.

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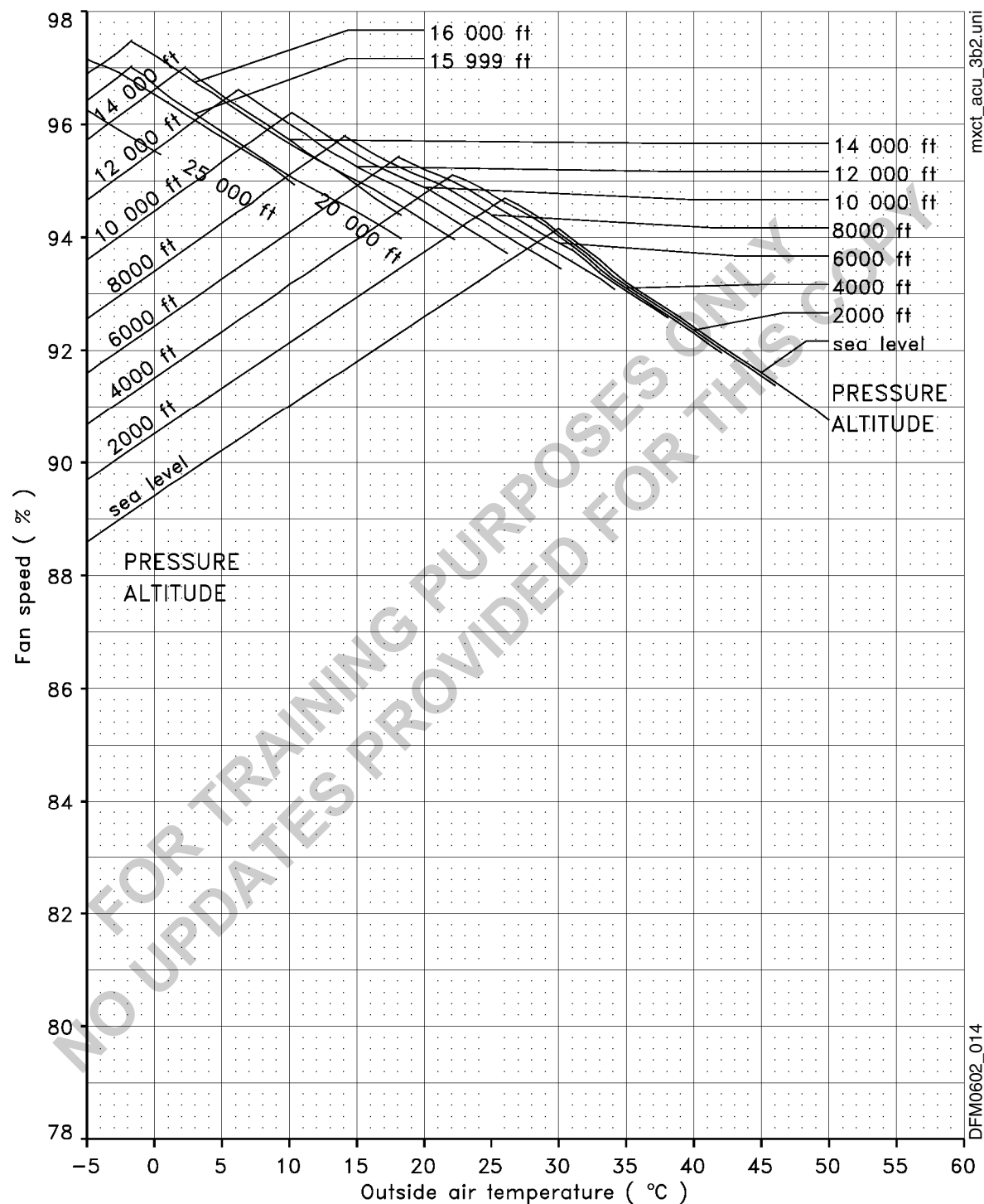
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1. THRUST SETTINGS (CONT'D)



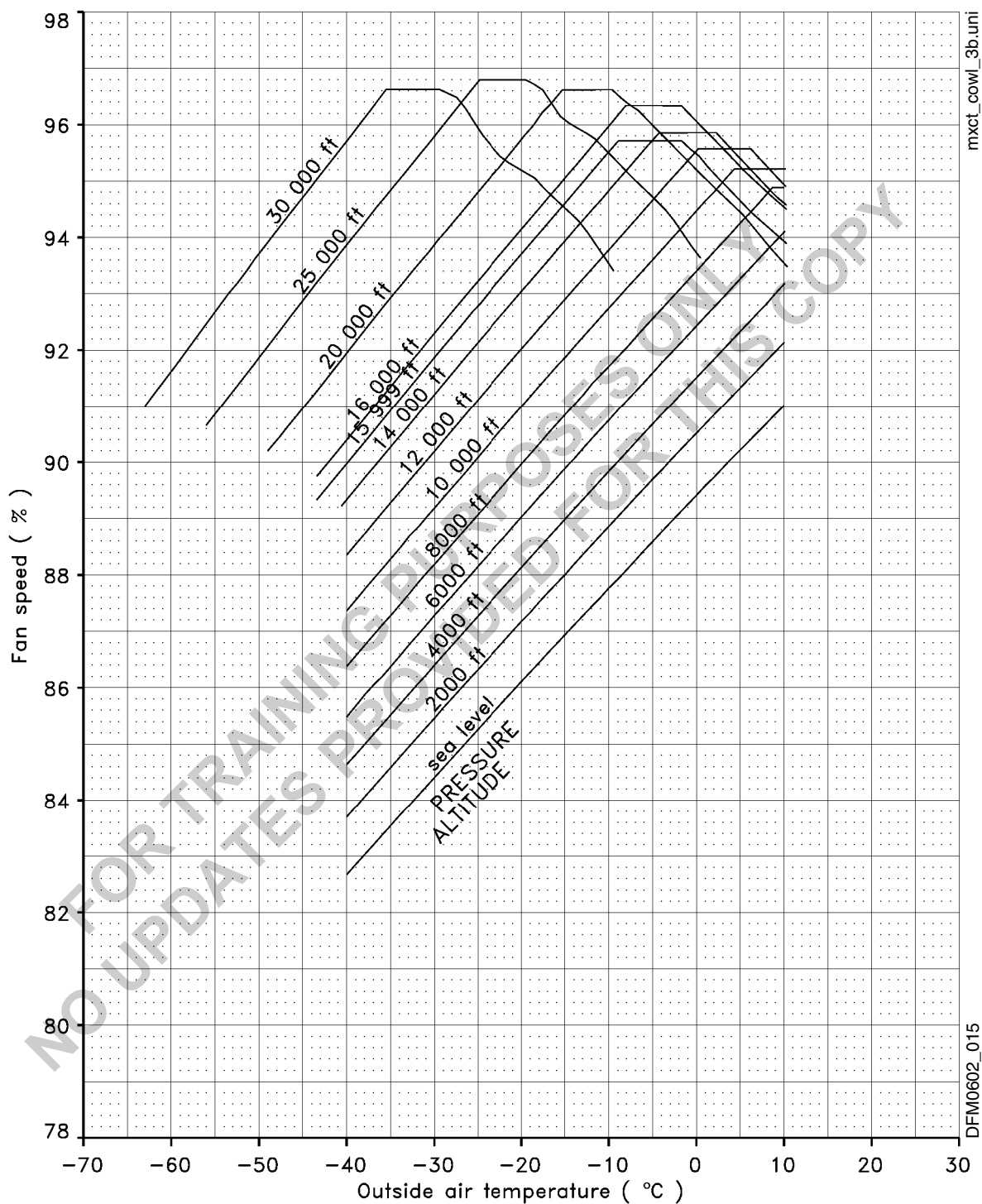
Maximum Continuous Thrust (Single Engine Climb) – 10th Stage Bleed Open/One Pack On
(Page 1 of 2)
Figure 06-02-9

1. THRUST SETTINGS (CONT'D)



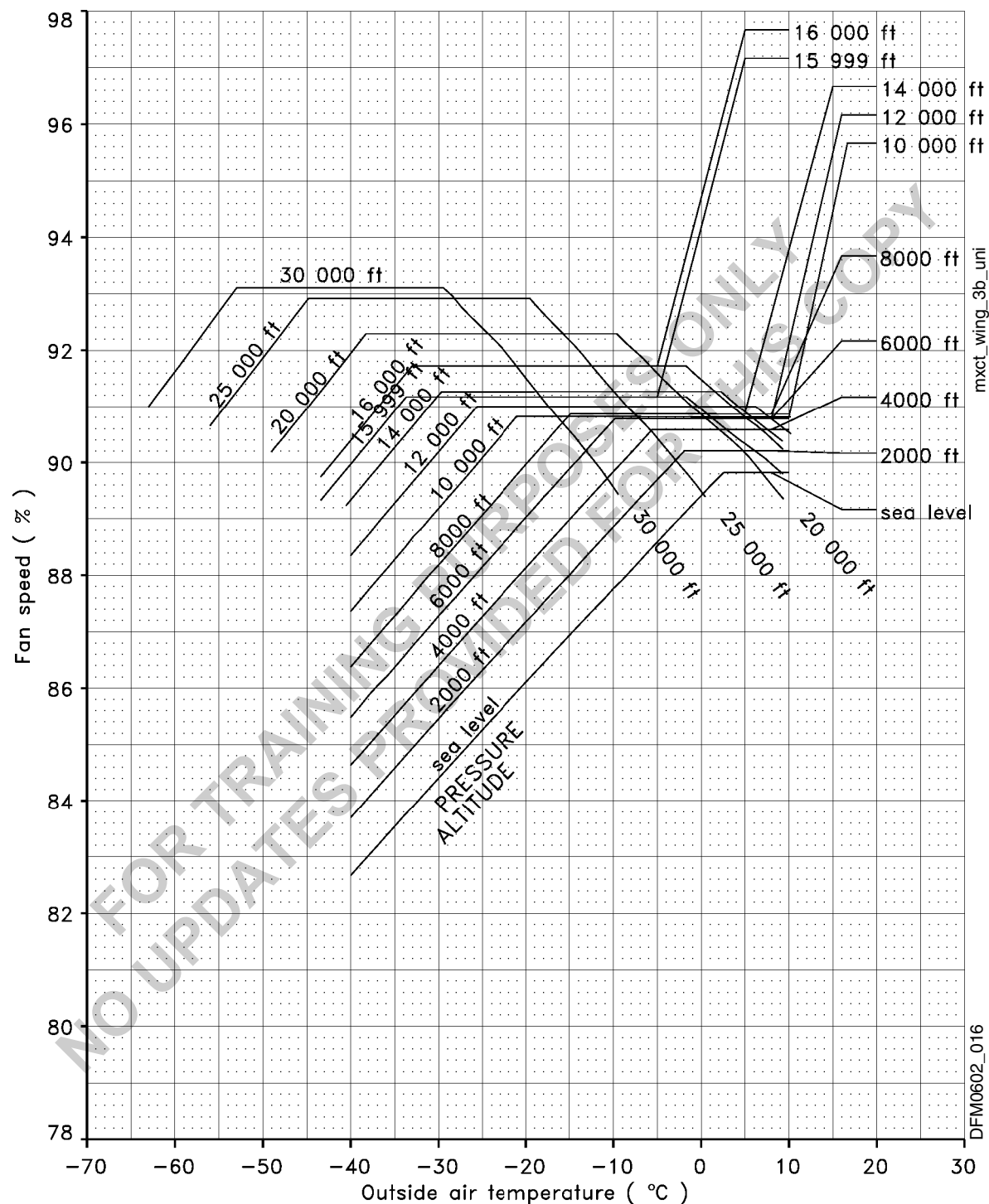
Maximum Continuous Thrust (Single Engine Climb) – 10th Stage Bleed Open/One Pack On
(Page 2 of 2)
Figure 06-02-9

1. THRUST SETTINGS (CONT'D)



Maximum Continuous Thrust (Single Engine Climb) – 10th Stage Bleed Open/One Pack On and
Engine Cowl Anti-Icing On
Figure 06-02-10

1. THRUST SETTINGS (CONT'D)



Maximum Continuous Thrust (Single Engine Climb) – 10th Stage Bleed Open/One Pack On/Engine Cowl and Wing Anti-Icing On
Figure 06-02-11

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-1
		Oct 27/15

1. INTRODUCTION

The Take-Off Performance section contains the charts used to obtain the take-off performance data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified take-off configurations.

2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT

The maximum allowable take-off weight is limited by the most restrictive of the following:

- Maximum approved take-off weight
(Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT),
- Runway length available,
- Climb requirements,
- Obstacle clearance requirements,
(First, verify if the second segment of the take-off flight path extends beyond a pressure height of 1,500 feet above the airport and consider if potential adjustments are to be made.)
- Maximum demonstrated brake energy,
- Maximum tire speed,
- Wheel brake cooling limitations
(Refer to PERFORMANCE – TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires, in this chapter).

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-2
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2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)

A. Take-Off Weight Limited By Climb Requirements – Flaps 20°

The maximum take-off weight limited by climb requirements for a flaps-20° take-off is determined from Figure 06-03-1, for varying conditions of ambient temperature and airport pressure altitude, taking into account the effects of different combinations of wing and cowl anti-icing and 10th stage bleeds, and also the use of the APR system.

The weights presented in Figure 06-03-1 are limited by second segment climb gradient requirements, as per FAR 25.121 (b). The first segment climb and final segment climb gradient requirements, as per FAR 25.121 (a) and FAR 25.121 (c), respectively, are not limiting.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 06-03-1, Anti-ice On), the maximum take-off weight limited by climb requirements is 21,600 kg (47,620 lb). The take-off weight in this case may be less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT), and may be considered limiting.

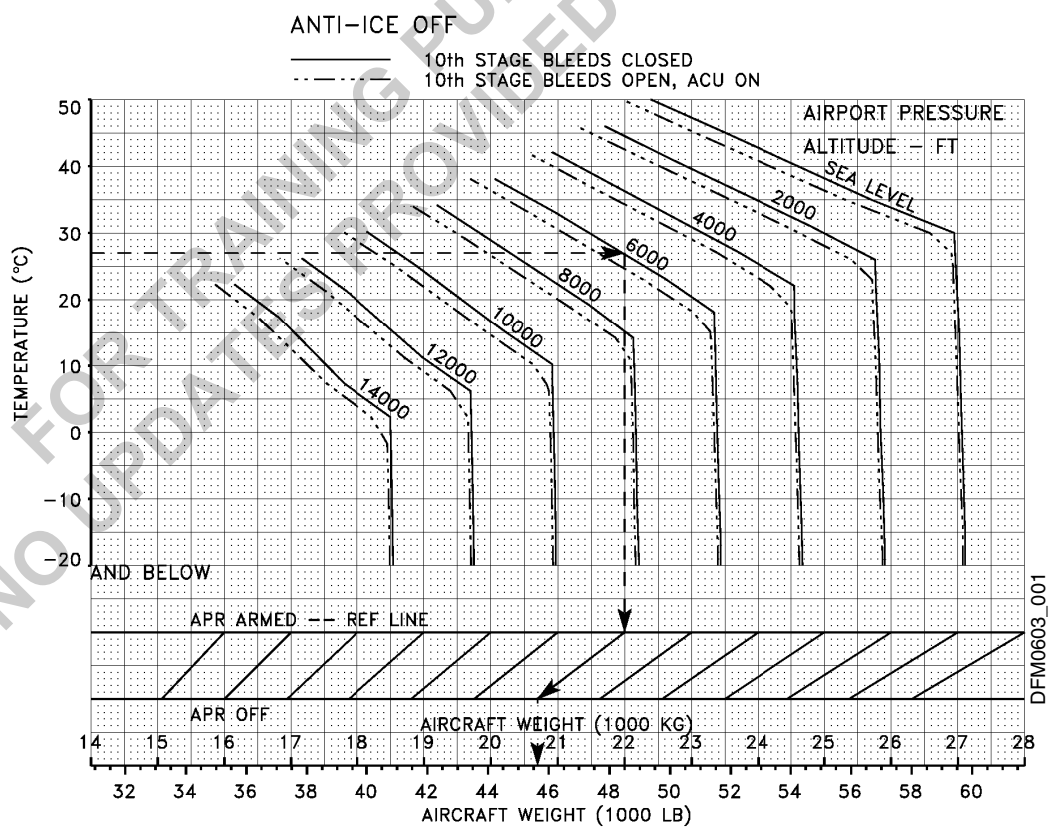
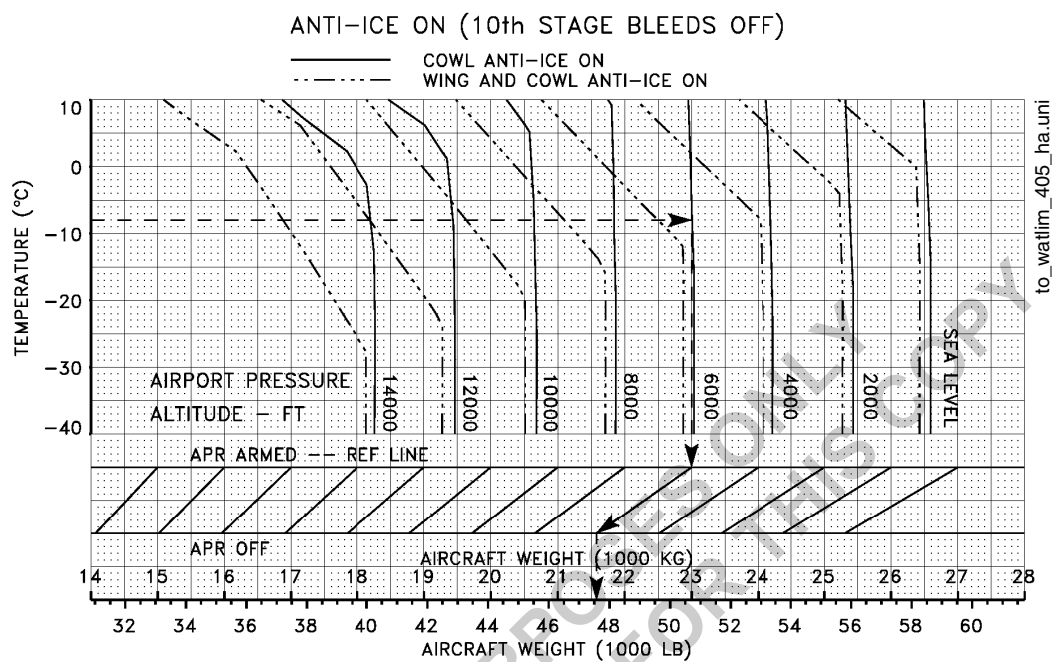
Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 06-03-1, Anti-ice Off), the maximum take-off weight limited by climb requirements is 20,700 kg (45,636 lb). The take-off weight in this case is less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT), and is considered limiting.

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2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)


Take-Off Weight Limited by Climb Requirements – Flaps 20°
 Figure 06-03-1

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-4
		Oct 27/15

2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)

FAA Certified Airplanes

The maximum take-off weight limited by climb requirements for a flaps-20° take-off is determined from Figure 06-03-1, for varying conditions of ambient temperature and airport pressure altitude, taking into account the effects of different combinations of wing and cowl anti-icing and 10th stage bleeds, and also the use of the APR system.

For take-offs with APR off, the weights presented in Figure 06-03-1 are limited by second segment climb gradient requirements, as per FAR 25.121 (b). For take-offs with APR armed, the weights presented in Figure 06-03-1 are limited by approach climb gradient requirements, as per FAR 25.121 (d), following a 15 minute flight in accordance with the requirements of FAR 25.1001 (a). The first segment climb and final segment climb gradient requirements, as per FAR 25.121 (a) and FAR 25.121 (c), respectively, are not limiting.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 06-03-1, Anti-ice On), the maximum take-off weight limited by climb requirements is 21,700 kg (47,840 lb). The take-off weight in this case may be less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT), and may be considered limiting.

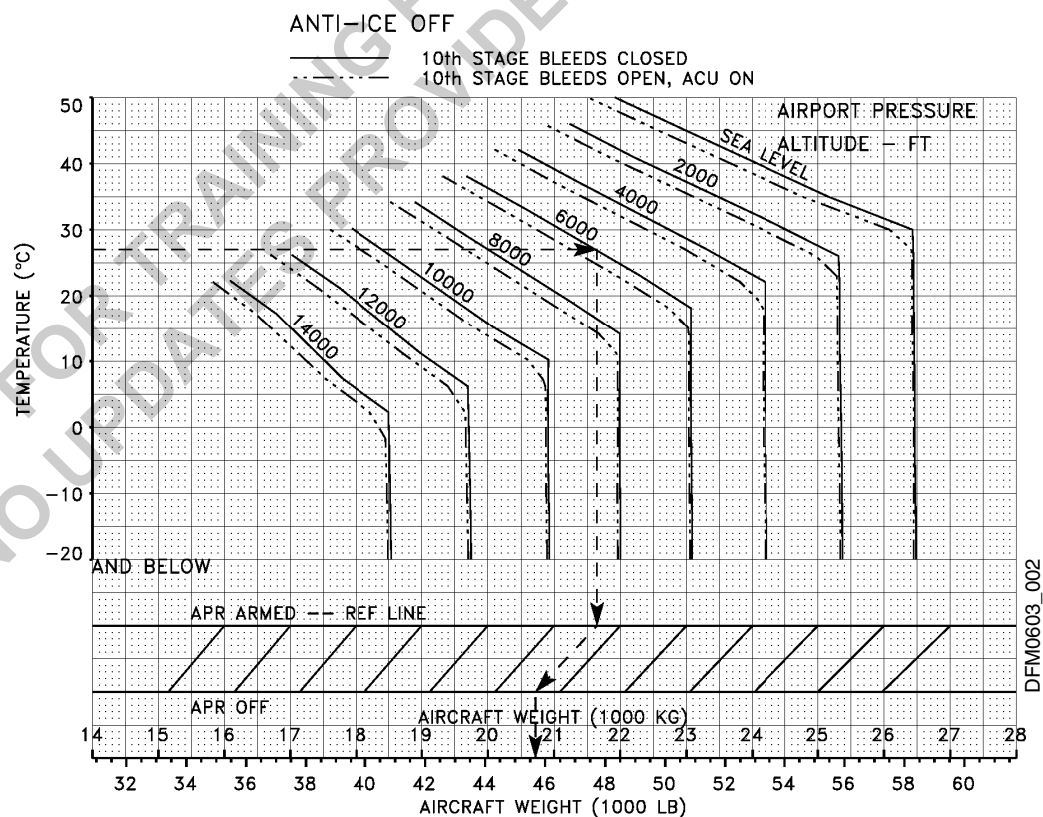
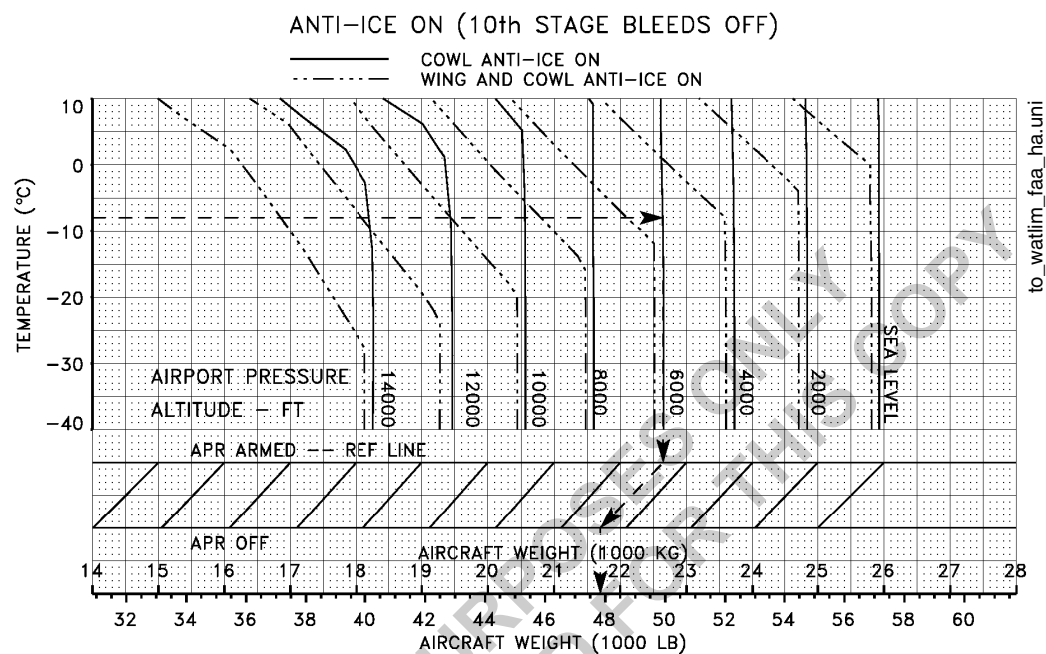
Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 06-03-1, Anti-ice Off), the maximum take-off weight limited by climb requirements is 20,700 kg (45,636 lb). The take-off weight in this case is less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT), and is considered limiting.

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2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)
FAA Certified Airplanes


Take-Off Weight Limited by Climb Requirements – Flaps 20°

Figure 06-03-1

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-6
		Oct 27/15

2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)

B. Take-Off Weight Limited By Brake Energy – Flaps 20°

The maximum take-off weight limited by brake energy for a flaps-20° take-off is determined from Figure 06-03-2, for varying conditions of ambient temperature and airport pressure altitude, wind and runway slope.

The weight derived from Figure 06-03-2 must be corrected for the following conditions:

- With 10th stage bleeds open and packs on, decrease the weight by 80 kg (176 lb);
- With only the engine cowl anti-icing on, decrease the weight by 100 kg (221 lb);
- With both engine cowl and wing anti-icing on, decrease the weight by 360 kg (794 lb);
- With the APR system off, decrease the weight by 320 kg (705 lb);

NOTE

1. The above decrements are cumulative except between the two anti-icing conditions. The 10th stage bleeds must be closed for take-off with cowl and/or wing anti-icing on.
2. Additional performance benefits may be obtained by referring to Chapter 7; SUPPLEMENT 13.

Example:

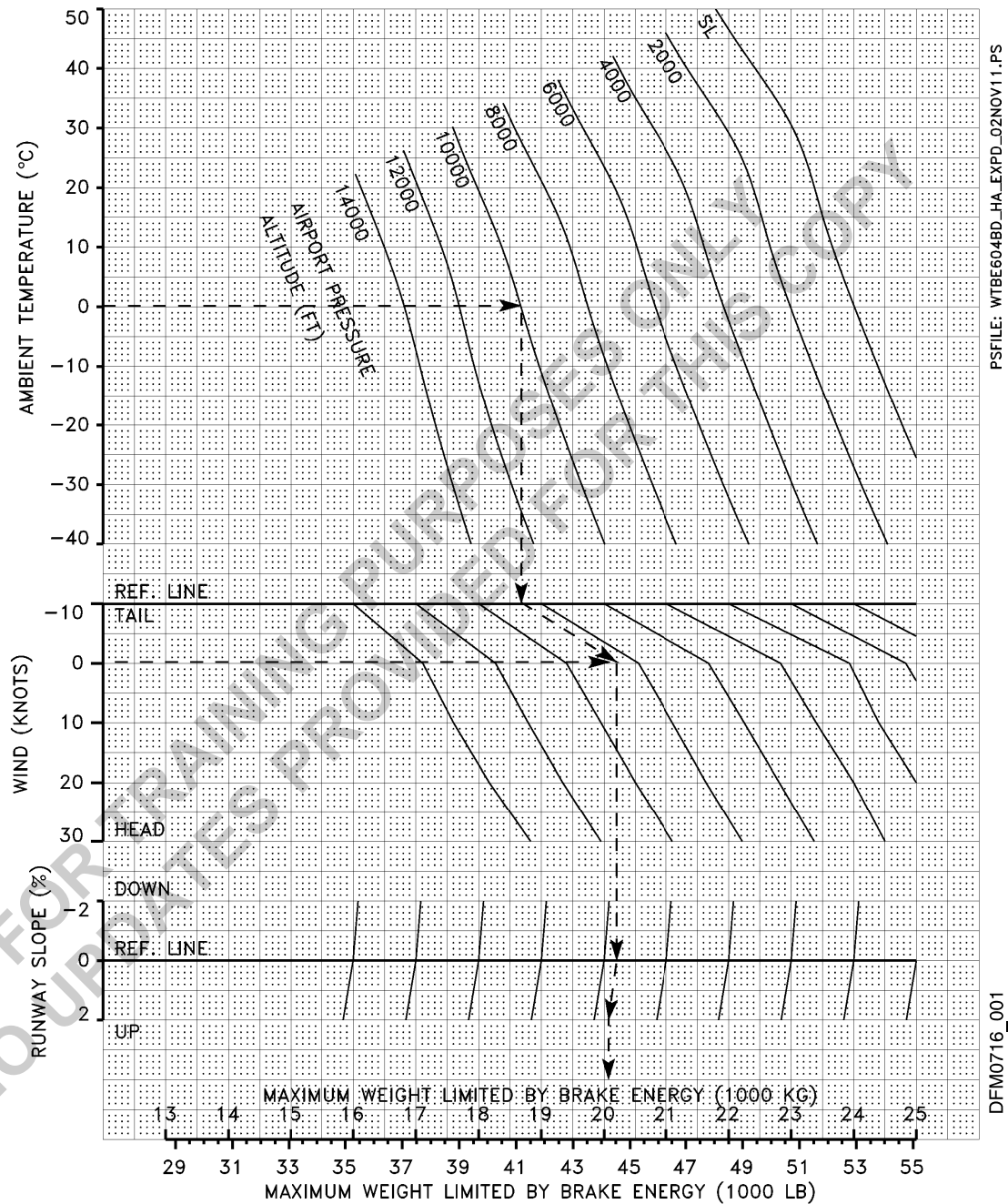
Associated Conditions:

Temperature	= 0°C
Airport pressure altitude	= 10,000 feet
Wind	= Calm
Runway slope	= +2%
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Armed

The example in Figure 06-03-2 shows that the maximum take-off weight limited by brake energy is 20,080 kg (44,269 lb).

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2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)



Take-Off Weight Limited By Brake Energy – Flaps 20°
Figure 06-03-2

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-8
		Oct 27/15

2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)

C. Take-Off Weight Limited by Maximum Tire Speed

The maximum take-off weight limited by the maximum tire speed of 210 mph (182 knots), for a flaps-20° take-off, in varying conditions of ambient temperature, airport pressure altitude and wind, is given by Figure 06-03-3.

The weights derived from Figure 06-03-3 are applicable for all engine bleed configurations.

NOTE

Additional performance benefits may be obtained by referring to Chapter 7; SUPPLEMENT 13.

Example:

Figure 06-03-3 shows that at an ambient temperature of 0°C, pressure altitude of 10,000 feet, calm wind conditions and APR on, the take-off weight limited by maximum tire speed is 20,830 kg (45,922 lb).

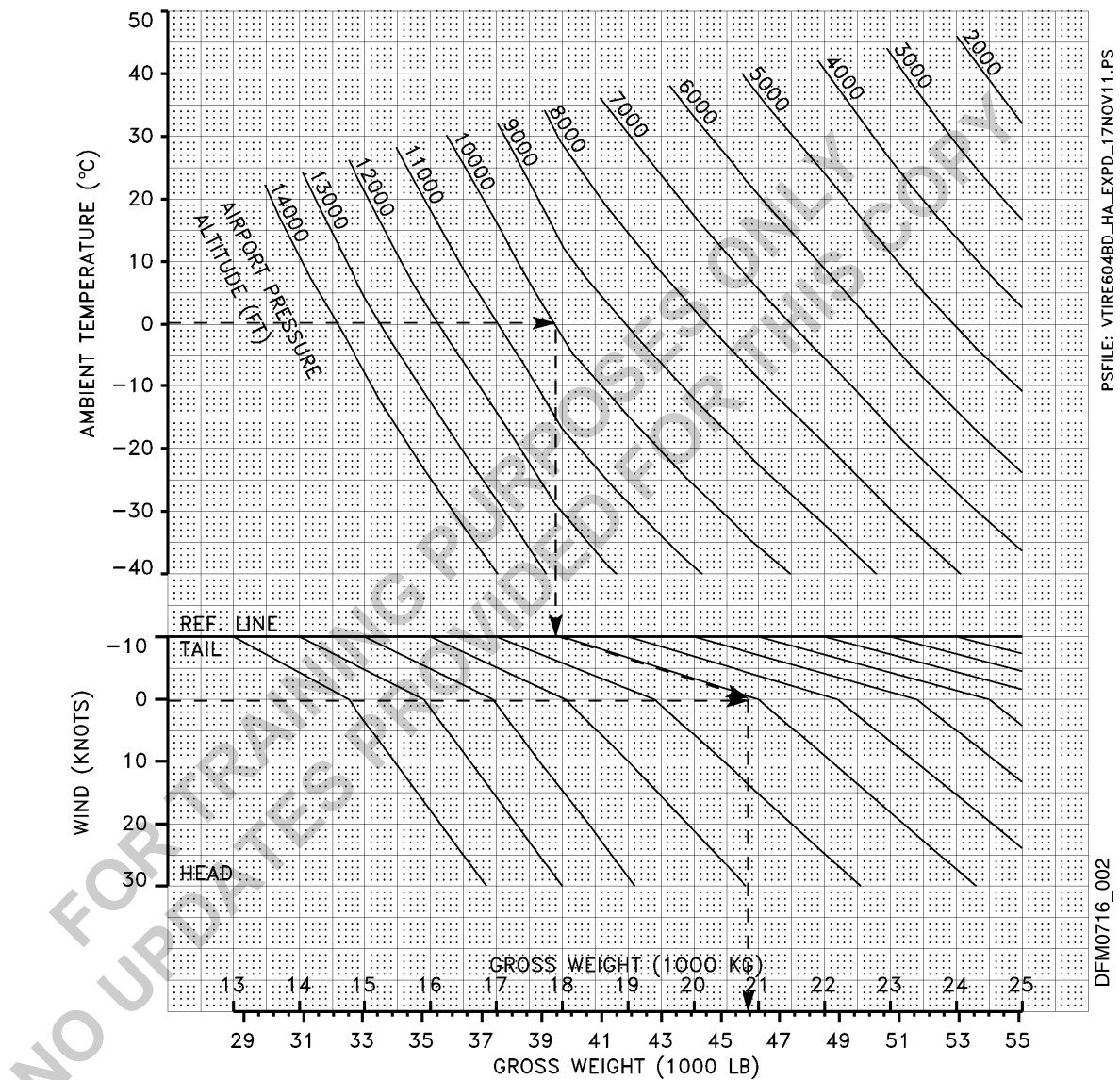
NOTE

If take-off is performed with the APR off, the weight derived from Figure 06-03-3 must be reduced by 200 kg (440 lb).

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2. MAXIMUM ALLOWABLE TAKE-OFF WEIGHT (CONT'D)



Take-Off Weight Limited by Maximum Tire Speed
Figure 06-03-3

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-10
		Oct 27/15

3. TAKE-OFF DISTANCE

The following charts are used to obtain the take-off distance for a given value of airplane weight, ambient temperature, airport pressure altitude, wind, runway slope, APR usage and various bleed configurations.

These charts present the most limiting of the following:

- Take-off distance with all engines operating, multiplied by a factor of 1.15;
- Take-off distance with one engine inoperative; and
- Accelerate-stop distance.

A. Take-Off Distance, Flaps 20° – Anti-Icing Off

The take-off distance for a flaps-20° take-off, at airport pressure altitudes of 10,000 feet and below, with anti-icing off, is given by Figure 06-03-4 (Page 1 of 2).

Example:

Associated Conditions:

Temperature	= 20°C
Airport pressure altitude	= 4,000 feet
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 19,000 kg (41,887 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed

The example in Figure 06-03-4 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 1,800 meters (5,905 feet).

The take-off distance for a flaps-20° take-off, at airport pressure altitudes above 10,000 feet, with anti-icing off, is given by Figure 06-03-4 (Page 2 of 2).

Example:

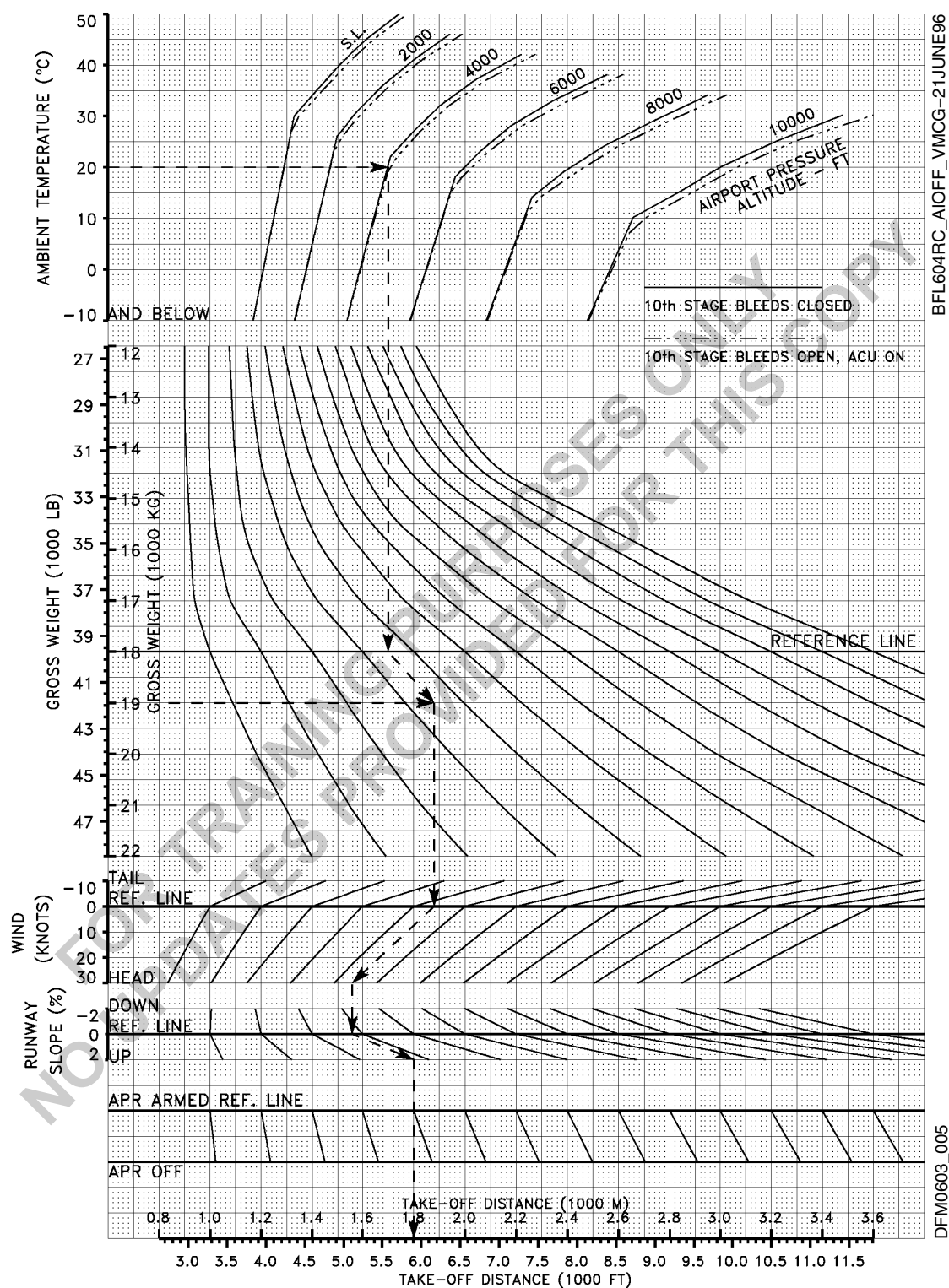
Associated Conditions:

Temperature	= 20°C
Airport pressure altitude	= 12,000 feet
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed

The example in Figure 06-03-4 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,880 meters (9,449 feet).

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3. TAKE-OFF DISTANCE (CONT'D)

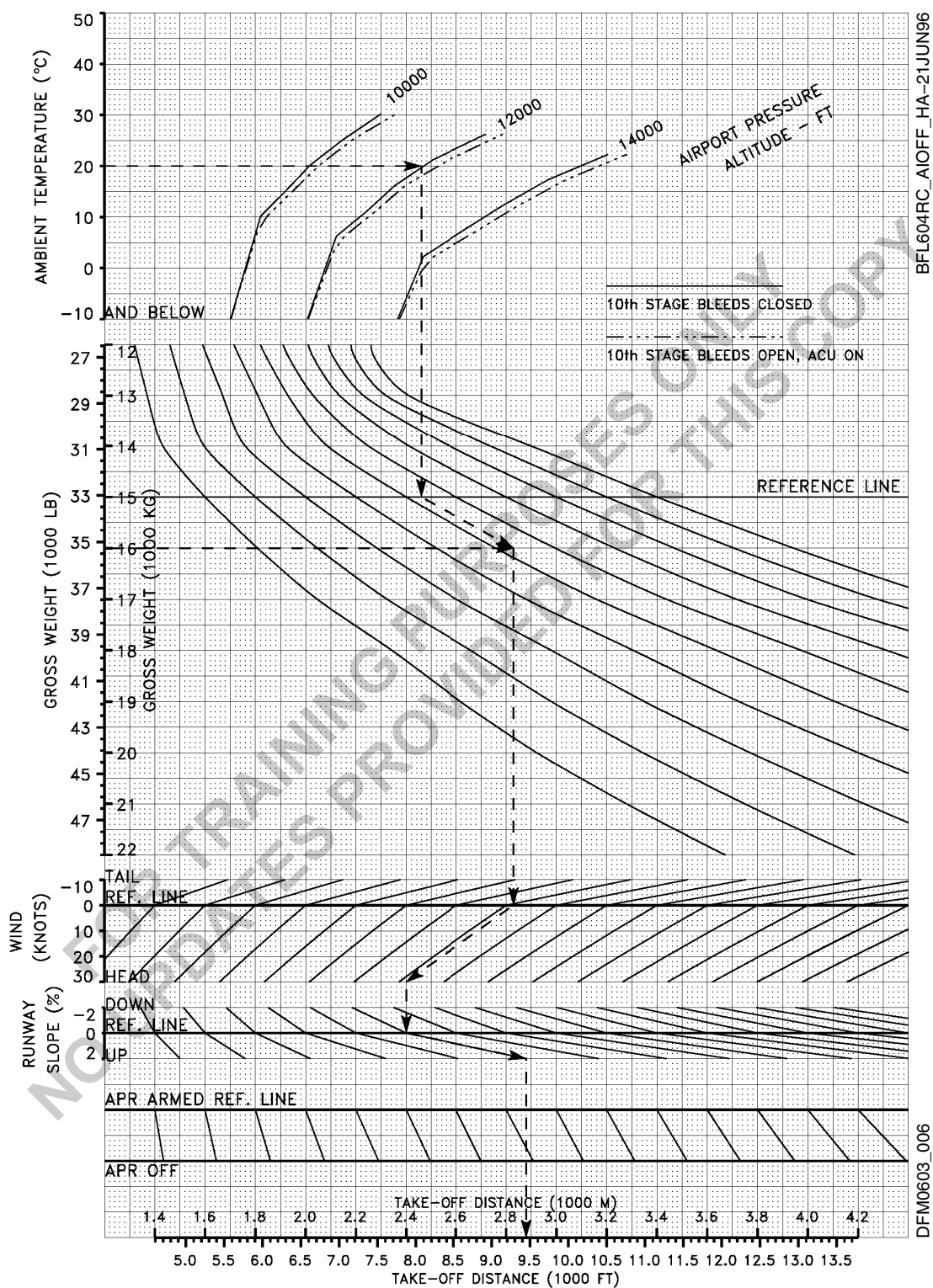


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Take-Off Distance, Flaps 20° – Anti-Icing Off (Page 1 of 2)
Figure 06-03-4

3. TAKE-OFF DISTANCE (CONT'D)

Take-Off Distance, Flaps 20° – Anti-Icing Off (Page 2 of 2)
Figure 06-03-4

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-13
		Oct 27/15

3. TAKE-OFF DISTANCE (CONT'D)

B. Take-Off Distance, Flaps 20° – Anti-Icing On

The take-off distance for a flaps-20° take-off, at airport pressure altitudes of 10,000 feet and below, with anti-icing on is given by Figure 06-03-5 (Page 1 of 2).

Example:

Associated Conditions:

Temperature	= -30°C
Airport pressure altitude	= 6,000 feet
10th stage bleeds	= Closed
Cowl anti-ice	= On
Gross weight	= 19,000 kg (41,887 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed

The example in Figure 06-03-5 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 1,800 meters (5,905 feet).

The take-off distance for a flaps-20° take-off, at airport pressure altitudes above 10,000 feet, with anti-icing on, is given by Figure 06-03-5 (Page 2 of 2).

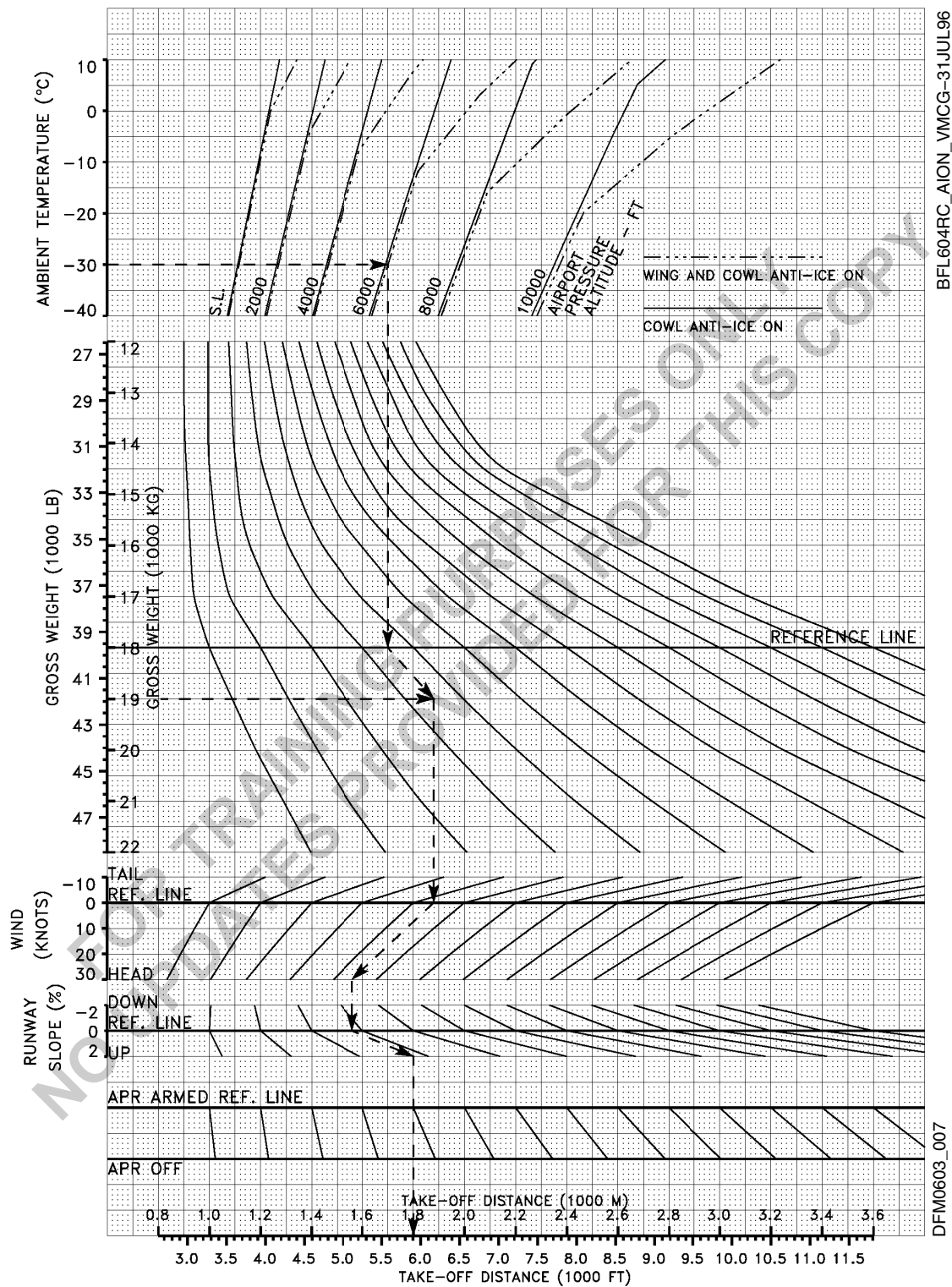
Example:

Associated Conditions:

Temperature	= -10°C
Airport pressure altitude	= 12,000 feet
10th stage bleeds	= Closed
Cowl anti-ice	= On
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed

The example in Figure 06-03-5 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,250 meters (7,382 feet).

3. TAKE-OFF DISTANCE (CONT'D)

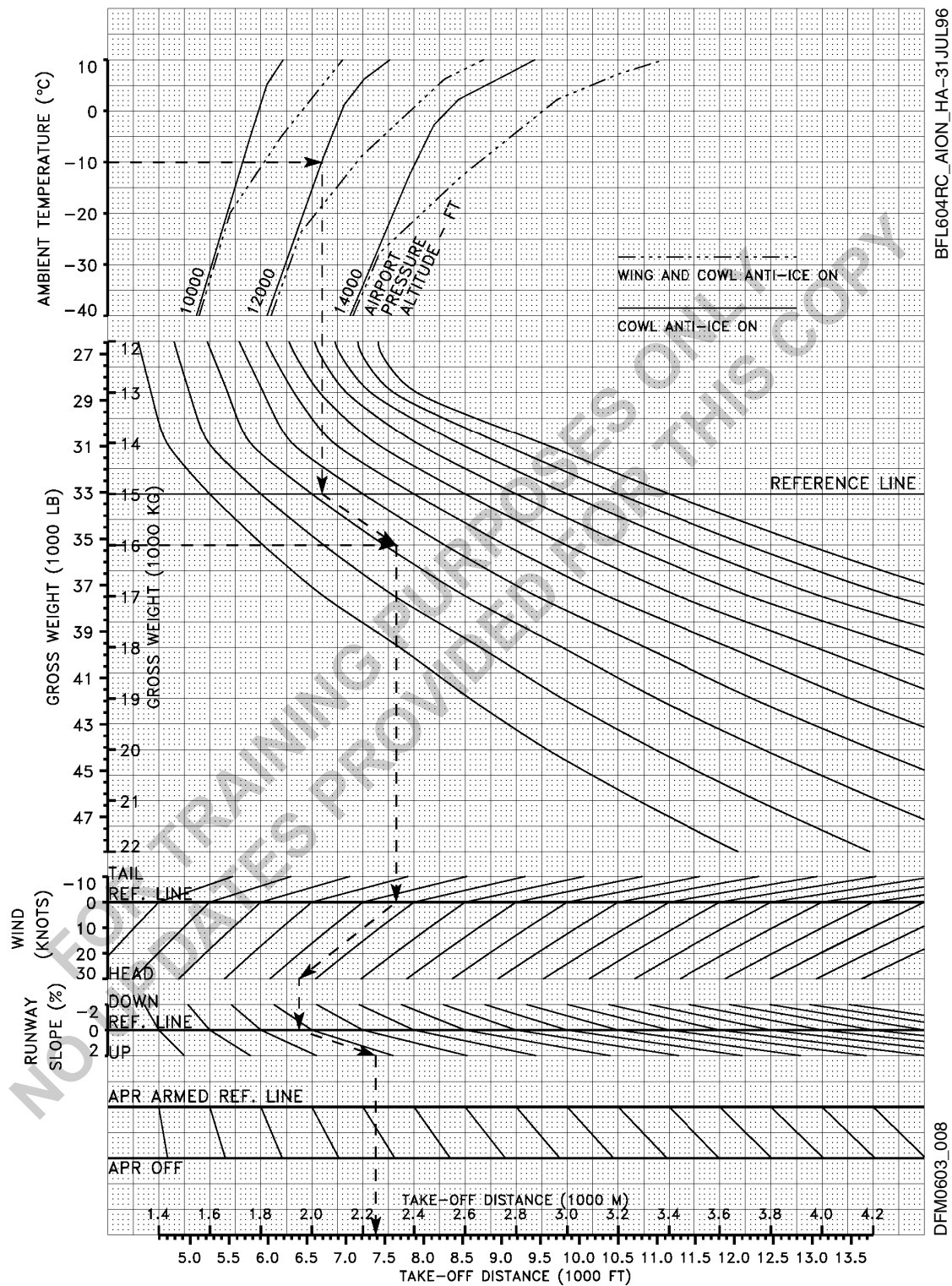


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Take-Off Distance, Flaps 20° – Anti-Icing On (Page 1 of 2)
Figure 06-03-5

3. TAKE-OFF DISTANCE (CONT'D)



Take-Off Distance, Flaps 20° – Anti-Icing On (Page 2 of 2)
Figure 06-03-5

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-16
		Oct 27/15

3. TAKE-OFF DISTANCE (CONT'D)

C. Rolling Take-Off Procedure

When using a rolling take-off procedure:

- The aircraft take-off distance is to be increased by 200 feet.
- For obstacle clearance calculation, use either obstacle distance from end of runway or obstacle distance from corrected take-off distance.

If the FMS take-off function is used to derive the performance information for a rolling take-off:

- The available runway length, reduced by 200 feet, must be manually entered on the CDU TAKEOFF REF pages.
- For obstacle clearance calculation, set OBST DIST to either of the following values on the CDU TAKEOFF REF pages:
 - Obstacle distance from end of runway (not reduced by 200 feet), or
 - Obstacle distance derived using the uncorrected take-off field length.

NOTE

When RWY LENGTH is manually entered, the FMS does not compute headwind/crosswind; these must be manually entered on the CDU.

4. TAKE-OFF SPEEDS

The following charts are used to determine the take-off speeds (V_1 , V_R and V_2).

V_1 can not be less than V_{1MCG} nor greater than V_R .

A. Minimum V_1 Limited By Control on the Ground – V_{1MCG}

Figure 06-03-6 is used to determine the minimum V_1 limited by control on the ground (V_{1MCG}) for varying conditions of take-off weight, ambient temperature, airport pressure altitude and runway slope.

The V_{1MCG} speeds derived from Figure 06-03-6 are applicable for all engine bleed configurations.

Example:

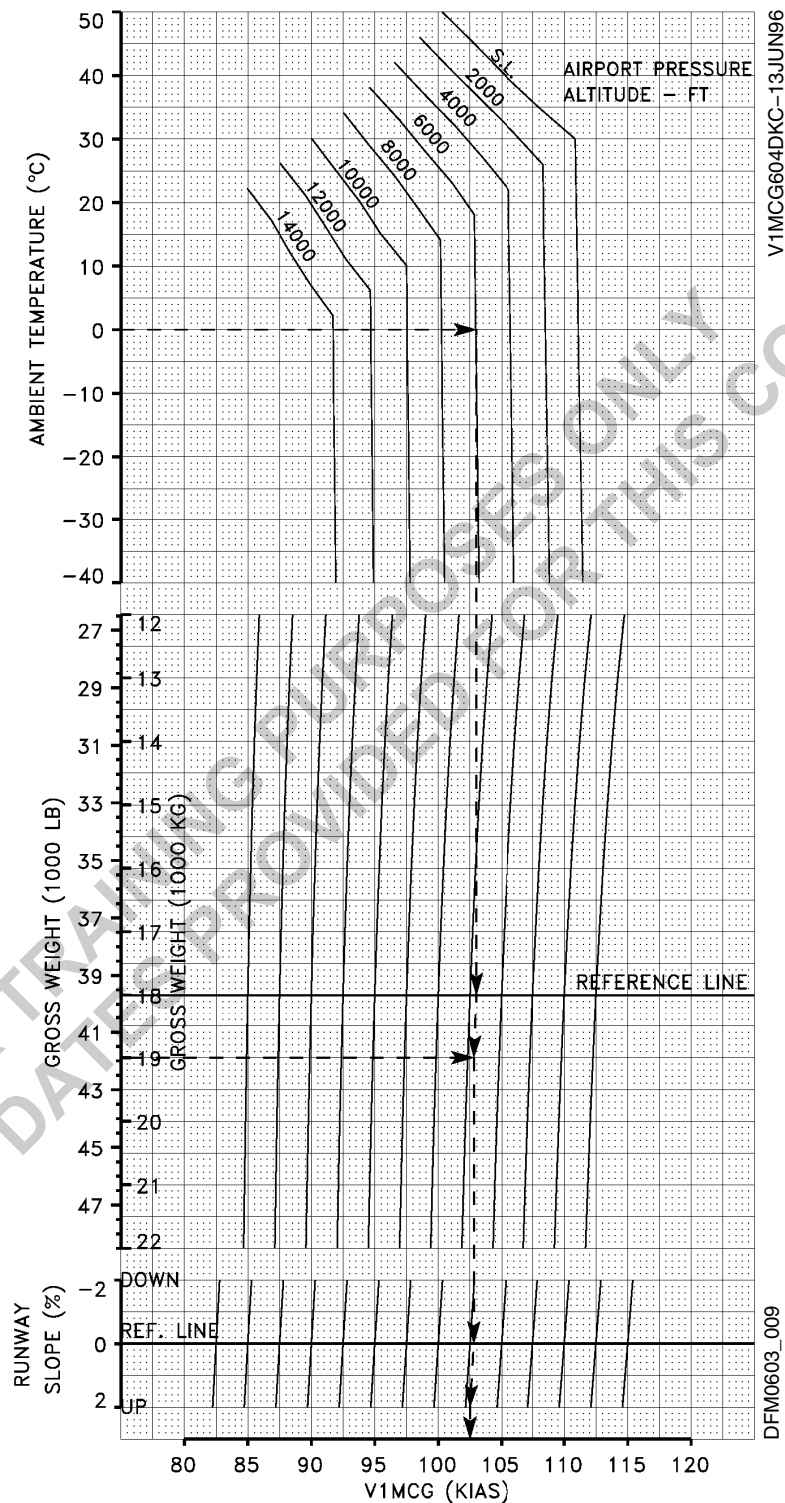
Associated Conditions:

Take-off weight	= 19,000 kg (41,887 lb)
Airport pressure altitude	= 6,000 feet
Ambient temperature	= 0°C
Runway slope	= +2%

The example in Figure 06-03-6 shows that for the above associated conditions, V_{1MCG} is equal to 102.5 KIAS.

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4. TAKE-OFF SPEEDS (CONT'D)



Minimum V₁ Limited by Control on the Ground – V_{1MCG}
Figure 06-03-6

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-18
		Oct 27/15

4. TAKE-OFF SPEEDS (CONT'D)

B. Take-Off Decision Speed, V_1 – Flaps 20° – Anti-Icing Off

Figure 06-03-7 is used to determine the V_1 speed for a flaps-20° take-off (balanced field length), for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope, APR usage and various bleed configurations.

Example:

Associated Conditions:

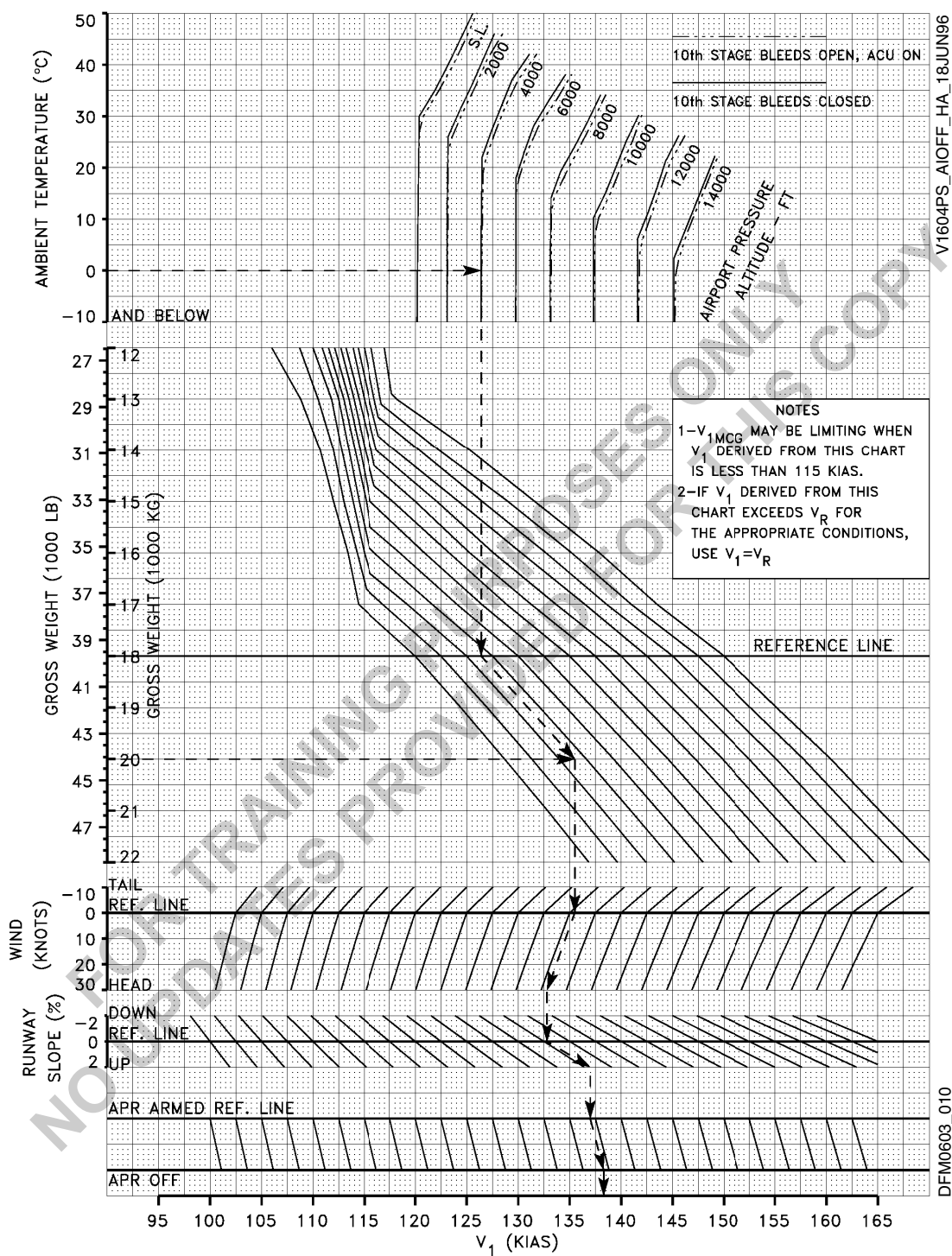
Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 4,000 feet
10th stage bleeds	= Closed
Ambient temperature	= 0°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off

The example in Figure 06-03-7 shows that for the above associated conditions, V_1 is equal to 138.3 KIAS.

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4. TAKE-OFF SPEEDS (CONT'D)



V_1 Speed – Flaps 20°, – Anti-Icing Off
Figure 06-03-7

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-20
		Oct 27/15

4. TAKE-OFF SPEEDS (CONT'D)

C. Take-Off Decision Speed, V_1 – Flaps 20° – Anti-Icing On

Figure 06-03-8 is used to determine the V_1 speed for a flaps-20° take-off (balanced field length), for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope, APR usage and use of anti-icing.

Example:

Associated Conditions:

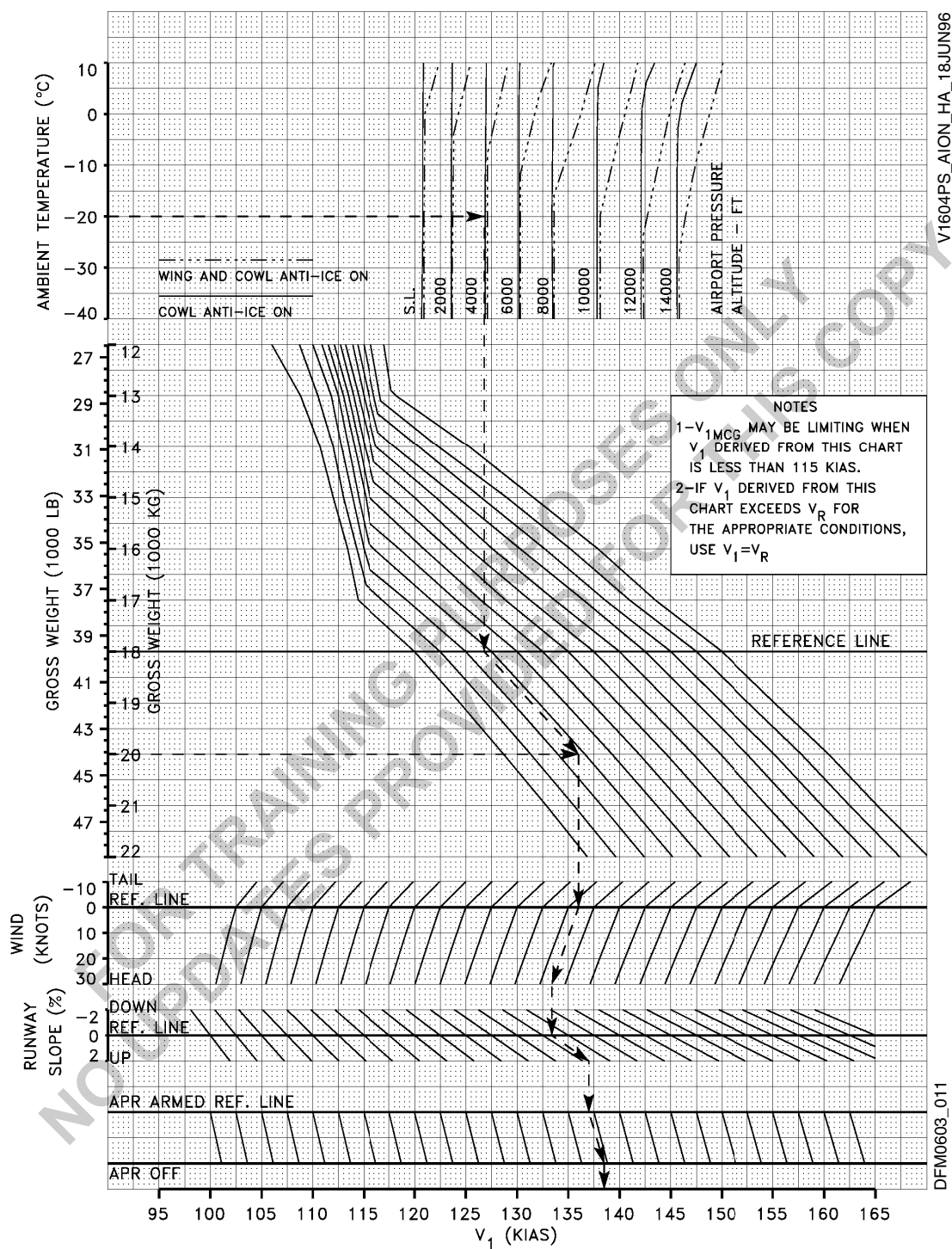
Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 4,000 feet
Cowl anti-icing	= On
Ambient temperature	= -20°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off

The example in Figure 06-03-8 shows that for the above associated conditions, V_1 is equal to 138.5 KIAS.

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4. TAKE-OFF SPEEDS (CONT'D)



V_1 Speed – Flaps 20°, – Anti-Icing On
Figure 06-03-8

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-22
		Oct 27/15

4. TAKE-OFF SPEEDS (CONT'D)

D. Rotation Speed, V_R , Flaps 20° – Anti-Icing Off

Figure 06-03-9 is used to determine the rotation speed (V_R) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude, bleed configurations, take-off weight, runway slope and use of APR.

V_R must not be less than 121 KIAS.

Example:

Associated Conditions:

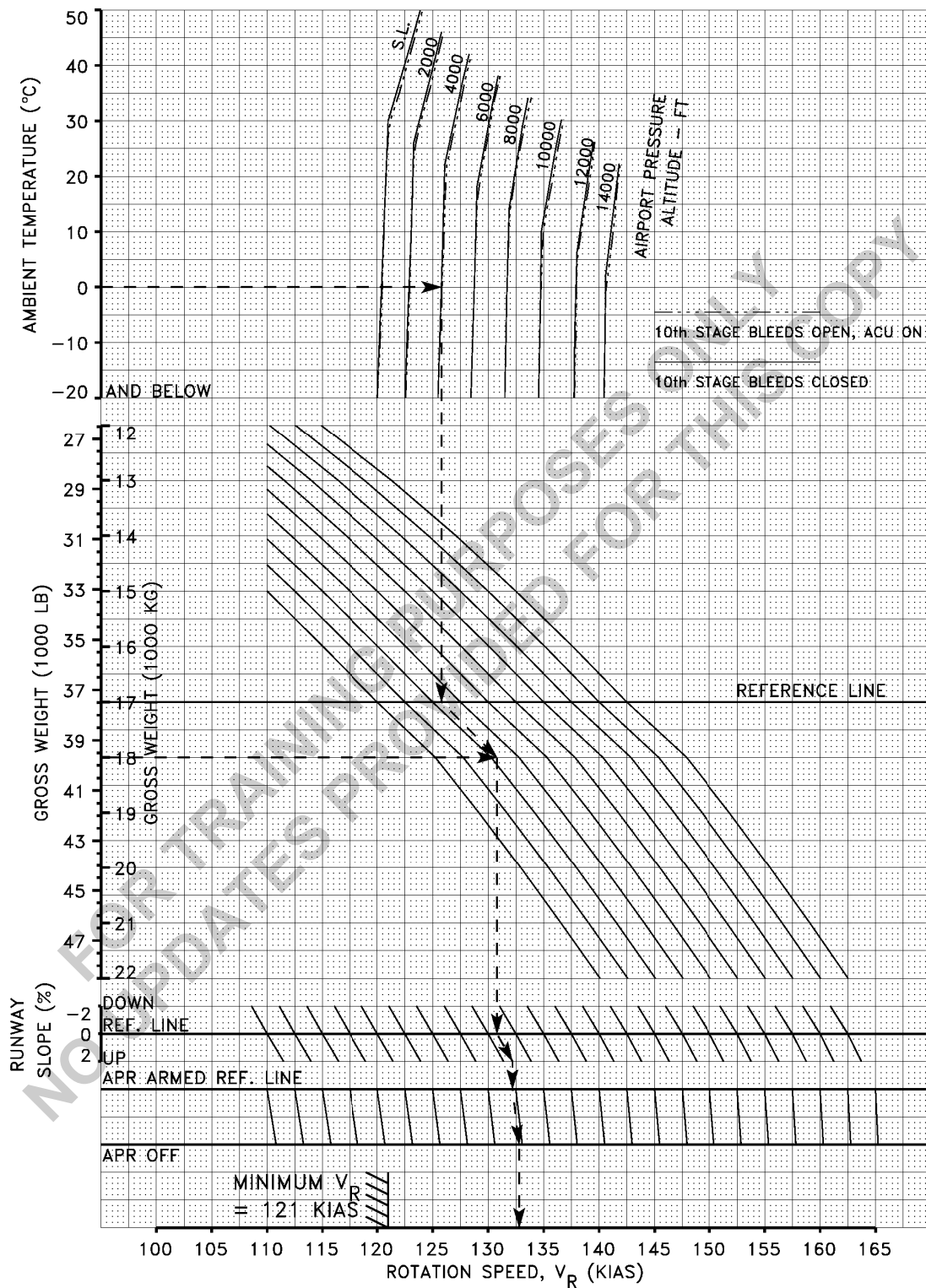
Ambient temperature	= 0°C
Airport pressure altitude	= 4,000 feet
10th stage bleeds	= Closed
Take-off weight	= 18,000 kg (39,683 lb)
Runway slope	= +2%
APR	= Off

The example in Figure 06-03-9 shows that for the above associated conditions, the rotation speed is 133 KIAS.

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4. TAKE-OFF SPEEDS (CONT'D)



Rotation Speed, V_R , Flaps 20° – Anti-Icing Off
Figure 06-03-9

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-24
		Oct 27/15

4. TAKE-OFF SPEEDS (CONT'D)

E. Rotation Speed, V_R , Flaps 20° – Anti-Icing On

Figure 06-03-10 is used to determine the rotation speed (V_R) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude, take-off weight, runway slope, APR usage and use of anti-icing.

V_R must not be less than 121 KIAS.

Example:

Associated Conditions:

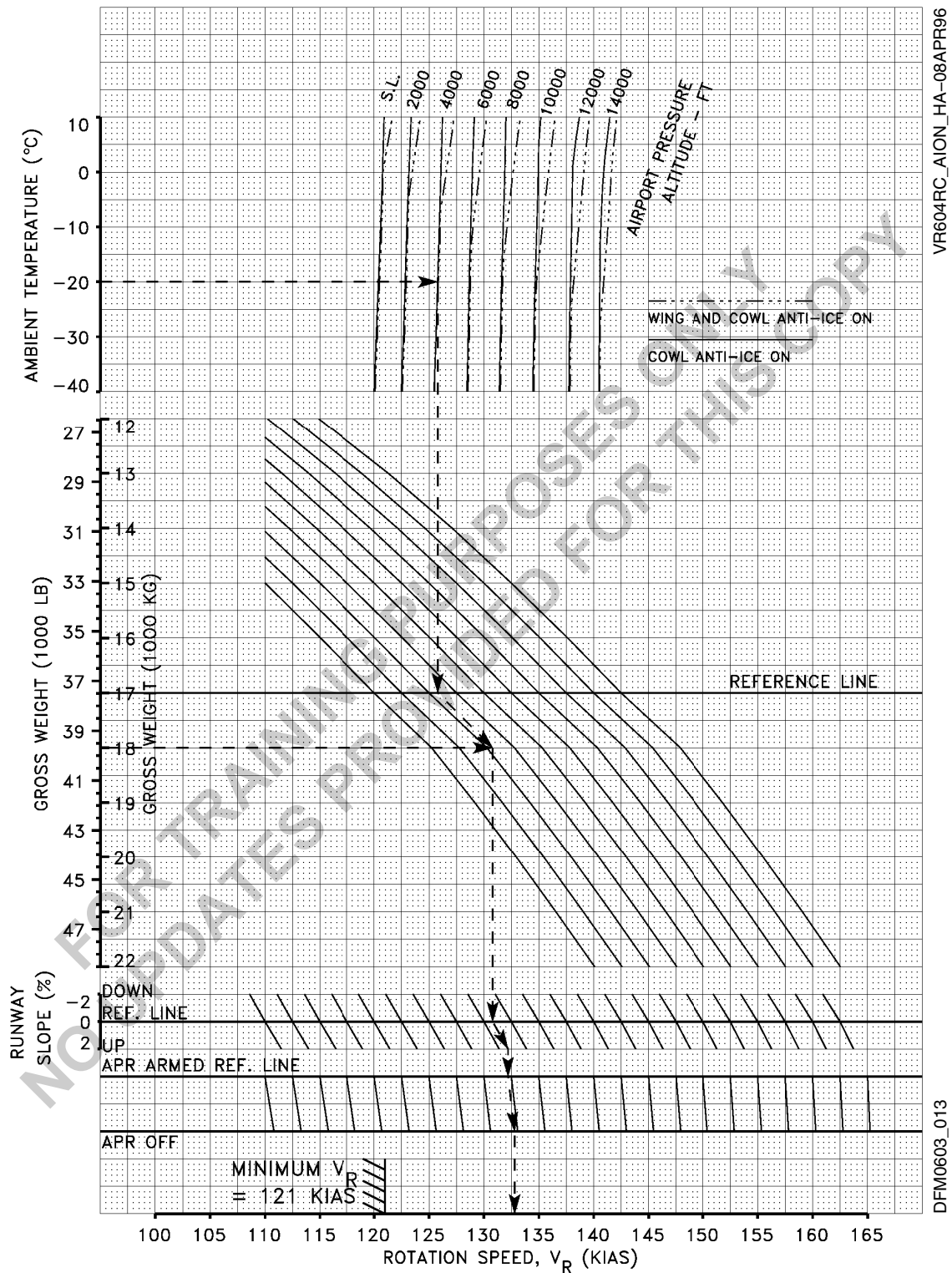
Ambient temperature	= -20°C
Airport pressure altitude	= 4,000 feet
Cowl anti-icing	= On
Take-off weight	= 18,000 kg (39,683 lb)
Runway slope	= +2%
APR	= Off

The example in Figure 06-03-10 shows that for the above associated conditions, the rotation speed is 133 KIAS.

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4. TAKE-OFF SPEEDS (CONT'D)



Rotation Speed, V_R , Flaps 20° – Anti-Icing On
Figure 06-03-10

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-26
		Oct 27/15

4. TAKE-OFF SPEEDS (CONT'D)

F. Take-Off Safety Speed, V_2 – Flaps 20°

Figure 06-03-11 is used to determine the take-off safety speed (V_2) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude and take-off weight.

NOTE

1. V_2 is the greater of the values obtained from Chart A and Chart B of Figure 06-03-11 .
2. V_2 from Chart A is not limiting at weights greater than 17,500 kg (38,581 lb).
3. V_2 from Chart B is not limiting at speeds less than 125 KIAS.

Example:

Associated Conditions:

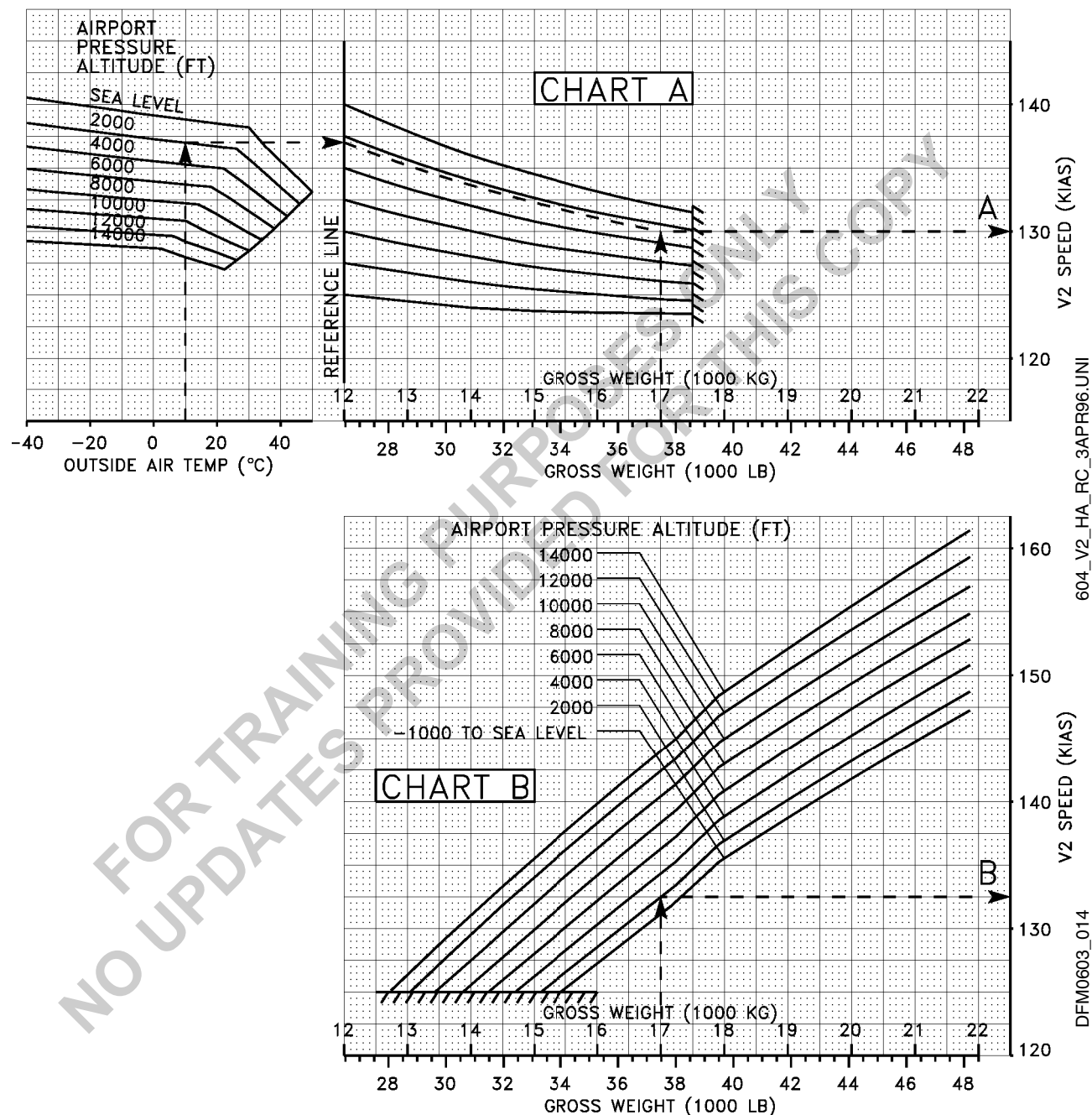
Outside air temperature	= 10°C
Airport pressure altitude	= 2,000 feet
Take-off weight	= 17,000 kg (37,478 lb)

Chart A of Figure 06-03-11 shows that for the above associated conditions, the take-off safety speed is 130.0 KIAS.

Chart B of Figure 06-03-11 shows that for the above associated conditions, the take-off safety speed is 132.5 KIAS.

Therefore, the take-off safety speed under these conditions is 132.5 KIAS.

4. TAKE-OFF SPEEDS (CONT'D)



Take-Off Safety Speed, V_2 – Flaps 20°
Figure 06-03-11

CHALLENGER 650	PERFORMANCE Take-Off Performance	06-03-28
		Oct 27/15

5. MAXIMUM ALLOWABLE BRAKE TEMPERATURE FOR TAKE-OFF

The minimum brake cooling time at the ramp specified in TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires, in this chapter, must be observed following a landing or a rejected take-off, to ensure that brake energy or brake temperature limits are not exceeded during the next take-off, or in the event that the next take-off is rejected.

6. ADJUSTMENTS FOR EXTENDED SECOND SEGMENT CLIMB

If the second segment of the take-off flight path extends beyond a pressure height of 1,500 feet above the airport (Refer to OBSTACLE CLEARANCE in this chapter), adjustments need to be made to the following items calculated from this section:

- V_1 , V_R and V_2 speeds;
- take-off distance;
- take-off weight limited by brake energy; and
- take-off weight limited by maximum tire speed.

Figure 06-03-12 provides these performance adjustments, for a flaps-20° take-off.

NOTE

Additional performance benefits may be obtained by referring to Chapter 7; SUPPLEMENT 13.

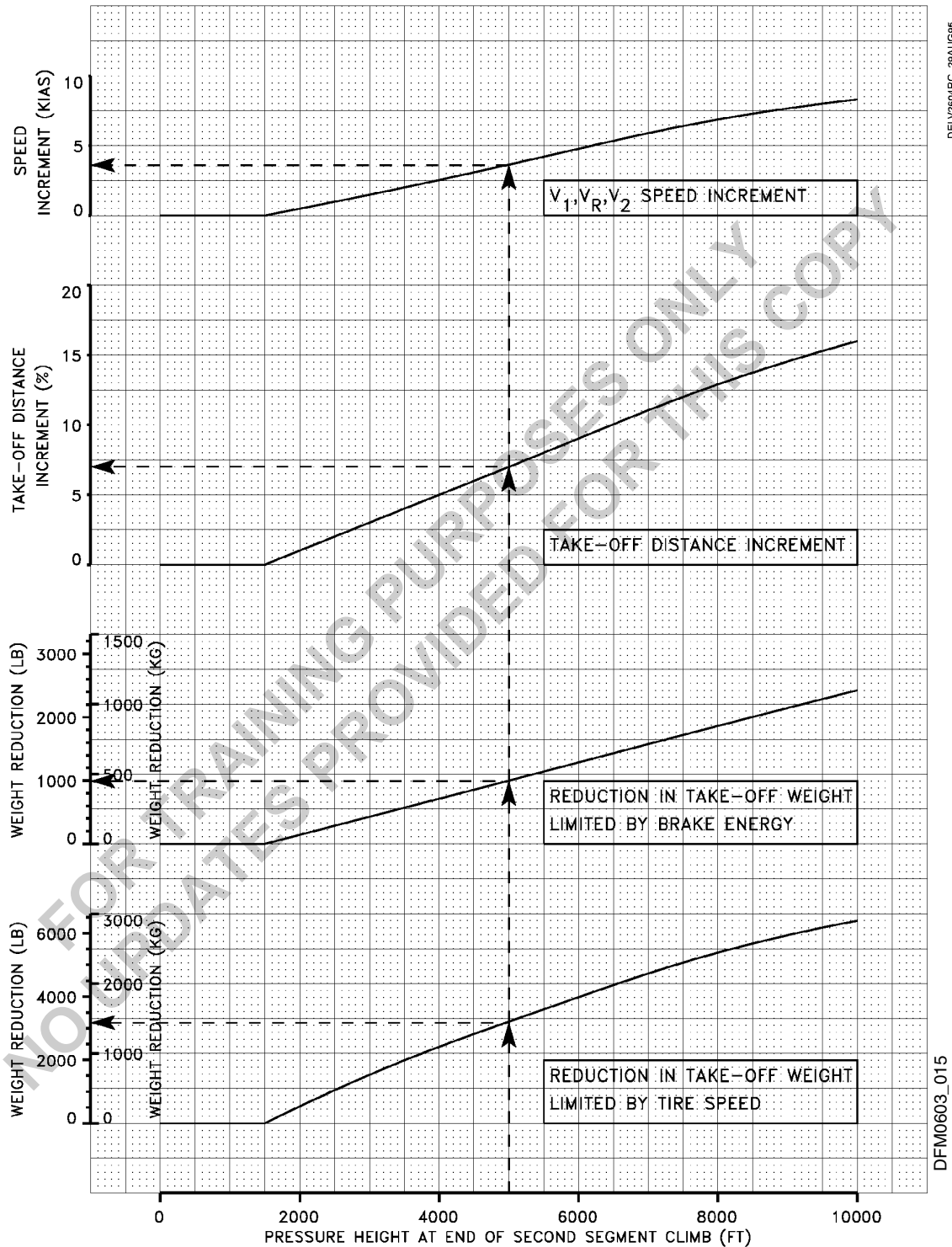
Example:

The example in Figure 06-03-12 shows that for a second segment climb ending at a pressure height of 5,000 feet above the airport, the following adjustments are needed:

- V_1 , V_R , V_2 speeds: increase by 3.5 KIAS
- Take-off distance: increase by 7.0%
- Take-off weight limited by brake energy: reduce by 450 kg (992 lb)
- Take-off weight limited by tire speed: reduce by 1,425 kg (3,142 lb)

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6. ADJUSTMENTS FOR EXTENDED SECOND SEGMENT CLIMB (CONT'D)

Adjustments for Extended Second Segment Climb – Flaps 20°
Figure 06-03-12

CHALLENGER ⁶⁵⁰	PERFORMANCE Take-Off Performance	06-03-30
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CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-1
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1. INTRODUCTION

The net take-off flight path and reference climb gradients required to clear a specific obstacle are provided by the charts in the following pages. The net take-off flight path and reference climb gradients obtained from these charts are used to determine the maximum allowable take-off weight limited by obstacle clearance requirements. An example describing the utilization of these charts is presented in section 3.F.

2. TAKE-OFF FLIGHT PATH

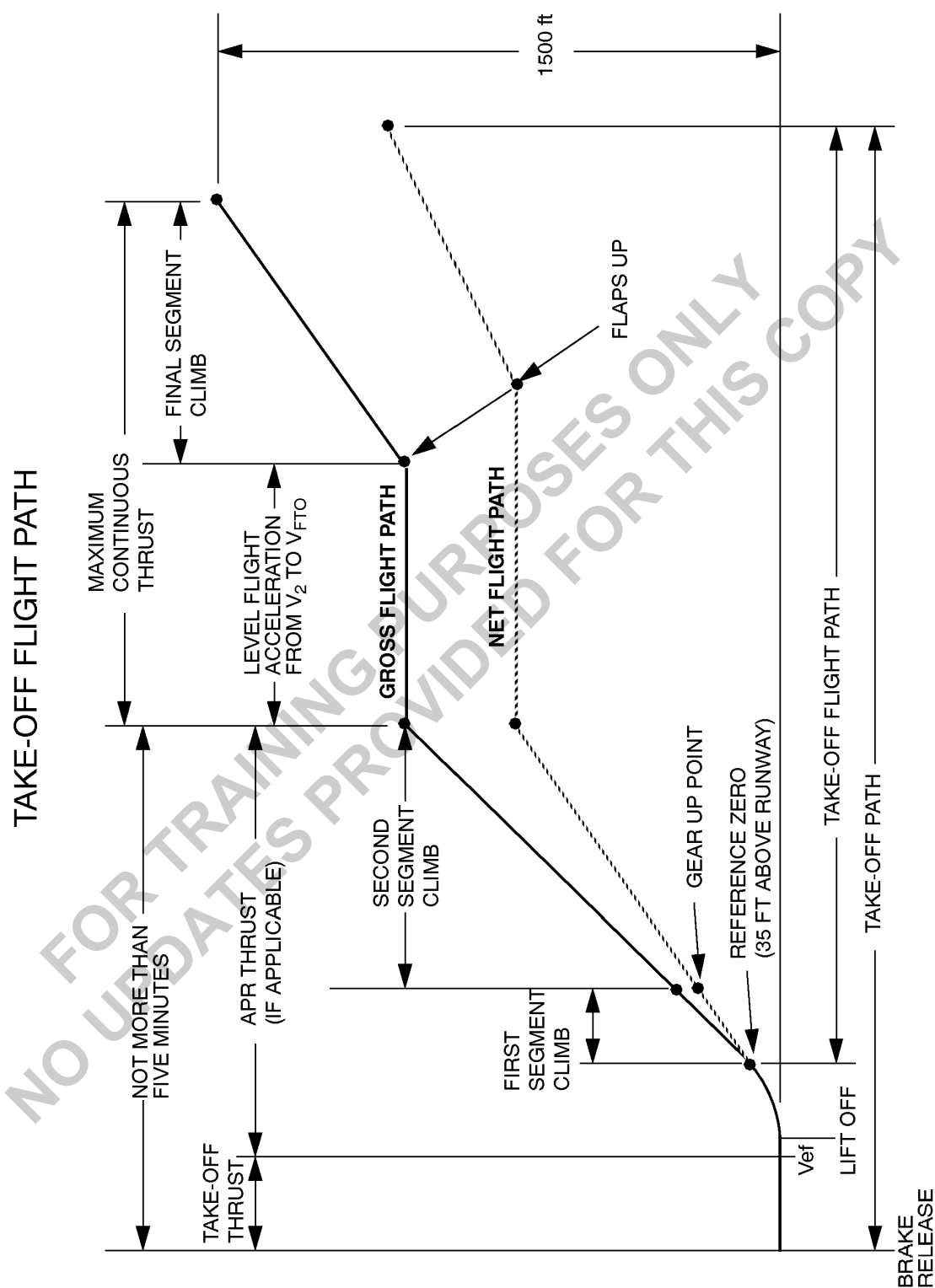
The gross and net take-off flight path segments are presented in Figure 06-04-1, for the case where the gross level-off height is less than 1,500 feet and in Figure 06-04-2 for the case where the gross level-off height is greater than 1,500 feet. The gross flight path represents the actual climb and acceleration performance of the airplane. The net flight path was obtained by reducing the gross climb performance by a climb gradient of 0.8% and by reducing the gross acceleration, during the level flight acceleration segment, by 0.008 g. The net take-off flight path must clear all obstacles vertically by a minimum of 35 feet.

The gross level-off height is selected before take-off. It must not be less than 400 feet and not greater than the gross height attained when the take-off power time limit expires. In addition, the pressure altitude corresponding to the gross level-off height must not exceed 15,000 feet.

Figure 06-04-1 and Figure 06-04-2 show that the take-off flight path starts at reference zero, which is the point at the end of the take-off distance, at 35 feet above the take-off surface. A speed of V_2 is maintained during the first and second segments. The second segment is completed within the take-off power time limit.

If the APR is armed, then the operating engine will be at APR thrust during the first and second segments. If the APR is off, then the operating engine will be at normal take-off thrust during the first and second segments. The level flight acceleration from V_2 to the final segment climb speed, V_{FTO} , is flown with the operating engine at maximum continuous thrust (MCT). The flaps are retracted during the level flight acceleration at a speed of $V_{FTO} - 5$ KIAS. The final climb segment is required if the gross level-off height is less than 1,500 feet. The final climb segment extends to a gross height of 1,500 feet and is flown at a speed of V_{FTO} with the operating engine at MCT.

2. TAKE-OFF FLIGHT PATH (CONT'D)

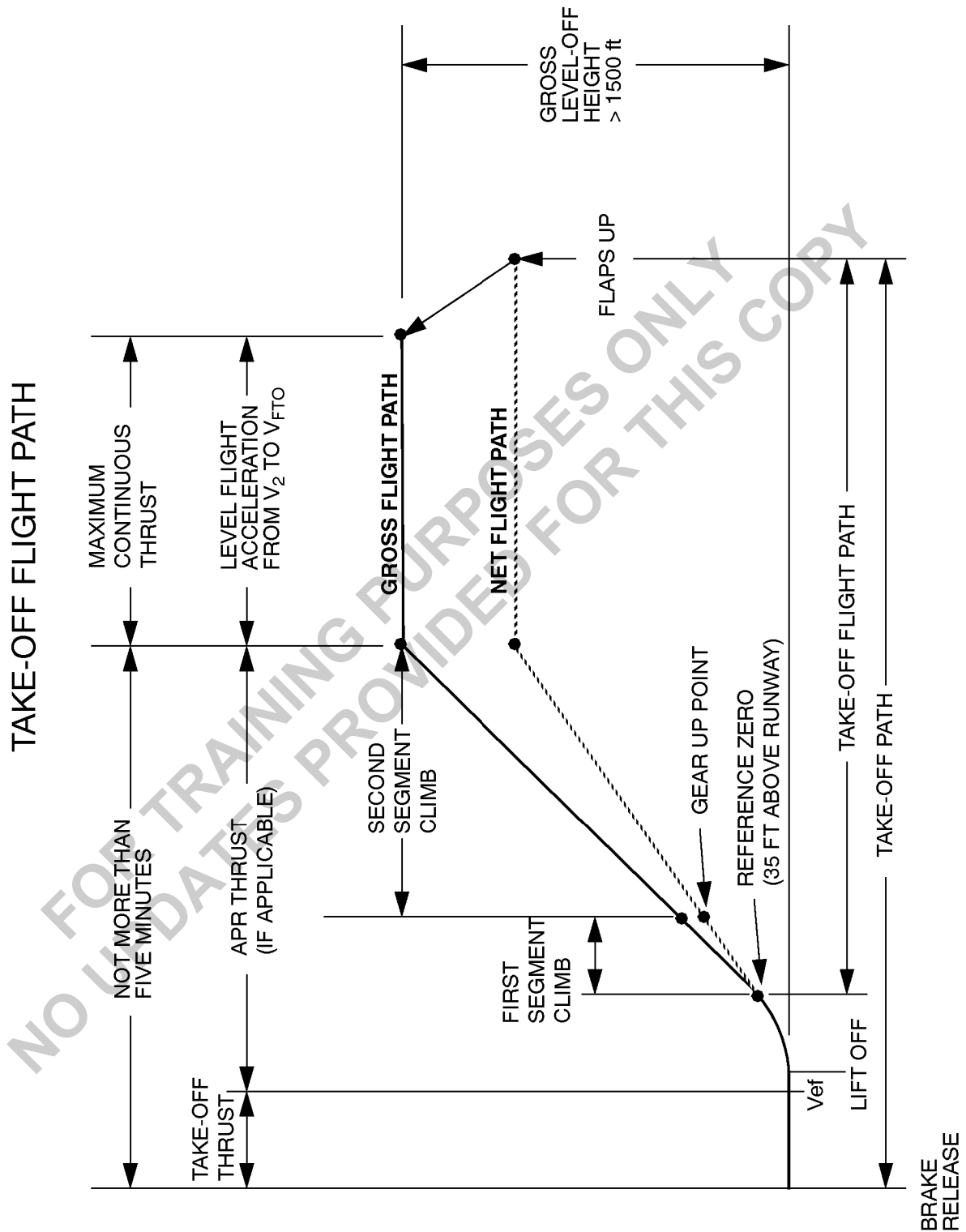


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Take-Off Flight Path – Gross Level-Off Height Less than 1,500 Feet
Figure 06-04-1

CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-3
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2. TAKE-OFF FLIGHT PATH (CONT'D)



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Take-Off Flight Path – Gross Level-Off Height Greater than 1,500 Feet
Figure 06-04-2

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CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-4
		Oct 27/15

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS

The following charts are used to calculate the net take-off flight path to determine the maximum take-off weight limited by obstacle clearance requirements:

Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing Off	= Figure 06-04-3
Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing On	= Figure 06-04-4
Net Take-Off Flight Path, Flaps 20° – Close-In Obstacles	= Figure 06-04-5
Net Take-Off Flight Path, Flaps 20° – Far Obstacles	= Figure 06-04-6
Geometric Height Above Airport to Pressure Height Above Airport Conversion	= Figure 06-04-7
Net Acceleration Distance	= Figure 06-04-8

A. Obstacle Clearance Reference Climb Gradient, Flaps 20°

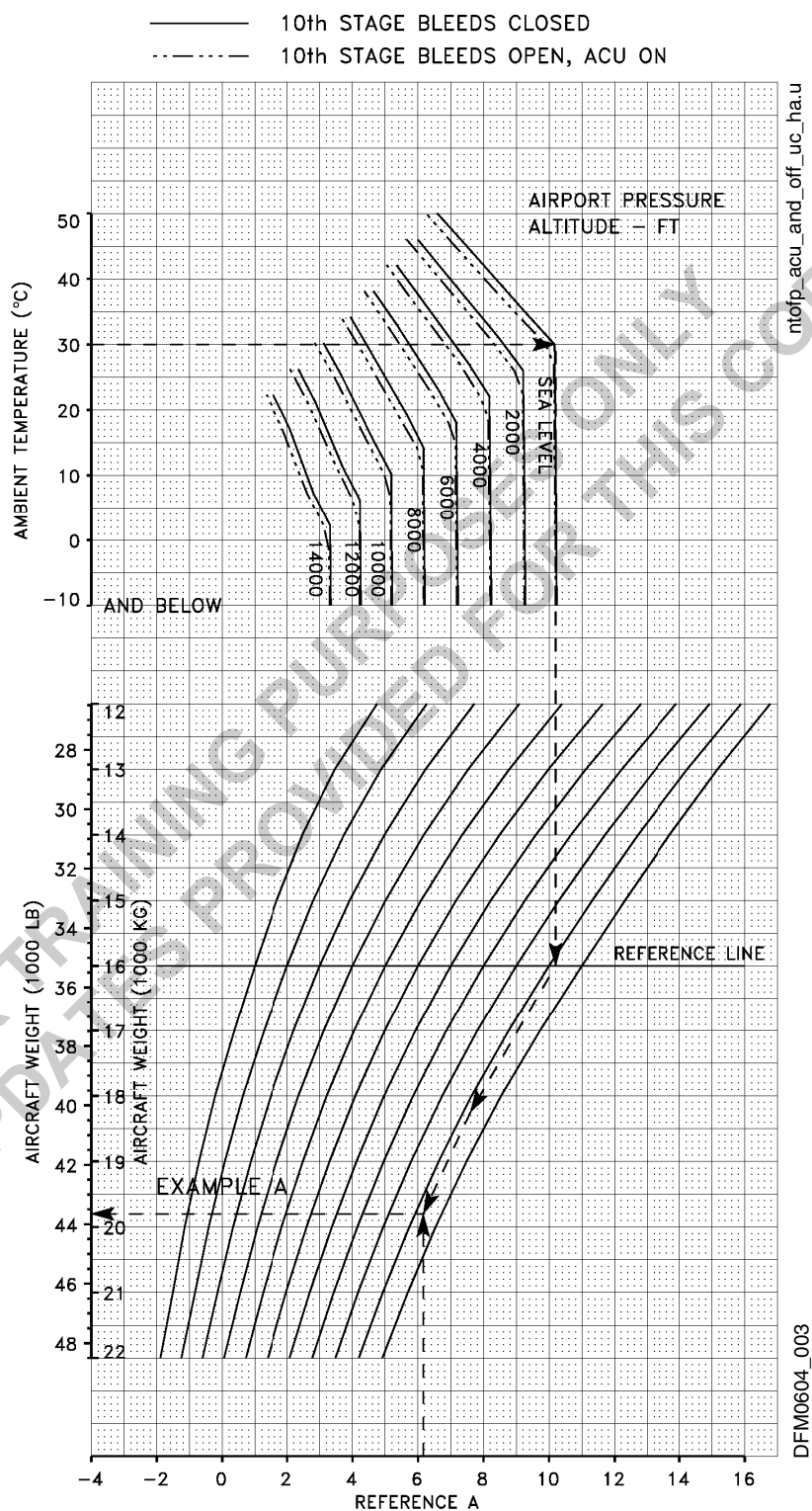
Figure 06-04-3 and Figure 06-04-4 are used to determine the obstacle clearance reference climb gradient for varying conditions of ambient temperature, airport pressure altitude, airplane weight, APR usage, wind condition and engine bleed configurations. These charts can also be used to determine the weight at which a specific value of obstacle clearance reference climb gradient is achieved.

The obstacle clearance reference climb gradient derived from Figure 06-04-3 and Figure 06-04-4 is equal to the average net gradient between the gear up point and the point where the airplane reaches the level-off height. The corresponding gross gradient can be obtained by adding 0.8% to the net gradient.

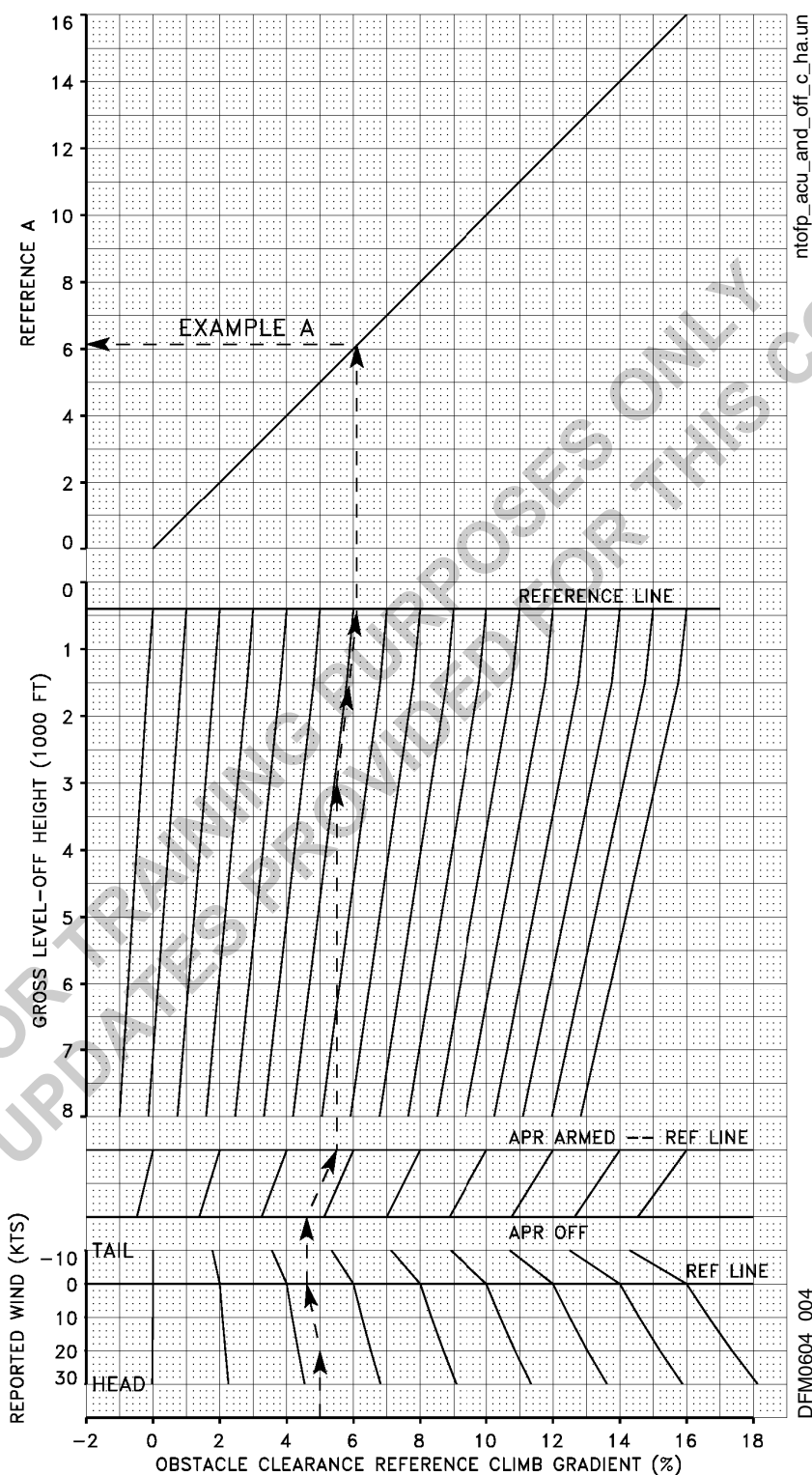
As the obstacle clearance reference climb gradient is a function of the gross level-off height, it is necessary to assume a value for the gross level-off height when using Figure 06-04-3 or Figure 06-04-4. For most obstacle clearance cases, best performance is obtained by selecting a gross level-off height which ensures that all obstacles will be cleared during the first or second segments. The minimum required gross level-off height can be conservatively evaluated from Figure 06-04-5 or Figure 06-04-6 for given values of obstacle height and distance.

Figure 06-04-3 provides the obstacle clearance reference climb gradient for a flaps-20° take-off with anti-icing off.

Figure 06-04-4 provides the obstacle clearance reference climb gradient for a flaps-20° take-off with anti-icing on.

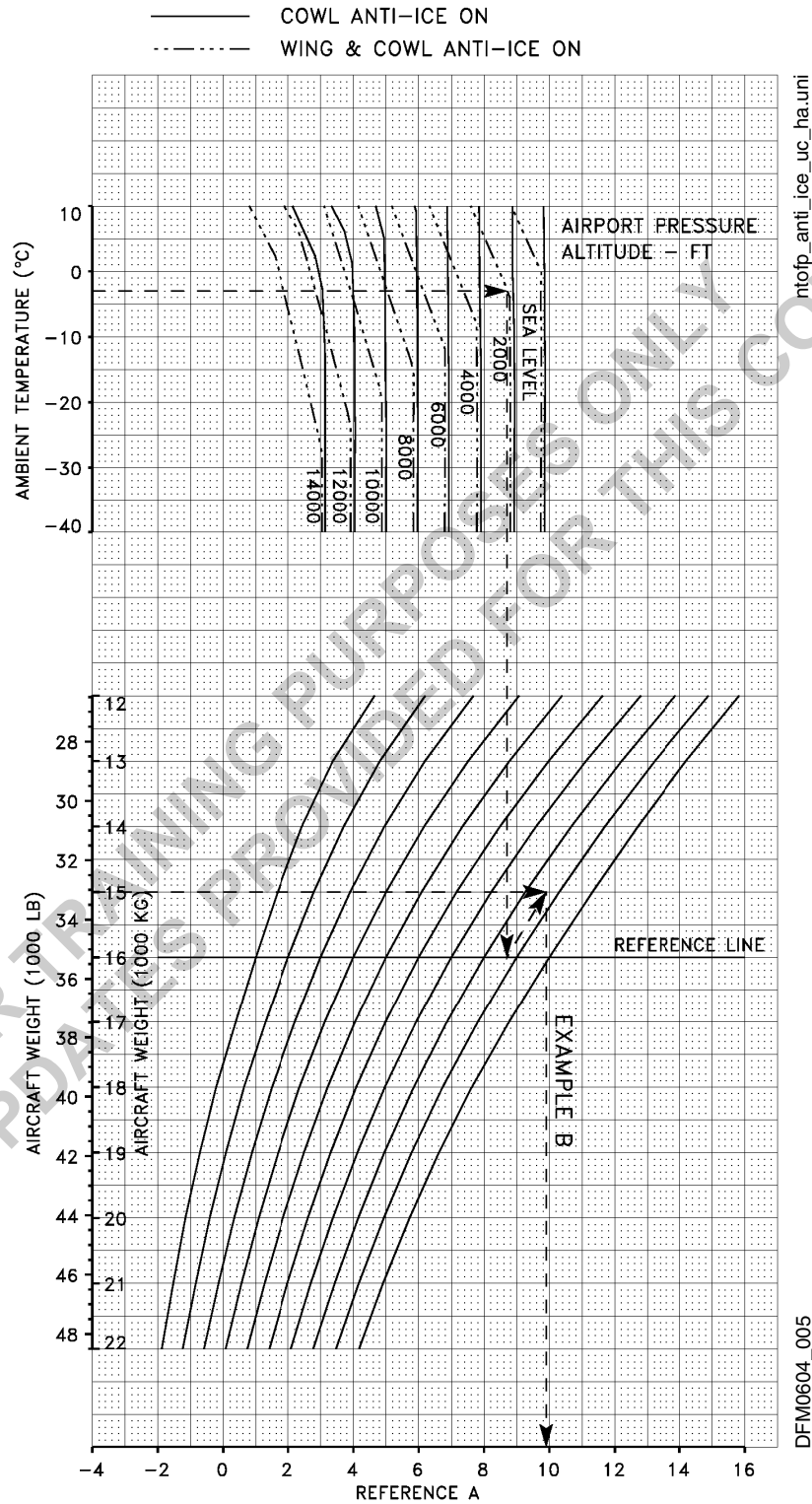
3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS
(CONT'D)Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing Off (Page 1 of 2)
Figure 06-04-3

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

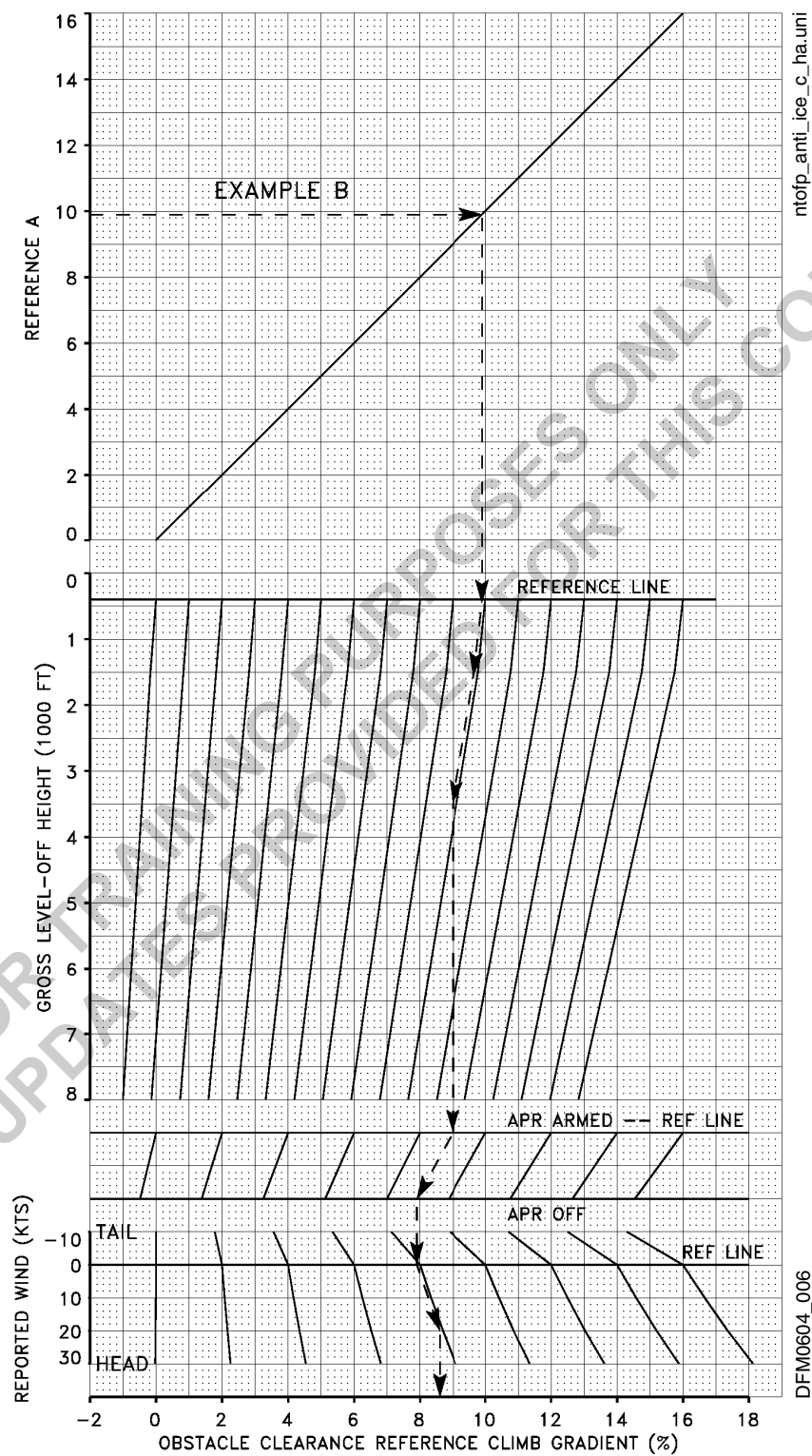


Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing Off (Page 2 of 2)
Figure 06-04-3

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)



Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing On (Page 1 of 2)
 Figure 06-04-4

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS
(CONT'D)Obstacle Clearance Reference Climb Gradient, Flaps 20° – Anti-Icing On (Page 2 of 2)
Figure 06-04-4

CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-9
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

B. Net Take-Off Flight Path

Figure 06-04-5 and Figure 06-04-6 present the net take-off flight path for various obstacle clearance reference climb gradients. These charts can be either used to:

- determine the reference climb gradient which provides a 35-foot margin between the net flight path and an obstacle, or
- determine the net take-off flight path for a given reference climb gradient.

These charts are also used to determine the gross level-off height.

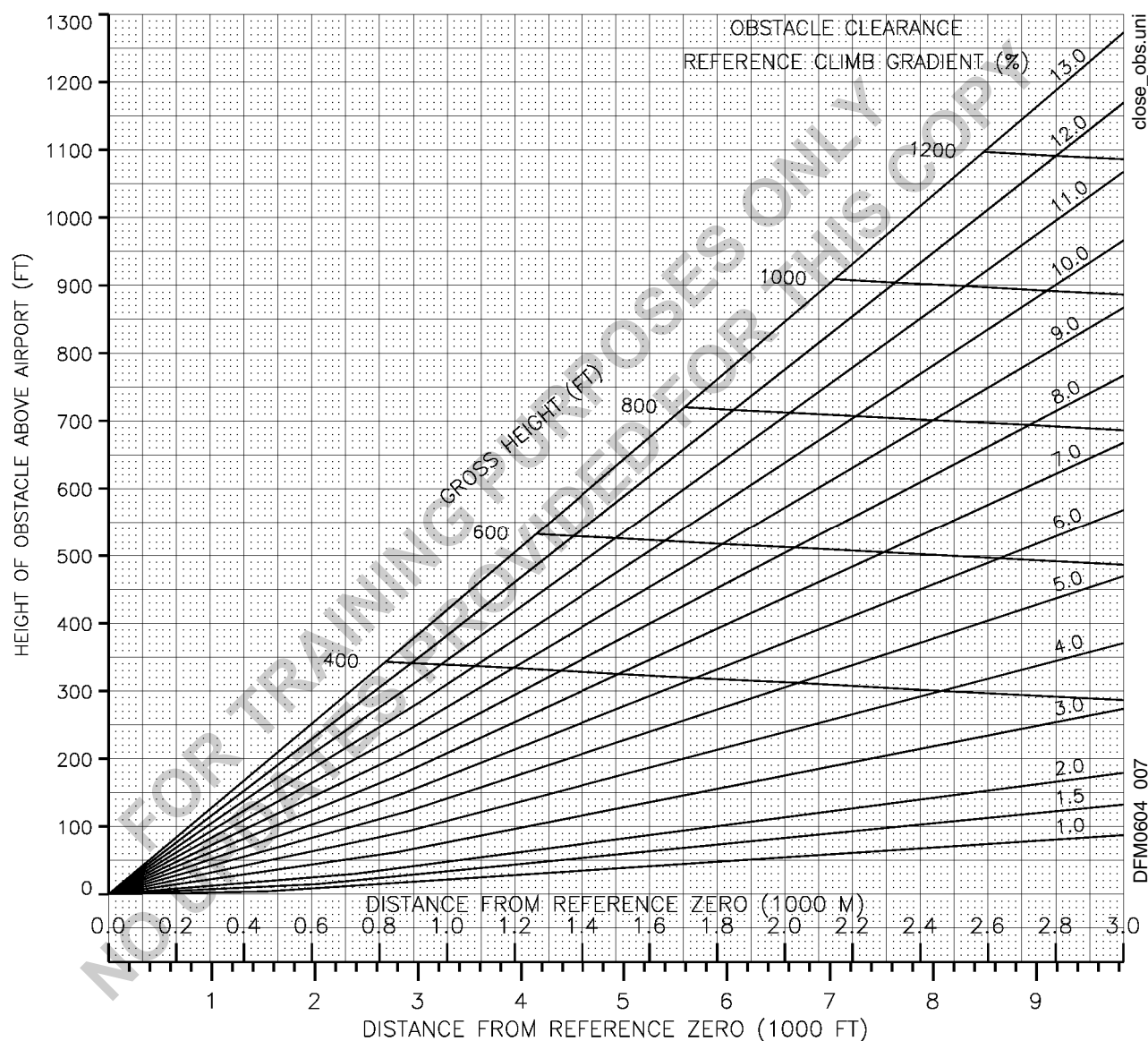
Figure 06-04-5 provides the net take-off flight path for a flaps-20° take-off, for obstacles at a distance of up to 3,000 meters (9,843 feet) from reference zero.

Figure 06-04-6 provides the net take-off flight path for a flaps-20° take-off, for obstacles starting at a distance of 3,000 meters (9,843 feet) from reference zero.

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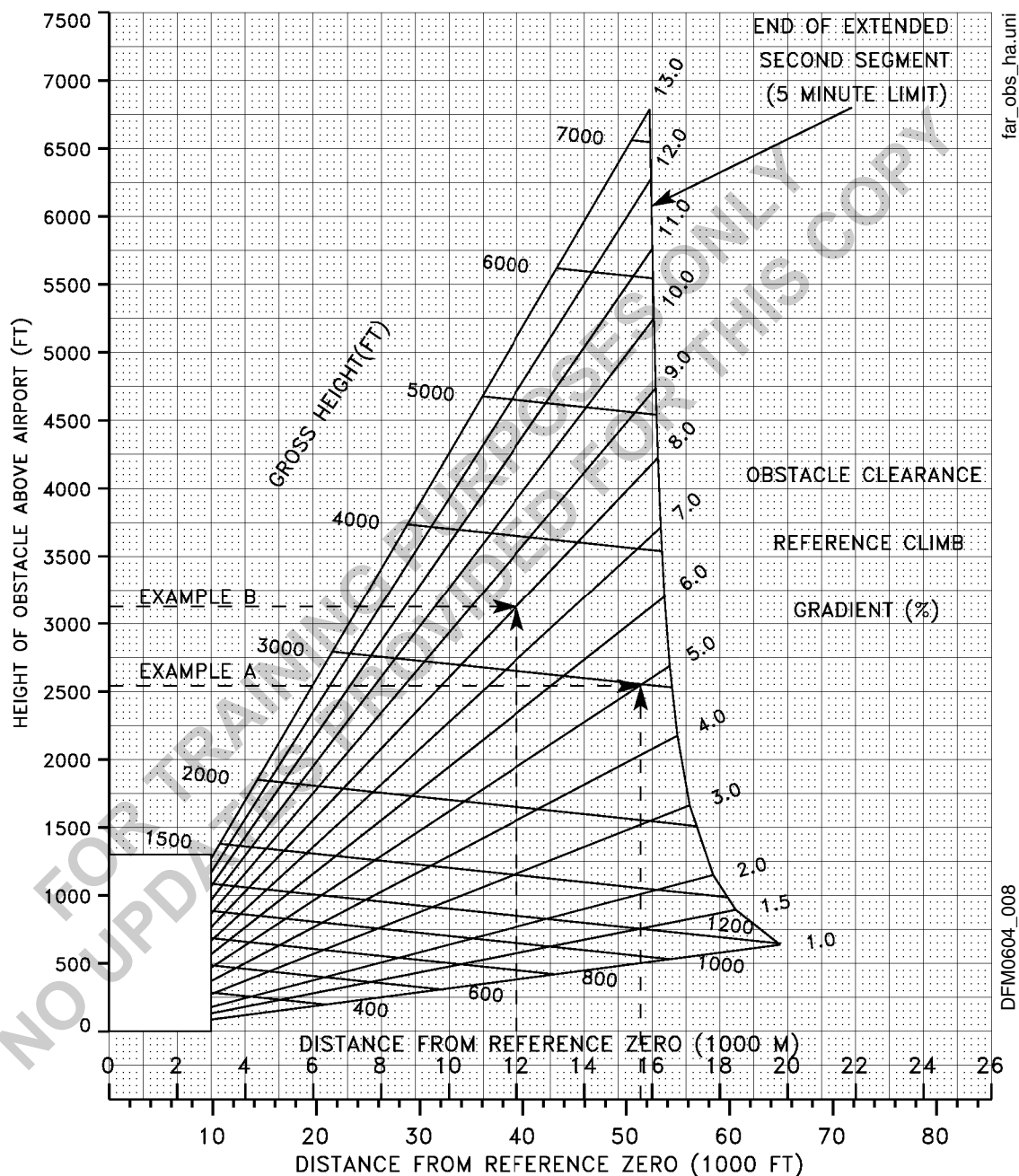
	CL-650 Airplane Flight Manual PSP 650-1	
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)



Net Take-Off Flight Path, Flaps 20° – Close-In Obstacles
Figure 06-04-5

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)



Net Take-Off Flight Path, Flaps 20° – Far Obstacles
Figure 06-04-6

CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-12
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

C. Conversion From Geometric Height To Pressure Height

Figure 06-04-7 is provided to convert geometric height above airport to pressure height above airport at different temperature deviations from standard.

The chart is entered with the gross level-off height (geometric height above airport) and the corresponding value of pressure height above airport is determined. The level-off pressure altitude is obtained by adding the pressure height above the airport, derived from Figure 06-04-7, to the airport pressure altitude.

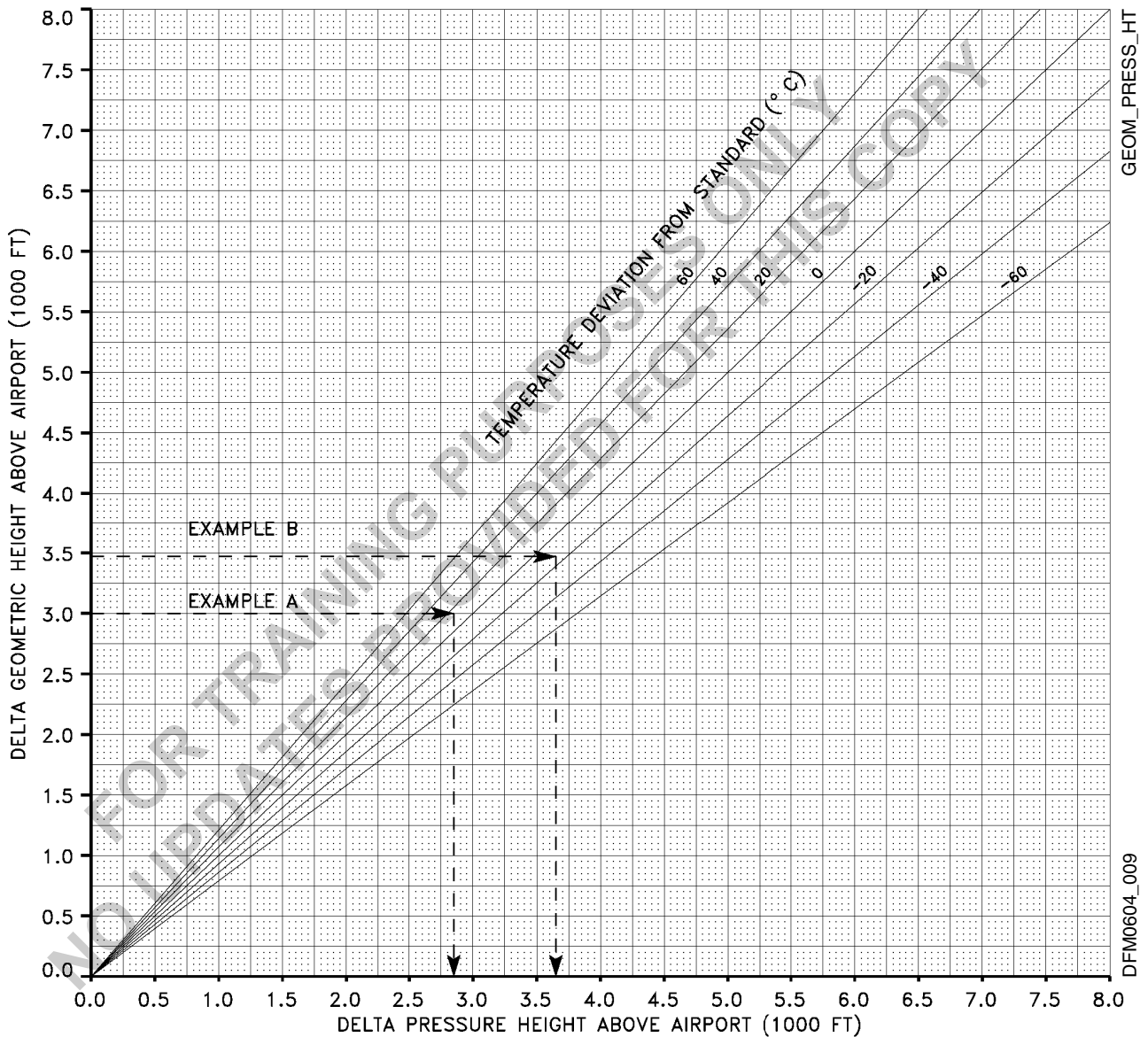
NOTE

Specific take-off performance adjustments, obtained from Figure 06-03-12 (Take-Off Performance section in this chapter) must be applied when the pressure height above the airport exceeds 1,500 feet at the end of the second segment climb.

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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)



Geometric Height Above Airport to Pressure Height Above Airport Conversion
Figure 06-04-7

CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-14
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

D. Net Acceleration Distance

The net acceleration distance, which is the horizontal distance of the level flight acceleration segment from the point where the second segment climb is completed, is provided in Figure 06-04-8.

E. Final Segment Net Climb Gradient

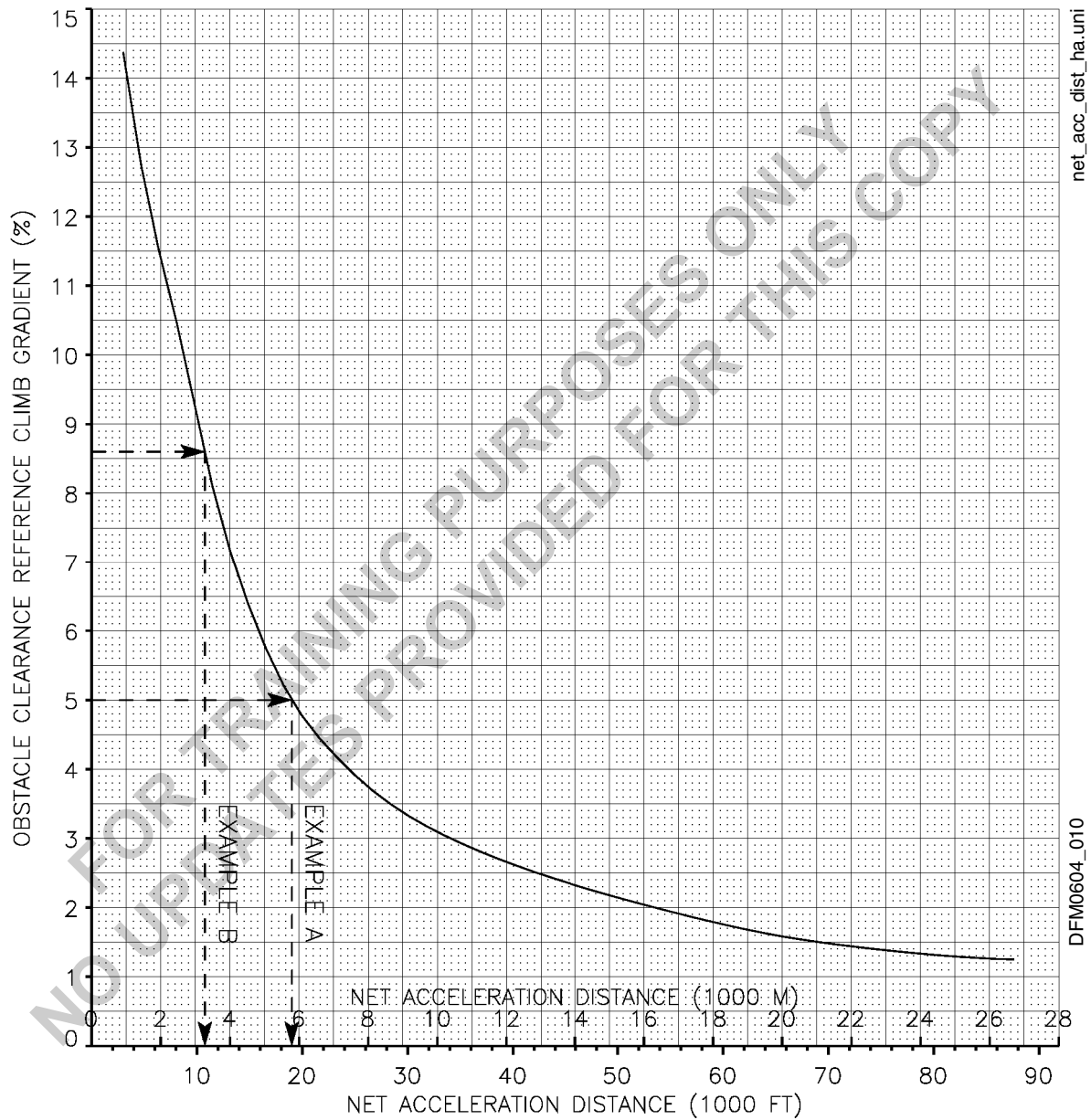
Figure 06-05-2 and Figure 06-05-3 (see next section) can be used to determine the final segment net climb gradient to a gross height of 1,500 feet, when the gross level-off height is less than 1,500 feet. In this case, the appropriate chart must be entered with the pressure altitude corresponding to the 1,500 feet gross height, as derived from Figure 06-04-7, and with a temperature 3°C lower than the ambient airport temperature.

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CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-15
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)



Net Acceleration Distance
Figure 06-04-8

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CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-16
		Oct 27/15

3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

F. Examples – Obstacle Clearance Calculations

The two methods of utilizing the preceding charts are given below. Example A shows the calculation of the maximum allowable take-off weight for a given obstacle. Example B, on the other hand, determines if the airplane with a given take-off weight can clear a specific obstacle.

To simplify the following example, it will be assumed that the take-off distance is equal to the runway length. This approach will result in a conservative value for the maximum take-off weight limited by obstacle clearance requirements.

Example: **DETERMINATION OF THE MAXIMUM ALLOWABLE TAKE-OFF WEIGHT FOR A GIVEN OBSTACLE**

Associated Conditions:

Airport altitude	= Sea level (0 ft)
Ambient temperature	= 30°C (ISA + 15°C)
Reported wind	= 20 knots (headwind)
Take-off flaps	= 20°
APR	= Off
Anti-ice	= Off
10th stage bleeds	= Closed
Obstacle height	= 2,540 feet (above the take-off surface at the end of the runway)
Obstacle distance	= 15,650 meters (51,348 feet) from the end of the runway

Example A of Figure 06-04-6 shows that the obstacle clearance reference climb gradient required for the above conditions is 5.0%. Figure 06-04-6 also shows that the gross height is equal to 3,000 feet when the obstacle is cleared, and that the gross height is equal to 3,220 feet at the end of the five-minute limit. The gross level-off height selected for take-off must therefore not be less than 3,000 feet and not greater than 3,220 feet. A gross level-off height of 3,000 feet will be used for the present example.

Figure 06-04-3 shows that the maximum weight to achieve a reference climb gradient of 5.0% is 19,800 kg (43,651 lb). Therefore, the maximum take-off weight limited by obstacle clearance requirements is 19,800 kg (43,651 lb).

For a gross level-off height of 3,000 feet, the pressure height above the airport for level acceleration, is obtained from Figure 06-04-7 and is found to be 2,860 feet, for an ISA + 15 day. Since the airport pressure altitude is 0 ft., the level-off pressure altitude is also equal to 2,860 feet.

The obstacle clearance reference climb gradient derived from Figure 06-04-6 is used to determine the net acceleration distance using Figure 06-04-8, which Example A shows to be 5,800 meters (19,030 feet).

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CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-17
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3. MAXIMUM TAKE-OFF WEIGHT LIMITED BY OBSTACLE CLEARANCE REQUIREMENTS (CONT'D)

The take-off weight limited by obstacle clearance requirements, determined in the previous example, can further be improved, if the take-off distance at that weight is less than the runway length. In this case, the previous calculation procedure is repeated using the obstacle height and distance relative to reference zero (end of the take-off distance). One or more iterations may be required to obtain the optimum weight.

Example: DETERMINATION OF THE NET TAKE-OFF FLIGHT PATH

Associated Conditions:

Airplane weight	= 15,000 kg (33,069 lb)
Airport altitude	= 2,000 feet
Ambient temperature	= -3°C (ISA -14°C)
Reported wind	= 20 knots (headwind)
Take-off flaps	= 20°
APR	= Off
Wing anti-ice	= On
Cowl anti-ice	= On
10th stage bleeds	= Closed
Obstacle distance	= 12,000 meters (39,372 feet) from reference zero (end of the take-off distance)
Obstacle height	= 3,125 feet above the airport

Figure 06-04-6 shows that an obstacle clearance reference climb gradient of not less than 8.0% is required in order to clear the obstacle. Figure 06-04-6 also shows that a gross level-off height of 3,475 feet can be used for take-off.

Figure 06-04-4 shows that for a gross level-off height of 3,475 feet and the associated conditions above, the actual obstacle clearance reference climb gradient is 8.6%. It can therefore be concluded that the airplane will clear the obstacle.

The gross level-off height (geometric height) of 3,475 feet derived from Figure 06-04-6 is used to determine the level flight acceleration pressure altitude using Figure 06-04-7, which for an ISA - 14 day is 3,650 feet + 2,000 feet = 5,650 feet.

The obstacle clearance reference climb gradient derived from Figure 06-04-6 is used to determine the net acceleration distance using Figure 06-04-8, which Example B shows to be 3,250 meters (10,663 feet).

CHALLENGER 650	PERFORMANCE Obstacle Clearance	06-04-18
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4. GRADIENT LOSS IN A STEADY TURN

The gradient loss in a steady turn is tabulated below for a 15° bank angle.

For bank angles less than 15°, the gradient loss may be considered proportional to bank angle.

FLAPS	GRADIENT LOSS (%)
0°	0.30
20°	0.38

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CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-1
		Oct 27/15

1. INTRODUCTION

The enroute performance section contains the charts used to determine the final take-off climb gradient and the single-engine enroute climb gradient of the airplane at flaps 0°.

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CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-2
		Rev 15, Mar 25/19

2. FINAL SEGMENT AND ENROUTE CLIMB SPEED

The final segment climb speed (V_{FTO}) and enroute climb speed (V_{ERC}) with flaps retracted, one engine inoperative and the other engine operating at maximum continuous thrust (MCT), is provided by Figure 06-05-1, for varying conditions of pressure altitude and airplane gross weight.

Example:

Associated Conditions:

Gross weight = 18,000 kg (39,683 lb)

Pressure altitude = 10,000 ft

The example on Figure 06-05-1 shows that for the above associated conditions, the final segment and enroute climb speed is 172 KIAS.

On airplanes 6050 to 6146 NOT incorporating Service Bulletin 650-34-012 or 650-34-015:

NOTE

The AFM must be used for the V_{ERC} data. The use of the V_{ERC} data on the THRUST LIMIT page is prohibited.

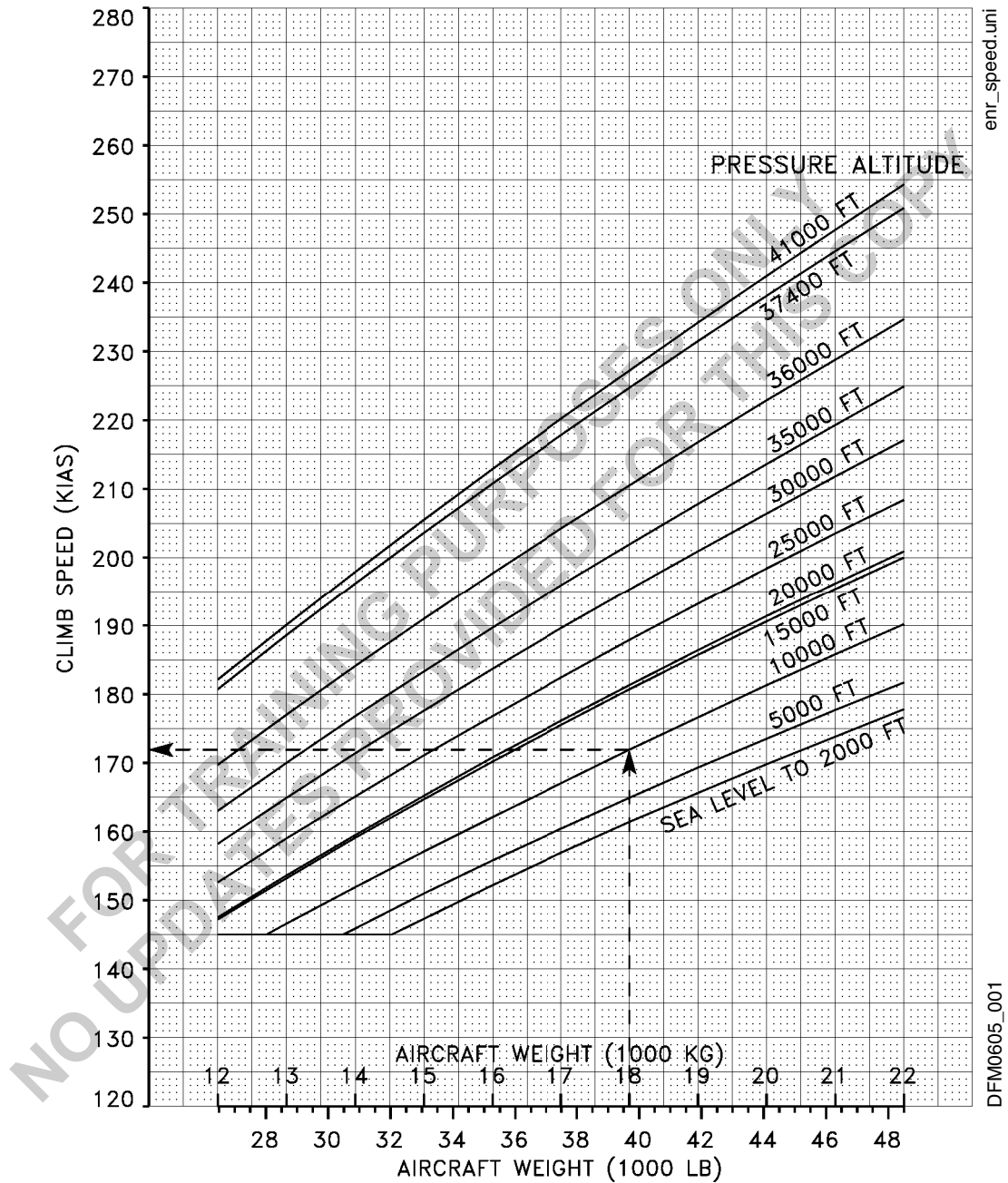
On airplanes 6147 and subs and airplanes incorporating Service Bulletin 650-34-012 or 650-34-015:

NOTE

The V_{ERC} data shown on the THRUST LIMIT page is advisory information only and must be cross checked with the AFM.

CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-3
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2. FINAL SEGMENT AND ENROUTE CLIMB SPEED (CONT'D)



Final Segment and Enroute Climb Speed – Flaps 0°
Figure 06-05-1

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CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-4
		Oct 27/15

3. FINAL SEGMENT AND ENROUTE CLIMB GRADIENT

A. Final Segment and Enroute Climb Gradient, Flaps 0° – Anti-Icing Off

The final segment and enroute climb gradient with one engine inoperative and the other engine operating at maximum continuous thrust (MCT), is provided by Figure 06-05-2, with 10th stage bleed open/one (1) pack on and anti-icing off.

NOTE

If ice accumulation is suspected on unprotected surfaces, subtract 0.9 percent from the gradient derived from Figure 06-05-2.

Example:

Associated Conditions:

Temperature	= 27°C
Pressure altitude	= 5,000 ft
Gross weight	= 15,000 kg (33,069 lb)

The example on Figure 06-05-2 shows that for the above associated conditions, the following climb gradients are obtained:

- Gross final segment and gross enroute climb gradients = 9.6%
- Net final segment climb gradient = 8.8%
- Net enroute climb gradient = 8.5%

CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-6
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3. FINAL SEGMENT AND ENROUTE CLIMB GRADIENT (CONT'D)

B. Final Segment and Enroute Climb Gradient, Flaps 0° – Anti-Icing On

The final segment and enroute climb gradient with one engine inoperative and the other engine operating at maximum continuous thrust (MCT), is provided by Figure 06-05-3, with 10th stage bleed closed and anti-icing on.

NOTE

If ice accumulation is suspected on unprotected surfaces, subtract 0.9 percent from the gradient derived from Figure 06-05-3.

Example:

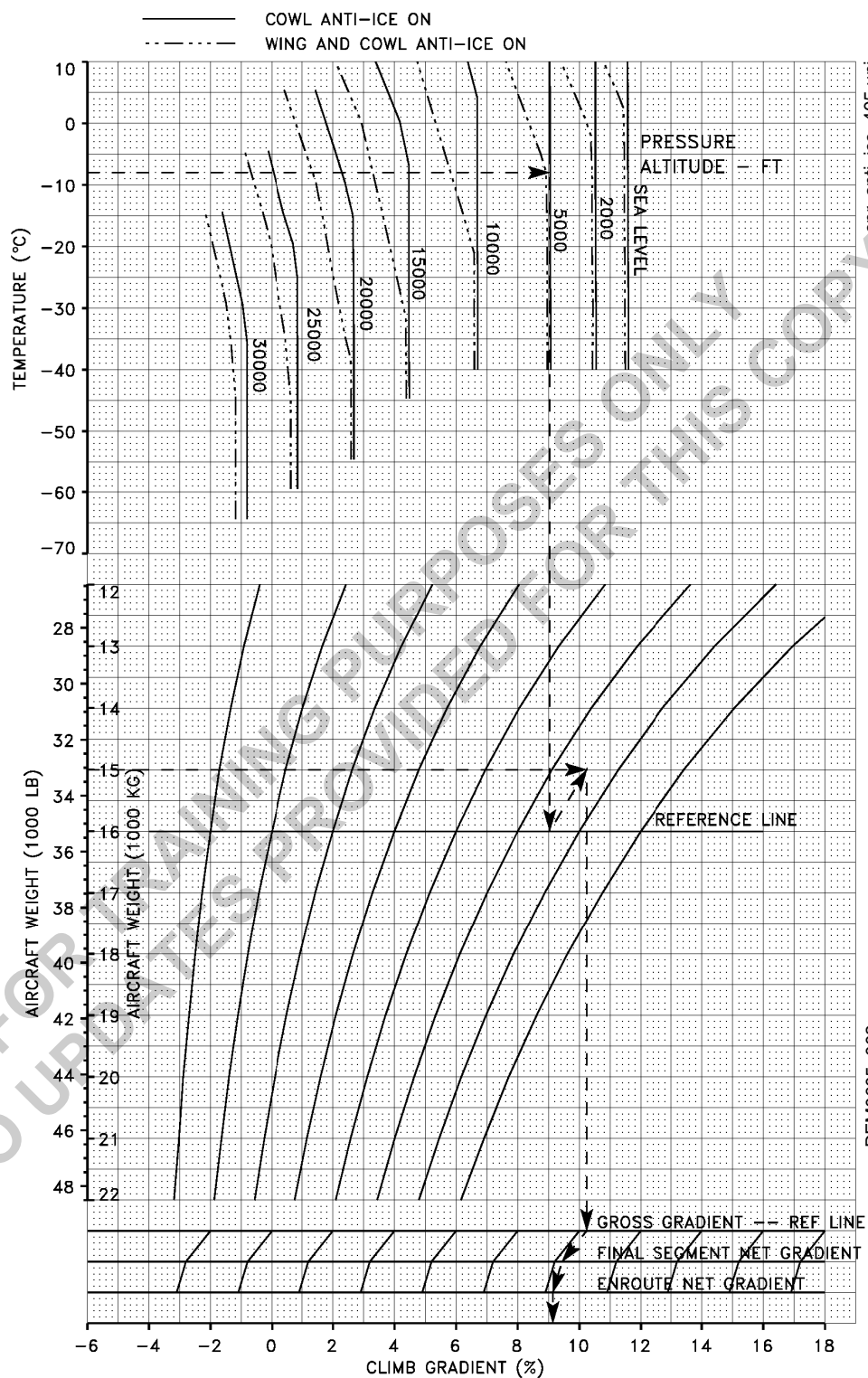
Associated Conditions:

Temperature	= -8°C
Cowl anti-icing	= On
Pressure altitude	= 5,000 ft
Gross weight	= 15,000 kg (33,069 lb)

The example on Figure 06-05-3 shows that for the above associated conditions, the following climb gradients are obtained:

- Gross final segment and gross enroute climb gradients = 10.2%
- Net final segment climb gradient = 9.4%
- Net enroute climb gradient = 9.1%

3. FINAL SEGMENT AND ENROUTE CLIMB GRADIENT (CONT'D)



Final Segment and Enroute Climb Gradient, Flaps 0° – Anti-Icing On
Figure 06-05-3

CHALLENGER 650	PERFORMANCE Enroute Performance	06-05-8
		Oct 27/15

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CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-1
		Oct 27/15

1. INTRODUCTION

The Approach and Landing Climb section contains the charts used to obtain the approach and landing climb gradients and speed schedules for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified approach and landing configurations.

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CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-2
		Oct 27/15

2. APPROACH CLIMB SPEED

A. Approach Climb Speed, Flaps 20°

The approach climb speed for varying conditions of airplane weight and airport pressure altitude, for an approach with flaps at 20°, is provided by Figure 06-06-1.

Example:

Associated Conditions:

Airplane weight = 18,000 kg (39,683 lb)

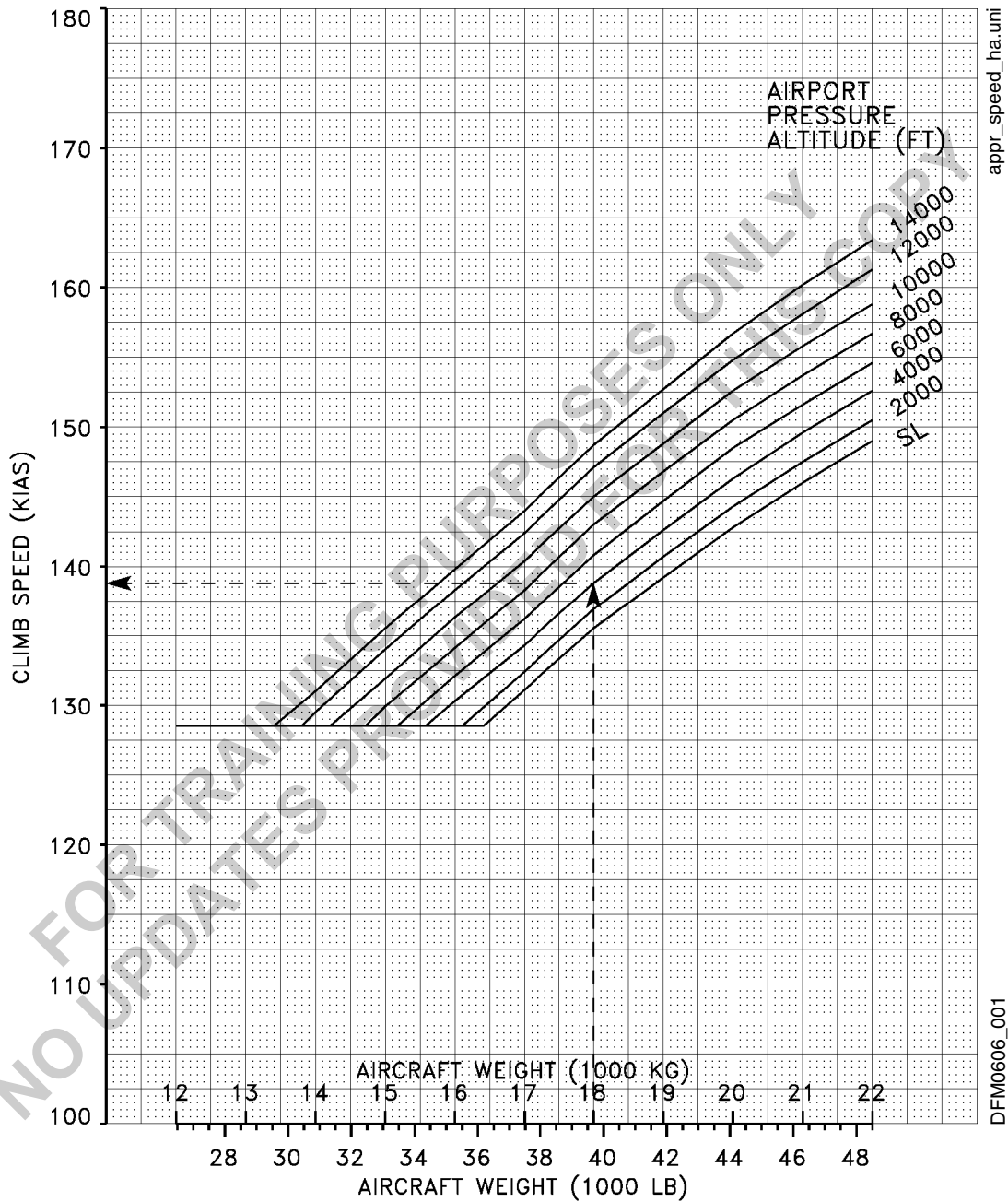
Airport pressure altitude = 4,000 feet

The example in Figure 06-06-1 shows that for the above associated conditions, the approach climb speed for an approach with flaps at 20° is 138.8 KIAS.

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2. APPROACH CLIMB SPEED (CONT'D)



Approach Climb Speed, Flaps 20°
Figure 06-06-1

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-4
		Oct 27/15

3. APPROACH CLIMB GRADIENT

A. Approach Climb Gradient, Flaps 20° – Anti-Icing Off

The approach climb gradient for varying conditions, for an approach with flaps at 20° with the anti-icing off, is provided by Figure 06-06-2.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the approach climb gradient values derived from Figure 06-06-2 by 0.8%.

Example:

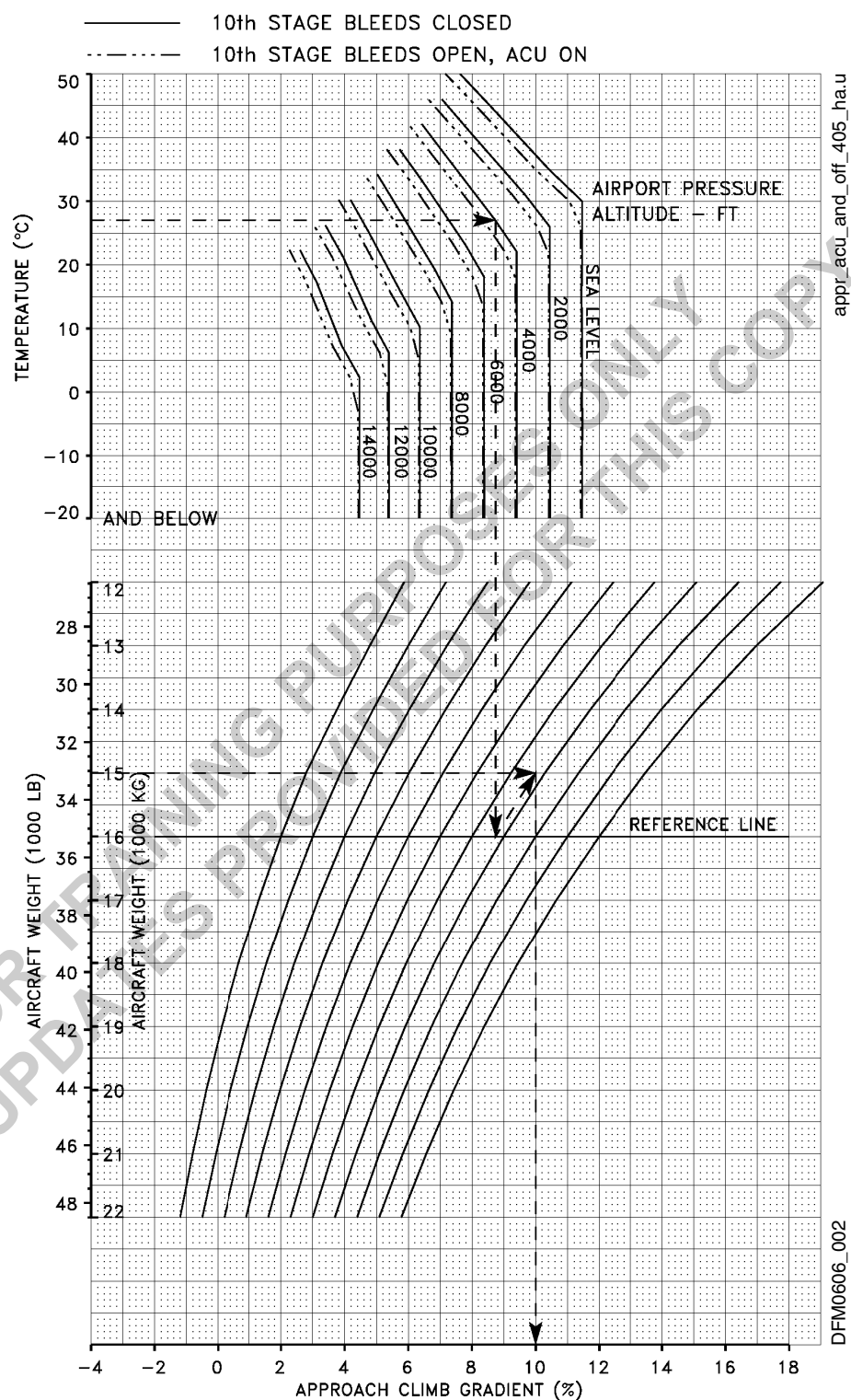
Associated Conditions:

Airport pressure altitude	= 4,000 feet
Temperature	= 27°C
Wing & Cowl Anti-Icing	= Off
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-2 shows that for the above associated conditions, the approach climb gradient for an approach with flaps at 20° is 10%.

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3. APPROACH CLIMB GRADIENT (CONT'D)


Approach Climb Gradient, Flaps 20° – Anti-Icing Off
 Figure 06-06-2

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-6
		Oct 27/15

3. APPROACH CLIMB GRADIENT (CONT'D)

FAA Certified Airplanes

The approach climb gradient for varying conditions, for an approach with flaps at 20° with the anti-icing off, is provided by Figure 06-06-2.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the approach climb gradient values derived from Figure 06-06-2 by 0.8%.

Example:

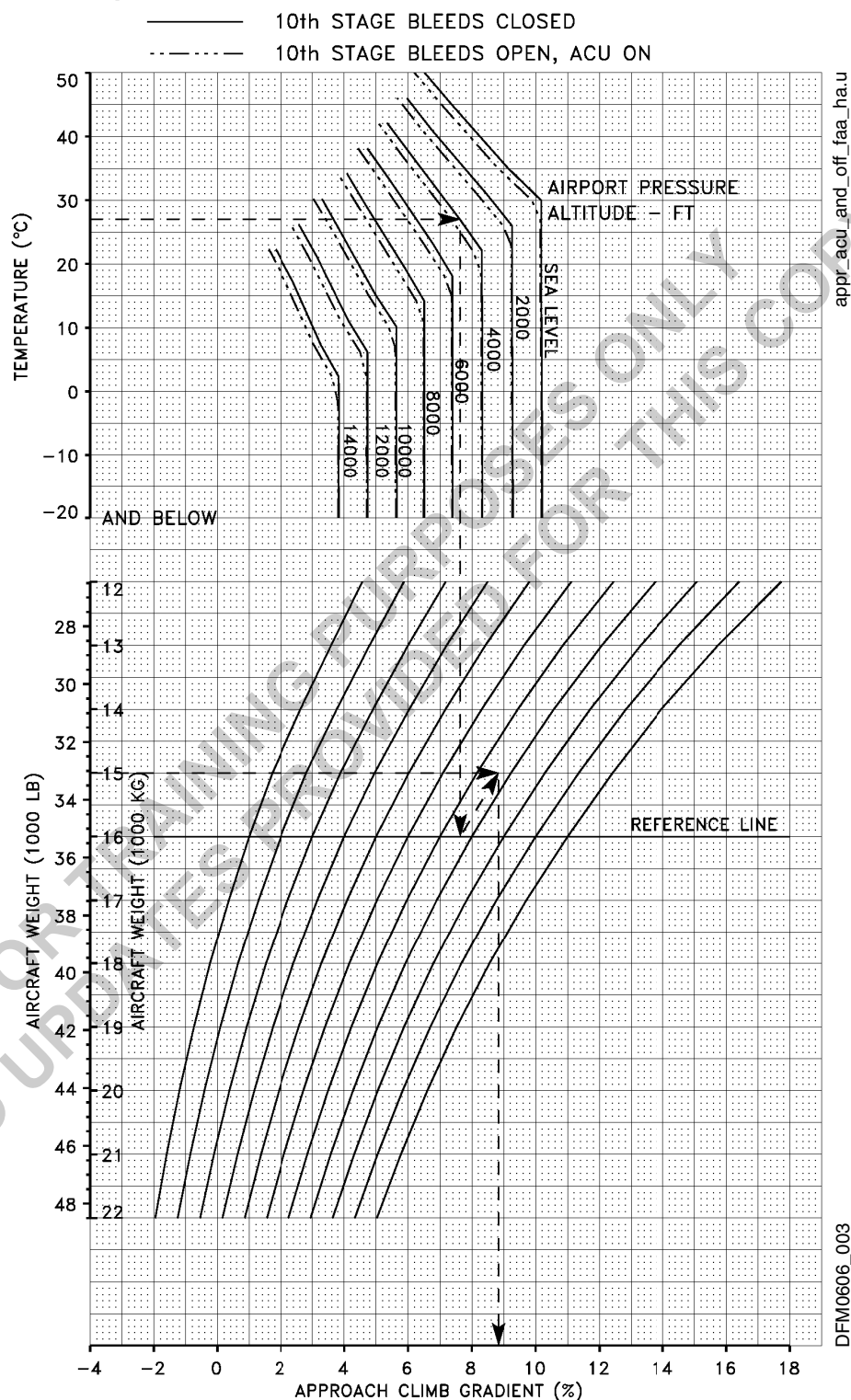
Associated Conditions:

Airport pressure altitude	= 4,000 feet
Temperature	= 27°C
Wing & Cowl Anti-Icing	= Off
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-2 shows that for the above associated conditions, the approach climb gradient for an approach with flaps at 20° is 8.8 %.

3. APPROACH CLIMB GRADIENT (CONT'D)

FAA Certified Airplanes

Approach Climb Gradient, Flaps 20° – Anti-Icing Off
Figure 06-06-2

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-8
		Oct 27/15

3. APPROACH CLIMB GRADIENT (CONT'D)

B. Approach Climb Gradient, Flaps 20° – Anti-Icing On

The approach climb gradient for varying conditions, for an approach with flaps at 20° with the wing and/or cowl anti-icing on, is provided by Figure 06-06-3.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the approach climb gradient values derived from Figure 06-06-3 by 0.8%.

Example:

Associated Conditions:

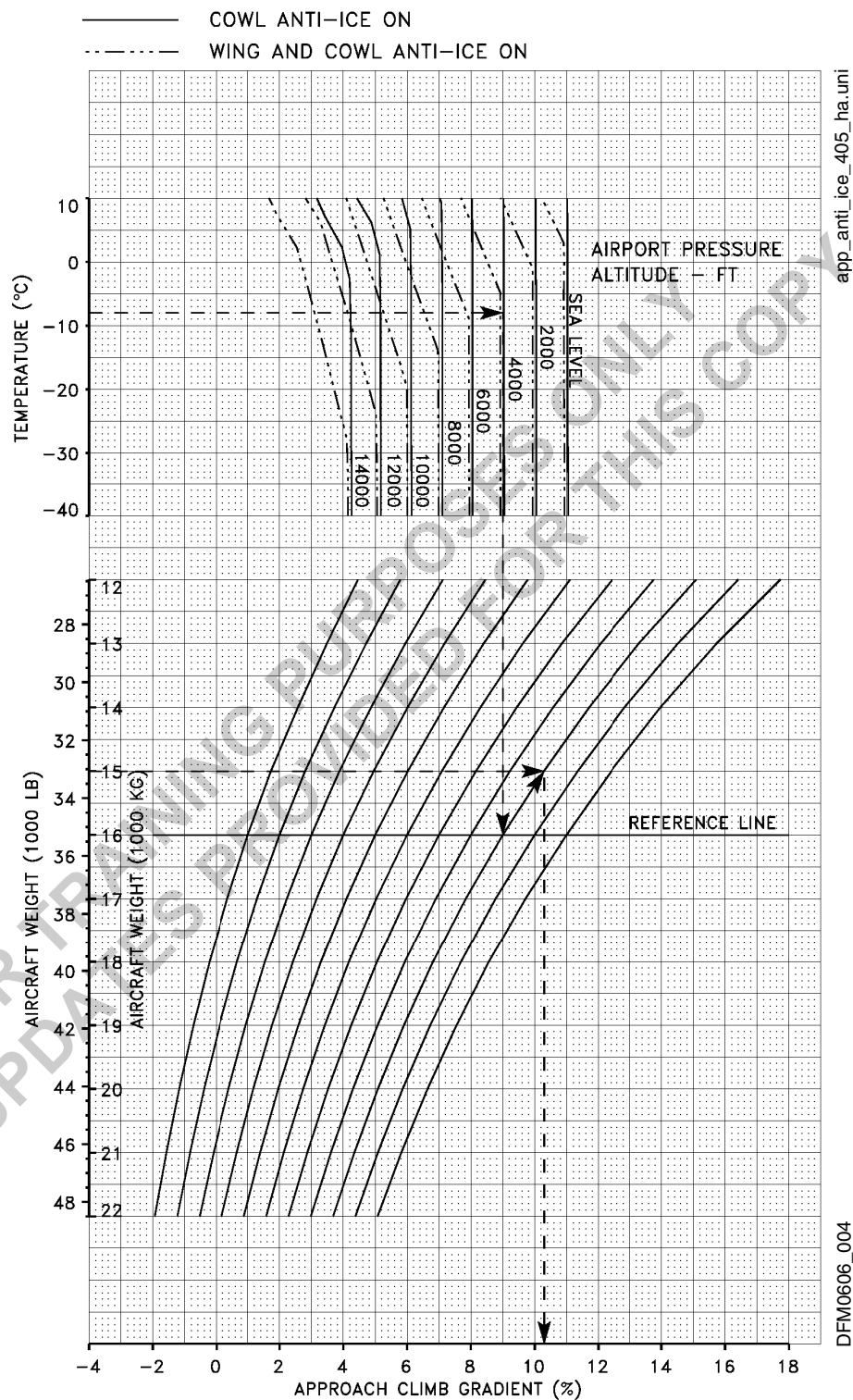
Airport pressure altitude	= 4,000 feet
Temperature	= -8°C
Cowl Anti-Icing	= On
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-3 shows that for the above associated conditions, the approach climb gradient for an approach with flaps at 20° is 10.3%.

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3. APPROACH CLIMB GRADIENT (CONT'D)



Approach Climb Gradient, Flaps 20° – Anti-Icing On
Figure 06-06-3

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-10
		Oct 27/15

3. APPROACH CLIMB GRADIENT (CONT'D)

FAA Certified Airplanes

The approach climb gradient for varying conditions, for an approach with flaps at 20° with the wing and/or cowl anti-icing on, is provided by Figure 06-06-3.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the approach climb gradient values derived from Figure 06-06-3 by 0.8%.

Example:

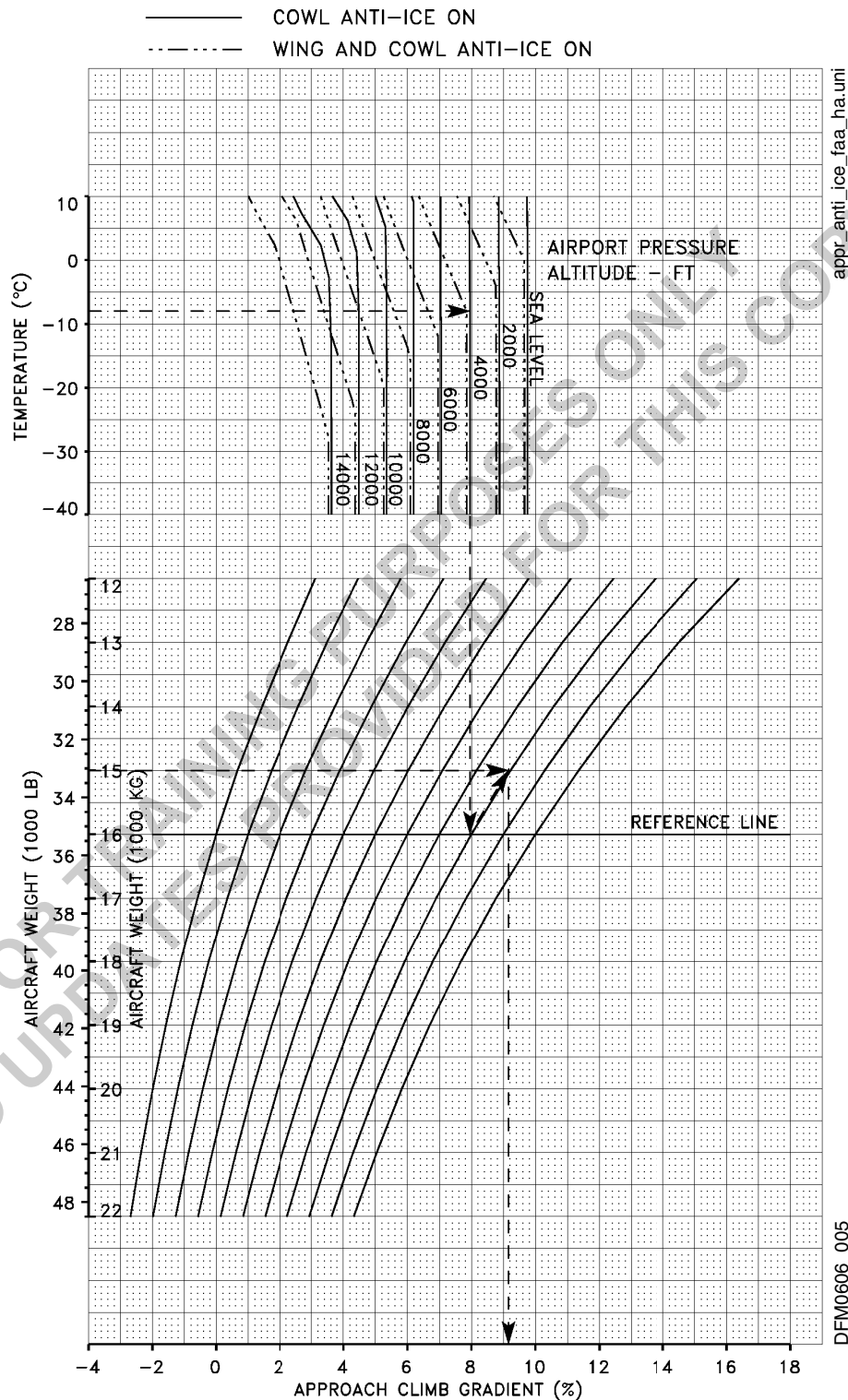
Associated Conditions:

Airport pressure altitude	= 4,000 feet
Temperature	= -8°C
Cowl Anti-Icing	= On
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-3 shows that for the above associated conditions, the approach climb gradient for an approach with flaps at 20° is 9.2 %.

3. APPROACH CLIMB GRADIENT (CONT'D)

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Approach Climb Gradient, Flaps 20° – Anti-Icing On
Figure 06-06-3

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-12
		Oct 27/15

4. LANDING CLIMB SPEED

A. Landing Climb Speed, Flaps 45°

The landing climb speed for varying conditions of airplane weight and airport pressure altitude, for a landing with flaps at 45°, is provided by Figure 06-06-4.

Example:

Associated Conditions:

Airplane weight = 18,000 kg (39,683 lb)

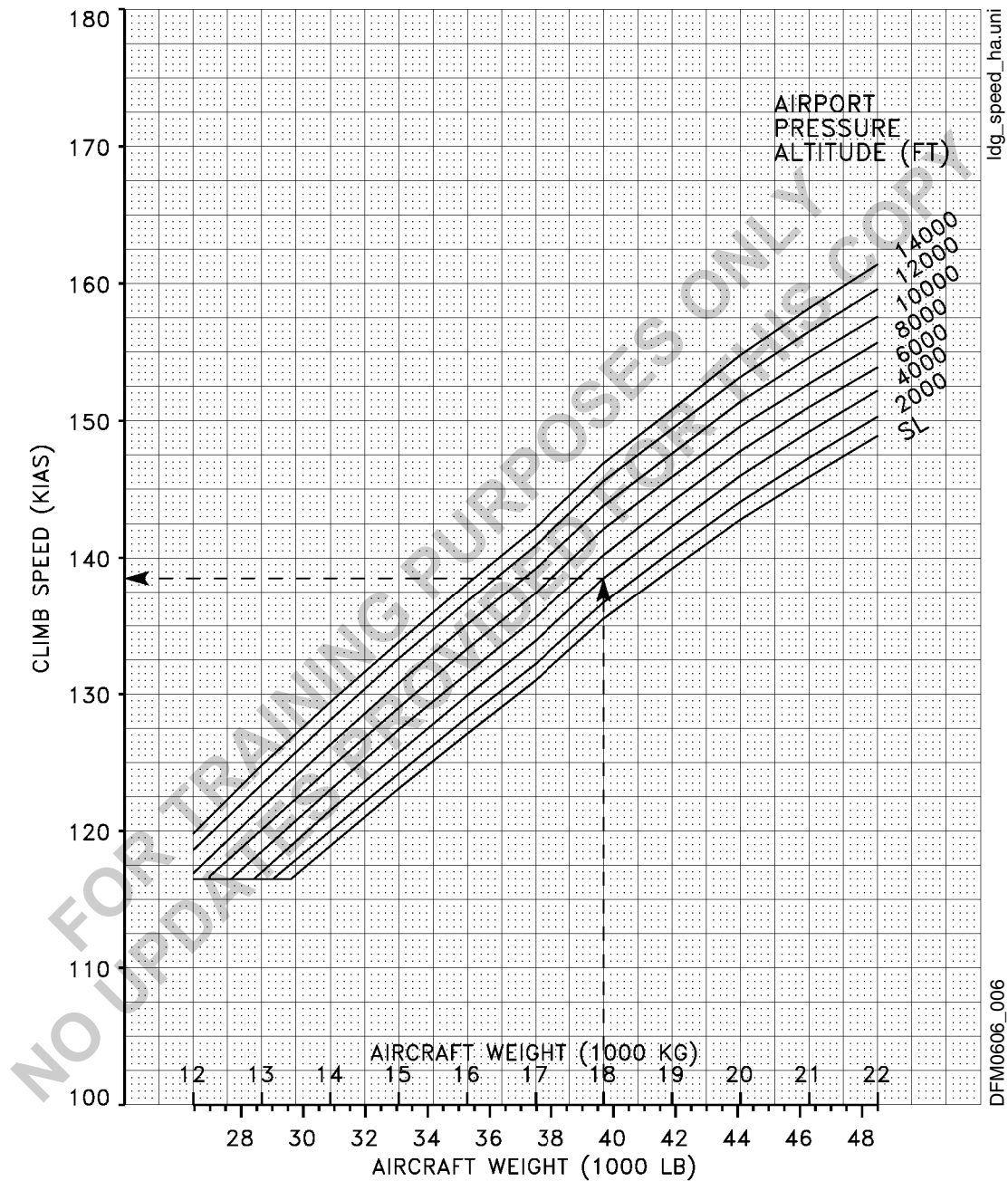
Airport pressure altitude = 4,000 feet

The example in Figure 06-06-4 shows that for the above associated conditions, the landing climb speed for a landing with flaps at 45° is 138.5 KIAS.

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4. LANDING CLIMB SPEED (CONT'D)



Landing Climb Speed, Flaps 45°
Figure 06-06-4

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-14
		Oct 27/15

5. LANDING CLIMB GRADIENT

A. Landing Climb Gradient, Flaps 45° – Anti-Icing Off

The landing climb gradient for varying conditions, for a landing with flaps at 45° with the anti-icing off, is provided by Figure 06-06-5.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the landing climb gradient values derived from Figure 06-06-5 by 3.1%.

Example:

Associated Conditions:

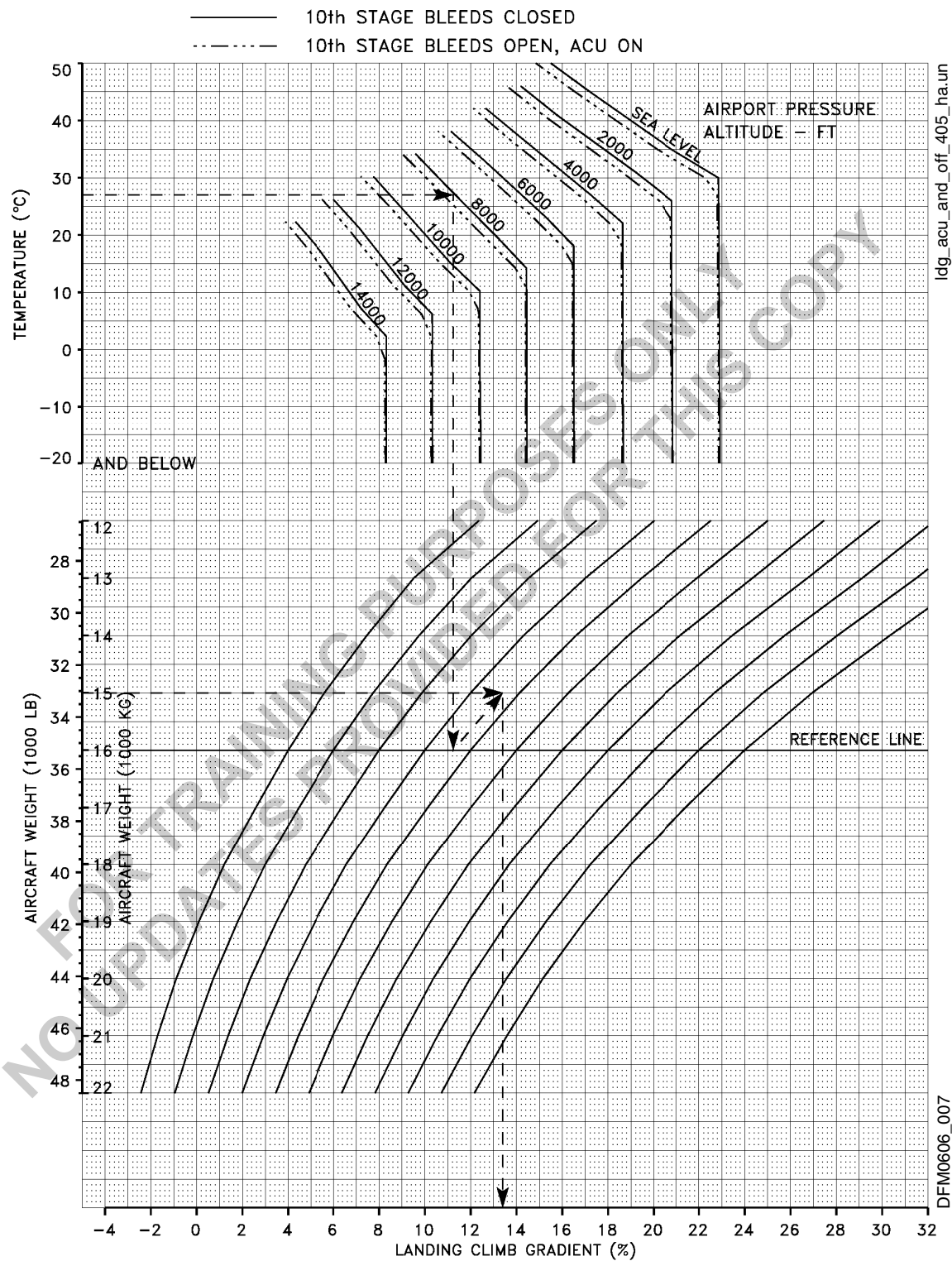
Airport pressure altitude	= 8,000 feet
Temperature	= 27°C
Wing & Cowl Anti-Icing	= Off
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-5 shows that for the above associated conditions, the landing climb gradient for a landing with flaps 45° is 13.4%.

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5. LANDING CLIMB GRADIENT (CONT'D)



Landing Climb Gradient, Flaps 45° - Anti-Icing Off
 Figure 06-06-5

CHALLENGER 650	PERFORMANCE Approach and Landing Climb	06-06-16
		Oct 27/15

5. LANDING CLIMB GRADIENT (CONT'D)

B. Landing Climb Gradient, Flaps 45° – Anti-Icing On

The landing climb gradient for varying conditions, for a landing with flaps 45° with the anti-icing on, is provided by Figure 06-06-6.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the landing climb gradient values derived from Figure 06-06-6 by 3.1%.

Example:

Associated Conditions:

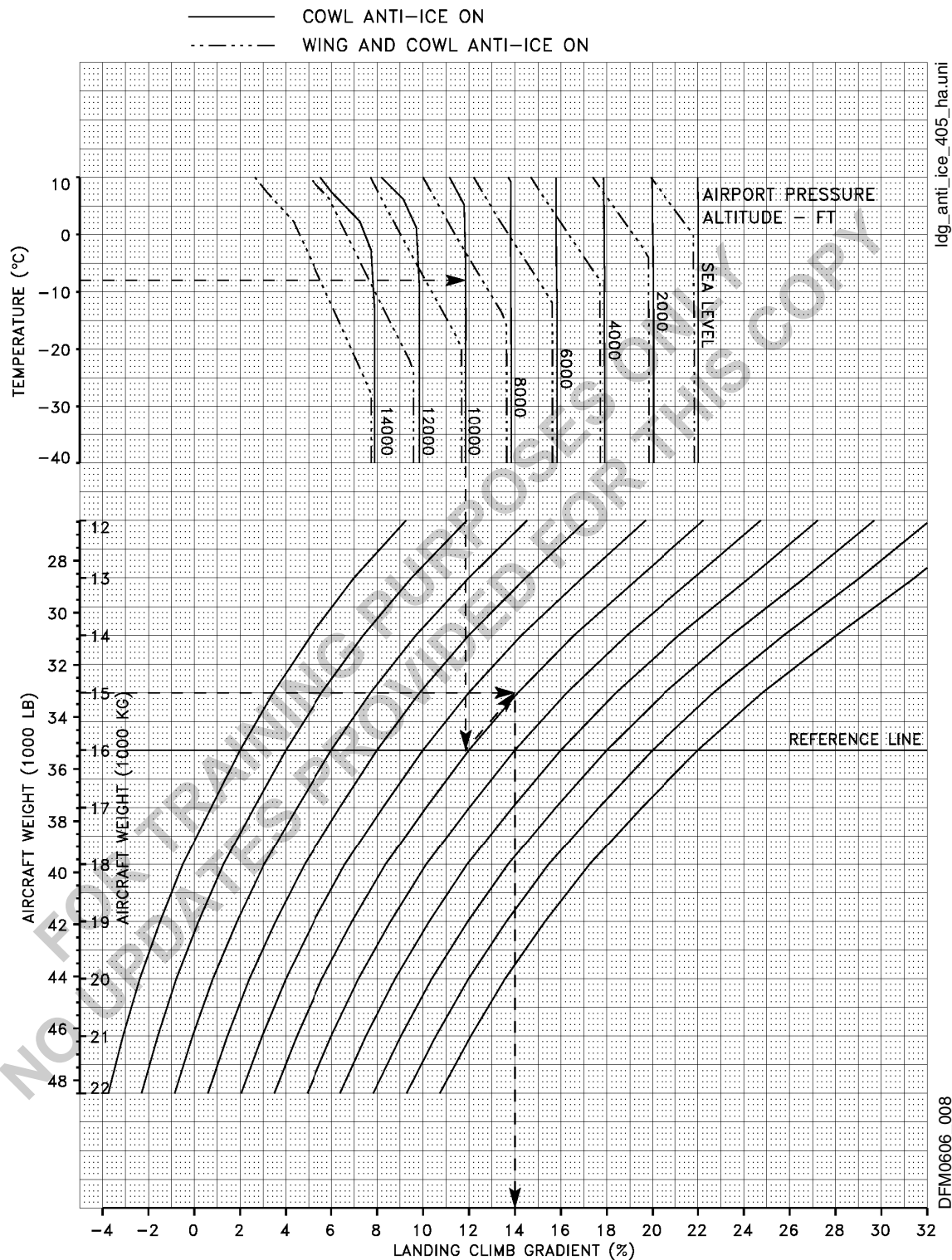
Airport pressure altitude	= 10,000 feet
Temperature	= -8°C
Cowl Anti-Icing	= On
10th Stage Bleeds	= Closed
Airplane weight	= 15,000 kg (33,069 lb)

The example in Figure 06-06-6 shows that for the above associated conditions, the landing climb gradient for a landing with flaps 45° is 14.0%.

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5. LANDING CLIMB GRADIENT (CONT'D)



Landing Climb Gradient, Flaps 45° - Anti-Icing On
Figure 06-06-6

CHALLENGER ⁶⁵⁰	PERFORMANCE Approach and Landing Climb	06-06-18
		Oct 27/15

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CHALLENGER 650	PERFORMANCE Landing Performance	06-07-1
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1. INTRODUCTION

The Landing Performance section contains the charts used to obtain the landing performance data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified approach and landing configurations.

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CHALLENGER 650	PERFORMANCE Landing Performance	06-07-2
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2. MAXIMUM ALLOWABLE LANDING WEIGHT

The maximum allowable landing weight is limited by the most restrictive of the following:

- Maximum approved landing weight
(Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT)
- Runway length available
- Climb requirements

A. Maximum Landing Weight Limited By Climb Requirements – Approach Flaps 20°/Landing Flaps 45°

The maximum landing weight limited by climb requirements for varying conditions of airport pressure altitude and temperature for the flaps 20° approach climb configuration and a subsequent landing flaps 45° configuration, are given in Figure 06-07-1. The chart incorporates all applicable anti-icing and 10th stage bleeds combinations.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the maximum landing weight values derived from Figure 06-07-1 by 800 kg (1,764 lb).

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 4,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed

As shown in the example (Figure 06-07-1, Anti-ice On), the maximum landing weight limited by climb requirements is 24,900 kg (54,895 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

Example: ANTI-ICE OFF

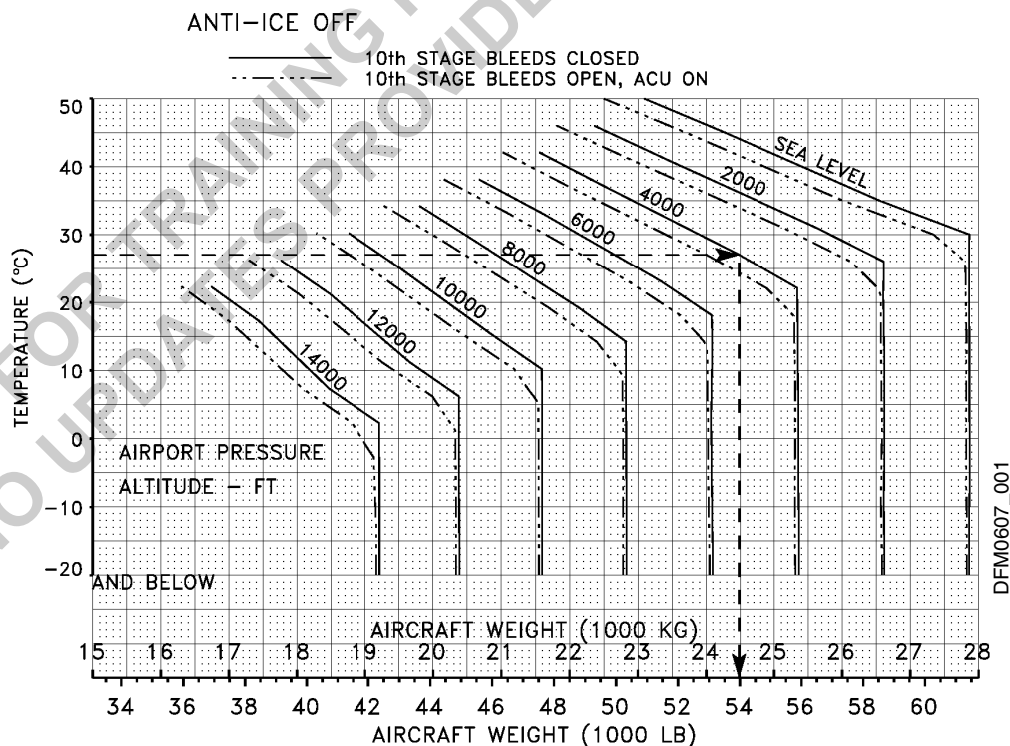
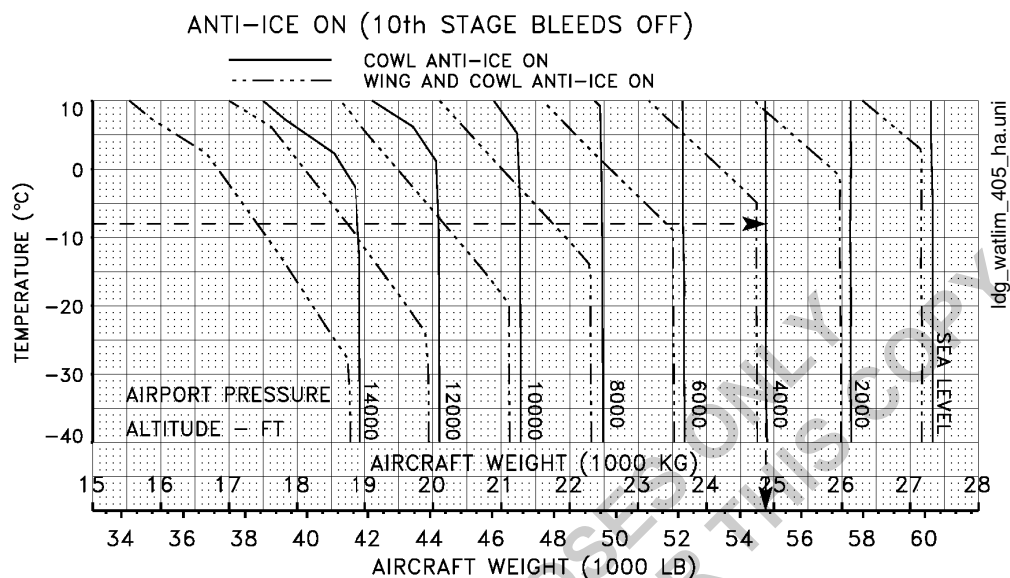
Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 4,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

As shown in the example (Figure 06-07-1, Anti-ice Off), the maximum landing weight limited by climb requirements is 24,500 kg (54,013 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

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2. MAXIMUM ALLOWABLE LANDING WEIGHT (CONT'D)



Landing Weight Limited by Climb Requirements – Approach Flaps 20°/Landing Flaps 45°
Figure 06-07-1

CHALLENGER 650	PERFORMANCE Landing Performance	06-07-4
		Oct 27/15

2. MAXIMUM ALLOWABLE LANDING WEIGHT (CONT'D)

FAA Certified Airplanes

The maximum landing weight limited by climb requirements for varying conditions of airport pressure altitude and temperature for the flaps 20° approach climb configuration and a subsequent landing flaps 45° configuration, are given in Figure 06-07-1. The chart incorporates all applicable anti-icing and 10th stage bleeds combinations.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the maximum landing weight values derived from Figure 06-07-1 by 800 kg (1,764 lb).

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 4,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed

As shown in the example (Figure 06-07-1, Anti-ice On), the maximum landing weight limited by climb requirements is 23,400 kg (51,588 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

Example: ANTI-ICE OFF

Associated Conditions:

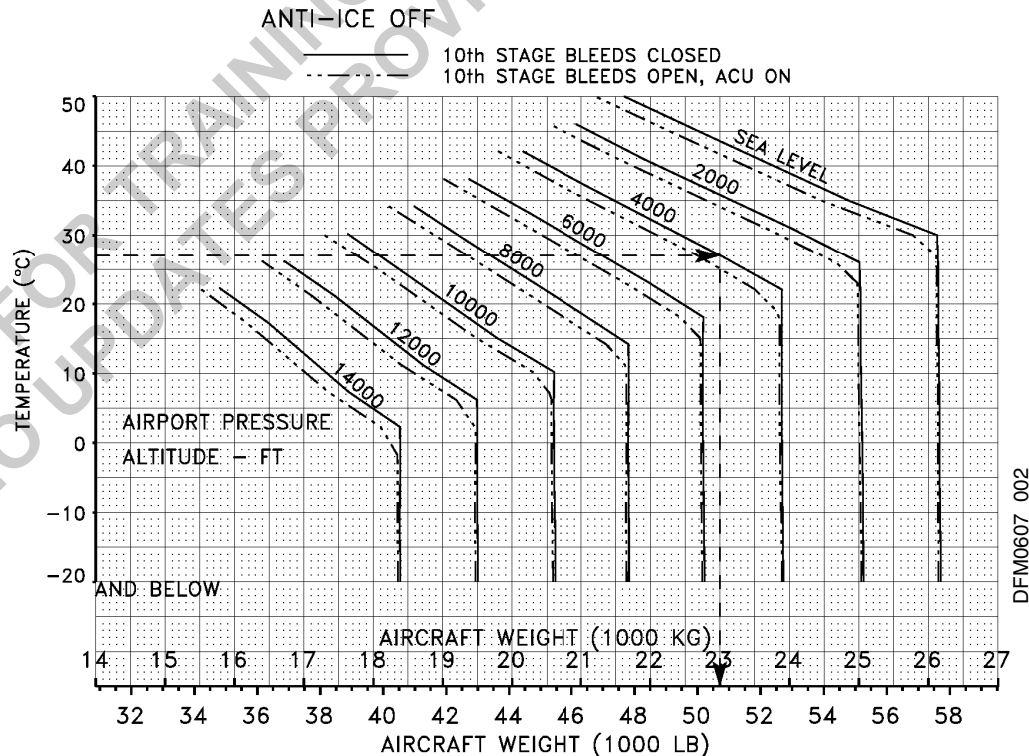
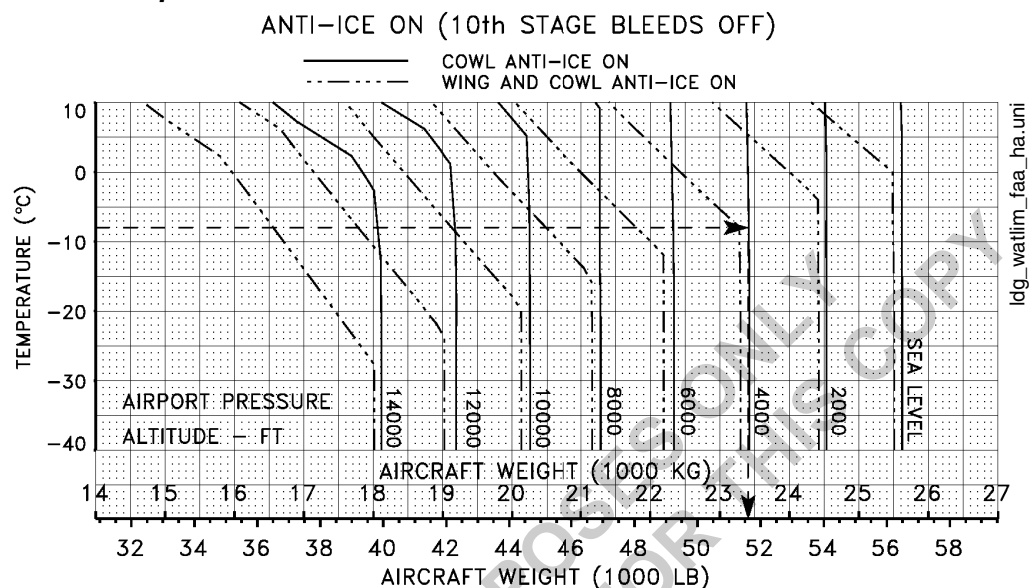
Temperature	= 27°C
Airport pressure altitude	= 4,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

As shown in the example (Figure 06-07-1, Anti-ice Off), the maximum landing weight limited by climb requirements is 23,000 kg (50,706 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

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2. MAXIMUM ALLOWABLE LANDING WEIGHT (CONT'D)

FAA Certified Airplanes



Landing Weight Limited by Climb Requirements – Approach Flaps 20°/Landing Flaps 45°
Figure 06-07-1

CHALLENGER 650	PERFORMANCE Landing Performance	06-07-6
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3. LANDING FIELD LENGTH AND SPEED

A. Landing Field Length and Landing Speed, Flaps 45°

The landing field length required for varying conditions of airplane landing weight, airport pressure altitude and wind is provided by Figure 06-07-2. The actual landing distance required is also given within this chart.

The landing approach speed (V_{REF}) for varying airplane landing weights and airport pressure altitude is also given by Figure 06-07-2.

The actual landing distance given on Figure 06-07-2 is the distance from a 50 foot height at V_{REF} with flaps 45°, to a full stop on a smooth, dry, level, hard-surfaced runway and ISA temperature.

NOTE

1. The landing field length derived from Figure 06-07-2 includes factors for operational rules which require use of 60% of the available runway (landing field length required).
2. Operational rules may require that for wet runway surfaces, the runway length available be at least 115% of the landing field length derived from Figure 06-07-2.

Example:

Associated Conditions:

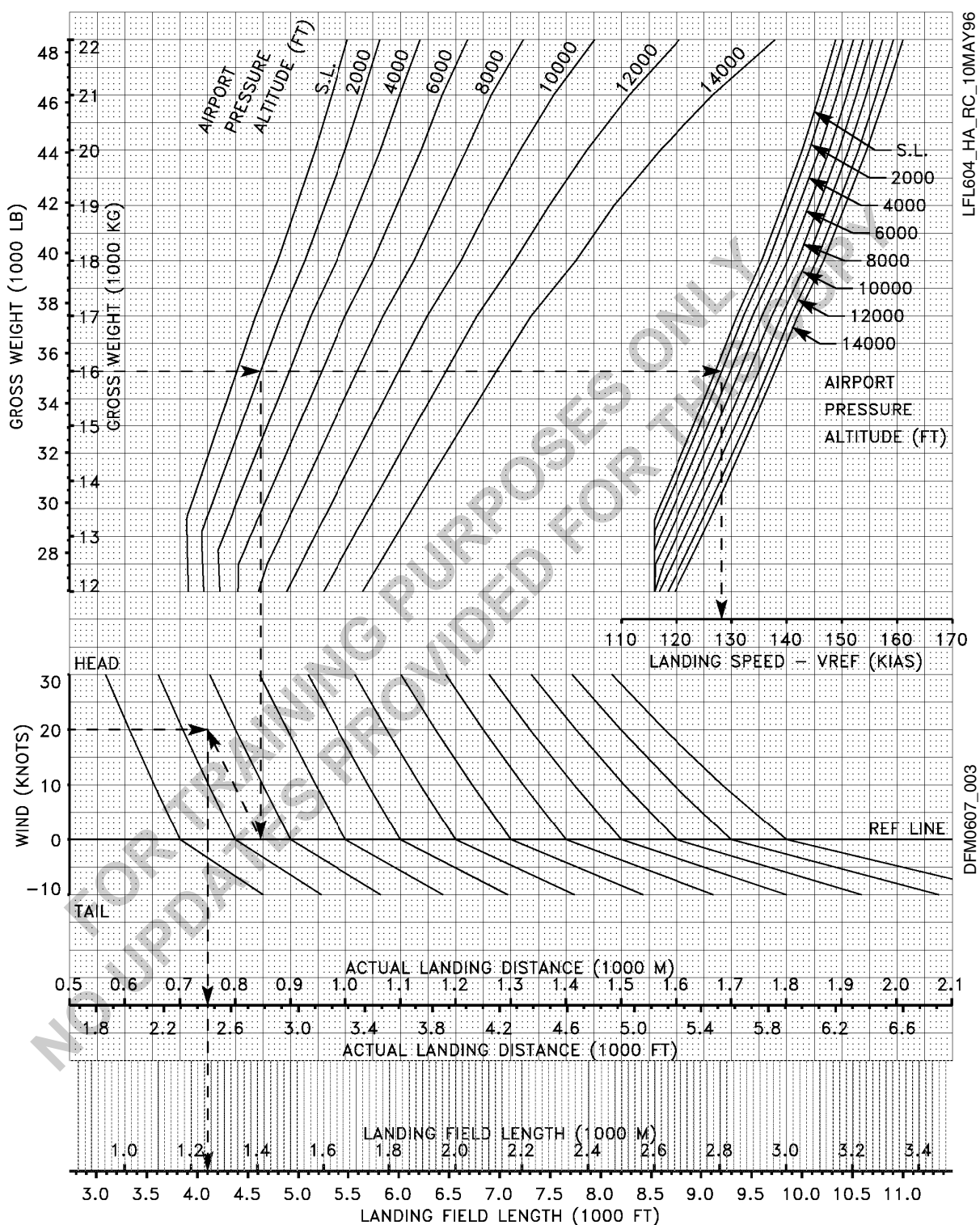
Landing gross weight	= 16,000 kg (35,274 lb)
Airport pressure altitude	= 2,000 feet
Wind	= 20 knots, headwind

The example in Figure 06-07-2, shows that for the above conditions, the actual landing distance is 750 meters (2,461 feet) and the landing field length is 1,250 meters (4,100 feet).

V_{REF} for the same conditions as above is 128 KIAS.

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3. LANDING FIELD LENGTH AND SPEED (CONT'D)



Landing Field Length and Landing Speed, Flaps 45°
Figure 06-07-2

CHALLENGER 650	PERFORMANCE Landing Performance	06-07-8
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4. MAXIMUM PERMISSIBLE QUICK TURN-AROUND LANDING WEIGHT

Figure 06-07-3 provides the maximum permissible quick turn-around landing weights for varying conditions of airport pressure altitude, ambient temperature, wind and runway slope.

After landing at a weight exceeding the quick turn-around landing weight, a minimum waiting time of 50 minutes must be observed, followed by an inspection of the wheels, brakes, tires, and landing gear in accordance with the Airplane Maintenance Manual.

After landing at a weight equal to or less than the quick turn-around landing weight, a brake cooling time, determined in accordance with the method described in TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires, in this chapter, must be observed before take-off.

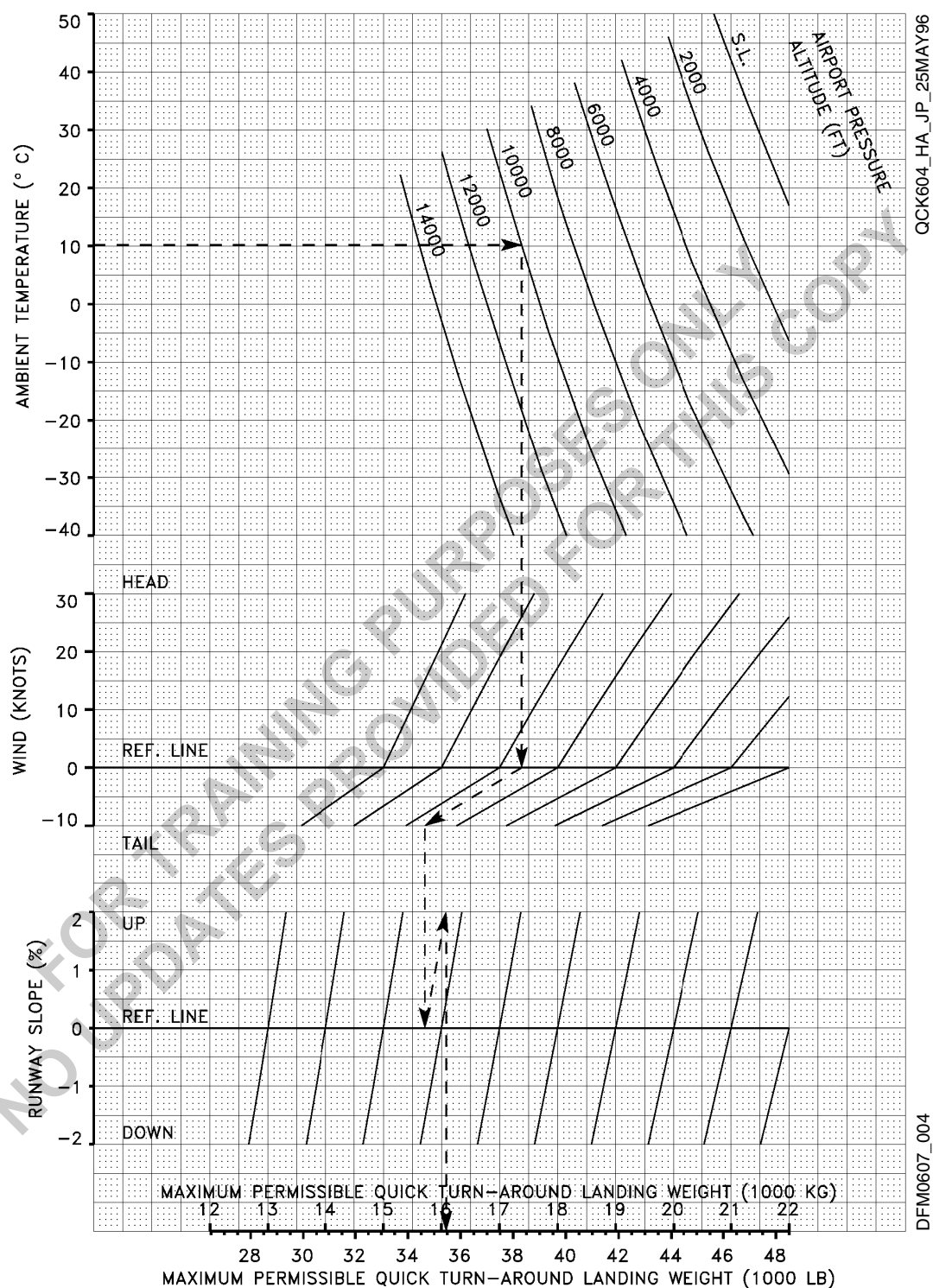
Associated Conditions:

Ambient temperature	= 10°C
Airport pressure altitude	= 10,000 feet
Wind	= 10 knots, tailwind
Runway slope	= +2%

The example in Figure 06-07-3 shows that for the above associated conditions, the maximum quick turn-around landing weight is 16,100 kg (35,494 lb).

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4. MAXIMUM PERMISSIBLE QUICK TURN-AROUND LANDING WEIGHT (CONT'D)



Maximum Permissible Quick Turn-Around Landing Weight
Figure 06-07-3

CHALLENGER ⁶⁵⁰	PERFORMANCE Landing Performance	06-07-10
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CHALLENGER 650	PERFORMANCE Turn-Around Time	06-08-1
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1. COOLING OF WHEELS, BRAKES AND TIRES

Brake cooling times derived from Figure 06-08-3, must be observed between a landing or a low energy rejected take-off (RTO) and a subsequent take-off, to ensure that brake energy or brake temperature limits are not exceeded during the next take-off or in the event that the next take-off is rejected. The cooling times provided in Figure 06-08-3 apply only to single landing/take-off turn-around.

It is assumed that the airplane is operated in the approved take-off or landing configuration.

A. Use of Brake Cooling Charts

Brake cooling times are obtained from Figure 06-08-3, after the landing (or previous RTO) brake energy and the required RTO brake energy for the subsequent take-off have been determined.

Landing brake energy is determined from Figure 06-08-1. When establishing the landing brake energy, a braking speed equal to the indicated airspeed (KIAS) at the time the wheel brakes were first applied, must be used.

The required RTO brake energy for the subsequent take-off is determined from Figure 06-08-2, using the V_1 speed appropriate for the take-off condition.

Figure 06-08-2 can also be used to determine the brake energy absorbed during a previous RTO. In this case, the indicated airspeed (KIAS) at the time the wheel brakes were first applied must be used instead of the V_1 speed.

B. Repeated Turn-Around Operations

For repeated turn-around operations, such as multi-segment flights or crew training, additional in-flight cooling may be required to avoid progressive temperature build-up of heat in the wheels, brakes and tires.

A ten minute in-flight cooling period with the landing gear extended prior to landing on the second and subsequent turn-around, will allow the turn-around times to be used on a repeated basis.

NOTE

In-flight cooling does not replace the ground cooling requirements specified in Figure 06-08-3, but is additional cooling required for multiple turn-around operations.

C. Systems Malfunction

If a brake or anti-skid system malfunction occurs during landing or RTO, or a landing is conducted at reduced flap angles, the brake energy expended derived from Figure 06-08-1 or Figure 06-08-2 (as appropriate) may be increased by a factor of up to two.

A minimum turn-around time of 50 minutes must be observed to ensure that the fusible plug limit has not been exceeded. The system's defect must be rectified prior to the next flight, and the gear assembly, wheels, brakes and tires visually inspected for damage.

D. Fusible Plug Release

If a fusible plug releases, the affected wheel, brakes and anti-skid wheel speed sensors must be inspected in accordance with the procedures outlined in the Airplane Maintenance Manual, PSP 605-2, and any damage rectified before the next flight.

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CHALLENGER 650	PERFORMANCE Turn-Around Time	06-08-2
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1. COOLING OF WHEELS, BRAKES AND TIRES (CONT'D)

E. Brake Cooling Time

(1) CALCULATION OF LANDING BRAKE ENERGY

Associated Conditions:

Airplane gross weight	= 16,500 kg (36,376 lb)
Braking speed	= 130 KIAS
Airport pressure altitude	= 5,000 feet
Ambient temperature	= 40°C
Runway slope	= +1%
Wind	= 20 knots, headwind
Flaps setting	= 45°

The example in Figure 06-08-1 shows that for the above associated conditions, the landing brake energy (A) is 27.5×10^6 ft. lb.

(2) CALCULATION OF REQUIRED RTO BRAKE ENERGY

Associated Conditions:

Airplane gross weight	= 14,500 kg (31,967 lb)
V ₁	= 130 KIAS
Airport pressure altitude	= 5,000 feet
Ambient temperature	= 40°C
Runway slope	= +1%
Wind	= 20 knots, headwind
Flaps setting	= 20°

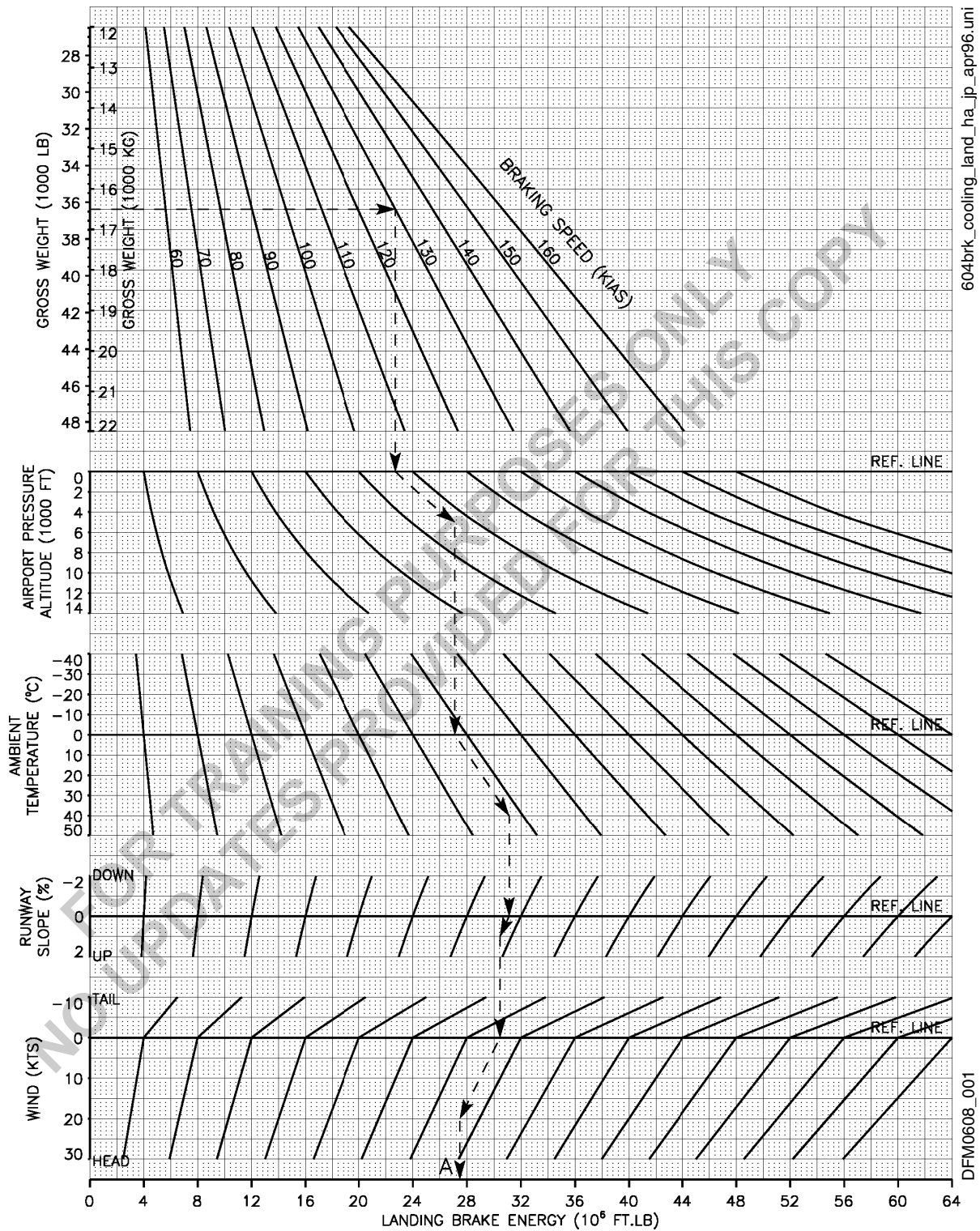
The example in Figure 06-08-2 shows that for the above associated conditions, the required RTO brake energy (B) is 31.2×10^6 ft. lb.

(3) CALCULATION OF BRAKE COOLING TIME

Using the values derived from Figure 06-08-1 (A) and Figure 06-08-2 (B), Figure 06-08-3 shows that the minimum brake cooling time required for a turn-around under these conditions is 9.0 minutes (C).

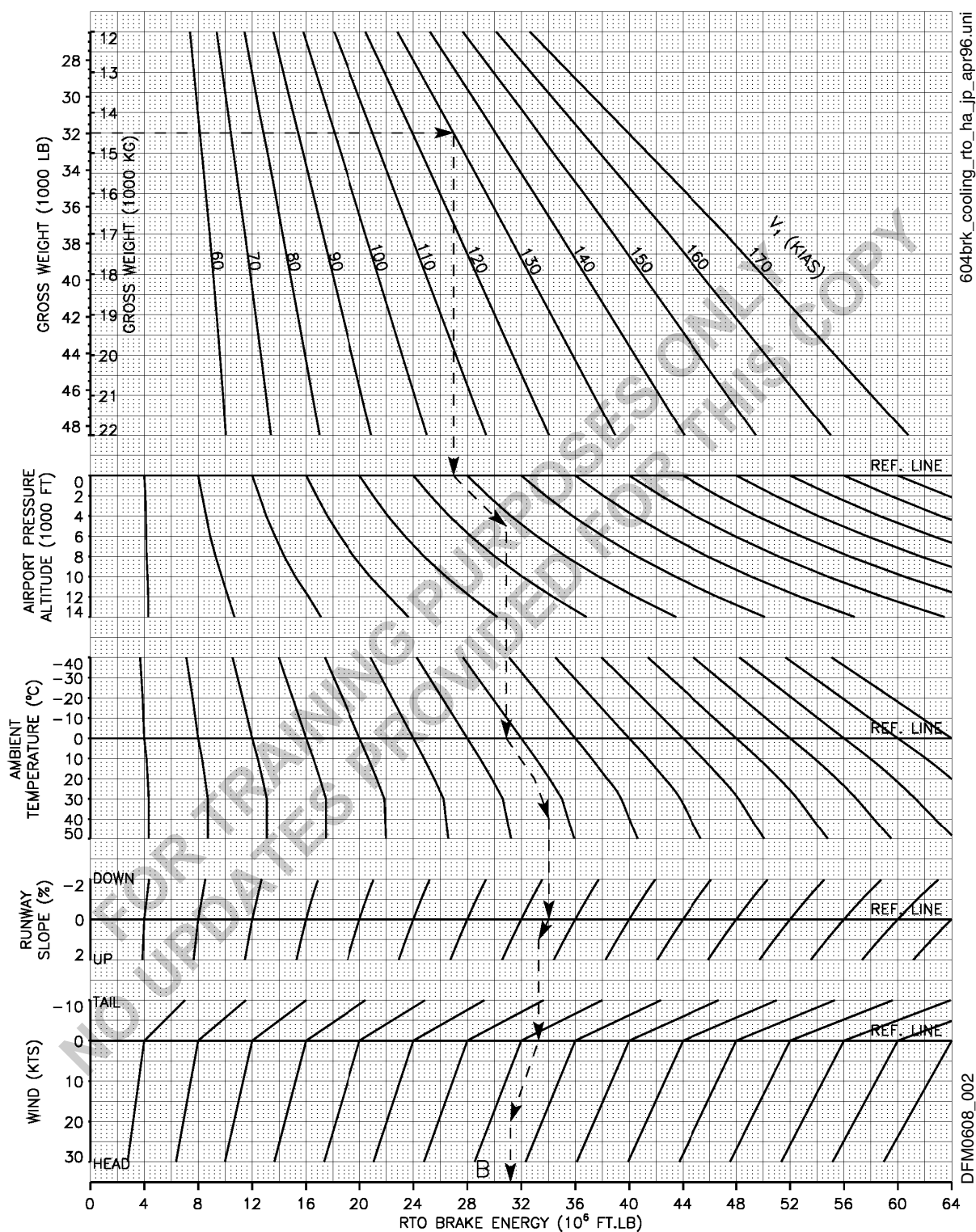
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1. COOLING OF WHEELS, BRAKES AND TIRES (CONT'D)



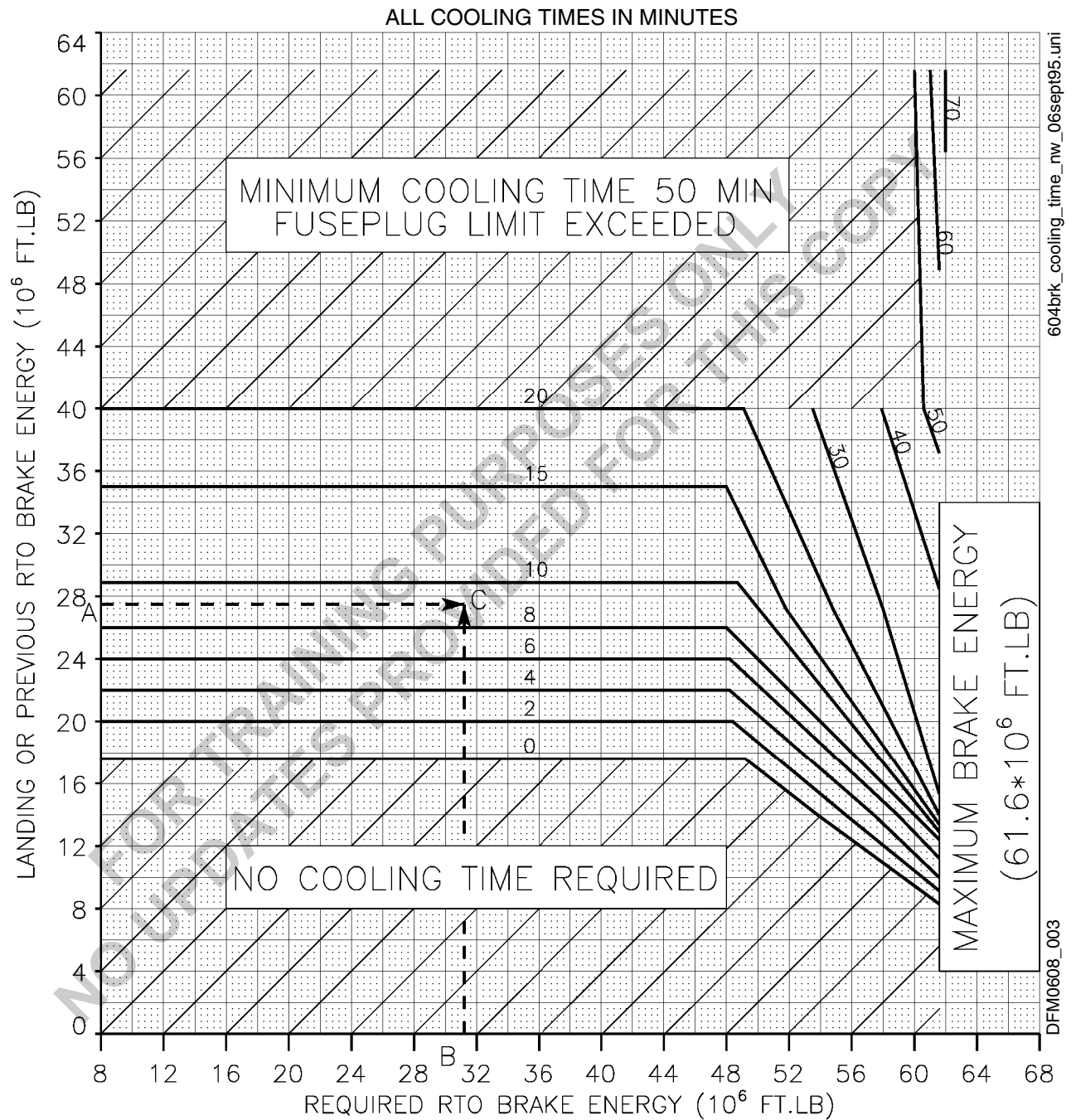
Landing Brake Energy, Flaps 45°
Figure 06-08-1

1. COOLING OF WHEELS, BRAKES AND TIRES (CONT'D)



Rejected Take-Off Brake Energy, Flaps 20°
Figure 06-08-2

1. COOLING OF WHEELS, BRAKES AND TIRES (CONT'D)



Mainwheel Brake Cooling Time
Figure 06-08-3

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CHALLENGER 650	SUPPLEMENT 1 Noise Characteristics	07-01-01-1
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1. INTRODUCTION

This supplement contains the noise characteristics data for the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent), which demonstrate compliance with the following:

- AWM Chapter 516, Change 516-10,
- FAR 36 Amendment 36-28, and
- ICAO Annex 16, Volume 1, Amendment 10.

The effect of this Supplement on the basic Airplane Flight Manual is given below:

A. Introduction

The general information in Chapter 1 is applicable.

B. Limitations

The limitations in Chapter 2 are applicable.

C. Emergency Procedures

The emergency procedures in Chapter 3 are applicable.

D. Normal Procedures

The normal procedures in Chapter 4 are applicable.

E. Abnormal Procedures

The abnormal procedures in Chapter 5 are applicable.

F. Performance

The performance data in Chapter 6 are applicable.

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2. CERTIFICATE AIRPLANE CONFIGURATION

Compliance has been demonstrated in the following configuration:

A. Take-Off and Sideline Noise Levels

Take-off and sideline noise levels were obtained in the following configuration:

- Maximum take-off weight 21,863 kg/48,200 lb.
- Climb speed $V_2 + 10$ KIAS
- Flaps setting 20 degrees
- APU Off
- Air-Conditioning packs Off
- Wing and cowl anti-ice Off
- Normal take-off thrust (both engines operating)

B. Approach Noise Levels

Landing approach noise levels were established as per the following configuration:

- Glideslope 3 degrees
- Landing gear Down
- Landing weight 17,237 kg/38,000 lb
- Approach speed $V_{REF} + 10$ KIAS
- Flaps setting 45 degrees
- APU On
- Air-Conditioning packs Off
- Wing and cowl anti-icing Off

3. CERTIFICATE NOISE LEVELS

The demonstrated effective perceived noise levels (EPNdB), noise limits and margins of compliance to Chapter 3 and Stage 3 are as listed below:

21,863 KG/48,200 LB MTOW			
Phase of Flight	Test Level EPNdB	Noise Limit (dB)	Margin (dB)
Take-Off Flyover	81.2	89.0	7.8
Sideline/Lateral	84.6	94.0	9.4
Approach	91.3	98.0	6.7

Compliance to ICAO Annex 16, Chapter 4 and FAR 36, Stage 4 has been demonstrated with a 13.8 dB margin. For FAA registered airplanes, refer to Supplement 1A.

No determination has been made such that the noise levels of this airplane are, or should be, acceptable or unacceptable for operation at, into, or out of, any airport.

These noise level values are stated for reference conditions of standard atmospheric pressure at sea level, 25°C (77°F) ambient temperature, 70% relative humidity, and zero wind.

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NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes on the FAA Registry.

1. INTRODUCTION

This supplement contains the noise characteristics data for the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent), when Stage 4 compliant.

The effect of this Supplement on the basic Airplane Flight Manual is given below:

A. Introduction

The general information in Chapter 1 is applicable.

B. Limitations

The limitations in Chapter 2 are applicable.

C. Emergency Procedures

The emergency procedures in Chapter 3 are applicable.

D. Normal Procedures

The normal procedures in Chapter 4 are applicable.

E. Abnormal Procedures

The abnormal procedures in Chapter 5 are applicable.

F. Performance

The performance data in Chapter 6 are applicable.

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2. CERTIFICATE AIRPLANE CONFIGURATION

Compliance has been demonstrated in the following configuration:

A. Take-Off and Sideline Noise Levels

Take-off and sideline noise levels were obtained in the following configuration:

- Maximum take-off weight 21,863 kg/48,200 lb.
- Climb speed $V_2 + 10$ KIAS
- Flaps setting 20 degrees
- APU Off
- Air-Conditioning packs Off
- Wing and cowl anti-ice Off
- Normal take-off thrust (both engines operating)

B. Approach Noise Levels

Landing approach noise levels were established as per the following configuration:

- Glideslope 3 degrees
- Landing gear Down
- Landing weight 17,237 kg/38,000 lb
- Approach speed $V_{REF} + 10$ KIAS
- Flaps setting 45 degrees
- APU On
- Air-Conditioning packs Off
- Wing and cowl anti-icing Off

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3. CERTIFICATE NOISE LEVELS

The following noise levels comply with part 36, Appendix B, Stage 4 maximum noise level requirements and were obtained by analysis of approved data from noise tests conducted under the provisions of part 36, Amendment 36-28. The noise measurement and evaluation procedures used to obtain these noise levels are considered by the FAA to be equivalent to the Chapter 4 noise level required by the International Civil Aviation Organization (ICAO) in Annex 16, Volume 1, Appendix 2, Amendment 7, effective March 21, 2002.

The demonstrated effective perceived noise levels (EPNdB) are as listed below:

21,863 KG/48,200 LB MTOW	
Phase of Flight	Level (EPNdB)
Take-Off Flyover	81.2
Sideline/Lateral	84.6
Approach	91.3

No determination has been made by the Federal Aviation Administration that the noise levels of this airplane are, or should be, acceptable or unacceptable for operation at, into, or out of, any airport.

These noise level values are stated for reference conditions of standard atmospheric pressure at sea level, 25°C (77°F) ambient temperature, 70% relative humidity, and zero wind.

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CHALLENGER 650	SUPPLEMENT 2 Operation on Wet and Contaminated Runways	07-02-01-1
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1. INTRODUCTION

A. General

This Supplement contains information and procedures for operation of the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) on wet runways and on runways contaminated by standing water, slush, loose snow, compacted snow or ice.

This information has been prepared by the manufacturer and approved as guidance material, to assist operators in developing suitable guidance, recommendations or instructions for use by their flight crews, when operating on wet and contaminated runway surface conditions.

The data have been prepared using reasonable estimates of the effects of wet and contaminated runway surface conditions on the accelerating ground roll and the braking ground roll. The effects of actual conditions may differ from those used to establish the data.

The level of safety is decreased when operating on contaminated runways, therefore, every effort should be made to ensure that the runway surface is cleared adequately of any significant precipitation.

B. Runway Conditions

(1) DRY RUNWAY

A runway is considered to be dry when the runway surface is not “wet” or “contaminated”.

(2) WET RUNWAY

A runway is considered to be wet when there is sufficient moisture on the runway surface to cause it to appear reflective, but without significant areas of standing water.

NOTE

A runway with standing water would be a contaminated runway.

(3) RUNWAY CONTAMINATED BY STANDING WATER, SLUSH OR LOOSE SNOW

A runway is considered to be contaminated when more than 25% of the runway surface area (whether in isolated areas or not), within the required length and width being used, is covered by more than 3 millimeters (1/8 inch) of standing water or slush, or by loose snow, equivalent to more than 3 millimeters (1/8 inch) of water.

(4) RUNWAY CONTAMINATED BY COMPACTED SNOW

A runway is considered to be contaminated by compacted snow, when covered by snow which has been compacted into a solid mass which resists further compression, and will hold together or break into lumps if picked up.

(5) RUNWAY CONTAMINATED BY WET ICE

A runway surface condition where braking action is expected to be very low, due to the presence of wet ice.

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1. INTRODUCTION (CONT'D)

C. Limitations

The limitations established in Chapter 2 of the basic Airplane Flight Manual are applicable, with the addition of the following:

- (1) The maximum depths of runway contaminants covering an appreciable part of the runway are:

CONTAMINANT	TAKE-OFF	LANDING
SLUSH/STANDING WATER	6.4 mm (0.25 in)	12.7 mm (0.5 in)
WET SNOW	12.7 mm (0.5 in)	25.4 mm (1.0 in)
DRY SNOW	30.5 mm (1.2 in)	61.0 mm (2.4 in)

- (2) The use of a take-off distance on a wet or contaminated runway that is shorter than the take-off distance on a dry runway is prohibited.
- (3) The use of actual landing distance on contaminated runways that is shorter than the dry runway landing field length is prohibited.
- (4) During taxi, do not use thrust reversers if movement area surfaces are covered with slush, ice, standing water or snow, except in the interest of safety.

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- (5) The maximum crosswind component for take-off and landing on a wet runway with water depth no more than 3.00 mm (0.125 inch), is defined in the following table for different values of the coefficient of friction:

COEFFICIENT OF FRICTION	MAXIMUM CROSSWIND COMPONENT
0.3 (poor braking)	10 knots (5 meters/sec)
0.4 (average braking)	15 knots (8 meters/sec)
0.5 (good braking, equivalent to dry)	24 knots (12 meters/sec)

- (6) The maximum crosswind component for take-off and landing on a contaminated runway is 10 knots (5 meters/sec).
- (7) Operation on runways with a coefficient of friction less than 0.3 is prohibited.

D. Emergency Procedures

The emergency procedures in Chapter 3 are applicable.

E. Normal Procedures

The normal procedures in Chapter 4 are applicable.

F. Abnormal Procedures

The abnormal procedures in Chapter 5 are applicable.

G. Performance

The performance data given in Chapter 6 are applicable, in addition to paragraphs 2. through 4. in the following pages.

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2. WET RUNWAY TAKE-OFF PERFORMANCE

A. General

Take-off performance data are calculated based on the assumption that maximum available reverse thrust is used down to a complete stop.

The following charts are used to obtain the take-off distance and V_1 speed data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified take-off conditions, for a wet runway.

If the take-off distance on a wet runway is less than the take-off distance on a dry runway, use the take-off distance for a dry runway.

The take-off distance charts herein present the most limiting of the following:

- (1) The distance from the start of the take-off roll to the point at which the airplane attains a height of 15 feet above the take-off surface, with a failure of the critical engine at V_{EF} , or
- (2) 115% of the distance from the start of the take-off roll to the point at which the airplane attains a height of 35 feet above the take-off surface, with all engines operating, or
- (3) The accelerate-stop distance.

B. Take-Off Distance, Flaps 20° – Anti-Icing Off – Wet Runway

The take-off distance for a flaps 20° take-off on a wet runway, at airport pressure altitudes of 10,000 feet and below, with anti-icing off, is given by Figure 07-02-1 (Page 1 of 2).

NOTE

When dispatching with one thrust reverser inoperative, the take-off distance derived from Figure 07-02-1 (Page 1 of 2) must be increased by 200 meters (656 feet).

Example:

Associated Conditions:

Ambient temperature	= 20°C
Airport pressure altitude	= 4,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 19,000 kg (41,887 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-02-1 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 1,750 meters (5,741 feet).

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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

The take-off distance for a flaps 20° take-off on a wet runway, at airport pressure altitudes above 10,000 feet, with anti-icing off, is given by Figure 07-02-1 (Page 2 of 2).

NOTE

When dispatching with one thrust reverser inoperative, the take-off distance derived from Figure 07-02-1 (Page 2 of 2) must be increased by 200 meters (656 feet).

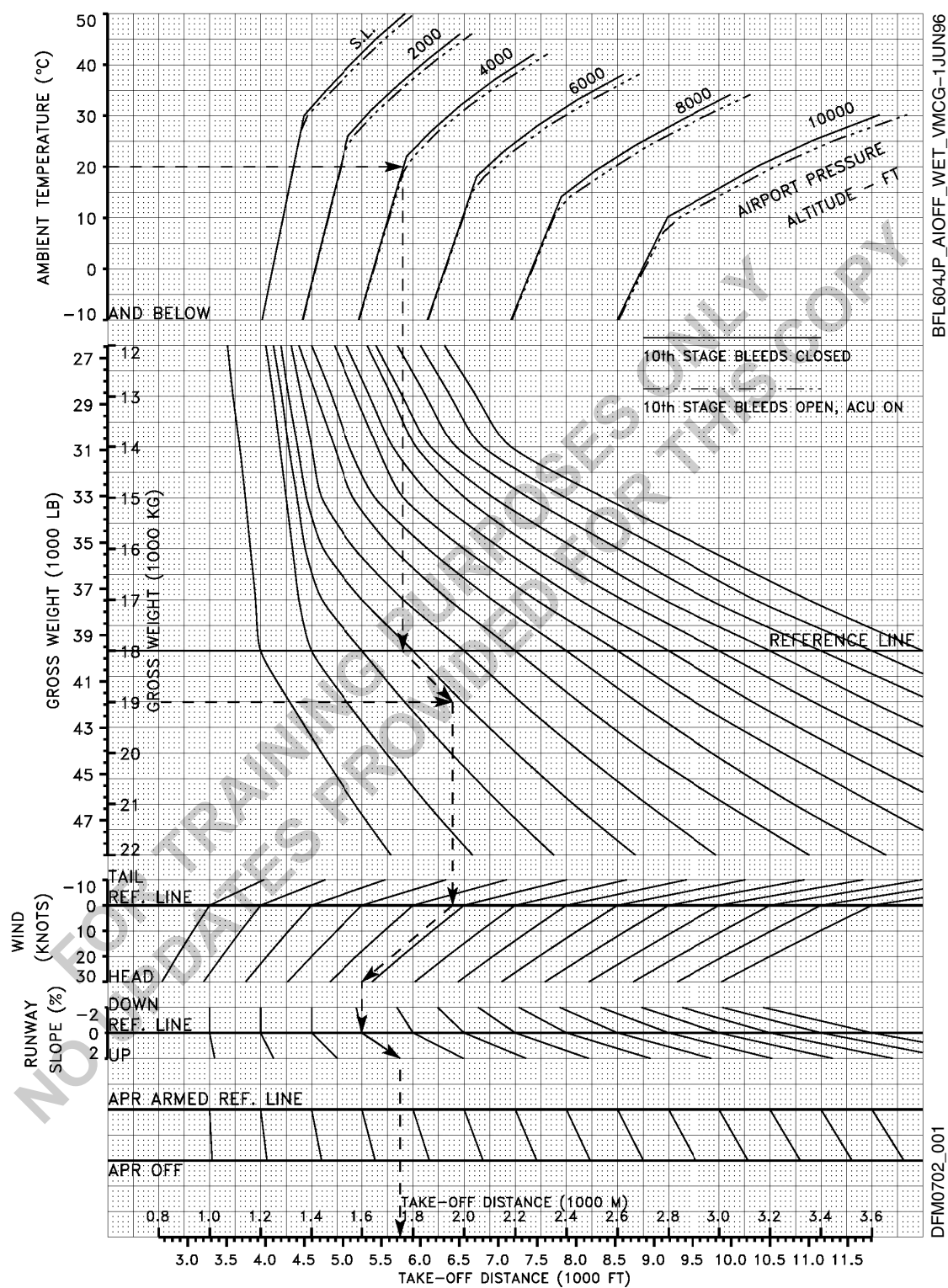
Example:

Associated Conditions:

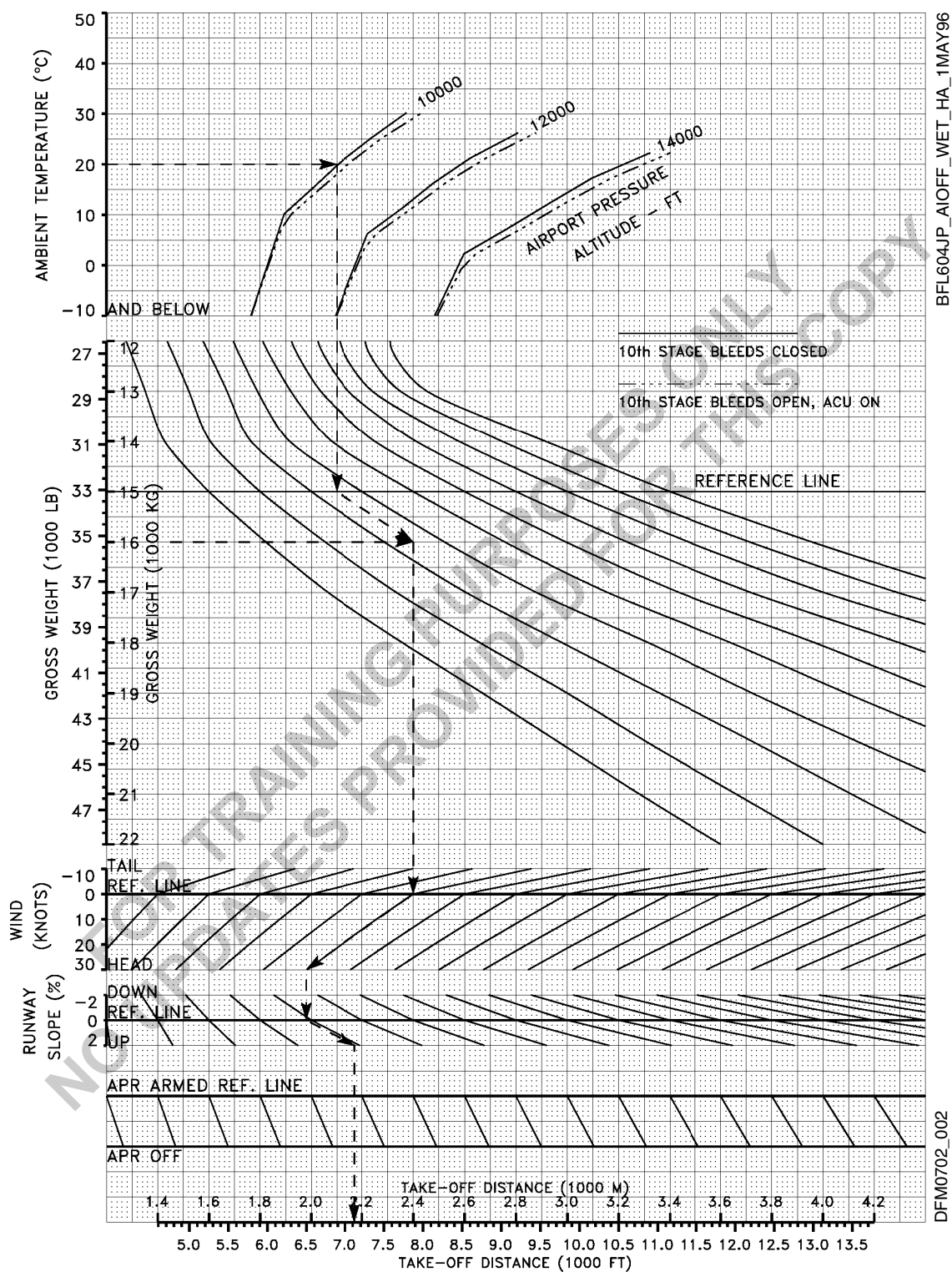
Ambient temperature	= 20°C
Airport pressure altitude	= 10,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-02-1 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,170 meters (7,119 feet).

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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Wet Runway (Page 1 of 2)
 Figure 07-02-1

2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Wet Runway (Page 2 of 2)
 Figure 07-02-1

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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

C. Take-Off Distance, Flaps 20° – Anti-Icing On – Wet Runway

The take-off distance for a flaps 20° take-off on a wet runway, at airport pressure altitudes of 10,000 feet and below, with anti-icing on is given by Figure 07-02-2 (Page 1 of 2).

NOTE

When dispatching with one thrust reverser inoperative, the take-off distance derived from Figure 07-02-2 (Page 1 of 2) must be increased by 200 meters (656 feet).

Example:

Associated Conditions:

Ambient temperature	= -30°C
Airport pressure altitude	= 6,000 ft
10th stage bleeds	= Closed
Cowl anti-ice	= On
Gross weight	= 19,000 kg (41,887 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-02-2 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 1,750 meters (5,741 feet).

The take-off distance for a flaps 20° take-off on a wet runway, at airport pressure altitudes above 10,000 feet, with anti-icing on is given by Figure 07-02-2 (Page 2 of 2).

NOTE

When dispatching with one thrust reverser inoperative, the take-off distance derived from Figure 07-02-2 (Page 2 of 2) must be increased by 200 meters (656 feet).

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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

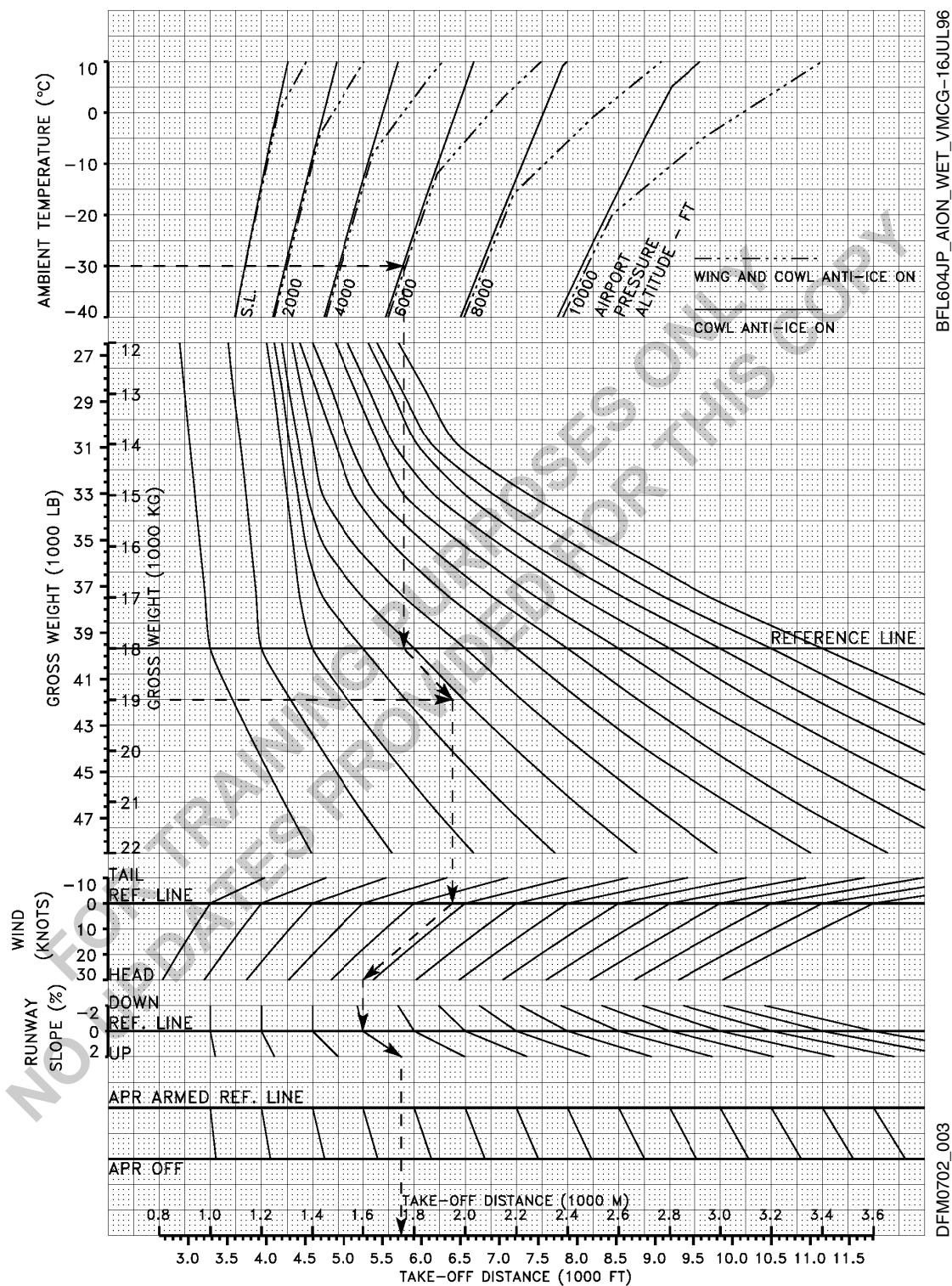
Example:

Associated Conditions:

Ambient temperature	= -30°C
Airport pressure altitude	= 12,000 ft
10th stage bleeds	= Closed
Cowl anti-ice	= On
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-02-2 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,030 meters (6,660 feet).

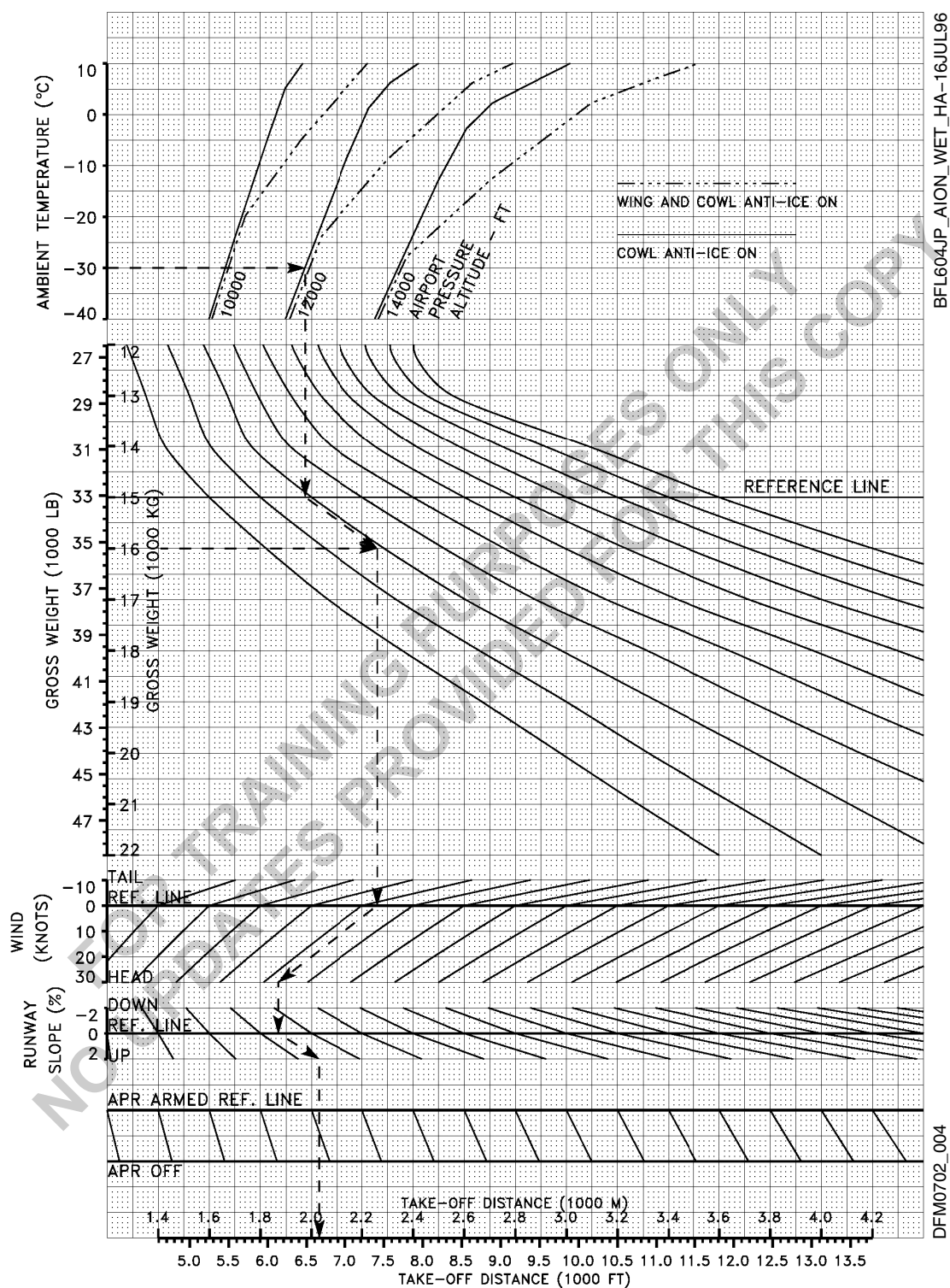
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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


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Take-Off Distance, Flaps 20° – Anti-Icing On – Wet Runway (Page 1 of 2)
 Figure 07-02-2

2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Wet Runway (Page 2 of 2)
 Figure 07-02-2

CHALLENGER 650	SUPPLEMENT 2 Operation on Wet and Contaminated Runways	07-02-01-11
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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

D. Take-Off Decision Speed, V_1 (Flaps 20°) – Anti-Icing Off – Wet Runway

Figure 07-02-3 is used to determine the V_1 speed for a flaps 20° take-off (balanced field length) on a wet runway, for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope and various bleed configurations.

NOTE

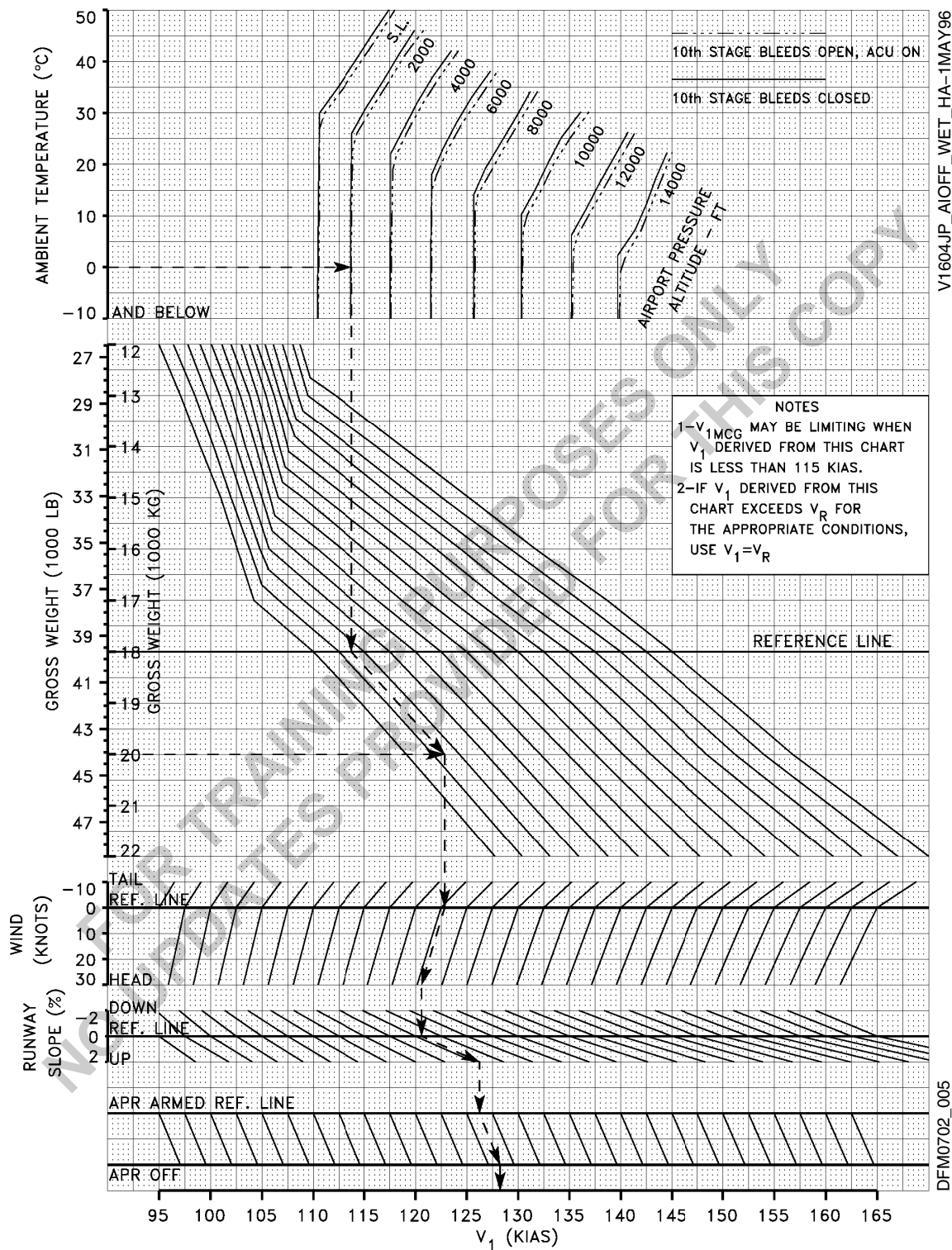
When dispatching with one thrust reverser inoperative, the take-off decision speed, V_1 , derived from Figure 07-02-3 must be reduced by 2.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

Example:

Associated Conditions:

Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 2,000 ft
10th stage bleeds	= Closed
Ambient temperature	= 0°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off
Thrust reversers	= Both available

The example in Figure 07-02-3 shows that for the above associated conditions, V_1 is equal to 128.3 KIAS.

2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

V1 Speed (Flaps 20°) – Anti-Icing Off – Wet Runway
 Figure 07-02-3

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2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

E. Take-Off Decision Speed, V_1 (Flaps 20°) – Anti-Icing On – Wet Runway

Figure 07-02-4 is used to determine the V_1 speed for a flaps 20° take-off (balanced field length) on a wet runway, for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope and use of anti-icing.

NOTE

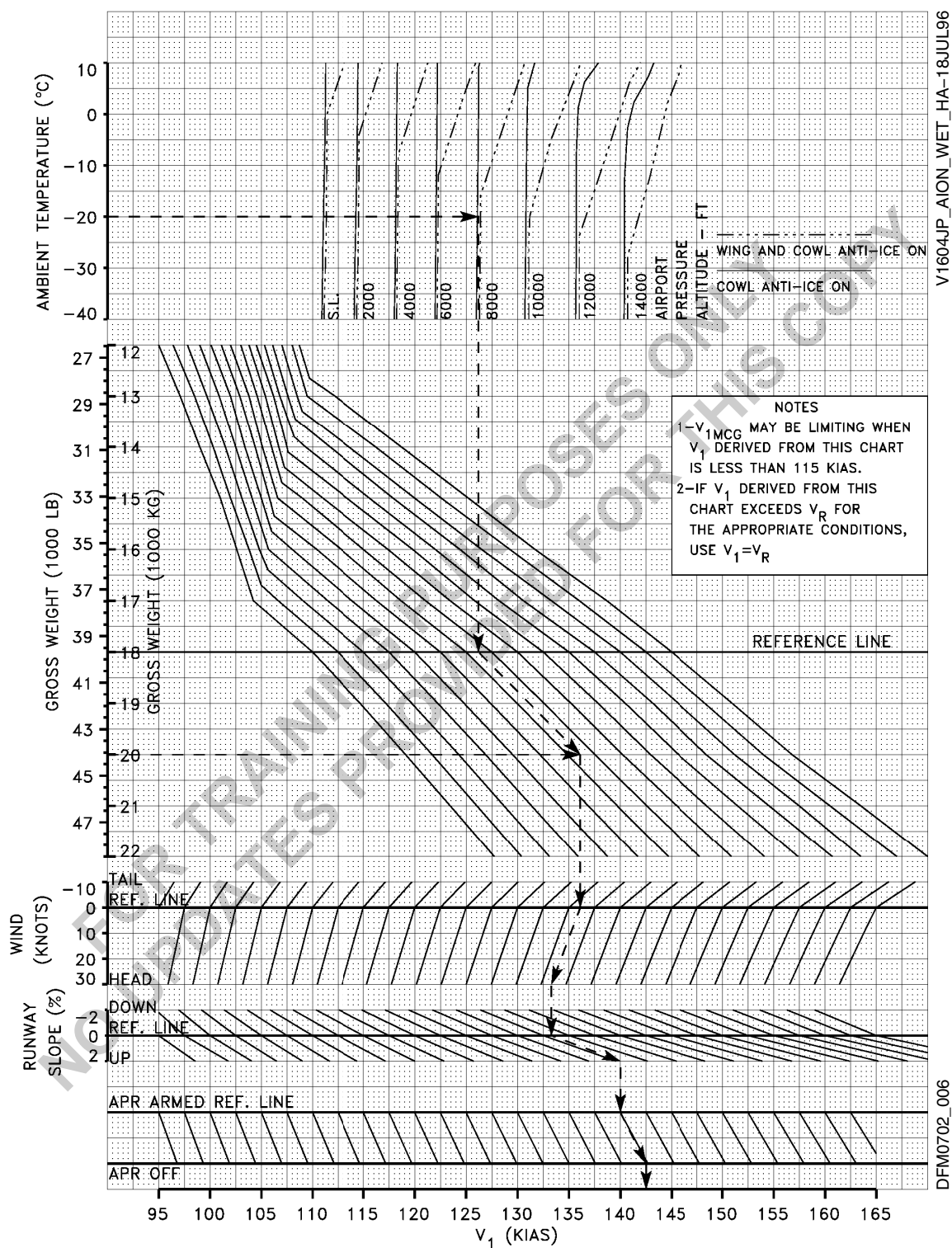
When dispatching with one thrust reverser inoperative, the take-off decision speed, V_1 , derived from Figure 07-02-4 must be reduced by 2.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

Example:

Associated Conditions:

Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 8,000 ft
Cowl anti-icing	= On
Ambient temperature	= -20°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off
Thrust reversers	= Both available

The example in Figure 07-02-4 shows that for the above associated conditions, V_1 is equal to 142.5 KIAS.

2. WET RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


V1 Speed (Flaps 20°) – Anti-Icing On – Wet Runway
 Figure 07-02-4

CHALLENGER 650	SUPPLEMENT 2 Operation on Wet and Contaminated Runways	07-02-01-15
		Rev 15, Mar 25/19

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE

A. General

Contaminated runway distances have been estimated assuming that the runway is completely contaminated. However, selection of the type and amount of contaminant should be based on the runway conditions where the high speed portion of the take-off will occur and, in the case of an ice covered runway, where braking would be used during a rejected take-off.

Take-off performance data are calculated based on the assumption that maximum available reverse thrust is used down to a complete stop.

The following charts are used to obtain the take-off performance data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified take-off conditions, for runways contaminated by ice, compacted snow or slush (or its equivalent, i.e. standing water or loose snow).

The take-off distance charts herein present the most limiting of the following:

- (1) The distance from the start of the take-off roll to the point at which the airplane attains a height of 15 feet above the take-off surface, with a failure of the critical engine at V_{EF} , or
- (2) 115% of the distance from the start of the take-off roll to the point at which the airplane attains a height of 35 feet above the take-off surface, with all engines operating, or
- (3) The accelerate-stop distance.

NOTE

If the take-off distance on a contaminated runway is less than the take-off distance on a dry runway, use the take-off distance for a dry runway.

B. Take-Off on a Runway Contaminated with Ice

NOTE

Dispatch with one thrust reverser inoperative on a runway contaminated with ice is not recommended.

(1) ANTI-ICING OFF

The take-off distance for a flaps-20° take-off with anti-icing off, on ice-contaminated runways, is given by Figure 07-02-5.

The V_1 speed for a flaps-20° take-off with anti-icing off, on ice-contaminated runways, is given by Figure 07-02-6.

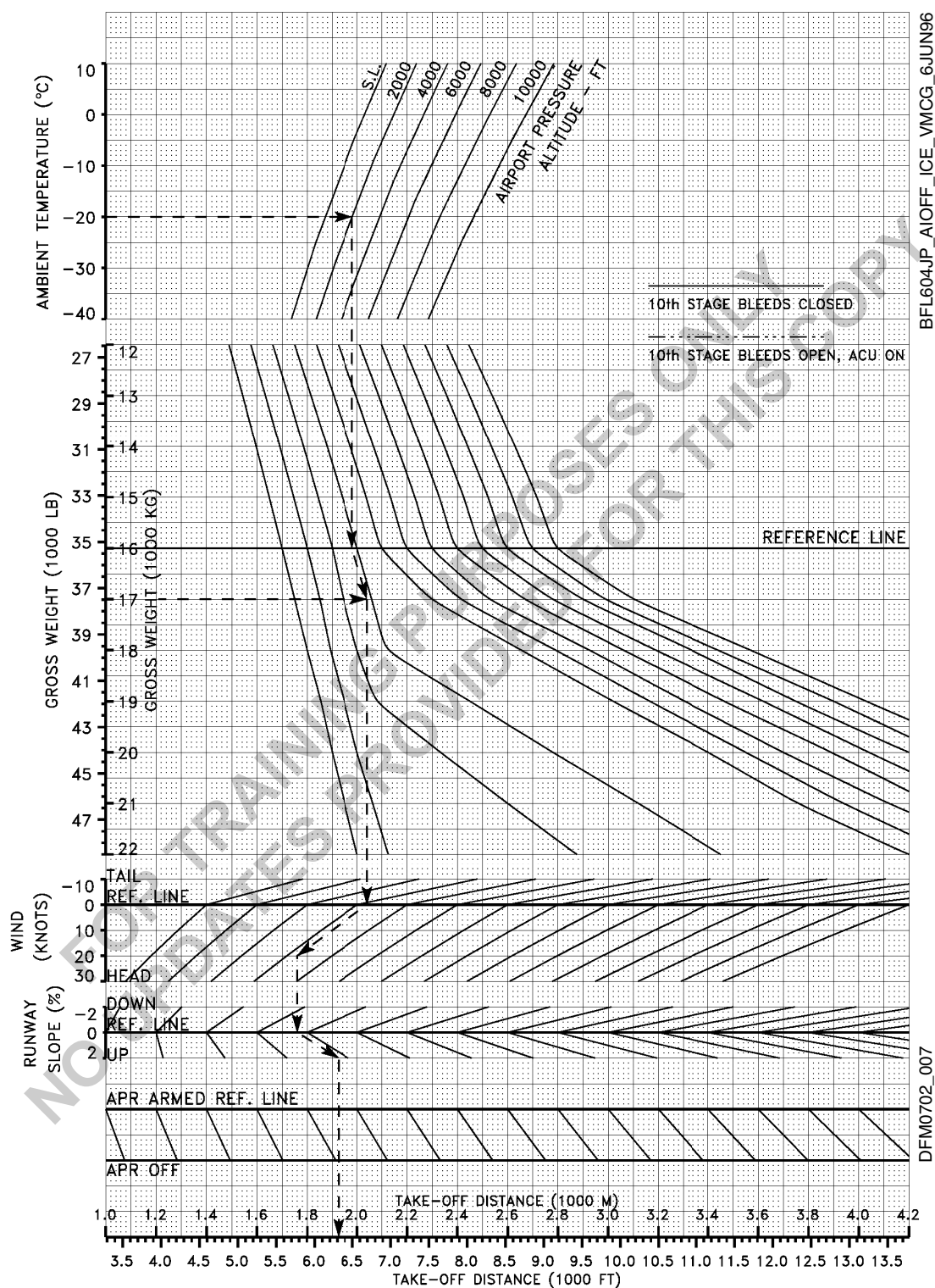
(2) ANTI-ICING ON

The take-off distance for a flaps-20° take-off with anti-icing on, on ice-contaminated runways, is given by Figure 07-02-7.

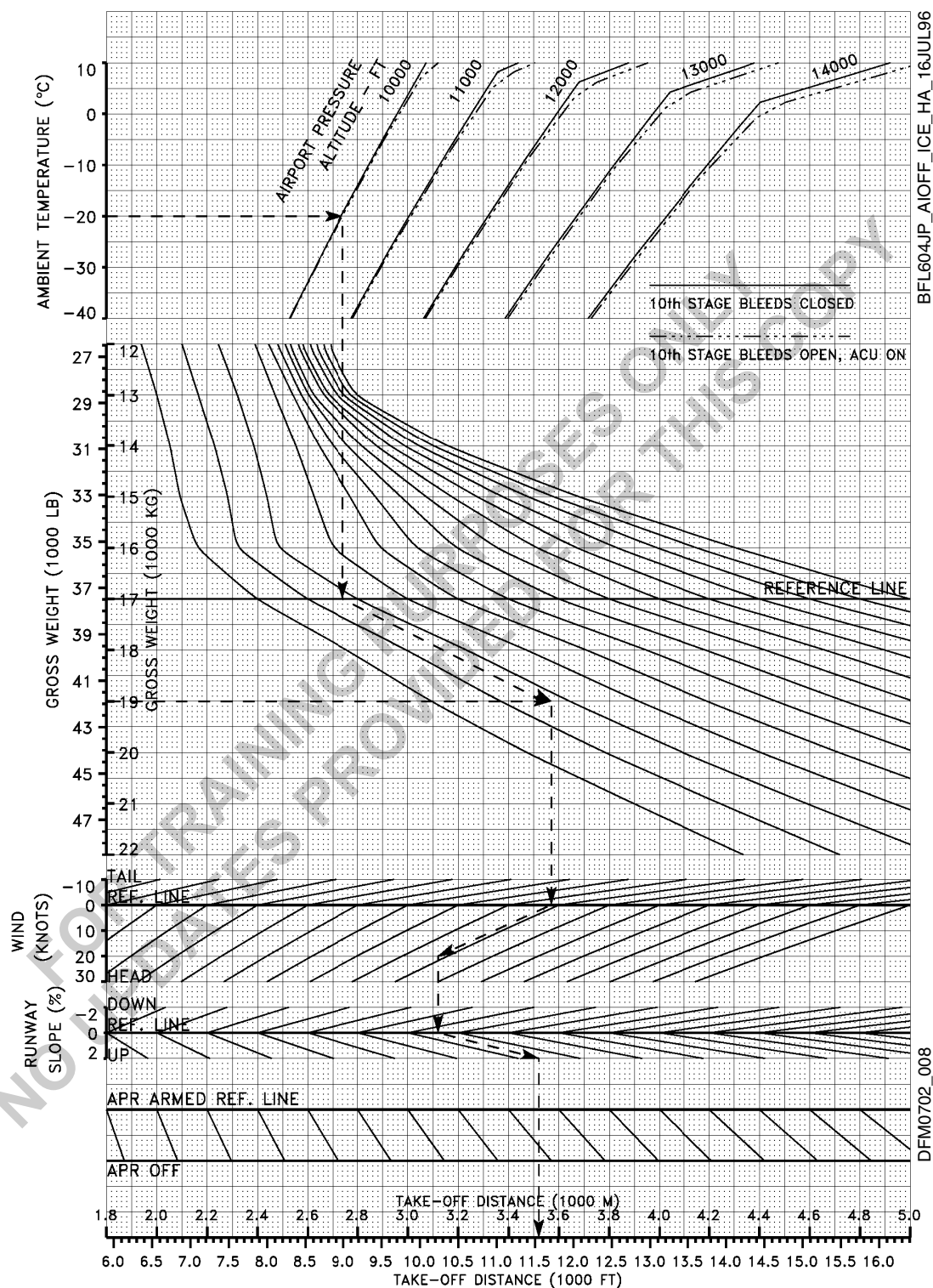
The V_1 speed for a flaps-20° take-off with anti-icing on, on ice-contaminated runways, is given by Figure 07-02-8.

The charts are utilized in the same manner as described in the previous examples for wet runways.

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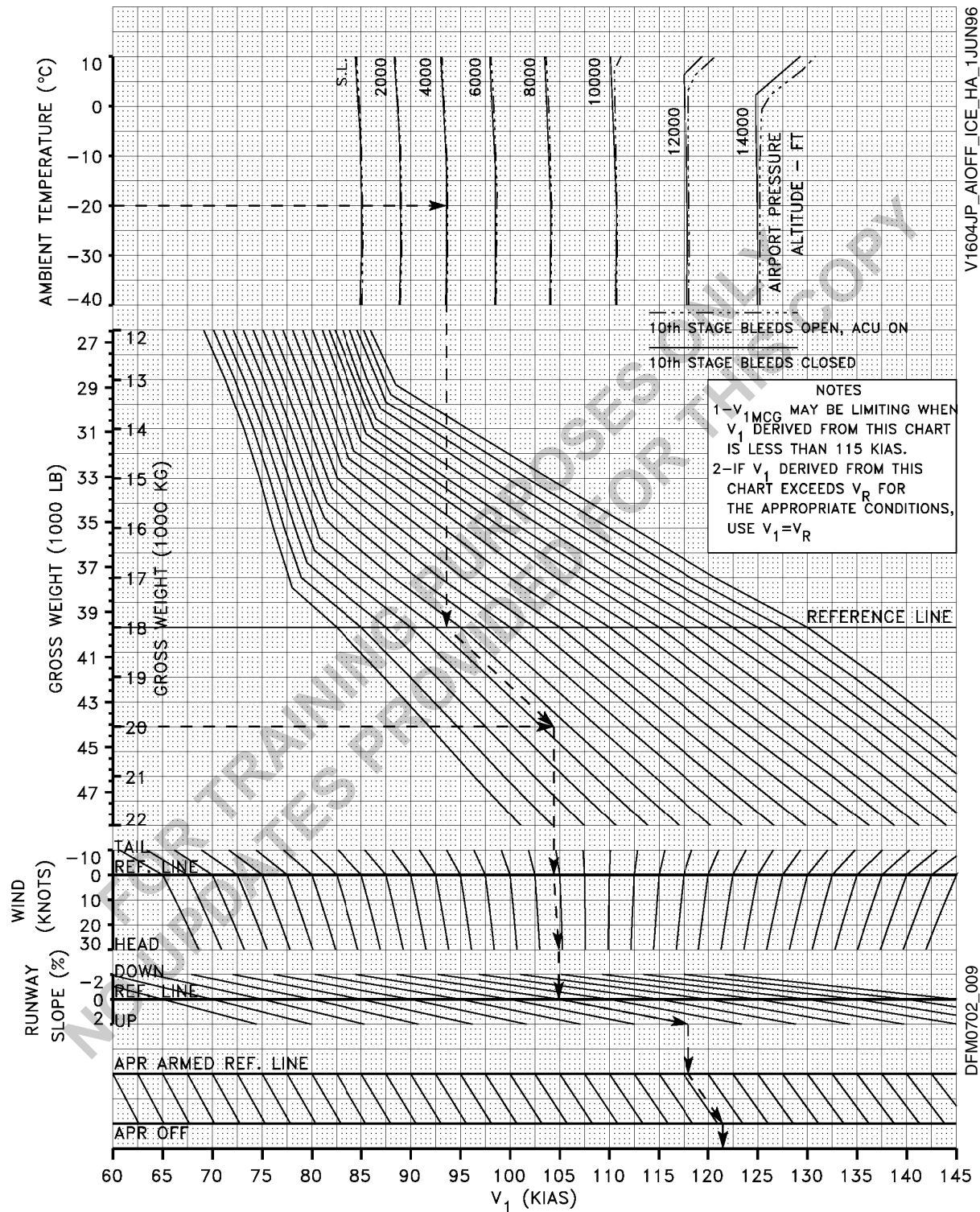
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Ice-Contaminated Runway (Page 1 of 2)
 Figure 07-02-5

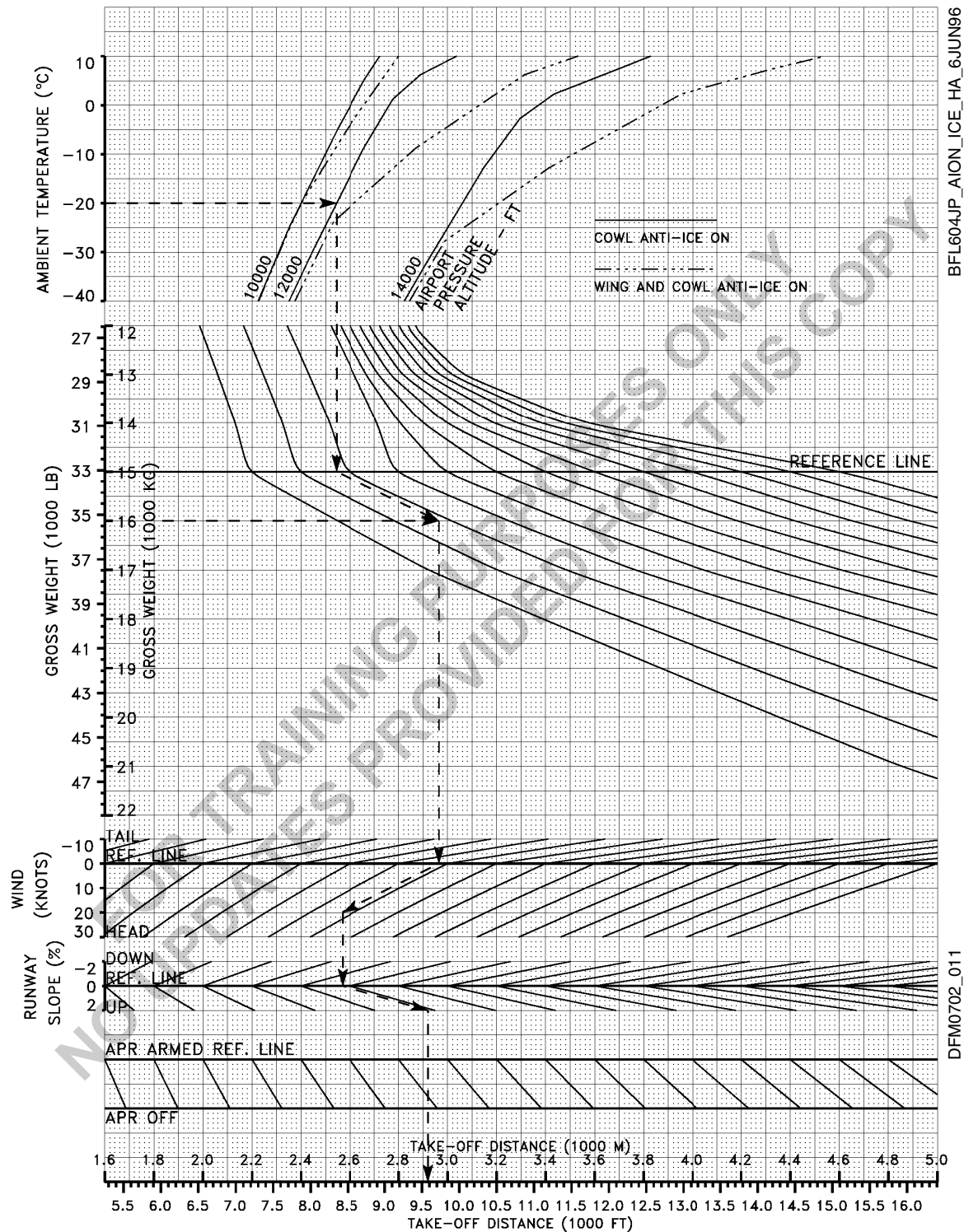
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Ice-Contaminated Runway (Page 2 of 2)
 Figure 07-02-5

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

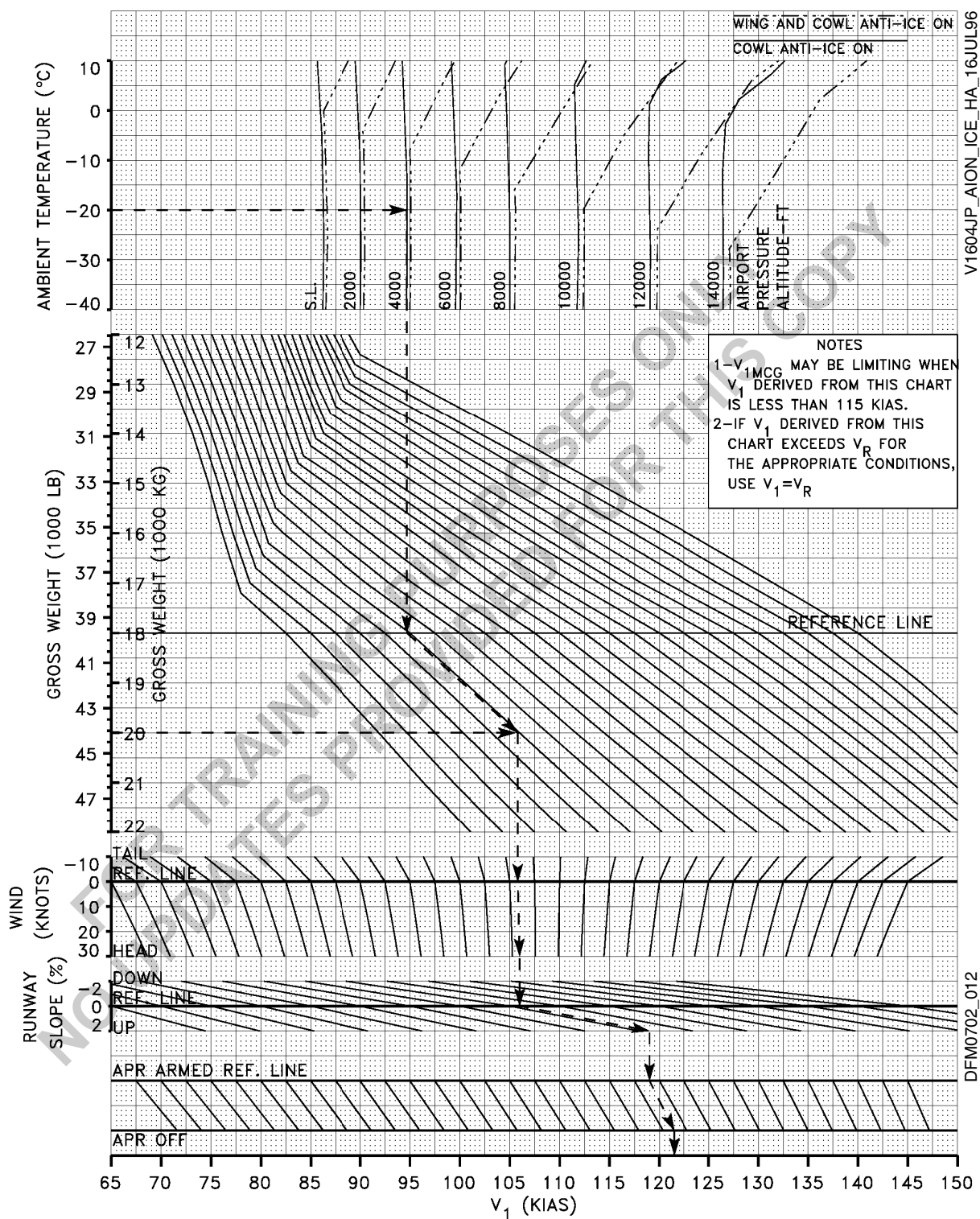


V1 Speed (Flaps 20°) – Anti-Icing Off – Ice-Contaminated Runway
 Figure 07-02-6

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Ice-Contaminated Runway (Page 2 of 2)
 Figure 07-02-7

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)



V1 Speed (Flaps 20°) – Anti-Icing On – Ice-Contaminated Runway
 Figure 07-02-8

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3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

C. Take-Off on a Runway Contaminated with Compacted Snow

NOTE

When dispatching with one thrust reverser inoperative on a runway contaminated with compacted snow, the following corrections must be applied:

- Increase take-off distance by 300 meters (984 feet);
- Reduce V_1 by 4.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

(1) ANTI-ICING OFF

The take-off distance for a flaps-20° take-off with anti-icing off, on runways contaminated with compacted snow, is given by Figure 07-02-9.

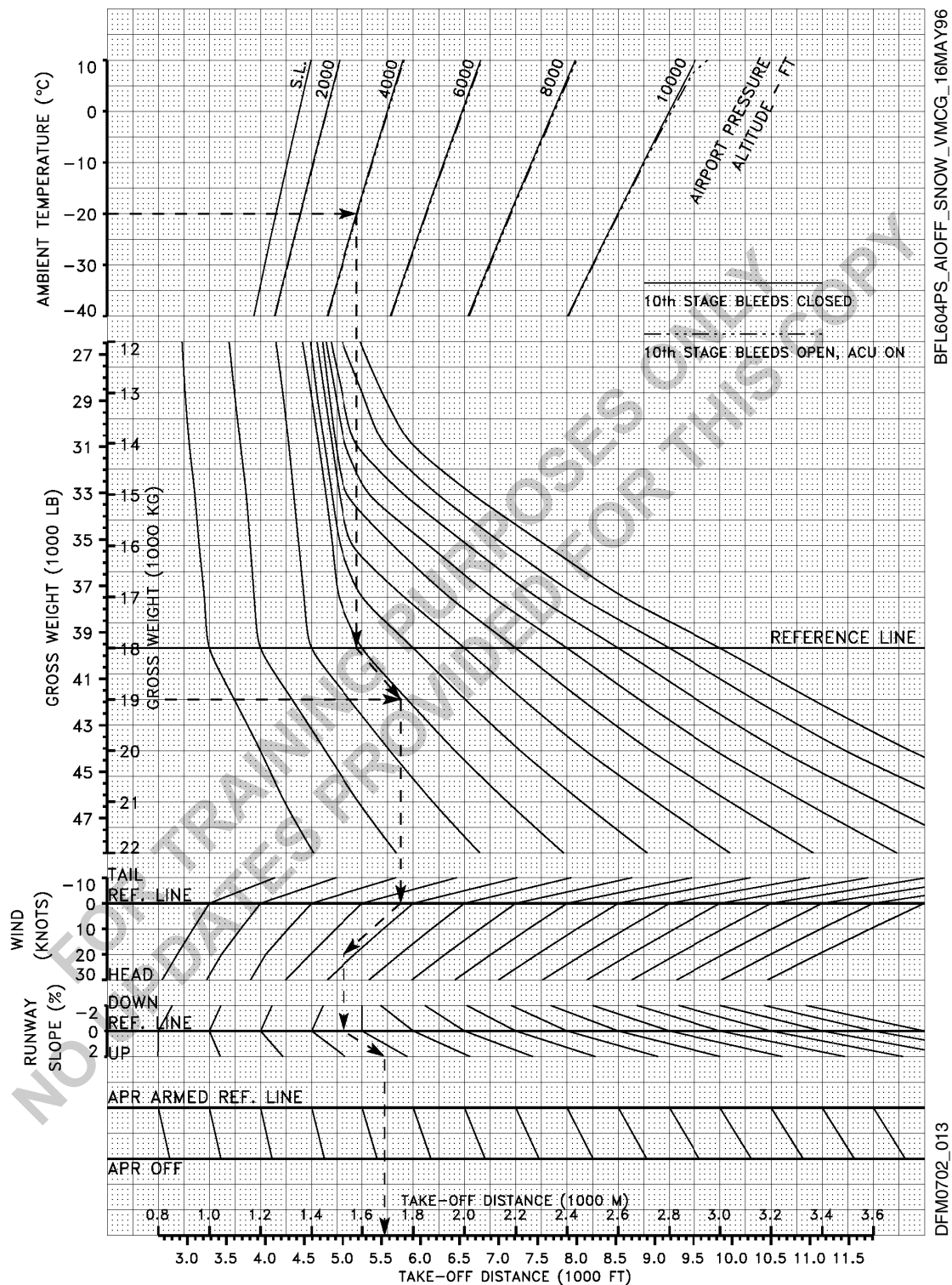
The V_1 speed for a flaps-20° take-off with anti-icing off, on runways contaminated with compacted snow, is given by Figure 07-02-10.

(2) ANTI-ICING ON

The take-off distance for a flaps-20° take-off with anti-icing on, on runways contaminated with compacted snow, is given by Figure 07-02-11.

The V_1 speed for a flaps-20° take-off with anti-icing on, on runways contaminated with compacted snow, is given by Figure 07-02-12.

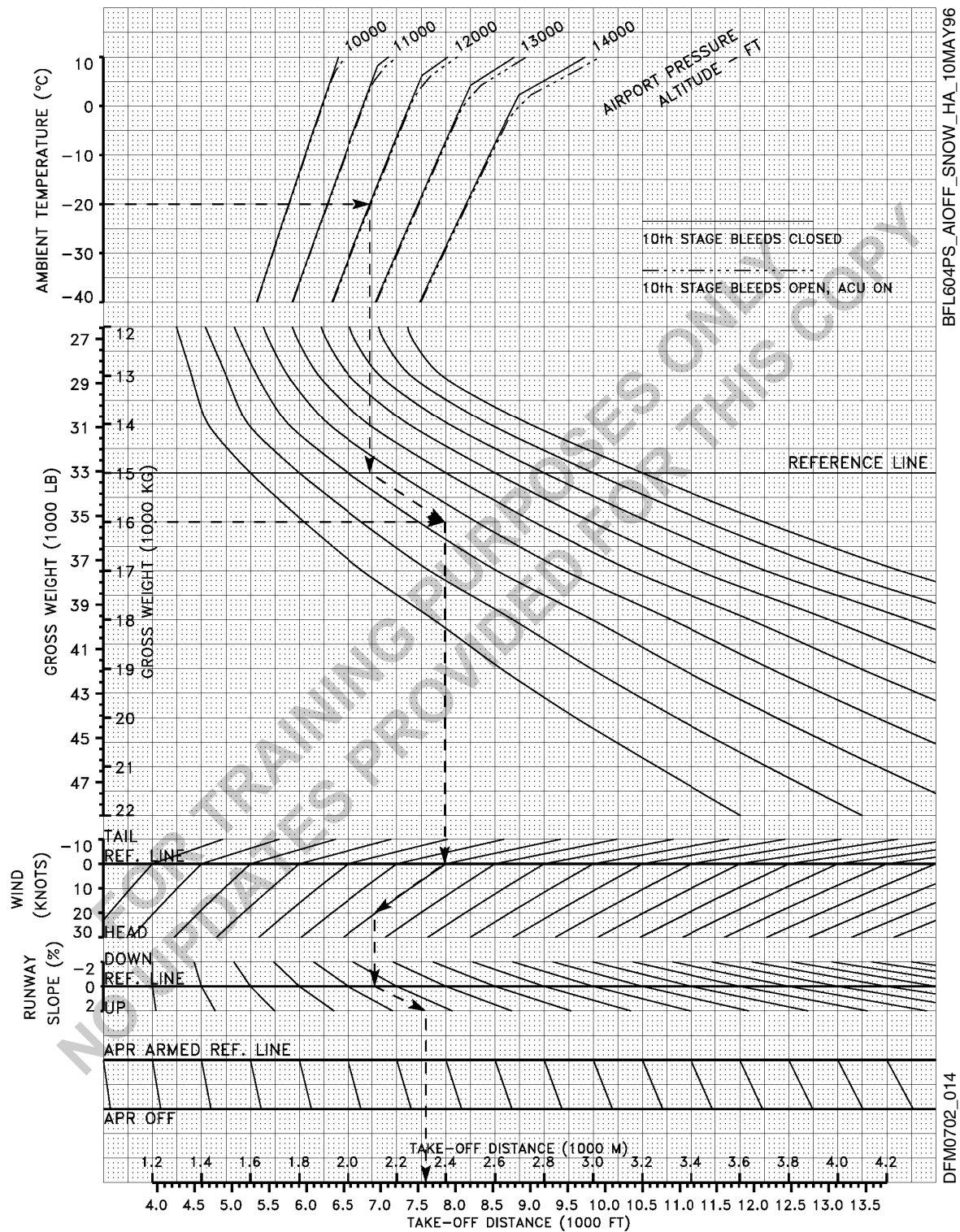
The charts are utilized in the same manner as described in the previous examples for wet runways.

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


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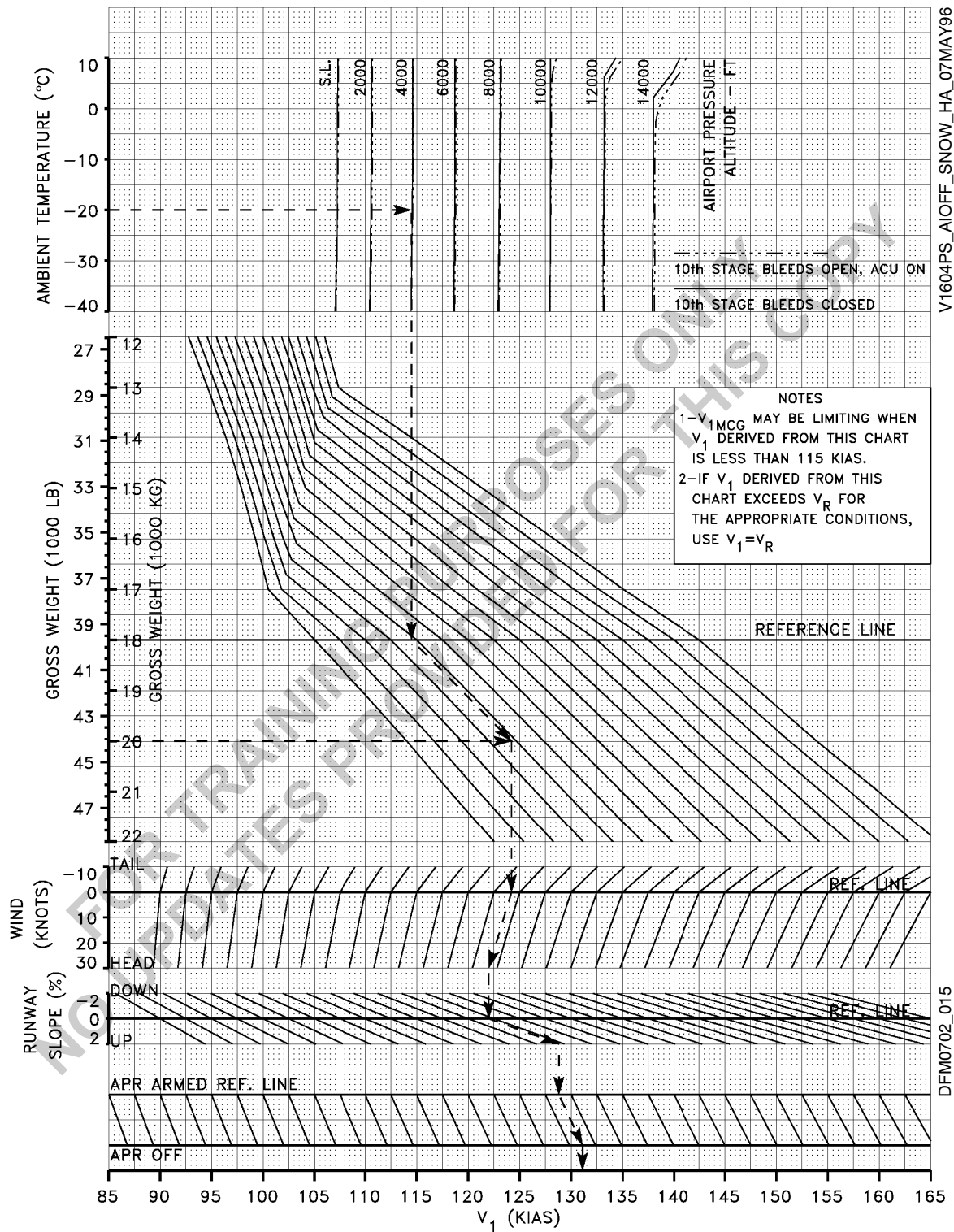
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Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Compacted Snow
 (Page 1 of 2)
 Figure 07-02-9

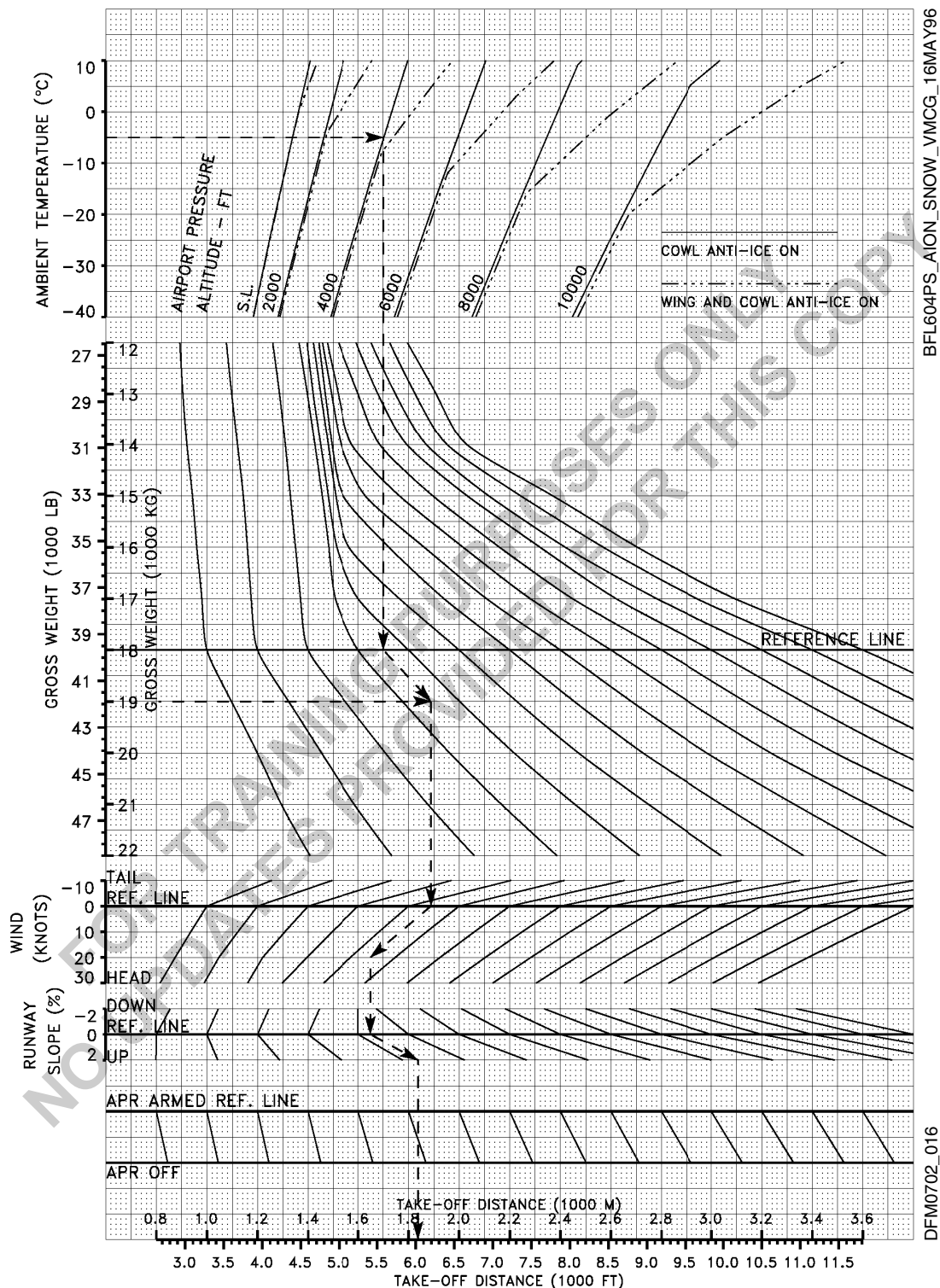
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Compacted Snow
 (Page 2 of 2)

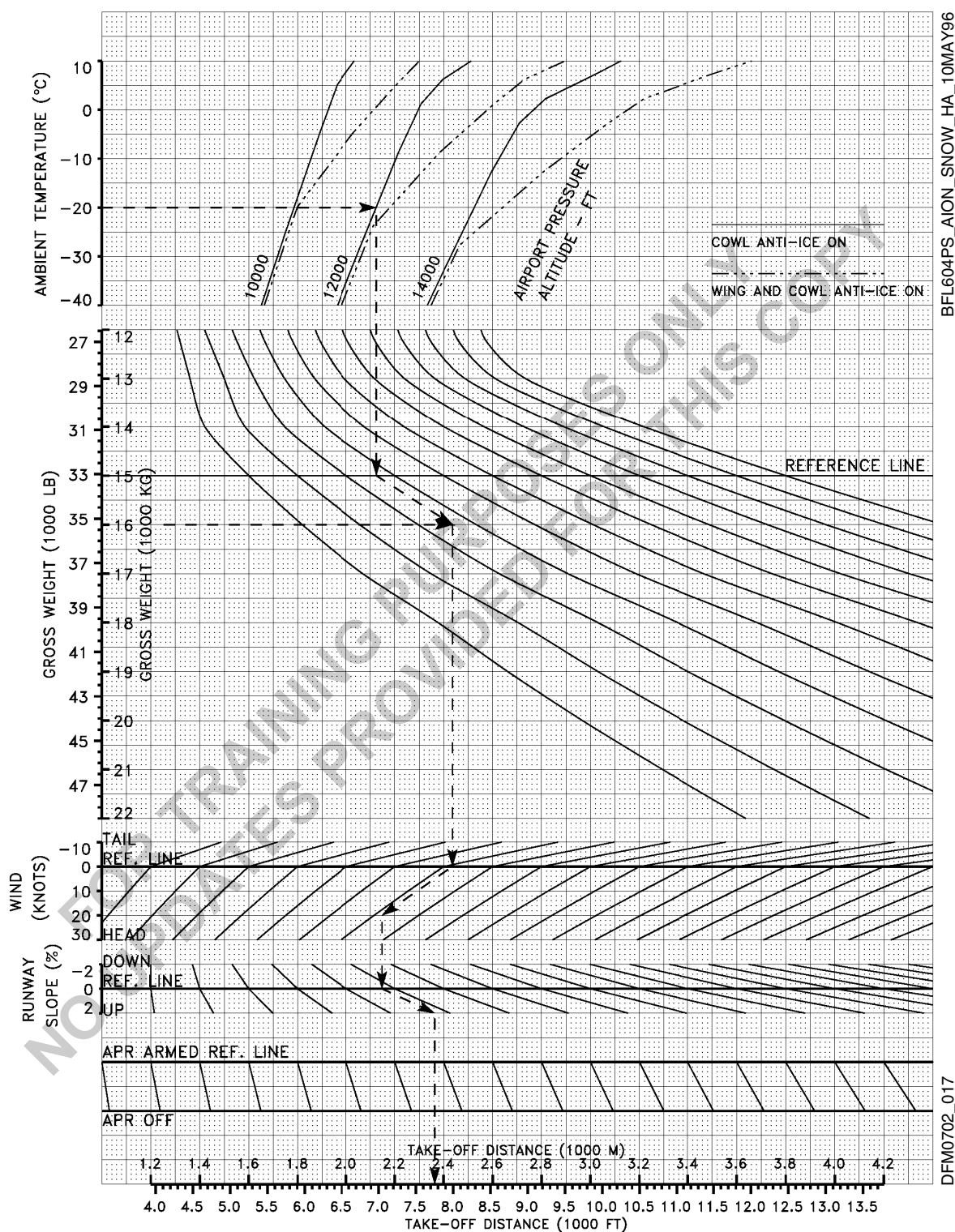
Figure 07-02-9

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


V1 Speed (Flaps 20°) – Anti-Icing Off – Runway Contaminated with Compacted Snow
 Figure 07-02-10

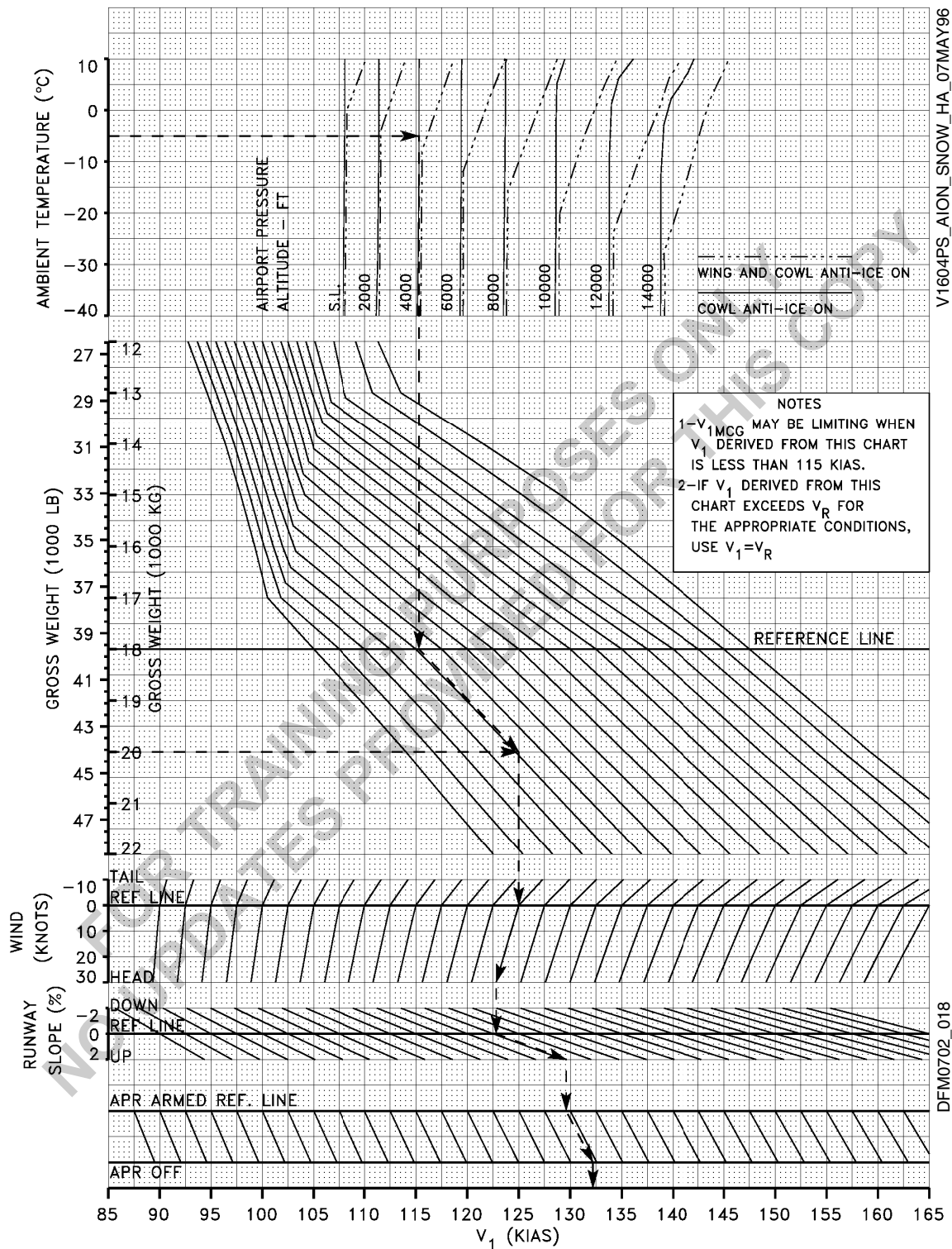
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Compacted Snow
 (Page 1 of 2)
 Figure 07-02-11

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Compacted Snow
 (Page 2 of 2)
 Figure 07-02-11

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)



V1 Speed (Flaps 20°) – Anti-Icing On – Runway Contaminated with Compacted Snow
 Figure 07-02-12

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3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

D. Take-Off on a Runway Contaminated with Slush (0.125 in) or Equivalent

NOTE

When dispatching with one thrust reverser inoperative on a runway contaminated with slush (0.125") or its equivalent, the following corrections must be applied:

- Increase take-off distance by 600 meters (1,968 feet);
- Reduce V_1 by 6.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

(1) ANTI-ICING OFF

The take-off distance for a flaps-20° take-off with anti-icing off, on runways contaminated with 3.2 mm (0.125 in) of slush or equivalent [3.2 mm (0.125 in) of standing water; or 6.4 mm (0.25 in) of wet snow; or 15.2 mm (0.6 in) of dry snow], is given by Figure 07-02-13.

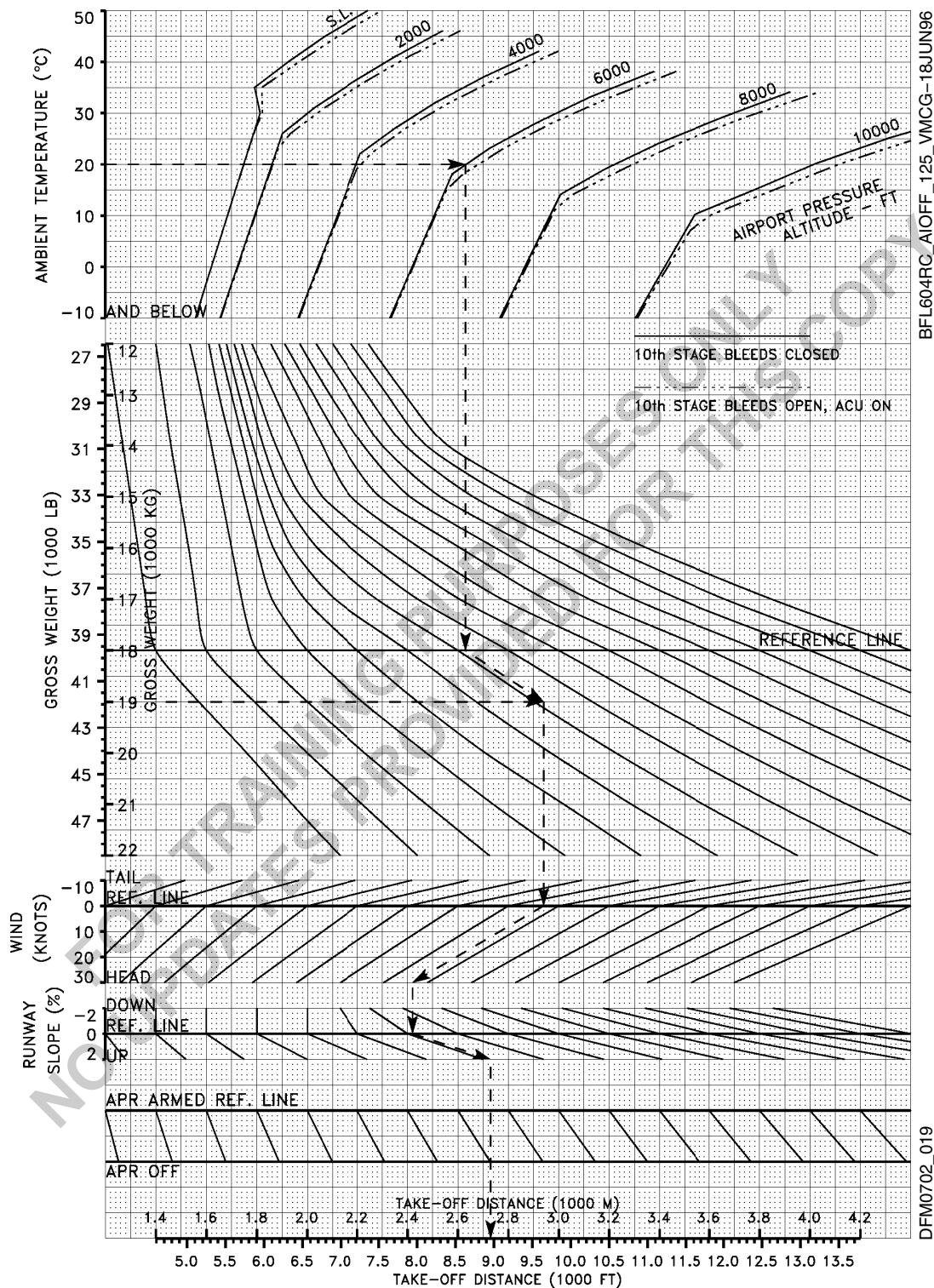
The V_1 speed for a flaps-20° take-off with anti-icing off, on runways contaminated with 3.2 mm (0.125 in) of slush or equivalent [3.2 mm (0.125 in) of standing water; or 6.4 mm (0.25 in) of wet snow; or 15.2 mm (0.6 in) of dry snow], is given by Figure 07-02-14.

(2) ANTI-ICING ON

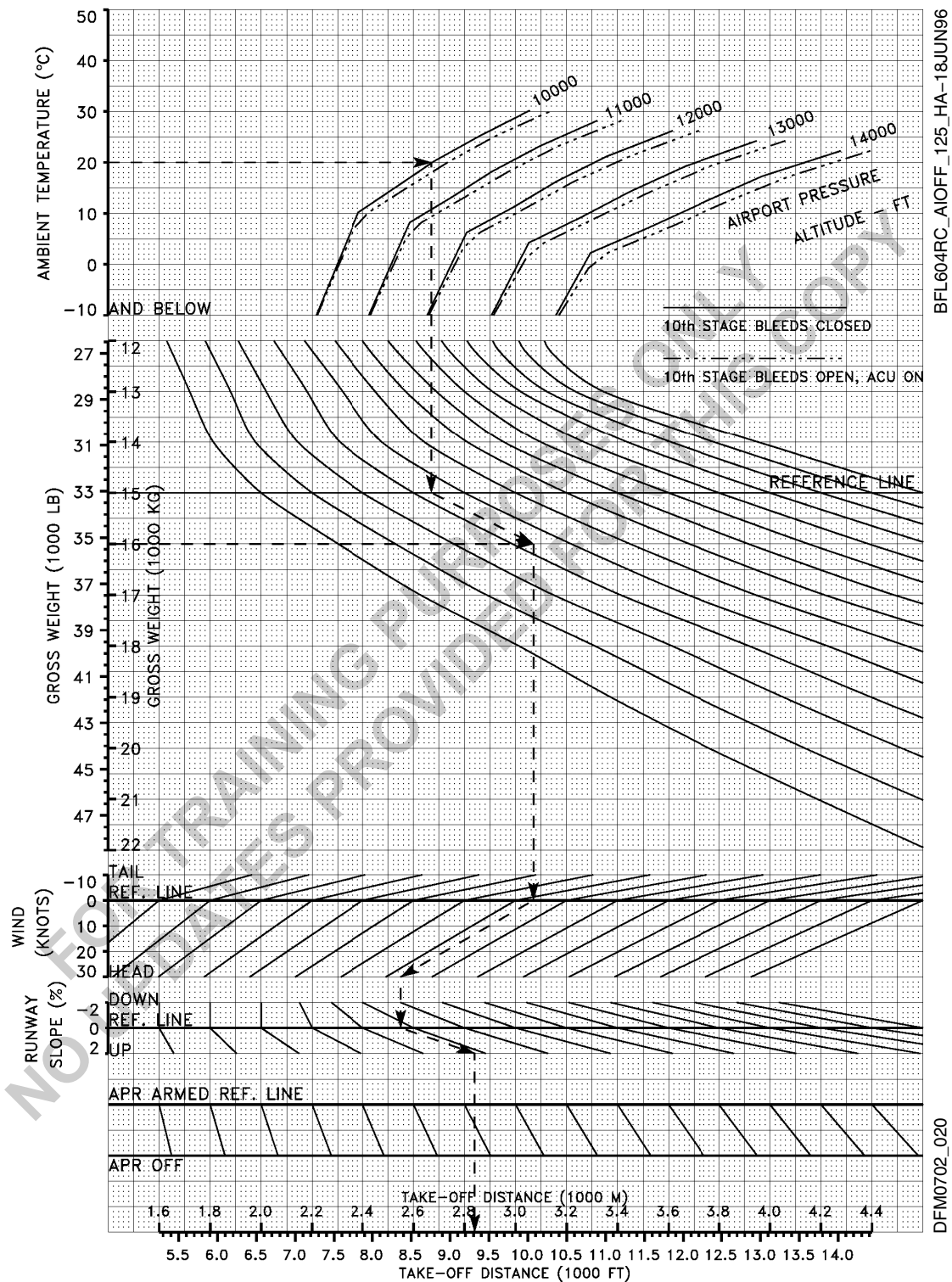
The take-off distance for a flaps-20° take-off with anti-icing on, on runways contaminated with 3.2 mm (0.125 in) of slush or equivalent [3.2 mm (0.125 in) of standing water; or 6.4 mm (0.25 in) of wet snow; or 15.2 mm (0.6 in) of dry snow], is given by Figure 07-02-15.

The V_1 speed for a flaps-20° take-off with anti-icing on, on runways contaminated with 3.2 mm (0.125 in) of slush or equivalent [3.2 mm (0.125 in) of standing water; or 6.4 mm (0.25 in) of wet snow; or 15.2 mm (0.6 in) of dry snow], is given by Figure 07-02-16.

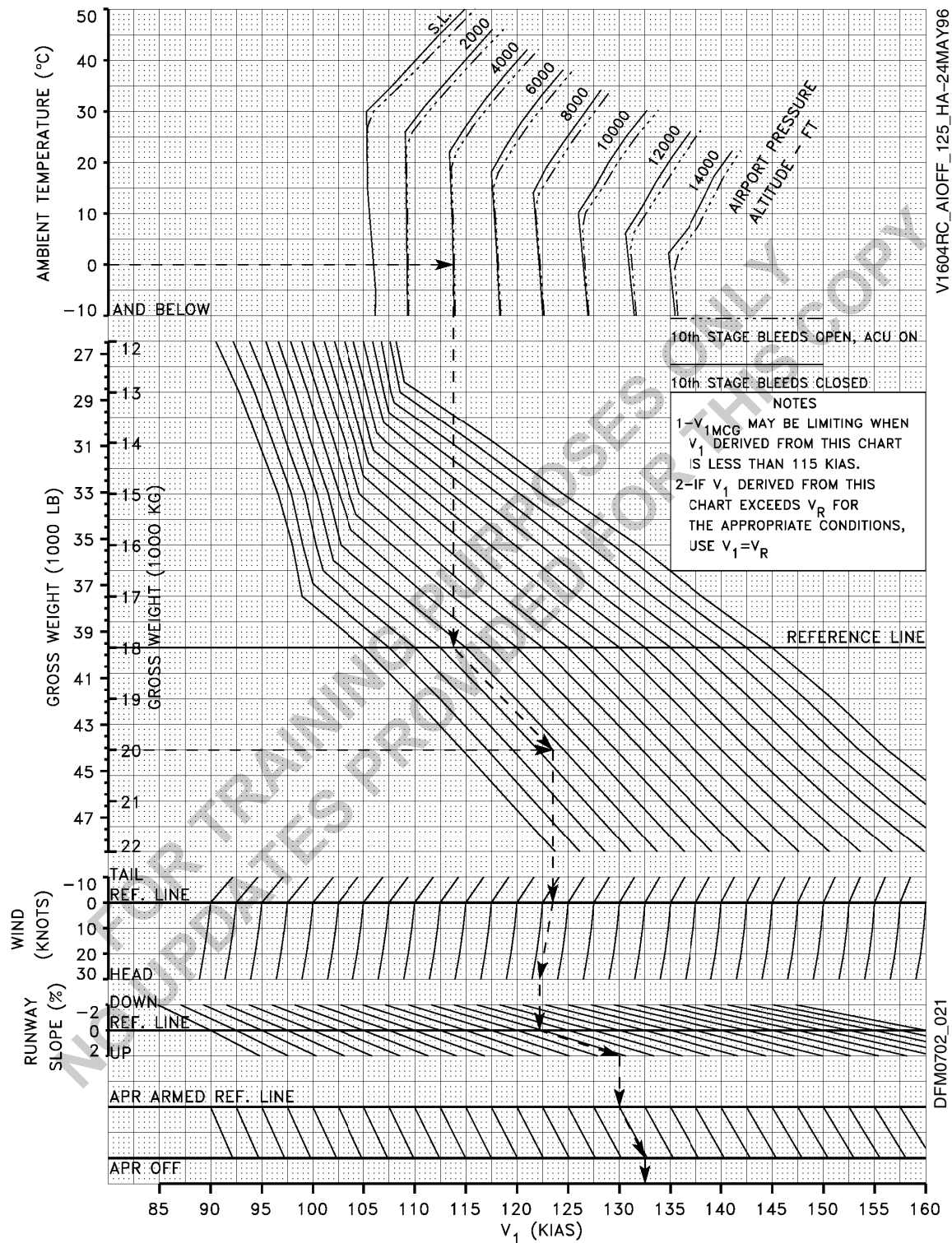
The charts are utilized in the same manner as described in the previous examples for wet runways.

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


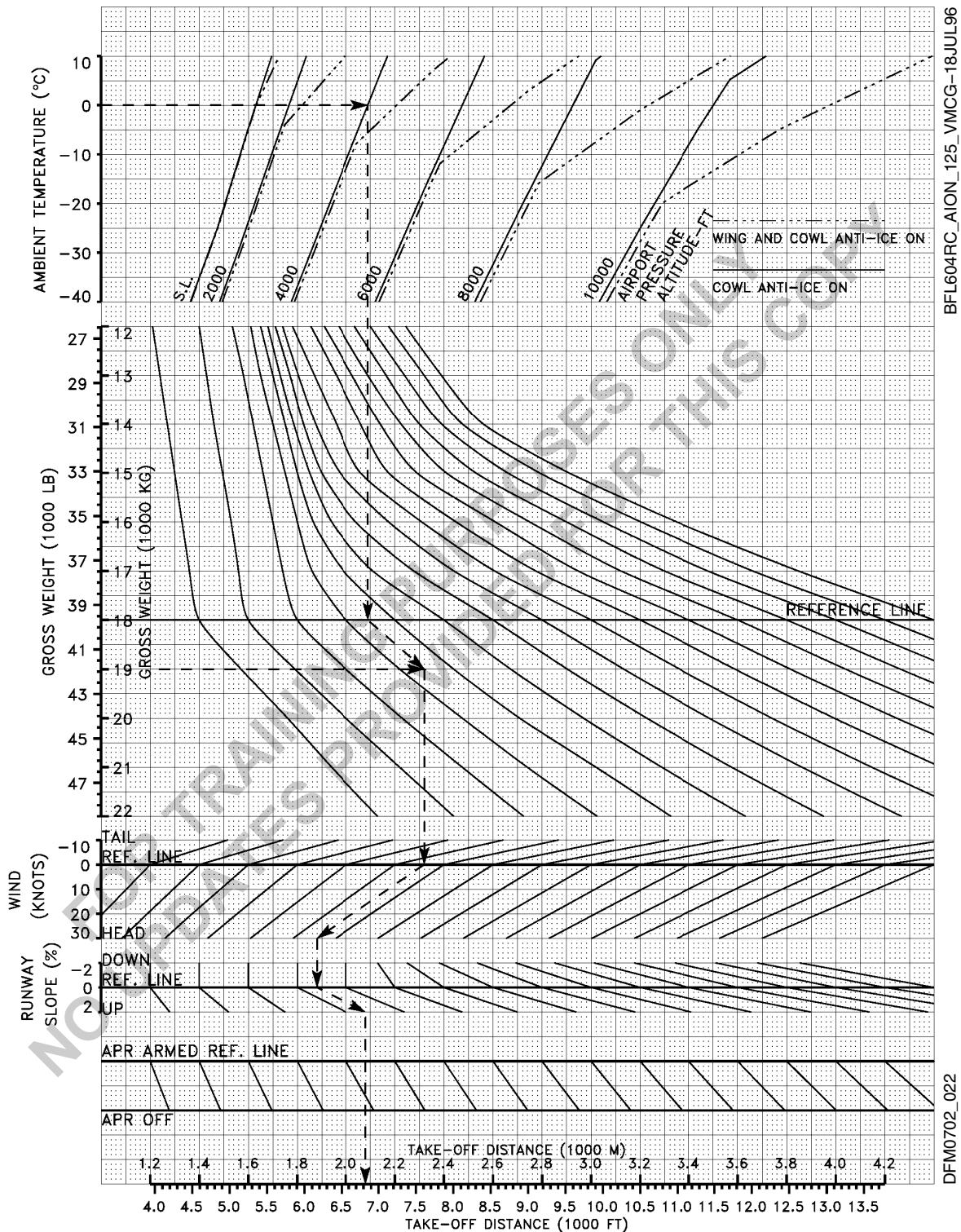
Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Slush (0.125") or Equivalent (Page 1 of 2)
 Figure 07-02-13

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


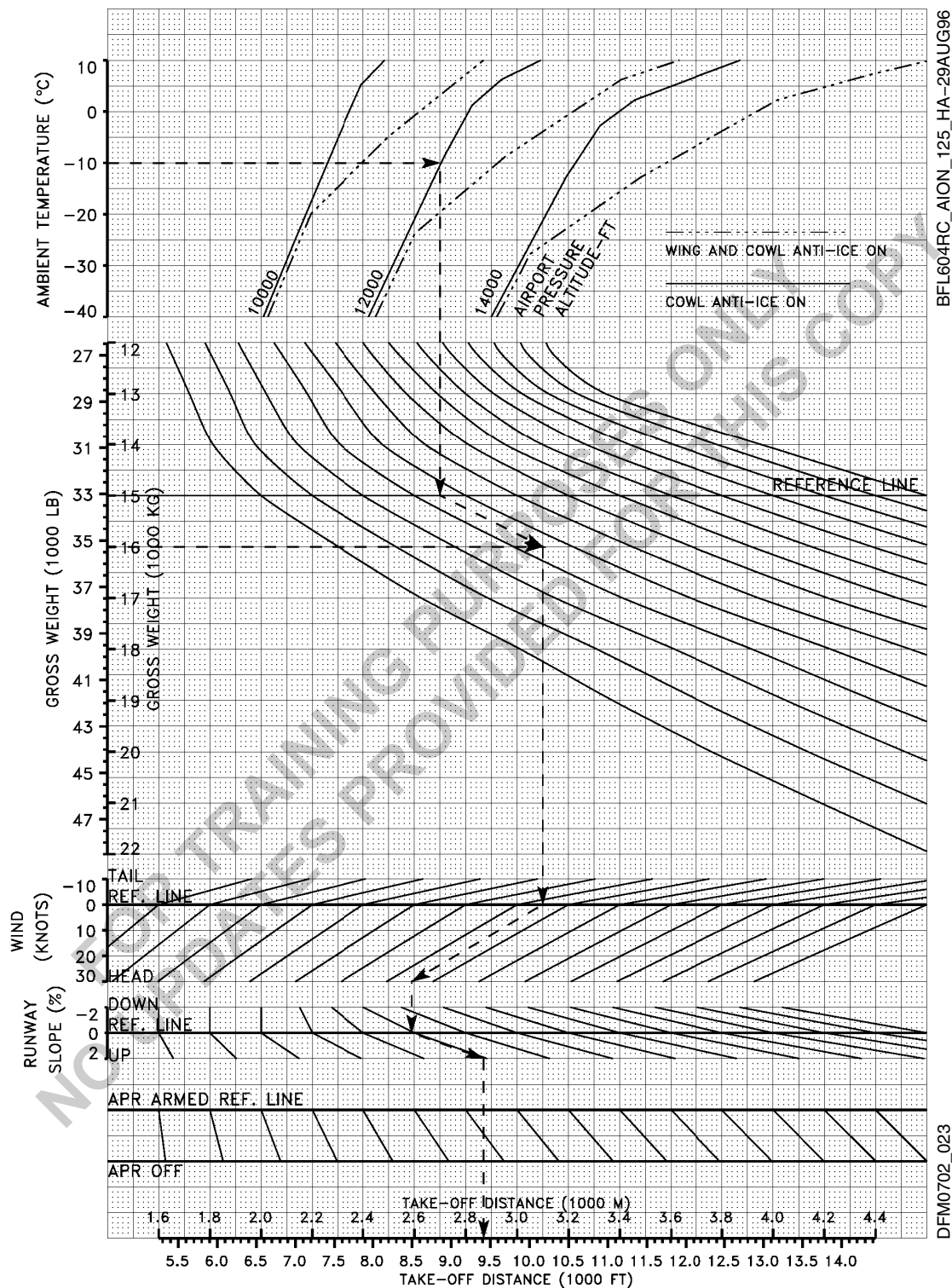
Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Slush (0.125") or Equivalent (Page 2 of 2)
 Figure 07-02-13

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


V1 Speed (Flaps 20°) – Anti-Icing Off – Runway Contaminated with Slush (0.125") or Equivalent
 Figure 07-02-14

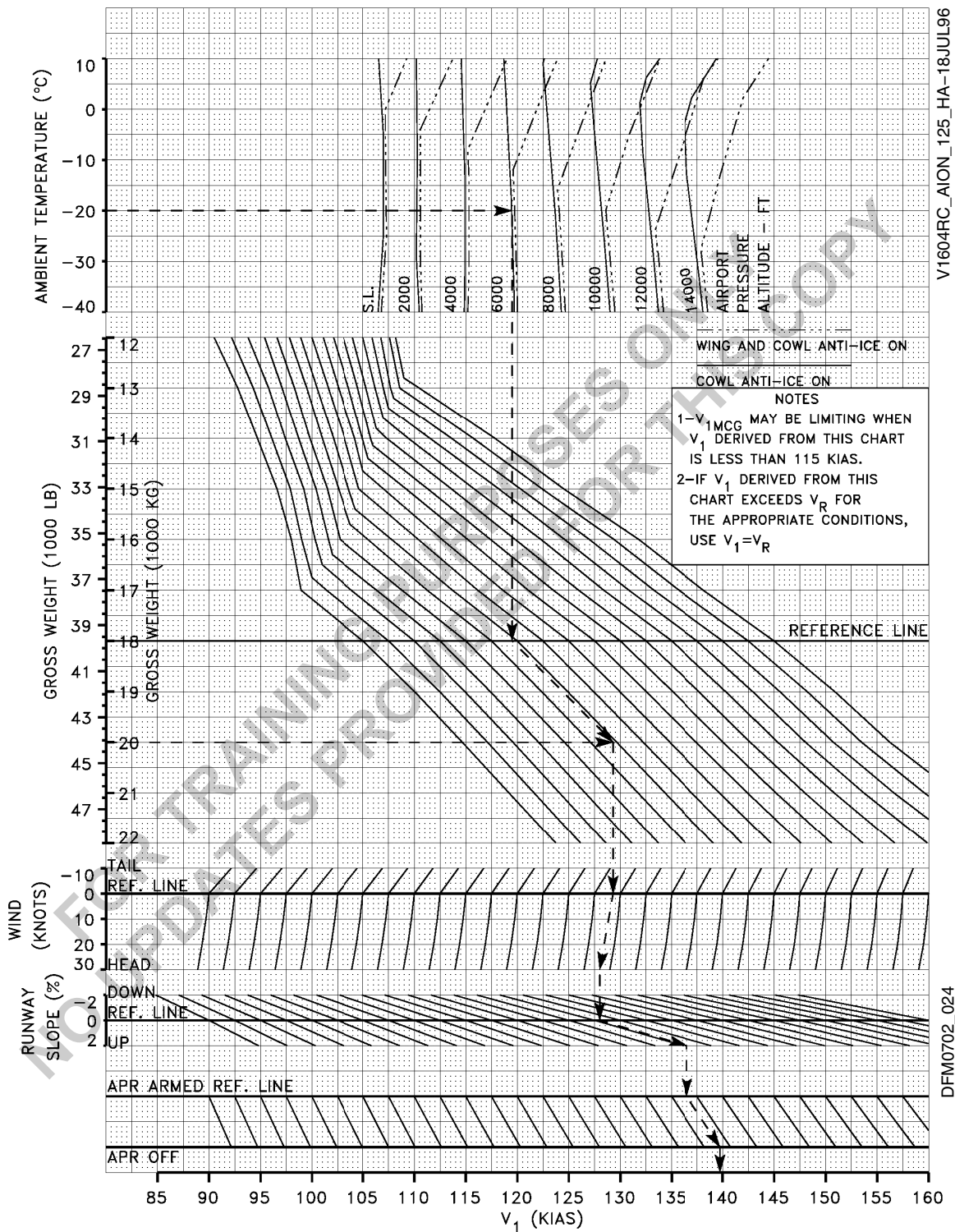
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Slush (0.125") or Equivalent (Page 1 of 2)
 Figure 07-02-15

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Slush (0.125") or Equivalent (Page 2 of 2)
 Figure 07-02-15

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)



V1 Speed (Flaps 20°) – Anti-Icing On – Runway Contaminated with Slush (0.125") or Equivalent
 Figure 07-02-16

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3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)

E. Take-Off on a Runway Contaminated with Slush (0.25 in) or Equivalent

NOTE

When dispatching with one thrust reverser inoperative on a runway contaminated with slush (0.25") or its equivalent, the following corrections must be applied:

- Increase take-off distance by 600 meters (1,970 feet);
- Reduce V_1 by 3.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

(1) ANTI-ICING OFF

The take-off distance for a flaps-20° take-off with anti-icing off, on runways contaminated with 6.4 mm (0.25 in) of slush or equivalent [6.4 mm (0.25 in) of standing water; or 12.7 mm (0.50 in) of wet snow; or 30.5 mm (1.2 in) of dry snow], is given by Figure 07-02-17.

The V_1 speed for a flaps-20° take-off with anti-icing off, on runways contaminated with 6.4 mm (0.25 in) of slush or equivalent [6.4 mm (0.25 in) of standing water; or 12.7 mm (0.50 in) of wet snow; or 30.5 mm (1.2 in) of dry snow], is given by Figure 07-02-18.

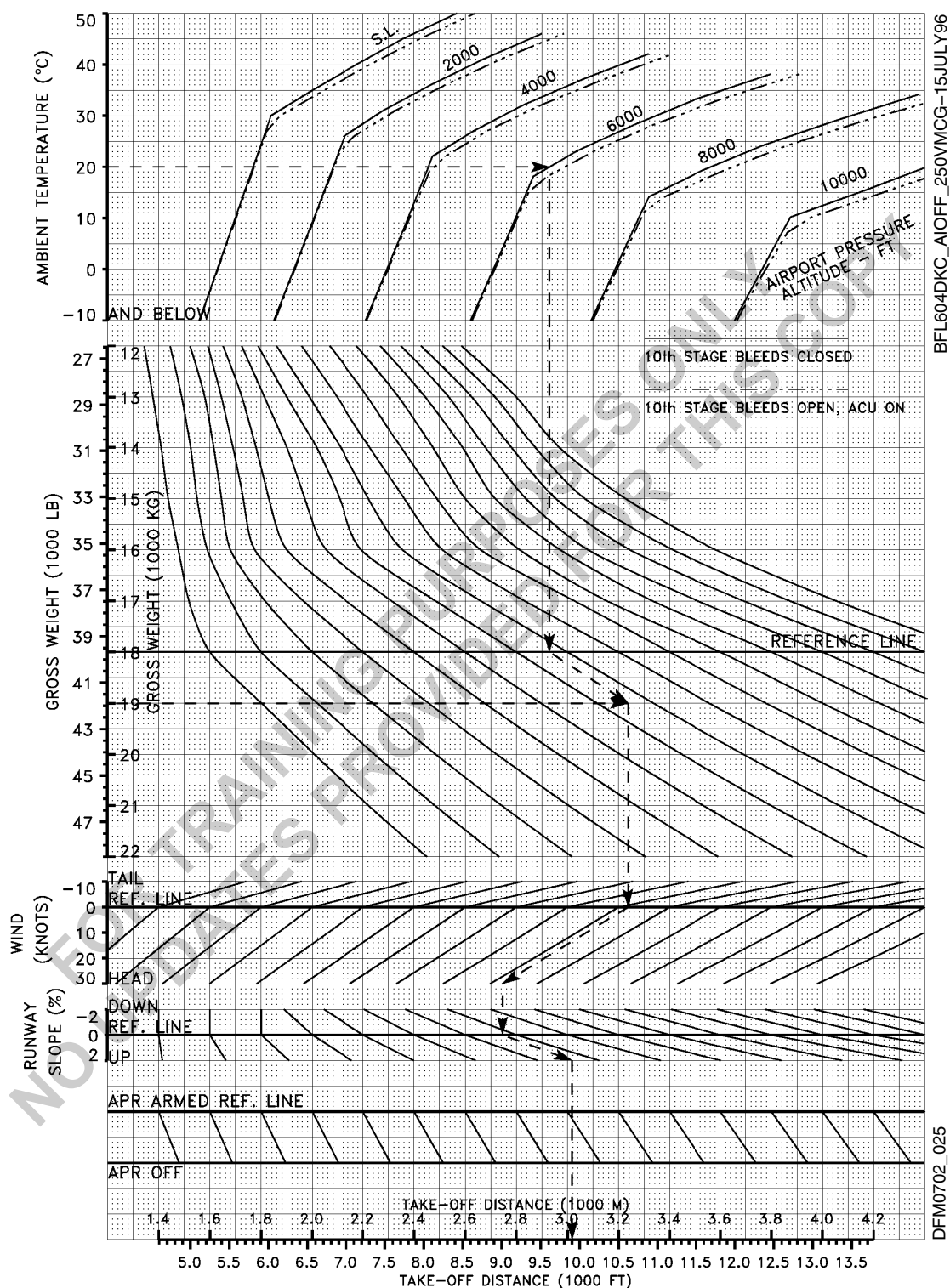
(2) ANTI-ICING ON

The take-off distance for a flaps-20° take-off with anti-icing on, on runways contaminated with 6.4 mm (0.25 in) of slush or equivalent [6.4 mm (0.25 in) of standing water; or 12.7 mm (0.50 in) of wet snow; or 30.5 mm (1.2 in) of dry snow], is given by Figure 07-02-19.

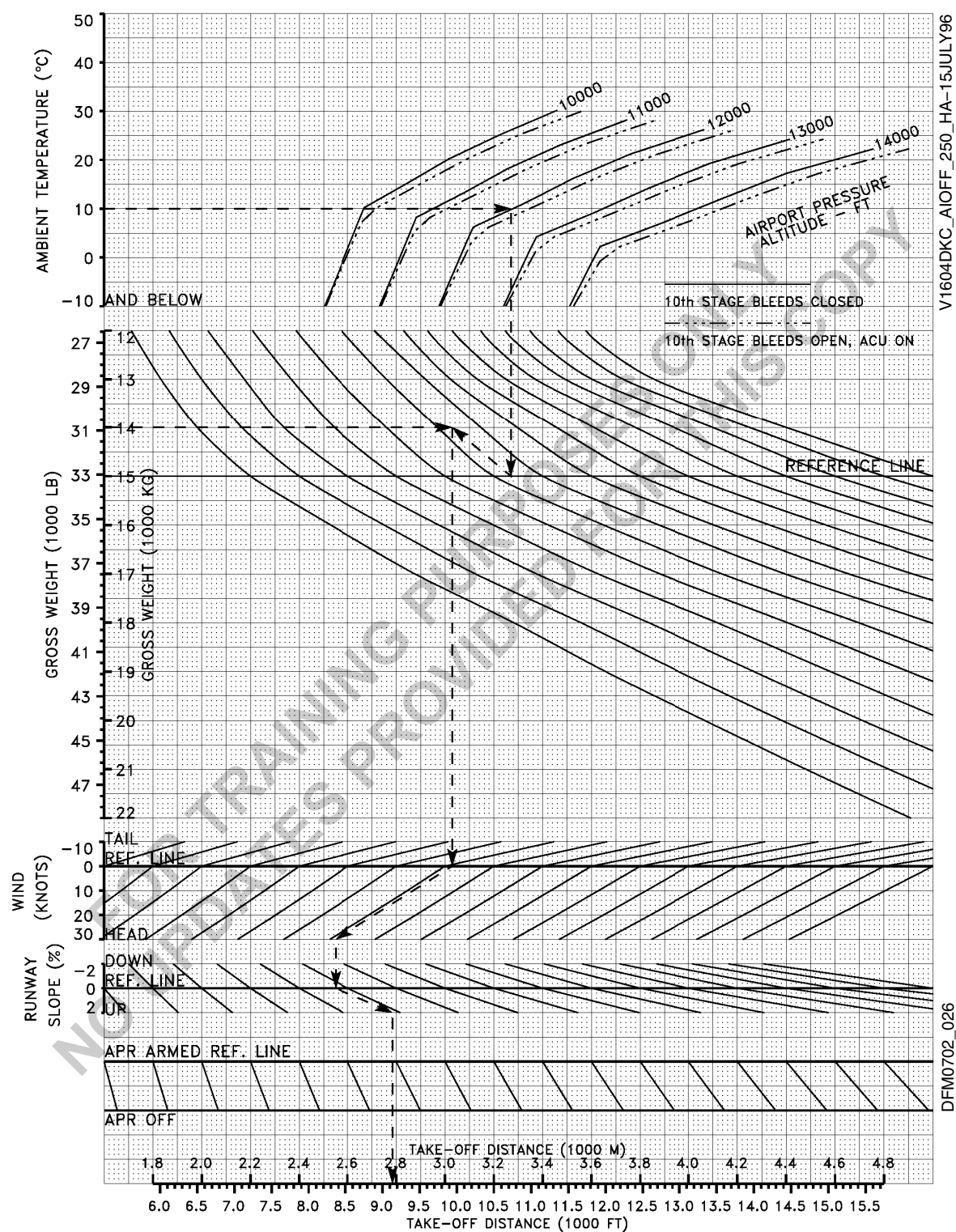
The V_1 speed for a flaps-20° take-off with anti-icing on, on runways contaminated with 6.4 mm (0.25 in) of slush or equivalent [6.4 mm (0.25 in) of standing water; or 12.7 mm (0.50 in) of wet snow; or 30.5 mm (1.2 in) of dry snow], is given by Figure 07-02-20.

The charts are utilized in the same manner as described in the previous examples for wet runways.

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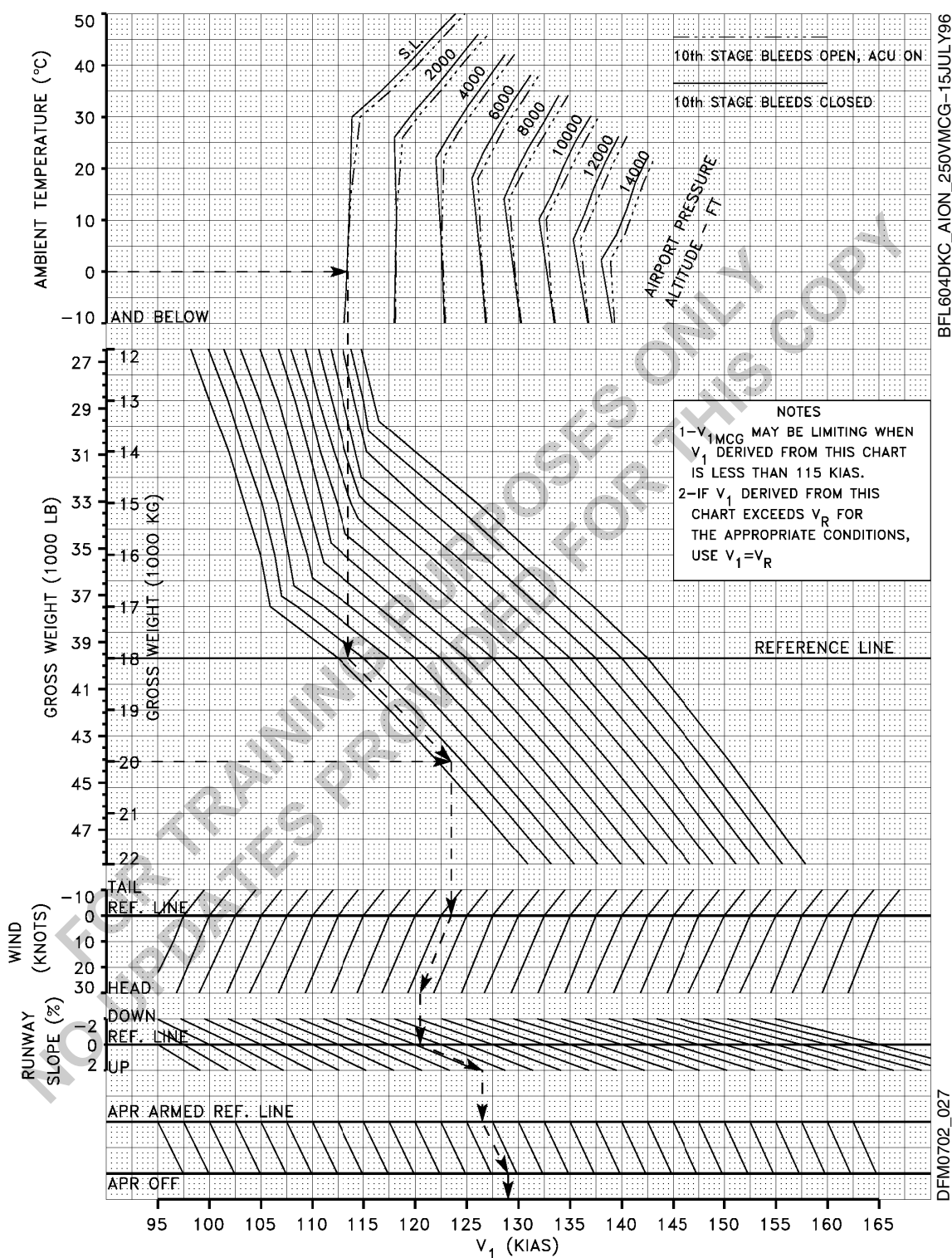
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Slush (0.25") or Equivalent (Page 1 of 2)
 Figure 07-02-17

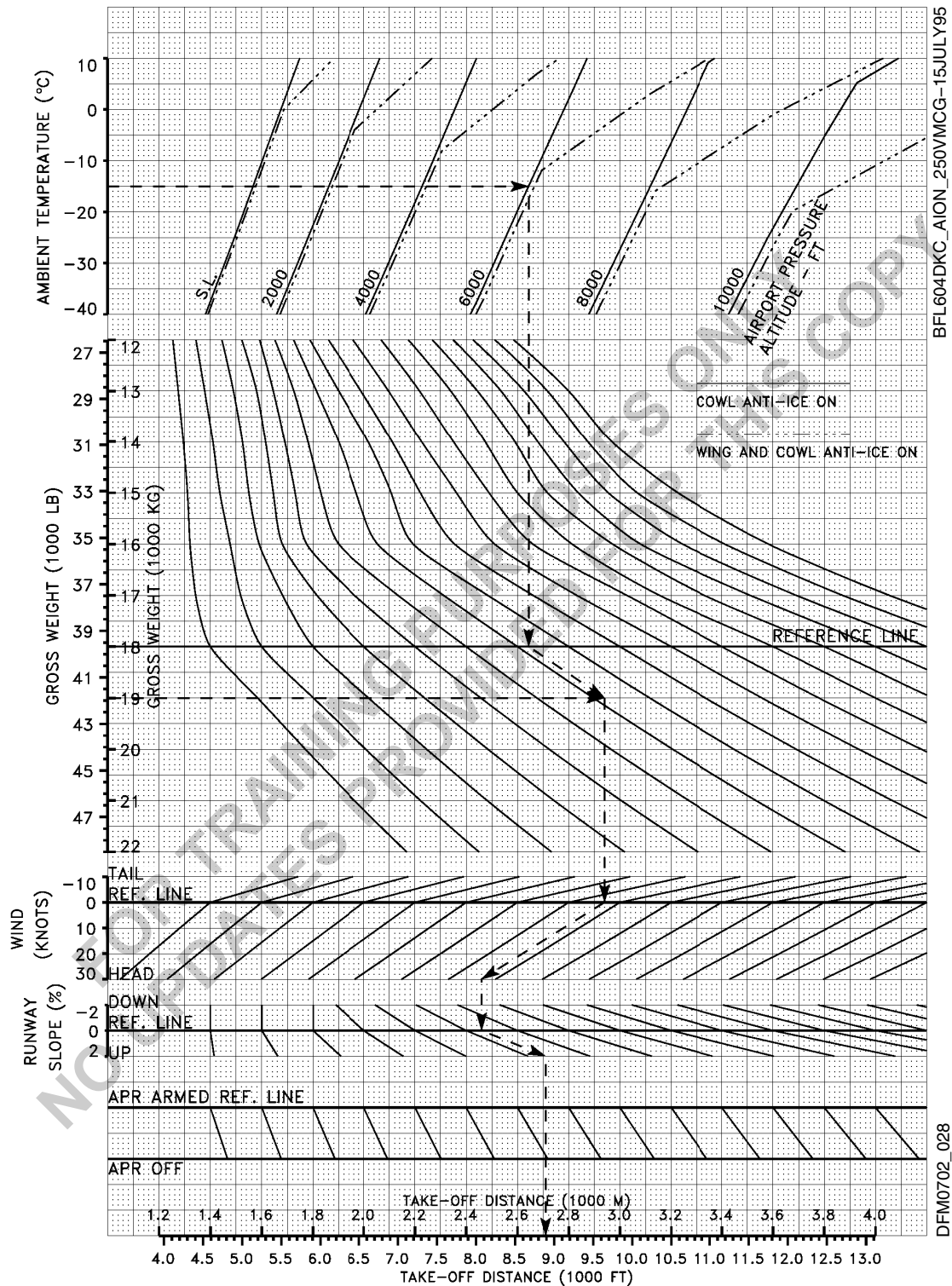
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing Off – Runway Contaminated with Slush (0.25") or Equivalent (Page 2 of 2)
 Figure 07-02-17

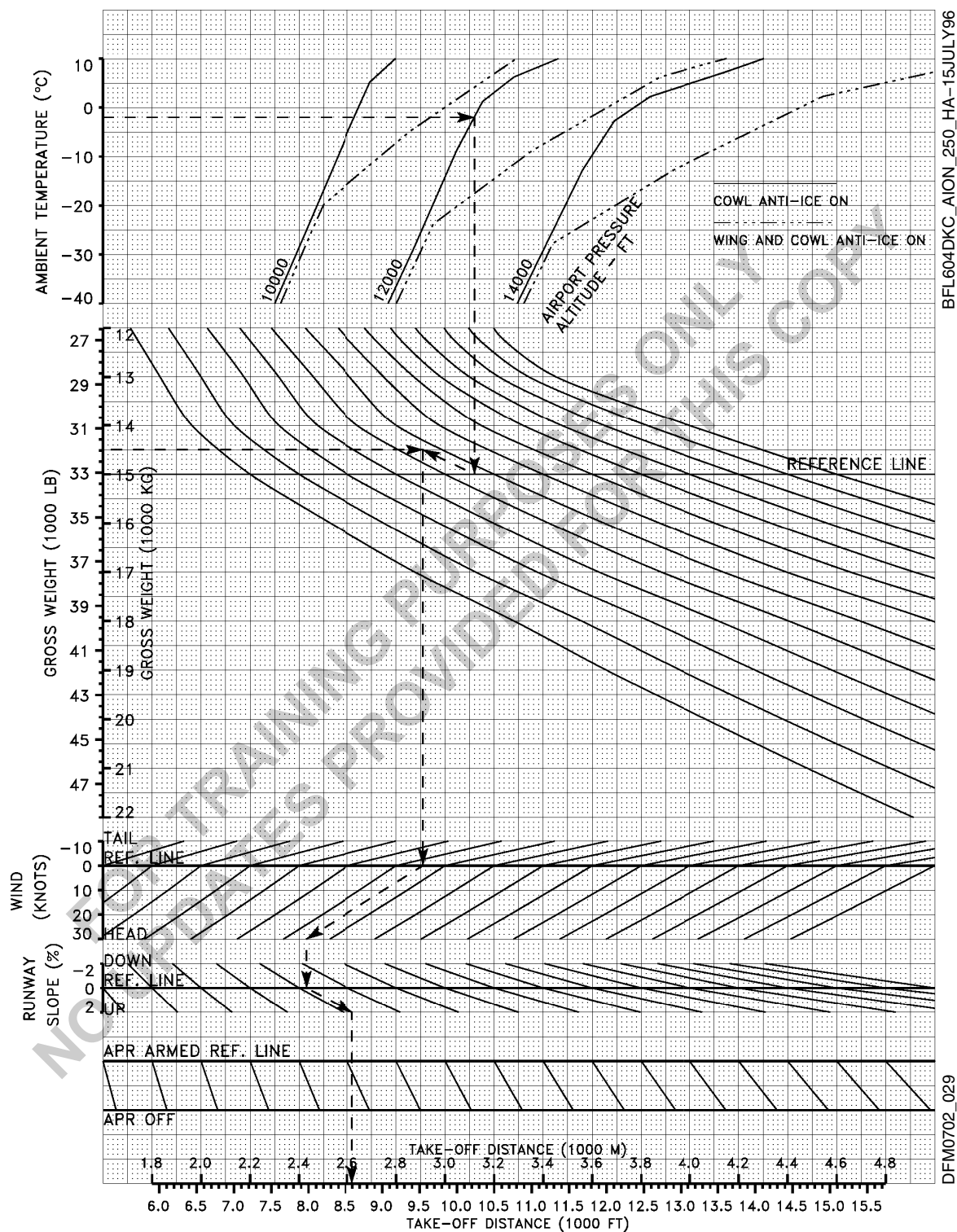
3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)



V₁ Speed (Flaps 20°) – Anti-Icing Off – Runway Contaminated with Slush (0.25") or Equivalent
 Figure 07-02-18

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


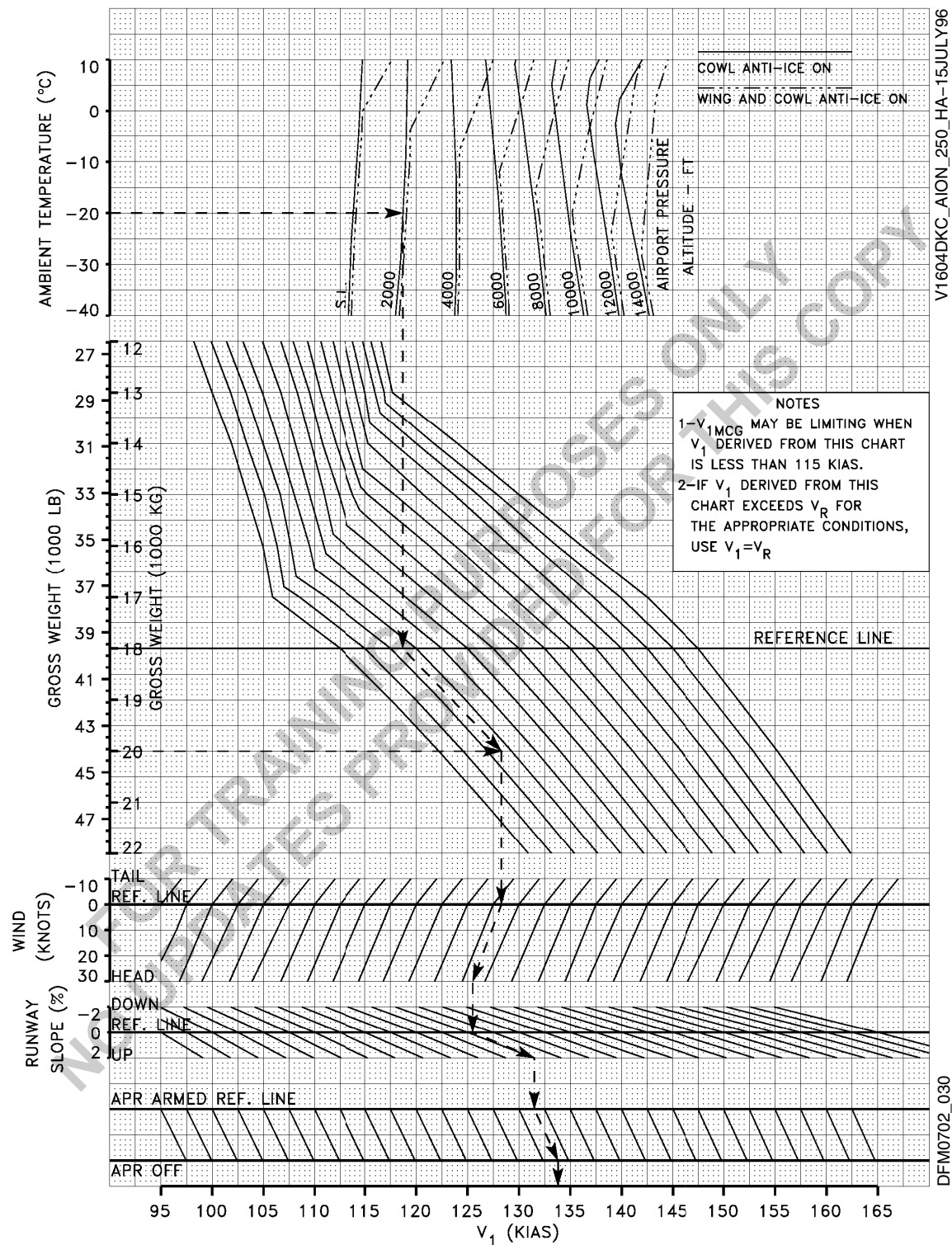
Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Slush (0.25") or Equivalent (Page 1 of 2)
 Figure 07-02-19

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


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Take-Off Distance, Flaps 20° – Anti-Icing On – Runway Contaminated with Slush (0.25") or Equivalent (Page 2 of 2)
 Figure 07-02-19

3. CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE (CONT'D)


V1 Speed (Flaps 20°) – Anti-Icing On – Runway Contaminated with Slush (0.25") or Equivalent
 Figure 07-02-20

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4. LANDING ON CONTAMINATED RUNWAYS

A. General

Landing performance data are calculated based on the assumption that maximum available reverse thrust is used down to a speed of 60 KIAS, and that 60% N_1 reverse thrust is used below 60 KIAS.

NOTE

If the landing field length on a dry runway is greater than the landing distance on the associated contaminated runway, use the landing field length.

B. Actual Landing Distance on Contaminated Runways – Flaps 45°

The actual landing distance on contaminated runways, for a flaps-45° landing, with both engines operating, for speeds at the runway threshold from V_{REF} to $V_{REF} + 10$ KIAS, are presented in the following charts:

- Figure 07-02-21 provides the actual landing distance for a runway contaminated with ice;
- Figure 07-02-22 provides the actual landing distance for a runway contaminated with compacted snow;
- Figure 07-02-23 provides the actual landing distance for a runway contaminated with loose snow; and
- Figure 07-02-24 provides the actual landing distance for a runway contaminated with slush/standing water.

Example:

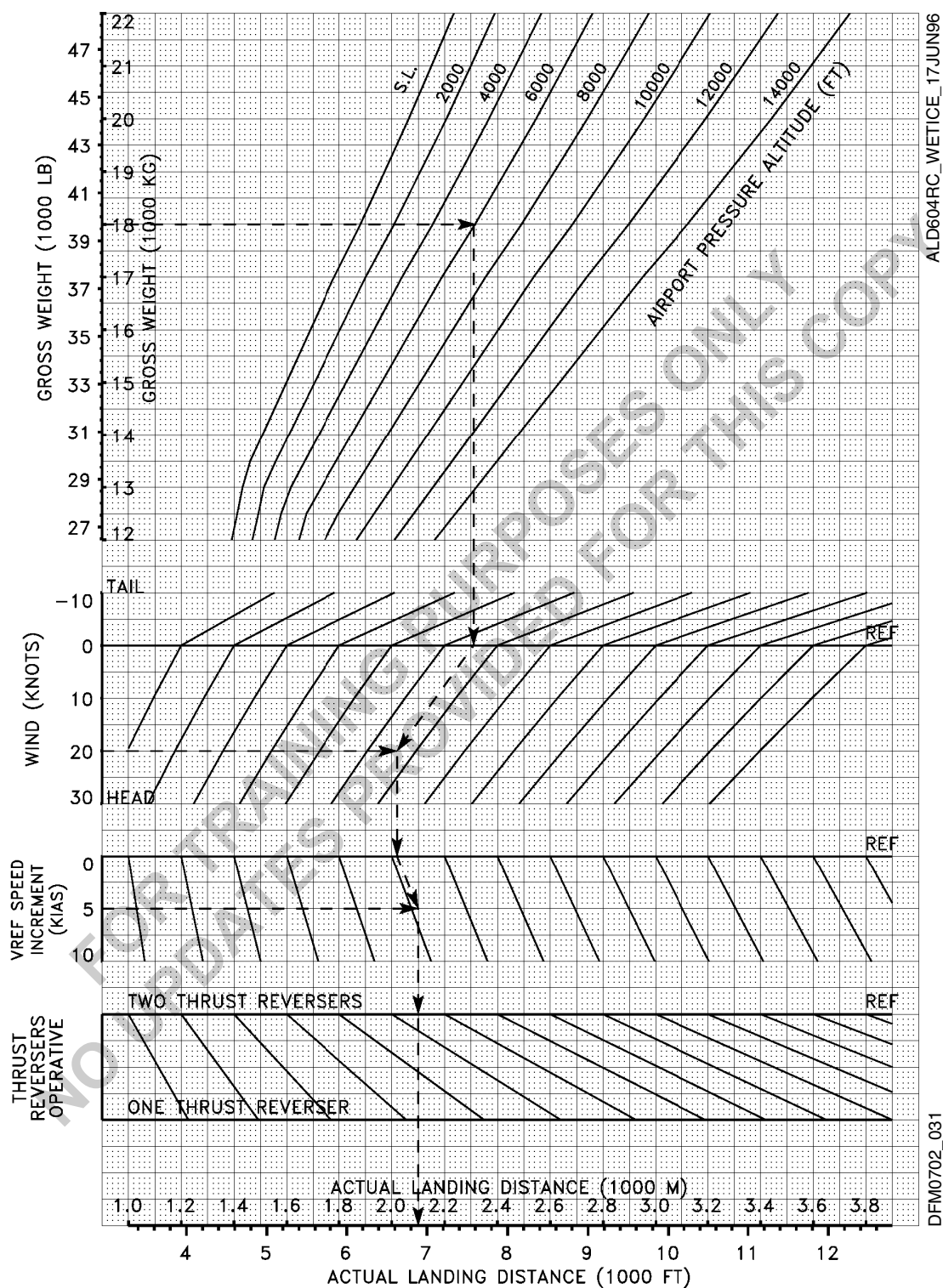
Associated Conditions:

Contaminant	= Ice
Airplane gross weight	= 18,000 kg (39,682 lb)
Pressure altitude	= 6,000 feet
Wind condition	= 22 knots at 26 degrees relative to runway (20 knots headwind; 10 knots crosswind)
ΔV_{REF}	= +5 KIAS
Thrust reversers	= Both operating

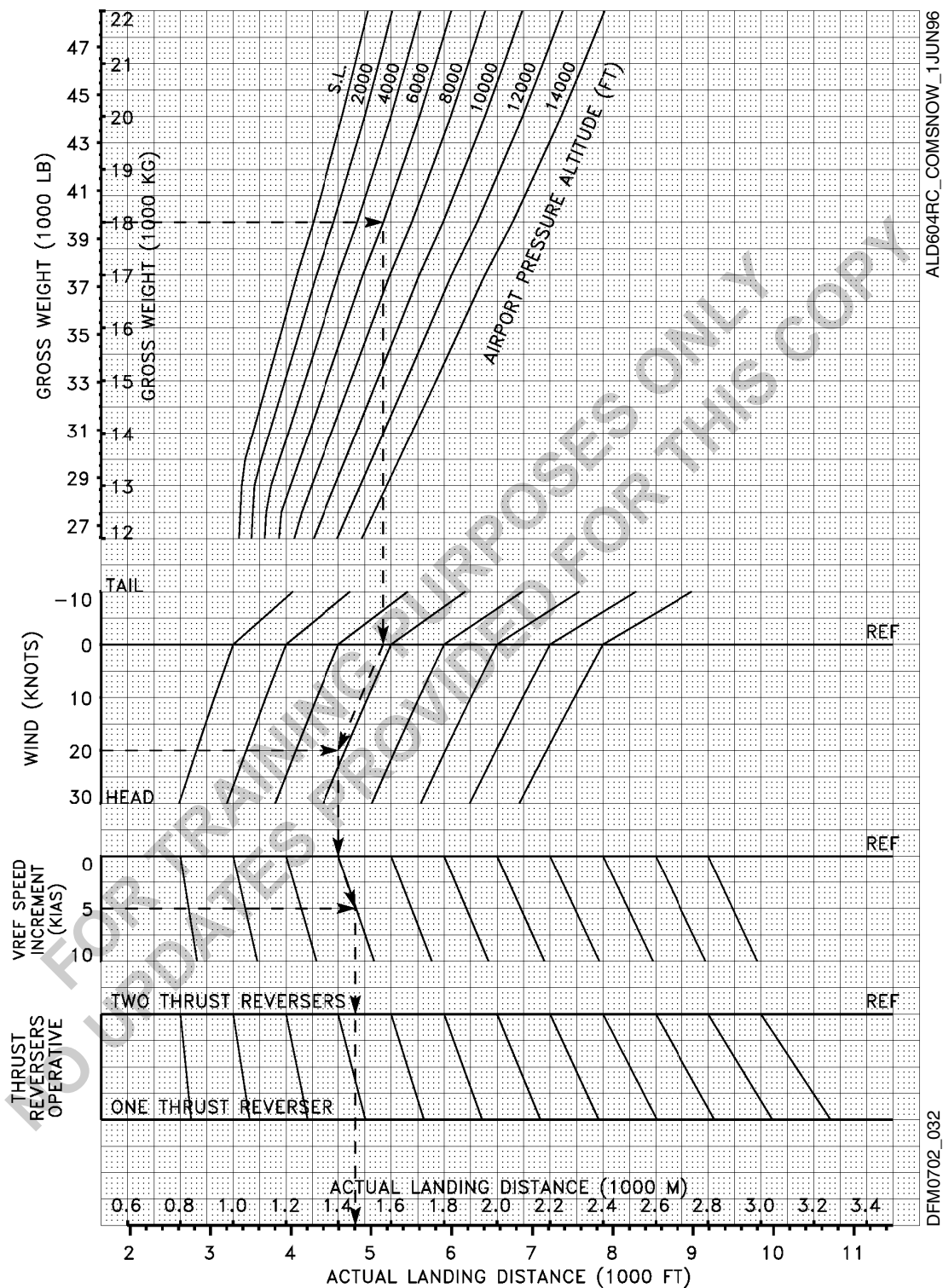
From the associated conditions stated above, the actual landing distance on a runway contaminated with ice, as shown in the example on Figure 07-02-21, is found to be 2,100 meters (6,890 feet).

The other charts are utilized in the same manner as the above example.

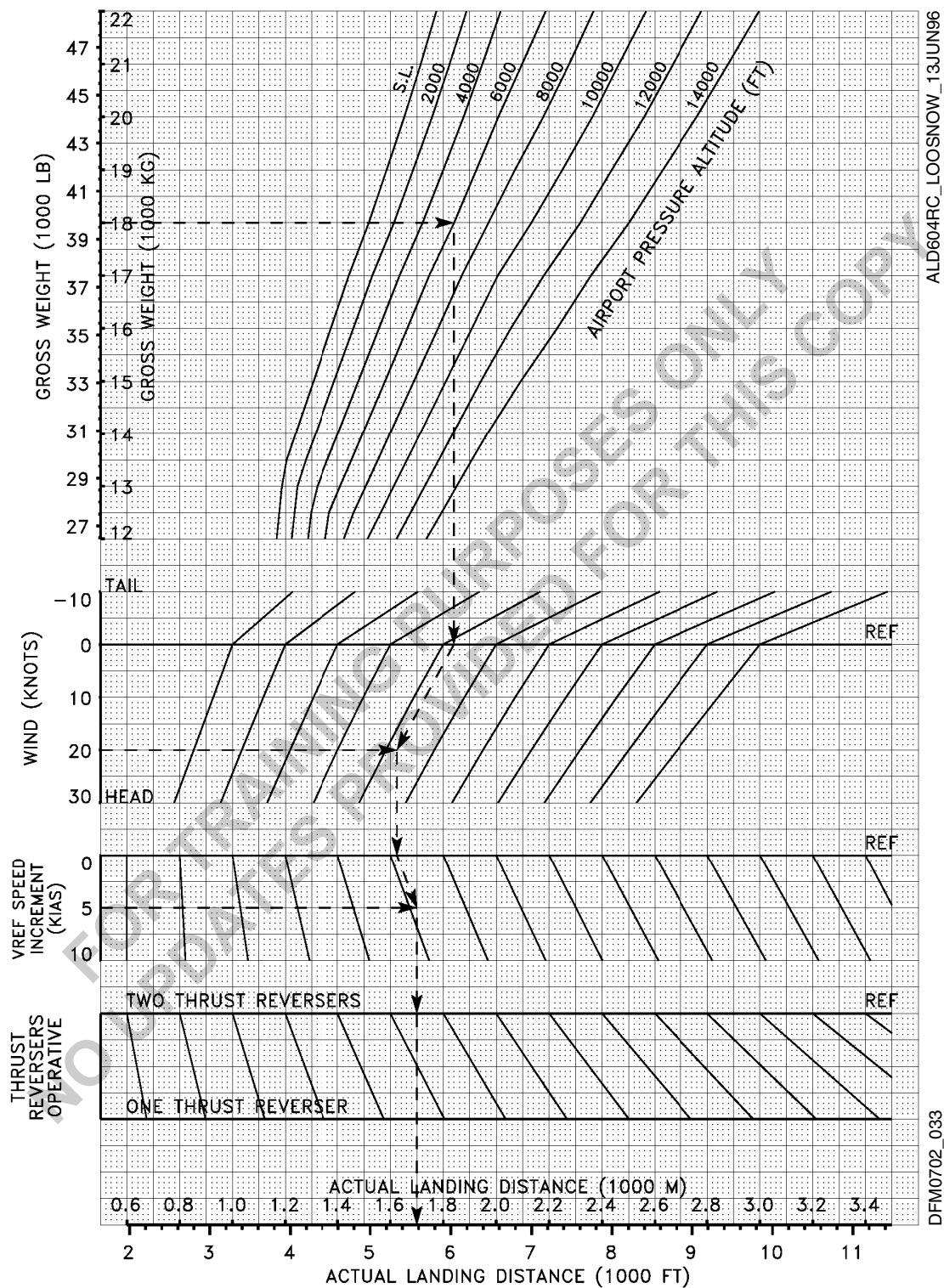
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4. LANDING ON CONTAMINATED RUNWAYS (CONT'D)


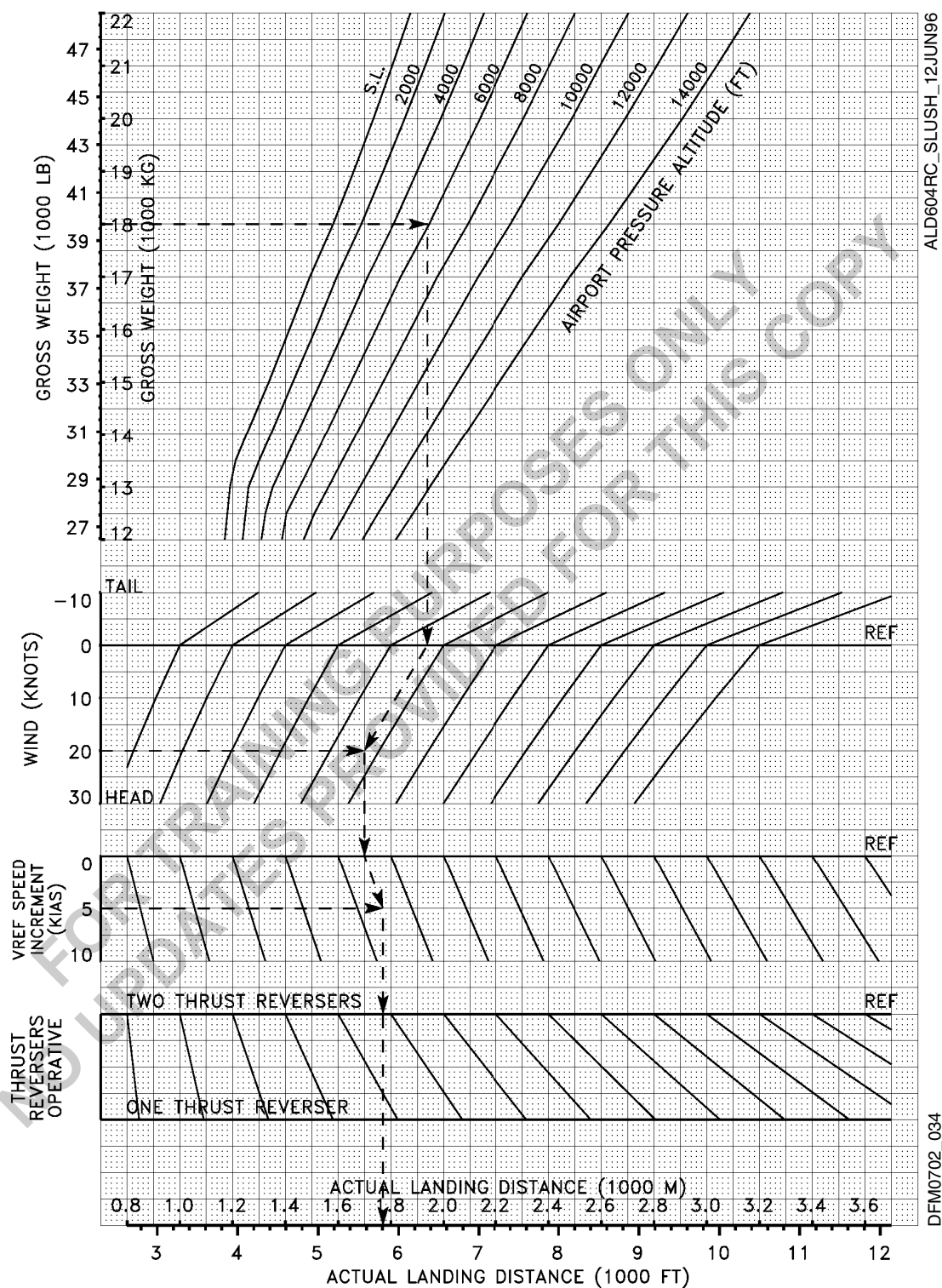
Actual Landing Distance on Runways Contaminated With Ice – Flaps 45°
 Figure 07-02-21

4. LANDING ON CONTAMINATED RUNWAYS (CONT'D)


Actual Landing Distance on Runways Contaminated With Compacted Snow – Flaps 45°
 Figure 07-02-22

4. LANDING ON CONTAMINATED RUNWAYS (CONT'D)


Actual Landing Distance on Runways Contaminated With Loose Snow – Flaps 45°
 Figure 07-02-23

4. LANDING ON CONTAMINATED RUNWAYS (CONT'D)


Actual Landing Distance on Runways Contaminated With Slush/Standing Water – Flaps 45°
 Figure 07-02-24

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		Rev 13, Aug 23/18

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CHALLENGER 650	SUPPLEMENT 3 Operation at Airport Elevations up to 14,000 ft.	07-03-01-1
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1. INTRODUCTION

The data given in this Supplement must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) when operating from airport pressure altitudes above 10,000 feet up to 14,000 feet. These data complement or supersede some data contained in the basic Airplane Flight Manual. This Supplement must, therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given below:

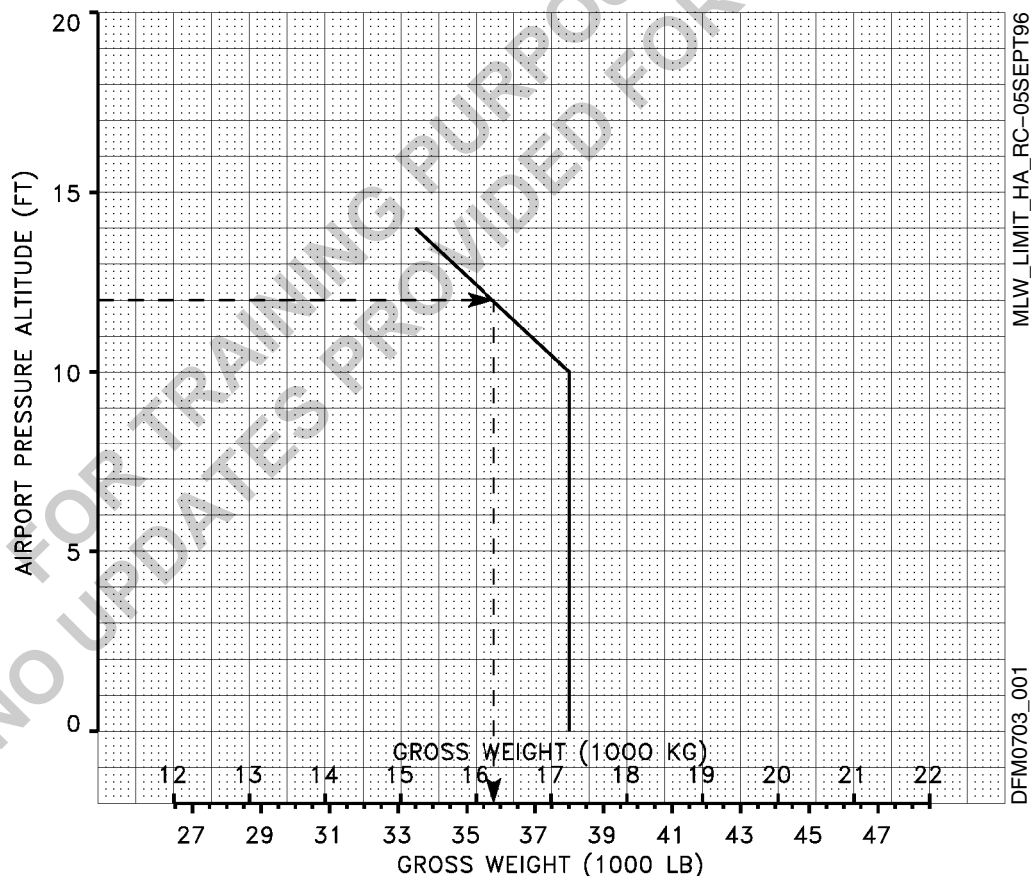
2. GENERAL

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations established in Chapter 2 of the basic Airplane Flight Manual are applicable, except as modified by the following:

A. The maximum landing weight limit is shown in Figure 07-03-1.



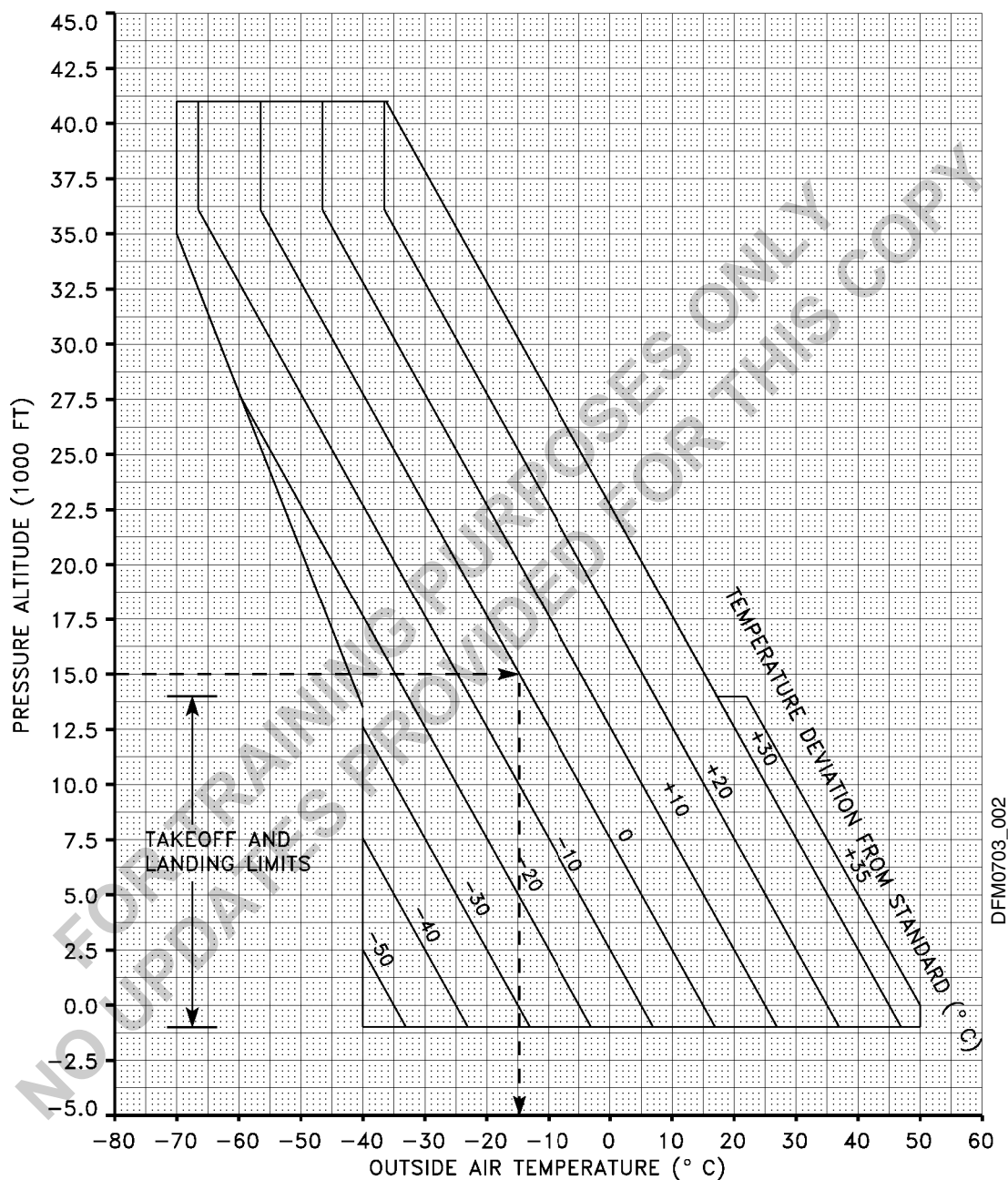
Maximum Landing Weight Limit
Figure 07-03-1

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3. LIMITATIONS (CONT'D)

B. The maximum altitude and temperature operating limits are shown in Figure 07-03-2.

- Maximum airport pressure altitude for take-off and landing is 14,000 feet.



Altitude and Temperature Operating Limits
Figure 07-03-2

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures established in Chapter 4 of the basic Airplane Flight Manual are applicable except as modified by the following:

A. Engine Start

The engine start procedures in the basic AFM are applicable in addition to the following:

NOTE

If difficulty is experienced in starting a cold engine using the APU, due to slow acceleration, start the other engine and perform a crossbleed start on the affected engine. If required, repeat sequence to obtain a successful start, and perform a crossbleed start for the opposite engine.

B. Climb/Descent Rate

With air-conditioning packs selected on and cabin pressurization rate set at PIP position, the minimum time required to climb to 25,000 feet or to descend from 25,000 feet without causing discomfort to passengers is as follows:

AIRFIELD ELEVATION	MINIMUM TIME
14,000 feet	10.0 minutes
13,000 feet	8.4 minutes
12,000 feet	6.7 minutes
11,000 feet	5.0 minutes
10,000 feet	3.4 minutes

NOTE

- During take-off, the minimum time may be reduced by selecting a higher cabin pressurization rate.
- During climb, do not exceed 25,000 feet until cabin pressure altitude has decreased to 8,000 feet or less.

C. Take-Off

Before take-off:

(1) CABIN PRESSURIZATION, LDG

ALT selector Set the following:

- Destination airfield altitude,
- Desired cabin pressure rate of change, and
- Barometric pressure.

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5. NORMAL PROCEDURES (CONT'D)

C. Take-Off (Cont'd)

- (2) Oxygen masks..... Don Crew will don oxygen masks and establish communications.

NOTE

1. The use of crew oxygen is required in accordance with the applicable operating regulations.
2. Carry out a high crosswind take-off procedure if there is an audible engine noise accompanied by engine indication instabilities when setting take-off power.

When cabin altitude is below 10,000 feet:

- (3) Oxygen masks..... Remove

D. Descent

- (1) CABIN PRESSURIZATION, LDG
ALT selector Check selected to the following:
- Destination airfield altitude,
 - Destination airfield barometric pressure.
 - Desired cabin pressure descent rate.
 - CABIN ALT WARN HI advisory message on.
- Check FAULT lamp out.

E. Landing

When cabin altitude is above 10,000 feet:

- (1) Oxygen masks..... Don Crew will don oxygen masks and establish communications.

NOTE

The use of crew oxygen is required in accordance with the applicable operating regulations.

Prior to landing:

- (2) Cabin altitudeCheck

Prior to opening passenger door:

- (3) Cabin differential pressure Ensure at zero.

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CHALLENGER 650	SUPPLEMENT 3 Operation at Airport Elevations up to 14,000 ft.	07-03-01-5
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6. ABNORMAL PROCEDURES

The abnormal procedures established in Chapter 5 of the basic Airplane Flight Manual are applicable except as modified by the following:

A. Flaps Failure

Operation at high airport elevations increases the probability of exceeding the maximum tire speed. The landing weight, resulting in a touchdown speed equal to the tire speed of 182 knots ground speed, for varying conditions of airport pressure altitude, wind and flap configurations, is given in Figure 07-03-3, Figure 07-03-4 and Figure 07-03-5. Landings at weights exceeding the weights derived in Figure 07-03-3, Figure 07-03-4 and Figure 07-03-5 will result in a tire speed greater than 182 knots ground speed at touchdown.

Example 1: FIGURE 07-03-3, FLAPS 0° (APPROACH SPEED = $V_{REF} + 30$ KIAS)

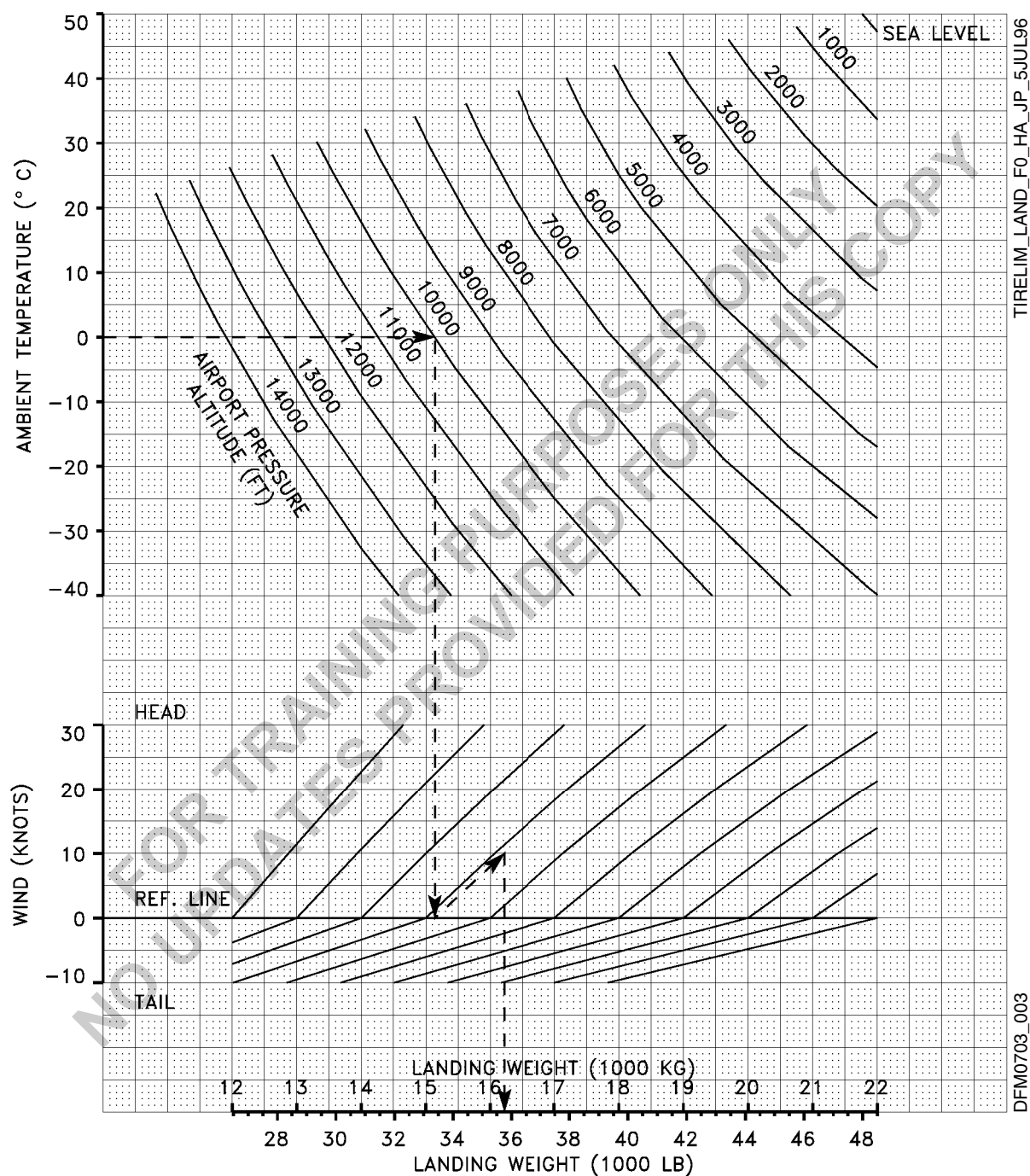
Associated Conditions:

Temperature	= 0°C
Airport pressure altitude	= 10,000 ft
Wind	= 10 knots, headwind

The example in Figure 07-03-3, shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 0° is 16,200 kg (35,715 lb).

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6. ABNORMAL PROCEDURES (CONT'D)

 APPROACH SPEED = $V_{REF} + 30$ KIAS

 Landing Weight Limited by Maximum Tire Speed, Flaps 0°
 Figure 07-03-3

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6. ABNORMAL PROCEDURES (CONT'D)

Example 2: FIGURE 07-03-4, FLAPS 20° (APPROACH SPEED = $V_{REF} + 14$ KIAS)

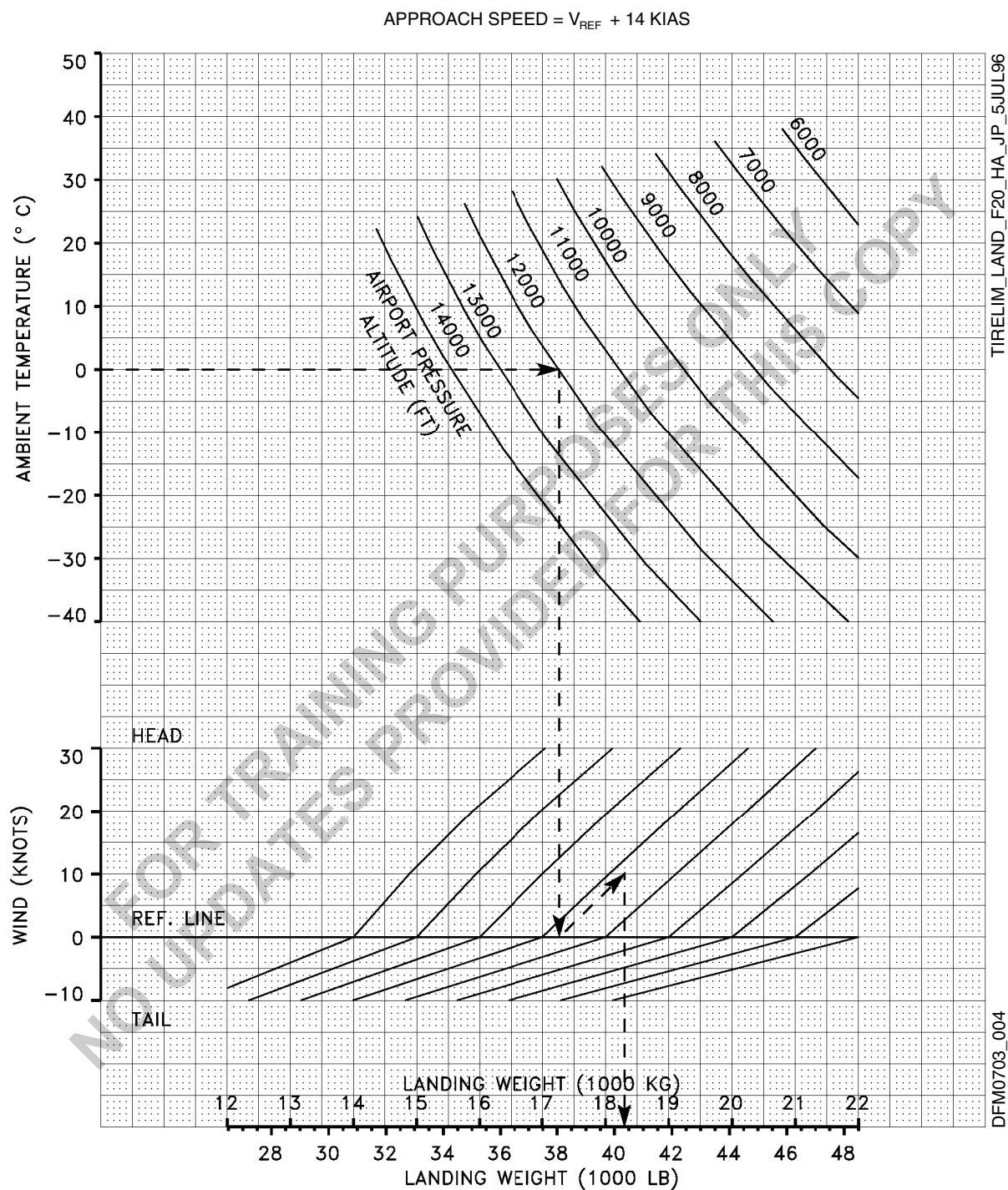
Associated Conditions:

Temperature	= 0°C
Airport pressure altitude	= 12,000 ft
Wind	= 10 knots, headwind

The example in Figure 07-03-4, shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 20° is 18,300 kg (40,345 lb).

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6. ABNORMAL PROCEDURES (CONT'D)


Landing Weight Limited by Maximum Tire Speed, Flaps 20°
 Figure 07-03-4

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6. ABNORMAL PROCEDURES (CONT'D)

Example 3: FIGURE 07-03-5, FLAPS 30° (APPROACH SPEED = $V_{REF} + 7$ KIAS)

Associated Conditions:

Temperature	= 0°C
Airport pressure altitude	= 13,000 ft
Wind	= 10 knots, headwind

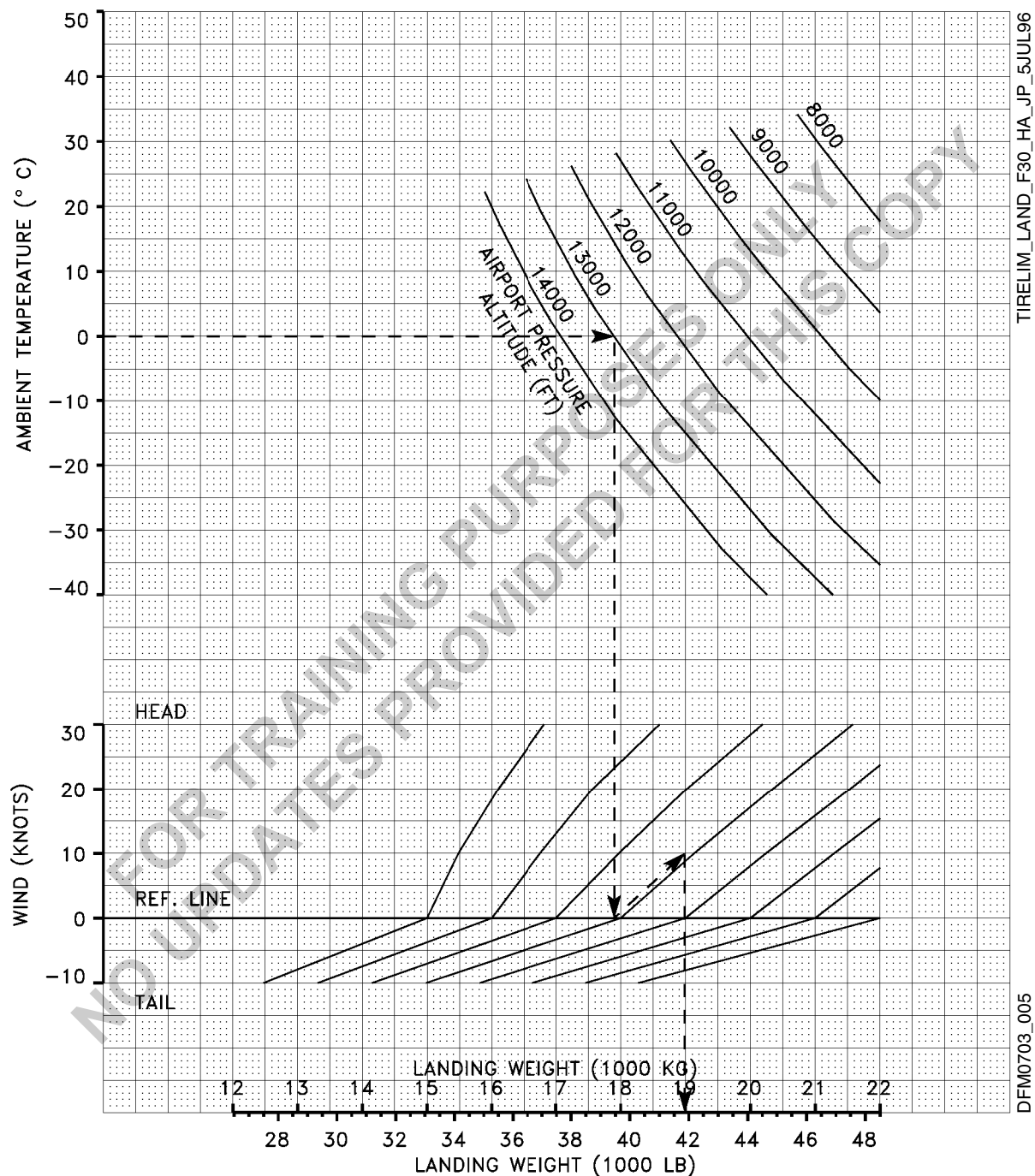
The example in Figure 07-03-5, shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 30° is 19,000 kg (41,888 lb).

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6. ABNORMAL PROCEDURES (CONT'D)

APPROACH SPEED = $V_{REF} + 7$ KIAS



Landing Weight Limited by Maximum Tire Speed, Flaps 30°
 Figure 07-03-5

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		Rev 13, Aug 23/18

7. PERFORMANCE

The performance data in Chapter 6 are applicable.

8. SUPPLEMENTS

The supplements in Chapter 7 are applicable except as specified by the following:

A. Supplement 13 – Additional Take-Off Performance Information

Use of Supplement 13 is prohibited.

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CHALLENGER ⁶⁵⁰	SUPPLEMENT 3 Operation at Airport Elevations up to 14,000 ft.	07-03-01-12
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CHALLENGER 650	SUPPLEMENT 4 Category II Operations	07-04-01-1
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NOTE

This Airplane Flight Manual Supplement does not constitute approval to conduct Category II operations.

1. GENERAL

The CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) has been shown to meet the airworthiness requirements for Category II Operations contained in Appendix 1 of AC 120-29.

The data in this supplement must be used when conducting Category II operations.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must, therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

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3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

- An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the Category II Required Equipment List below, and their ground installations, are operating satisfactorily.

CATEGORY II REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
VHF NAV 1 AND 2	BOTH MUST BE OPERATIONAL AND DISPLAYED ONSIDE.
PFD 1 AND 2	ONE (1) PFD AVAILABLE AND OPERATIONAL FOR EACH SIDE.
STAB CHANNEL 1 AND 2	ONE (1) CHANNEL MUST BE OPERATIONAL.
RADIO ALTIMETER	ONE MUST BE OPERATIONAL, WITH DISPLAY ON BOTH SIDES.
IRS	ANY TWO (2) MUST BE OPERATIONAL.
ADC 1 AND 2	BOTH MUST BE OPERATIONAL.
FD 1 AND 2	BOTH MUST BE OPERATIONAL.
EFIS COMPARATOR MONITORS	MUST BE OPERATIONAL.
AFCS PITCH TRIM	MUST BE OPERATIONAL.
AUTOPILOT	MUST BE OPERATIONAL.
HYDRAULICS (3)	ALL SYSTEMS MUST BE ON AND OPERATIONAL.
ELECTRICS	TWO (2) GENERATORS ON AND SHARING LOAD (SEE NOTE).

NOTE

Approach to Category II minima may be continued with one generator, if an engine fails during final approach (refer to ABNORMAL PROCEDURES).

- Operation of the autopilot is prohibited below 80 feet AGL.
- The approved flap configurations for approach are;
 - 45° for all engines operating, and
 - 20° for one engine inoperative.
- The minimum decision height for Category II operations is 100 feet AGL.
- Wind limits for autopilot approaches are;
 - 8 knots crosswind, and
 - 9 knots tailwind.
- The maximum ILS glidepath angle for Category II operations is 3.5 degrees.
- ATS must be disengaged.

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4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or AFCS MSGS FAIL warning message, "CAVALRY CHARGE" aural and/or flashing AP (red) message on the primary flight displays.

Continue approach if the following can be accomplished above 500 feet AGL:

- (1) Autopilot Disconnect
- (2) AFCS SEL 1(2) switch (centre pedestal)..... Select to operative AFCS channel.
Check AFCS 1 (2) INOP status message on.
- (3) Autopilot Engage and monitor operation.
- (4) Approach Continue to Category II minima.

If below 500 feet AGL, on a stabilized approach:

- (1) Autopilot Disconnect
- (2) Manual control Resume
- (3) Approach Continue to Category I minima.

If below 200 feet AGL and non-visual:

- (1) Go-around Initiate

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. Prior to Approach:

Airplane must be properly configured prior to approach:

- (1) Landing weightCheck within limits. See Chapter 6; PERFORMANCE – LANDING PERFORMANCE.
- (2) NAV SOURCE..... Set to onside ILS (green) or armed (cyan) by FMS for both PFDs.
- (3) IRS and ADC source selectors..... Set to NORM.
- (4) RA MINIMUMS..... Set on both PFDs.
- (5) BARO MINIMUMS..... Set as a barometric DH, in case RA fails.
- (6) MIN ALERT Set to RA on both PFDs.
- (7) Runway Course Set on both PFDs, or armed (cyan) by FMS.
- (8) EFISCheck that no comparator flags are displayed.
- (9) CASCheck that the following messages are not displayed:
 - AFCS MSGS FAIL warning message,
 - AP PITCH TRIM, EFIS COMP INOP and EFIS MISCOMP caution messages,
 - FD 1 and 2 FAIL status messages.
- (10) Stabilizer and elevator positionCheck valid on EICAS or F/CTL synoptic page.
- (11) Approach mode Arm
- (12) Expanded lateral deviation indicatorVerify displayed on both PFDs at 600 feet AGL.

B. Before Landing:

- (1) AutopilotDisengage at an altitude not less than 80 feet AGL.

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6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Single Engine Approach and Landing

- (1) Approach Conduct under Category I operations.

B. Engine Failure During Final Approach

Continue approach if the following can be accomplished above 800 feet AGL:

- (1) Autopilot Disconnect
- (2) Operating engine Increase thrust as required.
- (3) FLIGHT SPOILER lever Ensure retracted.
- (4) FLAPS Select to 20° for landing.
- (5) Airspeed Increase to $V_{REF} + 14$ KIAS.
- (6) Airplane Retrim
- (7) Autopilot Re-engage
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a landing without the use of reverse thrust.
- (9) TAWS WARNING, FLAPS OFF switch/light Press in to mute flap aural warning.

At glideslope intercept:

- (10) Airplane Retrim directionally as required.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

If above 800 feet AGL, and the above procedure cannot be accomplished:

- (11) Approach Conduct under Category I operations.

If below 800 feet AGL and non-visual:

- (12) Go-Around Initiate

After landing or during go-around, at a safe altitude:

- (13) Engine shutdown procedure Accomplish If applicable, refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown procedure.

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6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures:

SYSTEM	INDICATION
VHF NAV 1 or 2	ILS receiver, on applicable side, is inoperative.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) NAV SOURCE Select X-SIDE. (2) Approach Continue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side NAV: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side NAV: (1) Go-around Initiate
SYSTEM	INDICATION
PFD 1 or 2	Primary flight display, on applicable side, goes blank.
	ACTION
	Failure during approach: (1) Display reversionary panel selector switch Select to MFD REV. Applicable MFD defaults to primary flight display.
SYSTEM	INDICATION
STAB CH 1 or 2	Horizontal stabilizer trim, on applicable side, is inoperative. STAB CH 1 (2) INOP status message, displayed on EICAS.
	ACTION
	Single channel failure: (1) Approach Continue Dual channel failure (STAB TRIM caution message on: autopilot may disengage): (1) Go-around Initiate Dual channel failure during final approach (in stabilized trim condition): (1) Approach Continue to Category I minima.

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6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
RADIO ALTIMETER	False or missing radio altitude on both PFDs, or red or amber RA indication on both PFDs.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate

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6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
IRS	Erratic PFD data or ATT annunciation, and HDG/ROL/PIT annunciation on PFDs.
	ACTION
	First failure during the approach (AP will disconnect): If above 500 feet AGL, on a stabilized approach: (1) Manual controlResume (2) IRS sourceSelect an operative IRS. (if IRS 3 is installed, AP will be available). (3) ApproachContinue to Category I minima, if only one IRS available. If below 500 feet AGL and non-visual: (1) Manual controlResume (2) Go-aroundInitiate
SYSTEM	INDICATION
EFIS Comparator	EFIS COMP INOP (EFIS comparator failure) caution message or EFIS MISCOMP caution message displayed on EICAS and any of the ALT/HDG/IAS/PIT/ROL/LOC/GS annunciations on PFDs.
	ACTION
	Failure during approach: (1) Go-aroundInitiate (2) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – INSTRUMENTS SYSTEM.

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
ADC 1 or 2	Erratic PFD data or ADC FAIL annunciation, and ALT/IAS annunciation on PFDs.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) ADC Select 1 or 2. (2) Approach Continue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side ADC: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side ADC: (1) Go-around Initiate

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
FD 1 or 2	Flight director on applicable side is inoperative. FD 1 (2) FAIL displayed on EICAS.
	ACTION
	Continue approach, if the following is accomplished above 500 feet AGL: (1) Autopilot..... Disengage (2) AFCS SEL 1 (2)..... Select to operative AFCS channel. (3) FDs Engage (4) Autopilot..... Engage (5) Approach Continue to Category II minima. If below 500 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage (2) Manual control Resume (3) Approach Continue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-around Initiate

SYSTEM	INDICATION
AFCS PITCH TRIM	Autopilot pitch trim failure; AP PITCH TRIM caution message displayed on EICAS.
	ACTION
	During coupled approach: If above 200 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage Retrim, then re-engage. If caution message goes out: (2) Approach Continue If caution message persists: (2) Manual control Resume (3) Approach Continue to Category I minima. If below 200 feet AGL and non-visual: (1) Go-around Initiate

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
AUTOPILOT DISCONNECT	Autopilot fails to disengage when intentionally disconnected.
	ACTION
	Autopilot fails to disconnect using AP/SP DISC switch on control wheel during approach to landing: (1) Autopilot..... Disconnect using AP ENG switch on FCP or AP DISC switch on FCP, or TOGA switch or AP/SP DISC switch on other control wheel. (2) Airplane Trim manually using stabilizer trim switch. If unable to disconnect autopilot, expect higher than normal control feel force loads during landing.

SYSTEM	INDICATION
ILS LOCALIZER or GLIDESLOPE	ILS deviation error is greater than Category II "window". Flashing localizer or glideslope pointer.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
HYDRAULICS	Any hydraulic system(s) failure(s) annunciated as caution messages displayed on EICAS (See Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER).
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER. (2) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
MAIN GENERATOR	Generator failure; GEN 1 (2) OFF caution message displayed on EICAS.
	ACTION
	NOTE AP will disconnect if AFCS 2 is active. YD 1 or YD 2 may disengage. (1) ApproachContinue to Category I minima, or initiate go-around.

7. PERFORMANCE

The performance data in Chapter 6 are applicable.

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NOTE

The data of this Supplement is to be used when JAA/EASA operators are conducting Category II operations under JAR-AWO at Amendment 3.

NOTE

This Airplane Flight Manual Supplement does not constitute operational approval to conduct Category II operations. Such authorization must be obtained by the operator from the appropriate authorities.

1. GENERAL

The CL600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) has been shown to meet the airworthiness requirements for Category II Operations contained in Appendix 1 of AC 120-29 and Subpart 2 of JAR AWO, Amendment 3.

The data in this supplement must be used when conducting Category II operations.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must, therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

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3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

- An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the Category II Required Equipment List below and their ground installations, are operating satisfactorily.

CATEGORY II REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
VHF NAV 1 AND 2	BOTH MUST BE OPERATIONAL AND DISPLAYED ONSIDE.
PFD 1 AND 2	ONE (1) PFD AVAILABLE AND OPERATIONAL FOR EACH SIDE.
STAB CHANNEL 1 AND 2	ONE (1) CHANNEL MUST BE OPERATIONAL.
RADIO ALTIMETER	ONE MUST BE OPERATIONAL WITH DISPLAY ON BOTH SIDES.
IRS	ANY TWO (2) MUST BE OPERATIONAL.
ADC 1 AND 2	BOTH MUST BE OPERATIONAL.
FD 1 AND 2	BOTH MUST BE OPERATIONAL.
EFIS COMPARATOR MONITORS	MUST BE OPERATIONAL.
AFCS PITCH TRIM	MUST BE OPERATIONAL.
AUTOPILOT	MUST BE OPERATIONAL.
HYDRAULICS (3)	ALL SYSTEMS MUST BE ON AND OPERATIONAL.
ELECTRICS	TWO (2) GENERATORS ON AND SHARING LOAD (SEE NOTE).

NOTE

Approach to Category II minima may be continued with one generator, if an engine fails during final approach (refer to ABNORMAL PROCEDURES).

- Operation of the autopilot is prohibited below 80 feet AGL.
- The approved flap configurations for approach are;
 - 45° for all engines operating, and
 - 20° for one engine inoperative.
- The minimum decision height for Category II operations is 100 feet AGL.
- Wind limits for autopilot approaches are;
 - 8 knots crosswind, and 9 knots tailwind.

NOTE

Demonstrated headwind is 16 knots.

- The maximum ILS glidepath angle for Category II operations is 3.5 degrees.
- ATS must be disengaged.
- Use of FMS computed approach performance is prohibited.

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4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or AFCS MSGS FAIL warning message, "CAVALRY CHARGE" aural and/or flashing AP (red) message on the primary flight displays.

Continue approach if the following can be accomplished above 500 feet AGL:

- (1) Autopilot Disconnect
- (2) AFCS SEL 1(2) switch (centre pedestal)..... Select to operative AFCS channel.
Check AFCS 1 (2) INOP status message on.
- (3) Autopilot Engage and monitor operation.
- (4) Approach Continue to Category II minima.

If below 500 feet AGL, on a stabilized approach:

- (1) Autopilot Disconnect
- (2) Manual control Resume
- (3) Approach Continue to Category I minima.

If below 200 feet AGL and non-visual:

- (1) Go-around Initiate

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5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. Prior to Approach:

Airplane must be properly configured prior to approach:

- (1) Landing weight Check within limits. See Chapter 6; PERFORMANCE – LANDING PERFORMANCE.
- (2) NAV SOURCE Set to onside ILS (green) or armed (cyan) by FMS for both PFDs.
- (3) IRS and ADC source selectors Set to NORM.
- (4) RA MINIMUMS Set on both PFDs.
- (5) BARO MINIMUMS Set as a barometric DH, in case RA fails.
- (6) MIN ALERT Set to RA on both PFDs.
- (7) Runway Course Set on both PFDs, or armed (cyan) by FMS.
- (8) EFIS Check that no comparator flags are displayed.
- (9) CAS Check that the following messages are not displayed:
 - AFCS MSGS FAIL warning message,
 - AP PITCH TRIM, EFIS COMP INOP and EFIS MISCOMP caution messages,
 - FD 1 and 2 FAIL status messages.
- (10) Stabilizer and elevator position Check valid on EICAS or F/CTL synoptic page.
- (11) Approach mode Arm
- (12) Expanded lateral deviation indicator Verify displayed on both PFDs at 600 feet AGL.

B. Before Landing:

- (1) Autopilot Disengage at an altitude not less than 80 feet AGL.

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6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Single Engine Approach and Landing

- (1) Approach Conduct under Category I operations.

B. Engine Failure During Final Approach

Continue approach if the following can be accomplished above 800 feet AGL:

- (1) Autopilot Disconnect
- (2) Operating engine Increase thrust as required.
- (3) FLIGHT SPOILER lever Ensure retracted.
- (4) FLAPS Select to 20° for landing.
- (5) Airspeed Increase to $V_{REF} + 14$ KIAS.
- (6) Airplane Retrim
- (7) Autopilot Re-engage
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a landing without the use of reverse thrust.
- (9) TAWS WARNING, FLAPS OFF switch/light Press in to mute flap aural warning.

At glideslope intercept:

- (10) Airplane Retrim directionally as required.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

If above 800 feet AGL and the above procedure can not be accomplished:

- (11) Approach Conduct under Category I operations.

If below 800 feet AGL and non-visual:

- (12) Go-Around Initiate

After landing or during go-around at a safe altitude:

- (13) Engine shutdown procedure Accomplish If applicable (Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown procedure).

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6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures:

SYSTEM	INDICATION
VHF NAV 1 or 2	ILS receiver, on applicable side, is inoperative.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) NAV SOURCE Select X-SIDE. (2) Approach Continue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side NAV: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side NAV: (1) Go-around Initiate
SYSTEM	INDICATION
PFD 1 or 2	Primary flight display, on applicable side, goes blank.
	ACTION
	Failure during approach: (1) Display reversionary panel selector switch Select to MFD REV. Applicable MFD defaults to primary flight display.
SYSTEM	INDICATION
STAB CH 1 or 2	Horizontal stabilizer trim, on applicable side, is inoperative. STAB CH 1 (2) INOP status message, displayed on EICAS.
	ACTION
	Single channel failure: (1) Approach Continue Dual channel failure (STAB TRIM caution message on: autopilot may disengage): (1) Go-around Initiate Dual channel failure during final approach (in stabilized trim condition): (1) Approach Continue to Category I minima.

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
RADIO ALTIMETER	False or missing radio altitude on both PFDs, or red or amber RA indication on both PFDs.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
IRS	Erratic PFD data or ATT annunciation, and HDG/ROL/PIT annunciation on PFDs.
	ACTION
	First failure during the approach (AP will disconnect): If above 500 feet AGL, on a stabilized approach: (1) Manual controlResume (2) IRS sourceSelect an operative IRS. (if IRS 3 is installed, AP will be available). (3) ApproachContinue to Category I minima, if only one IRS available. If below 500 feet AGL and non-visual: (1) Manual controlResume (2) Go-aroundInitiate
SYSTEM	INDICATION
EFIS Comparator	EFIS COMP INOP (EFIS comparator failure) caution message or EFIS MISCOMP caution message displayed on EICAS and any of the ALT/HDG/IAS/PIT/ROL/LOC/GS annunciations on PFDs.
	ACTION
	Failure during approach: (1) Go-aroundInitiate (2) Applicable procedureAccomplish Refer to Chapter 5: ABNORMAL PROCEDURES – INSTRUMENTS SYSTEM.

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
ADC 1 or 2	Erratic PFD data or ADC FAIL annunciation, and ALT/IAS annunciation on PFDs.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) ADC Select 1 or 2. (2) Approach Continue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side ADC: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side ADC: (1) Go-around Initiate

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
FD 1 or 2	Flight director, on applicable side is inoperative. FD 1 (2) FAIL displayed on EICAS.
	ACTION Continue approach, if the following is accomplished above 500 feet AGL: (1) Autopilot..... Disengage (2) AFCS SEL 1 (2)..... Select to operative AFCS channel. (3) FDs Engage (4) Autopilot..... Engage (5) Approach Continue to Category II minima. If below 500 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage (2) Manual control Resume (3) Approach Continue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-around Initiate
SYSTEM	INDICATION
AFCS PITCH TRIM	Autopilot pitch trim failure; AP PITCH TRIM caution message displayed on EICAS.
	ACTION During coupled approach: If above 200 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage Retrim, then re-engage. If caution message goes out: (2) Approach Continue If caution message persists: (2) Manual control Resume (3) Approach Continue to Category I minima. If below 200 feet AGL and non-visual: (1) Go-around Initiate

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
AUTOPILOT DISCONNECT	Autopilot fails to disengage when intentionally disconnected.
	ACTION
	Autopilot fails to disconnect using AP/SP DISC switch on control wheel during approach to landing: (1) Autopilot..... Disconnect using AP ENG switch on FCP or AP DISC switch on FCP, or TOGA switch or AP/SP DISC switch on other control wheel. (2) Airplane Trim manually using stabilizer trim switch. If unable to disconnect autopilot, expect higher than normal control feel force loads during landing.
SYSTEM	INDICATION
ILS LOCALIZER or GLIDESLOPE	ILS deviation error is greater than Category II "window". Flashing localizer or glideslope pointer.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) Approach Continue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-around Initiate

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6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures: (Cont'd)

SYSTEM	INDICATION
HYDRAULICS	Any hydraulic system(s) failure(s) annunciated as caution messages displayed on EICAS (See Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER).
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER. (2) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate

SYSTEM	INDICATION
MAIN GENERATOR	Generator failure; GEN 1 (2) OFF caution message displayed on EICAS.
	ACTION
	NOTE AP will disconnect if AFCS 2 is active. YD 1 or YD 2 may disengage. (1) ApproachContinue to Category I minima, or initiate go-around.

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7. PERFORMANCE

The performance data in Chapter 6 are applicable, except as modified by the following:

A. Maximum Landing Weight Limited By Climb Requirements – Approach Flaps 20°/Landing Flaps 45°

The maximum landing weight limited by climb requirements for varying conditions of airport pressure altitude and temperature for the flaps 20° approach climb configuration and a subsequent landing flaps 45° configuration, are given in Figure 07-04A-1. The chart incorporates all applicable anti-icing and 10th stage bleeds combinations.

NOTE

If ice accumulation is suspected on unprotected surfaces, then reduce the maximum landing weight values derived from Figure 07-04A-1 by 800 kg (1,764 lb).

Example: ANTI-ICE ON

Associated Conditions:

Temperature = -8°C
 Airport pressure altitude = 4,000 ft
 Cowl anti-ice = On
 10th stage bleeds = Closed

As shown in the example Figure 07-04A-1, Anti-ice On), the maximum landing weight limited by climb requirements is 24,100 kg (53,131 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

Example: ANTI-ICE OFF

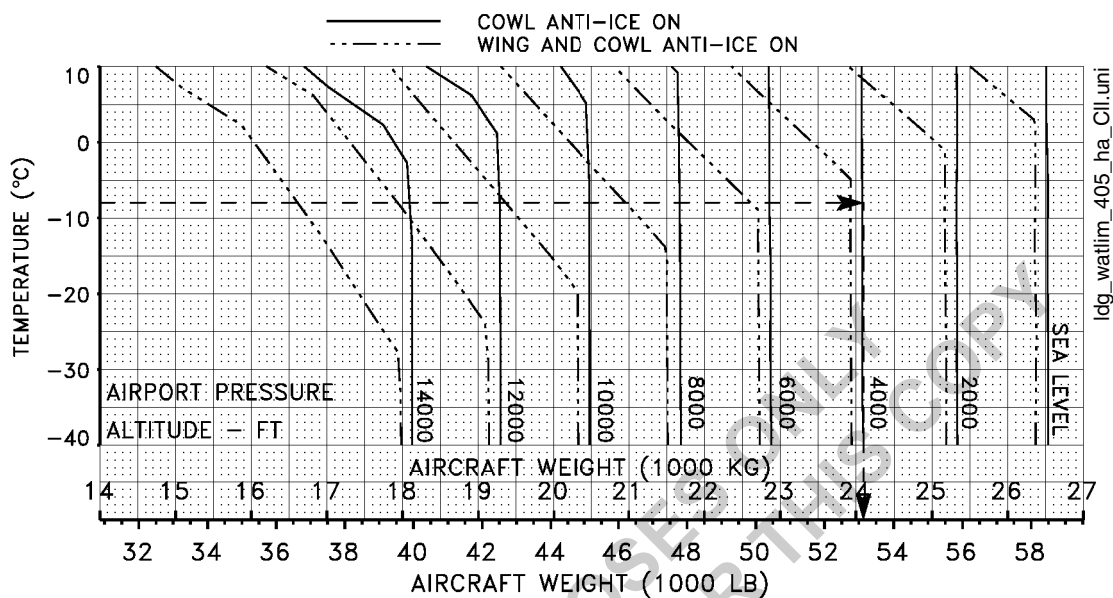
Associated Conditions:

Temperature = 27°C
 Airport pressure altitude = 4,000 ft
 Wing and cowl anti-ice = Off
 10th stage bleeds = Closed

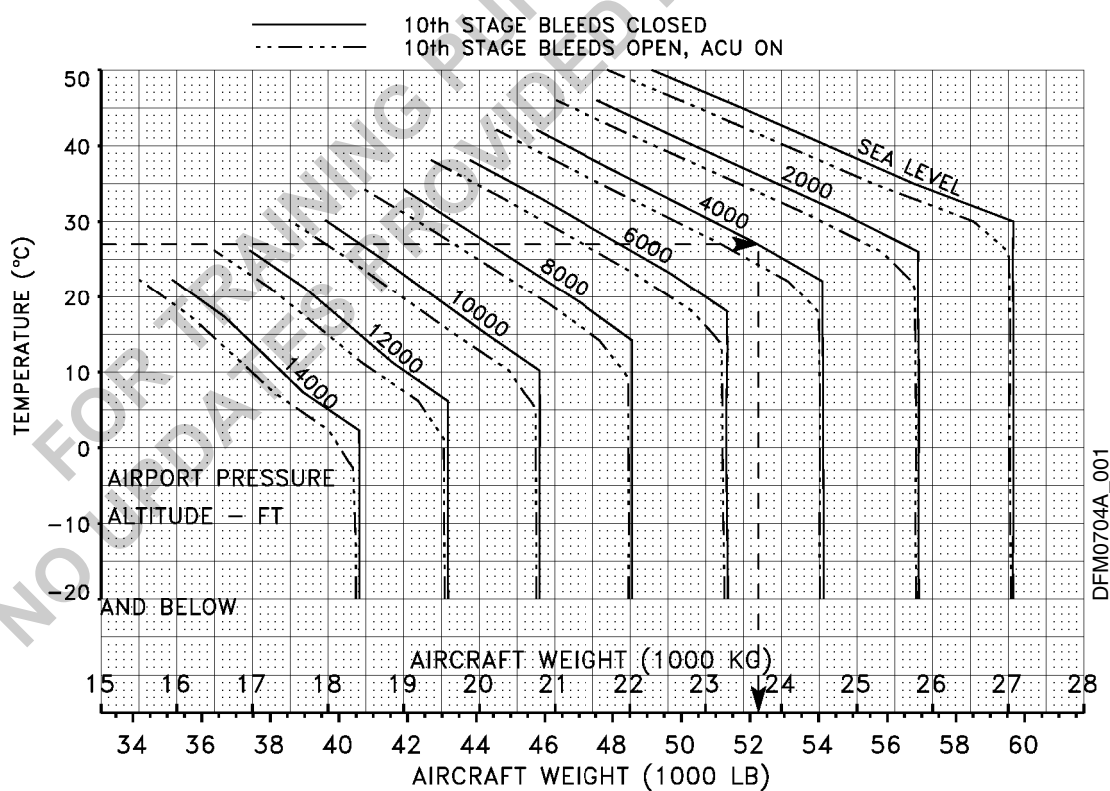
As shown in the example Figure 07-04A-1, Anti-ice Off), the maximum landing weight limited by climb requirements is 23,700 kg (52,250 lb). The landing weight in this case is greater than the maximum approved landing weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is not considered limiting.

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ANTI-ICE ON (10th STAGE BLEEDS OFF)



ANTI-ICE OFF



Landing Weight Limited by Climb Requirements – Approach Flaps 20°/Landing Flaps 45°
Figure 07-04A-1

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CHALLENGER 650	SUPPLEMENT 6 Flight With Landing Gear Down	07-06-01-1
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1. GENERAL

The data in this Supplement must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) when operating with fixed landing gear down.

These data complement or supersede data contained in the basic Airplane Flight Manual. This Supplement must, therefore, be used in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (7).

This supplement is applicable only when used in conjunction with a Minimum Equipment List approved by the appropriate authority.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

- Airspeed must not exceed V_{LE} (250 KIAS).
- Flight altitude above 20,000 ft is prohibited.
- Flight in known or anticipated icing conditions is not permitted.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. PERFORMANCE

The performance data in Chapter 6 are applicable, except as modified by the following:

A. Take-Off Performance

The take-off speeds (V_1 , V_R , and V_2) must be increased by 2 knots. The field length limited take-off weight must be reduced by 2%. For a given take-off weight, the take-off field length must be increased by 4%. The take-off weight limited by tire speed must be reduced by 4%. The take-off weight limited by maximum brake energy must be reduced by 2%.

B. Weight Altitude Temperature Limits

The following penalty must be applied to the performance data as a weight decrement:

FLIGHT SEGMENT	WEIGHT DECREMENT (%)
Take-off – Flaps 20°	16%
Landing – Approach Flaps 20° – Landing Flaps 45°	13%

The take-off weight is limited by the second segment climb gradient requirement (2.4% gross gradient – one engine inoperative). In landing, the limiting requirement is the approach climb gradient (2.1% gross gradient, one engine inoperative).

C. Climb Gradients and Climb Speeds

The following gradient penalty must be subtracted from the climb gradient data:

CLIMB GRADIENT	GRADIENT PENALTY (%)
Obstacle Clearance Reference Climb Gradient (Flaps 20°)	2.4%
Final Segment/Enroute (Flaps 0°)	3.3%
Approach (Flaps 20°)	2.4%
Landing (Flaps 45°)	Not applicable

The approach climb speed in the flaps 20° configuration must be increased by 2 knots.

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1. GENERAL

This supplement is applicable only on the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) when used in conjunction with a Minimum Equipment List approved by the appropriate authority.

The data in this supplement provides limitations, procedures and performance corrections to be applied when dispatching with some specific airplane systems inoperative.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must, therefore, be used in conjunction with the basic Airplane Flight Manual.

The effect of this supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except per MMEL or as modified by the following:

A. Hydraulic AC Motor Pump Inoperative

APR must be selected off, when dispatching with one hydraulic AC motor pump (1B or 2B) inoperative.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

7. PERFORMANCE

The performance data in Chapter 6 are applicable, except as modified by the following:

A. Ground Spoilers Inoperative

With the ground spoilers inoperative:

Take-off distance for dry or wet runways	Multiply by a factor of 1.08 (8%)
Landing distance and landing field length	Multiply by a factor of 1.14 (14%)

NOTE

No correction to the take-off decision speed (V_1) is required.

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7. PERFORMANCE (CONT'D)

B. Cowl Anti-Ice Pressure Relief Blow-Off Valve Inoperative

If cowl or wing and cowl anti-ice is on with one cowl anti-ice pressure relief blow-off valve failed open:

- Use cowl **and** wing anti-ice thrust settings (N_1), and reduce as follows:

Thrust settings (N_1)	Subtract 1.4%
---------------------------	---------------

NOTE

N_1 thrust setting reduction is applicable to both engines.

- Use cowl **and** wing anti-ice performance, and adjust as follows:

Take-off distance for dry or wet runways	Multiply by a factor of 1.14 (14%)
Obstacle clearance reference climb gradient	Subtract 1.5%
Final segment and enroute climb gradient (gross)	Subtract 1.4%
Approach climb gradient	Subtract 1.5%
Landing climb gradient	Subtract 3.0%
Take-off weight limited by climb requirements	Multiply by a factor of 0.95 (–5%)
Landing weight limited by climb requirements	Multiply by a factor of 0.95 (–5%)

NOTE

No correction to the take-off decision speed (V_1) is required.

C. Nose Wheel Steering Inoperative

All registries except FAA:

With nose wheel steering inoperative:

Take-off distance	Multiply by a factor of 1.02 (2%)
-------------------	-----------------------------------

NOTE

- The maximum crosswind component for take-off and landing is 20 knots.
- Take-off and landing is not permitted on contaminated runways.
- Excessive use of asymmetric braking may lead to fusible plug release or failure.

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7. PERFORMANCE (CONT'D)

D. Flap Power Drive Unit Inoperative

With flaps operating at half speed:

- Reduce the maximum landing weight as follows:

Landing weight limited by climb requirements	Reduce by 3,500 lb
--	--------------------

- Increase the acceleration distance as follows:

Net take-off flight path	Increase by 4,000 ft
--------------------------	----------------------

E. Anti-Skid Inoperative



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

With anti-skid inoperative:

Take-off distance for dry or wet runways	Multiply by a factor of 1.90 (90%)
Decision speed (V_1)	-17 KIAS
Landing distance and landing field length	Multiply by a factor of 1.95 (95%)

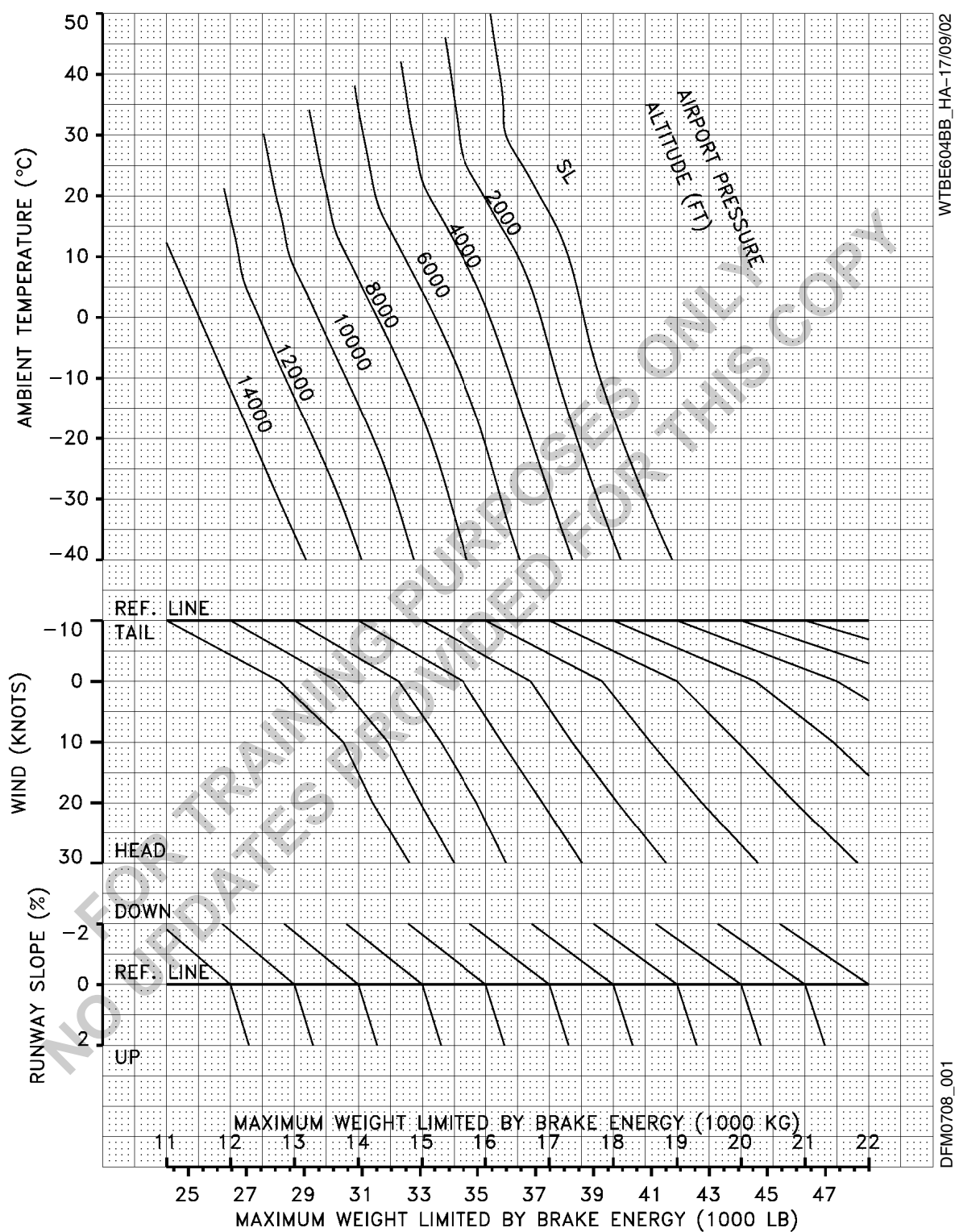
NOTE

- If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.
- Take-off and landing is not permitted on contaminated runways.
- Reduced thrust take-off is not permitted.

The maximum take-off weight limited by brake energy is determined from Figure 07-08-1.

A brake cooling time of 50 minutes must be observed between a landing or a low energy rejected take-off and a subsequent take-off.

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7. PERFORMANCE (CONT'D)

Take-Off Weight Limited By Brake Energy – Anti-Skid Inoperative
 Figure 07-08-1

7. PERFORMANCE (CONT'D)

F. Hydraulic AC Motor Pump Inoperative

(1) THRUST SETTINGS

All take-off performance must be calculated with APR selected OFF.

When dispatching with one hydraulic AC motor pump inoperative, determine the normal take-off thrust setting then apply the ΔN_1 correction for the dispatched bleed configuration as shown on Figure 07-08-2 to Figure 07-08-5.

Any readjustment to the thrust setting during the take-off climb must be adjusted by the appropriate ΔN_1 correction.

For approach climb, determine the thrust setting using the normal take-off thrust setting charts and then apply the appropriate ΔN_1 correction.

Altitude	Outside Air Temperature (°C)	ΔN_1
-1000 ft	-40	-3.8%
	32	-3.8%
	45	0.0%
	50	0.0%
0 ft	-40	-3.5%
	30	-3.5%
	41	0.0%
	50	0.0%
1000 ft	-40	-2.9%
	28	-2.9%
	37	0.0%
	48	0.0%
2000 ft	-40	-2.3%
	26	-2.3%
	33	0.0%
	46	0.0%
3000 ft	-40	-1.6%
	24	-1.6%
	29	0.0%
	44	0.0%
4000 ft	-40	-0.8%
	22	-0.8%
	24	0.0%
	42	0.0%
4900 ft and above	Any temperature	0.0%

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N_1 Corrections – Engine Bleeds Closed
Figure 07-08-2

7. PERFORMANCE (CONT'D)

Altitude	Outside Air Temperature (°C)	$\Delta N1$
-1000 ft	-40	-3.8%
	30	-3.8%
	43	0.0%
	50	0.0%
0 ft	-40	-3.5%
	28	-3.5%
	39	0.0%
	50	0.0%
1000 ft	-40	-2.9%
	27	-2.9%
	35	0.0%
	48	0.0%
2000 ft	-40	-2.2%
	25	-2.2%
	32	0.0%
	46	0.0%
3000 ft	-40	-1.5%
	23	-1.5%
	27	0.0%
	44	0.0%
4000 ft	-40	-0.8%
	22	-0.8%
	23	0.0%
	42	0.0%
4900 ft and above	Any temperature	0.0%

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N₁ Corrections – Packs On
Figure 07-08-3

7. PERFORMANCE (CONT'D)

Altitude	Outside Air Temperature (°C)	ΔN_1
-1000 ft	-40	-3.7%
	29	-3.7%
	42	0.0%
	50	0.0%
0 ft	-40	-3.5%
	28	-3.5%
	38	0.0%
	50	0.0%
1000 ft	-40	-2.8%
	26	-2.8%
	34	0.0%
	48	0.0%
2000 ft	-40	-2.2%
	24	-2.2%
	31	0.0%
	46	0.0%
3000 ft	-40	-1.5%
	22	-1.5%
	26	0.0%
	44	0.0%
4000 ft	-40	-0.7%
	20	-0.7%
	20	0.0%
	42	0.0%
4900 ft and above	Any temperature	0.0%

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N_1 Corrections – Packs OFF, Cowl Anti-Icing On
Figure 07-08-4

7. PERFORMANCE (CONT'D)

Altitude	Outside Air Temperature (°C)	ΔN1
-1000 ft	-40	-3.7%
	0	-3.7%
	26	0.0%
	50	0.0%
0 ft	-40	-3.4%
	-2	-3.4%
	22	0.0%
	50	0.0%
1000 ft	-40	-2.8%
	-3	-2.8%
	15	0.0%
	48	0.0%
2000 ft	-40	-2.2%
	-5	-2.2%
	11	0.0%
	46	0.0%
3000 ft	-40	-1.4%
	-7	-1.4%
	4	0.0%
	44	0.0%
4000 ft	-40	-0.6%
	-9	-0.6%
	-3	0.0%
	42	0.0%
4800 ft and above	Any temperature	0.0%

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N₁ Corrections – Packs OFF, Cowl and Wing Anti-Icing On
Figure 07-08-5

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7. PERFORMANCE (CONT'D)

(2) TAKE-OFF PERFORMANCE

(a) Take-Off Weight Limited By Climb Requirements

The maximum take-off weight limited by climb requirements when dispatching with one hydraulic AC motor pump inoperative is determined from Figure 07-08-6.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed
APR	= Off

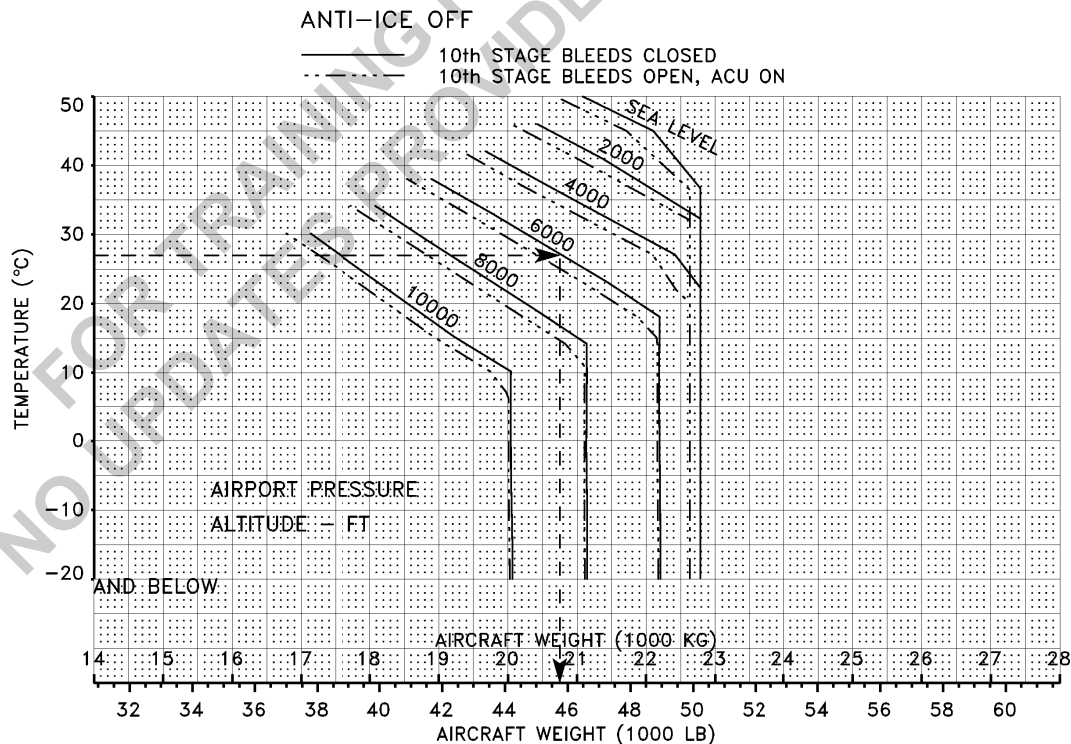
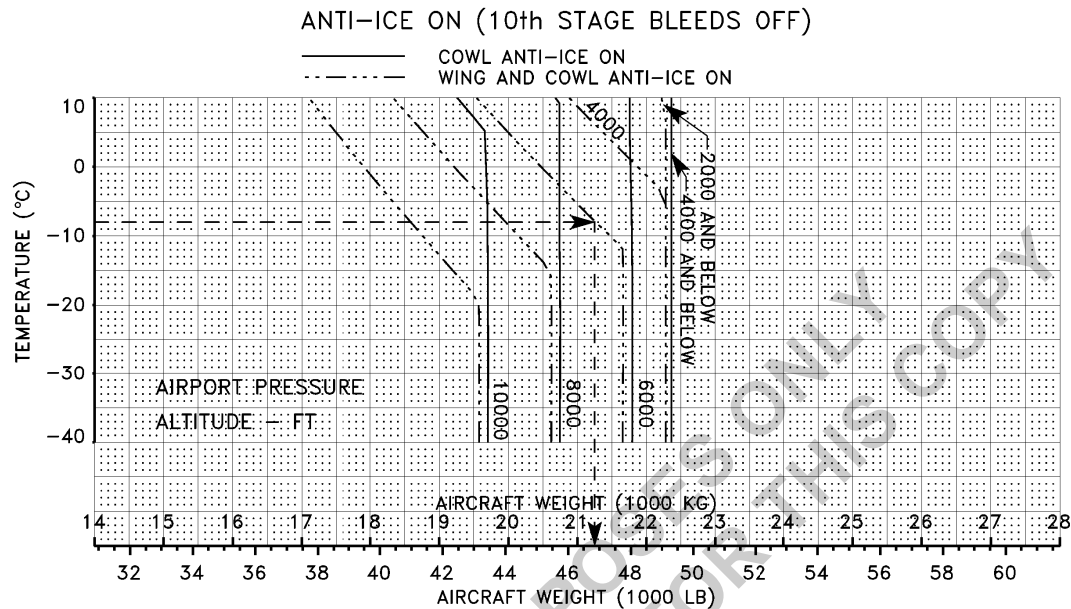
As shown in the example (Figure 07-08-6, Anti-ice On), the maximum take-off weight limited by climb requirements is 21,250 kg (46,848 lb).

Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 07-08-6, Anti-ice Off), the maximum take-off weight limited by climb requirements is 20,750 kg (45,746 lb).

7. PERFORMANCE (CONT'D)

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Take-Off Weight Limited by Climb Requirements
Figure 07-08-6

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7. PERFORMANCE (CONT'D)

(b) Take-Off Weight Limited By Brake Energy

When dispatching with hydraulic AC motor pump 2B inoperative, the maximum take-off weight limited by brake energy can be conservatively determined using Figure 07-08-1.

(c) Take-Off Distance

When dispatching with hydraulic AC motor pump 1B inoperative:

Take-off distance	Multiply by a factor of 1.3 (30%)
-------------------	-----------------------------------

When dispatching with hydraulic AC motor pump 2B inoperative:

Take-off distance	Multiply by a factor of 1.6 (60%)
-------------------	-----------------------------------

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7. PERFORMANCE (CONT'D)

(d) Take-Off Speeds

When dispatching with one hydraulic AC motor pump inoperative, and operating at or below 4,000 feet pressure altitude, apply V_R and V_2 corrections as shown on Figure 07-08-7 and Figure 07-08-8 respectively.

For V_1 , when dispatching with hydraulic AC motor pump 2B inoperative, use the highest V_{1MCG} or multiply the V_R from Figure 07-08-7 by a factor of 0.86 (-14%).

Altitude	Temperature	Weight	Correction
Sea level	ISA + 15 and below	below 37,000 lb	0
		37,000 and above	-1.5
	ISA + 20	below 37,000 lb	0.0
		37,000 and above	-1.0
2,000 ft	ISA + 25 and above	all weights	0
	ISA + 15 and below	below 37,000 lb	0
		37,000 and above	-1.0
	ISA + 20 and above	below 37,000 lb	0.0
		37,000 and above	-0.5
4,000 ft	all temperatures	all weights	0

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V_R Adjustment
Figure 07-08-7

Altitude	Temperature	Weight	Correction
Sea level	ISA + 15 and below	24,000 lb	-3.0
		30,000 lb	-2.5
		36,000 lb	-1.5
		37,000 and above	0
	ISA + 20	24,000 lb	-1.5
		30,000 lb	-1.0
		36,000 lb	-0.5
		37,000 and above	0
	ISA + 25 and above	all weights	0
2,000 ft	ISA + 15 and below	24,000 lb	-2.0
		30,000 lb	-1.5
		36,000 lb	-1.0
		37,000 and above	0
	ISA + 20 and above	all weights	0
4,000 ft	all temperatures	all weights	0

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V_2 Adjustment
Figure 07-08-8

7. PERFORMANCE (CONT'D)

(3) OBSTACLE CLEARANCE

When dispatching with one hydraulic AC motor pump inoperative, the normal net take-off flight path must be used (i.e. take-off thrust is to be used for acceleration segment).

When operating below 4,000 feet pressure altitude, apply the obstacle clearance reference climb gradient corrections as shown on Figure 07-08-9.

NOTE

Corrections are applicable to all bleed configurations.

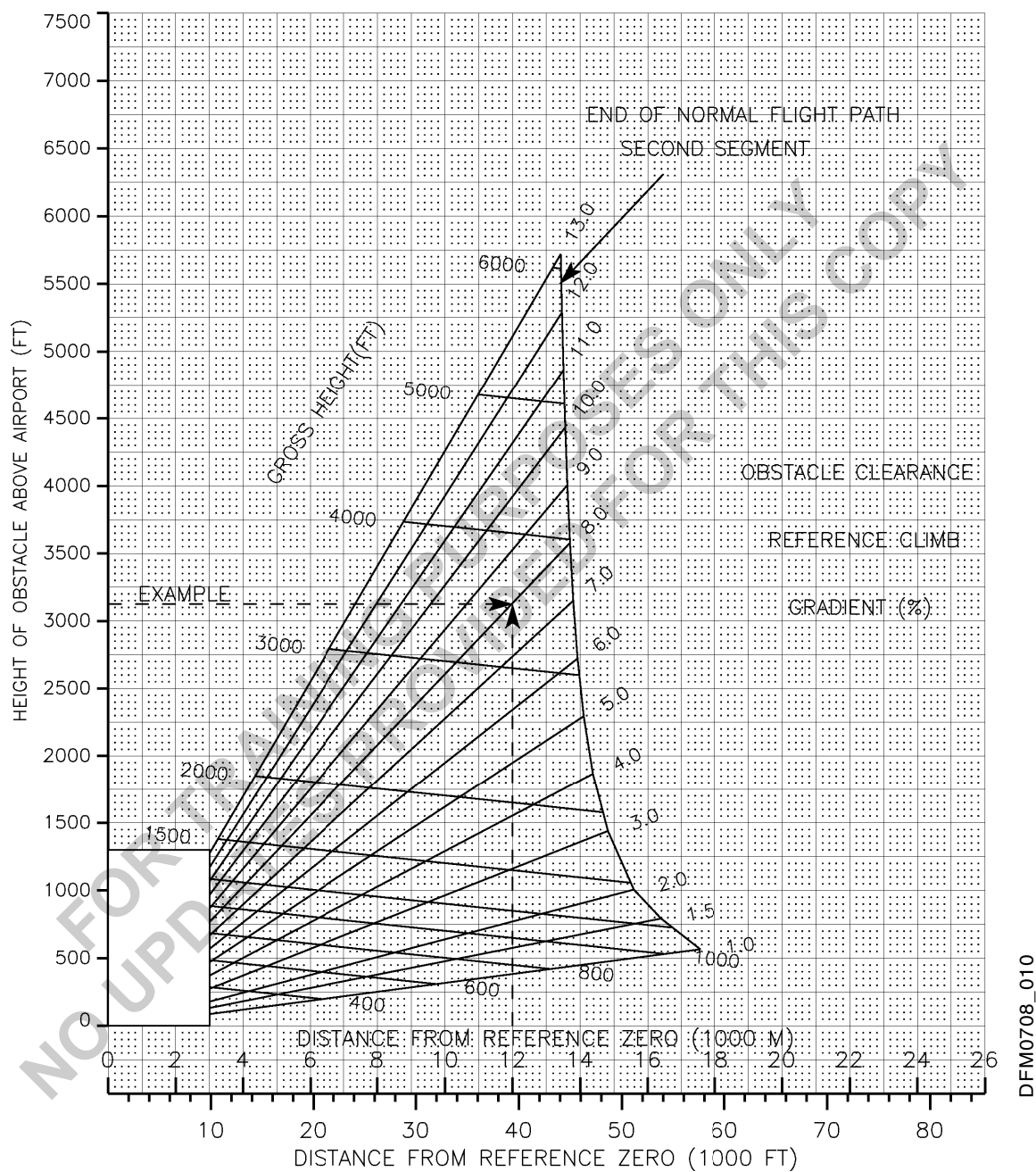
Pressure Altitude (ft)	0	2000	4000
Reference Gradient (%)			
1	-1.2	-0.7	-0.3
2	-1.3	-0.8	-0.3
3	-1.4	-0.9	-0.3
4	-1.5	-0.9	-0.3
5	-1.6	-1.0	-0.3
6	-1.7	-1.1	-0.3
7	-1.8	-1.1	-0.4
8	-1.9	-1.2	-0.4
9	-2.0	-1.2	-0.4
10	-2.1	-1.3	-0.4
11	-2.2	-1.3	-0.4
12	-2.3	-1.3	-0.4
13	-2.4	-1.4	-0.4
14	-2.4	-1.4	-
15	-2.5	-	-
16	-2.6	-	-

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Obstacle Clearance Reference Climb Gradient Corrections
Figure 07-08-9

7. PERFORMANCE (CONT'D)

The reduced maximum level-off height for a given obstacle clearance gradient is determined from Figure 07-08-10.



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Net Take-Off Flight Path
Figure 07-08-10

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7. PERFORMANCE (CONT'D)

Example: DETERMINATION OF THE MAXIMUM ALLOWABLE TAKE-OFF WEIGHT FOR A GIVEN OBSTACLE

Associated Conditions:

Airport altitude	= Sea level (0 ft)
Ambient temperature	= 30°C (ISA + 15°C)
Reported wind	= 20 knots (headwind)
Take-off flaps	= 20°
APR	= Off
Anti-ice	= Off
10th stage bleeds	= Closed
Obstacle height	= 3,125 feet above the airport
Obstacle distance	= 12,000 meters (39,372 feet) from reference zero (end of the take-off distance)

The example in Figure 07-08-10 shows that the required obstacle clearance reference gradient is 8% and the minimum level-off height is 3,475 feet, which is below the maximum level-off height of 3,950 feet for this reference gradient at the end of the second segment. The gross level-off height selected for take-off must therefore not be less than 3,475 feet and not greater than 3,950 feet. A gross level-off height of 3,475 feet will be used for the present example.

Entering the bottom end of Figure 06-04-3 from the basic Airplane Flight Manual with a reference gradient of 8%, correcting for a 20 knot headwind, adding the gradient correction of 1.9% from Figure 07-08-9 and correcting for APR selected OFF, gives a gradient value of 11.4%. Combining this value with the given ambient temperature of 30°C at sea level shows that the maximum weight to achieve a reference climb gradient of 8% is 15,100 kg (33,290 lb). Therefore, the maximum take-off weight limited by obstacle clearance requirements is 15,100 kg (33,290 lb).

7. PERFORMANCE (CONT'D)

(4) APPROACH AND LANDING CLIMB

When dispatching with one hydraulic AC motor pump inoperative, and operating at or below 4,000 feet pressure altitude, apply approach and landing climb gradient corrections as shown on Figure 07-08-11 and Figure 07-08-12 respectively.

NOTE

Corrections are applicable to all bleed configurations.

Pressure Altitude (ft)	0	2000	4000
Reference Gradient (%)			
1	-	-	-0.9
2	-2.0	-1.5	-1.0
3	-2.1	-1.6	-1.0
4	-2.3	-1.7	-1.1
5	-2.5	-1.9	-1.2
6	-2.6	-2.0	-1.3
7	-2.8	-2.1	-1.3
8	-3.0	-2.2	-1.4
9	-3.1	-2.4	-1.5
10	-3.3	-2.5	-1.6
11	-3.5	-2.6	-1.7
12	-3.7	-2.7	-1.7
13	-3.8	-2.9	-1.8
14	-4.0	-3.0	-1.9
15	-4.2	-3.1	-2.0
16	-4.3	-3.2	-2.0
17	-4.5	-3.4	-
18	-4.7	-3.5	-
19	-4.8	-	-

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Approach Climb Gradient Corrections
Figure 07-08-11

7. PERFORMANCE (CONT'D)

Pressure Altitude (ft)	0	2000	4000
Reference Gradient (%)			
3	-	-	-0.5
4	-	-1.5	-0.5
5	-2.4	-1.5	-0.6
6	-2.5	-1.6	-0.6
7	-2.6	-1.7	-0.6
8	-2.7	-1.8	-0.6
9	-2.9	-1.9	-0.7
10	-3.0	-1.9	-0.7
11	-3.1	-2.0	-0.7
12	-3.2	-2.1	-0.7
13	-3.3	-2.2	-0.8
14	-3.5	-2.2	-0.8
15	-3.6	-2.3	-0.8
16	-3.7	-2.4	-0.8
17	-3.8	-2.5	-0.9
18	-4.0	-2.5	-0.9
19	-4.1	-2.6	-0.9
20	-4.2	-2.7	-0.9
21	-4.3	-2.8	-1.0
22	-4.4	-2.9	-1.0
23	-4.6	-2.9	-1.0
24	-4.7	-3.0	-1.0
25	-4.8	-3.1	-1.1
26	-4.9	-3.2	-1.1
27	-5.1	-3.2	-1.1
28	-5.2	-3.3	-1.1
29	-5.3	-3.4	-1.1
30	-5.4	-3.5	-1.2
31	-5.5	-3.5	-1.2
32	-5.7	-3.6	-1.2

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Landing Climb Gradient Corrections
Figure 07-08-12

CHALLENGER 650	SUPPLEMENT 8 Operation With Airplane Systems Inoperative	07-08-01-18
		Oct 27/15

7. PERFORMANCE (CONT'D)

(5) LANDING PERFORMANCE

The maximum landing weight limited by climb requirements when dispatching with one hydraulic AC motor pump inoperative is determined from Figure 07-08-13.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= On
10th stage bleeds	= Closed

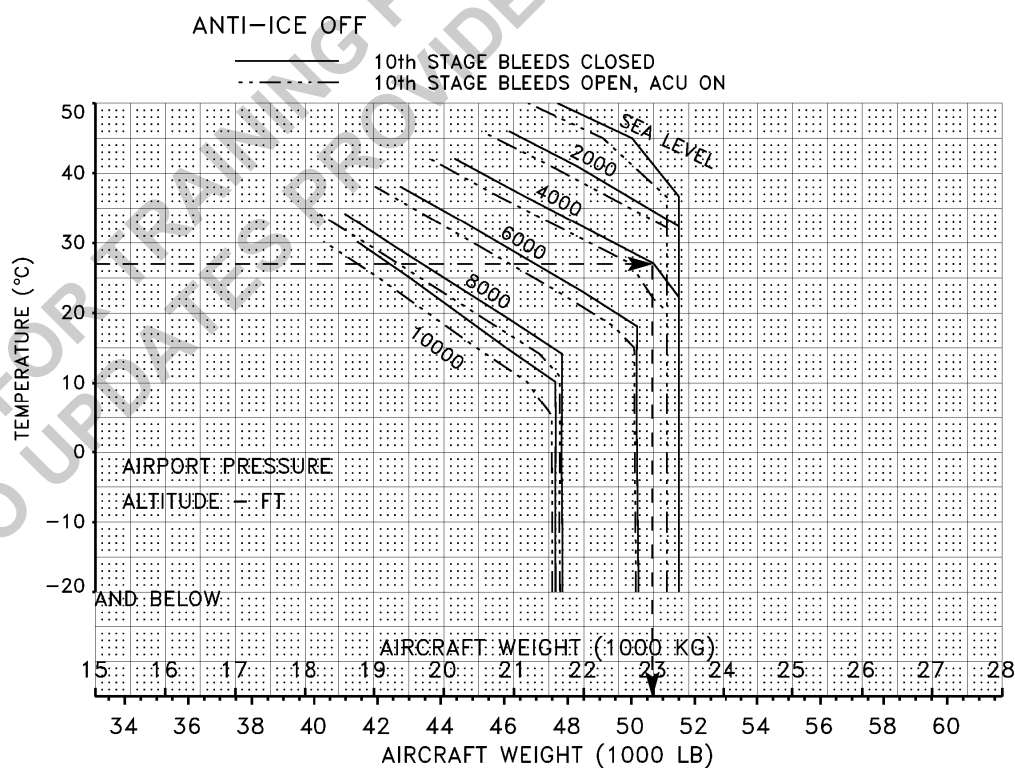
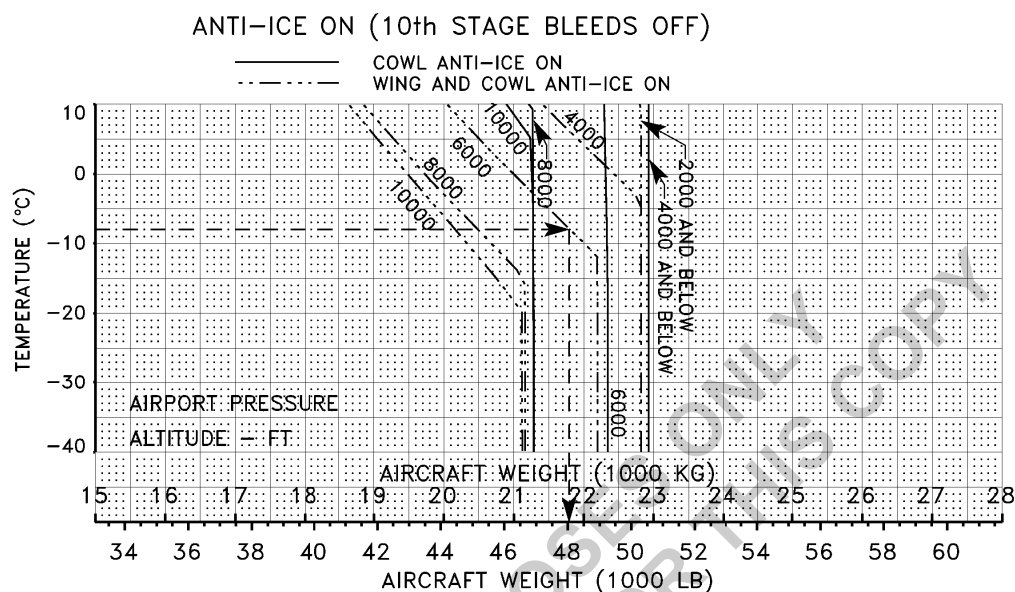
As shown in the example (Figure 07-08-13, Anti-ice On), the maximum landing weight limited by climb requirements is 21,800 kg (48,061 lb).

Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 4,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

As shown in the example (Figure 07-08-13, Anti-ice Off), the maximum landing weight limited by climb requirements is 23,000 kg (50,706 lb).

7. PERFORMANCE (CONT'D)


DFM0708_013

Landing Weight Limited by Climb Requirements
Figure 07-08-13

CHALLENGER 650	SUPPLEMENT 8 Operation With Airplane Systems Inoperative	07-08-01-20
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7. PERFORMANCE (CONT'D)

G. Cracked Windshield Faceply

One windshield faceply may be cracked provided the mission fuel requirements are increased by:

Fuel consumption	0.30%
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CHALLENGER 650	SUPPLEMENT 8 Operation With Airplane Systems Inoperative	07-08-01-21
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8. SUPPLEMENTS

The supplements in Chapter 7 are applicable, except as modified by the following:

A. Supplement 2 – Operation on Wet and Contaminated Runways

(1) LIMITATIONS

Operation on contaminated runways with hydraulic AC motor pump 2B inoperative is prohibited.

(2) TAKE-OFF DISTANCE

When dispatching with hydraulic AC motor pump 1B inoperative:

Take-off distance	Multiply by a factor of 1.3 (30%)
-------------------	-----------------------------------

When dispatching with hydraulic AC motor pump 2B inoperative:

Take-off distance	Multiply by a factor of 2.0 (100%)
-------------------	------------------------------------

(3) TAKE-OFF SPEEDS

When dispatching with one hydraulic AC motor pump inoperative, and operating at or below 4,000 feet pressure altitude, apply V_R and V_2 corrections as shown on Figure 07-08-7 and Figure 07-08-8 respectively.

For V_1 , when dispatching with hydraulic AC motor pump 2B inoperative, use the highest V_{1MCG} or multiply the V_R from Figure 07-08-7 by a factor of 0.86 (–16%).

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CHALLENGER 650	SUPPLEMENT 8 Operation With Airplane Systems Inoperative	07-08-01-22
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8. SUPPLEMENTS (CONT'D)

B. Supplement 4A – Category II Operations (JAA/EASA Operators)

The maximum landing weight limited by climb requirements when dispatching with one hydraulic AC motor pump inoperative is determined from Figure 07-08-14.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= On
10th stage bleeds	= Closed

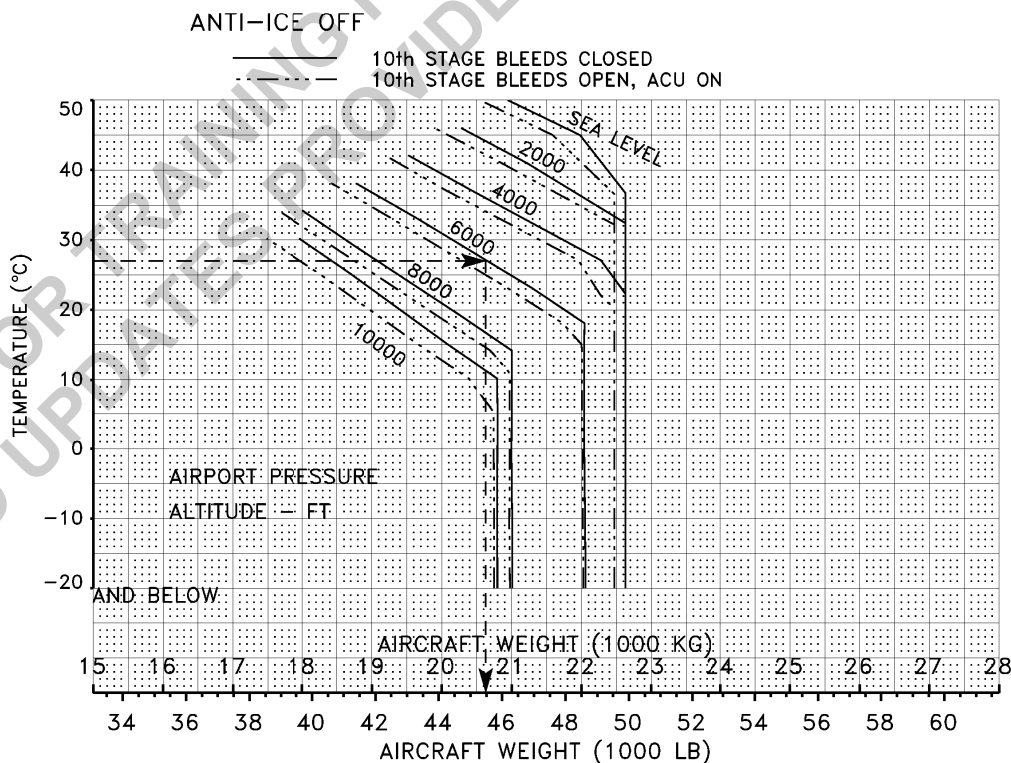
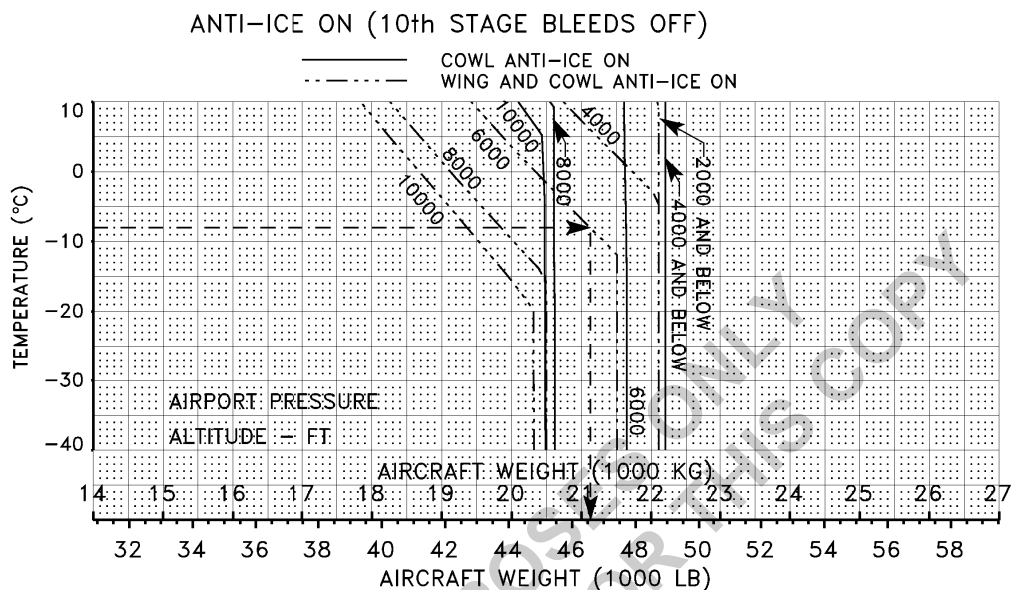
As shown in the example (Figure 07-08-14, Anti-ice On), the maximum landing weight limited by climb requirements is 21,100 kg (46,518 lb).

Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

As shown in the example (Figure 07-08-14, Anti-ice Off), the maximum landing weight limited by climb requirements is 20,600 kg (45,415 lb).

8. SUPPLEMENTS (CONT'D)


DFM0708_014

Landing Weight Limited by Climb Requirements – Category II Operations
 Figure 07-08-14

CHALLENGER 650	SUPPLEMENT 8 Operation With Airplane Systems Inoperative	07-08-01-24
		Rev 14, Nov 19/18

8. SUPPLEMENTS (CONT'D)

C. Supplement 6 – Flight with Landing Gear Down

(1) WEIGHT ALTITUDE TEMPERATURE LIMITS

When dispatching with one hydraulic AC motor pump inoperative, the following penalty must be applied to the performance data as a weight decrement:

FLIGHT SEGMENT	WEIGHT DECREMENT (%)
Take-off – Flaps 20°	16%
Landing – Approach Flaps 20° – Landing Flaps 45°	13%

(2) CLIMB GRADIENTS AND CLIMB SPEEDS

When dispatching with one hydraulic AC motor pump inoperative, the following gradient penalty must be subtracted from the climb gradient data:

CLIMB GRADIENT	GRADIENT PENALTY (%)
Obstacle Clearance Reference Climb Gradient (Flaps 20°)	2.9%
Final Segment / Enroute (Flaps 0°)	3.3%
Approach (Flaps 20°)	2.6%
Landing (Flaps 45°)	Not applicable

The approach climb speed in the flaps 20° configuration must be increased by 2 knots.

D. Supplement 19 – Operation on Wet Grooved Runways

When dispatching with anti-skid inoperative, use of the landing performance on a wet grooved runway is prohibited. Use the landing distance and landing field length factor of Supplement 8.

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CHALLENGER 650	SUPPLEMENT 11 Approaches below 4.5 Degrees	07-11-00-1
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CHALLENGER ⁶⁵⁰	SUPPLEMENT 11 Approaches below 4.5 Degrees	07-11-00-2
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CHALLENGER 650	SUPPLEMENT 11 Approaches below 4.5 Degrees	07-11-01-1
		Oct 27/15

1. GENERAL

The data in this Supplement must be used when conducting approaches above 3.5 degrees, but below 4.5 degrees.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Operating Limitations

(1) APPROACHES

On airplanes NOT incorporating Service Bulletin 650-34-008:

Operations with a glidepath angle exceeding the maximum shown in area A of Figure 07-11-1 are prohibited.

On airplanes incorporating Service Bulletin 650-34-008:

Operations with a glidepath angle exceeding the maximum shown in area B of Figure 07-11-1 are prohibited.

B. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

The minimum autopilot use height for precision approaches (ILS) with a glidepath angle that exceeds 3.5 degrees is 160 feet AGL.

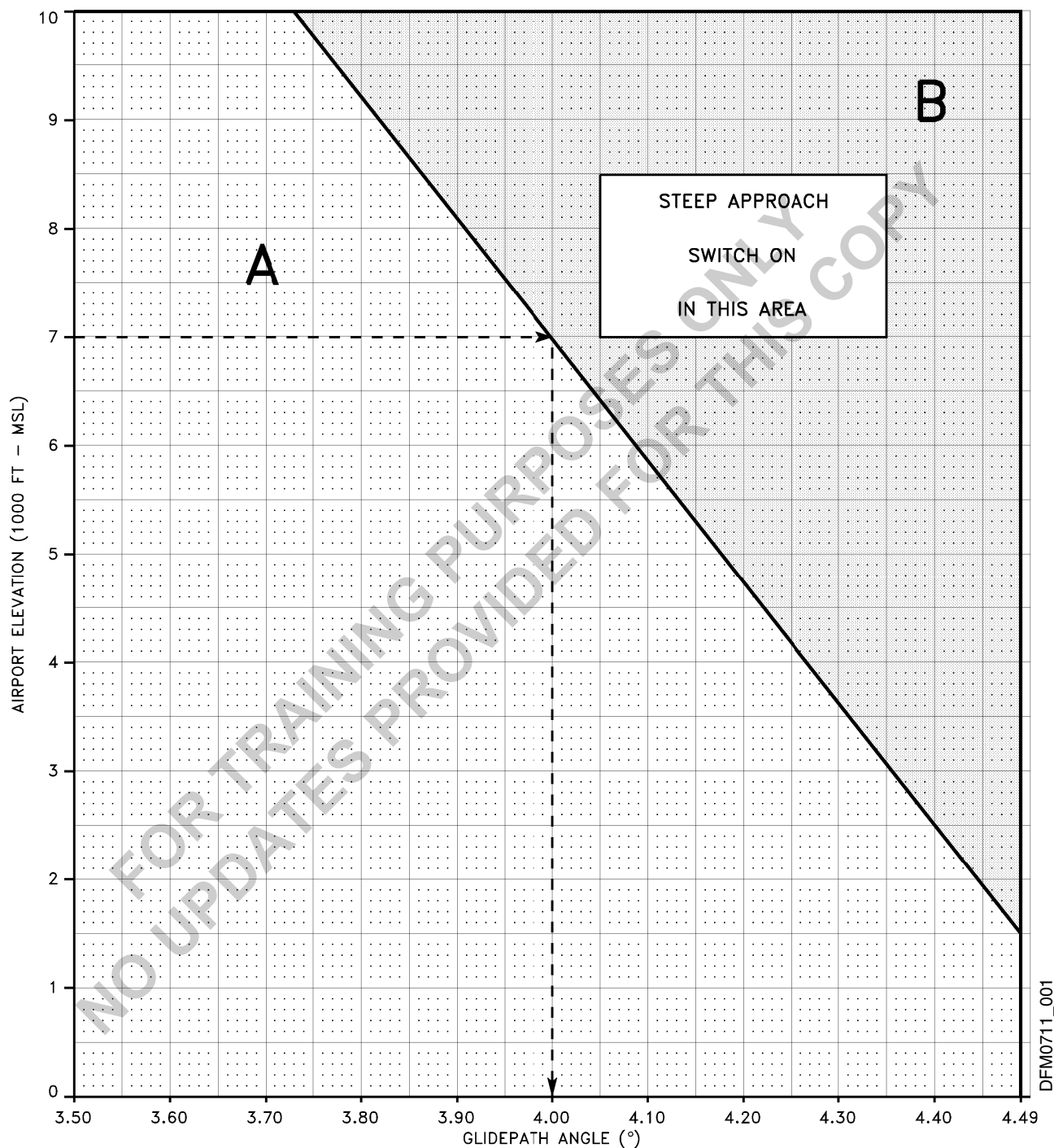
C. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

FMS approaches with LPV guidance are prohibited.

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3. LIMITATIONS (CONT'D)



Approach Glidepath Angle Operating Limit
 Figure 07-11-1

CHALLENGER 650	SUPPLEMENT 11 Approaches below 4.5 Degrees	07-11-01-3 Rev 10, Dec 04/17
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4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, in addition to the following:

A. Before Start Check

On airplanes incorporating Service Bulletin 650-34-008:

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

B. In Range Check

For operations in area B of Figure 07-11-1:

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press in Check ON light on.

C. Before Landing Check

NOTE

1. For approaches at angles above 4.0 degrees, the airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction).
2. Intermittent operation at idle when wing anti-ice and/or cowl anti-ice is ON is acceptable during final approach provided no associated EICAS caution messages are annunciated.

D. Go-Around

For operations in area B of Figure 07-11-1:

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

E. After Landing Check

On airplanes incorporating Service Bulletin 650-34-008:

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Flaps Failure

The following applies, in addition to the existing flap failure procedure:

If flaps fail at less than 20 degrees, divert to a suitable landing field with a glidepath angle of 3.5 degrees or less.

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CHALLENGER ⁶⁵⁰	SUPPLEMENT 11 Approaches below 4.5 Degrees	07-11-01-4
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7. PERFORMANCE

The performance in Chapter 6 is applicable.

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CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-1
		Rev 10, Dec 04/17

NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes incorporating Service Bulletin 650-34-008.

This Airplane Flight Manual Supplement is NOT applicable to FAA registered airplanes.

This Airplane Flight Manual Supplement does not constitute approval to conduct Steep Approach operations.

1. GENERAL

The data in this Supplement must be used when conducting steep approaches from 4.5 degrees to 5.5 degrees, inclusive.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (8).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Structural Weight

Maximum landing weight: 15,875 kg (35,000 lb)

B. Operating Limitations

(1) ALTITUDE AND TEMPERATURE OPERATING LIMIT

Maximum landing altitude is 5,000 feet.

(2) OPERATION IN ICING CONDITIONS

Steep approaches in known icing conditions are prohibited.

(3) WIND LIMITS

No tailwind component is permitted for steep approach landings.

The maximum crosswind component for steep approach landings is 20 knots.

Increasing landing reference speed by more than 10 KIAS for crosswind and/or gusts is prohibited. If wind conditions require a speed adder of more than 10 KIAS, divert to a suitable landing field that does not require a steep approach procedure.

(4) APPROACHES

Operations with a glidepath angle that exceeds 5.5 degrees is prohibited.

Deliberate one engine inoperative steep approaches are prohibited.

Partial flap steep approaches and landings are prohibited.

Steep approach landing on contaminated runways is prohibited.

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CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-2
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3. LIMITATIONS (CONT'D)

(5) LANDING DECISION POINT

The minimum height for go-around following an engine failure (Landing Decision Point) is 200 feet AGL.

C. Power Plant

(1) AUTOTHROTTLE (ATS)

Use of the ATS for steep approaches or landing is prohibited.

D. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

Use of autopilot for steep approaches is prohibited.

(2) ANTI-ICE SYSTEM

Approach and landing with wing and/or cowl anti-ice selected ON is prohibited.

(3) SYSTEMS INOPERATIVE

Landings at a steep approach airfield are prohibited if any of the following systems are inoperative:

- Flaps
- Ground Spoilers
- Anti-Skid
- Nose Wheel Steering
- Radio Altimeter
- Thrust Reverser

E. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

Use of the FMS approach performance function for steep approach is prohibited.

FMS approaches with LPV guidance are prohibited.

(2) TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

The TAWS steep approach switch/light must be selected ON as directed in the normal procedures of this supplement.

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CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-3
		Rev 10, Dec 04/17

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees,
- an increase in actual landing distance by a factor greater than 1.50 (50%) or
- a landing with gear up or unsafe.
- a landing with any of the systems inoperative as listed in the system limitations of this supplement.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

NOTE

Use of ILS or FMS VNAV guidance with or without the flight director is authorized.

A. Before Start Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

B. In Range Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press in Check ON light on.

C. Before Landing Check

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction) up to a maximum of $V_{REF} + 10$ KIAS.

D. Go-Around

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

E. After Landing Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

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CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-4
		Rev 9, Jun 16/17

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

I Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees or
- an increase in actual landing distance by a factor greater than 1.50 (50%).
- a landing with any of the systems inoperative as listed in the system limitations of this supplement.

A. Engine Failure During Approach

- I
- (1) Operating engine Increase thrust as required.
 - (2) FLIGHT SPOILER lever
(If extended) Select to RETRACT.
Check EICAS indicates spoilers retracted.
 - (3) FLAPS Select to 20°.
 - (4) Airspeed Increase to V_{REF} (Flaps 45°) + 14 KIAS.
 - (5) Airplane Retrim and go-around (Refer to Single Engine Go-Around in this section).
If required, autopilot may be engaged, if above 800 feet AGL.
Divert to a suitable landing field that does not require a steep approach procedure.

NOTE

The height loss during a single engine go-around is no more than 200 feet.

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CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-5
		Rev 7, Feb 23/17

6. ABNORMAL PROCEDURES (CONT'D)

B. Single Engine Go-Around

NOTE

- The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
- Single engine go-around capability is not assured with flaps greater than 20°.

- (1) Single Engine Climb Thrust (APR rating) Set immediately (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
 - (2) TOGA push switch Depress
 - (3) FLIGHT SPOILER lever (If extended) Select to RETRACT.
Check EICAS indicates spoilers retracted.
 - (4) Airplane Rotate smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB).
 - (5) FLAPS Select to 20°.
- When a positive rate of climb is achieved:**
- (6) LDG GEAR lever Select to UP.
- At a safe altitude (not below 400 feet AGL):**
- (7) FLAPS Select to 0° at a speed of not less than $V_{FTO} - 5$ KIAS.
 - (8) THRUST REVERSER, L and R switches Select to OFF.
Check L and R REV ARMED advisory messages out.
- If go-around was due to engine failure during approach:**
- (9) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5 – ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

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		Rev 15, Mar 25/19

7. PERFORMANCE

The performance in Chapter 6 is applicable, except as modified by the following:

The data in this supplement are applicable to steep approaches from 4.5 to 5.5 degrees (inclusive), except for this section, which is applicable only for approaches conducted at 5.5 degrees. For all other approach angles, the performance data in Chapter 6 is applicable.

A. Landing Performance

The reference screen height used for landing is 50 feet.

The reference steep approach glide slope angle is 5.5 degrees.

(1) LANDING FIELD LENGTH AND SPEED

The landing field length required for 5.5 degree approach operations for varying conditions of airplane landing weight, airport pressure altitude and wind is provided by Figure 07-12-1. The actual landing distance required is also given within this chart.

The landing approach speed (V_{REF}) for varying airplane landing weights and airport pressure altitude for steep approach operations is also given by Figure 07-12-1.

The actual landing distance given on Figure 07-12-1 is the distance from a 50 foot height at V_{REF} with flaps 45°, to a full stop on a smooth, dry, level, hard-surfaced runway and ISA temperature for steep approach operations.

NOTE

1. The landing field length derived from Figure 07-12-1 includes factors for operational rules which require use of 60% of the available runway (landing field length required).
2. Operational rules may require that for wet runway surfaces, the runway length available be at least 115% of the landing field length derived from Figure 07-12-1.

Example:

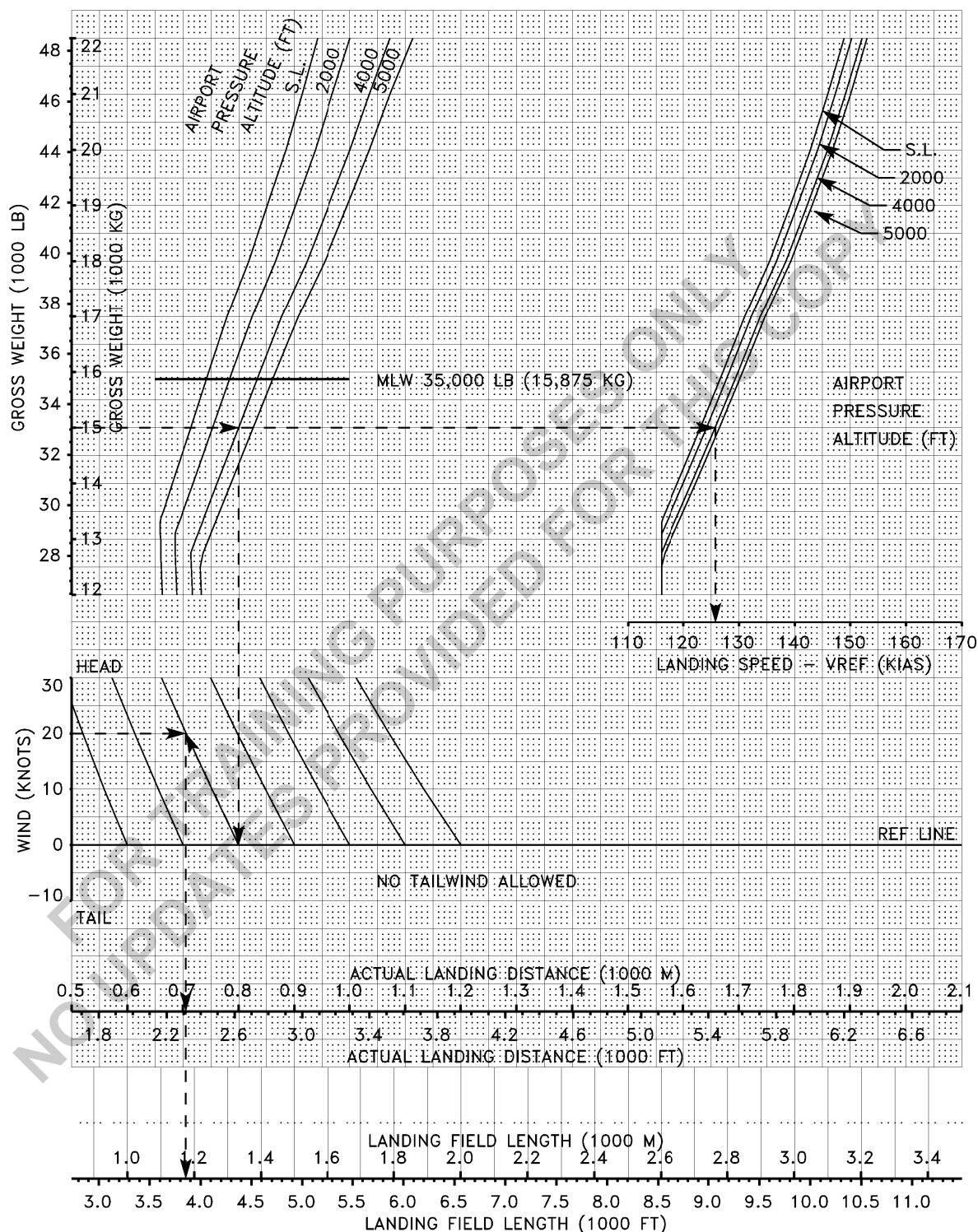
Associated Conditions:

Landing gross weight	= 15,000 kg (33,069 lb)
Airport pressure altitude	= 4,000 feet
Wind	= 20 knots, headwind

The example in Figure 07-12-1, shows that for the above conditions, the actual landing distance is 705 meters (2,313 feet) and the landing field length is 1,175 meters (3,855 feet).

V_{REF} for the same conditions as above is 126 KIAS.

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7. PERFORMANCE (CONT'D)

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Landing Field Length and Landing Speed, Flaps 45
 Figure 07-12-1

CHALLENGER 650	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-8
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7. PERFORMANCE (CONT'D)

(2) ADDITIONAL LANDING INFORMATION

The distance from the visual glide path system runway position to full stop for varying conditions of airplane landing weight, airport pressure altitude and wind is provided by Figure 07-12-2. This is the minimum stopping distance achievable on a dry runway past the intersection of the approach path and the runway surface. It is derived from an approach to landing glideslope of 5.5 degrees maintained to a threshold crossing height of 50 feet, a firm touchdown and maximum braking.

This distance is provided to cater for runway variability in position of the visual glide path systems and threshold crossing heights.

The most limiting distance from Figure 07-12-1 or Figure 07-12-2 is to be used.

Example:

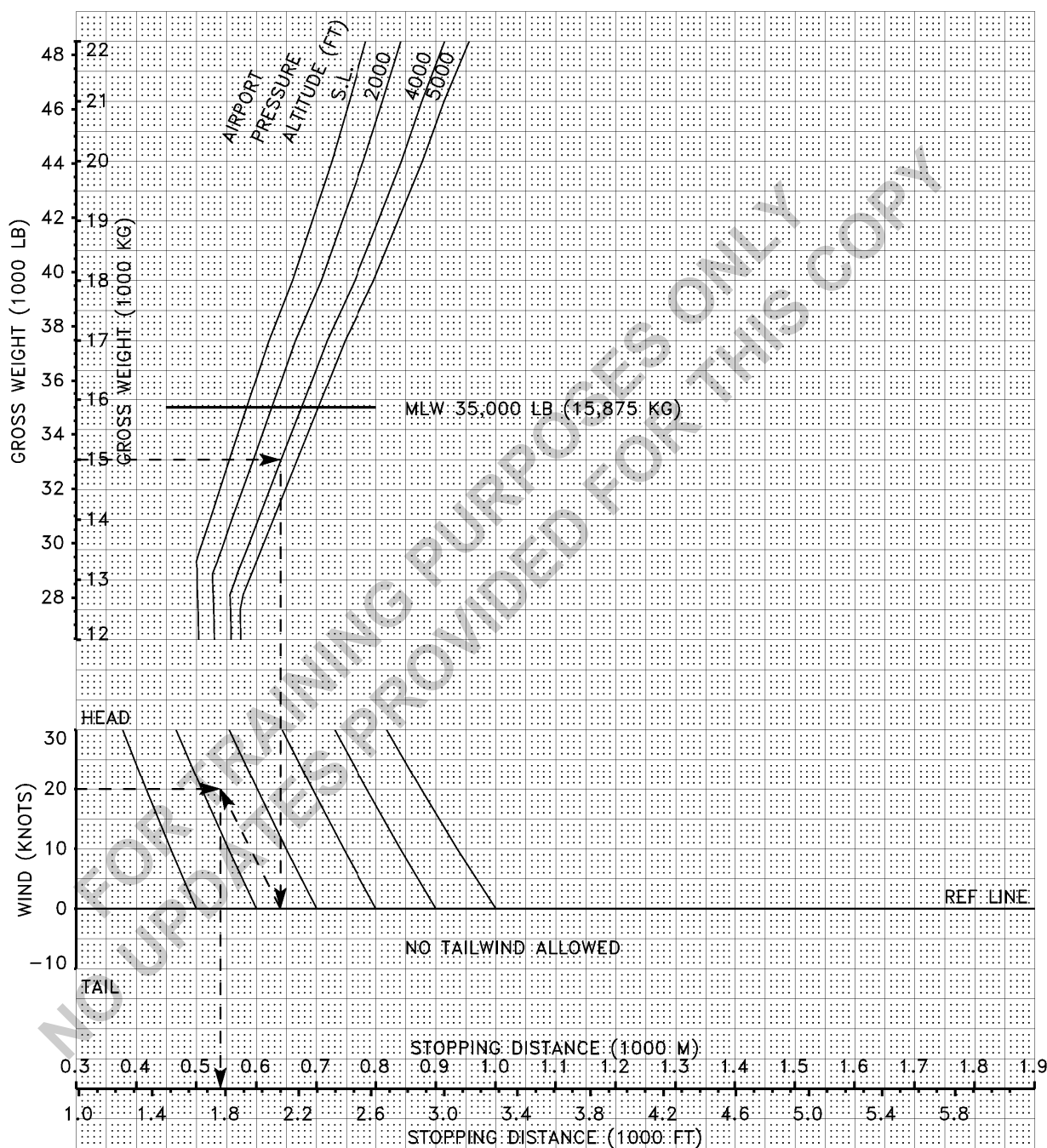
Associated Conditions:

Landing gross weight	= 15,000 kg (33,069 lb)
Airport pressure altitude	= 4,000 feet
Wind	= 20 knots, headwind

The example in Figure 07-12-2, shows that for the above conditions, the distance from glide path system position to full stop is 540 meters (1,772 feet).

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7. PERFORMANCE (CONT'D)

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Distance from Glideslope System Position on Runway to Full Stop
 Figure 07-12-2

8. SUPPLEMENTS

The Supplements in Chapter 7 are applicable except as modified by the following:

A. Supplement 19 – Operation on Wet Grooved Runways

(1) RUNWAY CONDITIONS

The data is applicable to concrete runway surfaces with transverse grooves constructed and maintained to FAA standard AC 150/5320-12C, or equivalent, and on which there are no areas of significant rubber deposit in sections where the high-speed portion of braking is expected.

(a) Wet Runway

A grooved runway is considered wet when there is sufficient moisture on the runway surface to cause it to appear partially reflective, but without significant areas of standing water (runway grooves not overflowing).

(2) LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

- The data may be used only for wet grooved runways.
- Non-performance related limitations and procedures that are applicable to a wet runway, are also applicable to a wet grooved runway.
- The use of this Supplement for landing performance requires both thrust reversers to be operational.
- The use of this Supplement for landing performance requires the anti-skid system to be fully operational.
- The use of this Supplement requires that :
 - the runway is maintained to FAA standard AC 150/5320-12C, or equivalent,
 - there are no areas of significant rubber deposit in sections where the high-speed portion of braking is to occur, and
 - there is no indication, via NOTAMs, PIREPs or braking action reports, that degraded braking exists on the runway.
- This Supplement is applicable only to grooved concrete surfaces.
- The use of a wet grooved runway corrected landing distance which is shorter than the dry runway landing field length is prohibited.

On EASA registered airplanes:

This Supplement is approved for use on concrete runways having a grooved surface that has been constructed and maintained to the FAA standard of AC 150/5320-12C or equivalent. Operational approval must be obtained in order to use the performance data presented in this Supplement.

The wet grooved landing distance determined during flight planning must account for the conditions expected at the time of arrival, including reported wind, runway slope and V_{REF} speed increment.

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		Rev 9, Jun 16/17

8. SUPPLEMENTS (CONT'D)

(3) EMERGENCY PROCEDURES

The emergency procedures in supplement 12 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

(4) ABNORMAL PROCEDURES

The abnormal procedures in supplement 12 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

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8. SUPPLEMENTS (CONT'D)

(5) PERFORMANCE

The performance in supplement 12 is applicable, except as modified by the following:

The landing distance required on a wet grooved runway for 5.5 degree approach operations for varying conditions of airplane weight, ambient temperature, airport altitude pressure, reported wind and runway slope is provided by Figure 07-12-3.

The landing field length from supplement 12 must also be determined when using the following data. If the landing field length on a dry runway is greater than the corrected landing distance on a wet grooved runway, use the landing field length.

The landing approach speed (V_{REF}) for varying airplane landing weights and airport pressure altitude for steep approach operations is also given by Figure 07-12-1.

The landing distance on wet grooved runways, for a flaps-45° landing, with both engines operating, for speeds at the runway threshold from V_{REF} to $V_{REF} + 10$ KIAS, is provided by Figure 07-12-3.

NOTE

The corrected landing distance on wet grooved runways includes a landing distance factor of 1.15 (15%).

Example:

Associated Conditions:

Landing gross weight	= 15,875 kg (35,000 lb)
Airport pressure altitude	= Sea Level
Ambient temperature	= 35°C
Wind	= 22 knots at 26 degrees relative to runway (20 knots headwind; 10 knots crosswind)
Runway slope	= +1%
ΔV_{REF}	= +5 KIAS

The example in Figure 07-12-3, shows that for the above conditions, the corrected landing distance is 1,074 meters (3,539 feet).

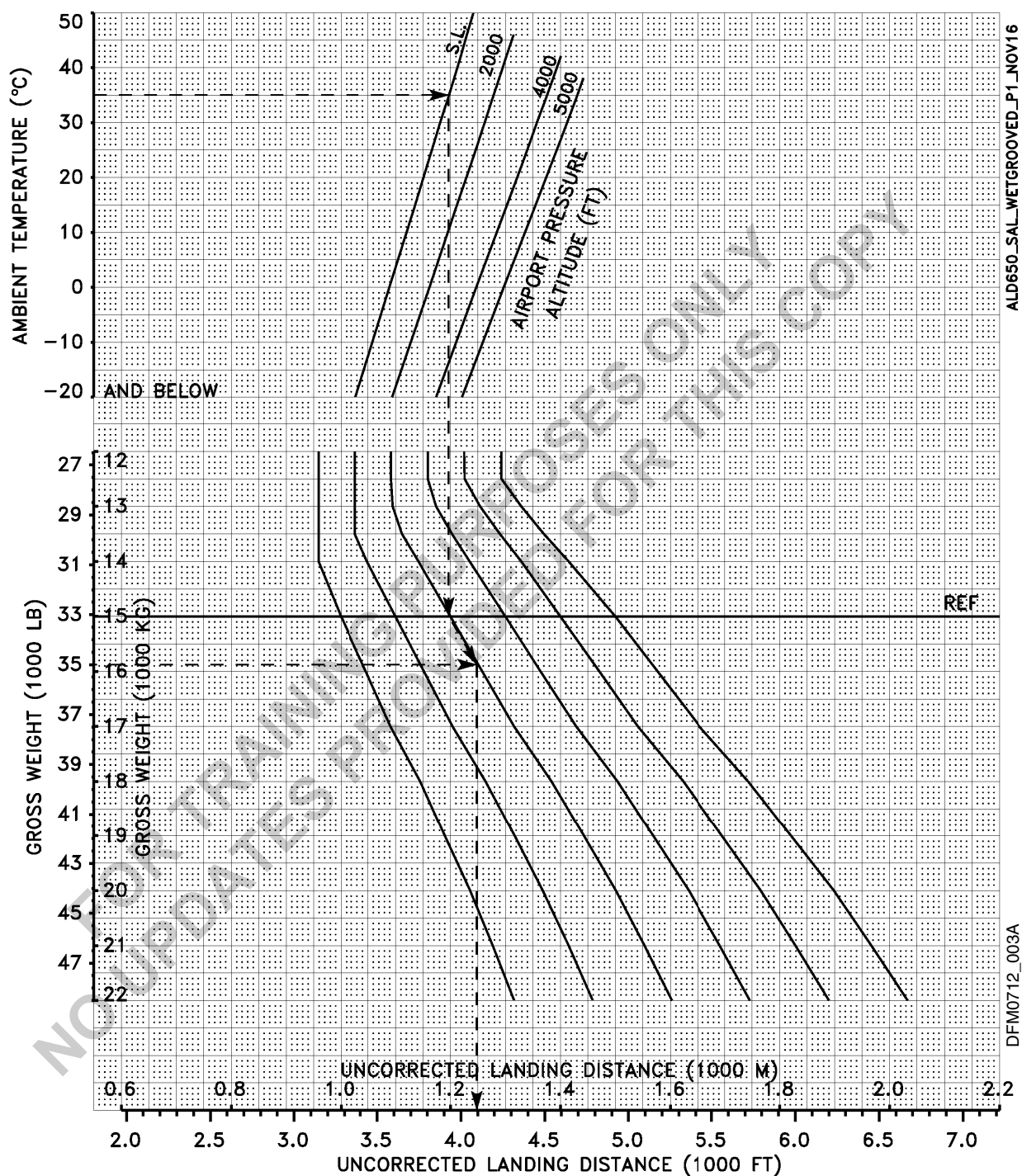
From the associated conditions stated above, the landing field length on a dry runway from Figure 07-12-1 is 1,087 meters (3,567 feet).

Since the corrected landing distance on a dry runway is greater than the landing field length on a wet grooved runway, the corrected landing distance on a dry runway must be used.

On EASA registered airplanes:

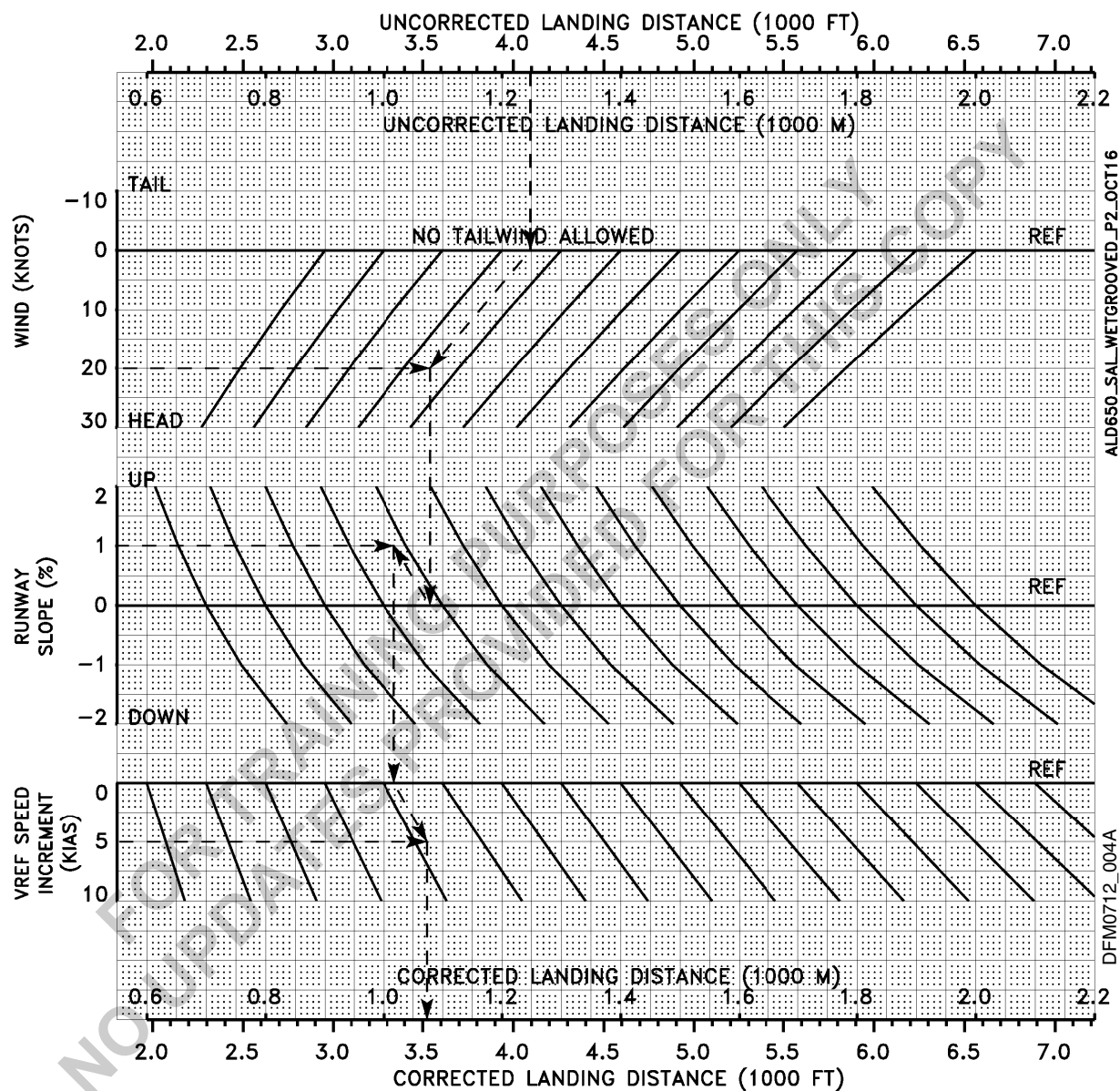
The wet grooved landing distance determined during flight planning must account for the conditions expected at the time of arrival, including reported wind, runway slope and V_{REF} speed increment.

8. SUPPLEMENTS (CONT'D)



Landing Distance – Wet Grooved Runway – Flap 45° (Page 1 of 2)
 Figure 07-12-3

8. SUPPLEMENTS (CONT'D)



Landing Distance – Wet Grooved Runway – Flap 45° (Page 2 of 2)
 Figure 07-12-3

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8. SUPPLEMENTS (CONT'D)

B. Supplement 20– FAA Steep Approach into London City Airport

Use of Supplement 20 is prohibited.

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CHALLENGER ⁶⁵⁰	SUPPLEMENT 12 Steep Approaches up to 5.5 Degrees	07-12-01-16
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1. GENERAL

This supplement presents take-off performance data in addition to the information contained in Chapter 6, Section 3; PERFORMANCE – TAKE-OFF PERFORMANCE.

This data provides additional flexibility relative to the Chapter 6 information by providing the required information to schedule the take-off decision speed V_1 such that the relevant performance requirements of a given airport are met.

Minimum and maximum values of V_1 are provided for a given set of take-off conditions (weight, altitude, temperature and runway slope). The minimum V_1 value is the greater of either a ratio of 0.8 of the take-off rotation speed (V_R) or the minimum V_1 limited by airplane controllability on the ground (V_{1MCG}). The maximum V_1 is the lesser of the take-off rotation speed (V_R) or the maximum V_1 limited by the brake system maximum kinetic energy rating (V_{1MBE}). Therefore, the Take-Off Weight Limited by Brake Energy chart in Chapter 6 is not applicable to this supplement.

The additional information allows the airplane to be operated to its fullest capabilities in situations where the resulting take-off weight is either limited by brake energy (Figure 06-03-2), or maximum tire speed (Figure 06-03-3), or by the adjustments associated with the extended second segment climb (Figure 06-03-12).

This supplement provides expanded data using the maximum V_1 limited by brake energy (V_{1MBE}) chart (Figure 07-13-6) and the actual runway length reduction associated with take-off weight limitations (all and one engine operating).

This supplement also provides an expanded chart for the maximum take-off weight limited by the tire speed (Figure 07-13-2) to better assess take-off weight limitations due to tire speed, especially during adjustments for extended second segment climb.

Balanced take-off field length data, which is contained in Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, presents take-off field length where the data presented is the greater of either the factored all engines operating take-off distance, the one engine inoperative take-off distance, the all engines operating accelerate-stop distance or the one engine inoperative accelerate-stop distance. The pilot does not know which is the most limiting case.

The scheduling of the V_1 is associated with the unbalanced take-off field length data contained in this supplement, which presents information for both all engines operating take-off field length and one engine inoperative take-off/accelerate-stop distance. The pilot must determine which is the most limiting case. This data also permits a clearway to be used to achieve the required 35 foot height on climbout and a stopway to be used to stop the aircraft.

Stopway

A stopway is defined as a rectangular area on the ground at the end of the runway, in the direction of take-off, prepared as a suitable area in which an airplane can be stopped in the case of a rejected take-off.

Clearway

A clearway is defined as a rectangular area on the ground under the control of the appropriate authority, selected or prepared as a suitable area over which an airplane may make a portion of the initial climb to a specified height.

The data in this supplement also allows operators to take credit for clearways and stopways when determining the take-off weight limited by field length requirements. Another use of the performance data of this supplement is to reduce the (unbalance) take-off decision speed V_1 to the point where the airplane would not be limited by its brake kinetic energy capability while still meeting both accelerate-go and accelerate-stop distances.

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1. GENERAL (CONT'D)

FMS take-off Vs speeds (V_1 , V_R , V_2 and V_{FTO}) must be manually entered and selected for view on the displays.

NOTE

The TAKEOFF function of the FMS Vspeed does not support the calculations associated with this supplement. Vspeed, maximum take-off weight and take-off distance must be derived from this supplement.

The CHECK TAKEOFF PERF message may be posted on the FMS CDU.

This supplement is an all-inclusive replacement for the take-off performance section of Chapter 6, Section 3 only.

The remaining sections of Chapter 6 are to be consulted for the remaining areas of performance.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (8).

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2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable with the addition of the following:

A. Flight Management System

FMS take-off Vspeed data must be manually entered when using data provided in this supplement.

B. Supplement 2– Operation on Wet and Contaminated Runways

Wet data is available. No unbalanced data is available when operating on a contaminated runway.

C. Supplement 3– Operation at Airport Elevations up to 14,000 Feet

Take-off performance is limited to 10,000 feet maximum airport pressure altitude.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable with the addition of the following:

FMS take-off Vspeeds (V_1 , V_R , V_2 and V_{FTO}) must be manually entered and selected for view on the displays.

NOTE

The TAKEOFF function of the FMS Vspeed does not support the calculations associated with this supplement. Vspeed, maximum take-off weight and take-off distance must be derived from this supplement.

The CHECK TAKEOFF PERF message may be posted on the FMS CDU.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. PERFORMANCE

The performance in Chapter 6 is applicable, except as modified by the following:

FMS take-off Vspeeds (V_1 , V_R , V_2 and V_{FTO}) must be manually entered and selected for view on the displays.

NOTE

The TAKEOFF function of the FMS Vspeed does not support the calculations associated with this supplement. Vspeed, maximum take-off weight and take-off distance must be derived from this supplement.

The CHECK TAKEOFF PERF message may be posted on the FMS CDU.

A. Take-Off Performance

The Take-Off Performance section contains the charts used to obtain the take-off performance data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified take-off configurations.

(1) MAXIMUM ALLOWABLE TAKE-OFF WEIGHT

The maximum allowable take-off weight is limited by the most restrictive of the following:

- Maximum approved take-off weight
(Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT),
- Runway length available,
- Climb requirements,
- Obstacle clearance requirements,

(First, verify if the second segment of the take-off flight path extends beyond a pressure height of 1,500 feet above the airport and consider if potential adjustments are to be made.)

NOTE

If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10) as applicable before proceeding.

- Maximum demonstrated brake energy,
- Maximum tire speed,
- Wheel brake cooling limitations

(Refer to Chapter 6; PERFORMANCE – TURN-AROUND TIME).

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CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-5
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7. PERFORMANCE (CONT'D)

(a) Take-Off Weight Limited By Climb Requirements – Flaps 20°

The maximum take-off weight limited by climb requirements for a flaps-20° take-off is determined from Figure 07-13-1, for varying conditions of ambient temperature and airport pressure altitude, taking into account the effects of different combinations of wing and cowl anti-icing and 10th stage bleeds, and also the use of the APR system.

The weights presented in Figure 07-13-1 are limited by second segment climb gradient requirements, as per FAR 25.121 (b). The first segment climb and final segment climb gradient requirements, as per FAR 25.121 (a) and FAR 25.121 (c), respectively, are not limiting.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 07-13-1, Anti-ice On), the maximum take-off weight limited by climb requirements is 21,600 kg (47,620 lb). The take-off weight in this case may be less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and may be considered limiting.

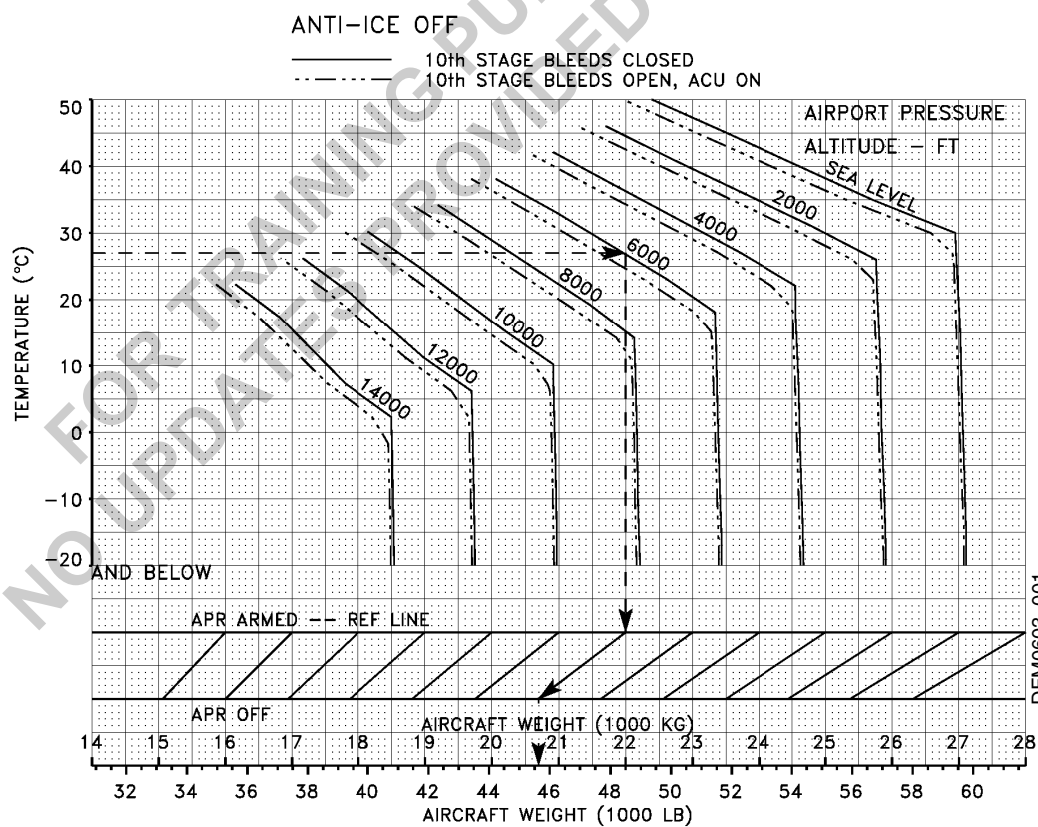
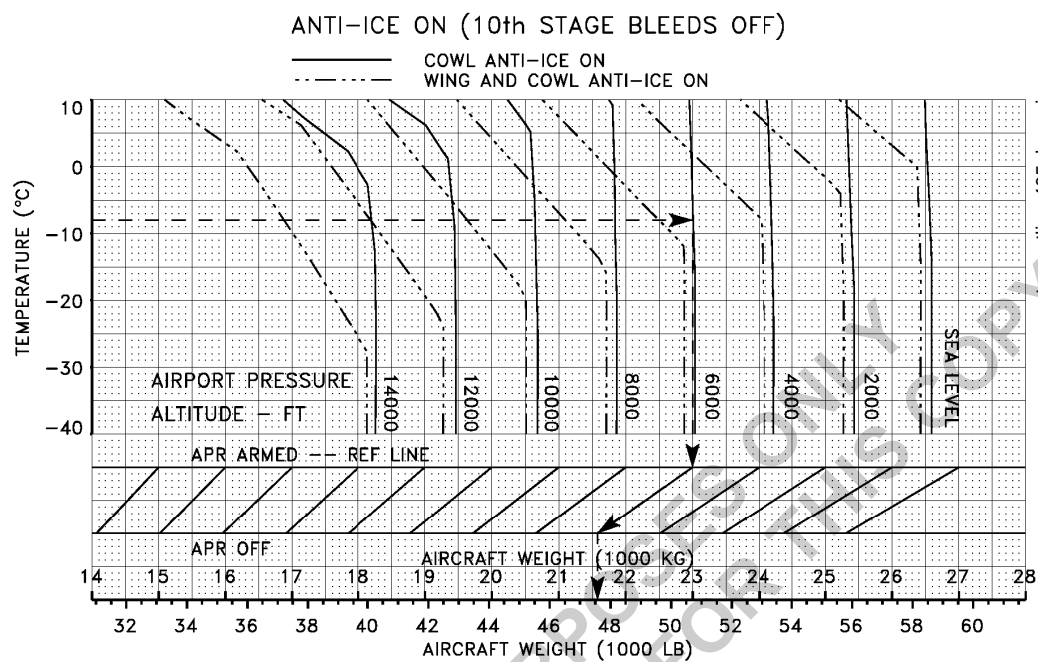
Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 07-13-1, Anti-ice Off), the maximum take-off weight limited by climb requirements is 20,700 kg (45,636 lb). The take-off weight in this case is less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is considered limiting.

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7. PERFORMANCE (CONT'D)


Take-Off Weight Limited by Climb Requirements – Flaps 20°
 Figure 07-13-1

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7. PERFORMANCE (CONT'D)

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The maximum take-off weight limited by climb requirements for a flaps-20° take-off is determined from Figure 07-13-1, for varying conditions of ambient temperature and airport pressure altitude, taking into account the effects of different combinations of wing and cowl anti-icing and 10th stage bleeds, and also the use of the APR system.

For take-offs with APR off, the weights presented in Figure 07-13-1 are limited by second segment climb gradient requirements, as per FAR 25.121 (b). For take-offs with APR armed, the weights presented in Figure 07-13-1 are limited by approach climb gradient requirements, as per FAR 25.121 (d), following a 15 minute flight in accordance with the requirements of FAR 25.1001 (a). The first segment climb and final segment climb gradient requirements, as per FAR 25.121 (a) and FAR 25.121 (c), respectively, are not limiting.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Cowl anti-ice	= On
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 07-13-1, Anti-ice On), the maximum take-off weight limited by climb requirements is 21,700 kg (47,840 lb). The take-off weight in this case may be less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and may be considered limiting.

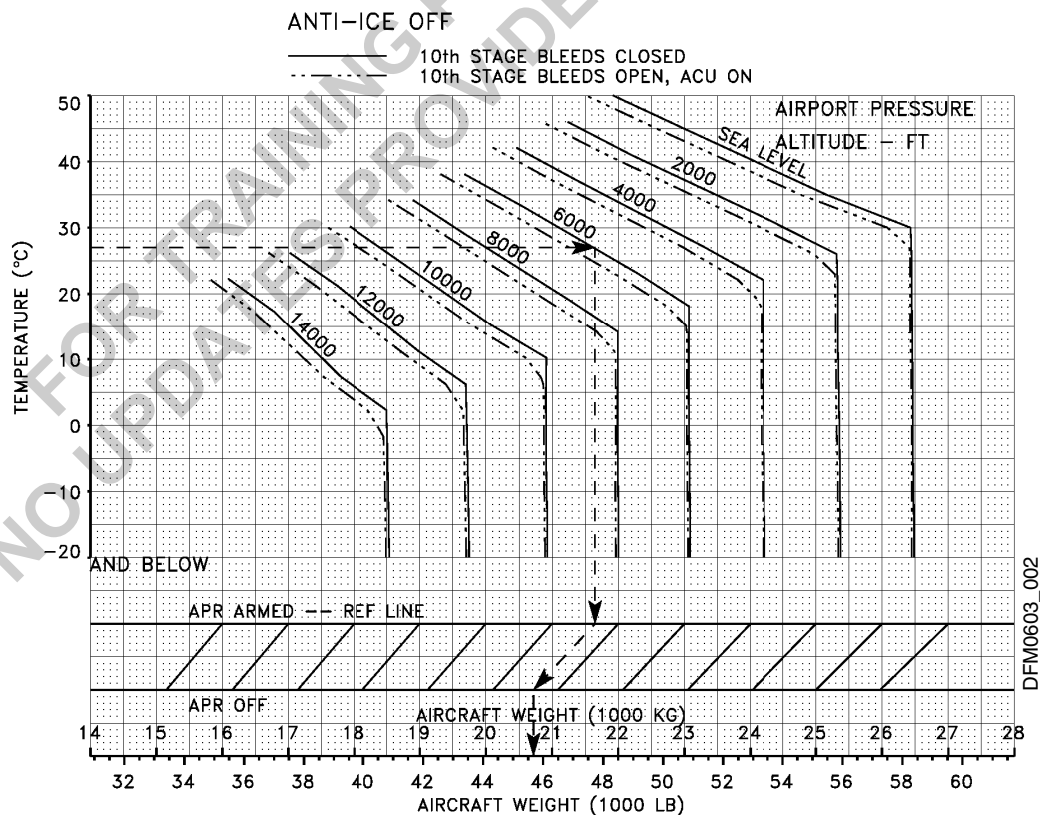
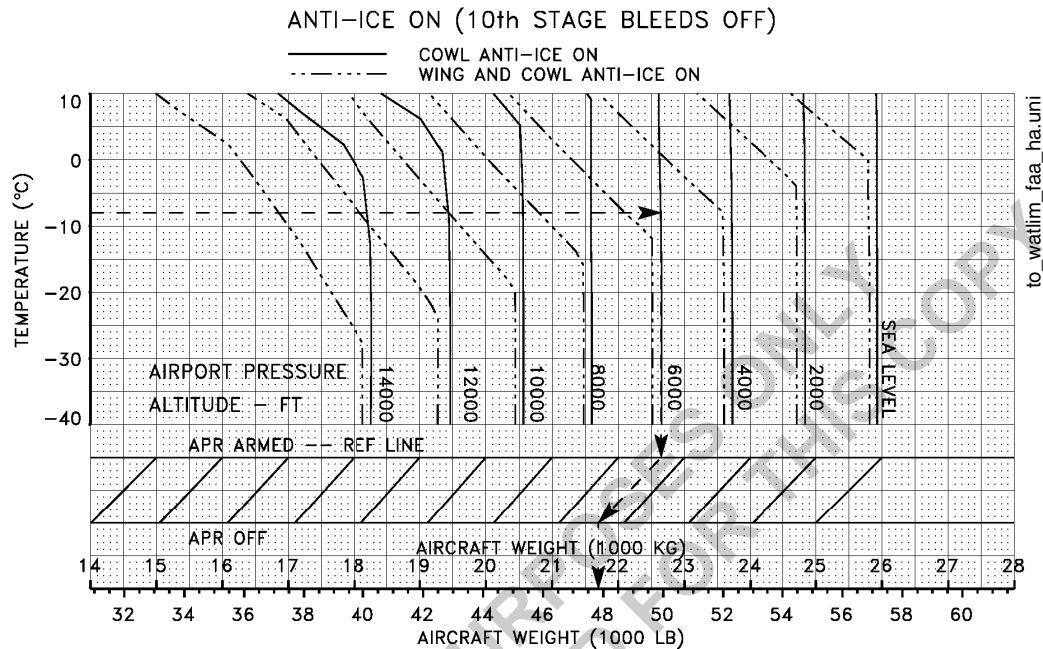
Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Off

As shown in the example (Figure 07-13-1, Anti-ice Off), the maximum take-off weight limited by climb requirements is 20,700 kg (45,636 lb). The take-off weight in this case is less than the maximum approved take-off weight (Refer to Chapter 2; LIMITATIONS – STRUCTURAL WEIGHT) and is considered limiting.

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7. PERFORMANCE (CONT'D)**FAA Certified Airplanes**

Take-Off Weight Limited by Climb Requirements – Flaps 20°
 Figure 07-13-1

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7. PERFORMANCE (CONT'D)

(b) Take-Off Weight Limited by Maximum Tire Speed

The maximum take-off weight limited by the maximum tire speed of 210 mph (182 knots), for a flaps-20° take-off, in varying conditions of ambient temperature, airport pressure altitude and wind, is given by Figure 07-13-2.

The weights derived from Figure 07-13-2 are applicable for all engine bleed configurations.

NOTE

If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

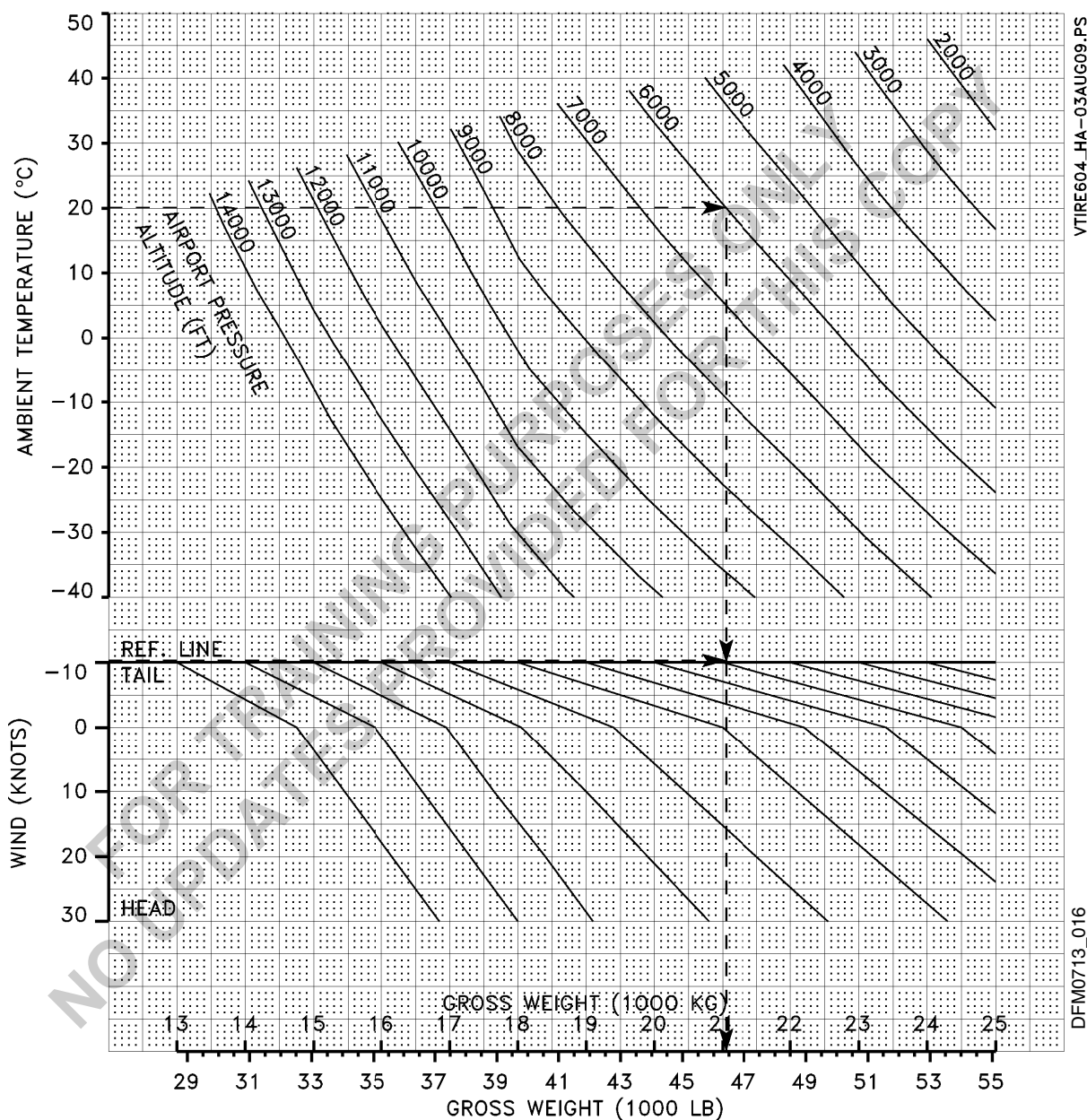
Figure 07-13-2 shows that at an ambient temperature of 20°C, pressure altitude of 6,000 feet, a tailwind of 10 knots and APR on, the take-off weight limited by maximum tire speed is 21,050 kg (46,408 lb).

NOTE

If take-off is performed with the APR off, the weight derived from Figure 07-13-2 must be reduced by 200 kg (440 lb).

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7. PERFORMANCE (CONT'D)

Take-Off Weight Limited by Maximum Tire Speed
 Figure 07-13-2

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7. PERFORMANCE (CONT'D)

(c) Take-Off Weight Limited by Field Length Requirements – One Engine Inoperative

The maximum take-off weight limited by field length requirements, with one engine inoperative, for a flaps-20° take-off, is given by Figure 07-13-3. The following charts take into account the accelerate-stop distance available including the stopway, the actual length of the runway and the clearway, the effects of runway slope, and prevailing wind conditions, for varying conditions of ambient temperature, airport pressure altitude, APR usage and various bleed configurations.

NOTE

If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

Associated Conditions:

Actual runway length	= 3,048 meters (10,000 feet)
Available accelerate-stop distance	= 3,048 meters (10,000 feet)
Clearway	= 305 meters (1,000 feet)
Runway slope	= -1%
Wind	= 10 knots tailwind
Temperature	= 20°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed
APR	= Armed

Page 1 of Figure 07-13-3 is used to obtain Reference A values. Enter the chart from the actual runway length then apply the correction for maximum allowable clearway, runway slope and reported wind to obtain a Reference A value.

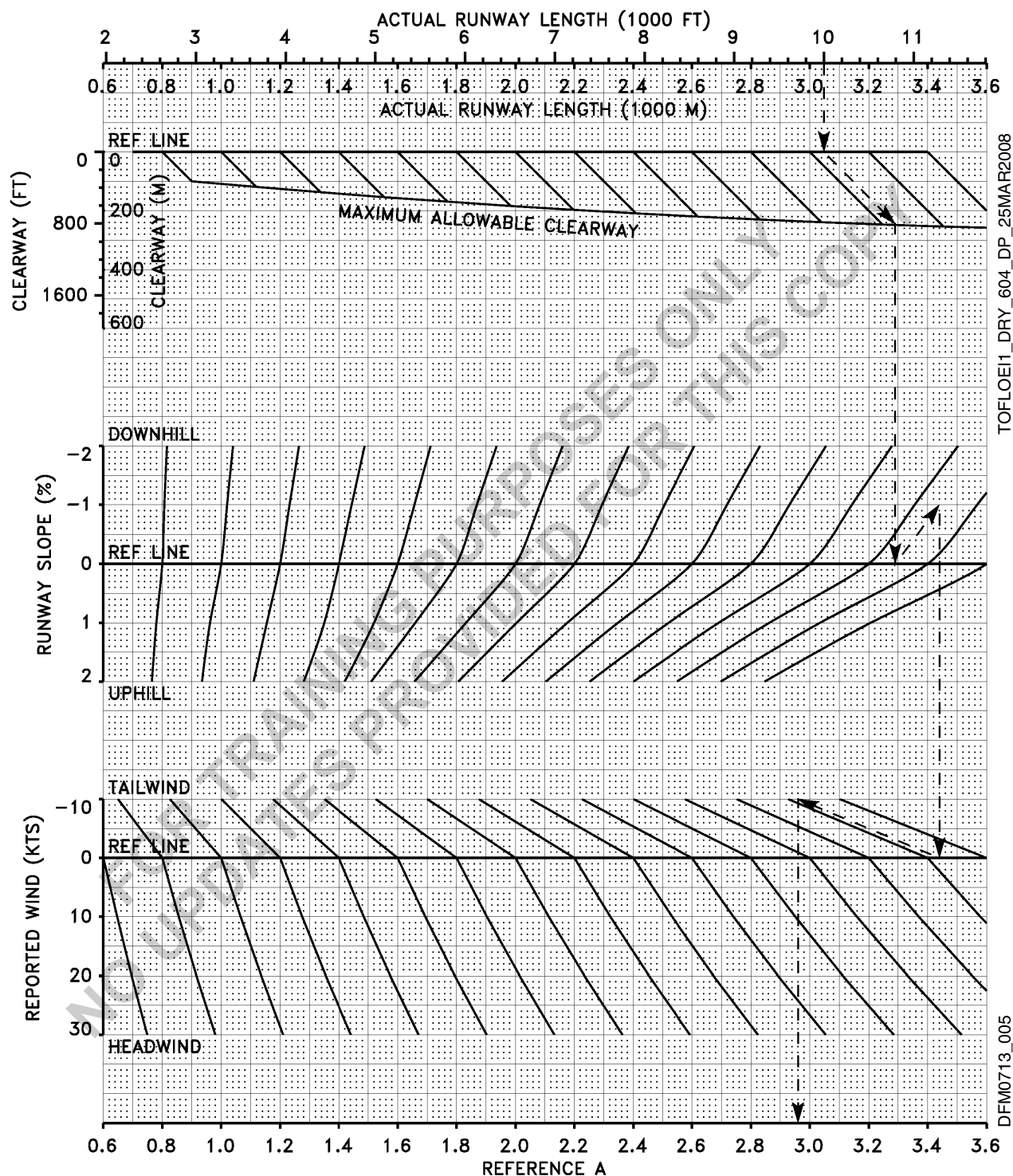
Page 2 of Figure 07-13-3 is used to obtain Reference B values. Enter the chart from the available accelerate-stop distance including the stopway, then apply the correction for runway slope and reported wind to obtain a Reference B value.

Page 3 of Figure 07-13-3 is used to obtain V_1/V_R ratio and Reference C value. Enter the chart using the Reference A and Reference B values to obtain a Reference C value and a V_1/V_R ratio of 0.97 as shown on the example.

Page 4 of Figure 07-13-3 is used to obtain Reference D values. Enter the chart at the temperature scale below the appropriate anti-ice and 10th stage bleed configuration to arrive at the Reference D value.

Finally, page 5 of Figure 07-13-3 is used to obtain the airplane take-off weight. Enter the chart at the Reference D scale and the Reference C value as shown on the example, to arrive at the take-off weight of 20,560 kg (45,327 lb).

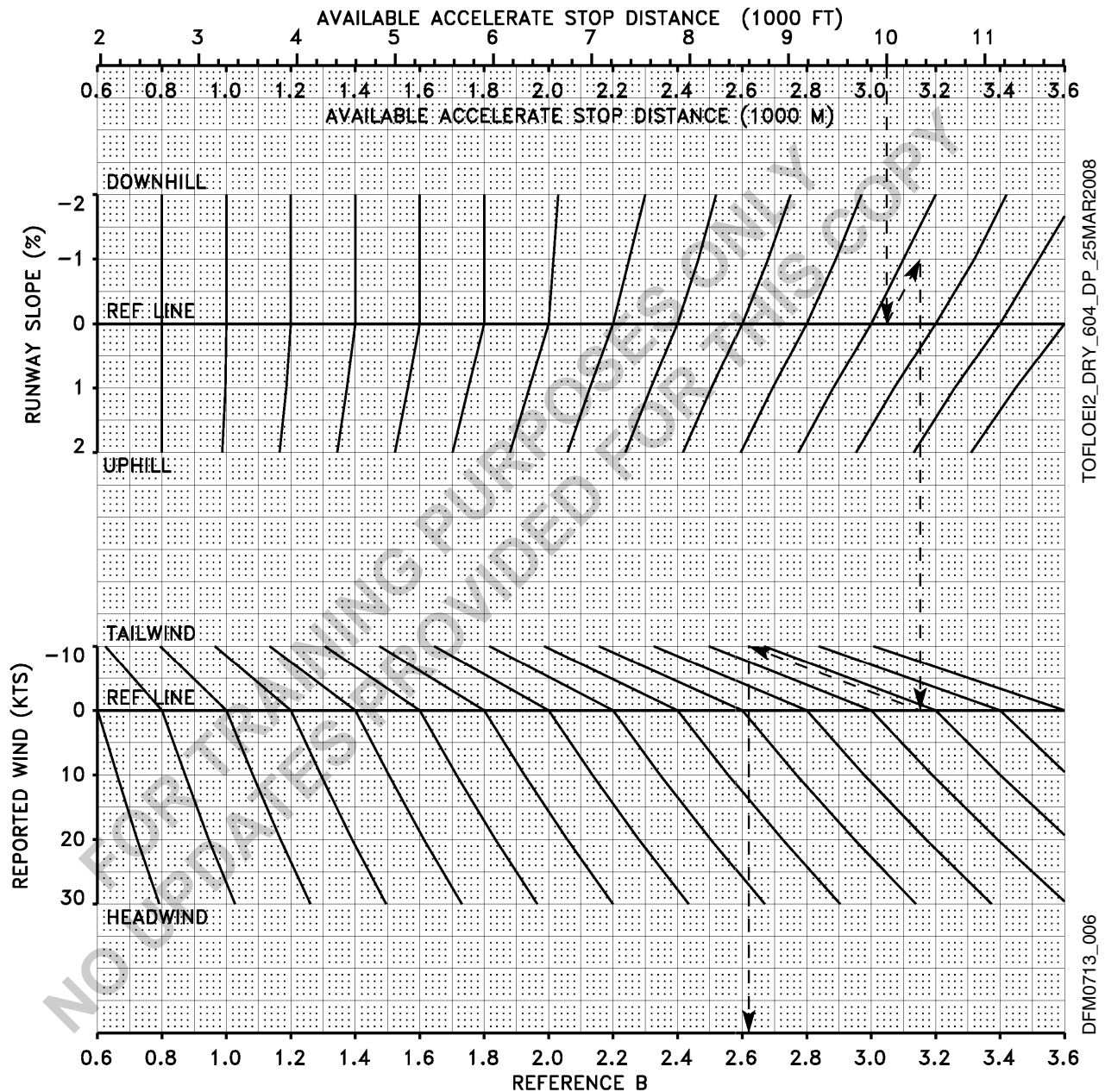
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7. PERFORMANCE (CONT'D)

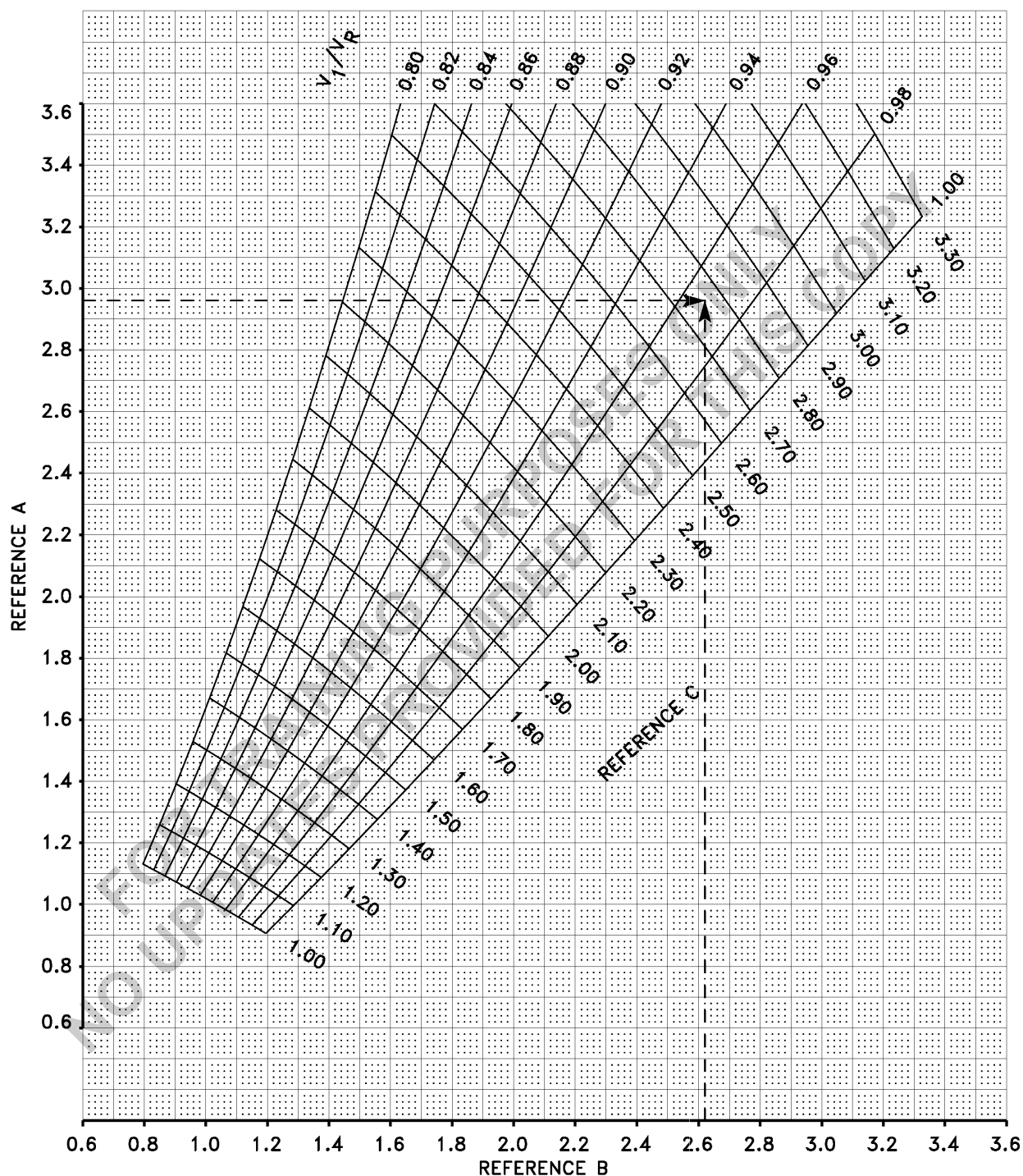
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Take-Off Weight Limited by Field Length Requirements (One Engine Inoperative) (Page 1 of 5)
 Figure 07-13-3

7. PERFORMANCE (CONT'D)

Take-Off Weight Limited by Field Length Requirements (One Engine Inoperative) (Page 2 of 5)
 Figure 07-13-3

7. PERFORMANCE (CONT'D)

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Take-Off Weight Limited by Field Length Requirements (One Engine Inoperative) (Page 3 of 5)
 Figure 07-13-3

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7. PERFORMANCE (CONT'D)

(d) Take-Off Weight Limited by Field Length Requirements – All Engines Operating

The maximum take-off weight limited by field length requirements, with both engines operating, for a flaps-20° take-off, is given by Figure 07-13-4. The following charts take into account the actual length of the runway and the clearway, the effects of runway slope, and prevailing wind conditions, for varying conditions of ambient temperature, airport pressure altitude, and various bleed configurations.

NOTE

If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

Associated Conditions:

Actual runway length	= 3,048 meters (10,000 feet)
Clearway	= 305 meters (1,000 feet)
Runway slope	= -1%
Wind	= 10 knots tailwind
Temperature	= 20°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

Page 1 of Figure 07-13-4 is used to obtain Reference A values. Enter the chart from the actual runway length then apply the correction for maximum allowable clearway, runway slope and reported wind to obtain a Reference A value.

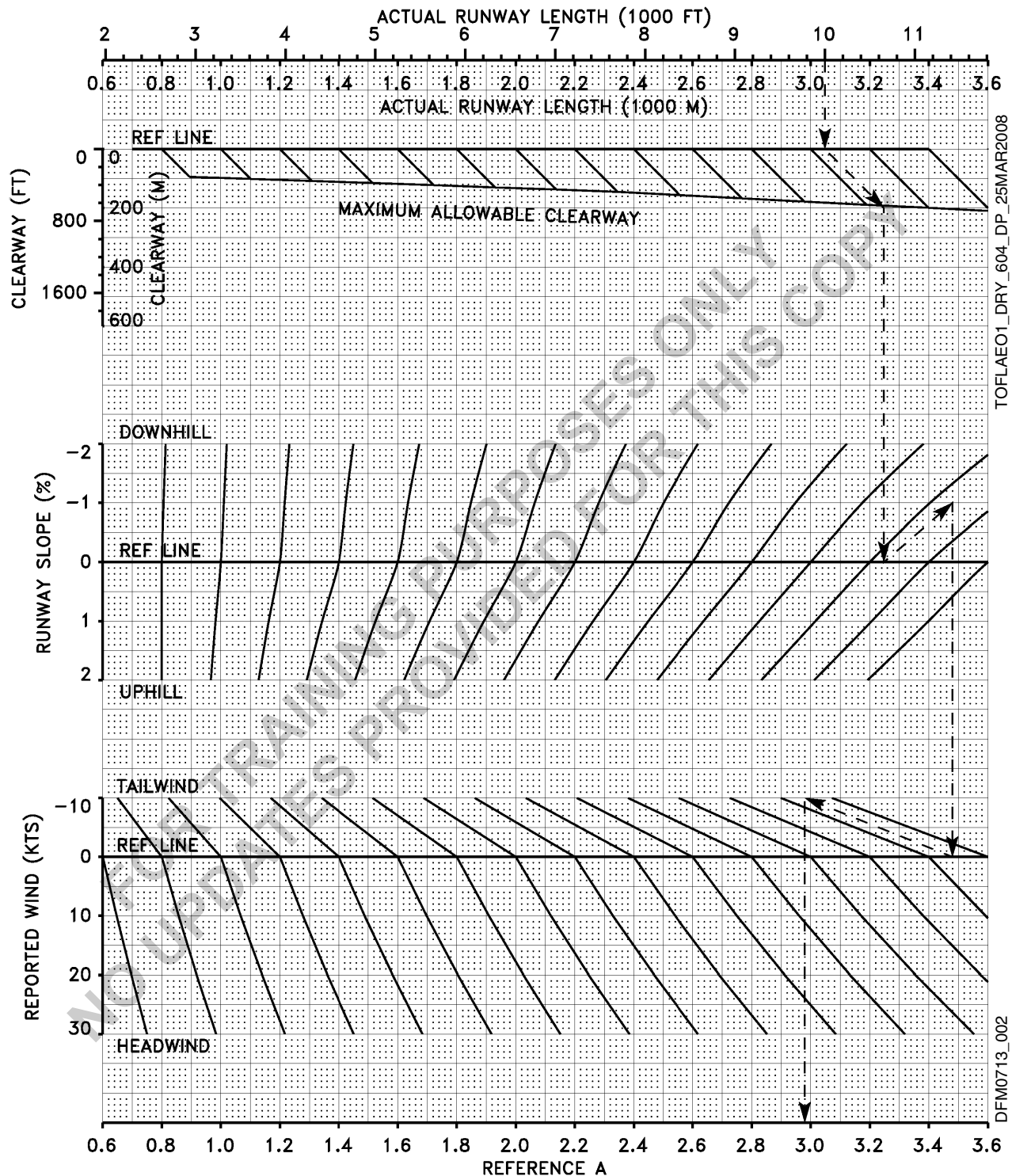
Page 2 of Figure 07-13-4 is used to obtain Reference B values. Enter the chart at the temperature scale below the appropriate anti-ice and 10th stage bleed configuration to arrive at the Reference B value.

Finally, page 3 of Figure 07-13-4 is used to obtain the airplane take-off weight. Enter the chart at the Reference B scale and the Reference A value as shown on the example, to arrive at the take-off weight of 22,200 kg (48,943 lb).

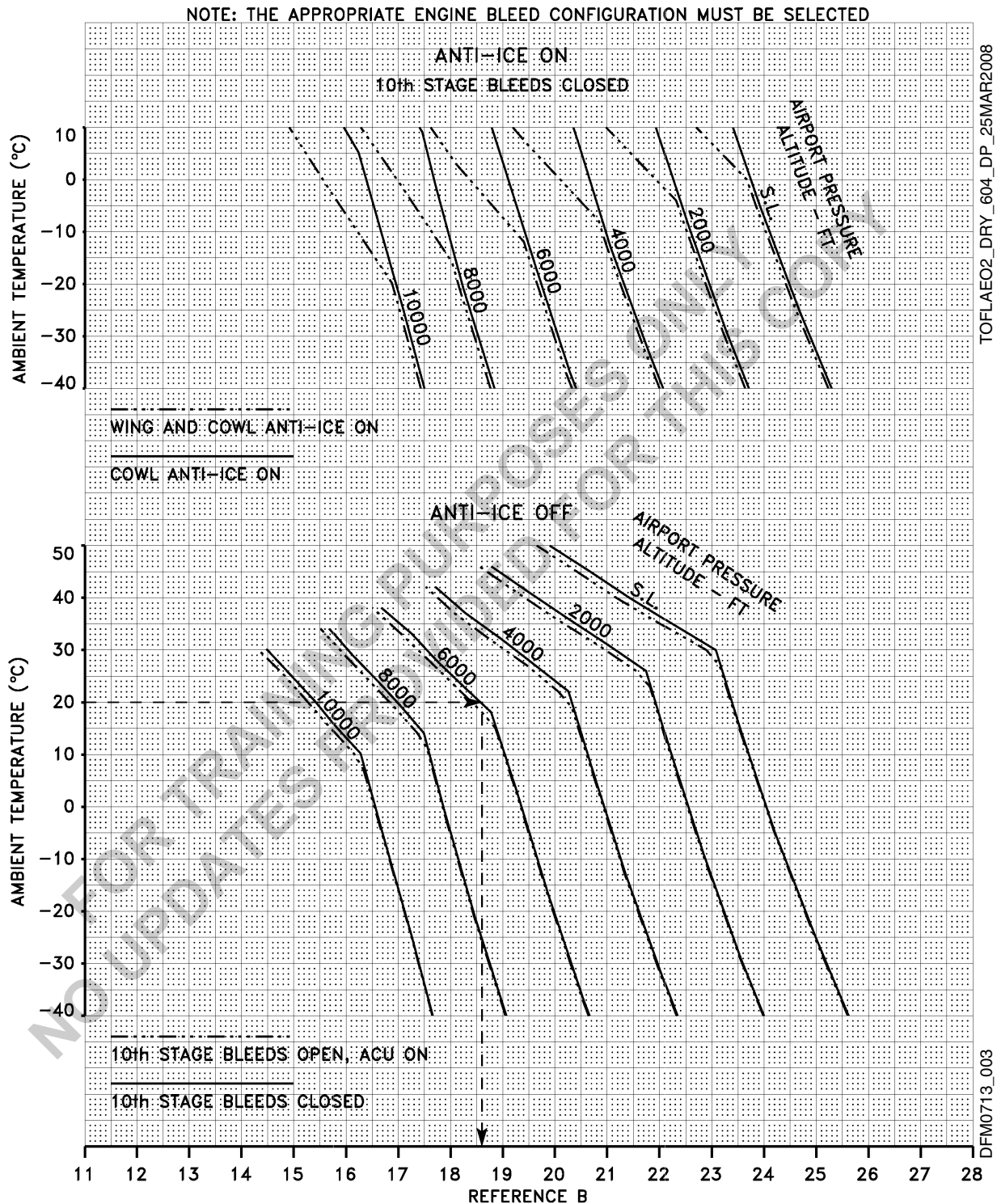
NOTE

The take-off weight limited by field length requirements on a dry runway is the lowest of the take-off weights obtained for the one engine inoperative and the all engines operating conditions.

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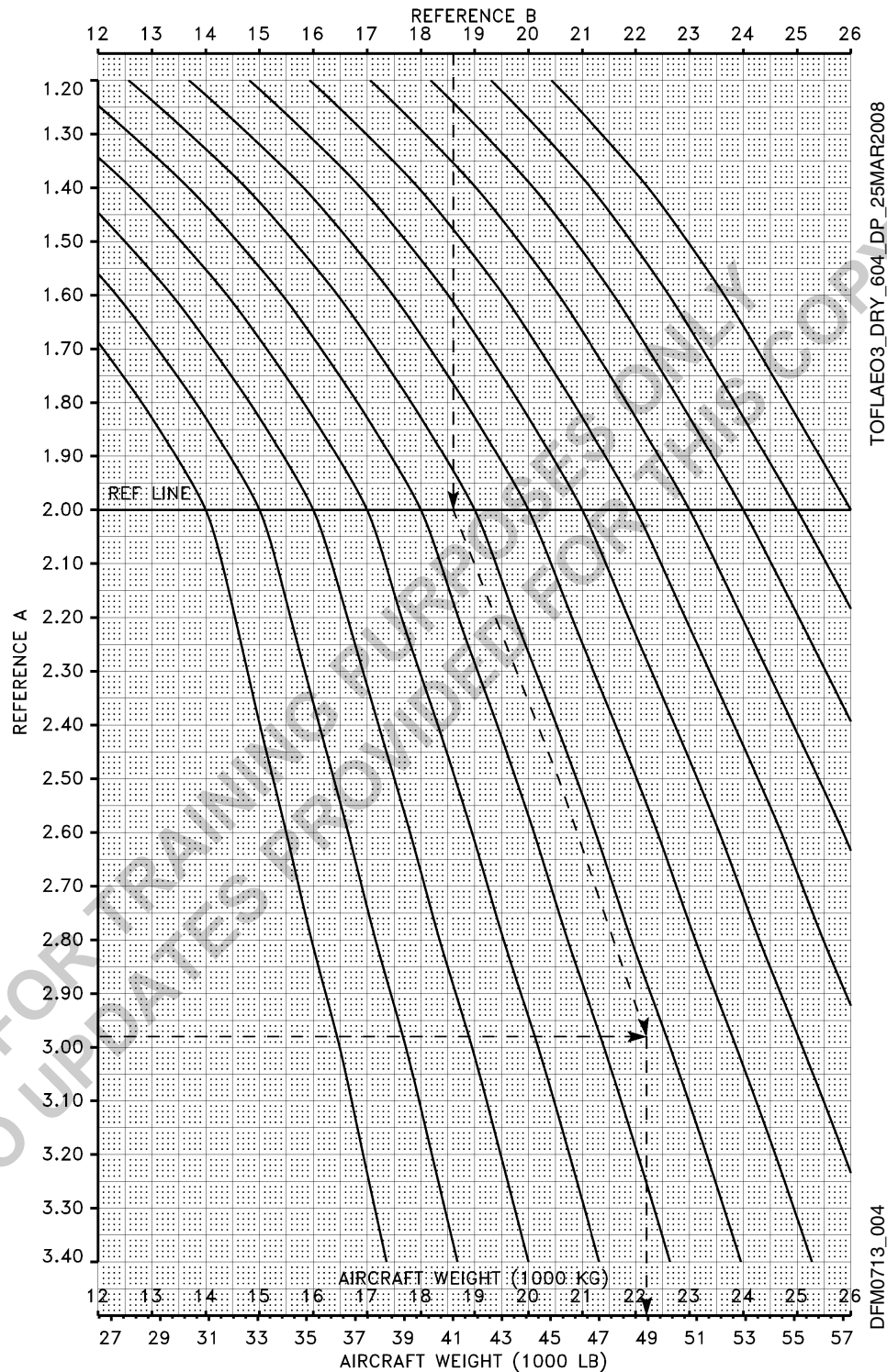
7. PERFORMANCE (CONT'D)

Take-Off Weight Limited by Field Length Requirements (All Engines Operating) (Page 1 of 3)
 Figure 07-13-4

7. PERFORMANCE (CONT'D)

Take-Off Weight Limited by Field Length Requirements (All Engines Operating) (Page 2 of 3)
 Figure 07-13-4

7. PERFORMANCE (CONT'D)



Take-Off Weight Limited by Field Length Requirements (All Engines Operating) (Page 3 of 3)
Figure 07-13-4

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7. PERFORMANCE (CONT'D)

(2) TAKE-OFF SPEEDS

The following charts are used to determine the take-off speeds (V_1 , V_R and V_2).

V_1 is obtained by multiplying V_R by V_1/V_R . The value of V_1/V_R is taken from Figure 07-13-3 or Figure 07-13-12, during the determination of the take-off weight limited by field length requirements with one engine inoperative.

V_1 can not be less than V_{1MCG} nor greater than V_{1MBE} . Otherwise, V_1 must be reduced or increased, and the resulting V_1/V_R ratio changes, causing a change in the take-off weight limited by field length requirements.

(a) Minimum V_1 Limited By Control on the Ground – V_{1MCG}

Figure 07-13-5 is used to determine the minimum V_1 limited by control on the ground (V_{1MCG}) for varying conditions of take-off weight, ambient temperature, airport pressure altitude and runway slope.

The V_{1MCG} speeds derived from Figure 07-13-5 are applicable for all engine bleed configurations.

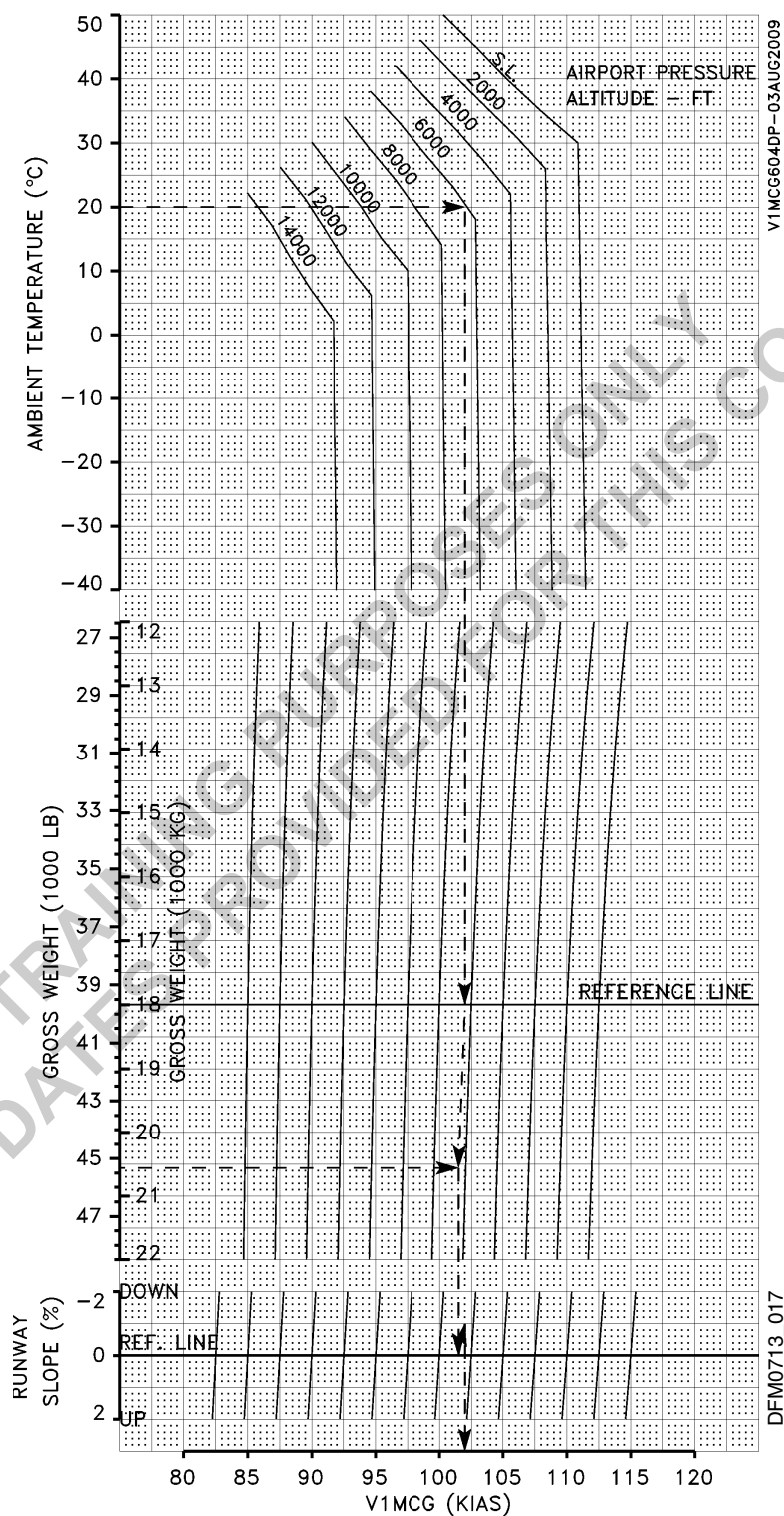
Example:

Associated Conditions:

Take-off weight	= 20,560 kg (45,327 lb)
Airport pressure altitude	= 6,000 feet
Ambient temperature	= 20°C
Runway slope	= -1%

The example in Figure 07-13-5 shows that for the above associated conditions, V_{1MCG} is equal to 102.0 KIAS.

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7. PERFORMANCE (CONT'D)


Minimum V_1 Limited by Control on the Ground – V_{1MCG}
 Figure 07-13-5

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7. PERFORMANCE (CONT'D)

(b) Maximum V_1 Limited By Brake Energy – V_{1MBE}

Figure 07-13-6 is used to determine the maximum V_1 limited by brake energy (V_{1MBE}) for varying conditions of take-off weight, ambient temperature, airport pressure altitude, prevailing wind conditions and runway slope.

NOTE

The V_{1MBE} speed derived from Figure 07-13-6 is for an airplane weight of 22,000 kg (48,500 lb). For weights lower than 22,000 kg (48,500 lb), increase the V_{1MBE} derived by this chart by 4 knots per 1,000 kg (2,200 lb).

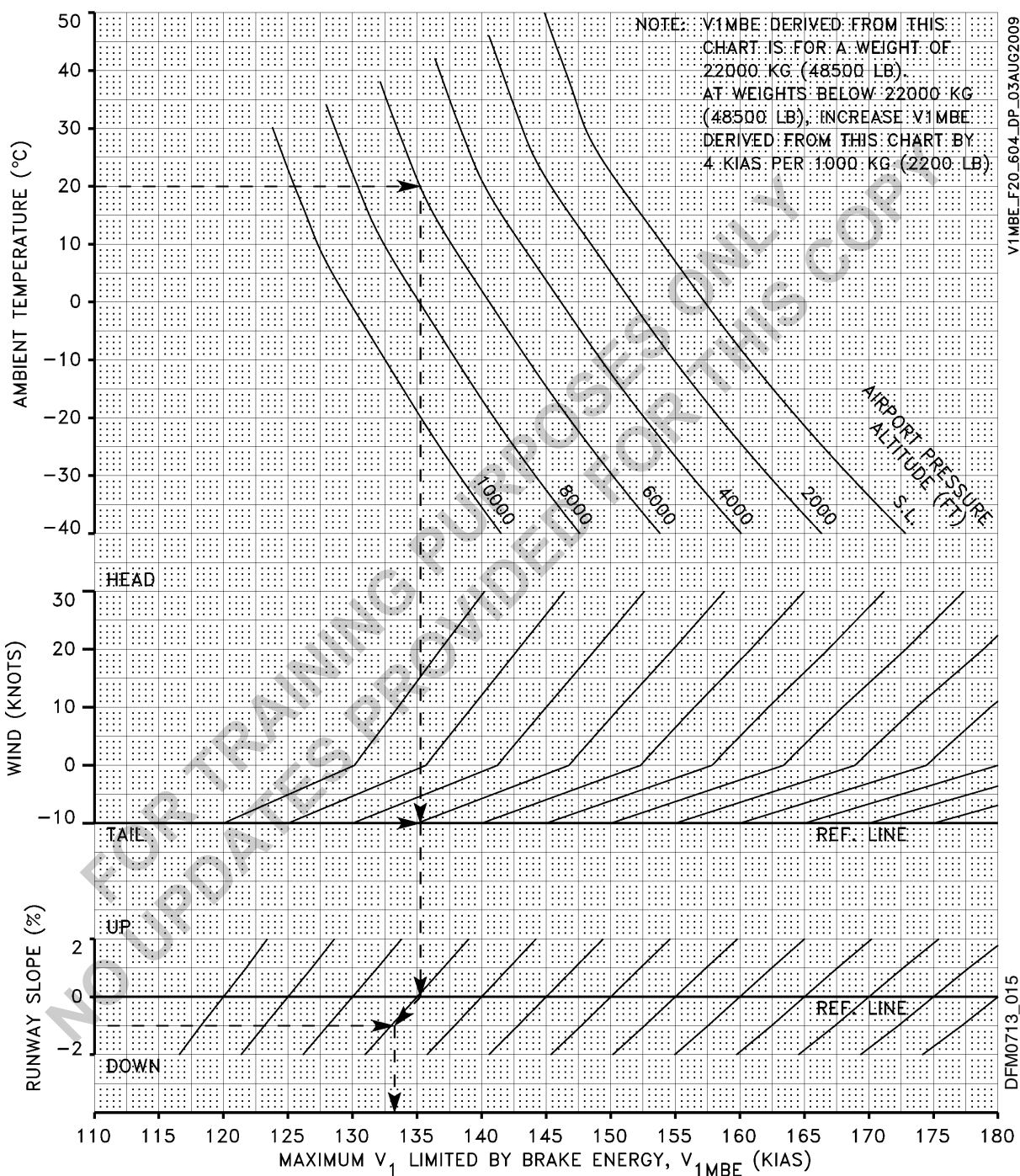
Example:

Associated Conditions:

Take-off weight	= 20,560 kg (45,327 lb)
Airport pressure altitude	= 6,000 feet
Ambient temperature	= 20°C
Wind	= 10 knots tailwind
Runway slope	= -1%

The example in Figure 07-13-6 shows that for the above associated conditions, V_{1MBE} is equal to:

$$\begin{aligned}
 V_{1MBE} &= 133 + [4 \times (22,000 - 20,560) \div 1,000] \\
 &= 139 \text{ KIAS}
 \end{aligned}$$

7. PERFORMANCE (CONT'D)


Maximum V₁ Limited by Brake Energy – V_{1MBE}
 Figure 07-13-6

CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-25
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7. PERFORMANCE (CONT'D)

(c) Rotation Speed, V_R , Flaps 20° – Anti-Icing Off

Figure 07-13-7 is used to determine the rotation speed (V_R) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude, bleed configurations, take-off weight, runway slope and use of APR.

V_R must not be less than 121 KIAS.

NOTE

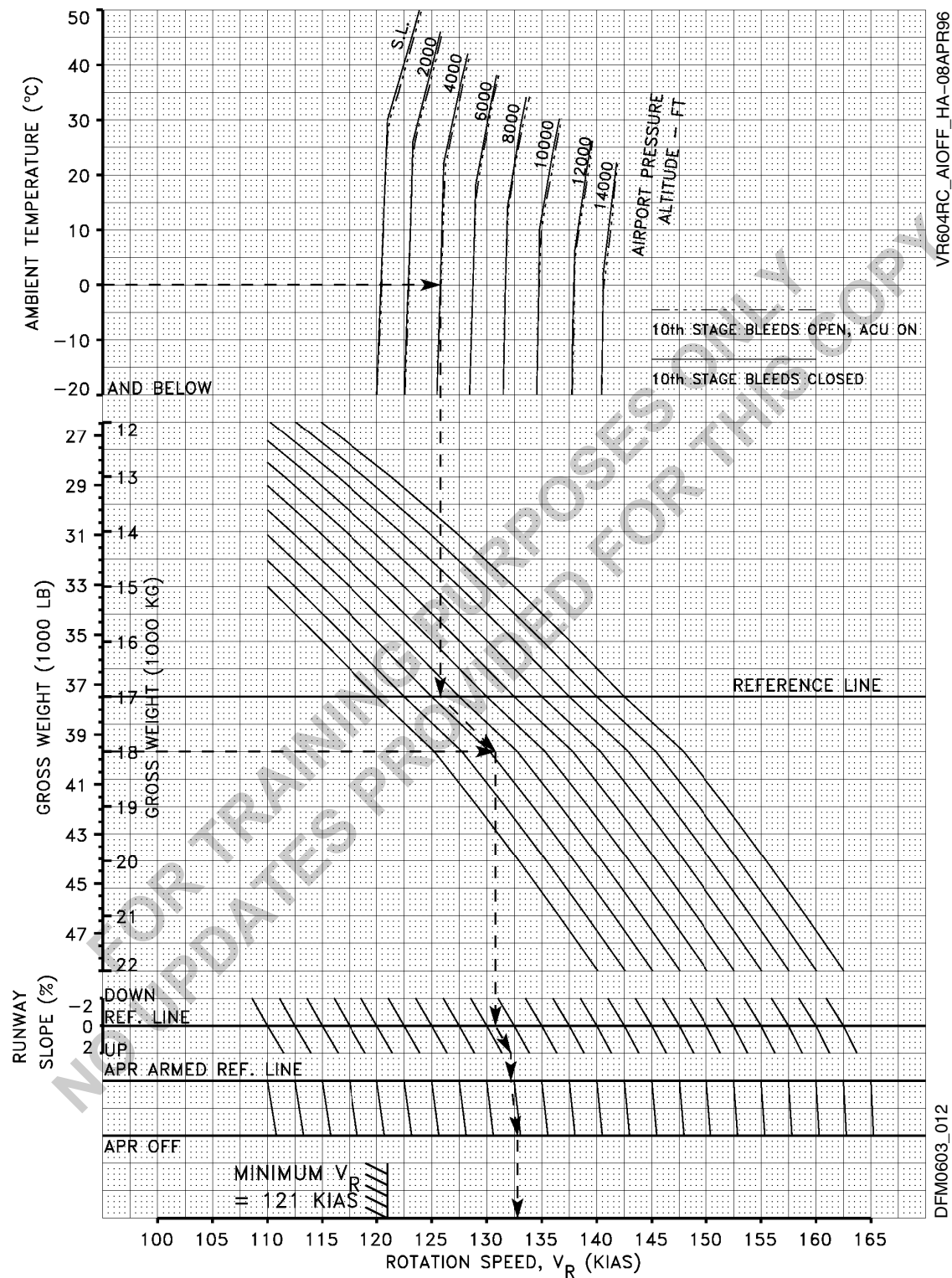
If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

Associated Conditions:

Ambient temperature	= 0°C
Airport pressure altitude	= 4,000 feet
10th stage bleeds	= Closed
Take-off weight	= 18,000 kg (39,683 lb)
Runway slope	= +2%
APR	= Off

The example in Figure 07-13-7 shows that for the above associated conditions, the rotation speed is 133 KIAS.

7. PERFORMANCE (CONT'D)

Rotation Speed, V_R , Flaps 20° – Anti-Icing Off
 Figure 07-13-7

CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-27
		Oct 27/15

7. PERFORMANCE (CONT'D)

(d) Rotation Speed, V_R , Flaps 20° – Anti-Icing On

Figure 07-13-8 is used to determine the rotation speed (V_R) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude, take-off weight, runway slope, APR usage and use of anti-icing.

V_R must not be less than 121 KIAS.

NOTE

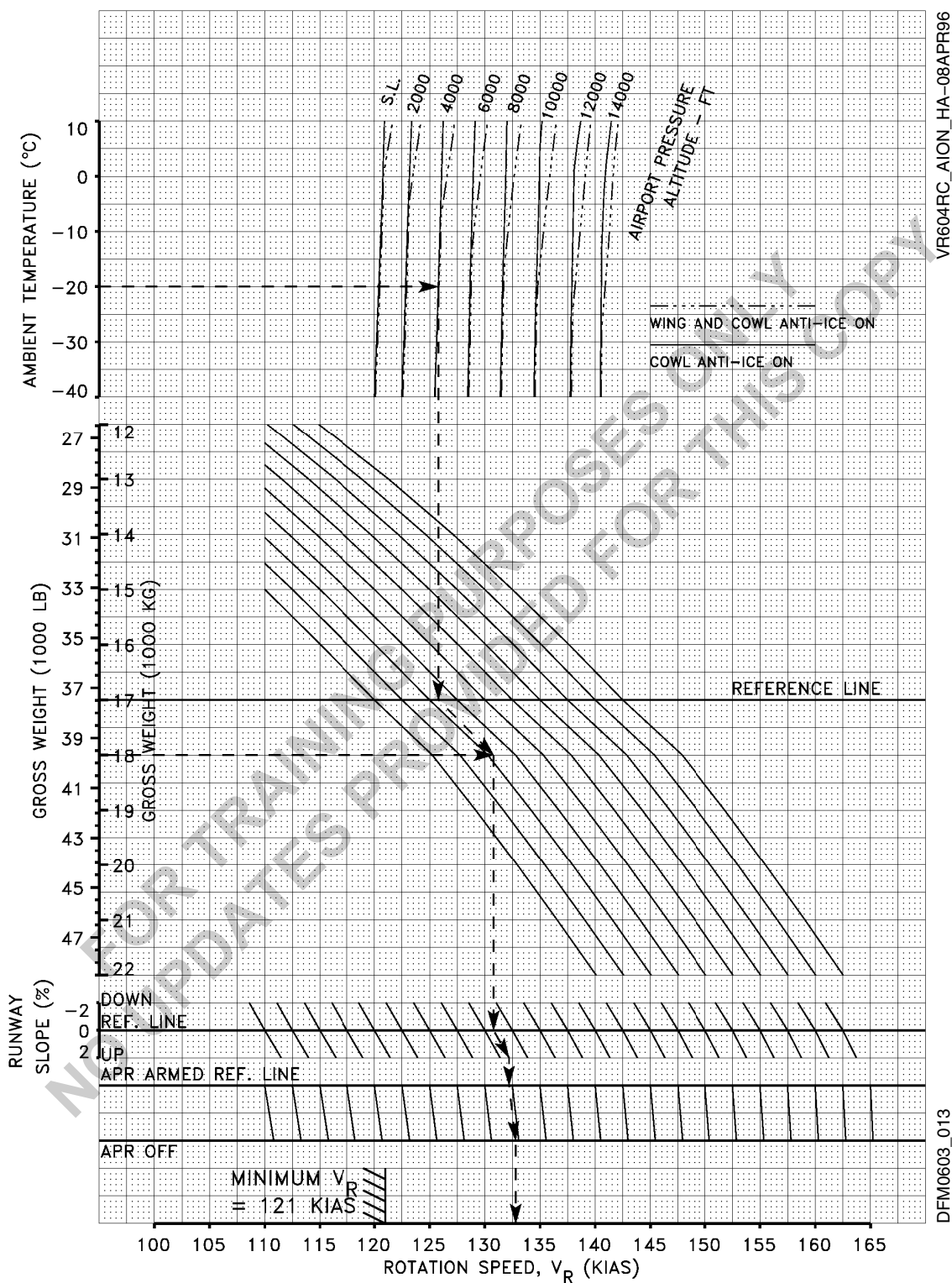
If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

Associated Conditions:

Ambient temperature	= -20°C
Airport pressure altitude	= 4,000 feet
Cowl anti-icing	= On
Take-off weight	= 18,000 kg (39,683 lb)
Runway slope	= +2%
APR	= Off

The example in Figure 07-13-8 shows that for the above associated conditions, the rotation speed is 133 KIAS.

7. PERFORMANCE (CONT'D)

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Rotation Speed, V_R , Flaps 20° – Anti-Icing On
 Figure 07-13-8

CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-29
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7. PERFORMANCE (CONT'D)

(e) Take-Off Safety Speed, V_2 – Flaps 20°

Figure 07-13-9 is used to determine the take-off safety speed (V_2) for a flaps-20° take-off, for varying conditions of ambient temperature, airport pressure altitude and take-off weight.

NOTE

1. V_2 is the greater of the values obtained from Chart A and Chart B of Figure 07-13-9 .
2. V_2 from Chart A is not limiting at weights greater than 17,500 kg (38,581 lb).
3. V_2 from Chart B is not limiting at speeds less than 125 KIAS.
4. If pressure height at the end of second segment climb is above 1500 feet, record the adjustments for extended second segment climb (Figure 07-13-10).

Example:

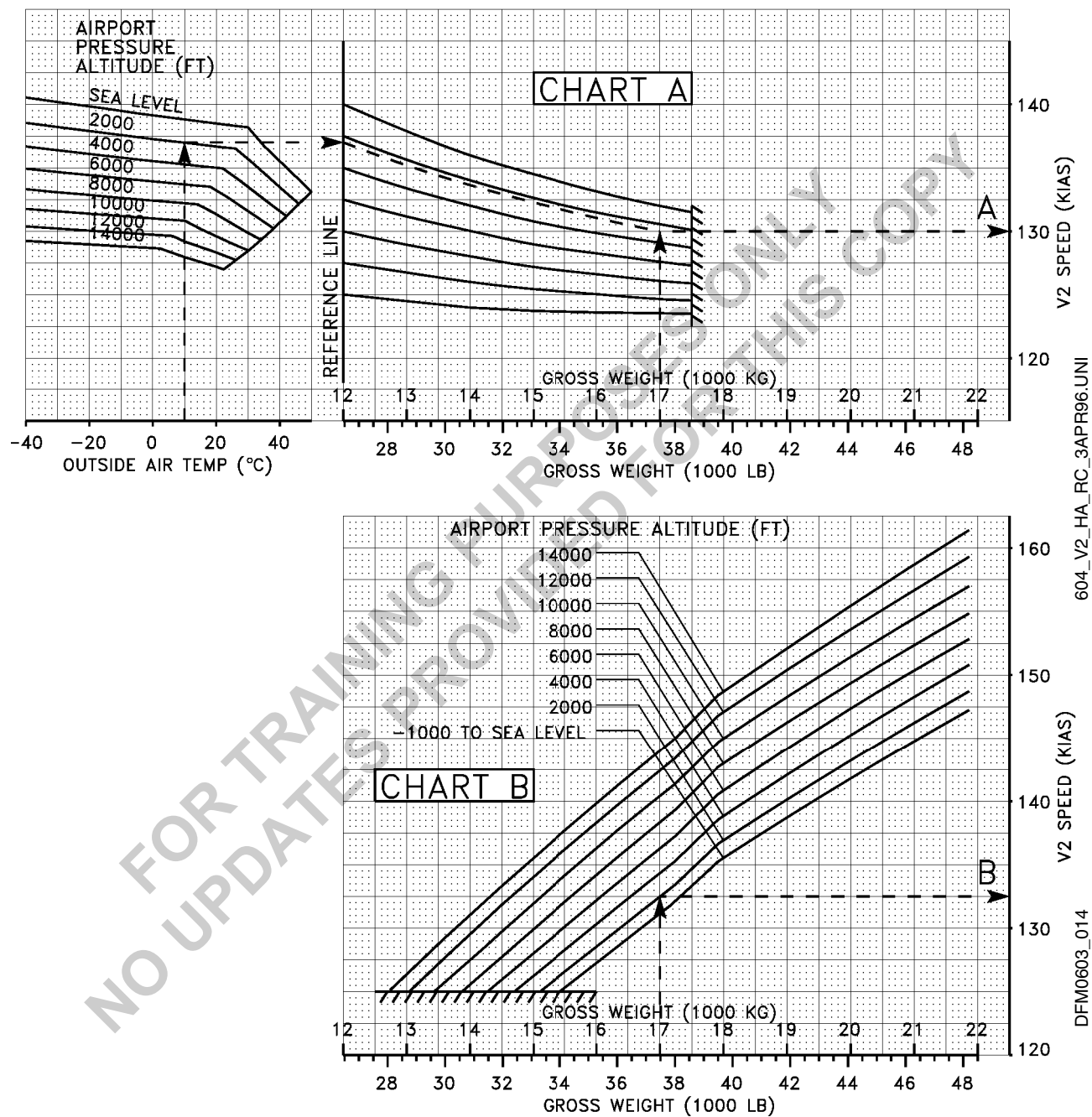
Associated Conditions:

Outside air temperature	= 10°C
Airport pressure altitude	= 2,000 feet
Take-off weight	= 17,000 kg (37,478 lb)

Chart A of Figure 07-13-9 shows that for the above associated conditions, the take-off safety speed is 130.0 KIAS.

Chart B of Figure 07-13-9 shows that for the above associated conditions, the take-off safety speed is 132.5 KIAS.

Therefore, the take-off safety speed under these conditions is 132.5 KIAS.

7. PERFORMANCE (CONT'D)

Take-Off Safety Speed, V_2 – Flaps 20°
 Figure 07-13-9

CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-31
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7. PERFORMANCE (CONT'D)

(3) MAXIMUM ALLOWABLE BRAKE TEMPERATURE FOR TAKE-OFF

The minimum brake cooling time at the ramp specified in Chapter 6; PERFORMANCE – TURN-AROUND TIME, must be observed following a landing or a rejected take-off, to ensure that brake energy or brake temperature limits are not exceeded during the next take-off or in the event that the next take-off is rejected.

(4) ADJUSTMENTS FOR EXTENDED SECOND SEGMENT CLIMB

If the second segment of the take-off flight path extends beyond a pressure height of 1,500 feet above the airport (Refer to Chapter 6; PERFORMANCE – OBSTACLE CLEARANCE), adjustments need to be made to the following items calculated from this section:

- V_R and V_2 speeds;
- runway length for a take-off weight limited by one engine inoperative;
- actual runway length for a take-off weight limited by all engines operating; and
- take-off weight limited by maximum tire speed.

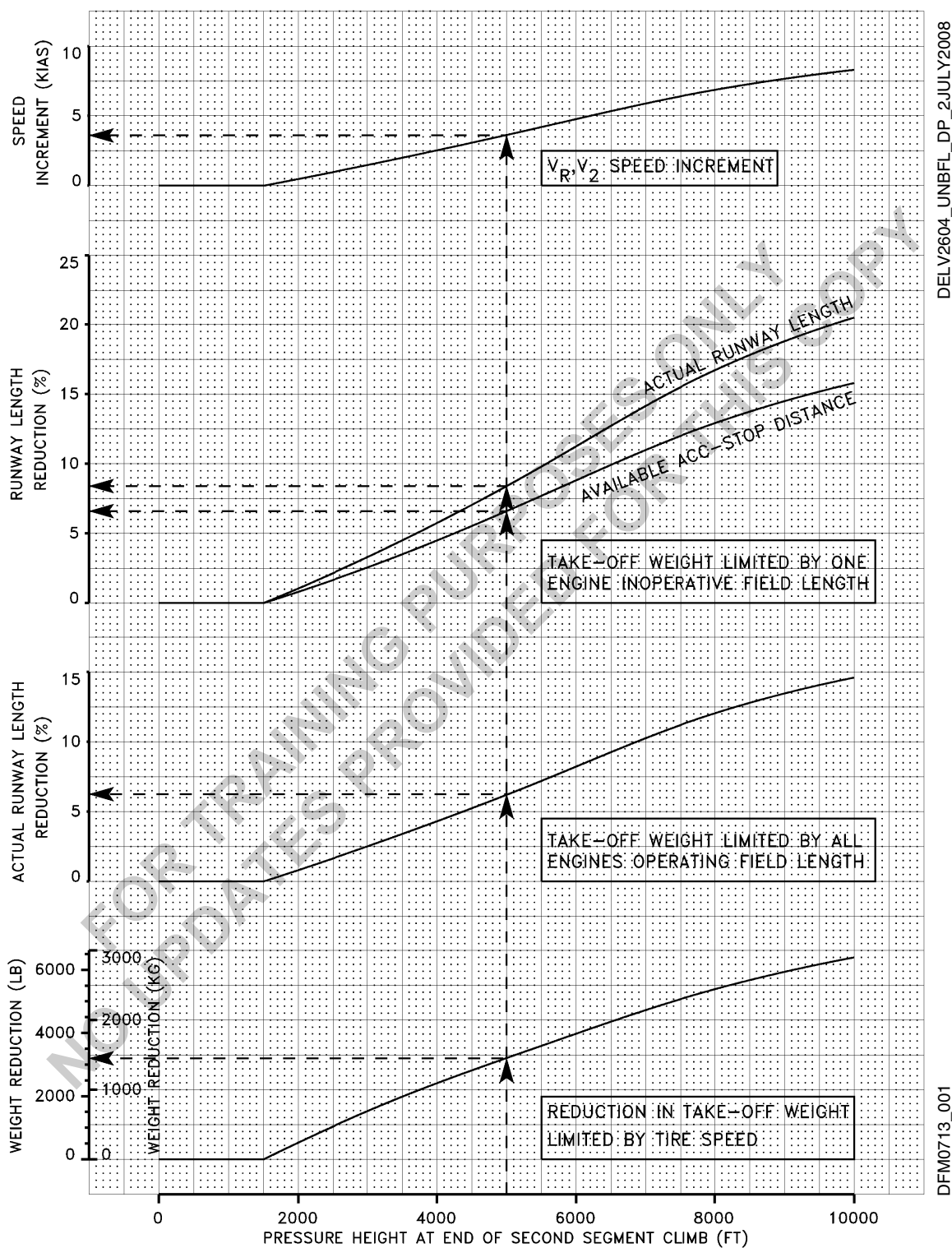
Figure 07-13-10 provides these performance adjustments, for a flaps-20° take-off.

Example:

The example in Figure 07-13-10 shows that for a second segment climb ending at a pressure height of 5,000 feet above the airport, the following adjustments are needed:

- V_R , V_2 speeds: increase by 3.5 KIAS
- Take-off weight limited by one engine inoperative: reduce the actual runway length by 8.25% and reduce the available accelerate-stop distance by 6.5%
- Take-off weight limited by all engine operating: reduce the actual runway length by 6.25%
- Take-off weight limited by tire speed: reduce by 1,450 kg (3,197 lb)

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7. PERFORMANCE (CONT'D)

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Adjustments for Extended Second Segment Climb
 Figure 07-13-10

CHALLENGER 650	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-33
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7. PERFORMANCE (CONT'D)

(5) FINAL SEGMENT AND ENROUTE CLIMB SPEED

The final segment climb speed (V_{FTO}) and enroute climb speed with flaps retracted, one engine inoperative and the other engine operating at maximum continuous thrust (MCT), is provided by Figure 07-13-11, for varying conditions of pressure altitude and airplane gross weight.

Example:

Associated Conditions:

Gross weight = 18,000 kg (39,683 lb)

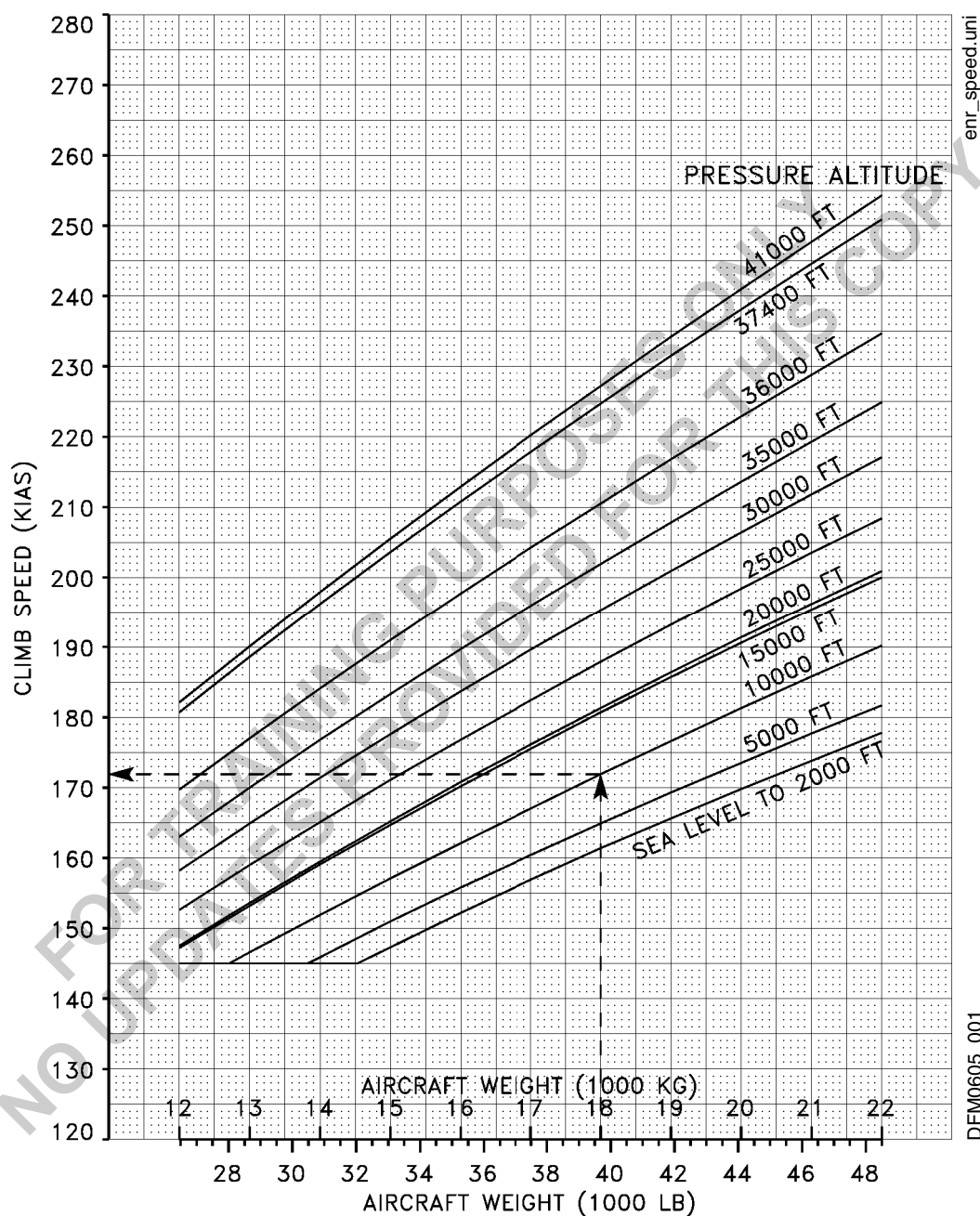
Pressure altitude = 10,000 ft

The example on Figure 07-13-11 shows that for the above associated conditions, the final segment and enroute climb speed is 172 KIAS.

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7. PERFORMANCE (CONT'D)



Final Segment and Enroute Climb Speed – Flaps 0°
Figure 07-13-11

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8. SUPPLEMENTS

The supplements in Chapter 7 are applicable, except as modified by the following:

A. Supplement 2– Operation on Wet and Contaminated Runways

(1) WET RUNWAY TAKE-OFF PERFORMANCE

(a) Maximum Allowable Take-Off Weight

The maximum take-off weight limited by field length requirements, with one engine inoperative, for a flaps-20° take-off on a wet runway, is given by Figure 07-13-12. The following charts take into account the accelerate-stop distance available, the actual length of the runway, the effects of runway slope, and prevailing wind conditions, for varying conditions of ambient temperature, airport pressure altitude, thrust reverser availability, APR usage and various bleed configurations.

Example:

Associated Conditions:

Actual runway length	= 3,048 meters (10,000 feet)
Available accelerate-stop distance	= 3,048 meters (10,000 feet)
Runway slope	= -1%
Wind	= 10 knots tailwind
Temperature	= 20°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Open, ACU On
Thrust reversers	= Both operable
APR	= Armed

Page 1 of Figure 07-13-12 is used to obtain Reference A values. Enter the chart from the actual runway length then apply the correction for runway slope and reported wind to obtain a Reference A value.

Page 2 of Figure 07-13-12 is used to obtain Reference B values. Enter the chart from the available accelerate-stop distance then apply the correction for runway slope and reported wind to obtain a Reference B value.

Page 3 of Figure 07-13-12 is used to obtain V_1/V_R ratio and Reference C value. Enter the chart using the Reference A and Reference B values to obtain a Reference C value and a V_1/V_R ratio of 0.92 as shown on the example.

Page 4 of Figure 07-13-12 is used to obtain Reference D values. Enter the chart at the temperature scale below the appropriate anti-ice and 10th stage bleed configuration to arrive at the Reference D value.

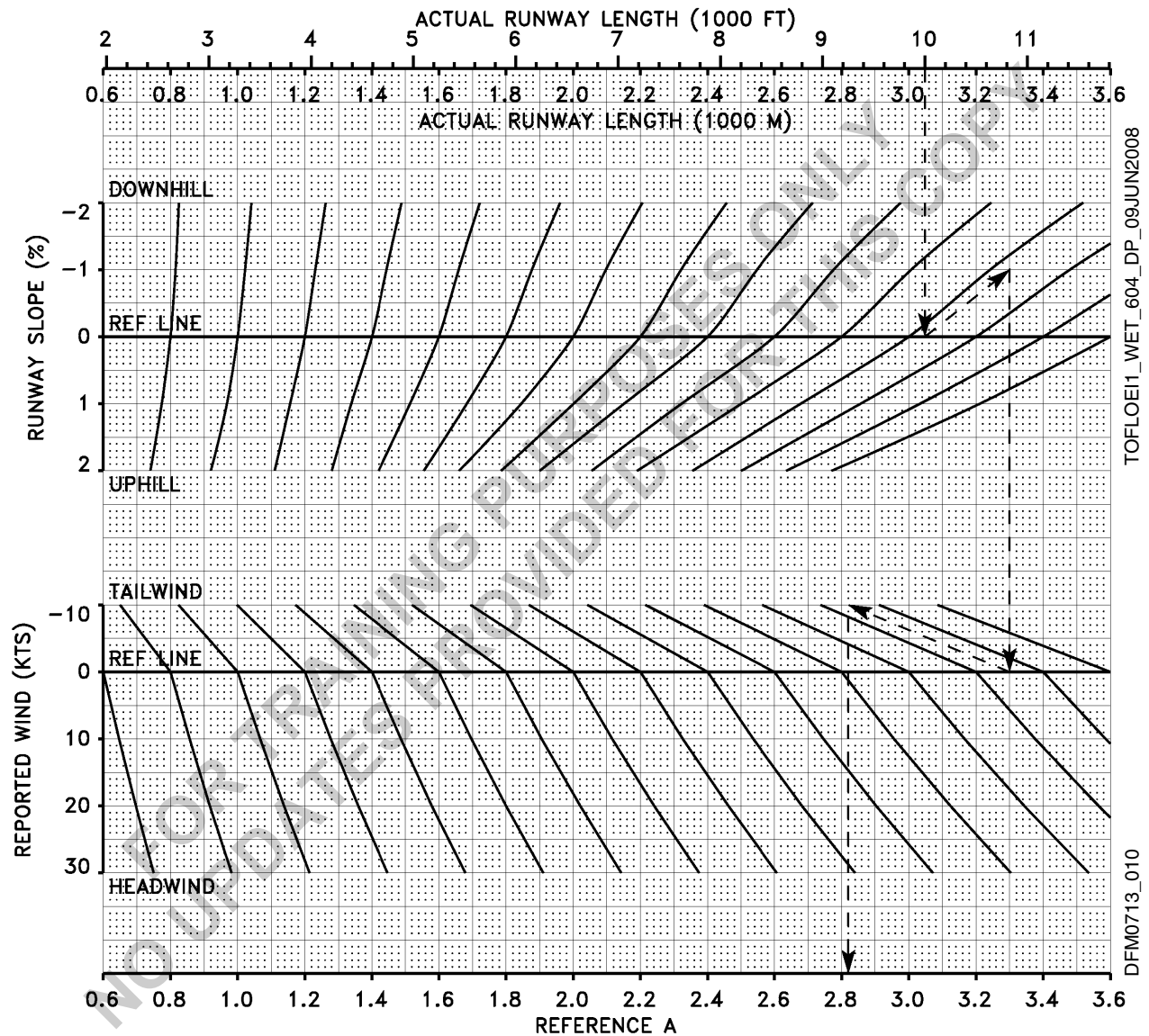
Finally, page 5 of Figure 07-13-12 is used to obtain the airplane take-off weight. Enter the chart at the Reference D scale and the Reference C value as shown on the example, to arrive at the take-off weight of 20,100 kg (44,313 lb).

NOTE

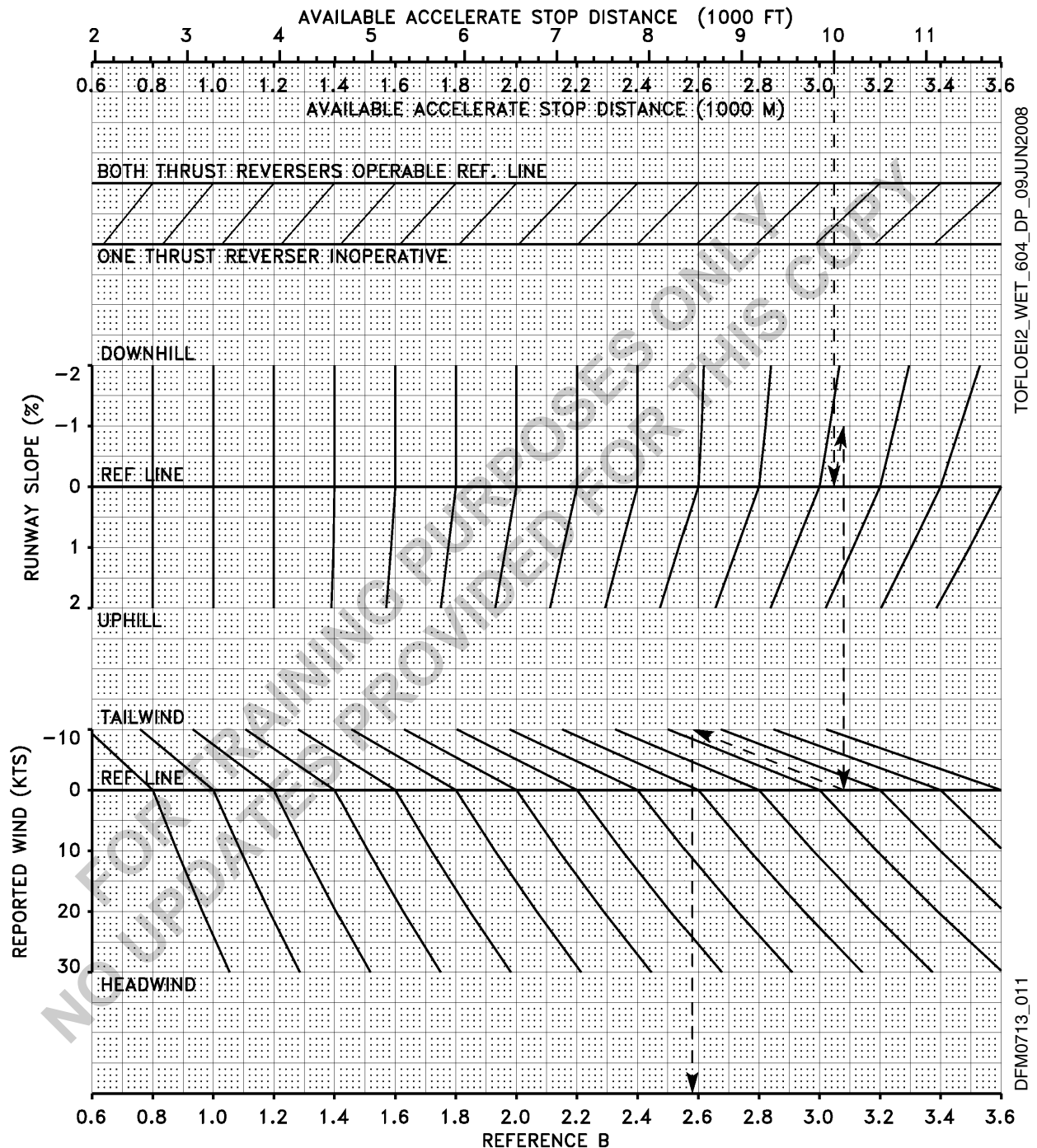
If the take-off weight calculated on a wet runway is greater than the corresponding take-off weight on a dry runway, then use the take-off weight for a dry runway.

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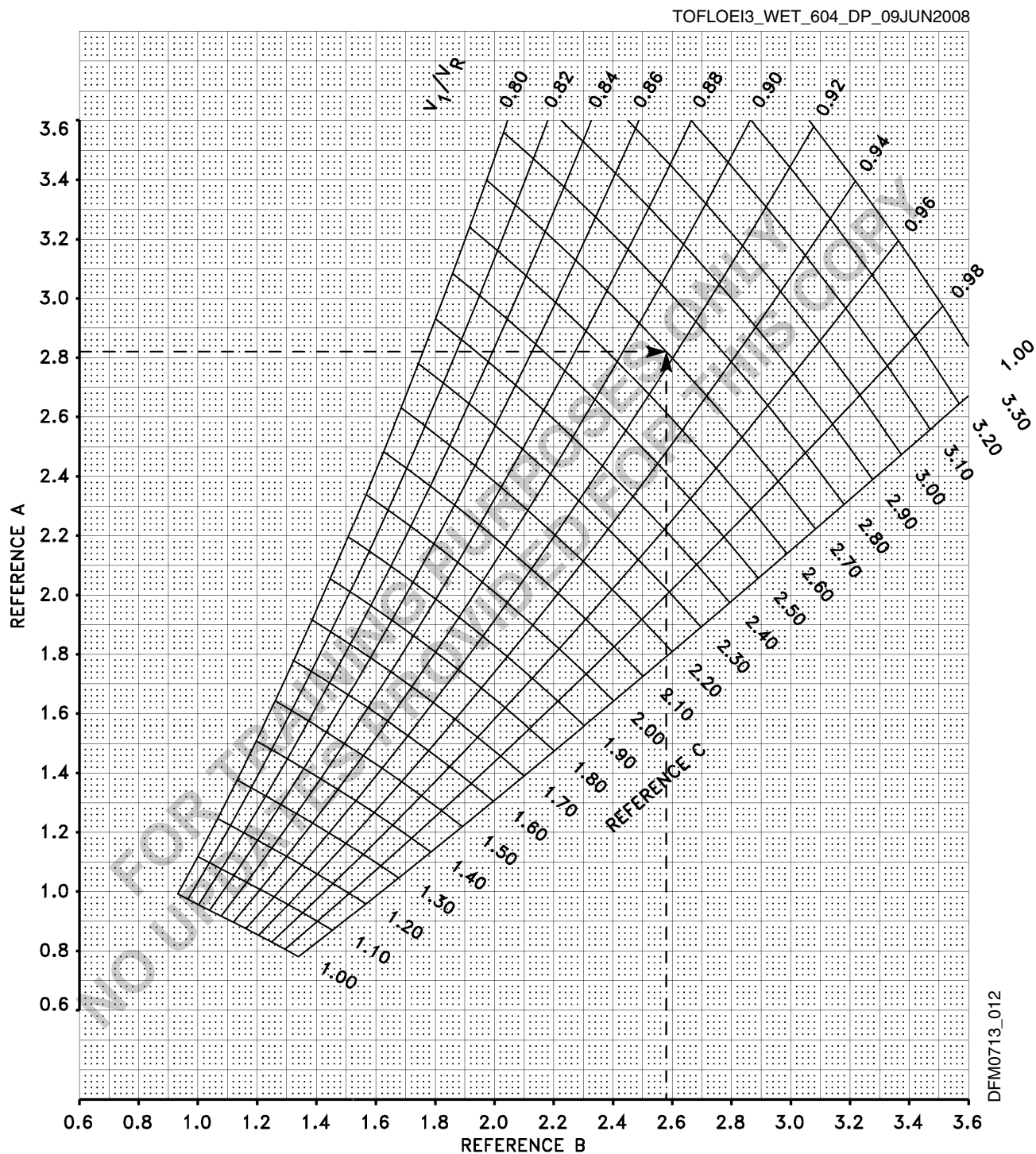
8. SUPPLEMENTS (CONT'D)



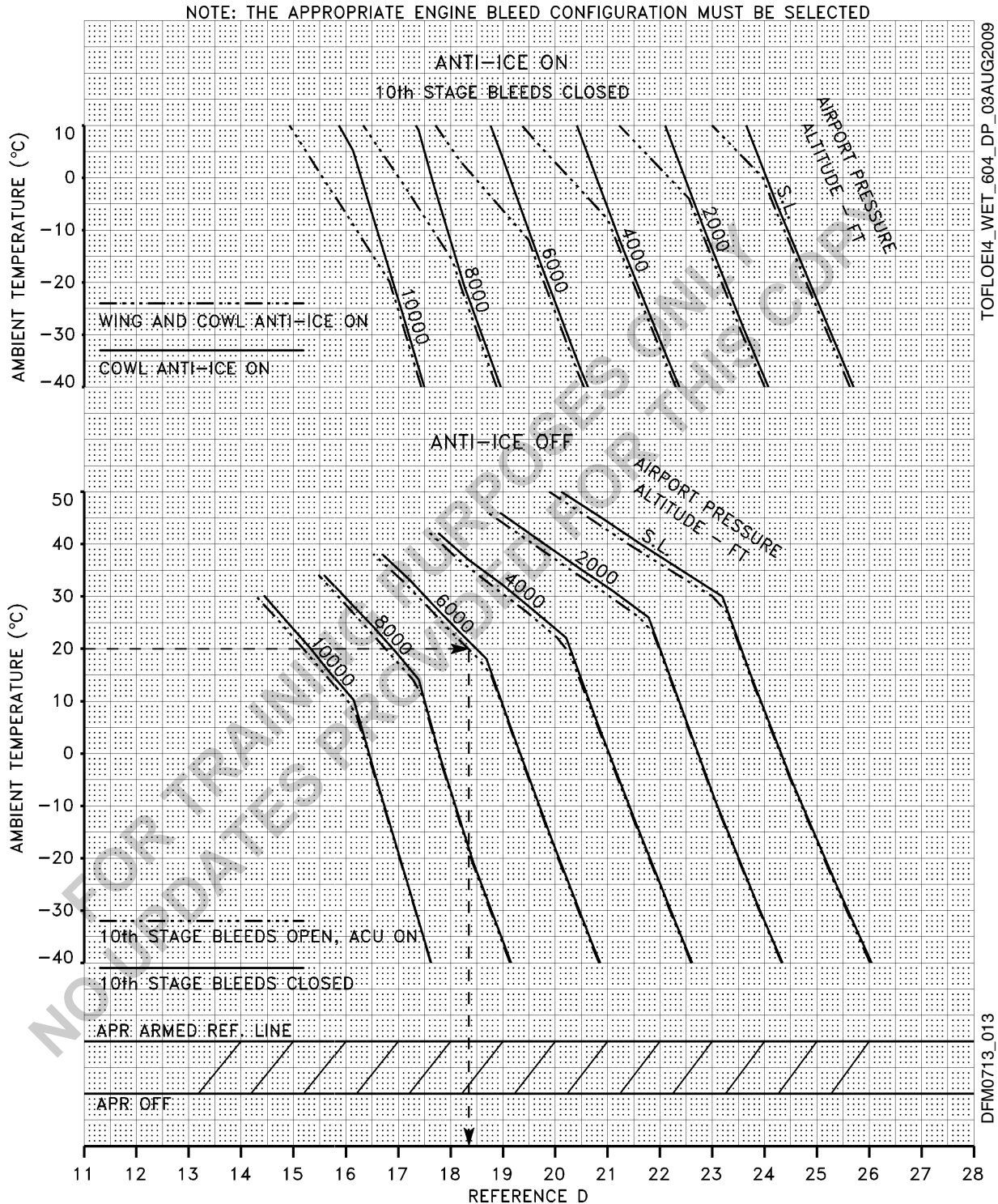
Take-Off Weight Limited by Field Length Requirements – Wet Runway (One Engine Inoperative)
 (Page 1 of 5)
 Figure 07-13-12

8. SUPPLEMENTS (CONT'D)

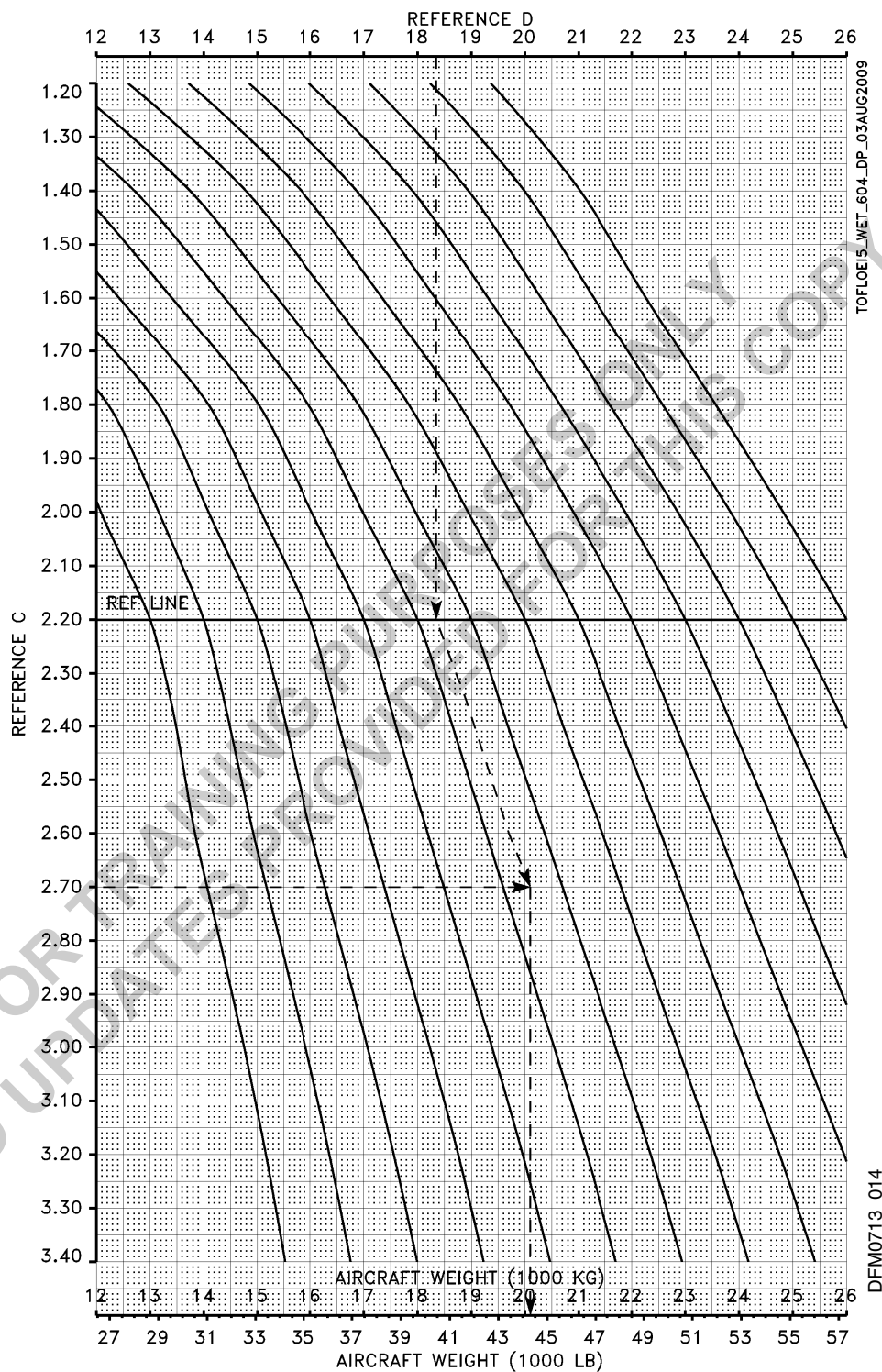
Take-Off Weight Limited by Field Length Requirements – Wet Runway (One Engine Inoperative)
 (Page 2 of 5)
 Figure 07-13-12

8. SUPPLEMENTS (CONT'D)

Take-Off Weight Limited by Field Length Requirements – Wet Runway (One Engine Inoperative)
 (Page 3 of 5)
 Figure 07-13-12

8. SUPPLEMENTS (CONT'D)

Take-Off Weight Limited by Field Length Requirements – Wet Runway (One Engine Inoperative)
 (Page 4 of 5)
 Figure 07-13-12

8. SUPPLEMENTS (CONT'D)

Take-Off Weight Limited by Field Length Requirements – Wet Runway (One Engine Inoperative)
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 Figure 07-13-12

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8. SUPPLEMENTS (CONT'D)

(2) CONTAMINATED RUNWAY TAKE-OFF PERFORMANCE

No unbalanced data is available when operating on a contaminated runway.

B. Supplement 3– Operation at Airport Elevations up to 14,000 Feet

Use of Supplement 3 is prohibited.

C. Supplement 19 – Operation on Wet Grooved Runways

No take-off performance credit for wet grooved runway operation is claimed for Supplement 13. When dispatching on a wet grooved runway and using Supplement 13, use the wet runway take-off data of Supplement 13.

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CHALLENGER ⁶⁵⁰	SUPPLEMENT 13 Additional Take-Off Performance Information	07-13-01-42
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CHALLENGER 650	SUPPLEMENT 14 Operational Capabilities	07-14-01-1
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1. GENERAL

This Supplement provides data for the operational capabilities and the applicable standards of certain airplane systems.

NOTE

Compliance with the standards noted below does not constitute an operational approval.

The effect of this Supplement on the basic Airplane Flight Manual is given below:

A. Introduction

The general information in Chapter 1 is applicable.

B. Limitations

The limitations in Chapter 2 are applicable.

C. Emergency Procedures

The emergency procedures in Chapter 3 are applicable.

D. Normal Procedures

The normal procedures in Chapter 4 are applicable.

E. Abnormal Procedures

The abnormal procedures in Chapter 5 are applicable.

F. Performance

The performance data in Chapter 6 is applicable.

G. Supplements

The supplements in Chapter 7 are applicable.

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CHALLENGER 650	SUPPLEMENT 14 Operational Capabilities	07-14-01-2
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2. NAVIGATION

A. RVSM

The airplane is certified capable of RVSM operations in accordance with FAA "Interim guidance material on the approval of operations/aircraft for RVSM operations" 91-RVSM, dated March 14, 1994, or later applicable revision.

RVSM operations must not be commenced or continued unless all of the required equipment specified below is operational:

RVSM REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR RVSM
AUTOPILOT (FCC, FCP, Elevator servo)	MUST BE OPERATIONAL.
ALTITUDE ALERTING SYSTEM	TWO (2) AFDs MUST BE OPERATIONAL DISPLAYING 2 PRIMARY ALTIMETERS
ALTITUDE REPORTING TRANSPONDER	ONE (1) MUST BE OPERATIONAL.
AIR DATA COMPUTER	TWO (2) MUST BE OPERATIONAL.

B. Traffic Alert and Collision Avoidance System (TCAS)

The TCAS installation is in accordance with JAA TGL No. 8.

Pilots are authorized to deviate from their Air Traffic Control (ATC) clearance in order to comply with a TCAS resolution advisory (RA) command.

AR Certified Airplanes

Pilots are authorized to deviate from their Air Traffic Control (ATC) clearance in order to comply with a TCAS resolution advisory (RA) command. The pilots shall inform ATC of the deviation from the ATC clearance following the response to the RA.

Maneuvers must not be based solely on information presented on the traffic display.

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2. NAVIGATION (CONT'D)

C. Flight Management System (FMS)

Approval Status

Provided the FMS is receiving adequate usable signals, the FMS has been demonstrated capable of and has been shown to meet the performance requirements for the following operations:

Lateral Navigation

- 1) VFR/IFR en route oceanic and remote, en route domestic, terminal operation, and instrument approach using WGS84 (or NAD83) coordinate reference datum in accordance with the performance/accuracy requirements of AC 20-138D.
- 2) Global Navigation Satellite System (GNSS) TSO C145a B3/AC 20-138D and Flight Management System (FMS) TSO-C115b and TSO-C146c D4/AC 20-138D: The installed equipment complies with AC 20-138D, AMC 20-27A and AMC 20-28; for navigation using GNSS and GNSS/SBAS (within the coverage of a satellite-based augmentation system complying with ICAO Annex 10) for enroute, terminal area, non-precision approach, and approach procedures with vertical guidance operations. Non-precision approach operations include those based on conventional navigation aids with or without GNSS in the title and those with GNSS and RNAV (GNSS) in the title to LNAV minimums. Approach procedures with vertical guidance include RNAV (GNSS) and LNAV/VNAV minimums as well as LPV approaches.

RNP Operations

RNP 10 (RNAV 10) – In accordance with AC 90-105A Appendix G, ICAO Doc 9613 PBN Volume II, Part B, Chapter 1, and AMC 20-12:

- the FMS installation with the IRS has been demonstrated to meet the criteria as a primary means of navigation for flight up to 6.2 hours in duration without updating. The determination of flight duration starts when the aircraft first begins to move after the system is placed in navigation mode.
- the FMS installation with the GNSS with RAIM has been demonstrated to meet the criteria as a means of navigation for flights without time limitations.

On airplanes incorporating Service Bulletin 650-34-009:

RNP-AR – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 6; AMC 20-26; and AC 20-138C, Appendix 2, the FMS has been demonstrated capable of primary navigation for RNP AR approach operations.

RNP-2 – In accordance with ICAO PBN document 9613 Edition 4, Volume II, Part C, Chapter 2, (Implementing RNP-2):

- the FMS has been demonstrated capable of primary navigation in RNP-2 for oceanic and remote airspace operations provided that two FMSs are receiving information from at least two GNSS and operating in GNSS navigation mode prior to initiating an RNP procedure.
- the FMS has been demonstrated capable of primary navigation in RNP-2 for continental operations provided that one FMS is receiving information from at least one GNSS and operating in GNSS navigation mode prior to initiating an RNP procedure, provided that an alternative means of navigation is available if required for the class of operation.

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2. NAVIGATION (CONT'D)

RNP-4 – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 1 and AC 90-105A Appendix F, the FMS has been demonstrated capable of primary navigation for RNP-4 Oceanic and remote area operations provided that two FMS systems are available and receiving information from two independent GNSS sensors and operating in GNSS navigation mode, when used in conjunction with a GNSS prediction program.

Basic RNP-1 – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 3 and AC 20-138D Chap 9 for FAA RNP 1 (Terminal) operations, the FMS has been demonstrated capable of primary navigation for RNP-1 operations provided that the FMS is receiving information from a GNSS sensor and operating in GNSS navigation mode.

RNP APCH – The FMS has been demonstrated capable of primary navigation for RNP APCH operations in accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 5, AC 20-138D for FAA RNP instrument approach with Baro-VNAV and AMC 20-27A for RNP APCH with Baro-VNAV Provided FMS is receiving information from a GNSS sensor and operating in:

- GNSS navigation mode; and
- RNP APPR annunciation is displayed at the Final Approach Fix.

LPV Approaches – The FMS and GNSS have been demonstrated capable of primary navigation for LPV operations in accordance with ICAO Doc 9613 PBN Volume II, Part B, Chapter 5, AC 20-138D for FAA LPV and AMC 20-28 for EASA LPV.

NOTE

LP approaches cannot be performed.

Baro-VNAV – Provided at least a single FMS, a single GNSS and single ADC are installed and operational and the NO APPR (on PFD) messages are NOT posted, this FMS installation has been demonstrated to meet the requirements of AC 20-138D for performing Baro-VNAV.

Vertical Navigation (VNAV) – The FMS has been demonstrated capable of and has been shown to meet the accuracy requirements of: VFR/IFR en route, terminal and approach VNAV operation within the contiguous United States and Alaska in accordance with the criteria of AC 20-138D. VNAV Temperature Compensation meets the requirements of TCCA AC 500.020 FMS Barometric VNAV Temperature Compensation and RTCA DO-236B MASPS RNP for Area Navigation, Appendix H.

Primary Means Oceanic and Remote – Use of the FMS with GNSS has been found to comply with the requirements of AC 20-138D, Appendix 1 for GNSS primary means of navigation in oceanic and remote airspace, with two operational GNSS receivers and with two operational FMS systems, when used in conjunction with a GNSS prediction program.

NAT MNPS – Provided dual FMS and any two independent long range navigation systems, consisting of any combination of GNSS or IRS are installed and operating, the FMS with GNSS inputs has been demonstrated capable of flight into North Atlantic (NAT) minimum navigation performance specification (MNPS) and has been shown to meet the accuracy specification in accordance with AC 120.33 or AC 91-49.

RNAV-1 / RNAV-2 – In accordance with AC90-100A, the FMS has been demonstrated capable of primary navigation in RNAV-1 and RNAV-2 airspace provided that the FMS is operating in GNSS or DME/DME navigation mode.

RNAV-5 / BRNAV – This FMS installation has been demonstrated to meet the BRNAV requirements of AC 90-96A and JAA AMJ 20X2 Leaflet 2, Revision 1.

2. NAVIGATION (CONT'D)

PRNAV – The FMS installation has been demonstrated to meet the PRNAV requirements of JAA TGL 10.

Compliance With AC 90-100A – When equipped with an operating FMS and GNSS, the aircraft meets the functional and the ± 1 nm accuracy requirements of AC 90-100A, U.S. Terminal and En Route Area Navigation (RNAV) Operations

D. FM Immunity

The VHF NAV radios installed comply with FM immunity requirements per ICAO Annex 10, and:

- ED-22B / DO-196 for VOR receivers
- ED-46B / DO-195 for LOC receivers

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3. SURVEILLANCE

A. Mode S Elementary Surveillance

The Mode S transponder has been certified to meet the requirements of Elementary Mode S Surveillance as defined by JAA TGL 13.

B. Enhanced Mode S Surveillance

The installed Mode S system satisfies the data requirements of ICAO Doc 7030/4, Regional Supplementary Procedures for SSR Mode S Enhanced Surveillance in designated airspace. The capability to transmit data parameters is shown in the following table:

PARAMETER	AVAILABLE/NOT AVAILABLE
MAGNETIC HEADING	AVAILABLE
INDICATED AIRSPEED	AVAILABLE
MACH NUMBER	AVAILABLE
VERTICAL RATE	AVAILABLE (SEE NOTE 1)
ROLL ANGLE	AVAILABLE
TRACK ANGLE RATE	AVAILABLE
TRUE TRACK ANGLE	AVAILABLE (SEE NOTE 2)
GROUND SPEED	AVAILABLE (SEE NOTE 2)
SELECTED ALTITUDE	AVAILABLE
BAROMETRIC PRESSURE SETTING	AVAILABLE

NOTE

1. Barometric rate of climb/descent as well as Inertial Rate of climb/descent are available as the aircraft is equipped with IRS.
2. Aircraft must be equipped with at least one functioning Flight Management Computer and FMS Control Display Unit.

C. Automatic Dependent Surveillance Broadcast (ADS-B)

The installed ADS-B Out system has been certified in accordance with the guidance provided in the FAA AC 20-165A, the EASA CS.ACNS.D.ADS and the Australian AC 21-45, and meets the equipment requirements of FAA 14 CFR 91.227 and those of CASA CAO 20.18.

ADS-B operations cannot be commenced or continued unless the required equipment specified below is operational:

ADS-B OUT REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR ADS-B OUT
ATC/MODE S TRANSPONDER	ONE (1) MUST BE OPERATIONAL.
AIR DATA COMPUTER	ONE (1) MUST BE OPERATIONAL.
GNSS SBAS RECEIVER	ONE (1) MUST BE OPERATIONAL.

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4. COMMUNICATIONS

A. FM Immunity

The VHF COM radios installed comply with FM immunity requirements per ICAO Annex 10, and ED-23B / DO-186 for VHF communication receivers.

B. Datalink

(1) ATN B1 CPDLC (CONTROLLER TO PILOT DATALINK COMMUNICATION)

On airplanes incorporating Service Bulletin 650-23-007:

The aircraft datalink system has been demonstrated to comply with the applicable safety, performance and interoperability requirements for continental ATN B1 datalink using VDL Mode-2.

The aircraft ATC datalink system does support multi-frequency operation as defined in ARINC Spec. 631-5.

(2) FANS 1/A+

On airplanes incorporating Service Bulletin 650-23-008:

The aircraft FANS 1/A+ system enables CPDLC and ADS-C capabilities in support of reduced aircraft separations in remote and oceanic areas, by using Satellite Communications and ACARS ground networks. The system complies with the interoperability requirements of RTCA DO-258A and the guidelines of ICAO GOLD.

The system adopts FAA AC 20-140A as an acceptable means of compliance for the following datalink capabilities:

Interoperability Designators:

- FANS 1/A+ with automation (including "push to load")
- ATN B1
- ACARS ATS

ATS Subnetworks:

- VDL Mode-A
- VDL Mode-2
- SATCOM Inmarsat Classic Aero
- SATCOM Iridium (Short Burst Data (SBD))

Aircraft-Allocated Performance:

- CPDLC: RCP 240
- ADS-C: RSP 180

CHALLENGER ⁶⁵⁰	SUPPLEMENT 14 Operational Capabilities	07-14-01-8
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CHALLENGER 650	SUPPLEMENT 15 Operation with Cracked Windshield Faceply	07-15-00-1
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CHALLENGER 650	SUPPLEMENT 15 Operation with Cracked Windshield Faceply	07-15-01-1
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NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes on the FAA registry.

1. GENERAL

This Supplement provides data for operation with a cracked windshield faceply.

The effect of this Supplement on the basic Airplane Flight Manual is given below:

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable with the addition of the following:

A. Systems Limitations

(1) WINDSHIELD AND SIDE WINDOWS

The faceply of one windshield may be cracked provided:

- Cracked faceply is repaired in accordance with Structural Repair Manual, Chapter 51,
- There is no FOD damage to the airframe, nacelles or engines, on initial discovery or if faceply pieces are missing,
- Vision is not impaired through remaining windows,
- Tape and cracked faceply is checked for general integrity prior to each flight,
- Visibility through the affected windshield is acceptable to the flight crew,
- Affected windshield heater is considered inoperative,
- Remaining windshield heater and both side window heaters are operative,
- Flight into known or forecast icing conditions, precipitation, thunderstorms or other conditions which could cause fogging or misting is prohibited and
- Airspeed is limited to 280 KIAS or less below 8,000 feet MSL.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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CHALLENGER 650	SUPPLEMENT 15 Operation with Cracked Windshield Faceply	07-15-01-2
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7. PERFORMANCE

The performance data in Chapter 6 is applicable with the addition of the following:

A. General

One windshield faceply may be cracked provided the mission fuel requirements are increased by:

Fuel consumption	0.30%
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8. SUPPLEMENTS

The supplements in Chapter 7 are applicable.

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CHALLENGER 650	SUPPLEMENT 17 Runway Awareness and Advisory System	07-17-00-1
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CHALLENGER 650	SUPPLEMENT 17 Runway Awareness and Advisory System	07-17-01-1
		Oct 27/15

NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes incorporating Service Bulletin 650-34-010.

1. GENERAL

This Supplement provides data for operation of the Runway Awareness and Advisory System (RAAS).

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore be used in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (8).

Functional Overview

The Runway Awareness Advisory System (RAAS) is a software enhancement hosted in the Terrain Awareness Warning System (TAWS) unit, which provides aural and visual advisories to supplement flight crew awareness of position during ground and flight operations. RAAS uses existing TAWS inputs to generate aural advisories in the flight deck. RAAS advisories are annunciated over the same audio system as the TAWS. Unique voices are used for RAAS advisories to allow adequate differentiation from the current TAWS Mode 1 – 7 and Terrain / Obstacle Awareness voice call outs. RAAS advisories are succinct by design. RAAS is designed to be fully compatible with normal aircraft operations. There is no requirement for pilot input to the system.

RAAS is configurable (during installation/activation) for voice gender, units (feet or meters) and most advisories may be configured ON or OFF. This flight manual supplement is arranged so that each configurable advisory is maintained in its own sub-section. This allows an operator to utilize this supplement by choosing each section that applies to them.

RAAS Implementation and Configuration

RAAS may be configured for a variety of advisory callout options via the RAAS configuration database.

The only customer selectable options are:

- Voice gender: Female/Male
- Units of measurement: Feet/Meter
- Altimeter Setting callout: ON/OFF.

A RAAS inhibit switch allows the crew to inhibit all RAAS visual indications and aural alerts. The white OFF switch/light illuminates when selected.

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CHALLENGER 650	SUPPLEMENT 17 Runway Awareness and Advisory System	07-17-01-2
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1. GENERAL (CONT'D)

The table below provides the settings of alert options certified on the aircraft. These options and configurations are set during installation and activation of the RAAS on the aircraft, and are NOT pilot selectable.

GROUP	ADVISORY/ ALERT	ENABLED	SETTINGS	AUDIO	VISUAL
	GPS Antenna Location		15 feet	–	–
	Advisory Suppress Window	YES	450 – 550 feet	–	–
RAAS	Approaching Runway – In Air/On Ground	YES		“Approaching XX” “Approaching Runways” (if more than one runway meets the qualifying condition)	APP XX (green) APP RWYS (green) (if more than one runway meets the qualifying condition)
	On Runway	YES		“On Runway XX” “On Runways” (if more than one runway meets the qualifying condition)	ON XX (green) ON RWYS (green) (if more than one runway meets the qualifying condition)
	Distance Remaining (Landing & Rollout)	YES	2,000 feet max	“ZZ Remaining”	NONE
	Runway End (< 100 feet)	YES		“One Hundred Remaining” (in feet) “Thirty Remaining” (in meters)	NONE
	Insufficient Runway Length (In Air)	YES	5,000 feet	“Approaching Runway XX, ZZ available”	APP XX ZZ (amber)
	Short Runway Landing	NO			
	Insufficient Runway Length (On Ground)	YES	6,000 feet	“On Runway XX, ZZ remaining”	ON XX ZZ (amber)
	Short Runway Takeoff	NO			
	Extended Hold On Runway Initial/Repeats	YES	120 sec/ 120 sec	“On Runway XX – On Runway XX”	ON XX (amber)
	Taxiway Takeoff	YES		“Caution, On Taxiway! On Taxiway”	ON TAXIWAY (amber) GND PROX alert on PFD also illuminated
	Distance Remaining (Rejected Takeoff)	YES	50% of Runway	“ZZ Remaining”	NONE
	Taxiway Landing	YES		“Caution, Taxiway! Caution, Taxiway!”	TAXIWAY (amber) GND PROX alert on PFD also illuminated
	Takeoff Flaps	YES	Min: 16°; Max: 24°	“On Runway XX, Flaps – Flaps”	FLAPS (amber) GND PROX alert on PFD also illuminated
Stability Approach Monitor	Landing Flaps (Not Aligned With Runway)	NO			
	Landing Flaps (Aligned With Runway)	YES	700 feet	“Flaps – Flaps”	FLAPS (amber)
	Too High (During Approach)	YES		“Too High – Too High”	TOO HIGH (amber)
	Too Fast (During Approach)	YES		“Too Fast – Too Fast”	TOO FAST (amber)
	Unstable (During Approach)	YES		“Unstable – Unstable”	UNSTABLE (amber) GND PROX alert on PFD also illuminated

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1. GENERAL (CONT'D)

GROUP	ADVISORY/ ALERT	ENABLED	SETTINGS	AUDIO	VISUAL
Long Landing Monitor	Long Landing Distance	YES	3,000 ft past approach end	"Long Landing – Long Landing"	LONG LANDING (amber) GND PROX alert on PFD also illuminated
	Long Landing Percentage	YES	67% remaining	"Long Landing – Long Landing"	LONG LANDING (amber) GND PROX alert on PFD also illuminated
	Long Landing Distance Remaining	YES	3,500 ft past approach end	"ZZ Remaining"	
	Long Landing Distance Remaining Percentage	YES	60% remaining	"ZZ Remaining"	
Corrected Altitude Monitor	Altimeter Setting (Below Transition Altitude)	YES		"Altimeter Setting"	ALTM SETTING (amber)
	Altimeter Setting (Above Transition Altitude)	YES		"Altimeter Setting"	ALTM SETTING (amber)

XX = Runway Identifier

ZZ = Available runway distance in feet or meters

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2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable with the addition of the following:

Refer to the Honeywell MK V-A Enhanced Ground Proximity Warning System (EGPWS), document PDS69000940-801 revision A or later applicable revision.

On airplanes 6050 to 6074:

Refer to the Honeywell MK V and MK VII Enhanced Ground Proximity Warning System (EGPWS) and Flight Safety Functions TSO C151b Class A TAWS Pilot Guide, document 060-4241-000 revision G or later appropriate revision for descriptive information.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

7. PERFORMANCE

The performance data in Chapter 6 is applicable.

8. SUPPLEMENTS

The supplements in Chapter 7 are applicable.

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CHALLENGER ⁶⁵⁰	SUPPLEMENT 18 RNP Authorization Required Approach Operations	07-18-01-1
		Oct 27/15

NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes incorporating Service Bulletin 650-34-009.

1. GENERAL

This Supplement contains information and procedures for Required Navigation Performance (RNP) Authorization Required (AR) – approach operations.

NOTE

Compliance with these standards noted below does not constitute approval to conduct RNP AR approach operations.

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (8).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

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3. LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

A. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

An RNP AR approach must not be commenced or continued unless the required navigation equipment, as specified in the RNP AR Approach Required Equipment List below, are operating satisfactorily.

RNP AR APPROACH – REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
YD 1 and 2	Two (2) yaw dampers must be available to commence the approach, one (1) yaw damper is required to continue the approach
PFD 1 and 2	One (1) PFD available and operational for each side
ADC 1 and 2	Two (2) ADCs must be available to commence and continue the approach
GNSS 1 and 2	Two (2) GNSS Sensors must be available to commence the approach, one (1) GNSS Sensor is required to continue the approach
FMS 1, 2 and 3	Two (2) FMSs must be available and operational for each side
IRS 1, 2 and 3 (ATT and HDG)	Two (2) IRSs must be available to commence the approach, one (1) IRS is required to continue the approach
TAWS	TAWS must be operational and displayed
FD 1 and 2	Two (2) must be operational and independently displayed

RNP AR approach operations beyond the IAF are prohibited unless RNP AR APPR is annunciated on the PFD.

RNP AR approach with RF Legs operations are approved for RNP greater than or equal to RNP 0.3 in the approach segment and RNP AR 1.0 in the missed approach segment.

RNP AR approach operations are approved with autopilot or flight director engaged only.

RNP AR approach operations are approved for autothrottle engaged or disengaged.

RNP AR approach operations are approved for any flap configuration up to the IF (Intermediate Fix) for all engines operating, or for flaps 0° and 20° with one engine inoperative, and with flaps 45° from IF to MAP (Missed Approach Point) for all engines operating, or for flaps 20° from IF to MAP with one engine inoperative.

RNP AR approach operations must be accomplished in accordance with instrument approach procedures that are retrieved from the FMS navigation database.

RNP AR approach operations are approved following failure of one engine.

RNP AR approach operations must be accomplished at or below published maximum speed limits for the approach segment depicted on the Instrument Approach Procedure. In the event of a missed approach the aircraft speed must not exceed the maximum speed for the segment of the approach.

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		Oct 27/15

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, with the addition of the following:

A. Approach

- (1) FMS navigation source..... Set independent source on each PFD.
- (2) IRS and ADC Set independent source on each PFD.
- (3) Flight director mode..... APPR LNV
- (4) RNP AR APPR annunciations..... Verify annunciated at IAF

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6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, with the addition of the following:

A. Loss of GNSS

Indication: **NO APPR** caution message on PFD.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

B. Excessive Lateral or Vertical Deviation

Indication: Amber flashing lateral or vertical deviation indication.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

C. EFIS Parameter Miscompare

Indication: EFIS MISCOMP caution message on and amber ALT annunciation on PFD.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

7. PERFORMANCE

The performance data in Chapter 6 is applicable.

8. SUPPLEMENTS

The supplements in Chapter 7 are applicable.

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CHALLENGER 650	SUPPLEMENT 19 Operation on Wet Grooved Runways	07-19-01-1
		Rev 5, Aug 30/16

1. GENERAL

A. General

This Supplement contains information and procedures for operation of the CL 600, Model 2B16 Challenger airplane (Serial No. 6050 and subsequent) on wet grooved runways.

This information has been prepared by the manufacturer and approved as guidance material, to assist operators in developing suitable guidance, recommendations or instructions for use by their flight crews, when operating on wet grooved runway surface conditions.

The data have been prepared using reasonable estimates of the effects of wet grooved runway surface conditions on the accelerating ground roll and the braking ground roll. The effects of actual conditions may differ from those used to establish the data.

B. Runway Conditions

The data is applicable to concrete runway surfaces with transverse grooves constructed and maintained to FAA standard AC 150/5320-12C, or equivalent, and on which there are no areas of significant rubber deposit in sections where the high-speed portion of braking is expected.

Non-performance related limitations and procedures in the Airplane Flight Manual that are applicable to a wet runway, are also applicable to a wet grooved runway.

(1) WET RUNWAY

A grooved runway is considered wet when there is sufficient moisture on the runway surface to cause it to appear partially reflective, but without significant areas of standing water (runway grooves not overflowing).

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CHALLENGER 650	SUPPLEMENT 19 Operation on Wet Grooved Runways	07-19-01-2
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2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

- The data may be used only for wet grooved runways.
- Non-performance related limitations and procedures that are applicable to a wet runway, are also applicable to a wet grooved runway.
- The use of this Supplement for landing performance requires both thrust reversers to be operational.
- The use of this Supplement for landing performance requires the anti-skid system to be fully operational.
- The use of this Supplement requires that :
 - the runway is maintained to FAA standard AC 150/5320-12C, or equivalent,
 - there are no areas of significant rubber deposit in sections where the high-speed portion of braking is to occur, and
 - there is no indication, via NOTAMs, PIREPs or braking action reports, that degraded braking exists on the runway.
- This Supplement is applicable only to grooved concrete surfaces.
- The use of a wet grooved runway take-off distance which is shorter than the dry runway take-off distance is prohibited.
- The use of a wet grooved runway corrected landing distance which is shorter than the dry runway landing field length is prohibited.
- FMS take-off Vspeed data must be manually entered when using data provided in this supplement.

On EASA registered airplanes:

This Supplement is approved for use on concrete runways having a grooved surface that has been constructed and maintained to the FAA standard of AC 150/5320-12C or equivalent. Operational approval must be obtained in order to use the performance data presented in this Supplement.

The wet grooved landing distance determined during flight planning must account for the conditions expected at the time of arrival, including reported wind, runway slope and V_{REF} speed increment.

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4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable with the addition of the following:

FMS take-off Vspeeds (V_1 , V_R , V_2 and V_{FTO}) must be manually entered and selected for view on the displays.

NOTE

The TAKEOFF function of the FMS Vspeed does not support the calculations associated with this supplement. Vspeed and take-off distance must be derived from this supplement.

The CHECK TAKEOFF PERF message may be posted on the FMS CDU.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. PERFORMANCE

The performance data in Chapter 6 are applicable, except as modified by the following:

A. Take-Off Performance

Take-off performance data are calculated based on the assumption that maximum available reverse thrust is used down to a complete stop.

The following charts are used to obtain the take-off distance and V_1 speed data for varying conditions of temperature, airport pressure altitude, and airplane weight for the specified take-off conditions, for a wet grooved runway.

The take-off distance from a dry runway from Chapter 6 must also be determined when using the data in this Supplement. If the take-off distance on a wet grooved runway is less than the take-off distance on a dry runway, use the take-off distance for a dry runway.

The take-off distance charts herein present the most limiting of the following:

- The distance from the start of the take-off roll to the point at which the airplane attains a height of 15 feet above the take-off surface, with a failure of the critical engine at V_{EF} , or
- 115% of the distance from the start of the take-off roll to the point at which the airplane attains a height of 35 feet above the take-off surface, with all engines operating, or
- The accelerate-stop distance.

(1) TAKE-OFF DISTANCE, FLAPS 20° – ANTI-ICING OFF

The take-off distance for a flaps 20° take-off on a wet grooved runway, at airport pressure altitudes of 10,000 feet and below, with anti-icing off, is given by Figure 07-19-1 (Page 1 of 2).

NOTE

When dispatching with one thrust reverser inoperative, V_1 must be reduced by 3 knots and the take-off distance derived from Figure 07-19-1 (Page 1 of 2) must be increased by 100 meters (328 feet).

Example:

Associated Conditions:

Ambient temperature	= 20°C
Airport pressure altitude	= 4,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 19,000 kg (41,888 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-19-1 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 1,670 meters (5,479 feet).

Since the take-off distance on a wet grooved runway is less than that derived from Chapter 6, the dry runway take-off distance of Chapter 6 must be used.

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7. PERFORMANCE (CONT'D)

The take-off distance for a flaps 20° take-off on a wet grooved runway, at airport pressure altitudes above 10,000 feet, with anti-icing off, is given by Figure 07-19-1 (Page 2 of 2).

NOTE

When dispatching with one thrust reverser inoperative, V_1 must be reduced by 3 knots and the take-off distance derived from Figure 07-19-1 (Page 2 of 2) must be increased by 100 meters (328 feet).

Example:

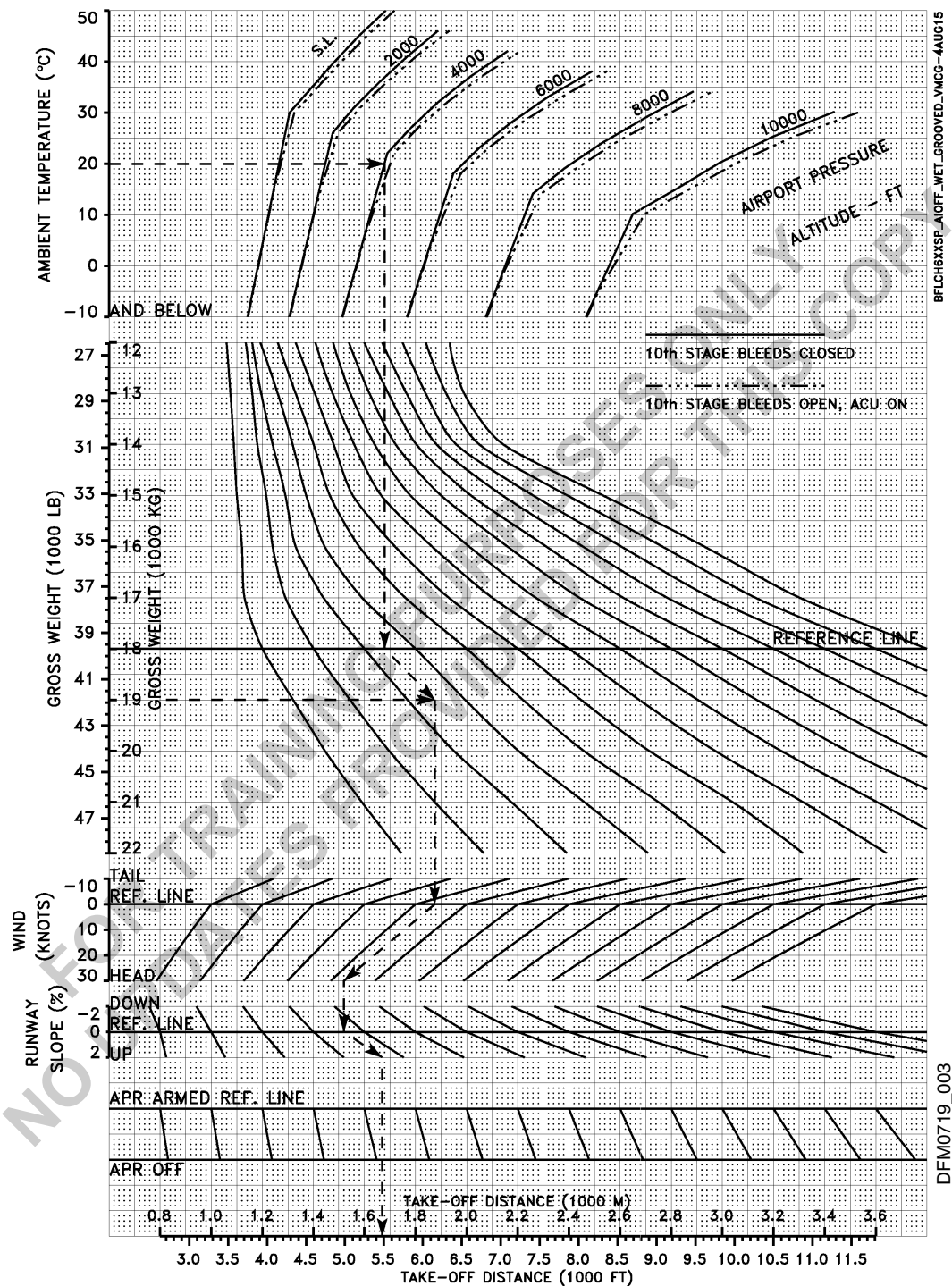
Associated Conditions:

Ambient temperature	= 20°C
Airport pressure altitude	= 12,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= Off
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

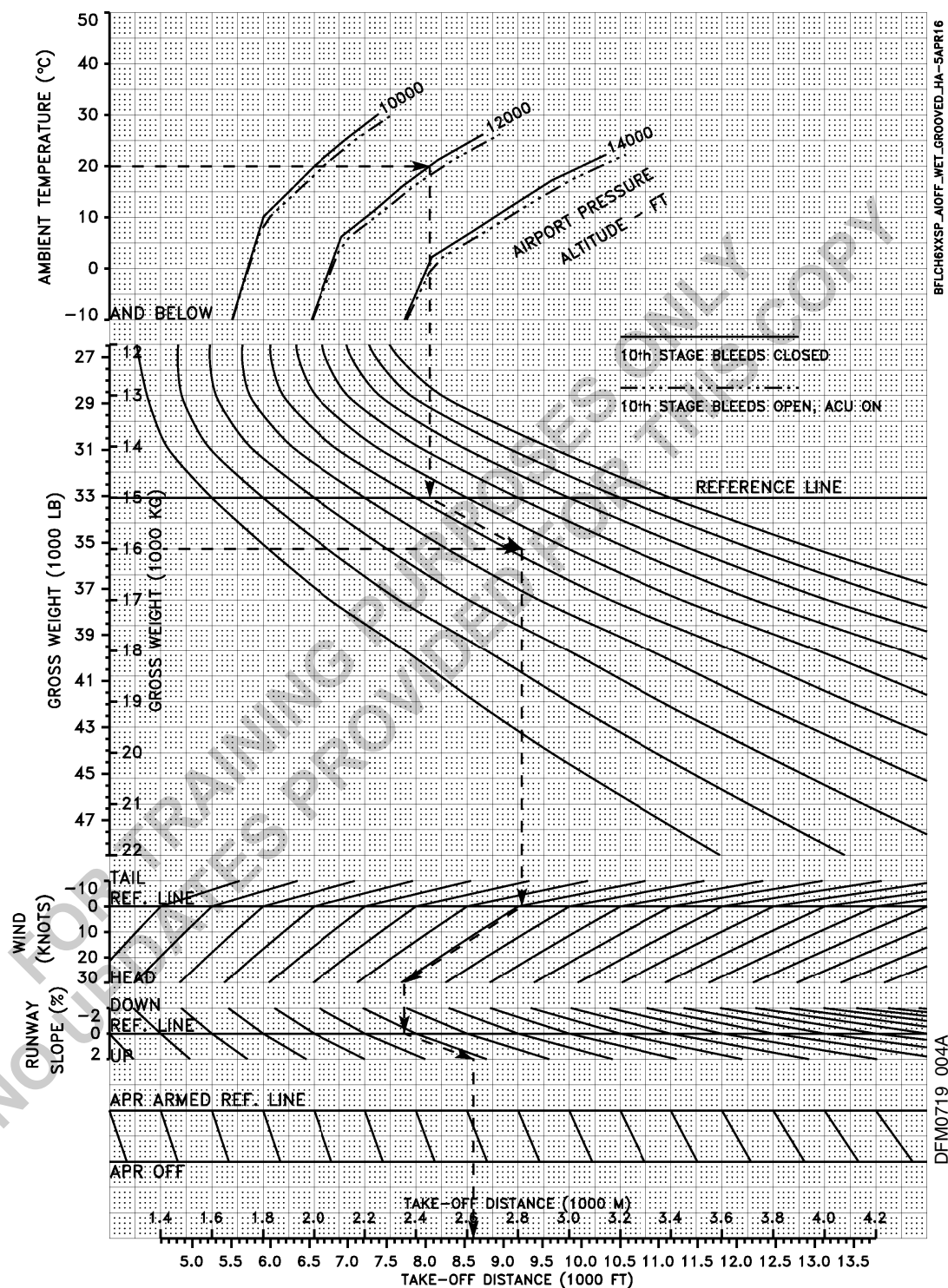
The example in Figure 07-19-1 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,625 meters (8,612 feet).

Since the take-off distance on a wet grooved runway is less than that derived from Chapter 6, the dry runway take-off distance of Chapter 6 must be used.

7. PERFORMANCE (CONT'D)



Take-Off Distance, Flaps 20° – Anti-Icing Off (Page 1 of 2)
 Figure 07-19-1

7. PERFORMANCE (CONT'D)

Take-Off Distance, Flaps 20° – Anti-Icing Off (Page 2 of 2)
 Figure 07-19-1

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7. PERFORMANCE (CONT'D)

(2) TAKE-OFF DISTANCE, FLAPS 20° – ANTI-ICING ON

The take-off distance for a flaps 20° take-off on a wet grooved runway, at airport pressure altitudes of 10,000 feet and below, with anti-icing on is given by Figure 07-19-2 (Page 1 of 2).

NOTE

When dispatching with one thrust reverser inoperative, V_1 must be reduced by 3 knots and the take-off distance derived from Figure 07-19-2 (Page 1 of 2) must be increased by 100 meters (328 feet).

Example:

Associated Conditions:

Ambient temperature	= 0°C
Airport pressure altitude	= 6,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= On
Gross weight	= 19,000 kg (41,888 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

The example in Figure 07-19-2 (Page 1 of 2) shows that for the above associated conditions, the take-off distance is 2,033 meters (6,670 feet).

Since the take-off distance on a wet grooved runway is less than that derived from Chapter 6, the dry runway take-off distance of Chapter 6 must be used.

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7. PERFORMANCE (CONT'D)

The take-off distance for a flaps 20° take-off on a wet grooved runway, at airport pressure altitudes above 10,000 feet, with anti-icing on is given by Figure 07-19-2 (Page 2 of 2).

NOTE

When dispatching with one thrust reverser inoperative, V_1 must be reduced by 3 knots and the take-off distance derived from Figure 07-19-2 (Page 2 of 2) must be increased by 100 meters (328 feet).

Example:

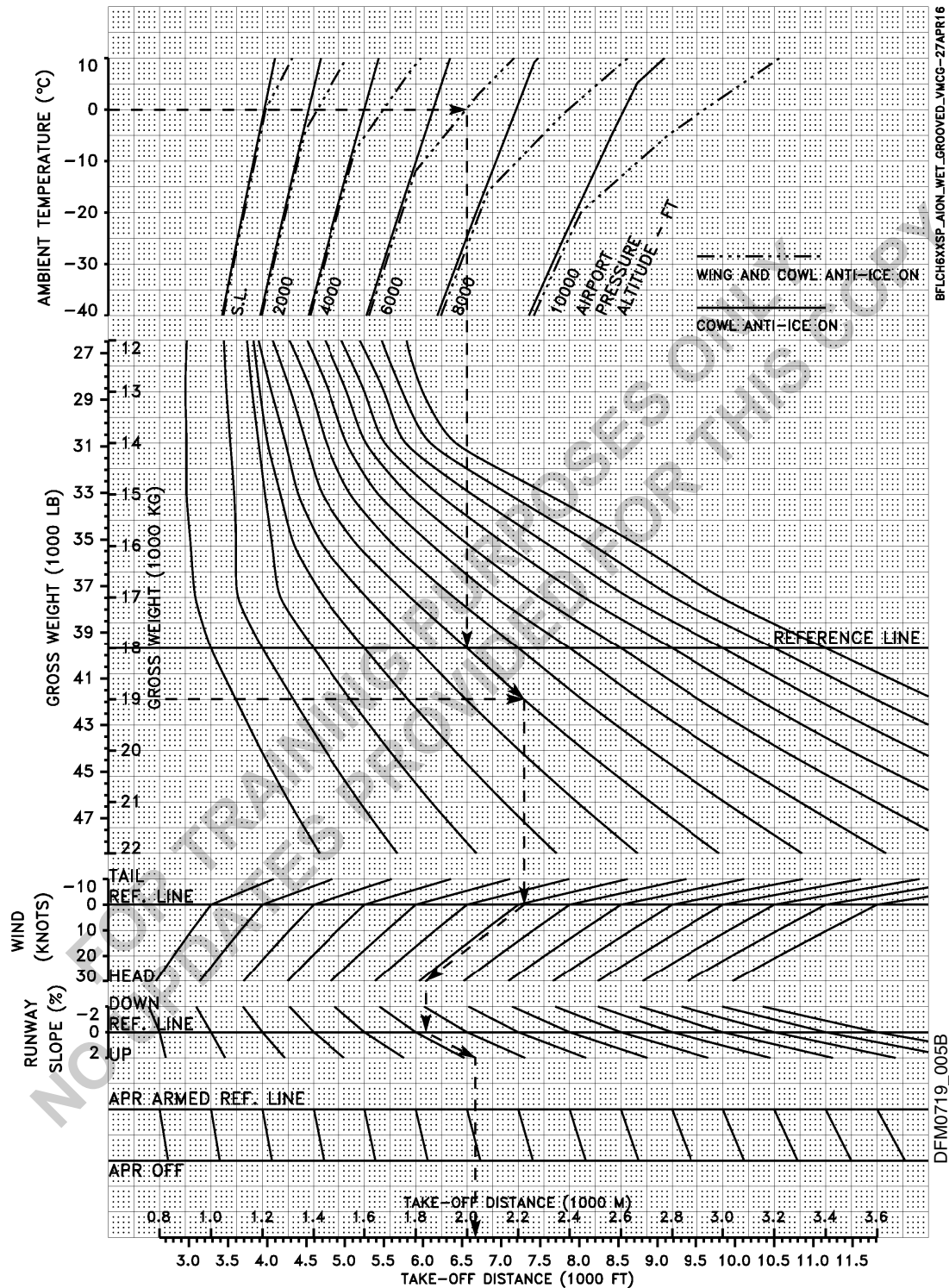
Associated Conditions:

Ambient temperature	= 0°C
Airport pressure altitude	= 12,000 ft
10th stage bleeds	= Closed
Wing and cowl anti-ice	= On
Gross weight	= 16,000 kg (35,274 lb)
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Armed
Thrust reversers	= Both available

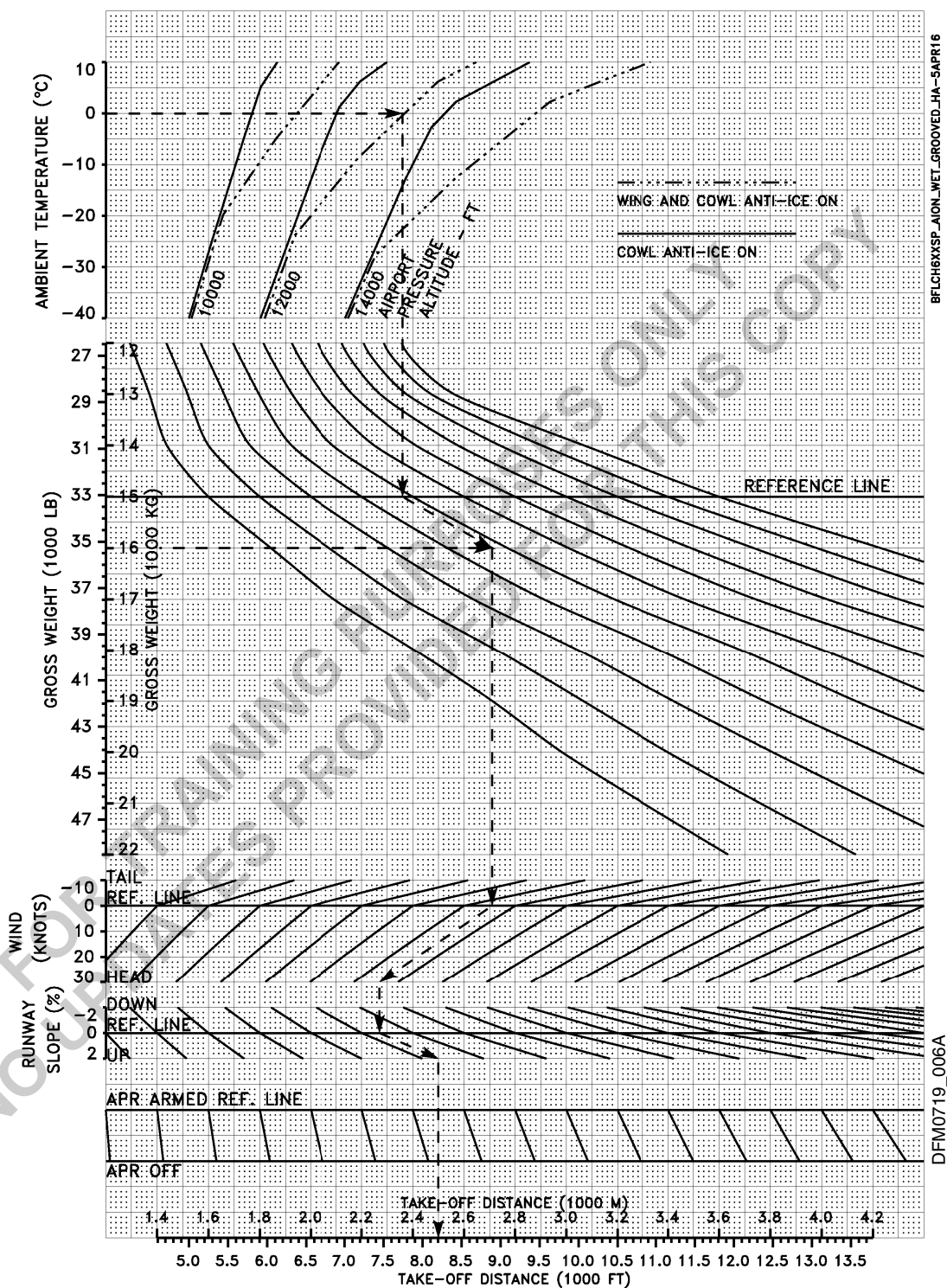
The example in Figure 07-19-2 (Page 2 of 2) shows that for the above associated conditions, the take-off distance is 2,510 meters (8,235 feet).

Since the take-off distance on a wet grooved runway is less than that derived from Chapter 6, the dry runway take-off distance of Chapter 6 must be used.

7. PERFORMANCE (CONT'D)



Take-Off Distance, Flaps 20° – Anti-Icing On (Page 1 of 2)
 Figure 07-19-2

7. PERFORMANCE (CONT'D)


Take-Off Distance, Flaps 20° – Anti-Icing On (Page 2 of 2)
 Figure 07-19-2

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7. PERFORMANCE (CONT'D)

(3) TAKE-OFF DECISION SPEED, V_1 – FLAPS 20° – ANTI-ICING OFF

Figure 07-19-3 is used to determine the V_1 speed for a flaps 20° take-off (balanced field length) on a wet grooved runway, for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope and various bleed configurations.

NOTE

When dispatching with one thrust reverser inoperative, the take-off decision speed, V_1 , derived from Figure 07-19-3 must be reduced by 2.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

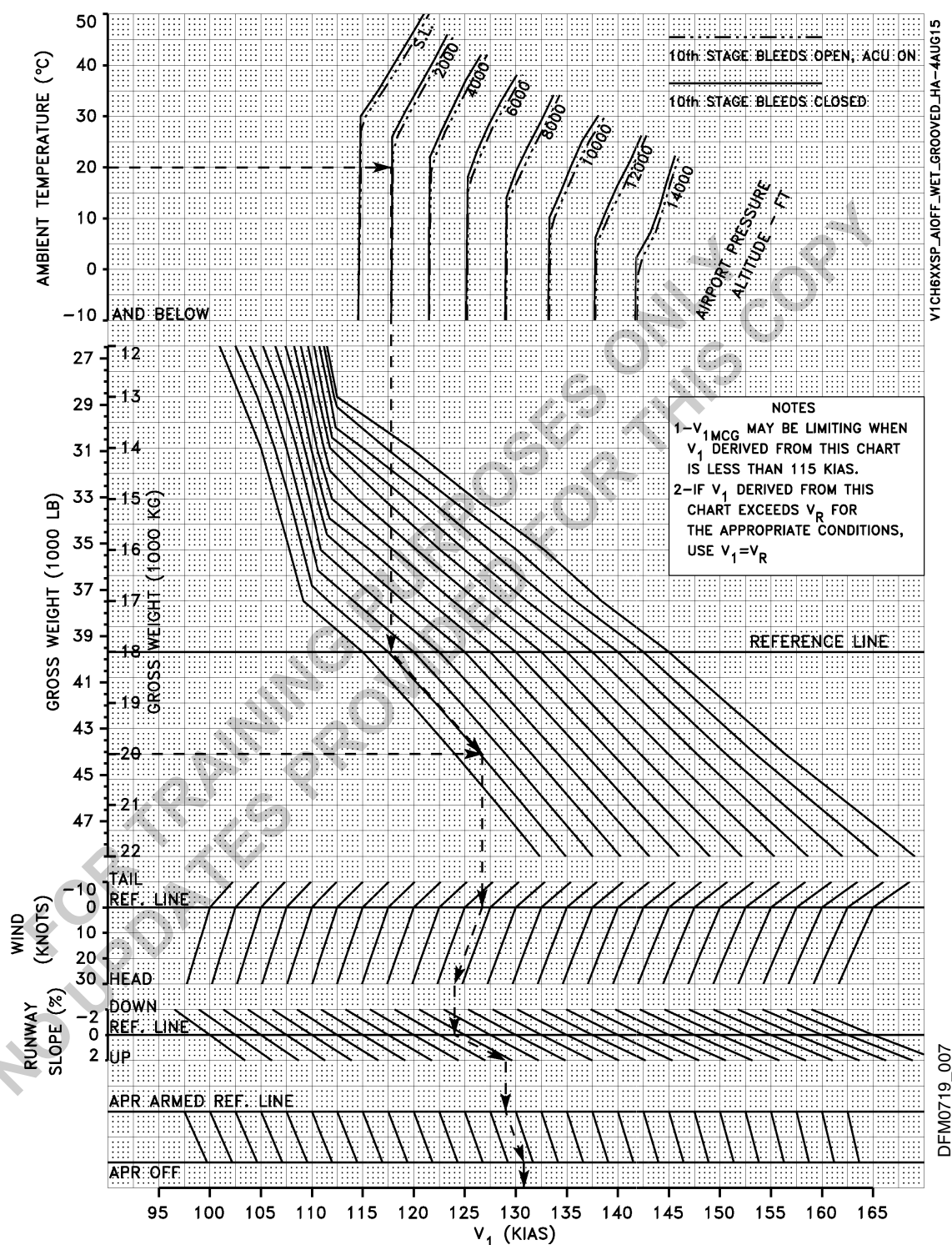
Example:

Associated Conditions:

Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 2,000 ft
10th stage bleeds	= Closed
Ambient temperature	= +20°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off
Thrust reversers	= Both available

The example in Figure 07-19-3 shows that for the above associated conditions, V_1 is equal to 130.8 KIAS.

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7. PERFORMANCE (CONT'D)

CHALLENGER 650	SUPPLEMENT 19 Operation on Wet Grooved Runways	07-19-01-14
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7. PERFORMANCE (CONT'D)

(4) TAKE-OFF DECISION SPEED, V_1 – FLAPS 20° – ANTI-ICING ON

Figure 07-19-4 is used to determine the V_1 speed for a flaps 20° take-off (balanced field length) on a wet grooved runway, for varying conditions of take-off weight, ambient temperature, airport pressure altitude, wind, runway slope and use of anti-icing.

NOTE

When dispatching with one thrust reverser inoperative, the take-off decision speed, V_1 , derived from Figure 07-19-4 must be reduced by 2.0 KIAS. If the reduced V_1 speed is lower than V_{1MCG} , use $V_1 = V_{1MCG}$.

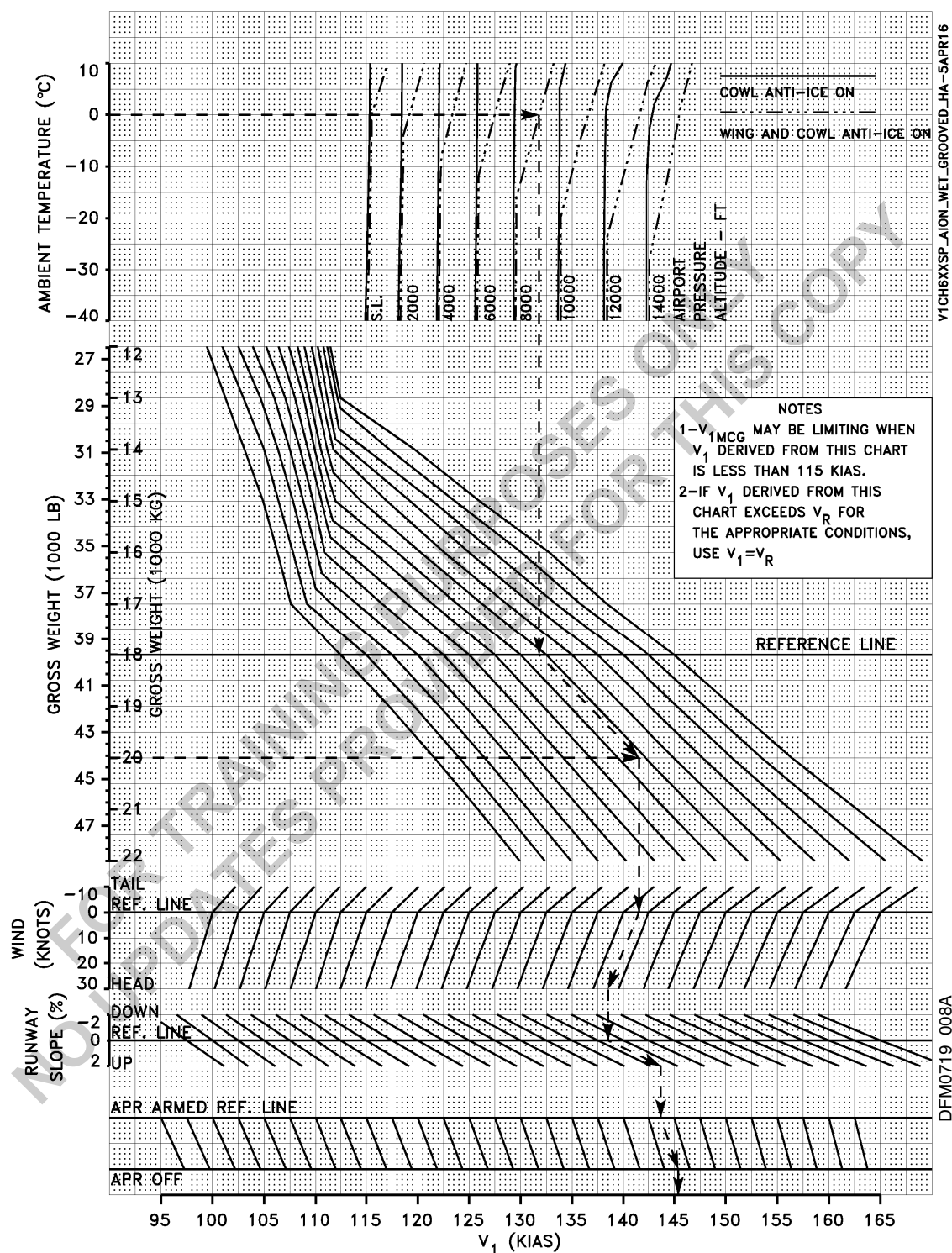
Example:

Associated Conditions:

Take-off weight	= 20,000 kg (44,092 lb)
Airport pressure altitude	= 8,000 ft
Wing and cowl anti-icing	= On
Ambient temperature	= 0°C
Wind	= 30 knots headwind
Runway slope	= +2%
APR	= Off
Thrust reversers	= Both available

The example in Figure 07-19-4 shows that for the above associated conditions, V_1 is equal to 145.4 KIAS.

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7. PERFORMANCE (CONT'D)

V_1 Speed – Flaps 20° – Anti-Icing On
 Figure 07-19-4

CHALLENGER 650	SUPPLEMENT 19 Operation on Wet Grooved Runways	07-19-01-16
		Rev 7, Feb 23/17

7. PERFORMANCE (CONT'D)

B. Landing Performance

NOTE

Compliance with the following landing performance does not constitute an operational approval.

The following chart is used to obtain the landing distance required for varying conditions of airplane landing weight, airport pressure altitude, temperature, wind and runway slope.

The landing field length from Chapter 6 must also be determined when using the data in this Supplement. If the landing field length on a dry runway is greater than the corrected landing distance on a wet grooved runway, use the landing field length.

The landing distance on wet grooved runways, for a flaps-45° landing, with both engines operating, for speeds at the runway threshold from V_{REF} to $V_{REF} + 10$ KIAS, is provided by Figure 07-19-5.

NOTE

The corrected landing distance on wet grooved runways includes a landing distance factor of 1.15 (15%).

Example:

Associated Conditions:

Landing gross weight	= 16,000 kg (35,274 lb)
Airport pressure altitude	= 2,000 feet
Ambient temperature	= +15°C
Wind	= 14 knots at 135 degrees relative to runway (10 knots tailwind; 10 knots crosswind)
Runway Slope	= 0%
ΔV_{REF}	= +5 KIAS

From the associated conditions stated above, the corrected landing distance on a wet grooved runway, as shown in the example on Figure 07-19-5, is found to be 2,049 meters (6,722 feet).

From the associated conditions stated above, the landing field length on a dry runway from Figure 06-07-2 is 1,678 meters (5,505 feet).

Since the corrected landing distance on a wet grooved runway is greater than the landing field length on a dry runway, the corrected landing distance on a wet grooved runway must be used.

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7. PERFORMANCE (CONT'D)

On EASA registered airplanes:

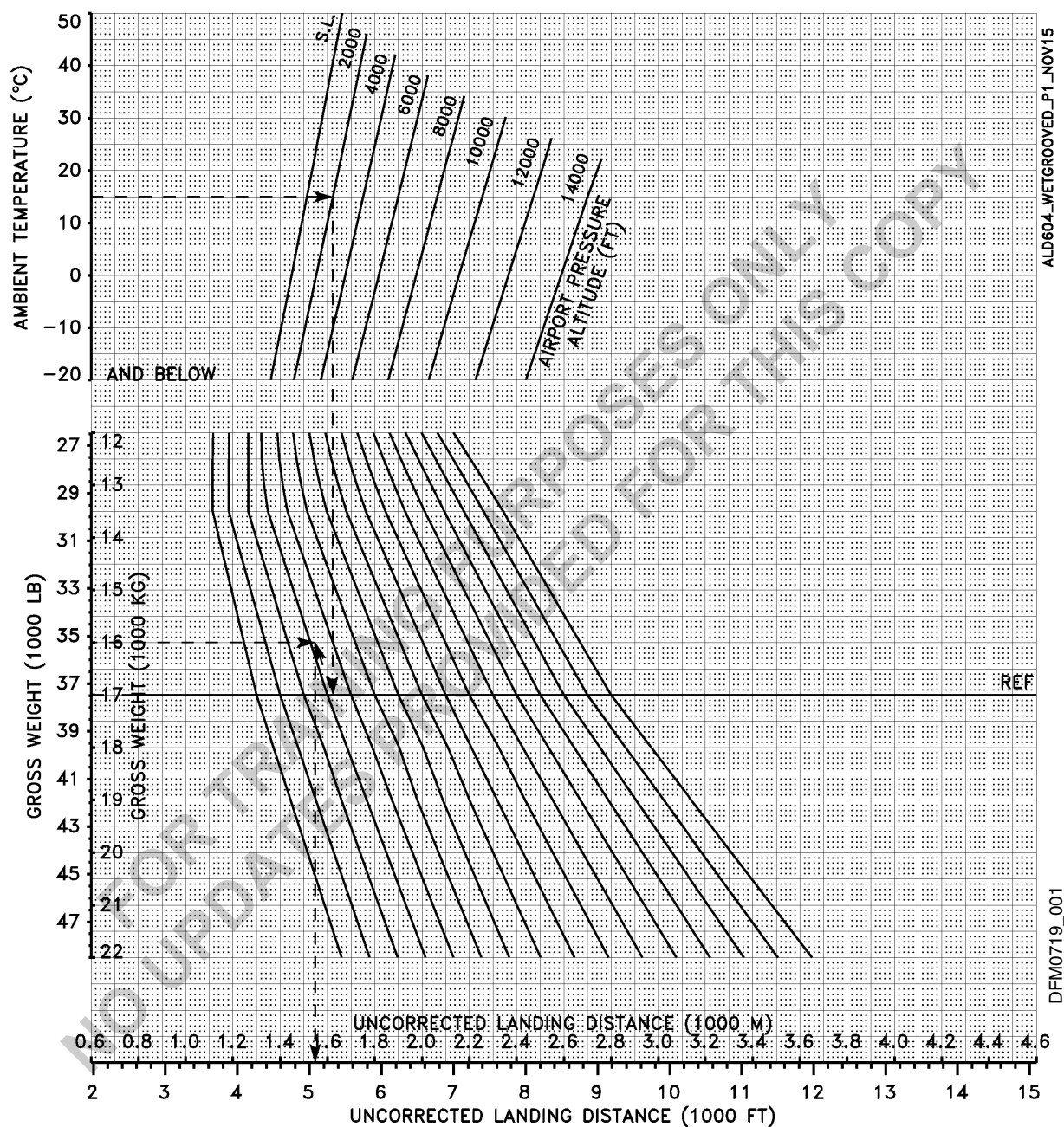
The wet grooved landing distance determined during flight planning must account for the conditions expected at the time of arrival, including reported wind, runway slope and V_{REF} speed increment.

For flight planning purposes, a V_{REF} speed increment of 10 Knots from Figure 07-19-5 (Page 2 of 2), must be used to determine the wet grooved landing distance required. The wet grooved landing distance must be re-assessed during flight for the actual conditions at the time of arrival.

The actual V_{REF} increment during landing may be less than 10 KIAS. The in-flight wet grooved landing distance assessment is based on the actual V_{REF} increment.

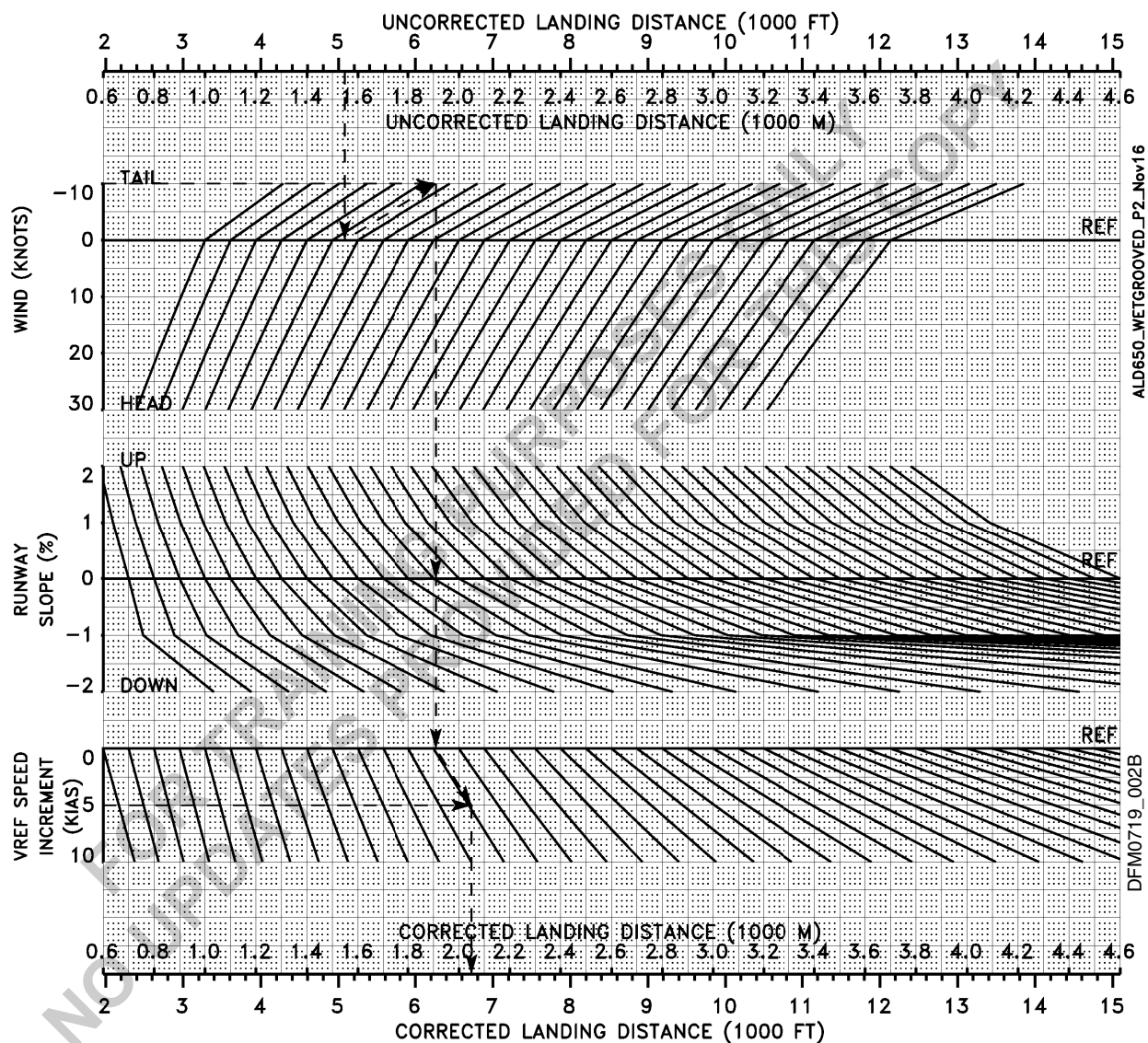
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7. PERFORMANCE (CONT'D)

Landing Distance – Flap 45° (Page 1 of 2)
 Figure 07-19-5

7. PERFORMANCE (CONT'D)

Landing Distance – Flap 45° (Page 2 of 2)
Figure 07-19-5

CHALLENGER 650	SUPPLEMENT 19 Operation on Wet Grooved Runways	07-19-01-20
		Rev 14, Nov 19/18

8. SUPPLEMENTS

The Supplements in Chapter 7 are applicable, except as specified by the following:

A. Supplement 12– Steep Approaches up to 5.5 Degrees

Use of Supplement 12 in conjunction with Supplement 19 is permitted provided that the limitations and procedures in Supplement 12, that pertain to Supplement 19, are followed.

B. Supplement 13– Additional Take-Off Performance Information

No take-off performance credit for wet grooved runway operation is claimed for Supplement 13. When dispatching on a wet grooved runway and using Supplement 13, use the wet runway take-off data of Supplement 13.

C. Supplement 20– FAA Steep Approach into London City Airport

Use of Supplement 20 in conjunction with Supplement 19 is permitted provided that the limitations and procedures in Supplement 20, that pertain to Supplement 19, are followed.

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		Rev 8, Mar 13/17

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CHALLENGER 650	SUPPLEMENT 20 FAA Steep Approach into London City Airport	07-20-01-1 Rev 10, Dec 04/17
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NOTE

This Airplane Flight Manual Supplement is only applicable to airplanes incorporating Service Bulletin 650-34-008.

This Airplane Flight Manual Supplement is **ONLY** applicable to FAA registered airplanes operating into London City Airport (LCY).

This Airplane Flight Manual Supplement does not constitute approval to conduct Steep Approach operations.

1. GENERAL

The data in this Supplement must be used when conducting steep approaches into London City Airport (LCY).

These data complement or supersede data contained in the basic Airplane Flight Manual. This supplement must therefore, be read in conjunction with the basic Airplane Flight Manual.

The effect of this Supplement on the basic Airplane Flight Manual is given in paragraphs (2) to (8).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Structural Weight

Maximum landing weight: 14,742 kg (32,500 lb)

B. Operating Limitations

(1) OPERATION IN ICING CONDITIONS

Steep approaches in known icing conditions are prohibited.

(2) WIND LIMITS

No tailwind component is permitted for steep approach landings.

The maximum crosswind component for steep approach landings is 20 knots.

Increasing landing reference speed by more than 10 KIAS for crosswind and/or gusts is prohibited. If wind conditions require a speed adder of more than 10 KIAS, divert to a suitable landing field that does not require a steep approach procedure.

(3) APPROACHES

Operations at airports other than London City Airport (LCY) are prohibited.

Deliberate one engine inoperative steep approaches are prohibited.

Partial flap steep approaches and landings are prohibited.

Steep approach landing on contaminated runways is prohibited.

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CHALLENGER 650	SUPPLEMENT 20 FAA Steep Approach into London City Airport	07-20-01-2
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3. LIMITATIONS (CONT'D)

C. Power Plant

(1) AUTOTHROTTLE (ATS)

Use of the ATS for steep approaches or landing is prohibited.

D. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

Use of autopilot for steep approaches is prohibited.

(2) ANTI-ICE SYSTEM

Approach and landing with wing and/or cowl anti-ice selected ON is prohibited.

(3) SYSTEMS INOPERATIVE

Landings at a steep approach airfield are prohibited if any of the following systems are inoperative:

- Flaps
- Ground Spoilers
- Anti-Skid
- Nose Wheel Steering
- Radio Altimeter
- Thrust Reverser

E. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

Use of the FMS approach performance function for steep approach is prohibited.

FMS approaches with LPV guidance are prohibited.

(2) TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

The TAWS steep approach switch/light must be selected ON as directed in the normal procedures of this supplement.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees,
- an increase in actual landing distance by a factor greater than 1.50 (50%) or
- a landing with gear up or unsafe.
- a landing with any of the systems inoperative as listed in the system limitations of this supplement.

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5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

NOTE

Use of ILS or FMS VNAV guidance with or without the flight director is authorized.

A. Before Start Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

B. In Range Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press in Check ON light on.

C. Before Landing Check

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction) up to a maximum of $V_{REF} + 10$ KIAS.

CHALLENGER 650	SUPPLEMENT 20 FAA Steep Approach into London City Airport	07-20-01-4
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5. NORMAL PROCEDURES (CONT'D)

D. Go-Around

- (1) Thrust levers/TOGA switchAdvance/Press
- (2) FLIGHT SPOILER lever
(if extended)Select to RETRACT.
Check EICAS indicates
spoilers retracted.
- (3) Pitch attitude..... Increase rapidly but smoothly to initial
pitch attitude of 10° at a speed
of not less than V_{REF} , then
adjust pitch attitude to
maintain approach climb
speed.

NOTE

1. Contact with the ground may be unavoidable if go-around is initiated from a low altitude or high sink rate. If the aircraft contacts the ground during the go-around, maintain the normal landing attitude or slightly higher. Aircraft configuration should not be changed until the aircraft is climbing if ground contact is made or not.
 2. A momentary GND SPLRS caution message may be displayed if ground contact occurs.
- (4) FLAPSSelect to 20°.

When a positive rate of climb is achieved:

- (5) LDG GEAR leverSelect to UP.

At a safe altitude (not below 400 feet AGL):

- (6) FLAPSSelect to 0° at a speed of not less
than $V_{FTO} - 5$ KIAS.
- (7) THRUST REVERSER, L and R
switches.....Select to OFF.
Check L and R REV ARMED
advisory messages out.
- (8) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

E. After Landing Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

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6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees or
- an increase in actual landing distance by a factor greater than 1.50 (50%).
- a landing with any of the systems inoperative as listed in the system limitations of this supplement.

A. Engine Failure During Approach

- (1) Operating engine Increase thrust as required.
- (2) FLIGHT SPOILER lever
(If extended) Select to RETRACT.
Check EICAS indicates
spoilers retracted.
- (3) FLAPS Select to 20°.
- (4) Airspeed Increase to V_{REF} (Flaps 45°) + 14 KIAS.
- (5) Airplane Retrim and go-around (Refer to
Single Engine Go-Around in
this section).
If required, autopilot may be
engaged, if above 800 feet
AGL.
Divert to a suitable landing
field that does not require a
steep approach procedure.

NOTE

1. The height loss during a single engine go-around is no more than 200 feet.
2. After an engine failure at or below 200 feet, single engine approach to landing is authorized with flaps at 45 degrees.

B. Single Engine Go-Around

NOTE

1. The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
2. Single engine go-around capability is not assured with flaps greater than 20°.

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6. ABNORMAL PROCEDURES (CONT'D)

B. Single Engine Go-Around (Cont'd)

- (1) Normal Take-off Thrust Set immediately (Refer to Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (2) TOGA push switch Depress
- (3) FLIGHT SPOILER lever
(If extended) Select to RETRACT.
Check EICAS indicates spoilers retracted.
- (4) Airplane Rotate rapidly but smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB).

NOTE

- Contact with the ground may be unavoidable if go-around is initiated from a low altitude or high sink rate. If the aircraft contacts the ground during the go-around, maintain the normal landing attitude or slightly higher. Aircraft configuration should not be changed until the aircraft is climbing if ground contact is made or not.
- A momentary GND SPLRS caution message may be displayed if ground contact occurs.

- (5) FLAPS Select to 20°.

When a positive rate of climb is achieved:

- (6) LDG GEAR lever Select to UP.

At a safe altitude (not below 400 feet AGL):

- (7) FLAPS Select to 0° at a speed of not less than $V_{FTO} - 5$ KIAS.

- (8) THRUST REVERSER, L and R switches Select to OFF.
Check L and R REV ARMED advisory messages out.

If go-around was due to engine failure during approach:

- (9) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5 – ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

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		Rev 15, Mar 25/19

7. PERFORMANCE

The performance in Chapter 6 is applicable, except as modified by the following:

The data in this supplement are applicable to steep approach (5.5 degrees) landings into London City Airport only.

A. Landing Performance

The reference screen height used for landing is 60 feet.

The reference steep approach glide slope angle is 5.5 degrees.

(1) LANDING FIELD LENGTH AND SPEED

The landing field length required for 5.5 degree approach operations for varying conditions of airplane landing weight, airport pressure altitude and wind is provided by Figure 07-20-1. The actual landing distance required is also given within this chart.

The landing approach speed (V_{REF}) for varying airplane landing weights and airport pressure altitude for steep approach operations is also given by Figure 07-20-1.

The actual landing distance given on Figure 07-20-1 is the distance from a 60 foot height at V_{REF} with flaps 45°, to a full stop on a smooth, dry, level, hard-surfaced runway and ISA temperature for steep approach operations.

NOTE

1. The landing field length derived from Figure 07-20-1 includes factors for operational rules which require use of 60% of the available runway (landing field length required).
2. Operational rules may require that for wet runway surfaces, the runway length available be at least 115% of the landing field length derived from Figure 07-20-1.

Example:

Associated Conditions:

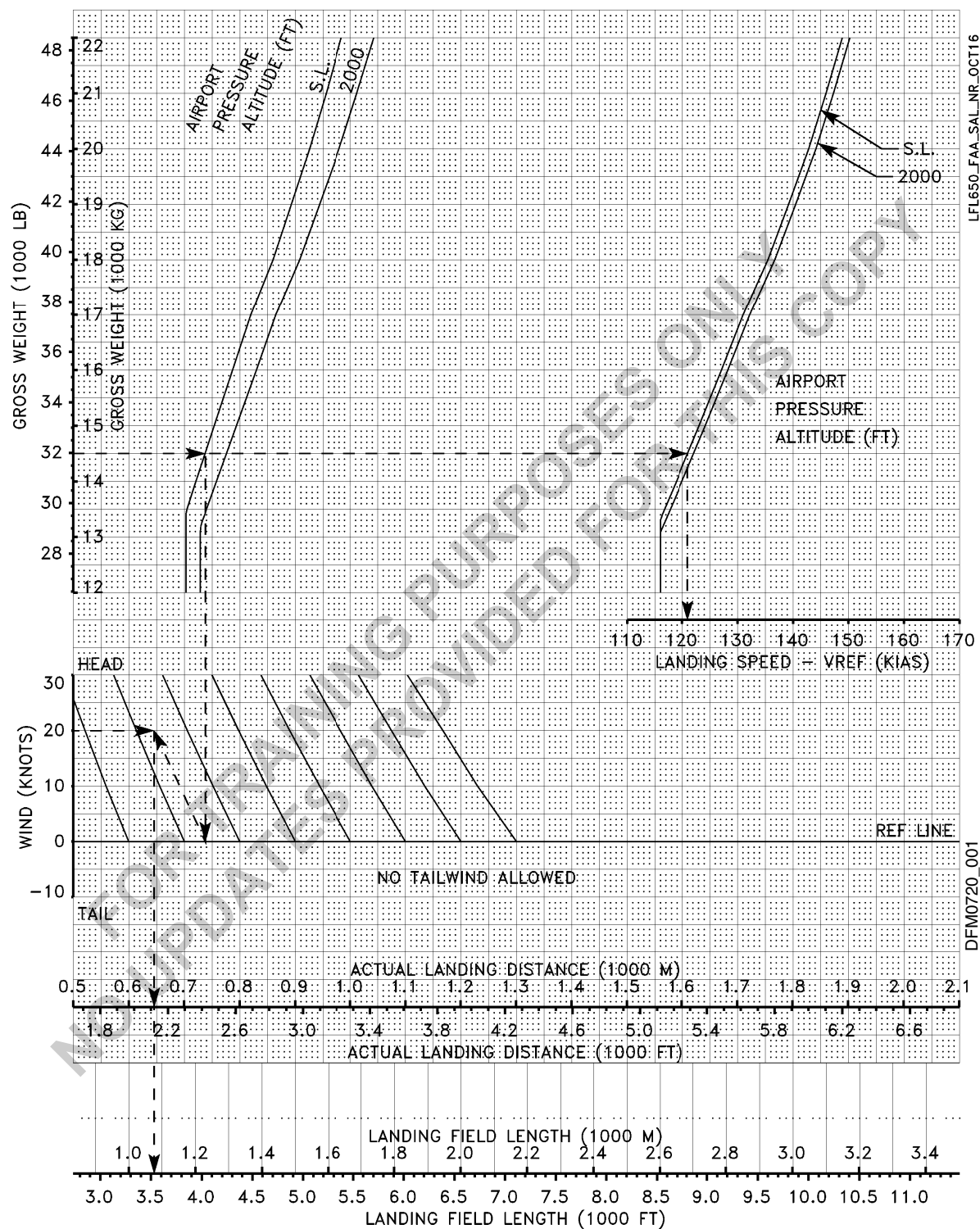
Landing gross weight	= 14,500 kg (31,967 lb)
Airport pressure altitude	= Sea Level
Wind	= 20 knots, headwind

The example in Figure 07-20-1, shows that for the above conditions, the actual landing distance is 646 meters (2,120 feet) and the landing field length is 1,077 meters (3,533 feet).

V_{REF} for the same conditions as above is 121 KIAS.

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7. PERFORMANCE (CONT'D)



Landing Field Length and Landing Speed, Flaps 45
Figure 07-20-1

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8. SUPPLEMENTS

The supplements in Chapter 7 are applicable, except as modified by the following:

A. Supplement 12– Steep Approaches up to 5.5 Degrees

Use of Supplement 12 is prohibited.

B. Supplement 19 – Operation on Wet Grooved Runways

(1) RUNWAY CONDITIONS

The data is applicable to concrete runway surfaces with transverse grooves constructed and maintained to FAA standard AC 150/5320-12C, or equivalent, and on which there are no areas of significant rubber deposit in sections where the high-speed portion of braking is expected.

(a) Wet Runway

A grooved runway is considered wet when there is sufficient moisture on the runway surface to cause it to appear partially reflective, but without significant areas of standing water (runway grooves not overflowing).

(2) LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

- The data may be used only for wet grooved runways.
- Non-performance related limitations and procedures that are applicable to a wet runway, are also applicable to a wet grooved runway.
- The use of this Supplement for landing performance requires both thrust reversers to be operational.
- The use of this Supplement for landing performance requires the anti-skid system to be fully operational.
- The use of this Supplement requires that :
 - the runway is maintained to FAA standard AC 150/5320-12C, or equivalent,
 - there are no areas of significant rubber deposit in sections where the high-speed portion of braking is to occur, and
 - there is no indication, via NOTAMs, PIREPs or braking action reports, that degraded braking exists on the runway.
- This Supplement is applicable only to grooved concrete surfaces.
- The use of a wet grooved runway corrected landing distance which is shorter than the dry runway landing field length is prohibited.

(3) EMERGENCY PROCEDURES

The emergency procedures in supplement 20 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

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8. SUPPLEMENTS (CONT'D)

(4) ABNORMAL PROCEDURES

The abnormal procedures in supplement 20 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

(5) PERFORMANCE

The performance in supplement 20 is applicable, except as modified by the following:

The landing distance required on a wet grooved runway for 5.5 degree approach operations for varying conditions of airplane weight, ambient temperature, airport altitude pressure, reported wind and runway slope is provided by Figure 07-20-2.

The landing field length from supplement 20 must also be determined when using the following data. If the landing field length on a dry runway is greater than the corrected landing distance on a wet grooved runway, use the landing field length.

The landing approach speed (V_{REF}) for varying airplane landing weights and airport pressure altitude for steep approach operations is also given by Figure 07-20-1.

The landing distance on wet grooved runways, for a flaps-45° landing, with both engines operating from a 60 foot height, for speeds at the runway threshold from V_{REF} to $V_{REF} + 10$ KIAS, is provided by Figure 07-20-2.

NOTE

The corrected landing distance on wet grooved runways includes a landing distance factor of 1.15 (15%).

Example:

Associated Conditions:

Landing gross weight	= 14,500 kg (31,967 lb)
Airport pressure altitude	= Sea Level
Ambient temperature	= 35°C
Wind	= 22 knots at 26 degrees relative to runway (20 knots headwind; 10 knots crosswind)
Runway slope	= -0.08%
ΔV_{REF}	= +5 KIAS

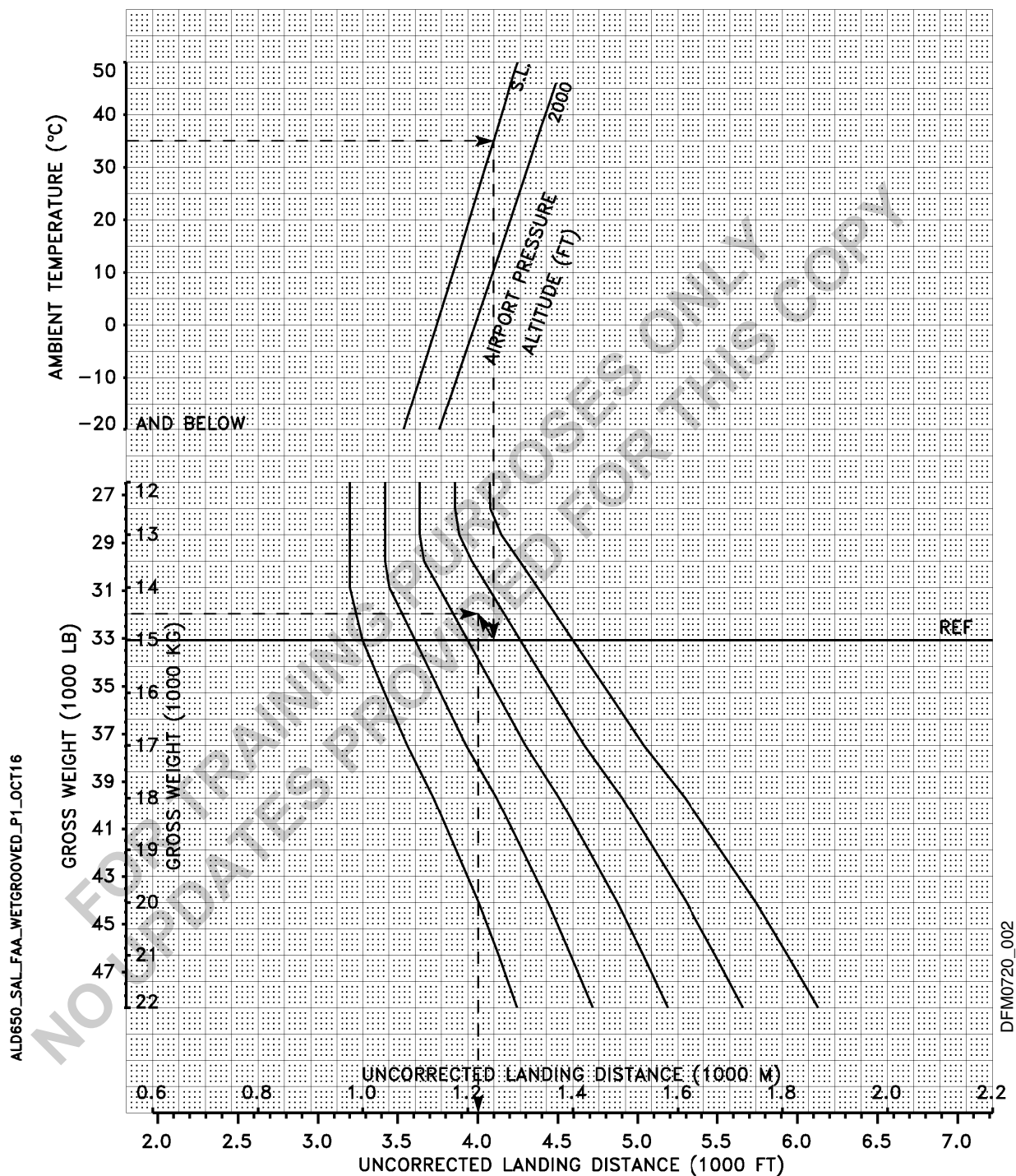
The example in Figure 07-20-2, shows that for the above conditions, the corrected landing distance is 1,115 meters (3,657 feet).

From the associated conditions stated above, the landing field length on a dry runway from Figure 07-20-1 is 1,077 meters (3,533 feet).

Since the landing field length on a dry runway is lower than the corrected landing distance on a wet grooved runway, the corrected landing distance on a wet grooved runway must be used.

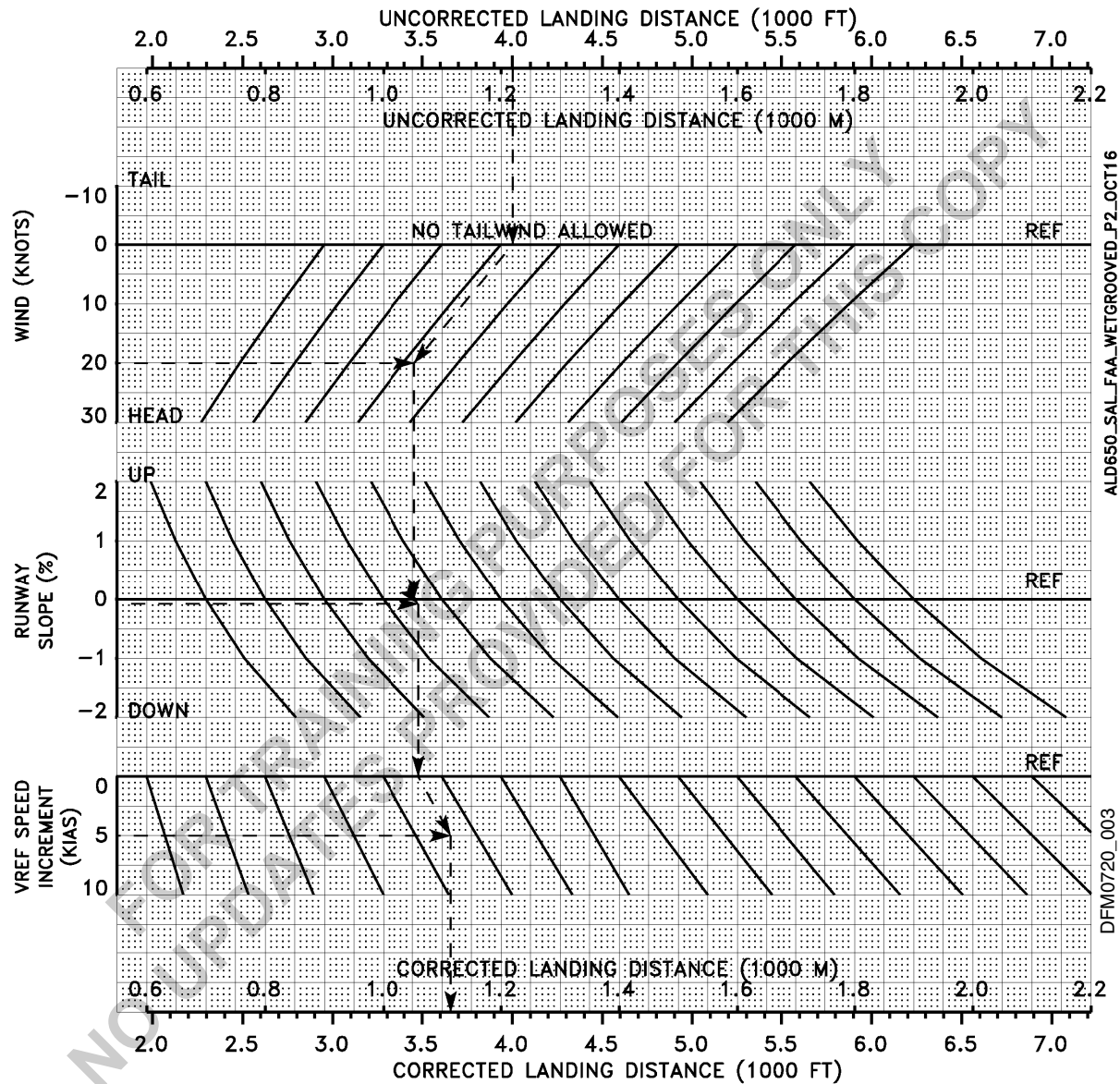
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8. SUPPLEMENTS (CONT'D)



Landing Distance – Wet Grooved Runway – Flap 45° (Page 1 of 2)
Figure 07-20-2

8. SUPPLEMENTS (CONT'D)



Landing Distance – Wet Grooved Runway – Flap 45° (Page 2 of 2)
 Figure 07-20-2

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CHAPTER 8 – APPENDICES

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APPENDIX 1 – CONFIGURATION DEVIATION LIST

08-01-00-1

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	CL-650 Airplane Flight Manual PSP 650-1	
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CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-01-1
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1. GENERAL LIMITATIONS

This Configuration Deviation List (CDL) contains additional limitations for operation of the CL600-2B16 Challenger airplane (Serial No. 6050 and subsequent), without certain secondary airframe and/or nacelle parts as listed herein. The Limitations in the Airplane Flight Manual are applicable except as amended in this appendix.

The associated limitations must be listed on a placard affixed in the flight compartment on the instrument panel in clear view of both pilots.

The pilot in command will be notified of each operation with a missing part(s) by listing the missing part(s) in the flight or dispatch release.

The operator will list in the airplane logbook an appropriate notation covering the missing part(s) on each flight.

If an additional part is lost in flight, the airplane may not depart the airport at which it landed following this event until it again complies with the limitations of this appendix. This, of course, does not preclude the issuance of a ferry permit to allow the airplane to be flown to a point where the necessary repairs or replacements can be made.

An asterisk (*) against a part indicates that only one flight is permitted to an airport when the necessary repairs or replacements can be made. This flight must not be carried out in known or forecast lightning conditions.

No more than one part for any one sub-system in this appendix may be missing unless specifically designated combinations are indicated herein. Unless otherwise specified herein, parts from different sub-systems may be missing. The performance penalties are cumulative unless specifically designated penalties for combination of missing parts are indicated.

Where performance penalties are listed as negligible, no more than three negligible items may be missing for take-off unless the following performance penalties are applied for each additional negligible item:

Take-Off and Landing Weight:	Reduce by 46 Kg (100 lb).
Enroute Climb Limited Weight:	Reduce by 46 Kg (100 lb).

Take-off performance decrements are applicable to take-off gross weight which are limited by structural limit, field length, first segment climb, second segment climb, final segment climb, or take-off flight path. The actual take-off weight must not exceed the normal take-off weight limit minus the CDL take-off weight penalty.

Enroute performance decrements are applicable to operations which are limited by enroute, one-engine-inoperative climb performance.

Landing performance decrements are applicable to landing gross weights which are limited by structural limit, landing field length, landing climb, or approach climb. The maximum allowable landing weight must not exceed the normal landing weight limit minus the CDL landing weight penalty.

The numbering and designation of systems in this appendix is based on ATA Spec. 100. The parts within each system are identified by functional description and, when necessary, by part numbers.

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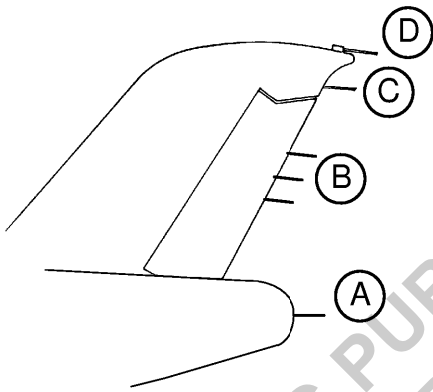
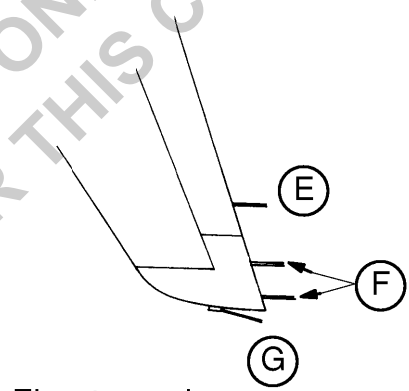
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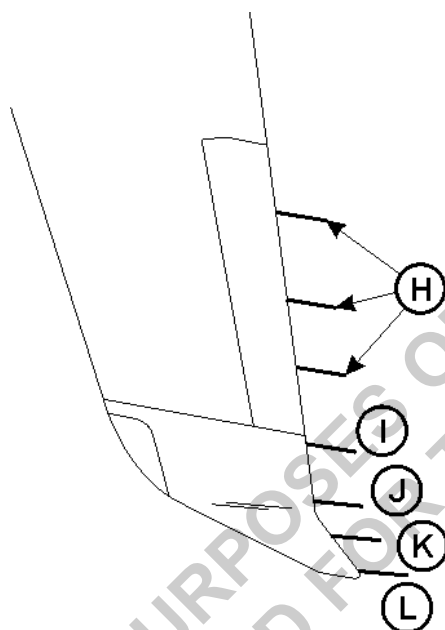
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CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-23-1
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SYSTEM 23 COMMUNICATIONS																										
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																								
		2. Remarks and/or Exceptions																								
23-61 Static Dischargers Items A to L		28 Certain static dischargers may be damaged or missing, as detailed in the deviation table.																								
 Vertical Stabilizer Tip, Rudder and Tail Cone  Elevator and Horizontal Stabilizer Tip <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Item</th><th>Quantity Installed</th><th>Deviation</th></tr> </thead> <tbody> <tr> <td>A</td><td>1</td><td>—</td></tr> <tr> <td>B</td><td>3</td><td>One may be damaged or missing.</td></tr> <tr> <td>C</td><td>1</td><td>—</td></tr> <tr> <td>D</td><td>1</td><td>—</td></tr> <tr> <td>E</td><td>2 (1 per side)</td><td>—</td></tr> <tr> <td>F</td><td>4 (2 per side)</td><td>One may be damaged or missing from each side.</td></tr> <tr> <td>G</td><td>2 (1 per side)</td><td>—</td></tr> </tbody> </table>			Item	Quantity Installed	Deviation	A	1	—	B	3	One may be damaged or missing.	C	1	—	D	1	—	E	2 (1 per side)	—	F	4 (2 per side)	One may be damaged or missing from each side.	G	2 (1 per side)	—
Item	Quantity Installed	Deviation																								
A	1	—																								
B	3	One may be damaged or missing.																								
C	1	—																								
D	1	—																								
E	2 (1 per side)	—																								
F	4 (2 per side)	One may be damaged or missing from each side.																								
G	2 (1 per side)	—																								

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SYSTEM 23 COMMUNICATIONS


Aileron and Wing Tip

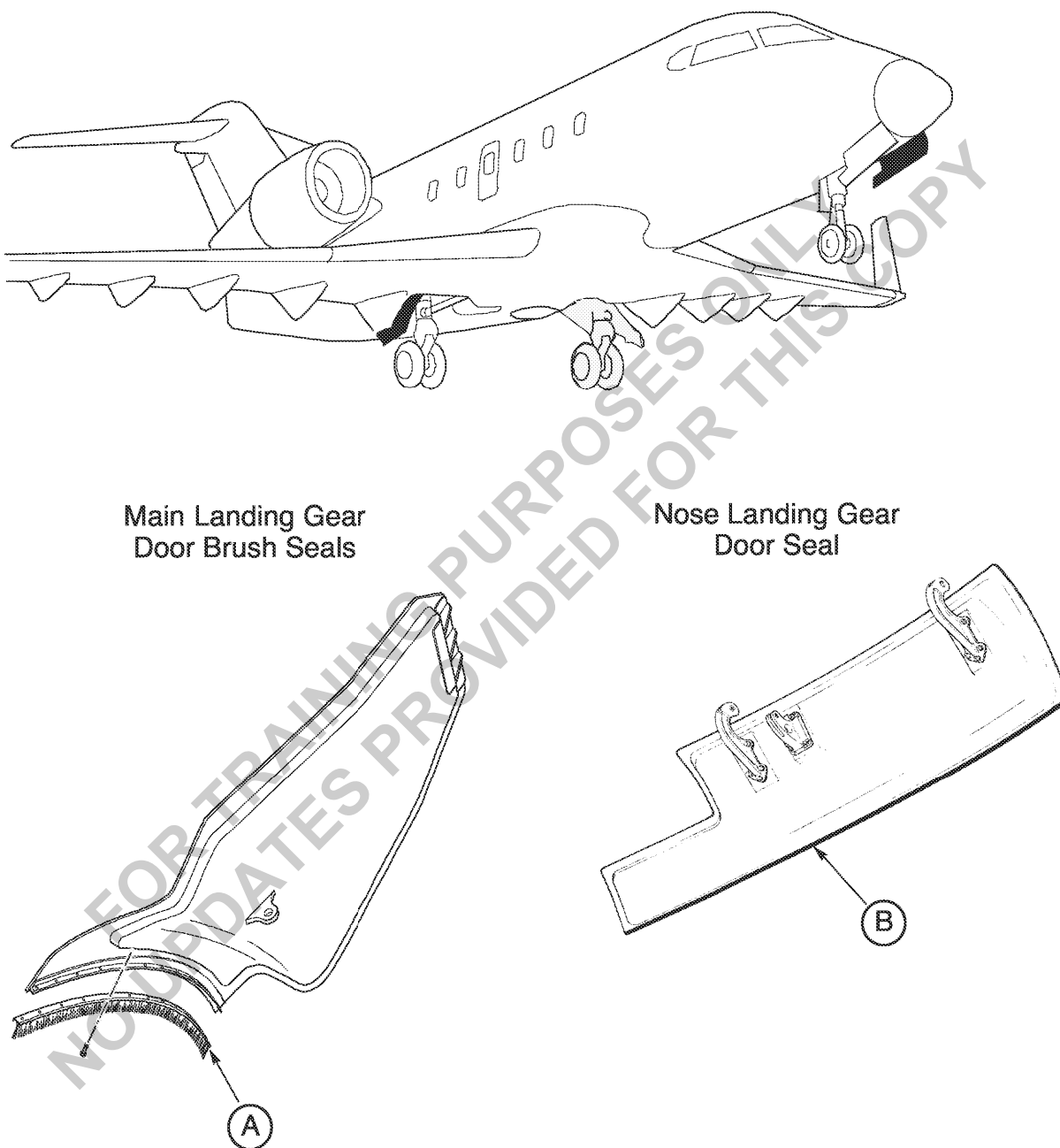
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Item	Quantity Installed	Deviation
H	6 (3 per side)	One may be damaged or missing from each side.
I	2 (1 per side)	One may be damaged or missing from each side. Item J and either item K or L must be intact.
J	2 (1 per side)	One may be damaged or missing from each side. Item I and either item K or L must be intact.
K	2 (1 per side)	One may be damaged or missing from each side. Item L and either item I or J must be intact.
L	2 (1 per side)	One may be damaged or missing from each side. Item K and either item I or J must be intact.

CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-32-1
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SYSTEM 32 LANDING GEAR																		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																
32-12 Main Landing Gear Door Brush Seal Item A	2	2. Remarks and/or Exceptions One or both brush seals may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–23 kg / brush seal</td><td>–50 lb / brush seal</td></tr></table><table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–16 kg / brush seal</td><td>–35 lb / brush seal</td></tr></table><table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–41 kg / brush seal</td><td>–90 lb / brush seal</td></tr></table> – and the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td>+0.4% / brush seal</td><td>+0.4% / brush seal</td></tr></table>	TAKE-OFF WEIGHT		–23 kg / brush seal	–50 lb / brush seal	LANDING WEIGHT		–16 kg / brush seal	–35 lb / brush seal	ENROUTE CLIMB		–41 kg / brush seal	–90 lb / brush seal	FUEL CONSUMPTION		+0.4% / brush seal	+0.4% / brush seal
TAKE-OFF WEIGHT																		
–23 kg / brush seal	–50 lb / brush seal																	
LANDING WEIGHT																		
–16 kg / brush seal	–35 lb / brush seal																	
ENROUTE CLIMB																		
–41 kg / brush seal	–90 lb / brush seal																	
FUEL CONSUMPTION																		
+0.4% / brush seal	+0.4% / brush seal																	
32-22 Nose Landing Gear Door Seal Item B	1	The seal of the nose landing gear door may be missing. The effect on the performance is negligible.																

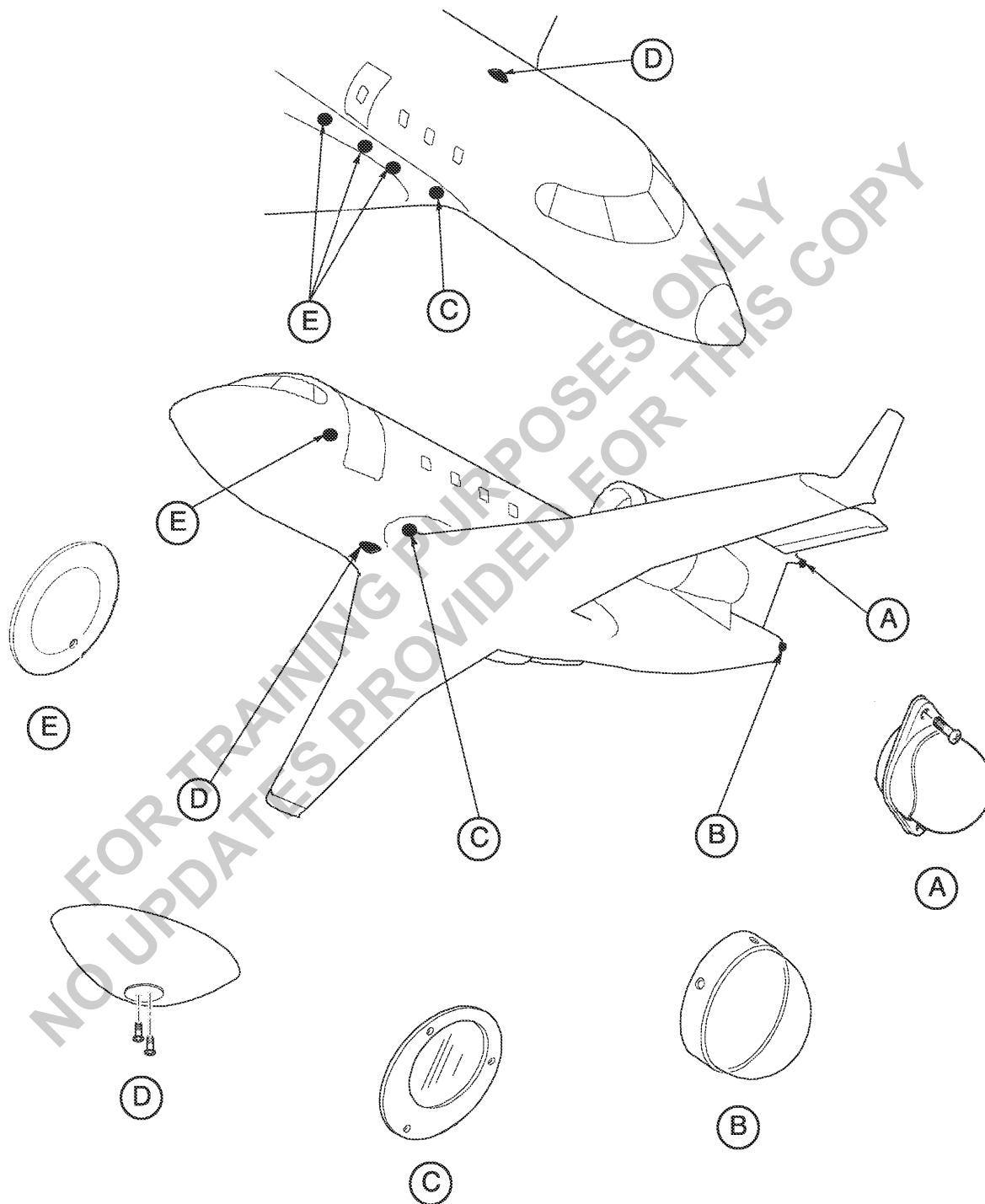
SYSTEM 32 LANDING GEAR



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* Only one flight is permitted, to an airport where the necessary repairs or replacements can be made. This flight must not be carried out in known or forecast lightning conditions.

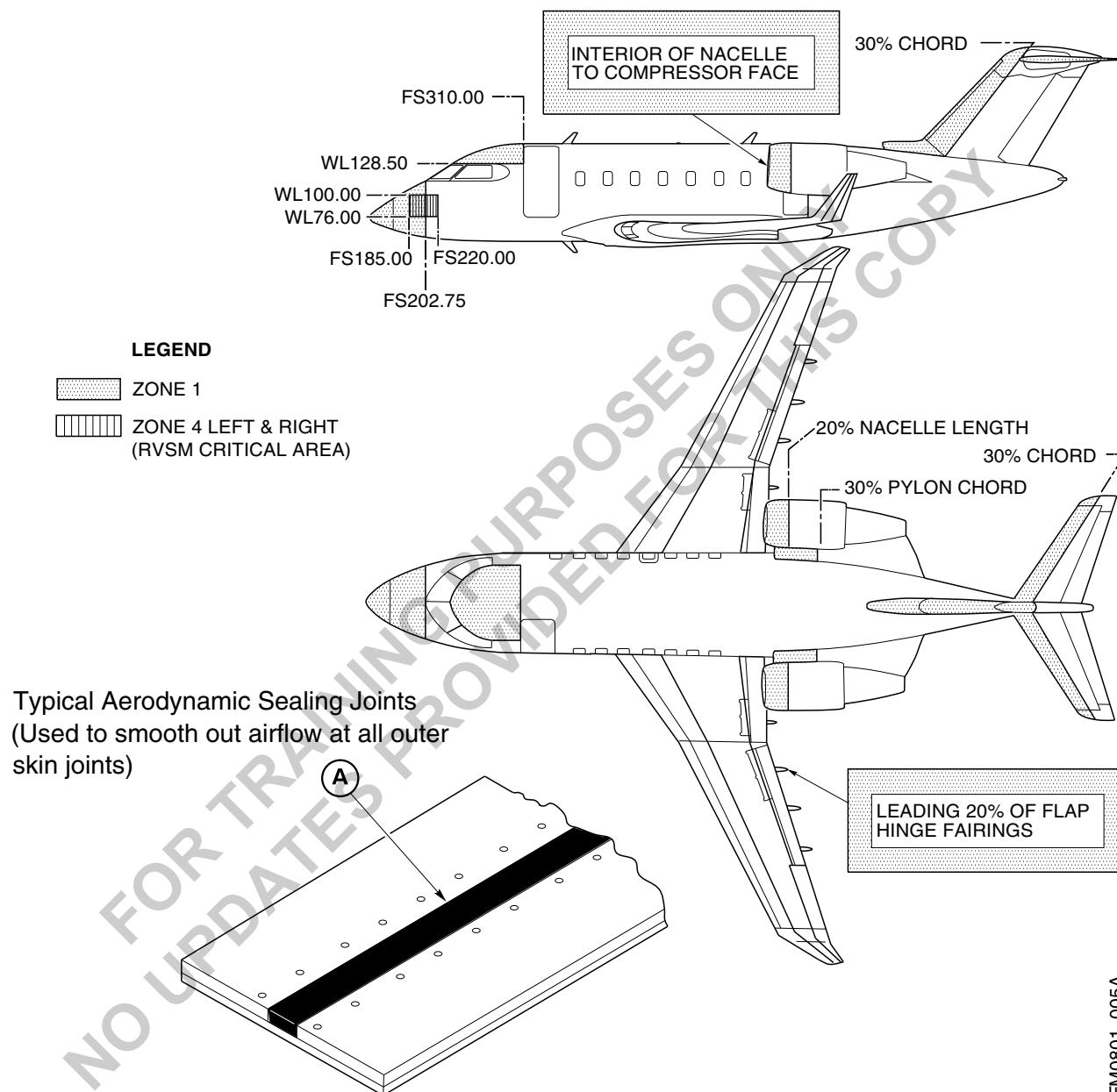
SYSTEM 33 LIGHTS



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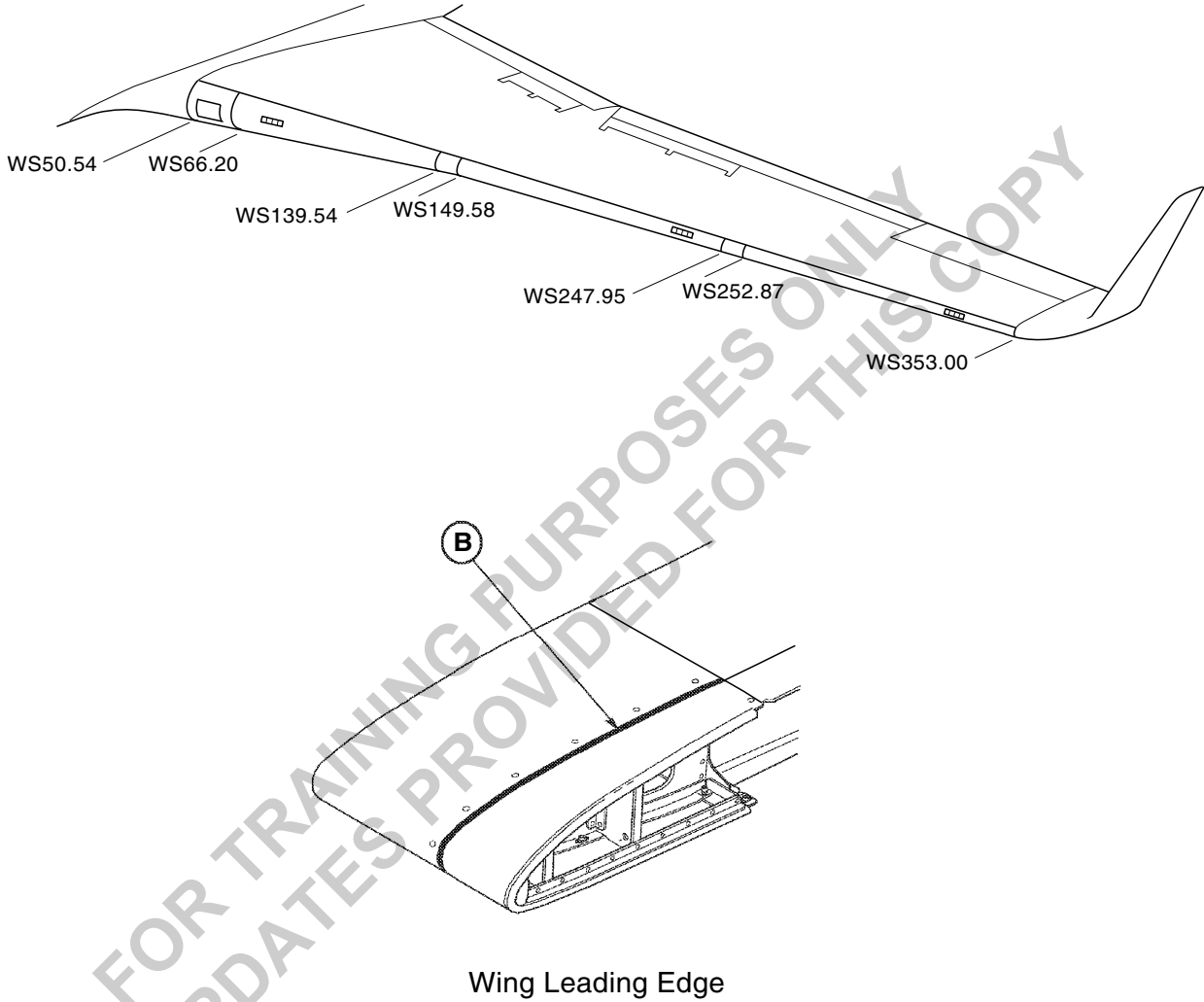
SYSTEM 51 STRUCTURES - GENERAL																	
SUB SYSTEM	ITEM																
51-00 Aerodynamic Sealant – General Item A	1. Number required for all flight conditions except as provided in column 2 2. Remarks and/or Exceptions – Any length of aerodynamic sealant may be missing in areas other than the wing leading edge / winglet, around pitot tubes and zones 1 and 4. In addition, the following areas may have aerodynamic sealant missing for a maximum of 100 flight hours: – Intersection of polished leading edge and adjacent skin on vertical stabilizer, – External portion of engine inlet cowl, – Scuff plate on leading edge of horizontal stabilizer, – Fuel cap on top of wing, – Light under each pylon and – Around tail tank relief valve on left side of vertical stabilizer. Aerodynamic sealant may be missing as specified above provided: – the performance and structurally limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–14 kg per every 10 ft or less missing</td><td>–30 lb per every 10 ft or less missing</td></tr></table><table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–11 kg per every 10 ft or less missing</td><td>–25 lb per every 10 ft or less missing</td></tr></table><table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–25 kg per every 10 ft or less missing</td><td>–55 lb per every 10 ft or less missing</td></tr></table> – the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td colspan="2">+0.25% per every 10 ft or less missing</td></tr></table>	TAKE-OFF WEIGHT		–14 kg per every 10 ft or less missing	–30 lb per every 10 ft or less missing	LANDING WEIGHT		–11 kg per every 10 ft or less missing	–25 lb per every 10 ft or less missing	ENROUTE CLIMB		–25 kg per every 10 ft or less missing	–55 lb per every 10 ft or less missing	FUEL CONSUMPTION		+0.25% per every 10 ft or less missing	
TAKE-OFF WEIGHT																	
–14 kg per every 10 ft or less missing	–30 lb per every 10 ft or less missing																
LANDING WEIGHT																	
–11 kg per every 10 ft or less missing	–25 lb per every 10 ft or less missing																
ENROUTE CLIMB																	
–25 kg per every 10 ft or less missing	–55 lb per every 10 ft or less missing																
FUEL CONSUMPTION																	
+0.25% per every 10 ft or less missing																	

SYSTEM 51 STRUCTURES - GENERAL



SYSTEM51 STRUCTURES - GENERAL																		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																
51-00 Aerodynamic Sealant – Wing Leading Edge Item B		2. Remarks and/or Exceptions – Any length of chord wise aerodynamic sealant may be missing from the wing leading edge for a maximum of 25 flight hours provided: – Sealant is missing only at one of WS50.54, WS66.20, WS139.54, WS149.58 or WS 353.00 on either or both wings; or sealant is missing at one or both of WS247.95 and/or WS252.87, on either or both wings. – The Maximum Takeoff Weight (MTOW) is reduced to 17,917 kg (39,500 lb). – The Maximum Landing Weight (MLW) is reduced to 14,923 kg (32,900 lb). Loose aerodynamic sealant on the wing leading edge is not permitted. Missing span wise aerodynamic sealant on the wing leading edge is not permitted. Aerodynamic sealant may be missing as specified above provided: – the performance limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–45 kg</td><td>–100 lb</td></tr></table><table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–36 kg</td><td>–80 lb</td></tr></table><table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–82 kg</td><td>–180 lb</td></tr></table> – the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td colspan="2">+1.0%</td></tr></table>	TAKE-OFF WEIGHT		–45 kg	–100 lb	LANDING WEIGHT		–36 kg	–80 lb	ENROUTE CLIMB		–82 kg	–180 lb	FUEL CONSUMPTION		+1.0%	
TAKE-OFF WEIGHT																		
–45 kg	–100 lb																	
LANDING WEIGHT																		
–36 kg	–80 lb																	
ENROUTE CLIMB																		
–82 kg	–180 lb																	
FUEL CONSUMPTION																		
+1.0%																		

SYSTEM 51 STRUCTURES - GENERAL



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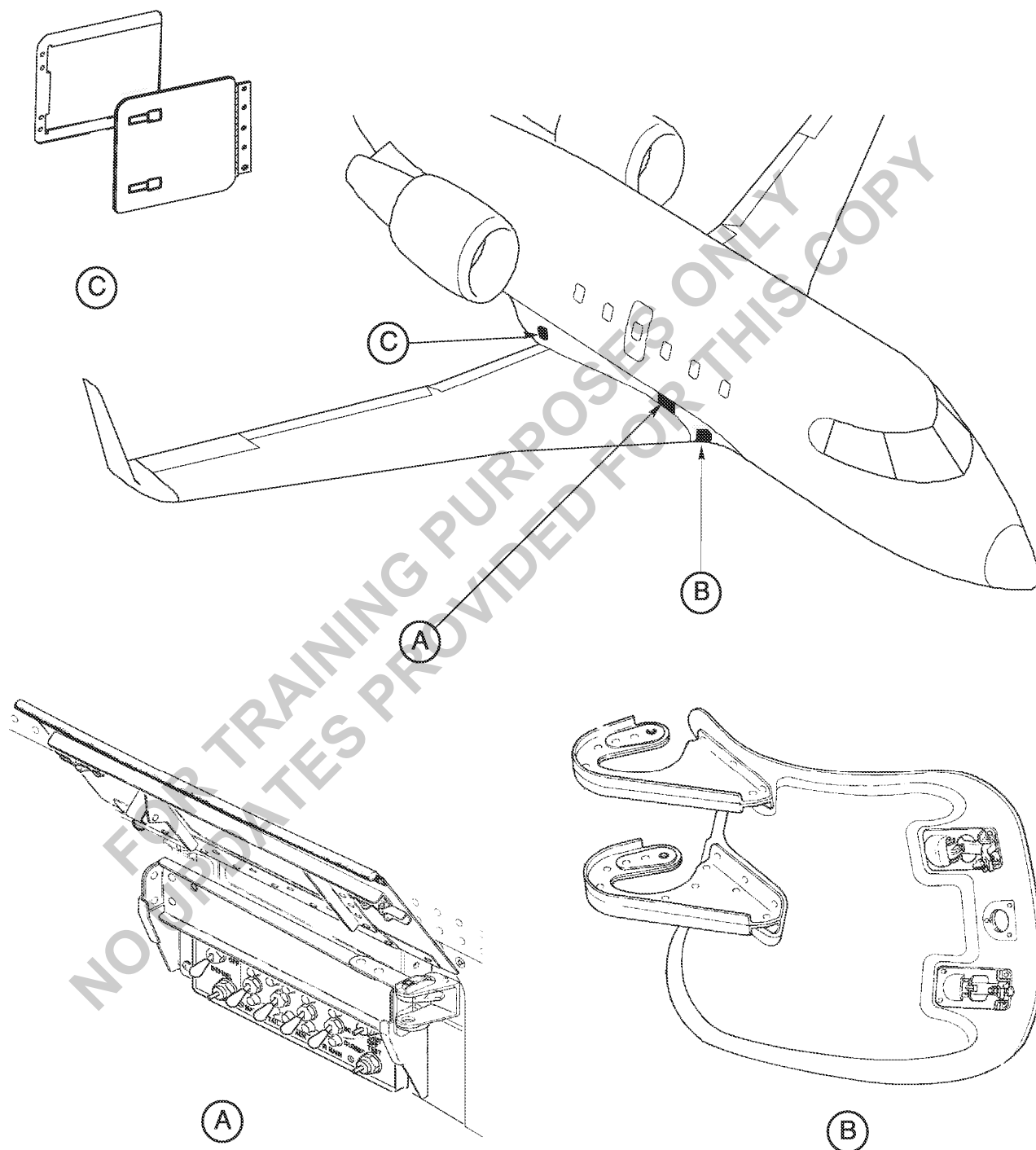
SYSTEM 52 DOORS	
SUB SYSTEM	ITEM
52-45 Main Pressure Refueling Control Panel Door *	Item A
52-45 Refueling Adaptor Door *	Item B
52-45 Hydraulic System Service Doors *	Item C

1. Number required for all flight conditions except as provided in column 2																	
2. Remarks and/or Exceptions																	
1	The main pressure refueling control panel door may be missing. The effect on the performance is negligible.																
1	The refueling adaptor door may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–226 kg</td><td>–500 lb</td></tr><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–171 kg</td><td>–375 lb</td></tr><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–421 kg</td><td>–930 lb</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+4.10%</td><td>+4.10%</td></tr></table>	TAKE-OFF WEIGHT		–226 kg	–500 lb	LANDING WEIGHT		–171 kg	–375 lb	ENROUTE CLIMB		–421 kg	–930 lb	FUEL CONSUMPTION		+4.10%	+4.10%
TAKE-OFF WEIGHT																	
–226 kg	–500 lb																
LANDING WEIGHT																	
–171 kg	–375 lb																
ENROUTE CLIMB																	
–421 kg	–930 lb																
FUEL CONSUMPTION																	
+4.10%	+4.10%																
2	One or both hydraulic system service doors may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–31 kg / door</td><td>–70 lb / door</td></tr><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–23 kg / door</td><td>–50 lb / door</td></tr><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–57 kg / door</td><td>–125 lb / door</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.55% / door</td><td>+0.55% / door</td></tr></table>	TAKE-OFF WEIGHT		–31 kg / door	–70 lb / door	LANDING WEIGHT		–23 kg / door	–50 lb / door	ENROUTE CLIMB		–57 kg / door	–125 lb / door	FUEL CONSUMPTION		+0.55% / door	+0.55% / door
TAKE-OFF WEIGHT																	
–31 kg / door	–70 lb / door																
LANDING WEIGHT																	
–23 kg / door	–50 lb / door																
ENROUTE CLIMB																	
–57 kg / door	–125 lb / door																
FUEL CONSUMPTION																	
+0.55% / door	+0.55% / door																

* Only one flight is permitted, to an airport where the necessary repairs or replacements can be made. This flight must not be carried out in known or forecast lightning conditions.

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SYSTEM 52 DOORS



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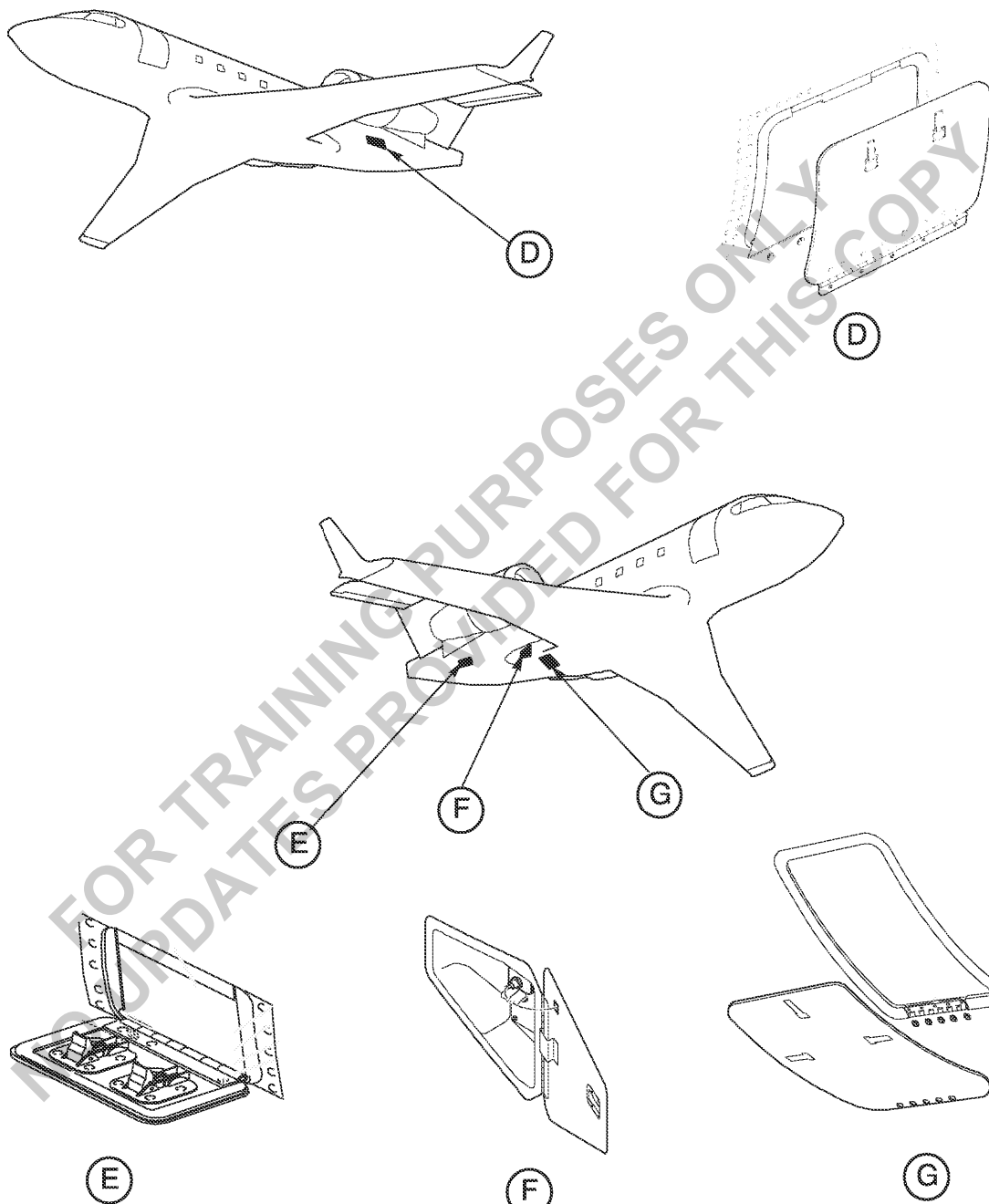
CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-52-3
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SYSTEM 52 DOORS		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2
		2. Remarks and/or Exceptions
52-45 APU Control and Ground Air Start Door *	Item D	1 The APU control and ground air start door may be missing. The effect on the performance is negligible.
52-45 DC Ground Power Door *	Item E	1 The DC ground power door may be missing. The effect on the performance is negligible.
52-45 Water Fill Access Door *	Item F	1 The water fill access door may be missing. The effect on the performance is negligible.
52-45 Toilet Service Compartment Door *	Item G	1 The toilet service compartment door may be missing. The effect on the performance is negligible.

* Only one flight is permitted, to an airport where the necessary repairs or replacements can be made. This flight must not be carried out in known or forecast lightning conditions.

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SYSTEM 52 DOORS



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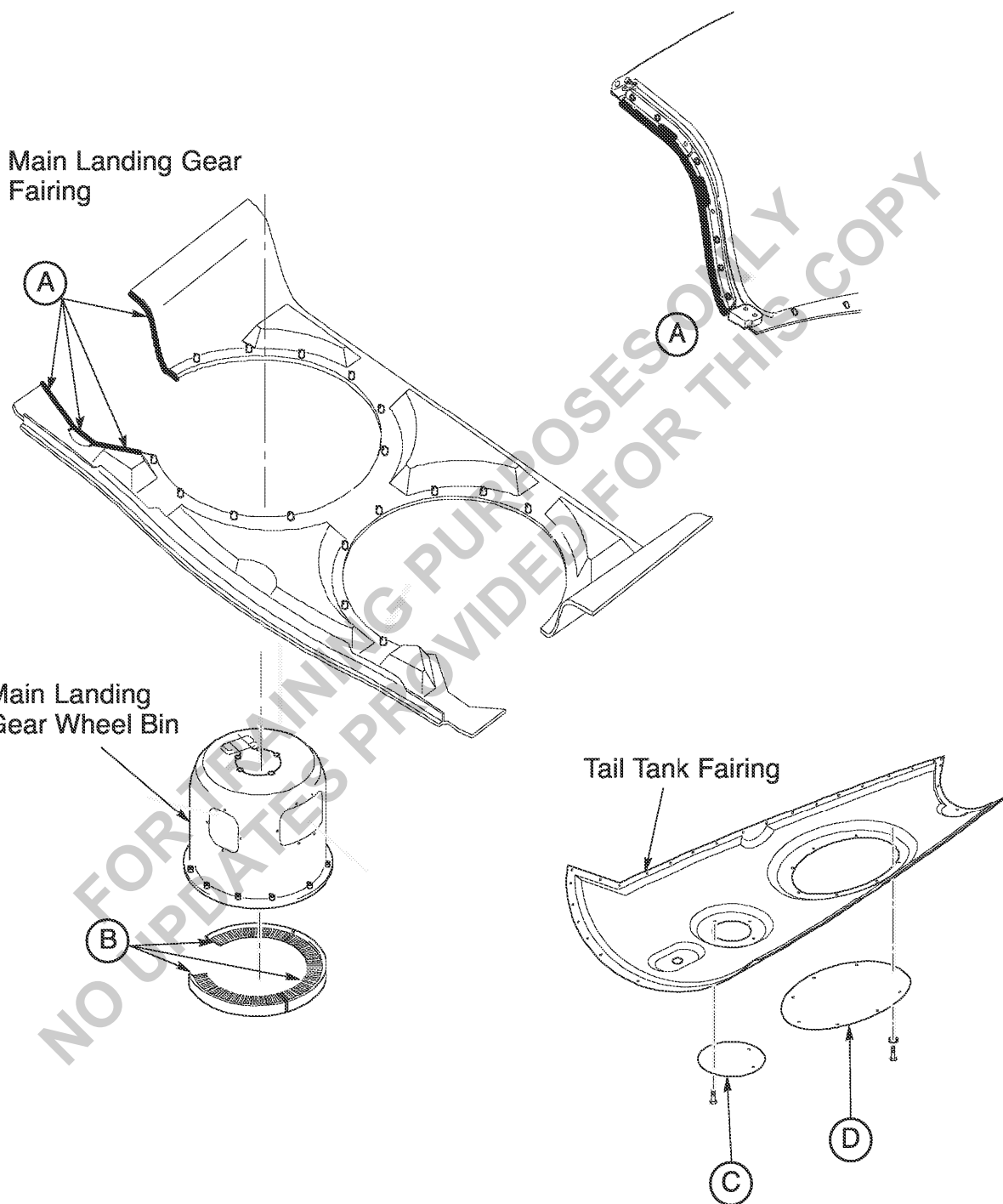
SYSTEM 53 FUSELAGE		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2
53–65	Main Landing Gear Door Seals Item A	8
53–65	Main Landing Gear Bin Brush Seals Item B	6
53–66	Tail Tank Fairing Access Panels: – Forward Panel * Item C – Aft Panel * Item D	1 1

2. Remarks and/or Exceptions	
One or any combination of main landing gear door seals may be missing. The effect on the performance is negligible.	
One or any combination of brush seals may be missing provided: – the performance and structurally limited weights are reduced by:	
TAKE-OFF WEIGHT	
–23 kg / brush seal	–50 lb / brush seal
LANDING WEIGHT	
–16 kg / brush seal	–35 lb / brush seal
ENROUTE CLIMB	
–41 kg / brush seal	–90 lb / brush seal
– and the mission fuel requirements are increased by:	
FUEL CONSUMPTION	
+0.4% / brush seal	+0.4% / brush seal

* Only one flight is permitted, to an airport where the necessary repairs or replacements can be made. This flight must not be carried out in known or forecast lightning conditions.

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SYSTEM 53 FUSELAGE



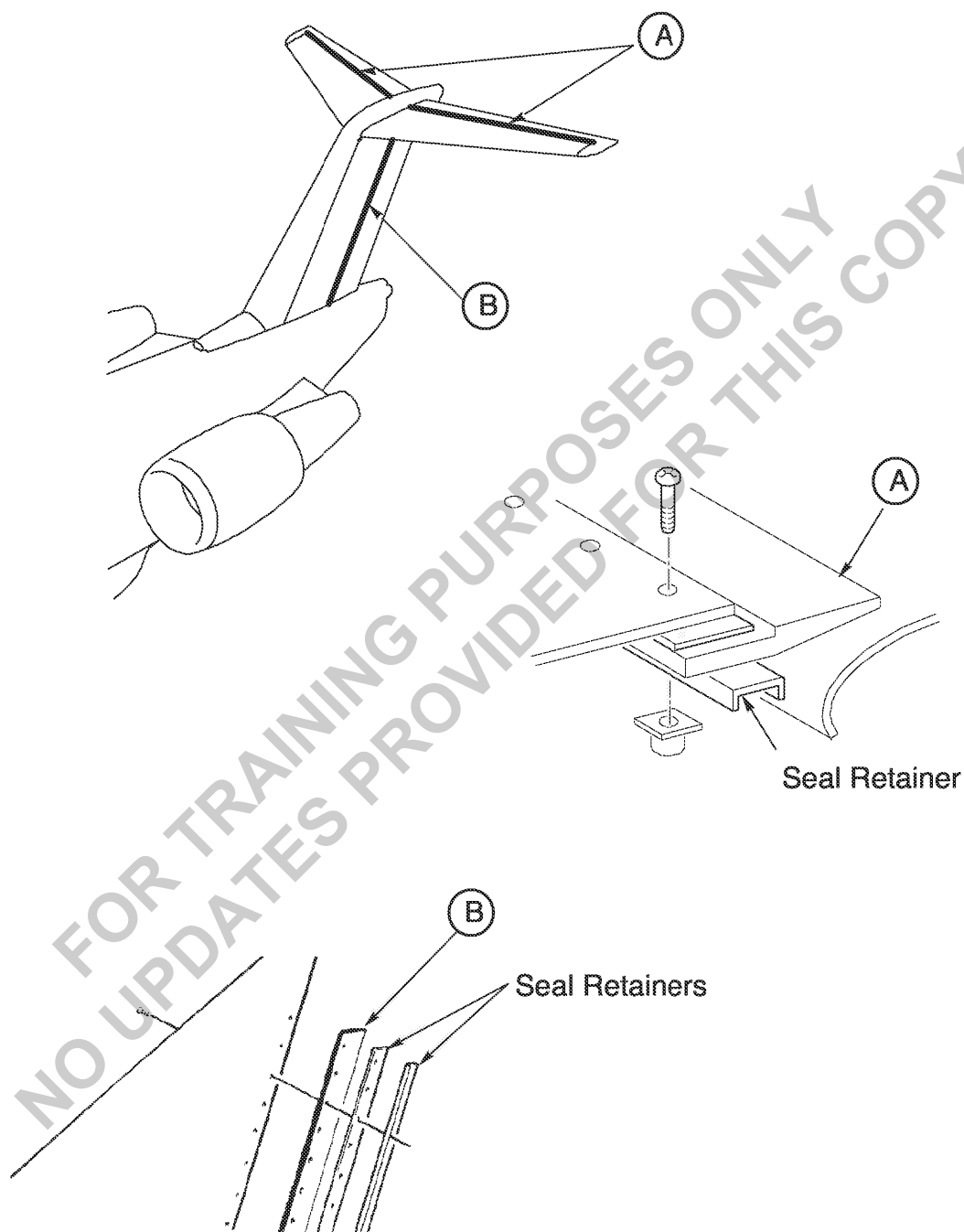
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SYSTEM 55 STABILIZERS																		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																
55–11 Seals between Elevator and Horizontal Stabilizer Upper Surface Item A	2	2. Remarks and/or Exceptions One or both seals between elevator and horizontal stabilizer upper surface may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–18 kg / seal</td><td>–40 lb / seal</td></tr></table> <table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–14 kg / seal</td><td>–30 lb / seal</td></tr></table> <table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–34 kg / seal</td><td>–75 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.35% / seal</td><td>+0.35% / seal</td></tr></table>	TAKE-OFF WEIGHT		–18 kg / seal	–40 lb / seal	LANDING WEIGHT		–14 kg / seal	–30 lb / seal	ENROUTE CLIMB		–34 kg / seal	–75 lb / seal	FUEL CONSUMPTION		+0.35% / seal	+0.35% / seal
	TAKE-OFF WEIGHT																	
–18 kg / seal	–40 lb / seal																	
LANDING WEIGHT																		
–14 kg / seal	–30 lb / seal																	
ENROUTE CLIMB																		
–34 kg / seal	–75 lb / seal																	
FUEL CONSUMPTION																		
+0.35% / seal	+0.35% / seal																	
55–31 Seals between Rudder and Vertical Stabilizer Item B	2	2. Remarks and/or Exceptions One or both seals between rudder and vertical stabilizer may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–20 kg / seal</td><td>–45 lb / seal</td></tr></table> <table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–16 kg / seal</td><td>–35 lb / seal</td></tr></table> <table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–36 kg / seal</td><td>–80 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.35% / seal</td><td>+0.35% / seal</td></tr></table>	TAKE-OFF WEIGHT		–20 kg / seal	–45 lb / seal	LANDING WEIGHT		–16 kg / seal	–35 lb / seal	ENROUTE CLIMB		–36 kg / seal	–80 lb / seal	FUEL CONSUMPTION		+0.35% / seal	+0.35% / seal
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FUEL CONSUMPTION																		
+0.35% / seal	+0.35% / seal																	

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SYSTEM 55 STABILIZERS



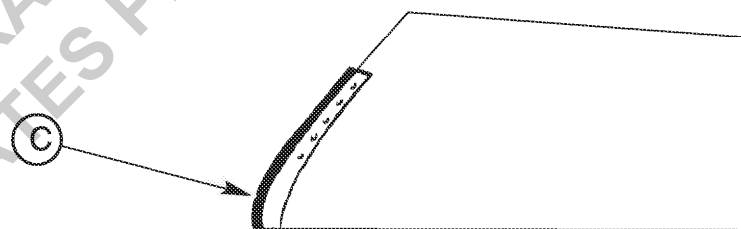
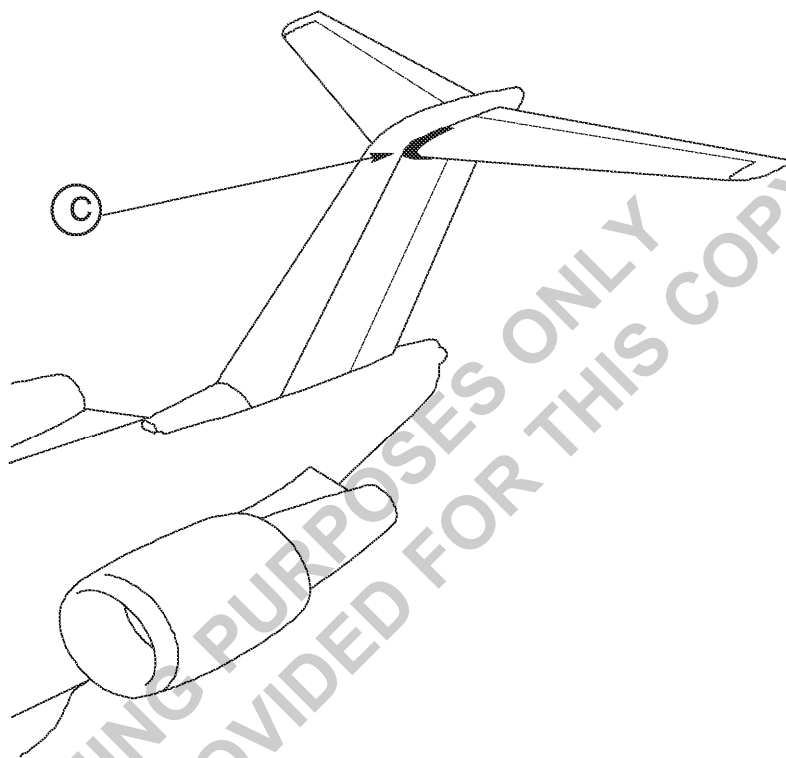
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CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-55-3
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SYSTEM55 STABILIZERS																		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																
55–32	Horizontal Stabilizer Forward Rubber Seal Item C	2 2. Remarks and/or Exceptions One or both horizontal stabilizer forward rubber seals may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–38 kg / seal</td><td>–85 lb / seal</td></tr></table><table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–29 kg / seal</td><td>–65 lb / seal</td></tr></table><table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–70 kg / seal</td><td>–155 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td>+0.75% / seal</td><td>+0.75% / seal</td></tr></table>	TAKE-OFF WEIGHT		–38 kg / seal	–85 lb / seal	LANDING WEIGHT		–29 kg / seal	–65 lb / seal	ENROUTE CLIMB		–70 kg / seal	–155 lb / seal	FUEL CONSUMPTION		+0.75% / seal	+0.75% / seal
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FUEL CONSUMPTION																		
+0.75% / seal	+0.75% / seal																	

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SYSTEM 55 STABILIZERS



Horizontal Stabilizer
Leading Edge

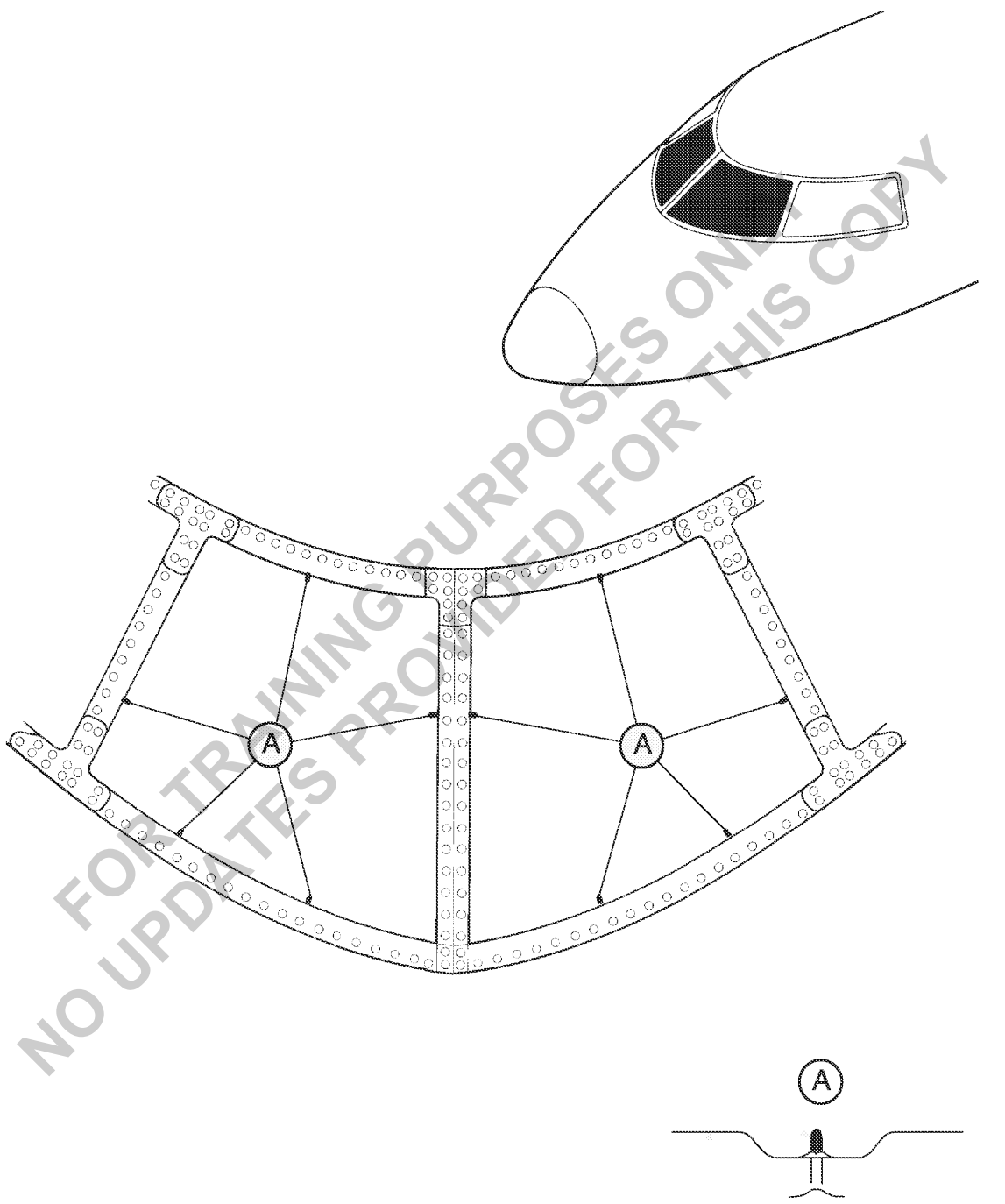
DFM0801_010

CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-56-1
		Oct 27/15

SYSTEM 56 WINDOWS		
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2
		2. Remarks and/or Exceptions
56-11 P-Static Probes. Item A		10 One probe may be missing with no performance penalty.

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SYSTEM 56 WINDOWS



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CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-57-1
		Oct 27/15

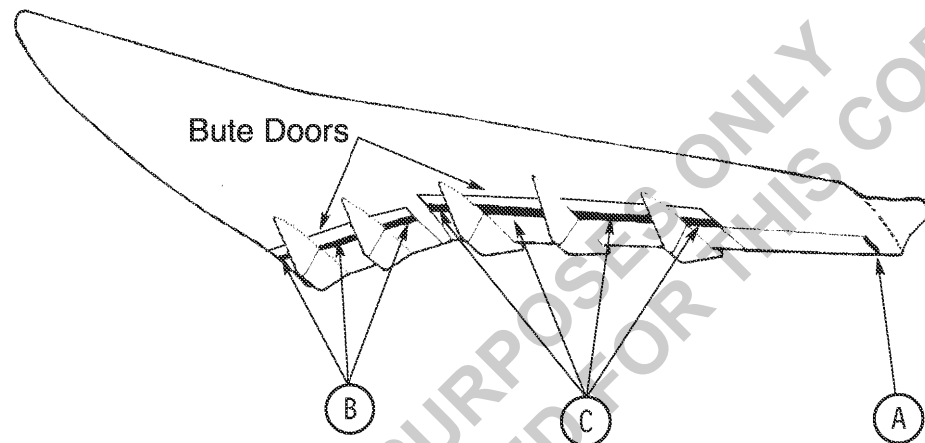
SYSTEM		57 WINGS																	
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																	
57-32 Seal between Aileron Side End and Wing Item A		2	2. Remarks and/or Exceptions One or both seals between aileron side end and wing may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–16 kg / seal</td><td>–35 lb / seal</td></tr></table> <table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–11 kg / seal</td><td>–25 lb / seal</td></tr></table> <table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–29 kg / seal</td><td>–65 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td>+0.3% / seal</td><td>+0.3% / seal</td></tr></table>	TAKE-OFF WEIGHT		–16 kg / seal	–35 lb / seal	LANDING WEIGHT		–11 kg / seal	–25 lb / seal	ENROUTE CLIMB		–29 kg / seal	–65 lb / seal	FUEL CONSUMPTION		+0.3% / seal	+0.3% / seal
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FUEL CONSUMPTION																			
+0.3% / seal	+0.3% / seal																		
57-51 Bute Door Seals – Inboard Flap Item B		6	One or any combination of bute door seals may be missing from the inboard flaps provided: – the performance and structurally limited weights are reduced by: <table><tr><th colspan="2">TAKE-OFF WEIGHT</th></tr><tr><td>–11 kg / seal</td><td>–25 lb / seal</td></tr></table> <table><tr><th colspan="2">LANDING WEIGHT</th></tr><tr><td>–9 kg / seal</td><td>–20 lb / seal</td></tr></table> <table><tr><th colspan="2">ENROUTE CLIMB</th></tr><tr><td>–20 kg / seal</td><td>–45 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><th colspan="2">FUEL CONSUMPTION</th></tr><tr><td>+0.2% / seal</td><td>+0.2% / seal</td></tr></table>	TAKE-OFF WEIGHT		–11 kg / seal	–25 lb / seal	LANDING WEIGHT		–9 kg / seal	–20 lb / seal	ENROUTE CLIMB		–20 kg / seal	–45 lb / seal	FUEL CONSUMPTION		+0.2% / seal	+0.2% / seal
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FUEL CONSUMPTION																			
+0.2% / seal	+0.2% / seal																		

CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-57-2
		Oct 27/15

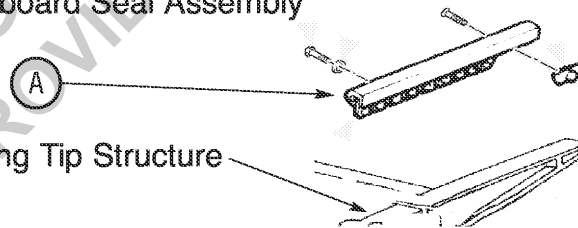
SYSTEM 57 WINGS																	
SUB SYSTEM	ITEM																
57-51 Bute Door Seals – Outboard Flap Item C	1. Number required for all flight conditions except as provided in column 2 2. Remarks and/or Exceptions 6 One or any combination of bute door seals may be missing from the outboard flaps provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–7 kg / seal</td><td>–15 lb / seal</td></tr></table><table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–5 kg / seal</td><td>–10 lb / seal</td></tr></table><table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–11 kg / seal</td><td>–25 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.1% / seal</td><td>+0.1% / seal</td></tr></table>	TAKE-OFF WEIGHT		–7 kg / seal	–15 lb / seal	LANDING WEIGHT		–5 kg / seal	–10 lb / seal	ENROUTE CLIMB		–11 kg / seal	–25 lb / seal	FUEL CONSUMPTION		+0.1% / seal	+0.1% / seal
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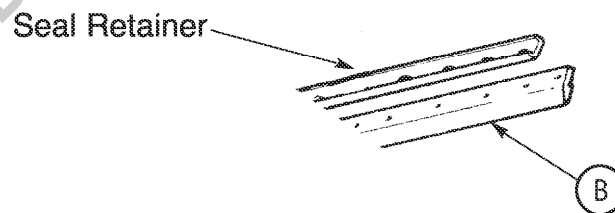
SYSTEM 57 WINGS



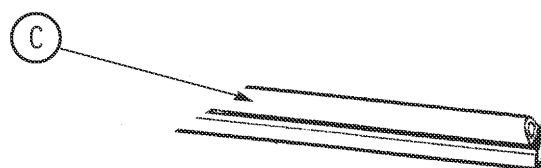
Typical Aileron Outboard Seal Assembly



Typical Inboard Flap Bute Door Seal Assembly



Typical Outboard Flap Bute Door Seal Assembly



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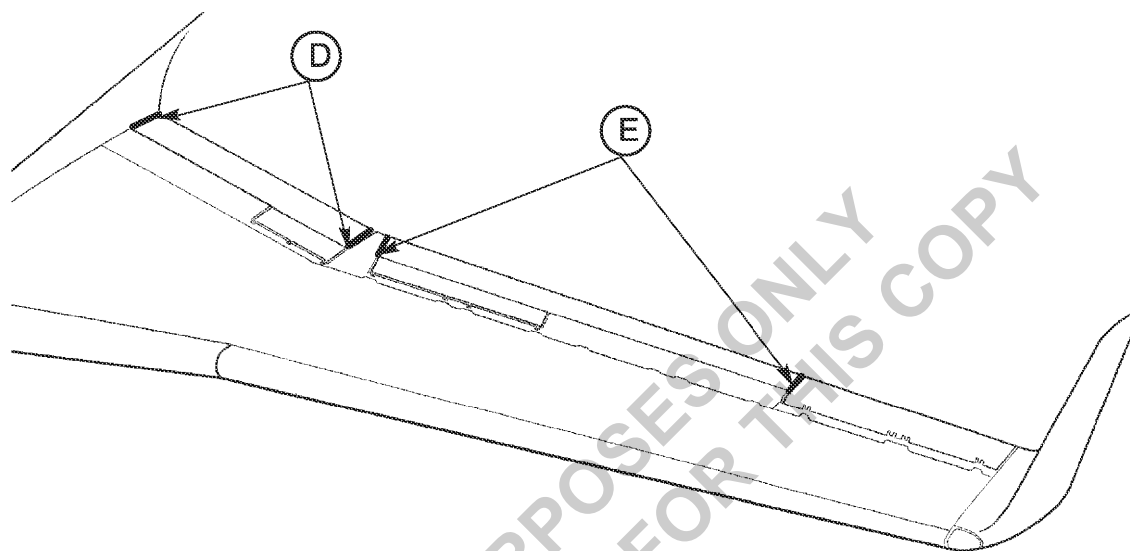
CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-57-4
		Oct 27/15

SYSTEM		57 WINGS																	
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																	
57-53 Inboard Flaps – Inboard Seals Item D		2	2. Remarks and/or Exceptions One or both seals between inboard flaps and fuselage fairing may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–39 kg / seal</td><td>–85 lb / seal</td></tr></table> <table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–29 kg / seal</td><td>–65 lb / seal</td></tr></table> <table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–70 kg / seal</td><td>–155 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.7% / seal</td><td>+0.7% / seal</td></tr></table>	TAKE-OFF WEIGHT		–39 kg / seal	–85 lb / seal	LANDING WEIGHT		–29 kg / seal	–65 lb / seal	ENROUTE CLIMB		–70 kg / seal	–155 lb / seal	FUEL CONSUMPTION		+0.7% / seal	+0.7% / seal
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57-53 Inboard Flaps – Outboard Seals Item D		2	One or both seals between inboard flaps and wing structure may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–36 kg / seal</td><td>–80 lb / seal</td></tr></table> <table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–27 kg / seal</td><td>–60 lb / seal</td></tr></table> <table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–68 kg / seal</td><td>–150 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.65% / seal</td><td>+0.65% / seal</td></tr></table>	TAKE-OFF WEIGHT		–36 kg / seal	–80 lb / seal	LANDING WEIGHT		–27 kg / seal	–60 lb / seal	ENROUTE CLIMB		–68 kg / seal	–150 lb / seal	FUEL CONSUMPTION		+0.65% / seal	+0.65% / seal
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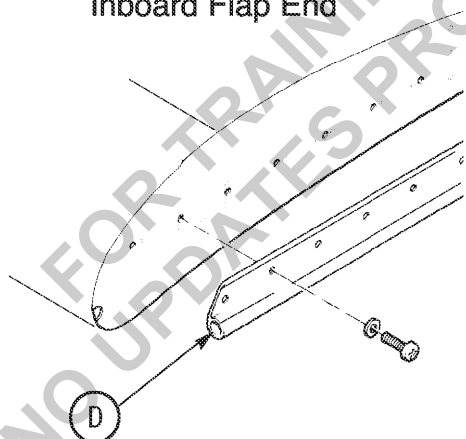
CHALLENGER 650	APPENDIX 1 Configuration Deviation List	08-01-57-5
		Oct 27/15

SYSTEM		57 WINGS																	
SUB SYSTEM	ITEM	1. Number required for all flight conditions except as provided in column 2																	
57-55 Outboard Flaps – Inboard Seals Item E		2	2. Remarks and/or Exceptions One or both seals between outboard flaps and wing structure may be missing provided: – the performance and structurally limited weights are reduced by: <table><tr><td colspan="2">TAKE-OFF WEIGHT</td></tr><tr><td>–30 kg / seal</td><td>–65 lb / seal</td></tr></table> <table><tr><td colspan="2">LANDING WEIGHT</td></tr><tr><td>–23 kg / seal</td><td>–50 lb / seal</td></tr></table> <table><tr><td colspan="2">ENROUTE CLIMB</td></tr><tr><td>–57 kg / seal</td><td>–125 lb / seal</td></tr></table> – the mission fuel requirements are increased by: <table><tr><td colspan="2">FUEL CONSUMPTION</td></tr><tr><td>+0.55% / seal</td><td>+0.55% / seal</td></tr></table>	TAKE-OFF WEIGHT		–30 kg / seal	–65 lb / seal	LANDING WEIGHT		–23 kg / seal	–50 lb / seal	ENROUTE CLIMB		–57 kg / seal	–125 lb / seal	FUEL CONSUMPTION		+0.55% / seal	+0.55% / seal
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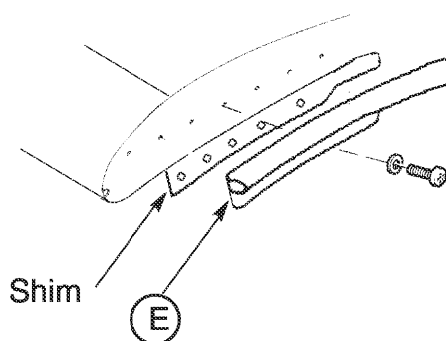
SYSTEM 57 WINGS



Inboard Flap End



Outboard Flap End



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